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Stefan Resch

User-Centered Design and Evaluation of Smart Devices for Foot Augmentation:

Interactive Mixed Reality Training and Feedback Applications

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User-Centered Design and Evaluation of Smart Devices for Foot Augmentation:
Interactive Mixed Reality Training and Feedback Applications

Author
Stefan Resch

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Approval by the Supervisors

Daniel Sánchez Morillo (Supervisor), Professor in the Department of Automation, Electronics, Architecture, and Computer Networks at the University of Cádiz, and Diana Völz (Co-Supervisor), Professor in the Department of Computer Science and Engineering at the Frankfurt University of Applied Sciences, in our capacity as supervisors of the Doctoral Thesis presented by Mr. Stefan Resch, entitled:

"User-Centered Design and Evaluation of Smart Devices for Foot Augmentation: Interactive Mixed Reality Training and Feedback Applications"

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Cádiz / Frankfurt am Main, April 07, 2026

Daniel Sánchez Morillo

D. Völz
Diana Völz

Author's Personal Statement

I, Stefan Resch, PhD candidate in the Doctoral Program in Manufacturing, Materials and Environmental Engineering at the University of Cádiz, in cooperation with the Frankfurt University of Applied Sciences, hereby declare that this dissertation, entitled:

"User-Centered Design and Evaluation of Smart Devices for Foot Augmentation: Interactive Mixed Reality Training and Feedback Applications"

constitutes my own original work. All sources of information and data have been properly acknowledged in the text and in the bibliography. This dissertation, in its present form, has not been submitted for the award of a degree or qualification at this or any other university. Parts of the work have been published in scientific articles, which are appropriately cited in the dissertation. I am fully aware that failure to comply with these standards of academic integrity may result in academic sanctions, without prejudice to any further actions that might be taken.

Frankfurt am Main, April 07, 2026

Stefan Resch

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"Learning is an experience. Everything else is just information."

— Albert Einstein —

Abstract

The global demand for assistive technologies, including orthotic devices such as foot orthoses and walking aids, is continuously increasing. While ankle-foot orthoses are essential for patients with foot injuries and impaired gait, these devices remain largely standardized and lack integrated sensing and feedback capabilities. As a result, complications may remain undetected, potentially prolonging recovery and impairing rehabilitation. Beyond these technological limitations, research has shown that users perceive the design of orthoses as a critical factor, which can lead to reduced acceptability. This may limit consistent device use, potentially compromising the healing process and reducing overall health benefits, underscoring the importance of actively involving users in the design process.

Recent developments in sensorized footwear, such as smart insoles with feedback capabilities, demonstrate potential in various areas. However, fully integrated solutions within standardized controlled ankle motion boots that support continuous monitoring, adaptive feedback, and cross-device communication are still limited. From an application perspective, first-time interaction with smart orthotic systems can be challenging. In this context, extended reality technologies offer promising interactive solutions for training and device interaction, but their integration with smart orthotic devices and their effects on user experience are not yet well understood. Moreover, while user-centered design approaches and generative artificial intelligence show promise for engaging patients in the design process, their effects on device acceptance and practical implementation in the orthotic footwear domain remain largely unknown. A better understanding of these aspects is necessary to guide the design of future interactions with smart orthotic devices and to overcome current limitations in adequately addressing user needs.

Addressing these gaps, this dissertation adopts a multidisciplinary perspective, focusing on technical smart device development, interactive virtual training applications, and user-centered design incorporating generative tools. The central research question is how a smart foot orthosis can be designed to enable active monitoring, adaptive feedback, and multi-device communication, while systematically involving users in the design process and leveraging extended reality technologies for intuitive interaction. This cumulative thesis is based on six publications, addressing open gaps in the areas of *Smart Devices*, *Extended Reality*, and *Artificial Intelligence in Design*.

A scoping review identified existing solutions, research gaps, and future opportunities in intelligent walking aids. Building on this, a comprehensive orthotic system was developed and empirically evaluated following an iterative user-centered design process. The system combines a smart foot orthosis, a smart walking aid, and a smartphone application, enabling gait and plantar pressure monitoring as well as haptic feedback across devices. Feasibility, usability, and feedback perception were validated, demonstrating the system's potential. Beyond technical development, immersive training scenarios were investigated in two user studies. The first study examined mixed reality avatar-based voice and gesture modalities for onboarding training using the developed smartphone app. The second study explored immersive gait training in mixed and virtual reality environments, using different visualizations of foot pressure while walking with the smart orthosis. Both studies provide new insights into immersive feedback interaction and offer concrete design recommendations for smart device training. Subsequently, the focus shifted to product design and user perception. Two empirical studies explored the impact of generative design as a supportive tool for orthotic footwear concept creation, highlighting effects on user acceptance. Based on these findings, a framework for systematically integrating generative tools into the design process was proposed to address both aesthetic and technical requirements and validated in a case study with user and expert feedback.

Overall, this thesis advances the field of human-computer interaction by providing empirical evidence of haptic feedback perception in smart orthotic devices, demonstrating the potential of immersive environments and feedback modalities for smart device interaction, and understanding the impact of designs created with generative tools. The findings in this thesis provide new insights into the design, evaluation, and user interaction of smart devices, contributing a validated integrated smart foot orthosis and a framework for user-centered design with generative artificial intelligence. Together, these contributions provide a deeper understanding of next-generation smart assistive devices from both technical and user-centered perspectives. It serves as a foundation for future research and offers guidance to researchers and practitioners in the design and implementation of intelligent orthotic systems.

Resumen

La demanda global de tecnologías de asistencia, que incluyen dispositivos ortésicos tales como ortesis de pie y auxiliares de marcha, experimenta un crecimiento constante. Si bien las ortesis tobillo-pie resultan esenciales para pacientes con lesiones podológicas y alteraciones de la marcha, estos dispositivos mantienen un carácter predominantemente estandarizado y carecen de capacidades integradas de detección y retroalimentación. Como resultado, diversas complicaciones pueden pasar inadvertidas, lo que podría prolongar el periodo de recuperación y comprometer la rehabilitación. Más allá de estas limitaciones tecnológicas, la investigación ha demostrado que los usuarios perciben el diseño de las ortesis como un factor crítico, cuya deficiencia puede derivar en una menor aceptación. Este fenómeno puede restringir el uso constante del dispositivo, lo que podría comprometer el proceso de curación y reducir los beneficios generales para la salud, subrayando la importancia de integrar activamente a los usuarios en el proceso de diseño.

Los avances recientes en calzado sensorizado, tales como las plantillas inteligentes con capacidades de retroalimentación, demuestran un potencial significativo en diversas áreas. No obstante, las soluciones plenamente integradas en botas de control de movimiento del tobillo (boots for controlled ankle motion) estandarizadas, que permitan la monitorización continua, la retroalimentación adaptativa y la comunicación entre dispositivos, siguen siendo limitadas. Desde una perspectiva aplicada, la interacción inicial con sistemas ortésicos inteligentes puede resultar compleja. En este contexto, las tecnologías de realidad extendida ofrecen soluciones interactivas alentadoras para la capacitación y el manejo de los dispositivos; sin embargo, su integración con sistemas ortésicos inteligentes y sus efectos sobre la experiencia del usuario todavía no se comprenden completamente. Asimismo, si bien los enfoques de diseño centrado en el usuario y la inteligencia artificial generativa resultan prometedores para involucrar a los pacientes en el proceso de diseño, su impacto en la aceptación del dispositivo y su implementación práctica en el ámbito del calzado ortésico aún se desconoce en gran medida. Una mejor comprensión de estos aspectos es necesaria para orientar el diseño de futuras interacciones con dispositivos ortésicos inteligentes y para superar las limitaciones actuales en la adecuada satisfacción de las necesidades de los usuarios.

Para abordar estas brechas, la presente tesis adopta una perspectiva multidisciplinaria, centrándose en el desarrollo técnico de dispositivos inteligentes, el uso de aplicaciones de entrenamiento virtual interactivo y el diseño centrado en el usuario mediante la incorporación de herramientas generativas. La pregunta de investigación central consiste en determinar cómo puede diseñarse una ortesis de pie inteligente que permita la monitorización activa, la retroalimentación adaptativa y la comunicación multidispositivo, involucrando sistemáticamente a los usuarios en el proceso de diseño y aprovechando las tecnologías de realidad extendida para una interacción intuitiva. Esta tesis por compendio de publicaciones se fundamenta en seis trabajos que abordan vacíos de conocimiento en las áreas de *Dispositivos Inteligentes*, *Realidad Extendida* y *Inteligencia Artificial Aplicada al Diseño*.

Una revisión de alcance (scoping review) permitió identificar las soluciones existentes, las brechas en la investigación y las oportunidades futuras en el ámbito de los auxiliares de marcha inteligentes. Sobre esta base, se desarrolló y evaluó empíricamente un sistema ortésico integral, siguiendo un proceso iterativo de diseño centrado en el usuario. Dicho sistema combina una ortesis inteligente para el pie, un auxiliar de marcha inteligente y una aplicación móvil, permitiendo la monitorización de la marcha y la presión plantar, así como la retroalimentación háptica entre dispositivos. Se validaron la viabilidad, la usabilidad y la percepción de la retroalimentación, demostrando el potencial del sistema. Más allá del desarrollo técnico, se investigaron escenarios de entrenamiento inmersivo en dos estudios de usuarios. El primero examinó modalidades de voz y gestos basadas en avatares de realidad mixta para la capacitación inicial mediante la aplicación móvil desarrollada. El segundo estudio exploró el entrenamiento de la marcha inmersivo en entornos de realidad mixta y virtual, utilizando diferentes visualizaciones de la presión del pie al caminar con la ortesis inteligente. Ambos estudios aportan nuevas perspectivas sobre la interacción con retroalimentación inmersiva y ofrecen recomendaciones de diseño concretas para la capacitación en el uso de dispositivos inteligentes. Posteriormente, el enfoque se desplazó hacia el diseño de producto y la percepción del usuario. Dos estudios empíricos exploraron el impacto del diseño generativo como herramienta de apoyo para la creación de conceptos de calzado ortésico, destacando sus efectos en la aceptación por parte del usuario. A partir de estos hallazgos, se propuso un marco de trabajo para la integración sistemática de herramientas generativas en el proceso de diseño, con el fin de abordar requisitos tanto estéticos como técnicos, el cual fue validado mediante un estudio de caso con retroalimentación de usuarios y expertos.

En conjunto, esta tesis hace avanzar el campo de la interacción persona-computadora (human-computer interaction) al proporcionar evidencia empírica sobre la percepción de la retroalimentación háptica en dispositivos ortésicos inteligentes, demostrar el potencial de los entornos inmersivos y las modalidades de retroalimentación para la interacción con dispositivos inteligentes y comprender el impacto de los diseños creados con herramientas generativas. Los hallazgos de esta tesis aportan nuevas perspectivas sobre el diseño, la evaluación y la interacción del usuario con dispositivos inteligentes, y contribuyen con una ortesis de pie inteligente integrada ya validada, así como con un marco de trabajo para el diseño centrado en el usuario mediante inteligencia artificial generativa. En su totalidad, estas contribuciones permiten una comprensión más profunda de los dispositivos de asistencia inteligentes de próxima generación, tanto desde una perspectiva técnica como centrada en el usuario. Este trabajo sirve como base para investigaciones futuras y ofrece una guía para investigadores y profesionales en el diseño e implementación de sistemas ortésicos inteligentes.

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List of Abbreviations

ABS	Acrylonitrile Butadiene Styrene
AE	Aesthetic Appeal
AFO	Ankle-Foot Orthosis
AI	Artificial Intelligence
ANN	Artificial Neural Network
ANOVA	Analysis of Variance
APK	Android Package Kit
AR	Augmented Reality
ART	Aligned Rank Transform
ATT	Attractiveness
BLE	Bluetooth Low Energy
bpm	Beats Per Minute
CAD	Computer-Aided Design
CoP	Center of Pressure
DFS	Diabetic Foot Syndrome
DL	Deep Learning
DOF	Degrees of Freedom
DRC	Design Rule Checks
ECAD	Electronic Computer-Aided Design
ECG	Electrocardiogram
EMG	Electromyography
ERC	Electrical Rule Check
ERM	Eccentric Rotating Mass Motor
FA	Focused Attention
FFF	Fused Filament Fabrication
FSM	Finite-State Machine
FSR	Force Sensing Resistor
GAN	Generative Adversarial Networks
GPS	Global Positioning System
GPT	Generative Pre-Trained Transformer
GUI	Graphical User Interface
HCI	Human-Computer Interaction
HMD	Head-Mounted Display
HMM	Hidden Markov Model
HQ	Hedonic Quality
HQ-I	Hedonic Quality Identity
HQ-S	Hedonic Quality Stimulation
IC	Integrated Circuit
IDE	Integrated Development Environment
IMU	Inertial Measurement Unit
IoT	Internet of Things

IR	Infrared
ISO	International Organization for Standardization
I²C	Inter-Integrated Circuit Communication
kNN	K-Nearest Neighbors
LCD	Liquid Crystal Display
LDR	Light-Dependent Resistor
LED	Light-Emitting Diode
LEQ	Learning Experience Questionnaire
LLM	Large Language Model
LRA	Linear Resonant Actuator
LSTM	Long Short-Term Memory
MCU	Microcontroller Unit
mHealth	Mobile Health
ML	Machine Learning
MoCap	Motion Capture System
MR	Mixed Reality
MRQ	Main Research Question
PCB	Printed Circuit Board
PLA	Polylactic Acid
PPG	Photoplethysmogram
PQ	Pragmatic Quality
PRISMA-ScR	Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews
PU	Perceived Usability
RF	Random Forest
RFID	Radio-Frequency Identification
RM	Repeated-Measures
RQ	Research Question
RTLX	Raw Task Load Index
RW	Reward Factor
SCM	Stereotype Content Model
SDK	Software Development Kit
SFO	Smart Foot Orthosis
SMD	Surface-Mounted Device
SRQ	Specific Research Question
STL	Stereolithography
SUS	System Usability Scale
SVM	Support Vector Machine
TAM	Technology Acceptance Model
TEQ	Tutorial Experience Questionnaire
THT	Through-Hole Technology
TPU	Thermoplastic Polyurethane
UART	Universal Asynchronous Receiver Transmitter
UES-SF	User Engagement Scale – Short Form
UI	User Interface
UTAUT	Unified Theory for Acceptance and Use of Technology
UX	User Experience

VDI	The Association of German Engineers
VR	Virtual Reality
WEAR	Wearable Acceptability Range
WHO	World Health Organization
XR	Extended Reality

CHAPTER 1

Introduction and Rationale

This chapter provides the foundation for the thesis by outlining the motivation, research questions, and objectives. It also describes the methodology and evaluation approach, presents the key contributions and publications, and concludes with an overview of the thesis structure.

1.1. Motivation

Assistive technologies play a crucial role in supporting mobility and rehabilitation. Worldwide, a growing number of patients rely on orthotic walking aids to compensate for impaired gait and reduced sensory feedback [6]. In particular, individuals with foot disorders caused by acute injuries or chronic diseases (such as neurological impairments and musculoskeletal conditions), are highly dependent on external device support. However, most commercially available controlled ankle motion boots (in this thesis referred to as "foot orthoses") remain standardized and offer only limited customization to individual biomechanics. This lack of adaptability can have serious consequences. For example, improper foot alignment or pressure overload may remain unnoticed in the absence of regular expert observation, which can contribute to secondary injuries and delayed rehabilitation. Furthermore, passive controlled ankle-foot orthoses can alter natural gait patterns, such as shifting hip and knee movements, which may lead to secondary complications during prolonged use [7]. In addition, these devices typically provide no active user feedback, so that wearers receive no information about their gait or foot health. In particular, for patients with impaired peripheral sensation (e.g., diabetic neuropathy), where excessive pressure, deformities, or open wounds may go unnoticed and lead to severe complications. These challenges highlight a clear need for assistive solutions that move beyond passive support and actively engage users in their rehabilitation process.

In this context, research has explored sensorized assistive technologies to address these limitations, covering a wide range of applications. These sensorized technologies (in this thesis referred to as "smart", "intelligent", or "instrumented" systems) enable real-time measurement, data processing, and wireless communication across devices, often embedded within Internet of Things (IoT) networks. Such systems are increasingly applied in human augmentation, demonstrating benefits in sports, rehabilitation, and healthcare [8]. This trend is reflected in the widespread adoption of wearable devices [9], such as smartwatches and activity trackers, which use embedded sensors to capture physiological signals and support continuous health monitoring, either directly or through connected devices (e.g., smartphones) [10]. Beyond passive sensing, many systems incorporate feedback mechanisms that provide users with immediate information through different output modalities such as auditory, visual, or haptic cues [11]. In physiological and biomechanical monitoring contexts, this approach is commonly referred to as biofeedback, where externally presented signals are derived from internally measured states [12]. In physical rehabilitation, biomechanical feedback can be used based on force, movement, or postural control measurements [12], which are relevant for human gait. For example, in the context of foot-worn health monitoring devices (e.g., smart socks, insoles, and shoes), various sensor modalities are integrated (e.g., accelerometer, gyroscope, pressure), depending on the application context [13, 8]. Building on this, the use of embedded sensors in orthopedic footwear shows high potential for data collection, supporting clinicians by enabling monitoring of foot health and providing feedback [14].

However, while several research prototypes have incorporated sensorized systems into ankle–foot orthoses [15, 16], fully developed orthotic gait systems with interactive feedback remain limited. In addition to this research gap, a literature review by Subramaniam et al. [13] highlighted current research challenges in foot health monitoring using smart insoles, such as sensor placement, system integration and validation, user comfort, usability, inter-system consistency. This emphasizes critical gaps that should be addressed in future development to overcome current limitations. Consequently, it remains unclear how a comprehensive smart orthosis should be designed to capture gait-relevant metrics, provide meaningful feedback, and interact effectively with companion devices, leading to the first research gap.

From an application perspective, Extended Reality (XR) technologies using touch-free interfaces and 3D visualizations offer new opportunities to address challenges in user–device interaction within healthcare contexts [17]. In this context, Augmented Reality (AR) and Virtual Reality (VR) environments enable interactive settings, with increasing exploration of immersive feedback applications (e.g., using virtual avatars) to support human gait [18] or balance training [19]. Furthermore, early-stage testing of device interactions in virtual training scenarios can provide insights into how users engage with a system and how feedback modalities affect task performance. XR encompasses a wide range of AR and VR applications, often in combination with wearable devices and multimodal feedback capabilities [20]. For example, smart insoles can augment foot pressure information and provide visual feedback during gait rehabilitation, for instance via floor projections from a projector mounted on a walking aid [21]. Beyond AR setups, fully immersive VR enables training in realistic scenarios, in which users can be confronted with everyday challenges under controlled conditions. This offers substantial potential for testing smart devices and understanding their impact on user behavior and performance early in the development process. In particular, issues such as expert dependency or incorrect device usage due to missing feedback could be directly addressed through interactive VR-based training applications. However, despite this technological potential, research on lower-limb applications remains significantly underrepresented in the field of rehabilitation using AR and VR technologies [22]. This reveals a second research gap, as the effects of immersive, feedback-based training applications on user interaction and performance with smart orthoses and companion systems remain unknown.

Beyond technical feasibility, related work identifies device acceptance as a critical barrier to successful adoption of orthotic footwear [23]. It has been shown that wearable devices attached to the ankle or foot are often perceived as less socially acceptable than those worn on other body locations, possibly due to their negative associations with medical or surveillance devices commonly worn at the ankle [24]. Similar challenges have been reported for conventional orthotic designs, where aesthetic shortcomings negatively affect wearing compliance and long-term use [25]. In addition, prior research emphasizes the need to improve overall user satisfaction and to actively involve end users in the design process of foot orthoses [26]. Recent advancements highlight the benefits of a user-centered design framework for personalized ankle-foot orthoses to ensure emotional well-being in combination with technical functionality [27]. Furthermore, the emerging trend of using generative Artificial Intelligence (AI) tools for rapid user-based image generation shows potential for footwear design [28, 29] and hand orthoses prototyping [30]. However, a systematic approach for incorporating AI into the design process of orthotic footwear is still missing. Together, these findings reveal a third research gap: the need for user-centered design approaches that systematically integrate user requirements, aesthetic considerations, and acceptance-related factors into the development of foot orthoses.

Bridging these gaps requires a comprehensive perspective that combines smart sensing technologies, immersive applications, and user-centered design approaches. In particular, it remains unclear how sensorized orthotic systems should be designed to provide effective interactive feedback. It is further unclear how XR environments influence user interaction with such systems and how device acceptance and aesthetic aspects can be systematically integrated into the development process.

Therefore, three main objectives were defined in the areas of technical development *(Smart Devices)*, application-oriented investigation *(Extended Reality)*, and user-centered product design *(AI in Design)*. An overview of these three themes is shown in Figure 1.1. Together, they form the overarching framework of this research, operationalized through seven Research Questions (RQs). Established HCI research methods were applied using a mixed-methods design to capture both objective performance measures and subjective user experience. Based on six published research articles, this cumulative dissertation comprises primary and secondary contributions, including empirical findings from user studies, the development and evaluation of functional research artifacts, as well as theoretical, methodological, and survey-based insights. In summary, this thesis aims to advance the field of HCI by:

1. Developing, implementing, and evaluating a comprehensive smart orthotic system, with a focus on feedback mechanisms, usability, and user interaction.

2. Evaluating immersive XR environments for training and interaction with smart devices, including a smartphone application and a smart foot orthosis.

3. Investigating design factors and acceptance-related perceptions of orthotic footwear using AI-supported generative design approaches.

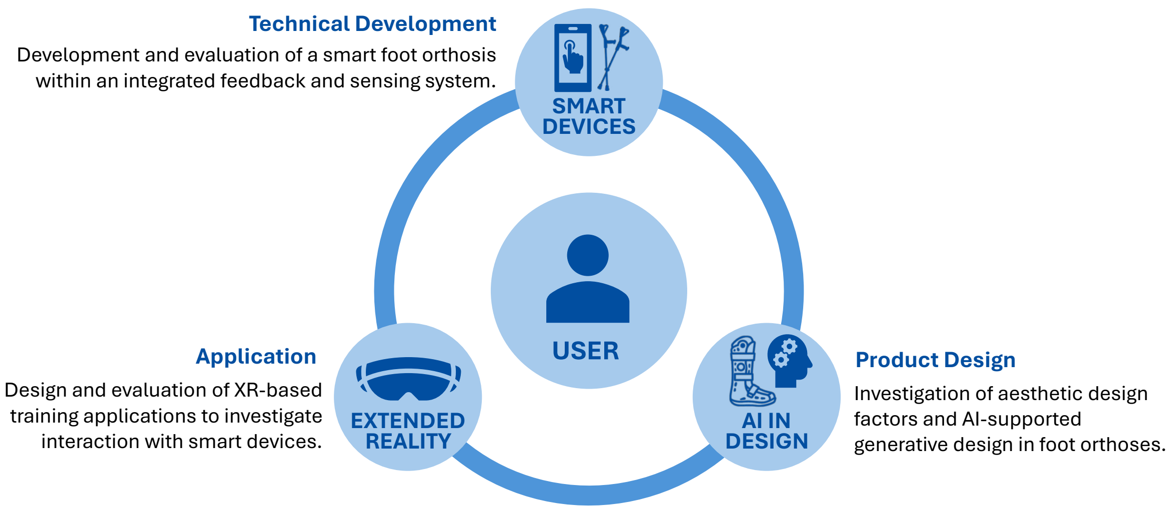


Figure 1.1: Schematic overview showing the three main themes of this thesis.

1.2. Research Questions

The identified research gaps were addressed by seven main RQs, which contribute to the defined research objectives in the three areas *Smart Devices*, *Extended Reality*, and *AI in Design*. The RQs are grouped according to these thematic clusters, which provide a coherent framework guiding the entire thesis.

Smart Devices explores the development and evaluation of sensorized technologies to support mobility. The corresponding RQs focus on identifying current trends and gaps in the field and exploring technological opportunities. Building on these insights, a new system was developed and tested to understand how users interact with such devices, including the effects of feedback modalities and system usability. This thematic cluster captures the state of the art and provides empirical evidence on how feedback integration affects smart device use, demonstrating how these devices can be designed to enhance user performance and acceptance.

Extended Reality addresses the application of VR and Mixed Reality (MR) training scenarios to examine user interaction and performance with smart devices in immersive settings. Based on the previously developed smart devices, this cluster provides new insights into how immersive environments and feedback modalities influence both objective and subjective outcomes, addressing empirical gaps in practical understanding of user adaptation and engagement in virtual training contexts. The results inform the design of future immersive training applications and demonstrate how virtual environments can be leveraged to enhance interaction with smart devices.

AI in Design investigates the role of AI in user-centered design processes for personalized orthotic devices. The RQs focus first on understanding how AI can influence product outcomes, including perceived social acceptability, and second on how AI can be integrated into a design framework to guide the development of functional and socially accepted devices. This cluster provides novel contributions by demonstrating the impact of AI on product perception and acceptance, as well as its application within the design process to support user-centered development.

An overview of the RQs is presented in Table 1.1. By addressing these questions, the thesis identifies key factors for the successful user-centered development and application of smart devices, ensuring both effective performance and positive user experience. It further derives theoretical and practical implications to inform the future development and evaluation of such technologies.

Table 1.1: Overview of the main research questions addressed in this thesis, together with their contributing topics and the chapters in which they are discussed.

Topic	**No.**	**Research Question**	**Chapter(s)**
Smart Devices	RQ1	What are the current technological trends, target user groups, and functional capabilities of smart walking aids, and which gaps remain in existing solutions that need to be addressed to improve mobility support and user-centered design?	No. 4.1
	RQ2	How can a smart orthosis and walking aid be developed as a comprehensive system, to ensure feasible sensor data acquisition and effective user feedback, while also enabling user-centered mobile application usability?	No. 4.2
	RQ3	How do different types of haptic feedback delivered via a smart orthosis and walking aid influence user perception?	No. 4.2
Extended Reality	RQ4	How can avatar-based feedback modalities in MR enhance usability and learning experience during smartphone app onboarding for a smart orthosis system?	No. 5.1
	RQ5	How do MR and VR environmental factors and visual foot pressure feedback influence objective gait parameters and subjective experience when walking with a smart foot orthosis?	No. 5.2
AI in Design	RQ6	How can AI-supported design approaches influence the perceived social acceptability of orthopedic footwear?	No. 6.1
	RQ7	How can AI be incorporated into a user-centered design framework to support the development of foot orthoses that address functional and aesthetic requirements?	No. 6.2

1.3. Methodology and Evaluation

This thesis combines primary and secondary research approaches. To obtain a comprehensive overview of the state of the art, a systematic literature review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) methodology. Building on these insights, functional prototypes of a smart foot orthosis, a sensorized walking aid, and an accompanying smartphone application were developed. The iterative development process followed a user-centered design approach: user requirements were derived through focus groups, followed by iterative usability testing and a structured prototyping and refinement process.

The primary research consists of prototype development and empirical evaluation through controlled laboratory studies and online surveys. Experimental user studies were designed using a mixed-method approach, integrating quantitative data with qualitative insights. All laboratory-based studies were conducted in the Mixed Reality Laboratory of the Frankfurt University of Applied Sciences under controlled conditions. Quantitative data were collected using objective measurement systems (e.g., motion capture, pulse sensors, instrumented treadmill data) as well as subjective assessments obtained through standardized questionnaires. These quantitative datasets were analyzed using descriptive and inferential statistical methods. Qualitative data were collected through semi-structured interviews and complementary open-ended feedback formats. These datasets were analyzed using thematic analysis following established procedures in the literature, as described in the corresponding chapters.

The studies received ethical clearance from the German Society for Nursing Science (No. 23-027). The approval document is provided in Appendix E. This research was funded by the Hessian Ministry of Science and Research, Arts and Culture (FL1, Mittelbau).

1.4. Research Contributions

The research presented in this thesis contributes to the broader field of HCI. Following the research contribution types in HCI proposed by Wobbrock and Kientz [31], the thesis makes contributions across multiple categories, which are described in the following. Table 1.2 provides a systematic overview of the contributions, grouped according to the three main themes: *Smart Devices*, *Extended Reality*, and *AI in Design*.

Empirical contribution: This type contributes to new knowledge through empirical studies based on quantitative and qualitative data. In the context of *Smart Devices* (Chapter 4.2), the developed system was validated for sensor data accuracy and tested for feedback perception and usability, providing new empirical insights on both measurement reliability and user experience with multimodal feedback. For *Extended Reality* applications (Chapter 5), two laboratory-based experimental user studies were conducted using a mixed-method approach. These studies investigated the effects of visual feedback and immersive environmental factors on gait, as well as the effectiveness of avatar-based guidance for interacting with smart devices and companion applications. In *AI in Design* (Chapter 6), two online surveys with a total of 134 participants contributed to understanding the impact of generative AI on orthotic design perception and social acceptability. Additionally, a case study evaluated a newly proposed AI-driven design framework from end-user and expert perspectives, highlighting its potential, limitations, and practical implications for integrating AI into the design process.

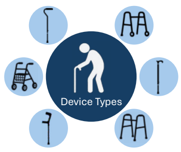
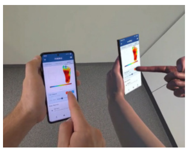
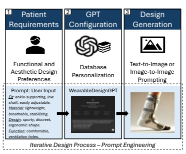
Artifact contribution: This thesis also provides technical contributions through the development of new prototypes and systems, which were tested in the context of empirical studies. In *Smart Device* development (Chapter 4.2), a complete system comprising hardware components and software architecture was implemented, enabling new explorations and insights. These systems were further extended and applied in *Extended Reality* applications (Chapter 5), where virtual training scenarios were developed, including novel visualizations and interaction techniques, to evaluate the smart devices in immersive settings.

Theoretical contribution: The thesis contributes theoretical knowledge through abstraction and formalization of design guidelines as final study outcomes. In *AI in Design* (Chapter 6.2), a novel framework was proposed, serving as a central theoretical contribution by providing a structured concept for user-centered AI-assisted design of orthoses.

Methodological contribution: Furthermore, this proposed framework additionally provides a methodological contribution through the operationalization of an eight-step process, aimed to enable reproducibility and guide the future design process. The approach was validated through an empirical case study, demonstrating its applicability as a method for integrating AI into user-centered product development.

Survey contribution: In *Smart Devices*, a systematic literature review was conducted following the PRISMA-ScR methodology (Chapter 4.1). By synthesizing the literature to identify trends, gaps, and current challenges in smart walking aids, this contribution provides a comprehensive overview, informs future developments, and highlights gaps for further research.

Table 1.2: Contributions grouped by theme, with description, method, and chapter.

	Study	Description	Type	Chapter
Smart Devices		A scoping review of 35 articles mapped sensor-based walking aids. The study highlights research gaps in clinical evaluation, sensor placement, and feedback modalities, and suggests directions for future user-centered development.	• Survey (Scoping Review)	No. 4.1
		A smart foot orthosis with an instrumented crutch was developed to provide real-time vibrotactile feedback. The system was validated for mobile gait assessment, usability, and device-specific differences in haptic feedback perception.	• Artifact (Prototype) • Empirical (User Study)	No. 4.2
Extended Reality		This study examined how multimodal avatars in MR guide users through a Mobile Health (mHealth) app for smart insoles. Voice and gestures improved tutorial guidance, usability, and learning, providing insights for MR-based onboarding.	• Artifact (Prototype) • Empirical (User Study)	No. 5.1
		Visual feedback from a smart insole was tested during treadmill walking in VR/MR. Results showed environment- and feedback-dependent effects on gait, engagement, and workload, informing design of immersive gait training.	• Artifact (Prototype) • Empirical (User Study)	No. 5.2
AI in Design		Two mixed-methods studies explored AI-generated foot orthosis designs and their social acceptability. AI designs, customized AI models, and targeted prompts significantly enhanced user acceptance.	• Empirical (User Study)	No. 6.1
		A user-centered framework integrating generative AI was applied to develop a personalized foot orthosis. The prototype was designed, fabricated, and evaluated, demonstrating the feasibility of AI-assisted workflows.	• Theoretical (Framework) • Method (Process) • Empirical (Case Study)	No. 6.2

In accordance with the CRediT contributor roles taxonomy [32], the thesis author initiated the presented work and was responsible for overall project administration. Personal contribution to the presented work: *Conceptualization, data curation, formal analysis, investigation, methodology, resources, software, validation, visualization, writing–original draft, and writing–review & editing.* Additional contributions by co-authors, including student theses and team projects, supported the roles *software* and *investigation* for the respective articles. These contributions were carried out under the supervision and coordination of the thesis author.

1.5. Publications

Several findings from the doctoral research have already been presented in peer-reviewed journals, as well as at national and international conferences. The publications presented in this thesis were conducted both independently and in collaboration with fellow researchers and students. Throughout these articles, the term "we" refers to the respective author teams, while the contributions of the thesis author are detailed in Section 1.4. The co-authors' consent for the included articles is listed in Appendix F.

The list of published articles is documented below:

Journal Publications *(see Appendix G for the full report)*

- S. Resch, A. Kousha, A. Carroll, N. Severinghaus, F. Rehberg, M. Zatschker, Y. Söyleyici, and D. Sanchez-Morillo, "Smart Device Development for Gait Monitoring: Multimodal Feedback in an Interactive Foot Orthosis, Walking Aid, and Mobile Application," Technologies, vol. 13, no. 12, Art. no. 588, 2025.
- S. Resch, A. Zirari, T. D. Q. Tran, L. M. Bauer, and D. Sanchez-Morillo, "Smart Walking Aids with Sensor Technology for Gait Support and Health Monitoring: A Scoping Review," Technologies, vol. 13, no. 8, Art. no. 346, 2025.
- S. Resch, J. Schauer, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Improving Social Acceptance of Orthopedic Foot Orthoses Through Image-Generative AI in Product Design," Applied Sciences, vol. 15, no. 8, Art. no. 4132, 2025.

Conference Proceedings *(including full papers, short papers, posters, and abstracts)*

- S. Resch, J.-G. Hanania, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Smart Insole Visual Foot Pressure Feedback in Mixed Reality Environments: Impact on Gait, Heart Rate, Workload, and Engagement in Treadmill Walking," in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, Association for Computing Machinery, pp. 1–8, 2026.
- S. Resch, Y. Söyleyici, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Enhancing mHealth App Onboarding Using a Multimodal Mixed Reality Avatar," in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, Association for Computing Machinery, pp. 1–9, 2026.
- S. Resch, "Designing Customized Wearable Devices: A Novel AI-Based Framework for Product Development," in *9th International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*, IEEE, pp. 1–6, 2025.
- S. Resch and A. Carroll, "Influence of Virtual Footwear on Gait Behavior, Embodiment, and Workload While Walking on a Treadmill in Virtual Reality," in *31st ACM Symposium on Virtual Reality Software and Technology (VRST '25)*, Association for Computing Machinery, pp. 1–2, 2025.
- S. Resch, J.-G. Hanania, D. Schor, E. Johannes, B.-J. Kovac, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Foot Placement Feedback in Physical Training: Effects of Spatial User Interfaces on Performance and Workload in Virtual Reality," in *ACM Symposium on Spatial User Interaction (SUI '25)*, Association for Computing Machinery, pp. 1–12, 2025.

- S. Resch, L. Damrau, D. Völz, and D. Sanchez-Morillo, "Personalized Foot Orthoses Through a Patient-Centered Design Approach Using Generative AI," *Abstracts of the 2025 Joint Annual Conference of the Austrian (ÖGBMT), German (VDE DGBMT) and Swiss (SSBE) Societies for Biomedical Engineering, Biomedical Engineering / Biomedizinische Technik*, De Gruyter, vol. 70, no. s1, p. 363, 2025.
- S. Resch, J.-G. Hanania, V. Schwind, D. Völz, and D. Sanchez-Morillo, "The Influence of Augmented and Virtual Reality Environments on Foot Positioning Success and Workload in Agility Ladder Exercises," in *Proceedings of the 2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, IEEE, pp. 679–686, 2025.
- S. Resch, L. Zimmermann, A. Kunkel, D. Gerstung, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Usability Evaluation of a Mobile Application for Foot Health Monitoring of Smart Insoles: A Mixed Methods Study," in *Adjunct Proceedings of the 26th International Conference on Mobile Human-Computer Interaction (MobileHCI '24 Adjunct)*, Association for Computing Machinery, pp. 1–7, 2024.
- S. Resch, A. Röll, A. Malech, V. Schwind, D. Sanchez-Morillo, and D. Völz, "Walking Aid with Haptic Feedback for Combined Use with a Smart Foot Orthosis," *Gerontechnology*, vol. 23, no. 2, p. 1–1, 2024.
- S. Resch, R. B. Tiwari, H. R. Vankawala, P. Singh, M. Rafati, V. Schwind, D. Völz, and D. Sanchez-Morillo, "The Impact of Visual Feedback and Avatar Presence on Balance in Virtual Reality," in *Proceedings of Mensch und Computer 2024 (MuC '24)*, Association for Computing Machinery, pp. 555–559, 2024.
- S. Resch, M. Rafati, A. Altomare, O. Raddi, A. Tahmas, V. Schwind, and D. Völz, "Correct Foot Positioning in Virtual Reality through Visual Agility Ladder Training," in *Proceedings of Mensch und Computer 2023 (MuC '23)*, Association for Computing Machinery, pp. 524–528, 2023.
- S. Resch, K. Zoufal, I. Akhouaji, M. A. Abbou, V. Schwind, and D. Völz, "Augmented Smart Insoles–Prototyping a Mobile Application: Usage Preferences of Healthcare Professionals and People with Foot Deformities," *Current Directions in Biomedical Engineering*, De Gruyter, vol. 9, no. 1, pp. 698–701, 2023.
- S. Resch, V. Schwind, and D. Völz, "Body Position in Virtual Reality—Balancing Support for Patients with Neuropathy," *Proceedings on Automation in Medical Engineering*, vol. 2, no. 1, p. 744, 2023.

Book Chapter

- D. Völz, J. Schneider, and S. Resch, "Generatives Produktdesign orthopädischer Hilfsmittel: Potenziale und Herausforderungen von KI im Produktentwicklungsprozess," in *Künstliche Intelligenz im Einsatz für die erfolgreiche Patientenreise: Innovation, Integration und Stärkung*, Wiesbaden: Springer Gabler, pp. 219–242, 2025.

1.6. Thesis Structure

This thesis comprises six main chapters, followed by the bibliography and appendices. Chapters 1, 2, and 3 provide the fundamentals (introduction and background). Chapters 4, 5, and 6 form the main corpus and are primarily based on six peer-reviewed publications, representing the main contributions of this work (see Figure 1.2). Chapter 4 introduces foundational concepts on *Smart Devices*, which serve as a basis for the investigations in Chapter 5 (*Extended Reality*) and Chapter 6 (*AI in Design*), together leading to the conclusions in Chapter 7.

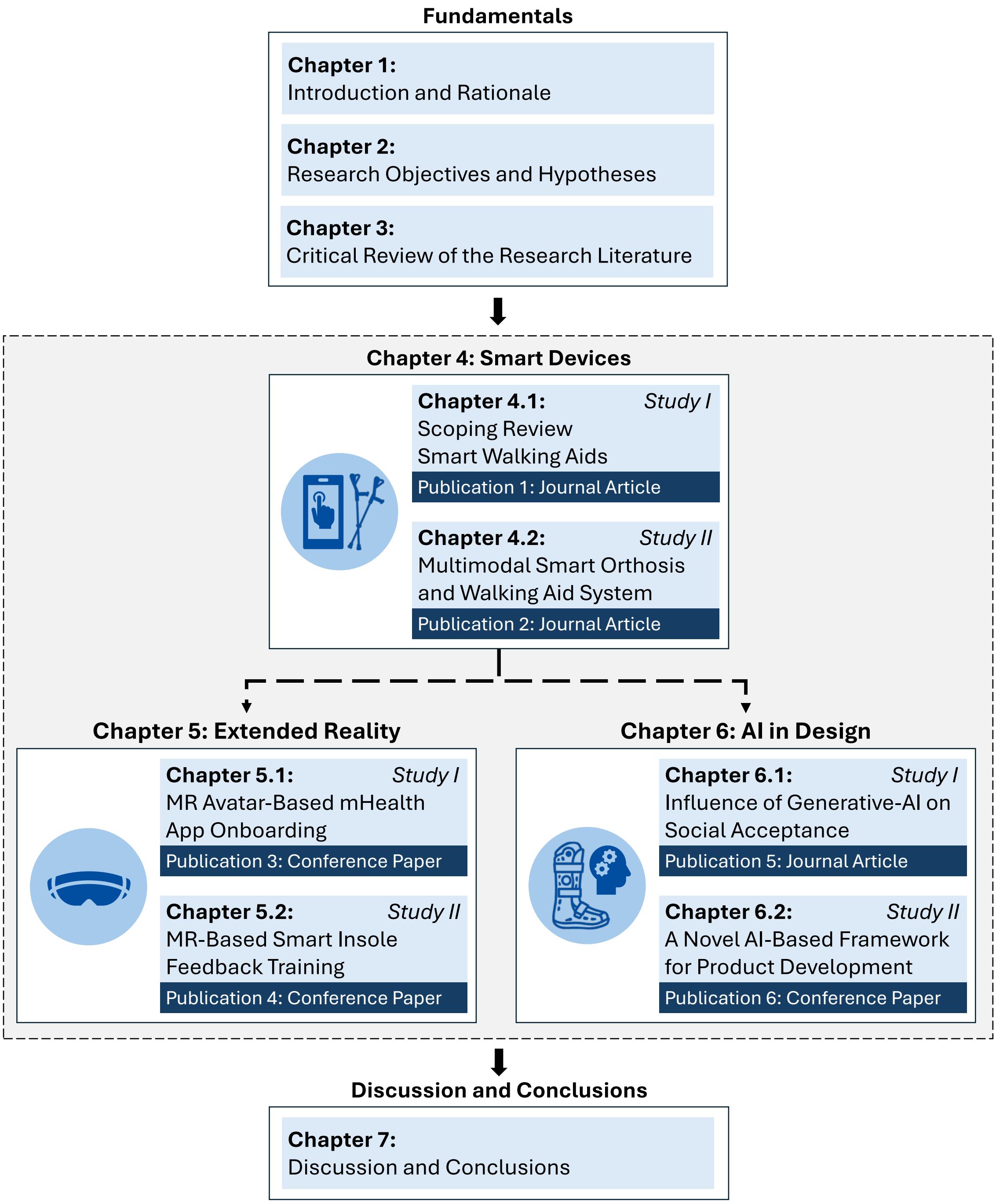


Figure 1.2: Overview of the thesis structure presented as a flowchart diagram.

The thesis is structured as follows:

Chapter 1: Introduction and Rationale describes the research motivation and the resulting research questions. It further outlines the methodology and evaluation approach, summarizes the research contributions, presents resulting publications, and provides an overview of the thesis structure.

Chapter 2: Research Objectives and Hypotheses presents the main objectives of the research and the corresponding hypotheses.

Chapter 3: Critical Review of the Research Literature summarizes definitions and key concepts related to *Assistive Technology and Orthotic Devices*, *Extended Reality*, and *User-Centered Design* to establish the necessary context for the research. Furthermore, it reviews the state of the art in *Smart Orthotic Devices and Sensorized Technologies*, *Extended Reality Applications*, and *AI in Product Design*, while identifying the research gaps that inform the objectives of this thesis.

Chapter 4: Smart Devices – Development and Validation presents the results of two journal articles, combining a systematic review with an experimental user study. The first section reports a scoping review conducted using the PRISMA-ScR methodology, analyzing 35 research articles to identify key sensor technologies, application domains, current challenges, and gaps in assistive mobility devices. Building on these insights, the second section describes the design and implementation of an integrated smart device system, comprising a smart foot orthosis, a sensorized walking aid, and a companion smartphone application. This article further evaluates system performance through experimental validation of gait data acquisition, haptic feedback perception, and usability.

Chapter 5: Extended Reality – Interaction and Evaluation of Smart Devices builds on the previous chapter by moving from device development to the application and evaluation of smart devices with end users. This chapter presents two experimental user studies that investigate how immersive VR and MR training scenarios influence interaction with smart devices. The first study examines the onboarding process for the previous developed smartphone application in MR, using avatar-based modalities, including voice and gesture, to provide insights into usability, learning experience, and interaction design. The second study evaluates MR/VR-based gait training with the smart foot orthosis on an instrumented treadmill, exploring how visual feedback and environmental factors affect objective performance and subjective experience.

Chapter 6: AI in Design – User-Centered Design of Smart Foot Orthoses builds on the previous chapters, highlighting that successful adoption of smart devices depends not only on functional performance but also on aesthetic and user-centered design aspects. This chapter presents two studies addressing the role of generative AI in orthotic design. The first section reports two online surveys investigating how AI-generated designs influence perceived social acceptability compared to conventional orthopedic products. The second section introduces and validates a patient-centered AI-driven design framework, exemplified with a smart foot orthosis, demonstrating the systematic integration of generative AI into product development.

Chapter 7: Discussion and Conclusions presents a summary of the research contributions, discusses implications and limitations, and future perspectives.

CHAPTER 2

Research Objectives and Hypotheses

This chapter presents the overall research objectives (Section 2.1) and the corresponding hypotheses (Section 2.2).

2.1. Research Objectives

The thesis pursues three overarching research objectives that jointly address the user-centered development, evaluation, and application of smart foot-worn assistive technologies across physical, virtual, and AI-supported design contexts. These main research objectives guided the examination in an iterative development and testing process and were defined as follows:

Main Research Objectives:

1. **Smart Devices:** To develop and experimentally evaluate an integrated smart foot orthosis system that enables the measurement of plantar pressure and gait characteristics, provides multimodal feedback, and supports real-time interaction through complementary devices and applications, including a connected walking aid and mobile application.
2. **Extended Reality:** To design immersive VR and MR training scenarios for foot augmentation and smart device interaction, in order to investigate how different feedback modalities and immersive environments affect user interaction, objective performance, and subjective experience.
3. **AI in Design:** To investigate how design factors influence the social acceptability of smart foot orthoses and to examine how generative AI can support the development of user-centered and socially accepted orthotic products, including its integration into the design process.

Furthermore, accompanying specific research objectives were defined to operationalize the main objectives and RQs.

Specific Research Objectives:

1. **Smart Devices:**
1.1 To systematically analyze existing intelligent walking aids concerning sensing technologies, feedback mechanisms, target user groups, and application domains, identifying current trends and research gaps.
1.2 To empirically evaluate the technical feasibility and measurement accuracy of the developed smart foot orthosis as a foundation for reliable real-time feedback.
1.3 To investigate user perception and usability across the integrated smart device ecosystem (including the orthosis, connected walking aid, and mobile application) focusing on feedback strategies, interaction, and application usability.

2. **Extended Reality:**
2.1 To investigate how immersive VR and MR environments influence objective measures (e.g., gait parameters, physiological responses) and subjective outcomes (e.g., workload, engagement) when interacting with a smart foot orthosis.
2.2 To examine how different immersive feedback and data visualization strategies affect user perception, usability, and engagement in virtual training scenarios.
2.3 To analyze how avatar-based instructional feedback modalities (e.g., visual and auditory cues) in MR influence the learning experience and usability during the onboarding process of the accompanying mobile application.

3. **AI in Design:**
3.1 To analyze how AI-generated orthotic designs influence perceived social acceptability compared to conventional orthopedic products across different device categories.
3.2 To investigate how generative AI models and prompt structures shape user perception and acceptance of personalized foot orthosis designs.
3.3 To develop and validate a patient-centered AI-driven design framework integrating functional requirements, aesthetic considerations, and user acceptance for personalized orthotic product development.

2.2. Hypotheses

Three overarching hypotheses were defined in accordance with the overall research objectives. These hypotheses provided a guiding framework for the investigations, while each study tested more specific aspects within this overarching framework.

Smart Devices: It is possible to demonstrate the feasibility, effectiveness, and usability of a smart orthosis system, including a connected walking aid and mobile application, while enhancing overall user experience through a closed-loop feedback system.

Extended Reality: It is possible to achieve enhanced interaction with the foot orthosis and companion smartphone application in VR/MR environments with integrated feedback modalities, as measured by objective and subjective metrics.

AI in Design: It is possible to positively influence perceived acceptability of orthotic footwear by incorporating generative AI into the design and methodological product development process.

CHAPTER 3

Critical Review of the Research Literature

This chapter provides an overview of the *Theoretical Background* (Section 3.1) and *Related Work* (Section 3.2) concerning the three central themes of this thesis. It introduces definitions and fundamental concepts within the research context and thereby establishes the conceptual foundation for the conducted studies. The review of related work offers a structured overview of prior research in these areas and identifies existing research gaps. Finally, a *Chapter Summary* (Section 3.3) synthesizes the central gaps in the literature and contextualizes them within the broader scope of the thesis.

3.1. Theoretical Background

The theoretical background section serves as a foundation for understanding the key terms and definitions necessary to contextualize this research, including *Assistive Technology and Orthotic Devices* (Subsection 3.1.1), *Extended Reality* (Subsection 3.1.2), and *User-Centered Design* (Subsection 3.1.3).

3.1.1 Assistive Technology and Orthotic Devices

According to the World Health Organization (WHO), the term *assistive technology* serves as an umbrella term covering "assistive products, systems, and services," which aim to support impaired individuals (e.g., older adults, children, or people suffering from health conditions or general disabilities) in their daily lives [6]. The developed technologies aim to provide assistance for visual, hearing, cognitive, mobility, or other impairments. In particular, the term *assistive products* is defined by the International Organization for Standardization (ISO) 9999:2022 [33] and the WHO [34] as "any external product (including devices, equipment, instruments, or software)" with the main purpose "to maintain or improve an individual's functioning and independence [...] and also to prevent impairments and secondary health conditions." Most such products are medical devices, including prosthetic/mobility devices, walking aids, or orthoses.

Lower-leg orthoses are a specific product class of assistive technology. According to the ISO, the terms related to orthotic footwear are defined in ISO 8549-3:2020-09 [35]. This standard describes an *orthotic device* as "an externally applied device used to compensate for impairments of the structure and function of the neuro-muscular and skeletal systems." The term *foot orthosis* is defined as an external device that "encompasses the whole or part of the foot." Foot orthoses are used by patients with various lower-leg conditions, e.g., acute foot fractures or ankle injuries. The form factor differs, as several types of foot orthoses exist, depending on the intended use. Controlled ankle motion boots are one specific type of orthotic footwear. These devices follow medical standards and are available in two formats (low and high-shaft), depending on the disease, as well as in five sizes (XS–XL) for unilateral use according to shoe size. By immobilizing the foot, they support the healing process while allowing walking outdoors with ground contact. Typically, a foot orthosis consists of an external enclosure (shell and front panel) made of hard plastic (mostly high-density polyethylene), including inner padding, which can be separated into the main cast (back part) and the enclosure (front part). To fit the shape of the lower leg, an air pump system with integrated air chambers is often implemented to ensure a tight fit. The main part (shell) of the walker boot, including inner padding, and the front panel can be fixed via Velcro straps.

In this thesis, a standardized lower-leg orthosis (AIRCAST® AIRSELECT™ Elite Walker [1]) from Enovis (USA) was used as a case example because this device is globally distributed and follows current standards. This device is used for immobilization (before or after surgery) to stabilize ankle or foot fractures, ankle sprains, and to ensure post-operative rehabilitation. Figure 3.1 shows the orthosis as well as the included air chamber system.

Figure 3.1: Three views of the AIRCAST® AIRSELECT™ Elite Walker (image by Enovis [1]): closed boot, sole, and inner air chamber system.

Additionally, patients who rely on a foot orthosis often require supplementary support from walking aids to stabilize and control their gait. Walking aids are a specific group of assistive products, including devices such as canes, walkers, and crutches. The choice of walking aid typically depends on the user's general mobility and the level of assistance required. In the case of foot injuries, two types of crutches are used: underarm and forearm crutches. While forearm crutches are commonly used in Europe, underarm crutches are more common in the US [36]. These devices are standardized for ergonomic handling and are adjustable in height to ensure proper arm positioning and optimal weight distribution during use.

Taken together, assistive products such as orthotic devices and walking aids complement each other to support patient mobility, rehabilitation, and daily activities, highlighting the importance of interoperability and coordination across devices.

3.1.2 Extended Reality

Extended Reality or "XR" serves as an umbrella term for summarizing immersive technologies such as AR, MR, and VR. These technologies can be systematically classified along the reality-virtuality continuum introduced by Milgram [2] in 1994, as illustrated in Figure 3.2. Within this continuum, MR describes the spectrum between the fully real/virtual environment and includes both AR and Augmented Virtuality. Accordingly, VR refers to fully immersive environments which exclude real elements, whereas AR is typically associated with semi-immersive environments that enable the overlay of virtual elements into the real world. In contrast, augmented virtuality describes environments in which only a few real elements are projected into a virtual world. However, this term is less commonly used in literature.

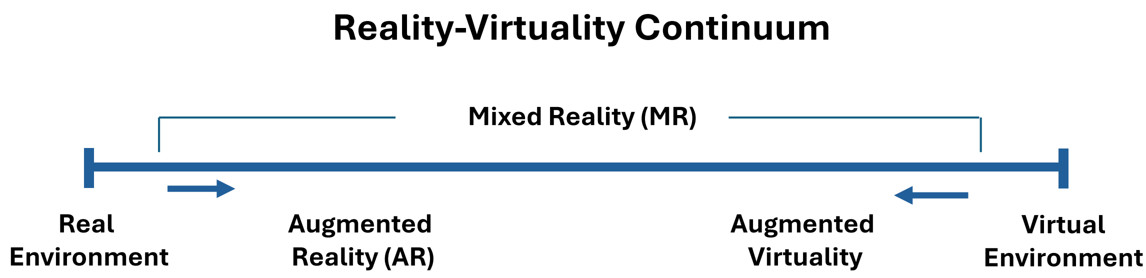


Figure 3.2: Schematic overview of the virtuality continuum introduced by Milgram [2], classifying the terms AR, MR, and VR.

In 1997, Ronald Azuma [37] identified three key characteristics of AR systems: the combination of real and virtual objects, real-time interaction, and three-dimensional perception. AR systems range from basic smartphone- or tablet-based applications to smart glasses or optical see-through Head-Mounted Displays (HMDs) (e.g., Microsoft HoloLens). These systems allow virtual elements to be displayed within the real environment, enabling users to interact with both physical and digital elements. In the context of this thesis, AR is realized by augmenting plantar foot pressure data and presenting it via a smartphone application.

MR is often associated with video-see-through HMDs, which enable the combination of real and virtual elements within the user's point of view. The real world can be captured via onboard cameras, allowing real-time visual overlay of virtual elements (such as 2D interfaces or 3D objects) onto the internal display. Recent developments of affordable standalone MR consumer devices have expanded application potential, allowing users to switch between fully immersive VR and pass-through MR interaction. For example, devices such as the Meta Quest 3 use inside-out tracking to achieve accurate positioning without external infrastructure, providing a basis for mobile training scenarios. In this thesis, the Meta Quest 3 was used to deliver HMD-based MR and VR interactions and to investigate differences in user perception and application potential.

VR is commonly described by three key characteristics: immersion, interaction, and imagination [38]. The virtual worlds are typically developed using 3D game engines and displayed to the user via HMDs. User interaction within the virtual environment is enabled via input devices such as controllers, connected trackers, or custom-developed devices and systems. These may include external tracking systems, wearable sensors, or integrated components such as cameras and microphones to support gesture or speech recognition. In terms of output modalities, the virtual scene includes visual and auditory cues, but could also include haptic feedback via controllers, haptic suits, or wearable devices. In this thesis, VR is used as a virtual training environment to test user interaction with the prototyped smart devices. The developed devices provide input signals, such as plantar foot pressure from the orthosis or interactions via the smartphone app, which can be used to evaluate different output feedback modalities.

To make the distinctions between these environments more tangible, the following study, conducted in the context of this doctoral research, illustrates their practical application in a training scenario [3]. Figure 3.3 shows the study investigating the effects of AR, VR, and real-world conditions on lower-limb agility ladder training. In AR (or MR), visual footprints were displayed on the ground via the Meta Quest 3 using passthrough mode, allowing participants to see the real world with virtual foot placement feedback. In VR, participants trained in a fully virtual environment without any real-world elements, wearing the same HMD. In the real-world condition, participants did not wear an HMD. To allow a fair comparison, visualizations were projected onto the floor, providing augmented feedback in the real environment without HMD influence, a setup typically infeasible without projection equipment. The study contributed to enhanced understanding of how environmental factors influence training and highlighted the potential of VR for lower-limb training, overcoming real-world limitations such as missing visual feedback.

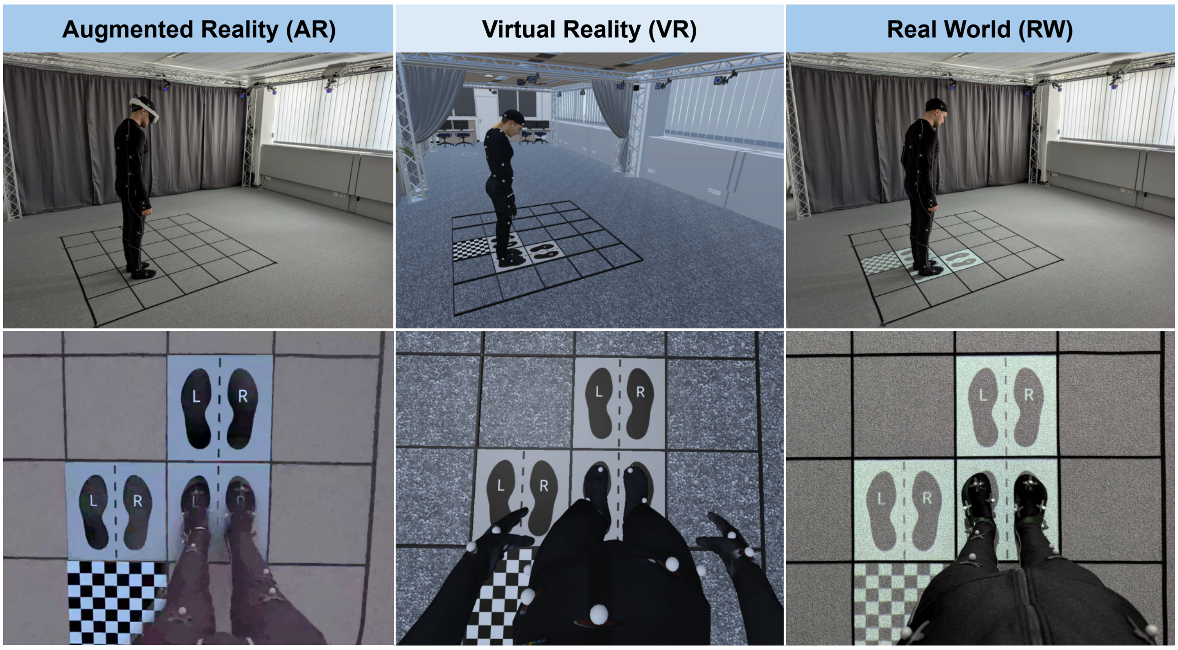


Figure 3.3: Study comparing the effects of AR, VR, and real-world environments on agility ladder training performance [3]. (©2025 IEEE)

Another example from this doctoral research demonstrates how immersive environments can be designed and how feedback modalities are implemented. As an illustrative case, a VR-based training environment focusing on balance board exercises was developed to demonstrate how immersive visual feedback can be integrated into a controlled training scenario (see Figure 3.4) [4]. In this setup, a virtual environment representing a replica of the real laboratory was created using the Unity game engine. A visual target provided color-coded feedback during balancing tasks. The virtual scene was streamed to an HMD to immerse the user in the virtual environment. An optical Motion Capture System (MoCap) from OptiTrack tracked reflective markers attached to the HMD, balance board, and participants' tracking suit, allowing motion data to be recorded using Motive software for subsequent analysis.

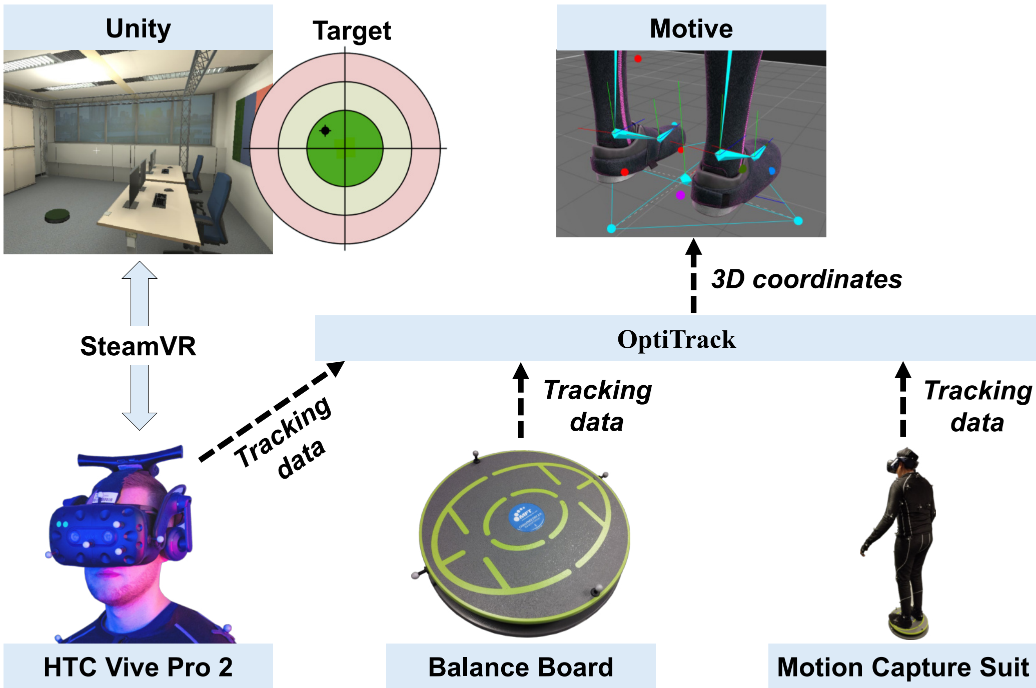


Figure 3.4: Technical overview of the created balance board training setup [4].

In summary, the previous examples illustrated how immersive XR technologies can be employed to create controlled training environments that integrate feedback and support performance assessment. In the context of smart device interaction and user-centered evaluation, XR tools enable early-stage testing of devices and feedback modalities, allowing controlled training in targeted scenarios. Compared to traditional non-immersive, screen-based technologies, immersive VR provides interaction in controlled environments without external distractions, which motivates the exploration of how these technologies can be effectively used to train users.

3.1.3 User-Centered Design

The concept of *user-centered design* was introduced by Donald A. Norman in 1986 [39]. It builds on the approach proposed by Gould and Lewis [40] in 1985, which emphasized the need to focus on users early in the design phase, empirically measure usability factors, and iteratively design, modify, and test. This methodology places the user at the center of the development process to achieve high usability. A key aspect is the continuous involvement of users throughout the development stages, ensuring that design decisions are informed by actual user needs. This iterative process allows the optimization of design solutions based on user input while considering human factors.

In HCI, the term *human-centered design* is defined by ISO 9241-210:2019 [41] and is often used synonymously with *user-centered design* [41]. In this thesis, the term user-centered design is specifically chosen to emphasize the focus on the end user in the context of smart devices and XR applications. The systems were developed for the intended target group who use assistive devices and interact within XR training environments. In contrast, the human-centered design process extends beyond the end user to also include stakeholders from a broader product development perspective.

According to the ISO definition [41], the process is organized into four key activities: (1) understand and specify the context of use; (2) specify the user requirements; (3) produce design solutions to meet these requirements; and (4) evaluate the designs against the requirements. This approach is essential for developing products (such as orthopedic devices) or systems (such as XR applications) in a systematic, user-targeted manner. Within design-based research, approaches such as the design thinking process [42] refine this structure by separating the third activity into solution development and prototype development, thereby distinguishing ideation and prototyping phases more explicitly. While purely technical products and systems often follow The Association of German Engineers (VDI) 2221 [43], user-centered design prioritizes understanding user–device interaction rather than focusing solely on technical abstraction.

Throughout all development phases, a diverse set of methods can be employed to empirically investigate and understand user behavior. For instance, formative evaluations are typically conducted in early development stages to explore design spaces and inform prototype creation, whereas summative testing is applied in later stages to assess prototypes and evaluate their impact. The choice of method depends on the development stage and the intended sample size. Options include focus groups, heuristic evaluations, field surveys, online surveys, think-aloud protocols, case or diary studies, as well as laboratory, online, or in-the-wild experiments. These empirical evaluations rely on both qualitative as well as quantitative measures and observations, each requiring appropriate assessment and analysis techniques.

In this thesis, empirical methods following a user-centered design approach were applied. For the development of a smartphone app for smart devices, focus groups with users and experts were used to gather requirements [44]. Mockups were then created and usability-tested [45], followed by the development of a fully functional prototype and a final impact evaluation [46]. Similarly, XR technology was employed as a tool for early device testing and for systematic laboratory studies assessing user behavior in controlled training environments. Beyond these established user-centered design approaches, emerging technologies such as generative AI offer new possibilities to enhance the design process. Specifically, AI tools can be used to directly involve users in generating design alternatives, thereby creating novel interactions that extend traditional user-centered workflows. Currently, the development of controlled ankle motion boots largely follows standardized procedures, focusing on technical aspects with minimal active user involvement in the design process. In the context of this thesis, the potential of AI was explored to support a more user-integrated orthosis design. Online surveys investigated its influence on design practices, and a case study combined AI tools with user feedback to examine their systematic application.

3.2. Related Work

The related work section provides an overview of three key areas: *Smart Orthotic Devices and Sensorized Technologies* (Subsection 3.2.1), *Extended Reality Applications* (Subsection 3.2.2), and *AI in Product Design* (Subsection 3.2.3). It also discusses the research gaps that arise from these fields.

3.2.1 Smart Orthotic Devices and Sensorized Technologies

In the following, an overview of the current state of research in smart orthotic devices is presented. By doing so, it serves as a foundation for the design and development of a sensorized foot orthosis and related companion devices for this research project.

S. Resch, A. Kousha, A. Carroll, N. Severinghaus, F. Rehberg, M. Zatschker, Y. Söyleyici, and D. Sanchez-Morillo, "Smart Device Development for Gait Monitoring: Multimodal Feedback in an Interactive Foot Orthosis, Walking Aid, and Mobile Application," Technologies, vol. 13, no. 12, Art. no. 588, 2025.

This section reviews related research in the areas of *Smart Footwear for Mobile Gait Analysis and Monitoring Health Parameters*, *Foot Augmentation with Multimodal Feedback for Patient Device Interaction*, and *Smart Walking Aids and Applied Sensor Technologies*. We conclude by summarizing current research gaps and outlining our contributions to the field.

Smart Footwear for Mobile Gait Analysis & Monitoring Health Parameters

The research field of sensor technology demonstrates that different types of sensors are used for measuring continuous physiological data depending on the application [13]. Based on the device type, sensor technologies can be integrated into various systems such as socks/textiles [47, 48, 49, 50, 51], shoes [52, 53, 54, 55, 56], insoles [57, 58, 59, 60, 61], or orthoses [15, 62, 16, 63, 64]. In particular, smart insoles are developed for a broad range of applications, ranging from ambulatory gait assessment and home-based monitoring [65, 66], to gait rehabilitation after stroke [67], fall risk detection [68], and activity classification using machine learning techniques [69, 70].

A systematic literature review by Subramaniam et al. [13] identified two primary health-related functions of smart insoles: (1) the measurement of plantar pressure distribution and (2) the analysis of gait and activity patterns. Plantar pressure data are typically acquired using integrated Force Sensing Resistors (FSRs) for the assessment of foot stability, balance, body weight distribution, and early detection of pressure overload conditions. A review by Razak et al. [71] recommended the use of 15 FSR sensors to ensure sufficient coverage of plantar pressure distribution according to the individual foot anatomy. This recommendation is based on an anatomical segmentation of the foot sole into 15 regions: heel (areas 1–3), midfoot (4–5), metatarsals (6–10), and toes (11–15), which support most of the body load [72]. However, the individual shoe size and foot anatomy need to be considered for proper sensor placement to prevent measurement errors [73].

Furthermore, the use of Inertial Measurement Unit (IMU) sensors (a combination of accelerometers and gyroscopes) enables the measurement of gait characteristics such as step length, cadence, walking speed, and swing phase duration [74]. These systems are used for performance tracking of athletes [13] or to enable mobile gait analysis [75]. A single IMU sensor embedded in the insole can provide sufficient data for this purpose [76]. However, the literature reports inconsistencies regarding the optimal sensor placement for precise data extraction, as signal drift and measurement errors may occur depending on the location of the sensor [77]. As a result, calibration remains a critical challenge in smart insole development [13]. To minimize mechanical load and improve signal quality, placement in the midfoot region is generally recommended [13].

Beyond IMU and FSR-based gait analysis, additional sensing modalities have been explored to extend the functionality of wearable systems. These include Global Positioning System (GPS) tracking for spatial gait analysis [78], foot temperature monitoring for early detection of Diabetic Foot Syndrome (DFS) [79, 80], and physiological measurements such as heart rate via Photoplethysmogram (PPG) sensors [81] and muscle activity using Electromyography (EMG) electrodes in textile-based wearables [48]. However, PPG and EMG-based approaches remain largely limited to research prototypes and have not yet been adopted in commercial devices. Other studies have also investigated sensing technologies for obstacle detection, including ultrasonic sensors, ToF cameras, and radar-based systems, particularly in applications for visually impaired users [82]. While these alternative sensing modalities offer promising extensions for specific use cases, the majority of current smart insole systems rely on the integration of IMU and FSR sensors. Due to its efficiency and widespread adoption, the combination is frequently used in smart insole systems [83]. This configuration enables accurate gait detection [84, 85] and simultaneous acquisition of kinetic (pressure) and kinematic (motion) data [86, 87]. Typically, data from IMU and FSR sensors are acquired using a portable microcontroller and transmitted wirelessly (e.g., via Bluetooth) to an external Graphical User Interface (GUI) [65, 88].

To ensure the reliability of gait-related measurements derived from wearable systems, their performance must be validated against established reference standards. A review by Hutabarat et al. [89] emphasized that such wearable gait systems need to be validated against proven standard gait measurement systems to ensure data accuracy. For this purpose, instrumented treadmills and MoCaps represent the gold standard for validating gait analysis systems [90, 89]. This approach has already been applied in several validation studies, for instance by comparing wearable systems to a stationary Zebris treadmill [91] or GAITRite force platforms [92, 93]. Overall, recent research highlights the importance of comparative validation studies with established gait analysis systems to strengthen the credibility and reliability of findings in wearable gait assessment [94]. In conclusion, despite considerable advances in this research field, key challenges remain regarding data accuracy, system validation, and participant-centered evaluation.

Foot Augmentation with Multimodal Feedback for Patient Device Interaction

Smart wearable systems such as insoles, socks, or orthoses are often connected to external devices (e.g., smartphone applications, smartwatches, or other interfaces) to enable continuous mobile health monitoring and interactive feedback [13]. A systematic review by Saboor et al. [95] highlighted the growing potential of smartphone applications for mobile gait analysis, particularly in remote health contexts, while also noting ongoing challenges related to accessible and user-friendly User Interface (UI) design. In this context, Resch et al. [44] developed a Mobile Health (mHealth) application for smart insoles on the basis of focus groups with patients and experts. In a follow-up study, functional mockups were evaluated in a usability test with 30 participants, which confirmed the user-friendliness and provided recommendations for future design improvements [45]. The resulting design recommendations, integrated features, and overall app concept provide a valuable basis for the continued development of smart insole applications and the integration of feedback modalities.

Beyond interface design, wearable systems increasingly incorporate real-time feedback mechanisms (through visual, acoustic, or haptic signals) typically via smartphone apps or connected devices to support user interaction. For instance, vibration cues have been applied to indicate pressure overload [96]. It was shown that vibration feedback applied within the shoe can significantly influence gait behavior in healthy participants [97], and vibrotactile biofeedback triggered by FSR input reduced overpronation duration by 50% [98]. Furthermore, auditory feedback has also been investigated, such as acoustic signals triggered by in-sock pressure sensors for gait correction [99], or auditory stimulation via insoles for patients with Parkinson's disease [100]. These approaches are particularly relevant for patients with impaired peripheral sensation, such as individuals with DFS, for whom smartwatch-based alerts and real-time gait correction via smart insoles have been proposed [101]. Several studies have applied multimodal feedback approaches and shown that they influence gait symmetry, posture and pressure distribution [102, 97, 98, 103]. In an experimental study, Redd et al. [102] compared visual, acoustic, and vibrotactile feedback modalities in healthy participants and found that both visual and haptic cues significantly improved gait symmetry. Despite the demonstrated potential of smartphone applications and feedback modalities, further validation is needed regarding user perception and system effectiveness.

Smart Walking Aids and Applied Sensor Technologies

Walking aids play a crucial role in daily mobility support for patients with walking impairments. Recent research has explored how these devices can be enhanced through sensor technologies to support health monitoring, user interaction, and feedback-based interventions [104, 105]. Various smart walking aid prototypes have been proposed, including instrumented sticks [106], canes [107], and forearm crutches [108]. Many of these systems have been developed specifically for users with visual impairments [109, 110, 111, 112]. These systems typically contain sensors such as IMUs for fall detection or for motion tracking [113, 114], FSRs embedded in handles to detect gait events [115], GPS modules for positional tracking [116], and ultrasonic sensors for obstacle detection [109, 110, 111].

Beyond data acquisition, several studies have explored the integration of feedback mechanisms to enhance user support [112, 117, 118]. For example, Merrett et al. [117] demonstrated the potential of biofeedback in forearm crutches using acoustic signals based on FSR data to support optimal weight distribution. Schuster et al. [118] showed that haptic feedback delivered via the handle of a cane can reduce knee adduction moments in individuals with knee osteoarthritis. In addition, a smart stick with integrated SOS functionality has been developed that connects to smartphone applications [119].

While individual sensor-based walking aids have demonstrated promising results, integrated systems that combine smart walking aids with intelligent footwear for foot augmentation remain rare and are typically limited to specific application scenarios. For instance, Suman et al. [120] proposed a system in which a smart cane communicates with smart footwear, using ultrasonic sensors in the shoe and auditory feedback via the cane handle. Another example is the use of force-sensitive crutches to assist lower-limb exoskeleton users by capturing ground force data to enable active support [121]. Furthermore, a robotic walking cane was developed that receives Center of Pressure (CoP) data from four FSRs integrated in an insole to support fall detection [113]. However, direct communication between walking aids and orthopedic devices, such as sensor-based foot orthoses, has not yet been realized. In this context, Resch et al. [122] proposed a conceptual system that enables a walking aid to communicate with a smart AFO through vibration feedback in the handle to indicate plantar pressure overload. Nevertheless, the proposed system remains a concept without implementation or empirical evaluation.

Overall, while sensor-based walking aids and feedback modalities offer high potential, most systems remain in the prototype stage and lack systematic validation or integration into broader rehabilitation frameworks [105].

Research Gaps

In summary, various systems for mobile foot health monitoring have been developed, primarily based on IMUs and FSRs, often in combination with wireless communication to external devices such as smartphone applications. While smart insoles have been extensively studied, their integration into orthopedic devices such as Ankle-Foot Orthosiss (AFOs) remains limited. Similarly, sensor-based walking aids show promising results for gait support and biofeedback, but most remain standalone prototypes without integration into more comprehensive assistive systems. Although haptic feedback has proven to be beneficial for improving user performance, there is no tested approach to date that combines it with sensor-based footwear and walking aids in a unified system. Related work lacks concepts and empirical evidence for the combined use of smart walking aids and foot monitoring technologies. However, this integration offers future potential for improving patient monitoring, supporting real-time gait analysis and enabling multimodal feedback in healthcare and everyday use. Building on these findings, our work addresses these gaps by developing and validating an integrated system that combines a smart foot orthosis, an instrumented walking aid, and a smartphone application for real-time gait monitoring and interactive feedback. By addressing these gaps, we contribute to the advancement of assistive device technologies, mobile gait analysis, and user-centered feedback systems.

3.2.2 Extended Reality Applications

Previous work has shown that immersive technologies such as VR and MR can support both motor learning processes and interaction with digital health systems. Therefore, this subsection provides an overview of research on XR-based training and feedback systems in the context of smart assistive devices. In particular, two areas are relevant: immersive environments for gait training with orthotic footwear, and MR technologies employing feedback strategies to support user interaction with mobile devices.

Accordingly, this subsection is structured into two parts. The first part reviews research on foot augmentation and virtual gait training approaches, while the second part examines immersive feedback and guidance strategies that support interaction with smartphone-based health applications. For each domain, the relevant literature is summarized and the remaining research gaps are discussed.

Foot Augmentation and Virtual Gait Training

To provide an overview of the relevant research landscape, related work in the areas of *Foot Augmentation and Immersive Feedback Modalities*, *Gait Training & Motor Rehabilitation in VR/MR*, and *Environmental Factors in Immersive Settings* is reviewed. Finally, the key findings are synthesized to identify open research gaps and to outline how this thesis aims to contribute to the field.

Foot Augmentation and Immersive Feedback Modalities

As previously shown in Section 3.2.1, wearables for foot augmentation include smart insoles [59, 123, 69], socks [50, 49], orthotic boots [46, 62], and instrumented shoes [124, 125, 126], which differ in configuration, form factor, and sensor inputs [8]. In healthcare, these sensorized devices enable mobile gait analysis and can provide biofeedback in therapy [124]. In particular, foot pressure measurements can be used as visual feedback (e.g., displaying pressure distribution or the CoP) or delivered via haptic cues such as vibration alerts in cases of high plantar pressure, making them suitable for immersive MR applications. As an example, Elvitigala et al. [103] introduced GymSole, a system worn inside a standard shoe to provide feedback on CoP while performing squats and deadlifts, thereby contributing to improved posture during exercise. Their study confirmed the usefulness of vibrotactile and visual screen feedback compared to no feedback. In a follow-up study, they proposed GymSole++ [127], a standalone system that provided vibrotactile CoP-based feedback via the insole while extending visual feedback through a Google Glass interface. Building on their use in lower-limb exercises, pressure-sensing insoles have further been integrated as real-time input devices in VR, supporting gait detection and locomotion techniques such as feeling ground surfaces while walking [128, 129]. Research further highlights that the gamification of VR applications can motivate patients to engage more actively in rehabilitation exercises [8]. More specifically, virtual training applications for foot and lower-limb augmentation highlight the benefits of combining visual feedback with gamified environments, for example in gait [130], mobility [131], balance [132], and agility training [133].

However, despite the demonstrated potential of foot augmentation and visual feedback in immersive environments, most applications have been limited to standard footwear such as shoes with insoles, which do not substantially affect gait. In contrast, orthopedic foot orthoses fundamentally modify gait and interaction, but remain unexplored in this context. This highlights the need to investigate feedback modalities specifically in gait-affecting footwear.

Gait Training and Motor Rehabilitation in VR/MR

Traditional gait training is commonly performed in laboratory settings, where clinical gait analysis allows precise assessment but limits feedback to that provided by experts. Therefore, the use of immersive environments opens new opportunities for more effective training, since it does not solely rely on the often inconsistent availability of therapist feedback [134]. As demonstrated by Jiao et al. [135], the use of VR shows high potential for gait training and motor rehabilitation, since patients can benefit from visual feedback and gamified elements, such as exergame tasks, which were positively assessed by both physiotherapists and post-stroke patients. In this context, the use of foot wearables as interactive interfaces in virtual environments offers the possibility to deliver real-time feedback and interaction, thereby extending conventional gait training methods. For example, Kim et al. [136] evaluated stroke patients in VR-based treadmill training using real-time CoP visualization feedback from in-shoe smart insoles, and compared the results with traditional gait training. They concluded that VR-based treadmill training with real-time insole feedback can positively impact balance, gait performance, and motor function in stroke patients. However, when developing virtual gait training systems, factors such as specialized treadmills and appropriate feedback positioning need to be considered to achieve effective outcomes when using an HMD [137]. For instance, Sloot et al. [138] showed that fixed-speed walking in VR improved gait patterns, whereas self-paced speed resulted in a more cautious gait.

In conclusion, virtual gait training can provide meaningful benefits when combined with real-time feedback. Consequently, it remains unclear how immersive environments and visual pressure augmentation influence gait adaptation and user experience when walking with a gait-modifying orthosis in VR/MR.

Environmental Factors in Immersive Settings

Environmental factors in semi- and fully-immersive settings can influence both subjective user perception during a task and objective performance outcomes. While conventional training is typically performed in laboratory settings, some VR studies have used virtual replicas of the same room to ensure better comparability and generalization of results [139]. However, the use of VR can also increase perceived cognitive load, which should be carefully considered for the respective task [3]. Therefore, when developing VR training applications, task-dependent cognitive load and stress levels should be minimized to preserve the benefits of immersive training. In this context, the influence of natural settings could be considered [140]. For example, Mucha et al. [141] demonstrated that walking in naturalistic VR environments, compared to urban environments, can relieve perceived pain. Furthermore, it has been shown that the use of natural landscapes in VR can support stress recovery [142]. As stress is often associated with changes in physiological activity, measures such as pulse rate are sometimes used as general indicators of physiological activation [143, 144], as well as physical fitness level [145]. In addition, the NASA Task Load Index [146] is frequently applied in HCI as a subjective measure of task-specific perceived cognitive load. Several studies have combined such physiological indicators with subjective workload measures to obtain a more comprehensive understanding of user experience and performance. For example, Winter et al. [147] investigated the usability of treadmill training with a VR HMD, while measuring heart rate and workload using the NASA Raw Task Load Index (RTLX). Overall, they reported comparable usability, as heart rate did not increase compared to screen-based or non-VR training; however, perceived mental and physical load differed.

Besides fully immersive VR, the use of MR offers a promising alternative, as it combines the realism of physical environments with the possibility of adding visual feedback. For example, the Meta Quest 3 can be used with its passthrough function as a cost-effective MR option, allowing users to maintain situational awareness while performing specific walking tasks [148]. It has also been tested in other training contexts, such as agility ladder training [3]. However, the impact of such environmental factors on gait training remains unknown. Overall, existing studies indicate that environmental context affects virtual training but also highlight challenges regarding user experience and workload.

Research Gaps

In summary, research has demonstrated the wide application of immersive treadmill training for patient rehabilitation. Recent work on foot interfaces and feedback modalities, such as pressure augmentation with smart insoles and CoP feedback, appears promising [103, 127, 136]. Studies further indicate that virtual gait training and environmental factors can affect both objective performance and subjective workload. However, most research examines feedback or environment in isolation and relies on standard footwear, leaving the combined effects of pressure augmentation, immersive environments, and gait-modifying orthoses unclear. The thesis addresses this gap by integrating a pressure-augmented orthosis into immersive gait training, examining combined feedback and environmental effects, and advancing HCI research on foot augmentation in VR/MR.

User Guidance and Interaction for Smart Device Applications

Beyond immersive gait training, MR technologies are also increasingly used to support user interaction with mobile devices. As discussed in Section 3.2.1, smart insole systems are often accompanied by dedicated smartphone applications. These companion applications play an important role in monitoring sensor data and providing user feedback. However, it was previously shown that usability is crucial because mHealth apps can introduce challenges during the initial user-device interaction, particularly during the onboarding phase. Therefore, immersive training and interactive feedback represent promising approaches to improve usability and learning during device interaction.

In the following, related work is reviewed in three areas: *Mobile Health Applications*, *Onboarding Guidance and Tutorials in Smartphone Apps*, and *Avatar Guidance in Mixed Reality*. Finally, the key findings are synthesized to identify remaining research gaps and to outline how this thesis contributes to addressing them.

Mobile Health Applications

The use of mHealth applications represents a rapidly growing field [149, 150]. They hold strong potential for monitoring physiological data, supporting prevention, and enabling rehabilitation in different medical domains [151, 152, 153, 154]. Recent developments have extended these applications to the area of foot health. In particular, smart insoles have emerged as promising tools for activity recognition, telemedicine, and rehabilitation, where sensor data are transmitted to external devices such as smartphones [69, 155, 156].

Several studies demonstrate the potential of such systems while underlining the importance of patient-centered design. To address this, as part of this doctoral research, focus groups were conducted with patients and experts to derive requirements for a foot health application [44], and the usability of the resulting app was subsequently evaluated in a follow-up study [45]. These results show that data interpretation and interaction can be challenging for users. However, the successful application and long-term use of such applications depend on their usability, which is a key factor [157]. For usability assessment of smartphone apps, the System Usability Scale (SUS) [158], a standardized questionnaire for interactive systems, is most frequently applied to evaluate usability of health apps [159, 149, 160]. Nevertheless, specific examples of foot health applications reveal persistent challenges in addressing user needs. For example, Moulaei et al. [161] tested a smart footwear app and reported several usability issues, particularly regarding the learning experience. Patients emphasized that instructional audio files and integrated tutorials would help them to understand how to use the application effectively during first-time use. Similarly, Ogrin et al. [162] highlighted that patients often struggle with self-education in foot health apps and mentioned the need for more interactive app-based education.

Therefore, new approaches should be developed to address these concerns and improve patient–device interaction. Overall, while mobile apps for smart insoles offer valuable patient feedback, their understanding, guidance, and intuitive usage remain limited and require additional interaction support.

Onboarding Guidance and Tutorials in Smartphone Apps

Building on the identified usability challenges in mHealth applications, onboarding guidance and interactive tutorials represent promising strategies to support users during the initial interaction with a mobile application. In this context, several studies highlight the importance of well-designed onboarding processes in smartphone applications. For example, Strahm et al. [163] investigated a mobile app onboarding process using minimalistic instruction, where interaction with prototypes supported users in step-by-step exploration and facilitated the creation of effective task flows. This was based on the minimalist instructional design proposed by Carroll [164].

Different approaches have been explored to enhance onboarding experiences in smartphone applications. For example, Palmquist et al. [165] explored a gamified onboarding process, demonstrating how playful elements can increase engagement during initial app use. To improve learning outcomes, the integration of visual and auditory cues can support users in acquiring knowledge more effectively [166]. One common example is voice assistants such as Siri or Google Assistant, where spoken instructions complement visual interaction on mobile devices [167]. In this context, Stoiber et al. [168] presented design guidelines for visual onboarding processes and highlighted challenges users face when interpreting visualizations for the first time. For evaluation of the onboarding experience, usability assessment using the SUS is frequently applied [169].

In conclusion, previous research highlighted that onboarding guidance can reduce barriers to adoption and improve user experience across diverse application fields. However, despite the demonstrated potential, the systematic implementation of onboarding tutorials into mHealth apps (e.g., foot health and smart insole systems) remains unexplored.

Avatar Guidance in Mixed Reality

A systematic literature review by Fu et al. [170] highlighted the potential of AR/VR/MR for gamified healthcare applications, with avatars frequently mentioned as a common element across diverse tasks. Furthermore, it was demonstrated that onboarding procedures can also be implemented in immersive environments such as AR, VR, and MR [171, 172].

For example, Chauvergne et al. [172] investigated onboarding practices in VR and showed that visual and auditory instruction modalities are frequently used. For this reason, they proposed design guidelines for combining both modalities in future applications. Similarly, Wu et al. [173] compared visual, audio, and combined cues for collaborative guidance in VR, demonstrating the advantage of multimodal instruction. In general, the integration of visual and auditory cues has been shown to improve user interaction, as demonstrated in studies using voice- and gesture-based input with the Microsoft HoloLens as an MR HMD [174].

In this context, avatars have been applied in different roles during onboarding and training. For example, avatar-based VR tutorials, such as the *Immersed* demo that uses hand gestures to teach interaction techniques [172], as well as MR collaboration studies exploring avatar representation in shared tasks [175], demonstrate the diverse roles of avatars. Moreover, Hajahmadi et al. [176] investigated voice-only and embodied conversational agents in MR puzzle tasks and assessed usability with the SUS, reporting comparable outcomes across modalities. To ensure realistic design, the perception of virtual avatar hands has also been studied extensively [177, 178, 179]. A systematic review by Yang et al. [180] showed that conversational avatar embodiment is often realized as female, life-sized, full-body representations, with most applications found in training and education contexts. Evaluation approaches range from qualitative interviews to standardized questionnaires, including the SUS for usability as well as specific instruments for social presence and engagement.

Overall, prior research suggests that avatar-based guidance in immersive environments can enhance onboarding through multimodal interaction. Despite promising findings in other domains, systematic evaluations of avatar-based onboarding in real mHealth applications are still lacking, particularly for contexts such as smart insole systems.

Research Gaps

In summary, research on mHealth apps for smart insoles shows that usability issues related to correct use and data interpretation remain an ongoing challenge in patient–device interaction. Onboarding guidance and tutorials in apps provide a beneficial approach to support users during the onboarding process. Furthermore, the use of MR has shown potential to assist users through conversational avatars and visible gesture interactions, which could serve as a personal assistant during the onboarding of a new health app. This aligns with related work by Weichbroth [181], who pointed out that future research in the usability of mobile apps should investigate MR multimodal interactions. However, despite the potential of these technologies, it remains an open question how user guidance in mHealth apps can be effectively improved. Therefore, the thesis aims to fill this gap by advancing user interaction research and contributing to the fields of usability in mHealth applications and user experience design through avatar-based interaction in MR.

3.2.3 AI in Product Design

Recently, generative AI has been increasingly applied in user-centered design contexts. As highlighted in the motivation, the limited social acceptability of aesthetic designs in orthotic footwear restricts user adoption. To address this challenge, generative AI offers the potential for participatory design approaches that actively involve end users in the design process. However, further research is required to understand its impact on user perception and to determine how such approaches can be systematically integrated into established design processes to align with user preferences.

> This subsection is based on the related work section of the following journal publication and contains text reproduced verbatim under the terms of the CC-BY-4.0 License:
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> S. Resch, J. Schauer, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Improving Social Acceptance of Orthopedic Foot Orthoses Through Image-Generative AI in Product Design," Applied Sciences, vol. 15, no. 8, Art. no. 4132, 2025.

This section examines generative AI techniques, particularly text-to-image and image-to-image generation, and explores their potential applications in product design. Following this, we discuss how design influences the social perception of wearable devices. Finally, we summarize the identified research gaps and outline how our studies intend to contribute to this research field.

Generative AI: Image-to-Image and Text-to-Image Creation

Generative AI is a subset of AI that uses trained models from deep learning, a key component of machine learning, to generate new content from collected data. It employs trained multi-layer artificial neural networks that can identify patterns in datasets and make decisions for data generation. Therefore, the use of AI image-generation tools enables rapid prototyping from a simple sketch to a 3D prototype, whether virtual or physical [182]. One example is the image generation tool DALL-E from OpenAI, which is a separate tool from a Generative Pre-Trained Transformer (GPT) chatbot—ChatGPT—based on an Large Language Model (LLM). DALL-E is a transformer model [183] and enables text-to-image or image-to-image generation via text or image prompts.

Text-to-image generation transforms textual information into pixel-based image data. This method can be used in product development to create initial concepts based on specific product requirements [184]. However, achieving appropriate outputs requires an iterative approach to prompt engineering. Therefore, White et al. [185] proposed a prompt pattern catalog to improve prompt engineering with ChatGPT. They stated that specifying personas and templates allows the model to adopt specific roles and ensures that output quality and consistency are tailored to user needs. Furthermore, they discussed prompt patterns for visualizations that enable ChatGPT to generate specific textual outputs, which can subsequently be used to create visualizations with tools like DALL-E. Additionally, a study by Liu and Chilton [186] highlighted the importance of prompt engineering in text-to-image generation and proposed guidelines to achieve better outcomes.

While text-to-image generation uses text input to create a visual output, image-to-image generation requires visual data from an existing image to create a visual output. The use of image-to-image generation is based on Generative Adversarial Networks (GAN) [187], which enable image generation through the interaction of two neural networks: a generator and a discriminator. This technology can be used to refine initial sketches or create design variations, improving the iterative design process.

Generative AI in Product Design

Image-generative AI enables new possibilities for designers in product development through creativity-expanding iterations of design sketches [188]. These tools could be used in the styling process for various applications [189, 190, 191, 192]. However, the application of image-generative AI tools in the development of orthotic devices has not yet been realized.

Nonetheless, initial research projects demonstrate promising results from using AI throughout the design process, from initial concept to final manufacturing via 3D printing [30, 193]. For example, Popescu [30] proposed *orthosis_GPT*, a custom version of ChatGPT trained with predefined configurations to support the preliminary design of a wrist–hand orthosis. Based on an iterative design process driven by prompt engineering, the wrist–hand orthosis was finally 3D-printed. This workflow illustrates the potential of using AI in the creation of orthosis designs and subsequent processing for 3D printing. Furthermore, Bartlett and Camba [193] presented examples of generative AI in product design, showcasing the creation of shoes with DALL-E. The generated image output was used by designer Kedar Benjamin to overlay the topology onto this image with additional software and to create a 3D model for 3D printing. Finally, a significant limitation identified was that although the printed versions appeared similar, manual input from designers was necessary to create a topologically optimized 3D model. Additionally, Chiou et al. [188] conducted a study with six designers to explore the collaboration between designers and generative AI in creating new ideas. It was shown that AI image generation offers new possibilities for artistic expression and inspiration with great potential for future contributions in the design field. However, the study also highlighted future challenges and raised questions about copyright and social justice. In conclusion, these studies established a novel approach for creating AI-based designs that meet user requirements and enable the progression from initial design to final 3D printed prototypes.

Influence of Design on the Social Perception of Wearable Devices

In the field of HCI, the influence of factors such as social perception and comfort on wearable devices is extensively researched [194, 195, 196, 197]. Research demonstrates across multiple studies that design is crucial for the acceptance and continued use of wearables [198, 199, 200]. For instance, a focus group study by Canhoto and Arp [199] investigated the influence of factors for the sustainable use of various IoT health and fitness wearables. The study showed that designing distinctive devices enhances visibility, which influences user acceptance and encourages them to continue wearing the device. In addition, Adapda et al. [201] found that social acceptance can vary greatly depending on the type of smart device as well as the user group. Furthermore, Liao et al. [202] showed that smart insoles placed in shoes received the highest comfort ratings compared to other electronic wearable devices that were directly attached to the body. However, social acceptance can be influenced by various factors, including gender and cultural aspects [195]. Therefore, designers must consider a wide range of user preferences to develop widely accepted product designs.

In HCI research, several standardized questionnaires exist for the measurement of social impact from interactive and ubiquitous wearable technologies (e.g., Technology Acceptance Model (TAM) [203], Unified Theory for Acceptance and Use of Technology (UTAUT) [204], Stereotype Content Model (SCM) [205], or the Wearable Acceptability Range (WEAR) [206]). For instance, Davis et al. [203] proposed the TAM model, which was later updated to TAM 3.0 by Venkatesh and Bala [207], an enhanced version for a more comprehensive user understanding of technology adoption. In order to summarize various competing models with differing acceptance factors, Venkatesh et al. [204] compared eight models for measuring technology acceptance and validated UTAUT as a unified theoretical model. However, the literature indicates that while acceptance models such as TAM or UTAUT are suitable for evaluating technological acceptance itself, they are not "clearly applicable" to technologies that are directly worn on the body [198].

For this reason, the WEAR scale developed by Kelly [206, 208] serves as an effective tool for measuring the social acceptability of wearable devices or prototypes. For example, Kelly and Gilbert [198] applied the WEAR scale in their study, demonstrating that medical necessity enhances the social acceptability of wearable devices, as it underscores the wearer's dependence on the device. However, they also highlighted the discrepancy in the literature between the stigmatization of medical devices and the recognized medical necessity of such assistive technologies. To address existing stereotypes and social perceptions, the SCM questionnaire by Fiske et al. [205] could be utilized, as it assesses these factors across two perceived dimensions: competence and warmth [209]. Previous work has applied this model to evaluate mobile devices [210] and has confirmed its effectiveness across different cultures [211]. A study by Sehrt et al. [24] utilized the SCM and WEAR scales to investigate the social acceptability of wearable devices based on their body location. The results showed that placement on the ankle received the lowest SCM ratings, indicating poorer acceptability compared to upper body locations. These findings support the statement that the social perception of wearable devices varies for different body locations [212, 213, 214]. Overall, the use of these questionnaires revealed current problems with the social acceptance of various devices and that better design is needed to achieve greater acceptance in the future.

Research Gaps

AI image generation opens up new possibilities for creativity, assisting designers in progressing from sketches to 3D models through text-to-image and image-to-image generation techniques. Initial studies have demonstrated that AI can be used effectively in product design by enabling the development of concepts such as wrist–hand orthoses and the subsequent 3D printing of prototypes. However, the impact of AI-generated aesthetic designs on the social acceptance of orthopedic foot orthoses remains unknown.

To evaluate stereotypes and social acceptance of wearables, the SCM and WEAR scales are often used as validated questionnaires. Previous research using these questionnaires has shown that wearables attached to the ankle typically receive lower acceptance rates compared to those worn on other body parts. Consequently, it remains unclear how orthopedic footwear is perceived by users and which aesthetic design adjustments are necessary to enhance acceptance for future smart developments.

By addressing these research gaps, we aim to contribute to the field of socially acceptable wearable orthopedic footwear design. Additionally, we aim to explore the potential of AI in future design processes to develop aesthetically acceptable design concepts. To this end, we conducted two studies to investigate the effects of AI-generated designs on the social acceptance of orthopedic footwear.

3.3. Chapter Summary

In summary, this chapter established the theoretical background in the areas of assistive technology and orthotic devices, XR technologies, and user-centered design. Key term definitions related to orthopedic footwear, VR, and MR were introduced and serve as a conceptual foundation for this thesis. Furthermore, existing user-centered design approaches were presented to outline relevant theoretical concepts and methodological frameworks.

Building on this foundation, the related work section provided an overview of recent research across three thematic areas. In the context of smart assistive devices, previous work has demonstrated the existence of several systems for foot augmentation, such as smart insole systems for measuring plantar pressure and gait metrics. However, current solutions are predominantly limited to non-gait-affecting footwear and are rarely integrated into comprehensive assistive systems.

From an application perspective, research on XR-based training and feedback systems for smart devices indicates considerable potential. In particular, immersive gait training scenarios using visual feedback modalities have been shown to influence user performance. However, the effects of such training approaches for gait-affecting footwear, such as foot orthoses, remain largely unexplored. Similarly, the combined influence of visual feedback and environmental factors on both objective performance and subjective training outcomes has not yet been sufficiently investigated. In addition, avatar-based guidance in immersive environments may enable embodied interaction and improve usability in interactions with digital health systems. In the context of MR-supported onboarding for mHealth applications, such approaches may enhance the user experience, although systematic research in this area remains limited. Recent work further highlights that multimodal interactions in MR should be explored in future usability research for mobile applications [181].

Moreover, recent studies have shown that generative AI enables rapid image generation and has already been explored in applications such as fashion footwear design and hand orthosis prototyping. Despite this potential, it remains unclear how AI-supported design influences user acceptance and perception, and how such approaches can be systematically integrated into product development processes for orthotic footwear.

Taken together, these findings highlight the current research landscape across the three thematic areas addressed in this thesis and reveal several open challenges that directly motivate the research contributions presented in the following chapters.

CHAPTER 4

Smart Devices: Development and Validation

Assistive technology, focusing on sensorized walking aids and footwear, represents a rapidly growing research field with broad applications in human-device interaction. However, a comprehensive overview of these devices is still missing. It also remains unclear how an integrated system of orthotic devices can be effectively designed with sensors and feedback mechanisms, and how such systems influence user interaction.

To address RQ1–3, this chapter summarizes the results of two journal publications. Section 4.1 provides a state-of-the-art overview of intelligent walking aids, addressing RQ1 to identify current trends, target user groups, functional capabilities, and gaps in the literature. These insights serve as a foundation for the subsequent development of novel smart device concepts. Building on these findings, Section 4.2 presents the development and validation of a comprehensive multimodal smart orthosis and connected walking aid system (see Figure 4.1). This study addresses RQ2 and RQ3, aiming to understand how a closed-loop biofeedback system across devices can influence user perception, usability, and interaction. An experimental user study was conducted to evaluate system feasibility, feedback effectiveness, and user experience. Taken together, this work demonstrates the potential of smart devices to enhance mobility and interaction, while providing empirical evidence to guide future device design and application in assistive technology.

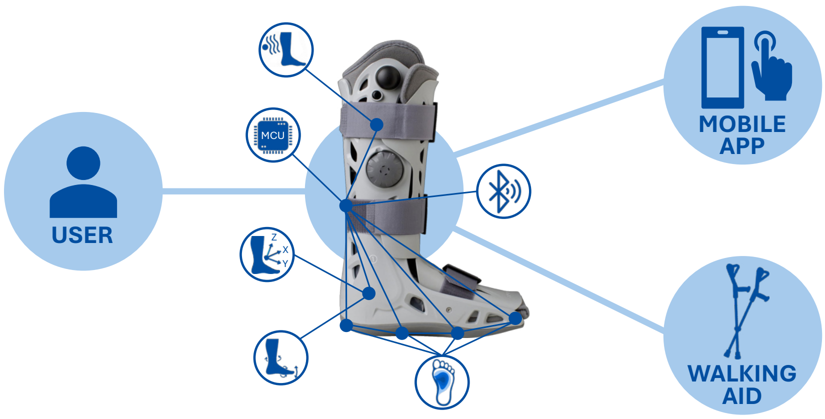


Figure 4.1: Schematic overview of the developed smart device system.

4.1. Study I: Scoping Review – Smart Walking Aids

The following section presents a literature review conducted in accordance with the PRISMA-ScR guidelines, providing a systematic overview of recent developments in sensorized walking aids, including device and sensor types as well as application fields. Figure 4.2 illustrates a graphical abstract that outlines the overall structure of the synthesis. The main findings identify current research trends and gaps, which serves as a basis for future developments in user-centered assistive technologies by highlighting key challenges and research opportunities.

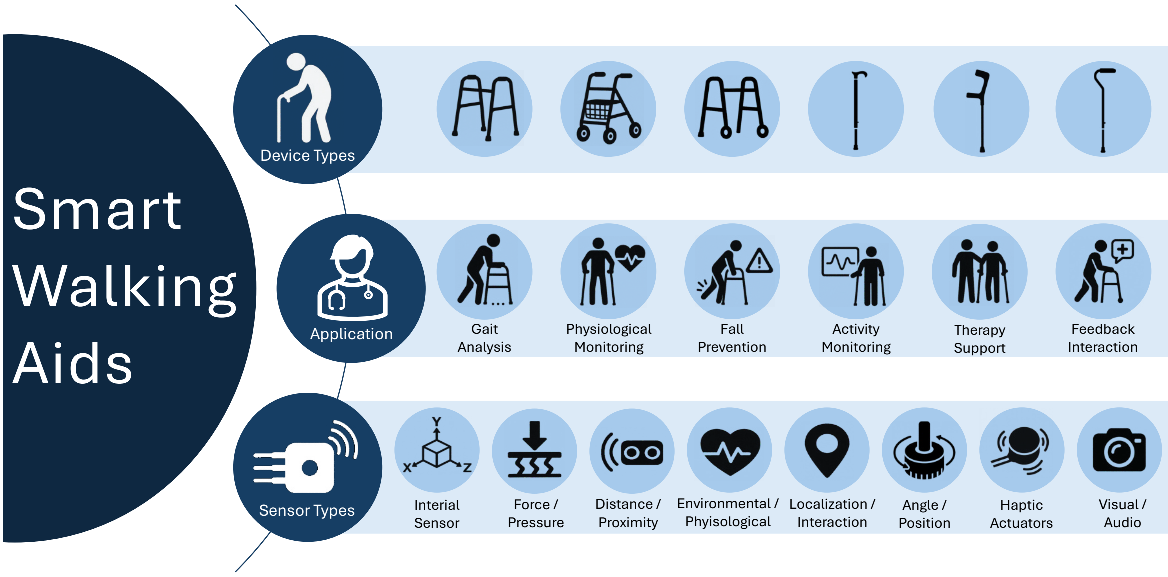


Figure 4.2: Graphical abstract of the review synthesis, showing the clustering of device types, application contexts, and sensor types.

This section is based on the following journal publication and contains text reproduced verbatim under the terms of the CC-BY-4.0 License:

S. Resch, A. Zirari, T. D. Q. Tran, L. M. Bauer, and D. Sanchez-Morillo, "Smart Walking Aids with Sensor Technology for Gait Support and Health Monitoring: A Scoping Review," Technologies, vol. 13, no. 8, Art. no. 346, 2025.

4.1.1 Introduction

Mobility impairments, particularly among older adults and individuals with long-term health conditions, are a growing global concern. The Global Report on Assistive Technology from the WHO and UNICEF [6] highlights that over 2.5 billion people (approximately one in three individuals globally) currently require at least one assistive device, a number that is expected to rise due to the aging population. The report emphasizes the urgent demand for accessible, technology-driven solutions and provides recommendations to foster research and innovation in this field.

Assistive technologies cover a wide range of products designed to maintain independence and improve health across diverse populations [6]. In the domain of mobility support, body-worn devices such as orthoses are often used together with external devices like canes, crutches, or walkers. Although these conventional walking aids are well-established in clinical and everyday settings, they are typically passive and lack adaptability to the individual needs of users.

Recent advancements in sensor technology, embedded systems, and AI have enabled the development of a new generation of smart walking aids that extend beyond basic support. Overall, current research highlights the benefits of smart walkers for monitoring balance and stability conditions, aiming to improve quality of life and prevent falls and injuries [215, 216]. A review by Martins et al. [217] emphasized the substantial potential of intelligent walkers for rehabilitation, as these systems can support various functions such as gait monitoring [218], obstacle detection [219], and real-time feedback [220]. Depending on the intended function, different sensor and actuator technologies are employed, such as FSRs for gait event detection [115], IMUs for motion sensing and fall detection [221], ultrasonic sensors for obstacle detection [222], and GPS modules for localization [223].

Based on the measured sensor input, active user feedback can be provided through a range of output modalities, including acoustic, haptic, and visual cues. Auditory and tactile feedback combinations are frequently used to support visually impaired users [224]. For example, haptic feedback can be provided via vibration in cane handles to guide postoperative weight-bearing [225]. In another case, visual cues such as a projected 2D laser line from a modified cane have been shown to improve gait initiation in individuals with Parkinson's disease [226].

Beyond local feedback, smart walking aids can be embedded into IoT infrastructures [227] to enable data transmission for enhanced gait monitoring and real-time alerts. In this context, AI-based methods such as Machine Learning (ML) have been applied for fall detection and motion classification based on accelerometer and gyroscope data [228]. For example, Wang et al. [229] proposed an intelligent walking stick capable of human motion analysis and imbalance classification, which communicates with a mobile application. Additionally, the interaction with other wearable systems, such as smart insoles [113] or foot orthoses [122], enables multimodal data fusion and cross-device feedback. For instance, Resch et al. [122] proposed a forearm crutch system that delivers vibration alerts triggered by pressure thresholds detected in a connected smart foot orthosis.

Although numerous studies have investigated specific aspects of smart walking aids for several use cases, there is currently no comprehensive synthesis of the field. In particular, there is a lack of systematic analysis regarding the types of sensors used, functional purposes, health-related contributions, user-specific applications, and the role of AI in extending device capabilities. However, such an overview is essential to inform evidence-based development, support cross-disciplinary innovation, and guide future implementation strategies. To address this gap and provide a structured overview, we conducted a systematic review of the literature using PRISMA-ScR [230].

The central aim of this review is to provide a structured overview of new trends in sensor-based walking aids developed to support the mobility of individuals with walking impairments. In particular, we aim to categorize the types of sensors used, identify their key functionalities, and examine the targeted user populations and domains of application. The Main Research Question (MRQ) guiding this review is:

- MRQ: What types of intelligent walking aids equipped with sensor technologies have been developed to support the mobility of individuals with walking impairments, and in which domains of application are they used?

In addition, we define two Specific Research Questions (SRQs):

- SRQ1: Which sensor technologies and AI-based approaches are integrated into smart walking aids to enable personalized functional support, health monitoring, and user-specific feedback mechanisms?
- SRQ2: Which target user groups are addressed by these devices, and what key functionalities do they offer to meet the specific mobility-related needs of these populations?

By synthesizing findings from the existing literature, we aim to identify technological trends, current advancements, and existing limitations in intelligent walking aids. We provide a comprehensive overview of sensor-based assistive systems and outline future directions to enhance mobility support for individuals with physical impairments. Based on our findings, we derive recommendations for researchers and practitioners. This work contributes to the fields of assistive technology, sensor technologies, and HCI.

4.1.2 Method

We conducted a systematic literature review using the PRISMA-ScR approach. The PRISMA-ScR checklist with 22 items was used to guide the identification, selection and synthesis of relevant articles.

Eligibility criteria

As eligibility criteria, the following inclusion criteria were defined: (1) The study must investigate smart, intelligent, or instrumented walking aids (e.g., canes, sticks, walkers, or crutches) with integrated sensor technology; (2) The walking aid must be user-controlled and serve a health-related purpose (e.g., fall prevention, gait rehabilitation, mobility or balance assistance); (3) Only English-language journal and conference papers were considered.

Exclusion criteria were as follows: (1) Non-original articles (e.g., literature reviews, meta-analyses, or editorials) and non-peer-reviewed conference papers (e.g., abstracts, posters) were excluded; (2) Studies focused solely on comfort, transportation, or navigation assistance and that did not directly relate to health outcomes were excluded. In addition, studies that exclusively targeted people with visual impairments were not included; (3) Autonomous systems without active user guidance (such as exoskeletons or humanoid robots), motorized systems, or wearable devices (such as orthotics or prosthetics) were excluded.

Database and Search Strategy

Three scientific databases were selected as information sources: ACM Digital Library, IEEE Xplore, and Web of Science. These were chosen because they comprehensively cover technical and interdisciplinary peer-reviewed research domains related to the development and evaluation of assistive technologies, particularly in engineering, computer science, and HCI. This focus aligns with the objective of this review, which is to investigate technological innovations and sensor-integrated system developments in the context of smart walking aids. The literature search in all three databases was conducted on 9 January 2025.

The search strategy was designed to capture studies at the intersection of assistive systems, sensor technology, walking aid devices, and health-related applications. To ensure thematic relevance, the search was explicitly restricted to health-related terms (e.g., walking impairments, mobility support, gait rehabilitation, patient support, foot conditions), as well as to device types such as walkers, canes, sticks, and crutches. In addition, keywords such as "sensor" and specific assistive technology terms (such as assistive device, assistive system, assistive technologies, and walking aid) were incorporated to identify studies focusing on these key categories of devices. Boolean operators were used to logically combine all relevant search terms. The search was applied to the full-text fields in all three databases. The specific search query for IEEE Xplore is documented below:

("Full Text Only": "assistive device" OR "assistive system" OR "assistive technolog" OR "walking aid") AND ("Full Text Only": sensor*) AND ("Full Text Only": walker* OR cane OR "walking stick" OR crutch) AND ("Full Text Only": "walking impairments" OR "mobility support" OR "gait rehabilitation" OR "patient support" OR "foot conditions").*

Full search strategies for ACM Digital Library and Web of Science are provided in the Appendix A.

Data Charting and Synthesis

All retrieved records were imported into the Zotero reference management software [231]. Duplicate entries were identified automatically and removed manually. The selection and data collection process was conducted by three reviewers (A.Z., T.D.Q.T., and L.M.B.), who divided the records into three equal parts. Each record was independently screened by two of the three reviewers, following the four-eyes principle, to ensure consistency and minimize bias. In cases of disagreement, the third reviewer counter-checked the record to resolve the conflict and reach consensus. After each screening phase, the first author (S.R.) independently reviewed all records to ensure methodological consistency, validate screening decisions, and document the process for transparency and accuracy. The inter-rater reliability was assessed using Cohen's κ and reached a substantial level ($\kappa = 0.72$), according to the definition by Landis and Koch [232].

The screening procedure followed a three-phase approach: First, titles and abstracts were screened based on the predefined eligibility criteria, with a focus on keyword relevance and study scope. Second, the included articles were screened for full text and examined in detail to determine whether they met the eligibility criteria. Third, relevant data were extracted from the included studies and cross-checked for accuracy.

For each included study, the following data items were extracted to enable a structured synthesis: device type, sensor technology (including sensor type, placement, and specifications), device communication, use of AI, target population, key functionality, sample size, objective, and results. In addition, general metadata were collected to support the descriptive analysis, including the first author, year of publication, and the country of origin of the study. No assumptions or simplifications were made beyond those explicitly stated in the original articles. Where necessary, unclear or missing data were noted accordingly in the table. No formal critical appraisal was conducted, as the aim of this scoping review was to map existing research rather than assess study quality.

For data synthesis, a deductive thematic analysis [233] was conducted. The initial synthesis process was performed by the first author (S.R.) and subsequently discussed and cross-checked with a second reviewer (D.S.-M.) to ensure consistency in the thematic interpretation. The predefined themes were derived from the research objectives and corresponded to the data items identified during extraction. The discussion follows a narrative synthesis structure, integrating the findings within this thematic framework. Structured summaries were provided for each theme and key study characteristics were presented in tabular format to facilitate comparison across studies. Descriptive statistics were applied to identify trends and to provide an overview of relevant study characteristics. However, no statistical synthesis was performed, following the methodological framework of scoping reviews.

4.1.3 Results

The selection process was documented using the PRISMA flow diagram [234], generated with the R package PRISMA2020 (version 1.1.1) proposed by Haddaway et al. [235], see Figure 4.3.

A total of 829 records were identified through database searches across the ACM Digital Library, IEEE Xplore, and Web of Science. After the removal of one duplicate, 828 records remained for title and abstract screening. During this screening phase, 719 records were excluded because they did not match the study scope or lacked relevant keywords in their titles or abstracts. Subsequently, 109 full-text articles were assessed for eligibility. A total of 74 articles were excluded for various reasons, as they did not meet the inclusion criteria, as shown in Figure 4.3. Finally, 35 studies met all inclusion criteria and were included in the synthesis.

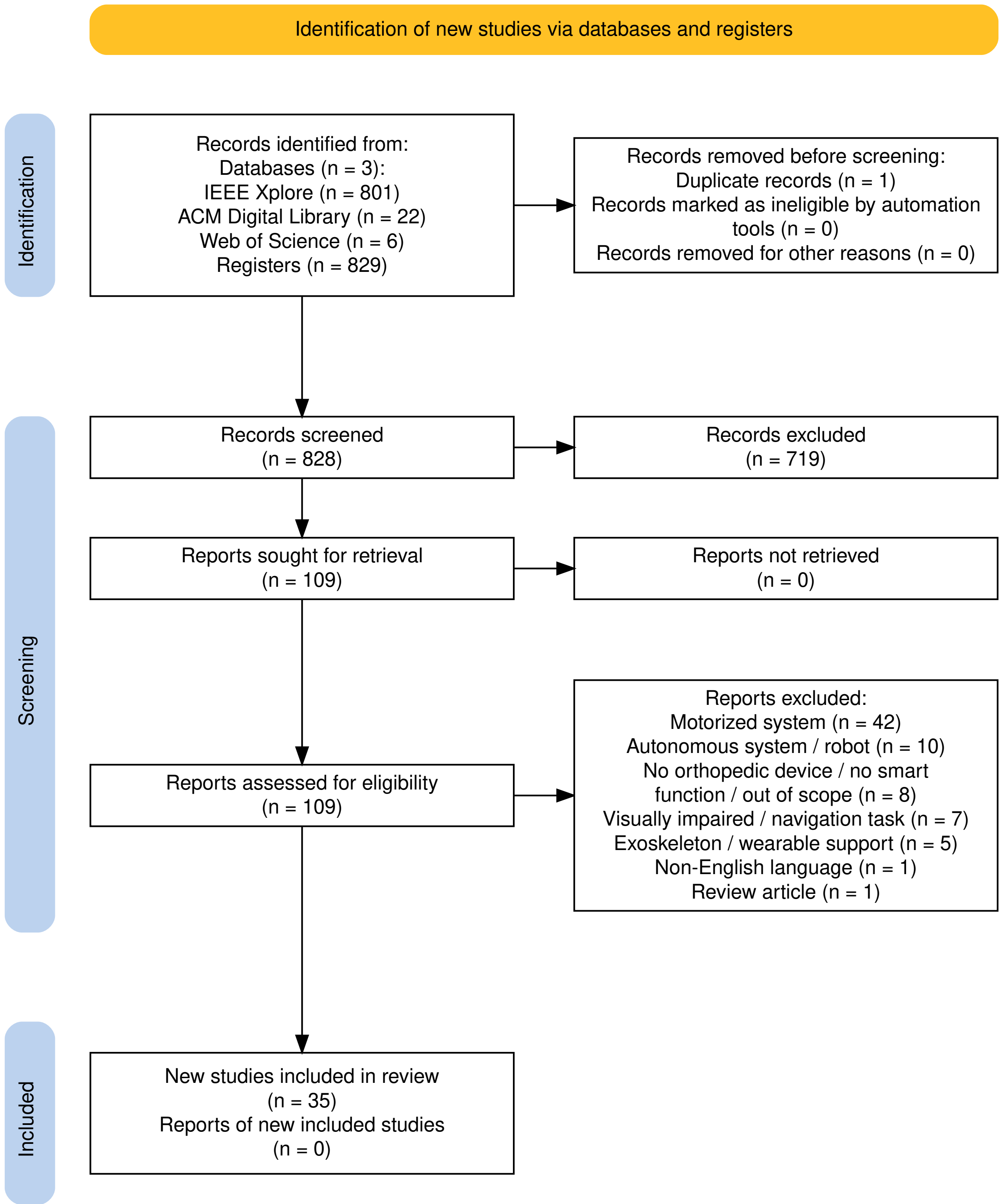


Figure 4.3: PRISMA flowchart of the article selection process, covering the identification, screening, and inclusion phases.

Study Characteristics

This section presents the main characteristics of the included studies, beginning with an overview of their application contexts, followed by a detailed synthesis of study populations, aims, and evaluation settings.

Application Context The main application areas covered by the included studies are summarized in Table 4.1. Several studies address multiple application domains and therefore appear in more than one category. The frequencies are presented descriptively, including the percentage of studies in which each application context was identified.

Table 4.1: Overview of application areas, their frequency in the included studies, and corresponding references, grouped by main application categories.

Application	Frequency	%	References
Gait Analysis	32	91.4	[236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266]
Fall Prevention / Detection	11	31.4	[239, 244, 245, 252, 262, 264, 265, 266, 267, 268, 269]
Therapy Support	8	22.9	[240, 241, 248, 250, 253, 260, 263, 265]
Feedback Interaction	7	20.0	[237, 244, 253, 265, 268, 269, 270]
Activity Monitoring	6	17.1	[239, 241, 247, 254, 255, 256]
Physiological Monitoring	5	14.3	[238, 248, 252, 258, 265]
Total	**69**		

Note: Some studies cover multiple application areas and appear in more than one category.

Synthesis of Study Characteristics The 35 included publications come from 16 different countries and were published between 2008 and 2024. Most studies were conducted in Europe, North America, and Asia. Table 4.2 summarizes the key characteristics of the selected studies, such as target population, key functionality, sample size, objective, and outcomes.

The primary aims across studies included gait monitoring and analysis, fall detection, force or pressure measurement, and the integration of feedback modalities such as haptic or remote monitoring. The most frequently targeted groups were older adults (10 studies) [246, 264, 251, 244, 238, 236, 239, 255, 266, 258], followed by a broad group of rehabilitation patients (9 studies) [265, 245, 268, 243, 240, 249, 237, 259, 270], and users with mobility or gait impairments (8 studies) [261, 256, 257, 241, 253, 254, 247, 252]. Additionally, specific use for physical therapy patients was reported in four studies [250, 263, 260, 248], and users at risk of falling were targeted in three articles [262, 267, 269]. Parkinson's patients were addressed in one study [242].

Sample sizes varied widely, ranging from single-case studies to group studies with up to 42 participants or technical validation without participants. The majority of studies included between 7 and 20 participants, often healthy, but also patients with several conditions. Demographic data varied widely, with mean participant ages spanning from 22 to over 80 years.

To further characterize the reviewed studies, we identified four types of research focus across the included studies:

- System development: [242, 245, 248, 252, 268, 265, 269, 258, 260]
  These articles present systems targeting a specific user group, but without comprehensive empirical validation involving participants. They typically focus on functionality demonstration and proof-of-concept implementations.
- Technical validation studies: [236, 237, 239, 241, 243, 250, 253, 270, 255, 256, 257, 261, 264]
  These studies focus on the development and validation of prototypes, often conducted in laboratory settings with healthy participants or a small number of users.
- Application-oriented studies: [266, 238, 246, 247, 267, 251, 254, 262, 263]
  These articles evaluate the developed systems with the intended user population, aiming to demonstrate feasibility and functionality in context.
- Patient-centered pilot studies: [240, 244, 249, 259]
  These studies assess the impact of the system within structured interventions involving affected users, and often serve as preliminary investigations toward clinical integration.

Table 4.2: Overview of study characteristics grouped by publication year.

Year	Authors	Country	Target population	Key functionality	Sample size	Objective	Results
2024	Anushree et al. [265]	India	Rehabilitation patients	To support balance, detect falls, monitor physiology, and provide emergency feedback	n = not specified	Develop a smart walker to support balance, posture, and feedback during physical therapy.	Prototype developed; no formal validation
2024	Postolache et al. [245]	Portugal	Rehabilitation patients	To detect and classify gait abnormalities	n = not specified (healthy volunteers)	Analyze gait patterns (normal/abnormal) using spectrogram-based metrics.	Successful gait abnormality classification using spectrogram-based features
2024	Cavagila et al. [262]	Italy, UK	Fall risk users	To detect falls, activate walker brakes, and alert caregivers	n = 6 (healthy, mean age 30 ± 8.3 y)	Design a multi-sensor system for fall and near-fall detection in smart walkers.	High fall detection accuracy; pre-fall prediction less reliable
2023	Arcobelli et al. [261]	Italy	Gait-impaired users	To detect stance phases and visualize data in real time	n = 1 (male, 29 y)	Demonstrate stance and force analysis with the mCrutch during walk tests.	Crutch stance segmentation accuracy: 94%
2023	Padmavathi et al. [268]	India	Rehabilitation patients	To correct posture, detect falls, and avoid obstacles via haptic feedback	n = 1 (not specified)	Develop a smart crutch with sensors for posture and safety monitoring.	Proof-of-concept; sensor data successfully validated
2022	Zhou et al. [246]	China	Older adults	To analyze gait (step count, stride, length, speed)	n = 6 (2 elderly with assistance; 4 healthy: 2 young, 2 elderly)	Implement a smart stick with 9-axis sensor for gait analysis.	Step count 100%; stride/step metrics > 94% accuracy
2022	Batoca et al. [250]	Portugal	Physical therapy patients	To assess gait and store data in the cloud with web-based visualization	n = 2 (healthy, 1m/1f, ages 24 & 25 y)	Collect and store gait and balance data via smart crutch during rehab.	Successful collection of force and orientation data
2022	Narváez et al. [243]	Spain	Rehabilitation patients	To identify and classify crutch gait patterns	n = 20 (healthy, 8f/12m)	Recognize gait patterns in crutch users using sensors and ML.	Gait classification accuracy: 88–89% (GPS, Artificial Neural Network (ANN))
2021	Valsangkar et al. [254]	Canada	Mobility-impaired users	To segment and analyze the Timed Up and Go (TUG) test	n = 16 (musculoskeletal injuries)	Assess sensor-based cane data for clinical mobility assessment.	High segmentation accuracy; low classification errors (Linear Discriminant Analysis/Artificial Neural Network (ANN))
2021	Ribeiro and Santos [264]	Portugal	Older adults	To detect abnormal gait and fall-related instability	n = 11 (healthy, mean age 24.2 ± 2.6 y, range 22–29 y)	Detect falls in real-time using instrumented cane with ML and Finite-State Machine (FSM).	Fall detection accuracy > 99%; phase classification ≈96.5%
2020	Mesanza et al. [256]	Spain	Gait-impaired users	To classify physical activities (e.g., walking, standing, stairs)	n = 11 (healthy, 4f/7m, age 24–48 y)	Classify physical activities from wearable sensor data using ML.	Feature selection achieved 92–97% classification accuracy
2020	Zambrano et al. [263]	Peru, Canada	Physical therapy patients	To monitor gait speed and handle pressure in real time	n = 10 (patients, not further specified)	Develop a low-cost walker to monitor gait parameters.	Session duration reduced from 25 to 5.2 minutes

Table 4.2: (continued)

Year	Authors	Country	Target population	Key functionality	Sample size	Objective	Results
2019	Ballesteros et al. [251]	Sweden, Spain	Older adults	To detect dynamic weight-bearing trends	n = 8 (elderly, cane users, 6m/2f, mean age 82.1 ± 6.0 y)	Validate sensor system for anomaly detection in cane load.	Sensor outputs correlated with user physical status
2018	Wade et al. [244]	USA	Older adults	To monitor axial load, grip pressure, and object proximity in real time	n = 9 + 18 (Study 1: 3m/6f; Study 2: 8m/10f; active cane users)	Assess feasibility of fall risk estimation via sensorized cane.	Significant correlation between grip pressure and gait performance
2018	Frango and Postolache [260]	Portugal	Physical therapy patients	To analyze force and orientation, and support therapists via mobile app	n = not specified (several healthy volunteers, various ages)	Support therapists with a mobile app for rehab progress monitoring.	Functional system; no formal validation conducted
2018	Seylan et al. [257]	Turkey	Gait-impaired users	To estimate 3D ground reaction forces	n = not specified	Estimate ground reaction forces using low-cost sensor system.	Prediction errors < 7% (static), < 8% (dynamic)
2018	Viegas et al. [238]	Portugal	Older adults	To assess gait using load and heart rate data	n = 1 (impaired gait, right lower limb injury)	Validate "Spy Walker" for gait monitoring during rehabilitation.	Step classification successful; potential for rehab assessment
2018	Ojeda et al. [236]	México, Spain	Older adults	To classify gait patterns and estimate walking age from force data	n = 42 (age range 22–94 y)	Predict walking age via unsupervised learning from gait data.	Clustering revealed four distinct gait types linked to age/speed/force
2018	Ballesteros et al. [267]	Spain	Fall risk users	To estimate fall risk from spatial and force sensor data	n = 10 (physical/neurological disabilities, 3m/7f, mean age 61.4 y, range 46–74 y)	Assess feasibility of smart rollator for fall risk prediction.	Fall risk score correlated significantly with Tinetti scores and speed
2018	Gill et al. [239]	Canada, India	Older adults	To monitor gait, activity levels, and walking environment	n = 10 (healthy, 8m/2f, mean age 22.9 ± 2.3 y)	Design IoT smart walker for monitoring gait and environmental data.	System identified gait events and periods of rest/activity
2017	Mekki et al. [242]	Italy, France	Parkinson patients	To extract gait parameters in real time (e.g., duration, asymmetry)	n = not specified (Parkinson participants)	Create an instrumented cane for gait analysis in Parkinson's patients.	Prototype developed; no formal validation
2017	Gill et al. [247]	Canada	Mobility-impaired users	To measure loading, mobility, and stability	n = 30 (healthy adults, 20m/10f, age 20–60 y, mean 24.2 ± 7.1 y)	Design smart cane for unobtrusive gait monitoring and event detection.	Gait event differences identified between normal and perturbed conditions
2017	Domingues et al. [252]	Brazil	Mobility-impaired users	To detect falls, monitor gait, and assess physiological signals	n = not specified	Develop sensor-integrated cane for mobility and health tracking	Strong correlation between behavior and sensor data
2017	Ballesteros et al. [240]	Spain	Rehabilitation patients	To estimate balance scales and extract spatiotemporal gait features	n = 19 (physical/cognitive disabilities, 6m/13f, mean age 68 ± 9.3 y, range 50–80 y)	Predict Tinetti scores using rollator sensor data.	Predicted vs. actual Tinetti scores: high correlation

Table 4.2: (continued)

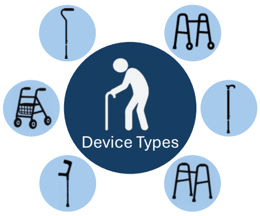
Year	Authors	Country	Target population	Key functionality	Sample size	Objective	Results
2017	Koziol et al. [269]	USA	Fall risk users	To deliver reminders and reduce fall risk	n = 0 (no subjects involved)	Evaluate "Reminding Walker" for posture detection and cueing	Proof-of-concept confirmed; reminder cues detected successfully
2016	Ballesteros et al. [249]	Spain	Rehabilitation patients	To estimate gait and balance using spatiotemporal parameters	n = 19 (physical/cognitive disabilities, 6m/13f, mean age 67.5 ± 9.7 y, range 46–80 y)	Estimate Tinetti scores in real-time from gait parameters.	Principal Component Analysis + Ridge Regression: R^2 = 0.92 (gait), 0.88 (balance)
2015	Wade et al. [241]	USA	Gait-impaired users	To assess timing, speed, acceleration, and angular velocity; therapist feedback	n = 7 (4f, mean age 27 ± 3.9 y; 3m, mean age 27.3 ± 4.5 y)	Support gait classification via sensor-based instrumented cane.	Gait classification accuracy > 95% (decision tree), 84% (Artificial Neural Network (ANN))
2015	Sardini et al. [237]	Italy	Rehabilitation patients	To measure axial/shear forces and provide vibratory feedback	n = 10 (healthy, mean age 30 y, range 28–55 y)	Quantify upper-limb input using wireless instrumented crutches.	Axial force error ∼9 N; shear ∼5 N; angular error ≈1°
2015	Ballesteros et al. [259]	Spain	Rehabilitation patients	To monitor gait parameters and detect abnormalities	n = 9 (physical/cognitive disabilities, 6f/3m, mean age 68.2 ± 14.6 y, range 45–86 y)	Estimate clinical gait parameters using sensorized systems.	Accurate extraction of cadence, gait time/length, and load metrics
2015	Postolache [248]	Portugal	Physical therapy patients	To monitor gait, posture, and force interaction via smart aids	n = not specified	Enable real-time rehab assessment using smart walking aids.	Feasibility of multi-sensor integration successfully demonstrated
2015	Gerena et al. [270]	USA	Rehabilitation patients	To measure upper-limb loading	n = 3 (2 healthy, 1 patient)	Monitor upper-limb load with low-cost instrumented walker.	±1 lb force accuracy; synchronized with motion capture and EMG
2014	Culmer et al. [253]	UK	Gait-impaired users	To assess kinematic and kinetic gait properties	n = 1 (female, 43 y, multiple sclerosis, walking aid on right)	Provide quantitative gait feedback via smart walking aid.	Accurate orientation/load data; aligned with clinical observations
2014	Chalvatzaki et al. [255]	Greece	Older adults	To recognize actions, gestures, and gait cycles	n = 10 (healthy subjects)	Design walker with user localization, pose, and intention recognition.	Accurate recognition of actions and gait segmentation (Hidden Markov Model (HMM))
2013	Kikuchi et al. [266]	Japan	Older adults	To support passive motion control via line-tracing navigation	n = 3 (disorders, ages 82, 90, 100; 2m/1f)	Evaluate line-tracing controller in an intelligent walker.	Stride width/length improved; positional errors reduced by controller
2008	Chan and Green [258]	Canada	Older adults	To monitor distance, speed, acceleration, handle force, and vital signs	n = 0 (no empirical study reported)	Enhance rollators with sensors for real-world gait monitoring.	Subsystems showed monitoring potential; complexity varied

Technological Aspects

This section introduces the categorized device types and corresponding sensor technologies, detailed in the subsequent subsections and concluded by a synthesis of key technological aspects.

Device Types The included studies covered a diverse range of smart assistive devices, which are descriptively summarized in Table 4.3.

Table 4.3: Overview of device types, their frequency in the included studies, and corresponding references.



Device Type	Frequency	%	References
Walkers	11	31.4	[238, 239, 245, 248, 262, 263, 265, 266, 268, 269, 270] *
Canes	8	22.9	[241, 242, 244, 247, 251, 252, 254, 264]
Rollators	8	22.9	[236, 240, 248, 249, 255, 258, 259, 267] *
Forearm crutches	8	22.9	[237, 243, 248, 250, 256, 257, 260, 261] *
Sticks	2	5.7	[246, 253]
Total	**37**		

** Note: Article [248] addresses three types of devices and is therefore listed in multiple categories.*

For analytical clarity, the devices were grouped into three main categories: walkers/rollators (n = 19), canes/sticks (n = 10), and forearm crutches (n = 8). Based on their form factor and intended use, walkers and rollators were combined into one category due to their comparable use cases and technical components. Similarly, canes and sticks were grouped based on shared sensor configurations and form factors. This categorization enables a structured comparison of technological components, feedback modalities, and target populations across different device types. The following paragraphs synthesize each category in detail, with a focus on the application context, technological components, and the use of AI for data processing.

Walkers and Rollators Eighteen of the included studies [238, 236, 240, 239, 248, 245, 262, 263, 265, 266, 268, 269, 270, 249, 255, 258, 259, 267] published between 2008 and 2024 focused on smart walkers or rollators designed to support individuals reliant on walking aids such as older adults, individuals with mobility limitations, or an increased risk of falling. The primary functions of these systems include gait monitoring, weight distribution analysis, and fall detection.

Ten studies investigated various walker configurations: five studies examined four-leg walkers [238, 265, 245, 268, 248], four studies focused on front-wheel walkers [239, 263, 270, 269], and one study investigated a four-wheel walker [262]. These devices are typically equipped with a combination of sensors, most commonly force or pressure sensors paired with IMUs. In addition, various other components are used, such as infrared or ultrasonic sensors, accelerometers, rotary encoders, GPS modules, or dry Electrocardiogram (ECG) electrodes integrated into the handgrips. Force sensors and load cells are commonly used to quantify physical load, grip forces, and weight distribution during ambulation. These signals are frequently combined with IMUs to generate a more holistic gait analysis. For instance, the combination of force sensors and accelerometers enables step time, stride length, and stop movements to be derived from linear acceleration signals. Such configurations facilitate basic gait event detection and activity classification and may be extended to assess gait symmetry, turning behavior, postural transitions, and stability.

All these systems enable wireless communication, using either Wi-Fi or Bluetooth-based solutions. Seven systems offer user feedback via visual, acoustic, or haptic cues. Visual feedback was provided in six systems (e.g., via smartphone apps, indicator lights, or Liquid Crystal Display (LCD) displays showing force distribution and pulse rate) [238, 265, 268, 269, 270, 248], acoustic feedback (e.g., sound alerts for nearby assistance) in three [265, 262, 269], and haptic feedback (e.g., vibration alerts for excessive handle force) in two [268, 265].

Six studies [266, 236, 267, 240, 249, 259] focused on the iWalker platform, which is based on previous publications and often referenced as a rollator-based system. These devices typically included rotary encoders mounted in the wheels and force sensors on the handlebars, as well as additional components depending on the specific application. Five of these devices did not incorporate user feedback mechanisms and did not specify the type of system communication. Only one article [266] referred to a cable-based connection to a mobile PC, which was also used to provide visual feedback. None of the iWalker-based systems applied AI methods, but instead relied on conventional techniques such as regression or clustering for basic prediction or classification tasks.

Two additional studies [255, 258] investigated rollators equipped with more extensive sensor arrays. One system [258] utilized force sensors, an accelerometer, a hall sensor, a pressure sensor, and a PPG sensor. The second system [255] integrated laser range sensors, MEMS microphones, a GoPro HD camera, two Kinect modules, rotary encoders, and 6-Degrees of Freedom (DOF) force/torque sensors. This device is the only walker-based system that explicitly employed ML, specifically Hidden Markov Models (HMMs) and Support Vector Machines (SVMs), for automatic activity recognition. In addition, three articles [265, 245, 268] mentioned the potential application of ML or Deep Learning (DL) methods in future work. In total, four studies [240, 249, 262, 236] have explored the use of predictive or regression-based methods for assessing gait patterns and estimating fall risk.

Forearm Crutches Smart crutches were described in eight [243, 237, 257, 260, 256, 250, 261, 248] of the included studies published between 2015 and 2024. These devices primarily focus on gait support within physical rehabilitation for patients with lower-limb impairments. The predominant functionality identified is gait assessment, aimed at enhancing mobile health monitoring for users. Five studies integrated IMU sensors, including two devices [256, 260] equipped with 9-DOF sensors, one [250] with a 10-DOF sensor, and two [243, 261] with unspecified DOF. Each study reported different sensor placements: two at the ground tip, one along the shaft, and two did not specify the exact positioning. Additionally, accelerometers were mentioned in two studies [257, 237], with placements at the ground tip and mid-shaft.

Furthermore, five studies utilized force sensors, with three [260, 250, 248] placing them on the handles and two at the ground tip [256, 257], with sensor counts varying from one to four. Load cells were consistently incorporated at the ground tip in two studies [260, 250]. Additionally, strain gauges were utilized in two studies:

one employing twelve sensors positioned on the lower shaft for measuring shear and axial forces [237], and another using eight sensors on the handles to measure axial forces [243]. Other sensor technologies, such as Radio-Frequency Identification (RFID) and ultrasonic sensors, were each mentioned once.

Four studies explicitly incorporated real-time user feedback: three [261, 260, 248] applied visual feedback via smartphone apps and Light-Emitting Diode (LED) lights, while one [237] implemented haptic vibratory biofeedback. Wireless communication methods were primarily based on Bluetooth, with only one study [250] reporting the use of Wi-Fi and MQTT protocols. Three articles explicitly mentioned connectivity to dedicated Android applications [261, 260, 248]. Machine learning algorithms, including SVM, Artificial Neural Network (ANN), Random Forest (RF), and K-Nearest Neighbors (kNN), were used in two studies [256, 243] for IMU-based physical activity classification.

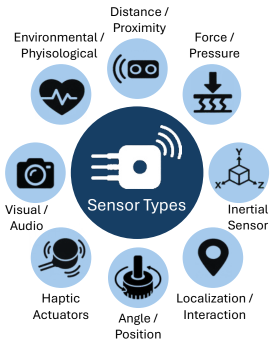
Walking Canes and Sticks A total of eight studies [254, 264, 251, 244, 242, 247, 241, 252] published between 2015 and 2021 addressed smart walking canes, while two [246, 253] additional studies focused on walking sticks (published in 2014 and 2022). These devices primarily target mobility-impaired users, including older adults and individuals with Parkinson's disease. The main purposes include the assessment of gait parameters, pressure distribution analysis, fall detection, and monitoring changes in gait behavior. Inertial sensors were the most commonly integrated technology, used in seven [246, 253, 241, 247, 242, 244, 264] of the ten studies. Two further studies [254, 252] employed discrete accelerometer and gyroscope sensors. Except for one study using two IMUs [241], all others utilized a single unit, positioned either at the handle or the ground tip. IMUs enable the measurement of linear acceleration, angular velocity, and orientation to support functionalities such as gait analysis and fall detection.

Force sensors were mentioned in four articles to detect grip force and ground contact. Two devices [244, 241] were equipped with eight FSRs each, placed in the handle and at the ground tip. One device [251] implemented two FSR sensors positioned at different heights in the shaft, while another [252] used a single sensor without specifying its placement. Additional sensor technologies included capacitive touch, ultrasonic, load cells, strain gauges, humidity sensors, and thermistors, each implemented to support specific applications.

Two devices applied visual feedback: one via a smartphone app [253], and the other via an OLED display combined with a real-time clock [252]. Furthermore, one study explicitly incorporated vibratory feedback via the cane handle for excessive force [244]. Communication methods were predominantly wireless via Bluetooth and Wi-Fi, though two studies also included a USB connection, and one did not specify the protocol. ML methods were reported in three studies [241, 264, 254], with algorithms such as ANN, SVM, and naive Bayes used for classifying gait-related activities. One study mentioned plans to incorporate ML in future work [244]. Moreover, two studies [246, 253] employed Kalman filtering for sensor data fusion, and one utilized a custom sensor fusion algorithm [247].

Sensor Technologies The reviewed studies highlight a diversity of sensor technologies integrated into smart walking aids. Table 4.4 presents an overview of the sensor types, grouped by their main categories, which are described in detail in the following paragraphs.

Table 4.4: Overview of sensor/actuator types, their frequency in the included studies, and corresponding references, grouped by sensor main categories.



Sensor/Actuator Type	**Frequency**	**%**	**References**
Force/pressure	29	82.9	[236, 237, 238, 240, 241, 242, 243, 244, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 265, 267, 268, 270]
Inertial sensors	25	71.4	[236, 237, 238, 239, 241, 242, 243, 244, 246, 247, 248, 250, 252, 253, 254, 256, 257, 258, 260, 261, 262, 264, 265, 268, 269]
Distance/proximity	13	37.1	[239, 244, 245, 248, 252, 255, 259, 262, 263, 265, 266, 268, 269]
Angle/position	7	20.0	[240, 249, 255, 258, 259, 262, 267]
Environmental/physiological	6	17.1	[238, 252, 256, 258, 265, 268]
Localization/interaction	5	14.3	[247, 250, 252, 265, 268]
Haptic actuators	4	11.4	[237, 244, 265, 268]
Visual/audio	4	11.4	[255, 262, 266, 267]
Total	**93**		

Note: The frequency indicates the number of studies that employed at least one sensor of the respective type, not the total number of sensors per device.

- **Force/Pressure Sensors, Load Cells, and Strain Gauges:** The most used sensor types across the device categories are those for measuring grip force, axial and shear forces, ground reaction forces, and weight-bearing compliance. In total, the use of force sensors is reported in twelve articles [265, 250, 236, 267, 240, 249, 259, 258, 268, 248, 256, 260]. Furthermore, seven articles explicitly mention the use of FSR integrated into the devices [251, 244, 241, 252, 257, 262, 263], while two others employed pressure sensors [258, 270], and one mentioned the use of a force/torque sensor [255]. Load cells were used in six studies [244, 253, 261, 250, 260, 238], where they play a crucial role in assessing ground reaction forces and ensuring correct weight distribution during gait training and rehabilitation. Strain gauges were reported in five articles [242, 247, 237, 243, 254]; three of these addressed canes (each equipped with four strain gauges), while the other two addressed forearm crutches (equipped with either eight or twelve strain gauges) for measuring shear and/or axial forces.
- **Inertial Sensors:** The second most frequently used sensor types across the device categories are IMUs, which typically comprise accelerometers, gyroscopes, and magnetometers. IMUs were used in 16 studies [264, 244, 242, 247, 241, 246, 253, 250, 256, 260, 243, 238, 262, 239, 269, 261]. IMU data were used for gait phase detection, posture monitoring, and movement analysis. Accelerometers (whether 3-DOF, 6-DOF, or 9-DOF) were mentioned in ten articles [248, 254, 244, 252, 265, 257, 237, 258, 268, 236], and gyroscopes were explicitly reported in three articles [254, 252, 236].

- **Distance and Proximity Sensors:** Distance and proximity sensors were incorporated in several studies to enable obstacle or user detection, enhance environmental awareness, and support navigation. A total of six studies used ultrasonic sensors [244, 252, 265, 268, 239, 269], three used laser rangefinders [255, 266, 263], and one applied a 2D laser scanner [259]. Additionally, two studies utilized microwave Doppler radar [245, 248], and two implemented Infrared (IR) sensors for user detection [269] and speed monitoring [263].
- **Angle and Position Sensor:** Rotary encoders for wheel position tracking were reported in six studies, all of which focused on rollators or walkers [259, 262, 267, 240, 249, 255]. A tilt sensor was also mentioned in one article [259], as well as one Hall effect sensor for rotary angle measurement [258].
- **Environmental and Physiological Sensors:** Environmental parameters were recorded using Light-Dependent Resistor (LDR) for light intensity in three studies [252, 268, 265], and a barometer for atmospheric pressure sensing in one study [256]. Furthermore, one device [252] included humidity and temperature sensors for ambient monitoring and embedded a thermistor in the handle to measure local temperature. Physiological sensors were each used once and included dry ECG electrodes [238], PPG sensors [258], and IR-based pulse monitoring [265].
- **Localization, Identification and Interaction Sensors:** GPS modules for outdoor localization were used in three studies [268, 252, 265]. Additionally, one device used RFID tags [250] for user identification, and one article implemented capacitive touch sensors [247] to detect user interaction.
- **Haptic Actuators:** Haptic feedback was implemented via vibration motors in four articles [244, 265, 237, 268]. The placement and number of vibration motors varied, including configurations such as four motors, a single motor in one handle, motors in both handles, or unspecified locations.
- **Visual and Audio Sensors:** Four articles mentioned the use of various camera systems, such as an RGB camera [267], a GoPro camera and Kinect [255], a webcam [266], and one unspecified camera [262]. As an audio sensor, MEMS microphones were employed in one article [255]. While not explicitly listed or categorized as sensors or actuators in this review, various studies implemented visual and auditory output interfaces to support user interaction, information display, or alert mechanisms. These include LCD [270, 268, 265] and OLED screens [252], LED indicators [269, 260], smartphone applications [248, 253, 260, 261] for feedback display, and audio components such as speakers [269] and alarms [265, 262].

Synthesis of Technological Aspects Table 4.5 presents the extracted technical data of the included studies, highlighting the device type, used sensor technology, and corresponding placement, feedback modality, communication, and use of AI/ML methods or conventional algorithms.

Table 4.5: Technical overview of smart walking aid systems, including device type, sensor specifications, feedback modality, communication, and AI use.

Ref.	Device Type	Sensor Type(s)	Sensor Placement	Feedback Modality	Communication	Algorithm, AI Method
[265]	Walker (Four-legs)	◦ 2x Force ◦ 1x Ultrasonic ◦ 1x Accelerometer ◦ 1x LDR ◦ 1x GPS ◦ 1x IR sensor	◦ Handle ◦ Cross bar (bottom) ◦ Top platform ◦ Top platform ◦ Top platform ◦ Handle	Haptic: Vibration motor Visual: LCD display, Panic SOS button, SMS with GPS on fall Acoustic: Sound alert	Global System for Mobile Communications (GSM) module, IoT (unspecified)	Algorithm: Fuzzy logic (fall detection, haptic feedback); Planned: AI/ML extension
[245]	Walker (Four-legs)	◦ 2x Doppler Radar	◦ Cross bar	–	Bluetooth to PC (LabView GUI)	Planned: DL classifiers (based on WVD spectrograms, MIF/MIB features)
[262]	Walker (Four-Wheels)	◦ 2x Force (FSR) ◦ 2x Rotary encoders ◦ 1x IMU (9-DOF) ◦ 1x IMU (6-DOF) ◦ 1x Camera ◦ 1x Laser rangefinder	◦ Handle ◦ Rear wheels ◦ Seat (bottom) ◦ Waist ◦ Backrest ◦ Cross bar	Acoustic: Alarm alert, caregiver notifications	Wireless (unspecified)	Algorithm: Threshold-based (e.g., Kangas, vertical velocity); fall/near-fall detection
[261]	Forearm Crutch	◦ 1x Load cell ◦ 1x IMU	◦ Ground tip	Visual: Smartphone app	Bluetooth to Android app (mCrutch)	Algorithm: Threshold-based segmentation; crutch stance detection
[268]	Walker (Four-legs)	◦ 2x Force ◦ 1x Ultrasonic ◦ 1x Accelerometer ◦ 1x LDR ◦ 1x GPS	◦ Handle ◦ Bottom ◦ Top platform ◦ Top platform ◦ Top platform	Haptic: Vibration motor Visual: LCD screen	Wi-Fi and Bluetooth	Algorithm: Fuzzy logic (fall detection); Planned: DL for camera module
[246]	Stick	◦ 1x IMU (9-DOF)	◦ Handle	–	Bluetooth to PC	Algorithm: Extended Kalman Filter
[250]	Forearm Crutch	◦ 1x Load cell ◦ 1x Force ◦ 1x IMU (10-DOF) ◦ 1x RFID	◦ Bottom ◦ Handle ◦ Not specified ◦ Not specified	–	Wi-Fi and MQTT to Cloud (IoT)	Planned: ML methods

Table 4.5: (continued)

Ref.	Device Type	Sensor Type(s)	Sensor Placement	Feedback Modality	Communication	Algorithm, AI Method
[243]	Forearm Crutch	∘ 8x Strain gauges (axial) ∘ 1x IMU	∘ Handle ∘ Not specified	–	Bluetooth to PC	Algorithm: Feature selection (Relief-F, mRMR, CFS); AI/ML: kNN, RF, SVM
[254]	Cane	∘ 1x Strain gauge ∘ 1x Accelerometer (3-DOF) ∘ 1x Gyroscope (3-DOF)	∘ Not specified	–	Not specified	AI/ML: LDA, SVM
[264]	Cane	∘ 1x IMU (9-DOF)	∘ Handle (box-mounted)	–	Local logging (USB)	AI/ML: LSTM, kNN, FSM, CNN, etc.
[256]	Forearm Crutch	∘ 1x IMU (9-DOF) ∘ *x Barometer ∘ *x Force	∘ Ground tip ∘ Ground tip ∘ Ground tip	–	Bluetooth Low Energy (BLE) to mobile phone	AI/ML: SVM, ANN, kNN
[263]	Walker (Front-Wheels)	∘ 1x IR sensor ∘ *x Force (FSR)	∘ Not specified ∘ Handle	–	Wireless to Android app	Not specified
[251]	Cane	∘ 2x Force (FSR)	∘ Shaft (low, multiple depths)	–	Bluetooth to external systems	Not specified
[244]	Cane	∘ 8x Force (FSR) ∘ 1x IMU (9-DOF) ∘ 1x Ultrasonic ∘ 1x Accelerometer (3-DOF) ∘ 1x Load cell	∘ Handle ∘ Handle ∘ Shaft (middle) ∘ Ground tip ∘ Ground tip	Haptic: Vibration motor	USB to PC	Planned: ML in future work (applied in previous work)
[260]	Forearm Crutch	∘ 3x Force (FlexiForce) ∘ 1x IMU (9-DOF) ∘ 1x Load cell	∘ Handle ∘ Shaft ∘ Ground tip	Visual: Smartphone app, LED light alert	Bluetooth to Android app	Algorithm: Kalman Filter Planned: Prediction models
[257]	Forearm Crutch	∘ 4x Force (FSR) ∘ 1x Accelerometer	∘ Ground tip ∘ Shaft (middle)	–	Bluetooth	Algorithm: Quadratic regression
[238]	Walker (Four-legs)	∘ 4x Load cells ∘ 2x Dry ECG electrodes ∘ 1x IMU (9-DOF)	∘ Ground tip ∘ Handle ∘ Cross bar	Visual: Spy Walker application	Bluetooth to PC (Spy Walker app)	Not specified
[236]	Rollator (i-Walker)	∘ 1x Accelerometer ∘ 1x Gyroscope ∘ *x Force	∘ Central box ∘ Central box ∘ Handle	–	Not specified	AI/ML: BOSS (feature extraction), Bayesian Gaussian Mixture Model (clustering)

**Note: Asterisks (*) indicate missing or unspecified numerical data in the original source.*

Table 4.5: (continued)

Ref.	Device Type	Sensor Type(s)	Sensor Placement	Feedback Modality	Communication	Algorithm, AI Method
[267]	Rollator (i-Walker)	∘ 2x Rotary encoders ∘ 1x RGB-D camera ∘ *x Force	∘ Wheels ∘ Cross bar ∘ Handle	–	Not specified	Not specified
[239]	Walker (Front-Wheels)	∘ 1x IMU (9-DOF) ∘ 1x Ultrasonic	∘ Cross bar ∘ Cross bar	–	BLE to external device (IoT)	Not specified
[242]	Cane	∘ 4x Strain gauges (axial) ∘ 1x IMU (9-DOF)	∘ Shaft (low) ∘ Shaft (low)	–	Bluetooth to PC (LabView GUI)	Not specified
[247]	Cane	∘ 4x Strain gauges ∘ 1x IMU (9-DOF) ∘ 1x Touch (capacitive)	∘ 2x cane curve, 2x handle ∘ Handle ∘ Handle	–	BLE 4.2 to Tablet app (later Bluetooth 5.0)	Algorithm: Gait segmentation algorithm
[252]	Cane	∘ 1x Force (FSR) ∘ 1x Accelerometer and Gyroscope (6-DOF) ∘ 1x Ultrasonic ∘ 1x Ambient temperature and humidity ∘ 1x Thermistor ∘ 1x LDR ∘ 1x GPS	∘ Not specified ∘ Shaft (external housing) ∘ Shaft (external housing) ∘ Shaft (external housing) ∘ Handle ∘ Shaft (external housing) ∘ Shaft (external housing)	Visual: OLED monitor, Real-time clock	Wi-Fi and Bluetooth	Planned: ML (intended but not implemented)
[240]	Rollator (i-Walker)	∘ 2x Rotary encoder ∘ *x Force	∘ Wheels ∘ Handle	–	Not specified	Algorithm: Regression models for prediction
[269]	Walker	∘ 1x Ultrasonic ∘ 1x IR sensor ∘ 1x IMU (3-DOF)	∘ Cross bar ∘ Cross bar ∘ Cross bar	Visual: LED lights Acoustic: Speaker output	Wireless (unspecified)	Not specified
[249]	Rollator (i-Walker)	∘ 2x Rotary encoder ∘ *x Force	∘ Wheels ∘ Handle	–	Not specified	Algorithm: Regression models for prediction
[241]	Cane	∘ 8x Force (FSR) ∘ 2x IMU (9-DOF)	∘ 7x Handle, 1x Shaft (base) ∘ 1x Handle, 1x Shaft (base)	–	Wireless to PC application	AI/ML: C4.5, ANN, SVM, Naive Bayes
[237]	Forearm Crutch	∘ 12x Strain gauges (axial/shear) ∘ 1x Accelerometer (3-DOF)	∘ Shaft (low) ∘ Ground tip	Haptic: Vibratory biofeedback	Bluetooth to PC (LabView GUI)	Not specified

**Note: Asterisks (*) indicate missing or unspecified numerical data in the original source.*

Table 4.5: (continued)

Ref.	Device Type	Sensor Type(s)	Sensor Placement	Feedback Modality	Communication	Algorithm, AI Method
[259]	Rollator (i-Walker)	∘ 2x Force (3-axis) ∘ 2x Force (1-axis) ∘ 2x Rotary encoders ∘ 1x Tilt sensor ∘ 1x 2D laser scanner	∘ Handle ∘ Hind legs ∘ Wheels ∘ Not specified ∘ Not specified	–	Not specified	Not specified
[248]	Walker/Rollator and Crutches	∘ 2x Doppler Radar ∘ 1x Accelerometer ∘ *x Force	∘ Cross bar ∘ Cross bar ∘ Handle, ground tip	Visual: Smartphone app	Bluetooth to mHealth app	Not specified
[270]	Walker	∘ 2x Pressure	∘ Handle	Visual: LCD screen	XBee RF to receiver	Not specified
[253]	Cane/Stick	∘ 1x Load cell (1-DOF) ∘ 1x IMU (5-DOF)	∘ Ground tip ∘ Shaft	Visual: Smartphone app	Bluetooth to Smartphone (LabView GUI)	Algorithm: Kalman Filter
[255]	Rollator	∘ 2x Laser rangefinder ∘ 2x Kinect ∘ 2x Rotary encoders ∘ 2x F/T sensors (6-DOF) ∘ 1x 8-mic MEMS ∘ 1x GoPro HD camera	∘ Back and front side ∘ Top cross bar ∘ Rear wheels ∘ Handle ∘ Top cross bar ∘ Top cross bar	–	Not specified	AI/ML: HMM, SVM
[266]	Walker (i-Walker)	∘ 1x Webcam ∘ 1x Laser rangefinder	∘ On the top ∘ Lower chassis	Visual: Mobile PC	USB to mobile PC	Not specified
[258]	Rollator	∘ 2x Force (6-DOF) ∘ 1x Accelerometer (3-DOF) ∘ 1x Hall effect sensor ∘ 1x Pressure ∘ 1x PPG	∘ Handle ∘ Not specified ∘ Wheel ∘ Rollator seat ∘ Finger	–	Bluetooth	Not specified

Note: *Asterisks (*) indicate missing or unspecified numerical data in the original source.*

4.1.4 Discussion

In this PRISMA-ScR review, we systematically examined 35 articles focusing on intelligent walking aids equipped with sensor technologies to support health monitoring and mobility in individuals with impairments. Our synthesis revealed a diverse range of device types (including canes, walkers, and crutches) augmented with various sensor modalities to enable gait assessment, fall detection, and user feedback through real-time data processing and communication strategies. These systems addressed a wide spectrum of application areas, including rehabilitation, fall prevention, and activity monitoring, and were targeted at different user groups such as older adults, physical therapy patients, or fall risk users. In the following sections, we discuss the current challenges and technological gaps identified across the included studies. We also highlight opportunities for future research directions, followed by implications for researchers and practitioners, and a reflection on the limitations of this review.

Current Challenges and Technological Gaps

Besides the demonstrated potential of the systems, some challenges and technological gaps were also identified, which are discussed below.

Sensor Selection and Placement A wide range of sensors was used in the included studies, which varied in terms of specifications, placement, and calibration approaches. However, inconsistent sensor placement practices represent a challenge for data comparability, as there is currently no consensus on recommended placement strategies that ensure sufficient and reliable data collection. Ballesteros et al. [251] highlighted that sensor placement can present a number of challenges: Integrating sensors into the handle may require major modifications to canes, while components attached to the shaft could alter ergonomics, potentially requiring extensive validation or additional certification processes. For example, the use of IMU sensor varied strongly across the studies, in terms of the used DOF and in placement (e.g., at the ground tip, handle, or shaft). This variability directly affects the type and quality of motion data captured, and thus influences the detection of gait parameters or fall events. Similarly, the number and positioning of force and pressure sensors differed widely. Some systems integrated a single sensor, while others used multiple sensors, positioned on the handle or ground tip, to capture load distribution or grip force. These differences complicate the direct comparison of outcomes between studies and underscore the need for standardized sensor configurations. Furthermore, several articles did not specify the exact type of sensors used or failed to provide detailed system information, making it difficult to reproduce study outcomes and assess the validity of reported results. These findings indicate the need for more transparent reporting practices and the creation of standardized design guidelines to ensure data quality, reproducibility, and clinical utility.

Feedback Modalities This review showed that most systems provide visual feedback, usually via smartphone apps [261, 248, 253, 260], mobile PCs [266], or LED indicators [269, 260]. Haptic feedback was typically integrated into device handles via vibration motors. It was primarily used as biofeedback to alert users when predefined thresholds were exceeded, such as in cases of excessive handle force [265, 244, 237, 268].

However, no consistent procedure for setting these thresholds was described across studies. Sardini et al. [237] highlighted that such threshold parameters should be individually defined by physiotherapists in order to address patient-specific conditions. One prototype provided multimodal feedback by combining visual displays (e.g., LCD screens for force distribution and pulse rate) with haptic and acoustic alerts [265]. However, the description of feedback implementations was often superficial: details on actuator specifications, placement, or feedback timing were frequently missing, limiting reproducibility and comparability. Furthermore, the effectiveness of these feedback modalities on user experience or behavioral outcomes was not evaluated, indicating a gap in current research. No comparative analysis using quantitative measures to assess the impact of different feedback modalities was found. This limits the generalizability of recommendations regarding suitable feedback types for specific tasks or user populations. In particular, the implementation of such features should be guided by user needs and integrated in a manner that enhances both device usability and functionality. Therefore, future research should not only report feedback implementations more transparently but also systematically evaluate their effectiveness to ensure both user acceptance and clinical relevance.

Data Communication Data communication approaches varied depending on the application context, but most systems relied on wireless transmission. Bluetooth or Bluetooth Low Energy (BLE) were used in around half of the studies and are therefore the most frequently used wireless standard. In addition, a smaller number of systems employed Wi-Fi to transmit sensor data to external devices, including personal computers, tablets, or smartphones. One study [250] incorporated cloud connectivity through protocols such as MQTT to IoT platforms, enabling remote monitoring and data analysis. In several cases, communication protocols and data transfer details were not explicitly specified [254, 236, 267, 240, 249, 259, 255] or relied on cable-based connections [244, 266, 264]. The diversity of communication approaches reflects the varying use cases for the data, ranging from real-time visualization on a corresponding smartphone app to a simple GUI for laboratory validation. Only one study [265] explicitly addressed data security and privacy concerns. They used the Thingspeak IoT platform with enhanced protection features, like a private mode that restricts access to authorized people only. This approach enables data sharing with trusted parties, which is particularly relevant for the approval procedures and regulatory compliance. In general, topics such as data security and privacy were not mentioned in the other reviewed studies, but should be considered in future developments to ensure safe integration with digital health platforms and mobile applications.

Use of Artificial Intelligence / Algorithm The use of AI, ML, and predictive algorithms was heterogeneous across studies to analyze gait parameters. Several systems [241, 256, 264, 254, 243, 255] employed established methods for data processing and prediction, including SVM, ANN, RF, kNN, or planned for future use [265, 245, 268, 250, 244, 252]. For example, Narváez et al. [243] tested three ML techniques (kNN, RF, and SVM) for gait pattern identification with smart crutches. The results showed that all three could be used to identify gait patterns with similar results: 88% accuracy for both kNN and SVM, as well as 89% accuracy for RF. Additionally, Mesanza et al. [256] applied an ML approach for physical activity classification using a forearm crutch and highlighted that SVM, ANN, and kNN can reach a success rate of 92–97%, which was higher than the results of previous studies (91%). In contrast, Wade et al. [241] developed an instrumented cane and found that SVM performed worse for user activity prediction with just 64% accuracy, while the C4.5 algorithm achieved the highest accuracy with over 95%, followed by ANN with 84%. Some approaches utilized more advanced methods, such as Long Short-Term Memory (LSTM) [264], HMM [255], Bayesian Gaussian Mixture Model for clustering [236], or regression techniques for prediction. For example, Riberio and Santos [264] investigated the most suitable ML models for three fall classification problems (fall event, fall phase, and fall category) using an instrumented cane. They evaluated eight classifiers (LSTM, kNN, SVM, CNN, etc.) and demonstrated that LSTM was the most appropriate for fall event classification (accuracy: 99.26%) and fall phase classification (accuracy: 95.91%), due to its better performance compared to kNN. In contrast, the kNN model achieved the best performance for fall category classification (accuracy: 74.54%). The authors claimed that ML can lead to better performance than an Finite-State Machine (FSM). However, further studies are needed to evaluate this in the context of real-time classification.

Feature extraction and sensor fusion algorithms, such as Kalman or extended Kalman filters, were also reported [246, 260, 253]. For example, Zhou et al. [246] applied an extended Kalman filter for gait analysis using data from a walking stick for older adults and individuals with disabilities. They reported a step count accuracy of 100% and average accuracies between 94% and 96% for stride duration, step length, and step speed. Notably, most systems do not use AI or ML but rely on traditional algorithmic approaches or do not use predictive analytics at all.

Nevertheless, AI methods or advanced algorithms hold potential for future implementations, as they have been successfully applied to the identification of gait events [241], gait abnormality classification [245], as well as fall detection and prevention [264, 262]. Building on these findings, follow-up studies can refine these approaches and implement them across diverse application domains. In particular, current implementations are often limited to post-processed data analysis, highlighting that real-time integration requires further development.

System Validation and User Testing Overall, there was no clear consensus regarding system validation and user testing across the included studies. In several articles, the testing procedures and the characteristics of involved participants were not specified, and demographic data were partially missing [265, 257, 252, 248, 245, 242, 260]. Furthermore, some articles focused solely on presenting prototypes without any technical validation or without involving user testing [258, 269]. In general, most studies involved participants (both healthy and with impairments) of varying ages, with partially gender-balanced samples. However, some studies were conducted with only one participant [253, 238, 261, 268] or exclusively with healthy participants, which did not cover the target user group [255, 239, 237, 256, 264, 250, 243, 247, 236, 262]. In total, one long-term study was identified, which covered a 10-week rehabilitation process with patients [240]. Overall, the review revealed a lack of long-term studies or in-depth case studies that could provide deeper insights into real-world usage and long-term effects. For better comparability between systems, there is a need for standardized procedures in system validation and user testing. Such standardization would facilitate the evaluation of study outcomes and support the refinement of assistive devices for broader clinical and practical application.

Opportunities and Future Research Directions

Future research should systematically investigate the influence of sensor placement and calibration on measurement quality, including signal shifts and interpretation errors. This is essential for the development of practical guidelines that support the selection and placement of sensors and therefore improve the comparability, reproducibility, and generalizability of the results of different studies. In addition, the device tests should be conducted with the actual target groups under real conditions rather than in a laboratory environment. Moreover, larger sample sizes, as well as case studies and long-term evaluations, are needed to verify the reliability, usability, and clinical effectiveness of these systems over time. Beyond addressing current limitations, expanding sensor configurations and feedback modalities offers the opportunity to extend device functionalities and embed them within broader ecosystems of smart assistive technologies. In this context, future research should also explore the integration of intelligent walking aids with other connected devices (e.g., wearables) or mobile health platforms. Furthermore, the increasing use of ML holds potential to improve the accuracy of gait classification and the prediction of clinically relevant parameters, enabling broader application across various domains and advancing the analytical capabilities of such systems.

Implications

The findings of this review emphasize the need to further develop sensor-based walking aids beyond the early prototype stage. For researchers and practitioners, this involves conducting user-centered validation studies that address not only technical performance but also long-term usability, clinical effectiveness, and patient-specific needs. Several developments demonstrated the potential for modular sensor architectures, enabling adaptable configurations across various types of assistive devices. However, sensor systems must be integrated without compromising ergonomics and user comfort, and the measured health data should be interpretable for both patients and therapists. To enable wider adoption, future systems should be developed with consideration for interoperability and in line with ethical, legal, and data protection requirements. By overcoming these challenges and limitations, the development and implementation of smart walking aids can take a step forward in both clinical and home care settings.

Limitations of the Review

This review is limited to the three selected databases and the applied keyword search strategy, which covered literature published until January 2025. While these sources were selected to capture studies on sensor-based assistive technologies, the inclusion of additional health-focused databases, particularly those indexing clinical trials, as well as gray literature containing unpublished innovations, could potentially have expanded the scope of our findings. As a result, certain perspectives may not be fully reflected in this review, which could introduce a selection bias. Another limitation is the targeted device categories and contexts of use that were included, as expanding to additional areas such as exoskeletons, wearable devices, or visually impaired users would lead to a broader range of studies. The quality and risk of bias of the included studies were not assessed, as the aim was to provide a comprehensive overview rather than a critical appraisal. Finally, heterogeneity in study designs, reported outcomes, and user testing procedures limit the comparability and synthesis of findings across studies.

4.1.5 Conclusion

This scoping review provides a structured overview of intelligent walking aids equipped with sensor technologies developed to support individuals with mobility impairments. Addressing the MRQ, we identified a broad spectrum of device types (including crutches, canes/sticks, walkers, and rollators) and examined their domains of application across rehabilitation, fall prevention, and gait monitoring.

In response to SRQ1, we found that various sensor technologies such as IMUs, force/pressure sensors, and environmental sensors are integrated to enable functional support, health monitoring, and interactive feedback. However, sensor placement and configuration varied widely across studies, which challenges the comparability of results and underscores the need for standardization. Regarding SRQ2, target user groups ranged from older adults to individuals with Parkinson's disease or physical therapy patients. While the intended functionalities often align with user-specific needs, empirical evidence on system reliability and clinical effectiveness remains limited.

Our synthesis revealed several technological and methodological gaps. Although many studies propose innovative approaches, most systems remain in early-stage prototyping or proof-of-concept phases. In particular, few studies included system validation with the target population or long-term evaluation. As a result, the practical impact, reliability, and usability of these technologies are still unknown. Additionally, the heterogeneity in sensor placement, configuration, and system objectives complicates direct comparison and limits generalizability.

To address these challenges, we offer recommendations for researchers and practitioners. Future research should focus on rigorous system validation and on extending the functionality of adaptive feedback mechanisms tailored to individual patient needs, to overcome current limitations and strengthen the evidence base. The insights gained from this review could serve as a basis for the development of next-generation assistive technologies that are personalized, interactive, and directly support health outcomes. Overall, this review contributes to the broader fields of assistive technology, sensor-based rehabilitation, and HCI by synthesizing current developments and outlining future research directions.

4.2. Study II: Development and Evaluation of a Multimodal Smart Orthosis and Walking Aid System

In previous work, patient and healthcare professional requirements were determined via focus groups [44], smartphone application mock-ups were evaluated for usability [45], and a system concept was proposed [122]. Building on these findings, this section focuses on the development and evaluation of a comprehensive multimodal assistive system. As highlighted in the scoping review (Section 4.1), existing sensorized walking aid systems frequently lack thorough system-level validation and the integration of feedback capabilities. Moreover, current research often presents early-stage prototypes or isolated solutions, highlighting the need for integrated systems that connect intelligent walking aids within a broader interaction and feedback framework.

To address these gaps, a multimodal system was developed, comprising a smart foot orthosis, a connected walking aid, and a smartphone app. This section evaluates the system's feasibility and validity for gait-related measurements, examines the perception and impact of haptic feedback across different system components, and assesses the usability of the smartphone app. The contributions of this work include the demonstration of a novel integrated assistive system, empirical evidence of system feasibility and measurement validity, insights into cross-device haptic feedback perception, and usability findings that inform the design of future user-centered assistive technologies.

S. Resch, A. Kousha, A. Carroll, N. Severinghaus, F. Rehberg, M. Zatschker, Y. Söyleyici, and D. Sanchez-Morillo, "Smart Device Development for Gait Monitoring: Multimodal Feedback in an Interactive Foot Orthosis, Walking Aid, and Mobile Application," Technologies, vol. 13, no. 12, Art. no. 588, 2025.

The following conference papers provide complementary preliminary results:

S. Resch, L. Zimmermann, A. Kunkel, D. Gerstung, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Usability Evaluation of a Mobile Application for Foot Health Monitoring of Smart Insoles: A Mixed Methods Study," in *Adjunct Proceedings of the 26th International Conference on Mobile Human-Computer Interaction (MobileHCI '24 Adjunct)*, Melbourne, VIC, Australia: Association for Computing Machinery, pp. 1–7, 2024.

S. Resch, A. Röll, A. Malech, V. Schwind, D. Sanchez-Morillo, and D. Völz, "Walking Aid with Haptic Feedback for Combined Use with a Smart Foot Orthosis," *Gerontechnology*, vol. 23, no. 2, pp. 1–1, 2024.

S. Resch, K. Zoufal, I. Akhouaji, M. A. Abbou, V. Schwind, and D. Völz, "Augmented Smart Insoles–Prototyping a Mobile Application: Usage Preferences of Healthcare Professionals and People with Foot Deformities," *Current Directions in Biomedical Engineering*, vol. 9, no. 1, pp. 698–701, De Gruyter, 2023.

4.2.1 Introduction

Gait rehabilitation plays an important role in the recovery process after acute foot injuries and in the management of chronic foot-related conditions. A variety of assistive devices are used to support these patients in their daily mobility, including orthopedic devices such as foot orthoses, insoles, bandages, and pressure-relieving footwear [271]. In addition, mobility aids such as crutches, walking sticks, or canes are commonly used for temporary or long-term walking impairments [272]. While these devices are essential in stabilizing gait and reducing mechanical stress [273, 274], they are typically standardized and require customization to achieve optimal recovery outcomes [275], as they can measurably affect gait [276]. These limitations underscore the need for more individualized and adaptive orthopedic solutions that combine health monitoring with interactive feedback.

In recent years, the integration of wearable electronics has contributed to various developments such as smart insoles, socks, and shoes equipped with sensor technologies for gait analysis and health monitoring [277, 58, 60]. In contrast to conventional stationary systems, these mobile solutions enable gait analysis outside clinical settings and open up new possibilities for continuous monitoring of spatial, temporal, and spatio-temporal parameters [89]. These systems frequently rely on FSRs and IMUs to assess parameters such as plantar pressure distribution and gait characteristics (e.g., step cadence and foot strike patterns) [86]. Such technologies show promise in areas such as fall risk assessment, biofeedback interventions, and the prevention of foot ulcers [79]. In particular, monitoring plantar pressure is crucial for patients with DFS, where elevated sole pressure can impair wound healing and increase the risk of ulcer formation [278]. They also hold potential for injury prevention and early pathology detection [13].

However, orthopedic devices such as AFOs, which are used to stabilize gait and support recovery, still rarely combine embedded sensing and feedback within an interactive system. Integrating feedback modalities (such as haptic alerts triggered by pressure overload) could enhance the benefits of Smart Foot Orthosis (SFO) development, particularly for individuals with foot conditions [122]. For example, diabetic neuropathy in the lower extremities impairs the sensation of pain and limits the detection of excessive plantar pressure [279], which can lead to unnoticed injuries and increase the risk of serious deformities such as Charcot foot [280]. Providing patients with intrinsic feedback could therefore positively impact gait recovery. While several conceptual prototypes have integrated electronics into orthoses [15, 62, 281], most solutions remain technically focused, without addressing how haptic feedback should be designed or delivered. There is a lack of empirical evidence on how feedback modality in smart footwear and walking aids influences user perception, perceived effectiveness, and intended use. This is critical, as unclear or intrusive feedback can reduce acceptance even when technically accurate.

To address the identified research gap, we developed a modular sensor system to investigate the influence of haptic feedback modality and device-dependent feedback on user perception in smart footwear and walking aids. The system comprises three key components: (1) an SFO equipped with IMU and FSR sensors as well as a vibration feedback module, (2) a sensor-integrated walking aid that communicates with the orthosis and includes its own IMU and haptic actuators, and (3) a mobile smartphone application for health monitoring and feedback interaction. The system architecture enables real-time communication across all components and provides the basis for evaluating closed-loop vibrotactile biofeedback strategies. To ensure a valid basis for user perception testing, we examined the system's feasibility for gait detection and the usability of the mobile application from a user interaction perspective. These validation objectives guided the following SRQs:

- SRQ1 (System Feasibility): How accurately does the developed SFO measure plantar pressure and gait parameters compared to a clinical reference system, as a foundation for reliable real-time multimodal feedback?
- SRQ2 (Application Usability): How do users evaluate the usability of the accompanying mobile application for interpreting gait and pressure information within the integrated system?

Based on this, the core objective of this work is to investigate feedback strategy design for user-centered biofeedback, addressing the MRQ:

- MRQ (Feedback Evaluation): How do different vibrotactile feedback types (continuous vs. pattern-based), delivered through the orthosis and the walking aid, influence user perception in terms of noticeability, intrusiveness, perceived usefulness, and intended use?

This work contributes to the field of assistive technology by introducing a novel closed-loop, sensor-driven haptic feedback mechanism in a sensorized foot orthosis and instrumented forearm crutch, enabling real-time feedback that responds to gait dynamics and pressure distribution. Our key research findings provide empirical insights that users significantly prefer pattern-based over simple vibrotactile feedback notifications and that device location influences user perception. These results offer concrete recommendations for designing effective, user-centered haptic rehabilitation systems and can serve as a basis for future patient-centered evaluations of multimodal feedback in HCI.

4.2.2 Method

This section describes the overall system architecture and the technical implementation of the smart orthosis, the instrumented walking aid, and the smartphone application. It is structured as follows: First, a system overview is provided. Then, the hardware architecture and concept development are described. Next, the software architecture and data processing for each subsystem are outlined. Finally, the experimental setup of the prototype validation study, including apparatus, procedure, and measurements with data analysis, is presented.

System Overview

Figure 4.4 provides an overview of the system architecture, showing the interaction between the SFO, the instrumented walking aid, and the mobile application for data visualization and feedback control. The SFO is based on a standardized lower-leg orthosis (AIRCAST® AIRSELECT™ Elite Walker [1] by ORMED GmbH, a company of Enovis, Freiburg, Germany), also known as a controlled ankle motion boot, which can be worn universally on either foot. A sensor concept consisting of multiple FSRs

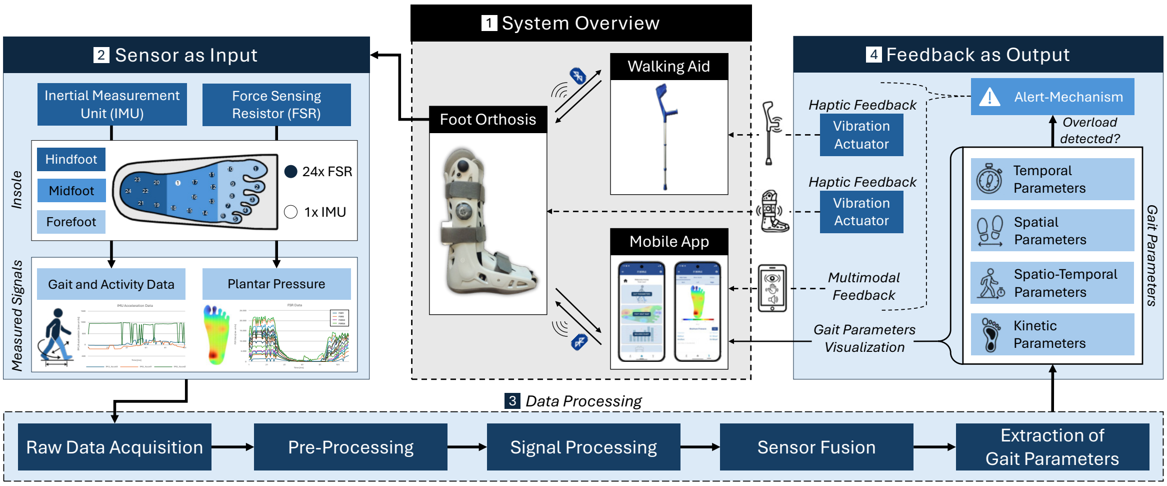


Figure 4.4: Overview of the sensor-based assistive system consisting of a smart orthosis with integrated FSR and IMU sensors, vibration feedback, and its communication with a walking aid and an mHealth application for interactive gait monitoring.

and a single IMU is integrated into a 3D-printed insole to capture plantar pressure and motion data. The insole is designed as a modular component, with adjustable size and shape to accommodate different orthotic devices and foot sizes.

Sensor fusion and real-time data processing enable the extraction of gait parameters and the detection of critical loading patterns. To support active user guidance, haptic feedback is provided via a vibration motor embedded in the orthosis, which is triggered in response to excessive plantar pressure. A custom-designed Printed Circuit Board (PCB), equipped with an on-board Microcontroller Unit (MCU), enables wireless communication via BLE with both the mobile application and the instrumented walking aid. The forearm crutch is based on a standardized design (ISO 11334-1:2007) [282] and includes a custom-designed PCB equipped with an MCU, an IMU sensor, two vibration modules, and a rechargeable battery. All components are embedded in the handle to enable synchronized haptic feedback and motion sensing. In addition, a corresponding mHealth application developed in Android Studio provides a graphical interface for visualizing gait parameters and plantar pressure data.

Hardware Architecture and Concept Development

Smart Orthosis A schematic overview of the SFO prototype is presented in Figure 4.5 and the assembled insole with integrated sensor components is shown in Figure 4.6. The SFO is based on a modular insole integrated into a standard lower-leg orthosis. An optional external vibration module can be placed at varying heights along the calf using the holes of the orthosis, allowing flexible positioning for haptic feedback. The insole includes multiple FSRs and an IMU sensor, connected via a flat cable to an external PCB, which is enclosed in a 3D-printed housing mounted on the posterior strap of the orthosis. The enclosure was designed using the Computer-Aided Design (CAD) software Autodesk Fusion 360 (v.2603.0.86) and fabricated from Polylactic Acid (PLA) with a 0.4 mm nozzle (HT extruder) using the Flashforge Creator 4 3D printer. The insole consists of a two-part structure, comprising a bottom and a top layer, both fabricated from flexible Thermoplastic Polyurethane (TPU) using the

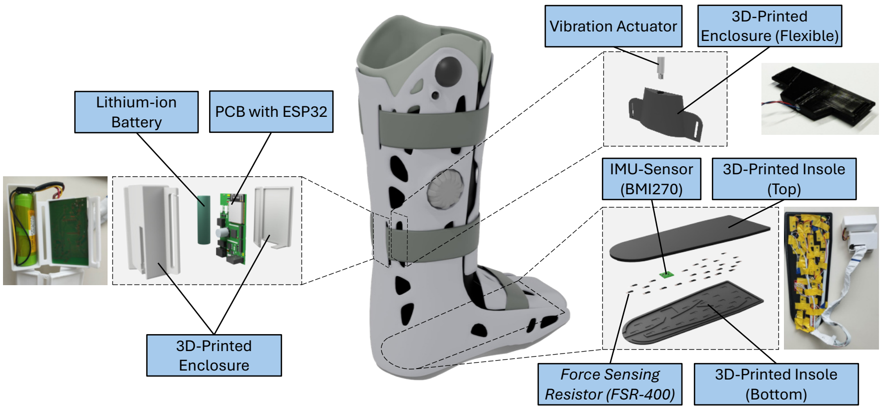


Figure 4.5: Overview of the orthosis prototype and assembled components.

Figure 4.6: Assembled prototype of the smart insole.

extruder F with 0.4 mm nozzle size. The geometry replicates the original insole of the AIRCAST® orthosis to ensure proper fit and alignment, without affecting wearing comfort. The design enables adaptation for the contralateral healthy foot by modifying the insole shape and enclosure mounting to fit standard footwear. It also supports future integration into other orthotic systems tailored to individual foot anatomies.

The modular insole was designed to include the recommended minimum of 15 FSRs [71] (Model 400 short tail, Interlink Electronics [283]) to capture plantar pressure distribution. The placement of the FSRs is based on the anatomical structure of the foot and covers key plantar regions, including the heel, forefoot, metatarsal arch, and toes. The layout supports extensions up to 24 FSRs, which may be beneficial for applications involving foot deformities or more detailed regional pressure mapping. Each sensor has an active area of 5.6 mm in diameter, a thickness of 0.3 mm, and a force sensitivity range of 0–20 N, offering fast response times and high reliability for dynamic pressure measurements [284].

For motion tracking, a 6-axis IMU (BMI270, Bosch [285]) was integrated into the insole. The IMU was positioned near the metatarsal arch to minimize pressure exposure. This sensor was selected based on key technical requirements for wearable gait analysis systems and is commonly used in fitness trackers and smart wearables, such as smart shoes for step counting and activity recognition [286]. The ultra-low power sensor combines a 3-axis accelerometer (sampling frequencies from 25 Hz to 6.4 kHz) and a 3-axis gyroscope (0.78 Hz to 1.6 kHz) within compact dimensions of 0.8 mm × 3 mm × 2.5 mm.

A custom PCB (1.55 mm thickness, 35 µm copper) with a total size of 60 × 35 mm was designed using Altium Designer (v.25.1.2) as Electronic Computer-Aided Design (ECAD) software. A four-layer design was chosen to achieve a compact and reliable layout. The top layer was used for signals and components, the inner layers for ground planes, and the bottom layer for additional signal routing, allowing clear separation of signal paths, power supply, and grounding. The SFO is powered by a 3.7 V, 2600 mAh lithium-ion battery (Emmerich ICR-18650NQ-SP) integrated with a fuse, a shutdown switch, and a battery management system (MCP73838). A buck-boost converter regulates the supply voltage to a stable 3.3 V. To enable Wi-Fi, Bluetooth, and BLE communication, a 2.4GHz ESP32-WROOM-32E-N8 MCU with an integrated PCB antenna from Espressif Systems was used, which is widely utilized in IoT applications. Additional components include a USB-C charging port and a multicolor LED, which indicates the connection state, battery level, and charging status. Electrostatic discharge protection for electronic components was implemented in accordance with IEC 61000-4-2 and IEC 61340-5, using protection diodes on critical inputs such as the USB interface. A 32-channel analog multiplexer was implemented to enable the integration of up to 24 FSRs, while keeping signal paths short and maintaining robust ground routing to preserve signal integrity.

An optional interface for vibration feedback was implemented using a transistor-based low-side switch controlled via pulse-width modulation. This simple and efficient solution was chosen over more complex driver ICs or Inter-Integrated Circuit Communication (I^2C)-based vibration modules to reduce system complexity and footprint. This approach allows for flexible use of an Eccentric Rotating Mass Motor (ERM). To generate a sufficiently strong tactile signal, an ERM vibration actuator (Model TC-13324656, TRU Components) operating at 2.4 V and 250 mA was selected.

To minimize space, components were placed and routed according to system requirements and manufacturer guidelines. The final design was verified using Electrical Rule Check (ERC) and Design Rule Checks (DRC). Board assembly was performed manually using solder paste and a heating plate for Surface-Mounted Device (SMD) soldering, followed by manual mounting of Through-Hole Technology (THT) components. The assembled board was visually inspected and functionally tested to ensure proper soldering and reliable system performance. Additionally, detailed battery tests confirmed the functionality of the integrated battery management system, including discharge and charging profiles. The estimated runtime without vibration feedback was approximately 35.5 h, with a full recharge time of 6–7 h depending on the initial state of discharge.

Smart Walking Aid To enable active haptic feedback through an assistive device, a forearm crutch was selected as the basis for sensor integration, as such walking aids are commonly used in combination with AFOs and offer a practical interface for feedback delivery. We developed a modular integration concept that can be transferred to various walking aids, including sticks, canes, or walkers, provided the handle design allows for modification without compromising ergonomics. This is essential, as major modifications to the handgrip require careful consideration of weight distribution and ergonomic design to avoid negative impact. For example, sensor placement along the shaft could alter the center of gravity, which may require new validation/certification of the walking aid [251]. Therefore, our design specifically targeted integration without major modifications to the handgrip. In this implementation, we used a standardized aluminum forearm crutch with a plastic grip and an integrated reflector. After removing the reflector, the inner cavity of the handle provided a usable space of 25 mm in diameter and 115 mm in length for the integration of the electronic components.

A custom-designed four-layer PCB (1.55 mm thickness, 35 µm copper) with a total size of 90 × 21 mm was developed to ensure compact placement and signal integrity. The board integrates a BMI270 IMU for motion sensing, an ESP32-WROOM-32E MCU (2.4 GHz Wi-Fi, Bluetooth) for wireless communication via BLE with the

SFO, and a Type-C USB connector for programming and battery charging. Power is supplied by an IFR 14500J AA lithium iron phosphate LiFePO4 battery (600 mAh, 3.2 V) from Keeppower, managed by a Microchip MCP73123 battery charge controller. A charge status LED was connected to a status pin for visual indication. To enable vibrotactile feedback, two DRV2605L haptic driver Integrated Circuits (ICs) from Texas Instruments were integrated, allowing the control of two ERM or Linear Resonant Actuator (LRA) vibration modules.

The USB connector, MCU, and battery holder were symmetrically arranged along the central axis of the PCB to ensure balanced layout and optimal use of space. The MCU and USB connector were placed on the front side, while the battery was positioned on the back side in accordance with the manufacturer's mounting guidelines. A Universal Asynchronous Receiver Transmitter (UART) interface was included to allow communication with an external PC and GUI during development. Before fabrication, ERC and DRC were performed to verify the circuit design and layout. The PCB was then assembled using a manual pick-and-place process, followed by reflow soldering in a three-zone convection oven (Mistral 260) for SMD components. Additional THT components were soldered manually in a subsequent step. After assembly, the boards underwent functional testing and visual inspection to ensure proper soldering and system performance. To ensure optimal positioning and mechanical stability within the handle, the assembled PCB was mounted inside a custom 3D-printed enclosure made of rigid PLA. To enable charging without disassembly, an access was added to the back of the crutch, providing external access to the USB-C port. An overview of the hardware architecture is provided in Figure 4.7, and the assembled prototype is shown in Figure 4.8.

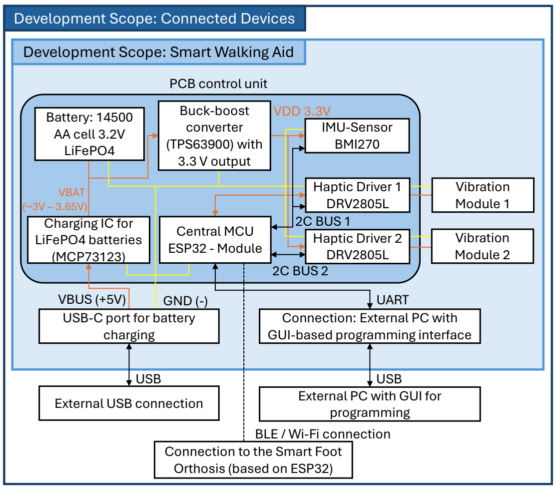


Figure 4.7: Hardware block diagram of the smart walking aid.

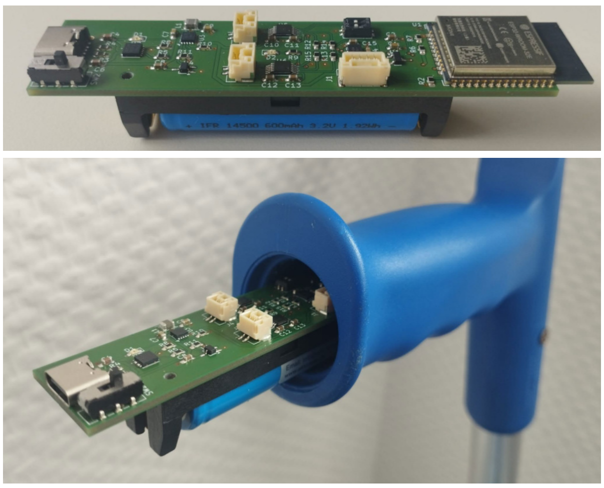


Figure 4.8: Final assembled prototype with integrated electronics.

Software Architecture and Data Processing

Smart Orthosis—Mobile Gait Analysis Figure 4.9 presents the SFO data processing architecture, structured into five stages: (1) raw data acquisition, (2) preprocessing, (3) signal processing, (4) sensor fusion, and (5) extraction of gait parameters. Each step is described in detail; see Appendix B.

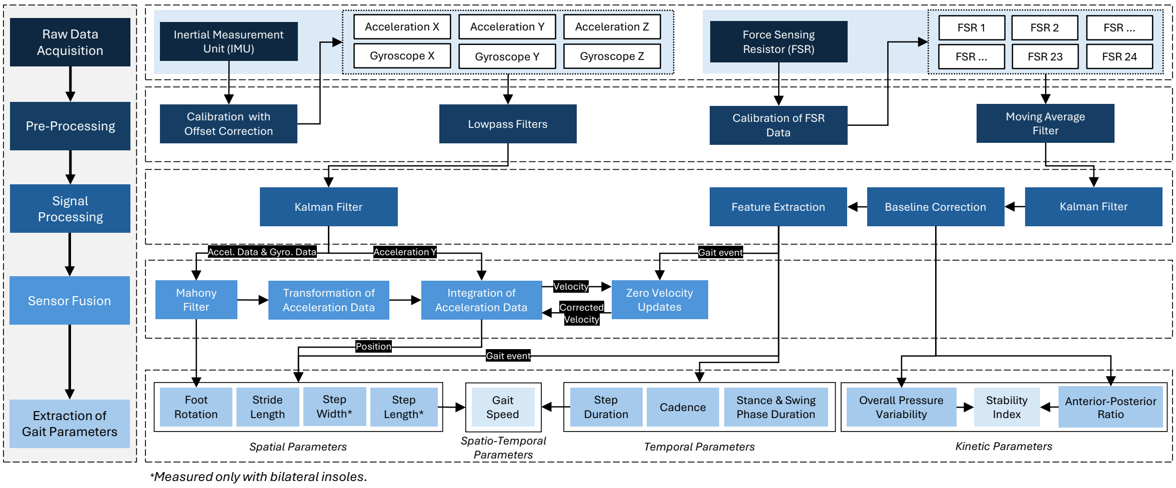


Figure 4.9: Overview of the data processing for gait data extraction of the SFO.

Smart Walking Aid—Haptic Feedback and Motion Data The ESP32 MCU receives processed gait event data from the SFO via wireless communication using either BLE or Wi-Fi. The firmware was developed in C/C++ using the Arduino Integrated Development Environment (IDE) and enables the control of two actuators to generate vibration patterns depending on pre-defined overloads. In the current prototype implementation, relative pressure thresholds were defined as fixed percentage values of the user's body weight to ensure consistent triggering conditions during system testing, similar to Mariani et al. [287], who used a threshold of 5% of body weight for gait phase detection. This constant threshold design served to demonstrate the feedback functionality under controlled conditions and was not intended to represent clinically calibrated loads. In future implementations, these thresholds could be dynamically personalized and adapted per gait phase (stance vs. swing) based on baseline load measurements obtained during standing and normal walking. However, such thresholds are individual to each patient and thresholds should be defined and validated by medical experts to ensure the detection of clinically relevant overload conditions [237]. Vibration signals are emitted using a configurable pulse-based pattern. Depending on the detected parameter both the intensity and interval of the vibration can be dynamically adapted. This customization allows targeted feedback for specific gait abnormalities, such as in-toeing or out-toeing patterns, where intensity can be increased with foot rotation angles. The integrated IMU, initialized with the same configuration as in the SFO, continuously records motion data for activity recognition. This data is also streamed to the smartphone application. To extend the functionality, these data could be used in future implementations for activity classification or fall detection.

Smartphone Application The design of the smartphone application builds upon the main functions proposed by Resch et al. [44] for a smart insole app dedicated to foot health monitoring. Building on these proposed functions, our application implements key features such as a heatmap visualization of plantar pressure based on the FSR sensors, real-time gait characteristics, and a walking diary displaying step counts and activity-related health status ratings. Additional modules include a knowledge database with a search function for articles on foot health, a training area featuring video tutorials for foot-related exercises, and a map for locating local medical specialists. The app also enables monitoring of the communication status between the SFO and the walking aid.

For pressure data visualization, a custom heatmap fragment shader was implemented. Its output was based on the number of FSRs used and colored using the Turbo colormap developed by Google [288]. The application was developed in Android Studio [289] (version 2024.2.1) using Kotlin as the main programming language. We utilized Jetpack Compose [290] and its integrated UI libraries from Material2 and Material3, enabling flexible design of forms, buttons, and icons. Wireless communication was implemented via Wi-Fi, Bluetooth, and MQTT protocols to ensure real-time data transmission between devices. The prototype was deployed as an Android Package Kit (APK) on an Android 14 device using Software Development Kit (SDK) version 35. Figure 4.10 shows selected screenshots of the main functionalities of the developed mHealth application.

Figure 4.10: Screenshots of the developed mHealth app showing key interface components, including account creation, main menu navigation, plantar pressure heatmap, gait diary, exercise tutorials, and access to foot health resources.

Prototype Validation Study: Experimental Setup

A technical feasibility and validation study was conducted with eight participants to evaluate the integrated system in terms of both objective measurement accuracy and subjective user experience. The primary goal of this proof-of-concept investigation was to verify the system's functional reliability and multimodal feedback performance under controlled laboratory conditions, rather than to focus on algorithmic novelty or benchmark testing of individual algorithms, or to conduct a clinical study. The study comprised two supporting validation objectives (system feasibility and application usability) and one primary objective (feedback evaluation).

1. System Feasibility: Validation of the SFO plantar pressure and spatial/temporal gait data against an instrumented treadmill.
2. Application Usability: User experience evaluation of the smartphone app based on the SUS [158].
3. Feedback Evaluation: Assessment of two feedback mechanisms applied within the SFO and forearm crutch using a customized questionnaire.

In the following subsections, the study design, apparatus, procedure and tasks, as well as the applied measures and data analysis, are documented.

Study Design The study was conducted in three parts, each with a specific validation objective. An overview of the experimental validation and evaluation setup is shown in Figure 4.11.

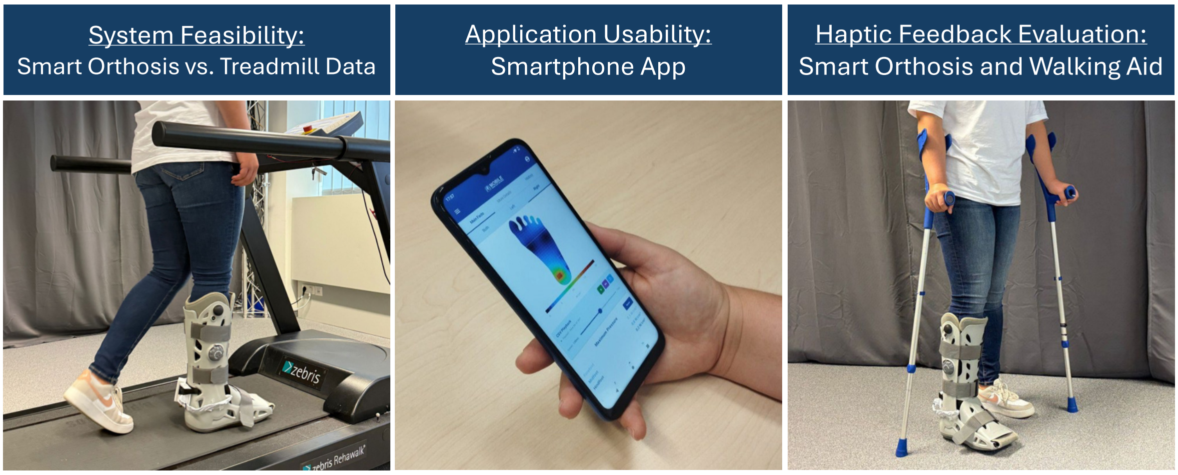


Figure 4.11: Visual overview of the experimental validation process including orthosis testing, usability assessment, and haptic feedback evaluation.

To evaluate the measurement accuracy of the SFO, an experimental validation was conducted against a clinical gait analysis reference system (Zebris FDM-T treadmill). The independent within-subject variable *Measurement System* included two levels: *SFO* and *Zebris Treadmill*. Dependent variables included key spatial (step and cycle length), temporal (stance, swing, and cycle phase durations), and kinetic (plantar pressure distribution) parameters.

The smartphone app was tested to assess perceived usability using the standardized SUS questionnaire. To evaluate the influence of haptic feedback on user perception, participants tested both the SFO and the crutch, each equipped with one vibration module. The study was based on a 2×2 factorial within-subject design with two independent variables: *System* (two levels: *Orthosis* and *Crutch*) and *Feedback Type* (two levels: *Continuous* and *Pulsed Pattern*). Dependent variables were based on subjective ratings regarding noticeability, intrusiveness, regular usage, and perceived usefulness. A 2×2 balanced Latin square design was applied to counterbalance the conditions and avoid sequence effects [291].

Apparatus The prototype systems (SFO, instrumented forearm crutch, and smartphone app) were used according to the previously described specifications and deployed in the study using MQTT for real-time data communication between the devices. The 3D-printed insole was designed to fit EU shoe sizes 38–43, covering the medium shoe size range in both women and men. To meet the minimum sensing requirements with optimal sensor placement, the insole of the orthosis was equipped with 15 FSRs. Furthermore, a single ERM 3.7V DC motor (model: JYCL0610R2540) from Jie Yi Electronics Limited was used as vibration actuator, integrated both into the orthosis and the handle of the crutch. This configuration was chosen to assess whether a single actuator can provide sufficient haptic feedback to be reliably perceived. The app was executed on a Motorola Moto G8 Power Lite 64 GB (Android 10).

For the evaluation of motion and pressure data acquired from the IMU and FSR sensors, we used an instrumented treadmill, which is considered the gold standard for validating gait analysis systems. Specifically, a RehaWalk® FDM-THPL-S-2i instrumented treadmill (Zebris Medical GmbH, Isny, Germany) was used to measure spatial and temporal gait parameters. The system is based on an h/p/cosmos pluto treadmill and integrates 3120 capacitive sensors (±0.05% measurement accuracy), arranged in an 80×39 grid layout, covering a sensor area of 101.6×49.5 cm. Data were recorded at a sampling rate of 100 Hz (similar to the SFO), within a sensor range of 1–120 N/cm². Zebris FDM software (version 2.0.4) was used to run gait analysis and export CSV raw data. The software ran on a Windows 10 Pro system with an AMD Ryzen 5900X 12-core processor, 3.70 GHz, a GeForce RTX 3070 graphics card, and 32 GB RAM.

Procedure and Tasks The study was conducted under controlled laboratory conditions without external influences. First, participants were informed about the study procedures and that they could withdraw or discontinue participation at any time. After providing informed consent, they completed a survey about their demographic data, foot conditions, previous experience with assistive devices, and mHealth applications. Furthermore, participants were asked to rate their familiarity with the use of smartphone healthcare apps. Following this, the SFO was provided to each participant as a fully preassembled system. An integrated vacuum-based mechanism was used to ensure optimal adaptation to the individual leg shape. After a brief instruction, each participant completed five walking trials on the Zebris treadmill at a fixed speed of 1.5 km/h. Each trial lasted 10 s to ensure the recording of at least one full gait cycle, typically capturing multiple cycles for averaging and comparison across multiple measurements. Participants were instructed to walk as normally as possible without holding onto the treadmill handles.

After the walking trials, the walking aid prototype was introduced and tested both while standing and walking, in combination with the SFO. Haptic feedback was evaluated for both systems under predefined overload conditions to demonstrate feedback alerts and to ensure identical vibration triggering conditions across all participants. Two feedback type modalities (continuous and pulsed patterns) were activated in randomized order for each participant. Following this, participants completed a questionnaire consisting of quantitative Likert items to assess noticeability, intrusiveness, regular usage, and perceived usefulness. Additionally, participants were asked to indicate their preferred vibration alert type in a single-choice question. Finally, participants completed a smartphone usability task and assessed the app using the SUS questionnaire. An optional certificate of participation was offered after the study.

Measurement and Data Analysis A quantitative measurement approach was chosen to evaluate the overall system and its individual subsystems. Descriptive and inferential statistical analyses were conducted using R (version 4.2.2) with the rstatix package [292]. The following paragraphs describe the corresponding measurements and data analysis in detail.

Smart Orthosis: Validation of Plantar Pressure and Gait Parameters Both systems recorded gait events and pressure-related data simultaneously. Manual triggering was performed during the swing phase of the right leg (foot orthosis) to synchronize both systems at the moment of initial right foot contact. Each trial lasted for 10 s, recorded at a sampling rate of 100 Hz in both systems, resulting in five datasets per participant. The Zebris FDM software enabled the export of pressure data and gait characteristics were aggregated as mean and standard deviation values per trial. The definitions of the respective gait parameters are provided in Table 4.6.

To enable a valid comparison, the raw data from the SFO were processed equally: gait characteristics were computed per trial and then aggregated on the participant level into mean and standard deviation values. Descriptive statistics were used to summarize the data, and inferential statistics were applied using paired t-tests to examine significant differences between both measurement systems.

Table 4.6: Overview of analyzed gait parameters grouped by parameter type.

Parameter Type	**Parameter**	**Unit**
Temporal	Stance Duration	s
	Swing Duration	s
	Cycle Duration	s
Spatial	Stride/Cycle Length	m
Kinetic	Mean Plantar Pressure	N/cm^2

For pressure data, the full time-series dataset was used for the SFO, while the treadmill provided aggregated data for one step. However, a direct quantitative comparison of force or pressure values between the treadmill and orthosis systems is not possible, due to fundamental differences in sensor configuration and measurement principles. The treadmill measures plantar pressure at the interface between the orthosis sole and the treadmill surface, while the orthosis records forces between the foot and the insole. As a result, the absolute values and spatial resolution of the pressure data differ between systems. Therefore, pressure data were visualized as heatmaps and force–time curves to qualitatively assess whether gait events could be reliably identified from the force signal progression.

Application Usability: Evaluation of Smartphone App Usability Usability of the smartphone application was assessed using the standardized SUS, which is frequently used to assess user experience of interactive systems such as smartphone apps. The questionnaire consists of ten items rated on a five-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Even-numbered items were reverse-scored and calculated according to the specifications by Brooke [158]. The total score was calculated and converted into a value between 0 and 100 as a quantifiable measure of overall usability. Descriptive statistics were used to analyze the SUS scores.

Feedback Evaluation: User Perception of Haptic Feedback (Orthosis and Walking Aid) To assess the user experience and perceived effectiveness of the haptic feedback system, the participants rated four statements on a 7-point Likert scale [293]. The items addressed noticeability, intrusiveness, regular usage, and perceived usefulness, with response options ranging from 1 (*strongly disagree*) to 7 (*strongly agree*).

The questionnaire items included:

- Noticeability: "*The haptic feedback was clearly noticeable during use.*"
- Non-Intrusiveness "*I found the feedback comfortable and non-intrusive.*"
- Regular Usage "*I could imagine using such feedback during everyday mobility tasks.*"
- Usefulness "*The feedback would be useful in real-life situations (e.g., alerting me of incorrect foot placement or walking posture).*"

Descriptive statistics were applied to summarize the participants' responses and identify trends in agreement levels. A two-factorial Repeated-Measures (RM) Analysis of Variance (ANOVA) was used to assess the effect of system and feedback type on noticeability, intrusiveness, regular usage, and perceived usefulness.

Participants The study was conducted in accordance with the data privacy regulations of our institution and received ethical approval from the German Society for Nursing Science (No. 23-027). Eight participants were recruited through institutional mailing channels and provided informed consent before participation. To participate in the study, individuals were required to meet the following inclusion criteria:

- Footwear size: European shoe size between 38 and 43.
- Current or past foot-related conditions, including either acute or chronic conditions
- Current or previous use of foot-related orthopedic products, such as shoe insoles, foot orthoses, bandages, or foot casts.
- Right foot affected or bilateral condition (but not solely left foot), as the sensor insole was configured specifically for the right foot.

Prior to study participation, all individuals completed a brief pre-screening to confirm that these criteria were met. In total, 8 participants (4 female, 4 male) were recruited. The age of the participants ranged between 24 and 72 years ($M = 39.75$, $SD = 17.92$). Participants' educational backgrounds included four with vocational training, three with a bachelor's degree, and one with a master's degree. Occupations were distributed as follows: services and sales ($n = 2$), engineering ($n = 2$), other fields ($n = 2$), clerical ($n = 1$), and health-related professions ($n = 1$). The shoe size ranged from EU 38 to EU 43 ($M = 40.37$, $SD = 1.85$). All participants reported having a current or past foot condition. Four had previous conditions (such as a broken leg, sprained ankle, pes planus, or torn ligament). Two participants reported chronic conditions (foot malalignment and arthrosis), one participant had an acute condition (sprained ankle), and one reported both an acute (sprained ankle) and chronic condition (torn ligament and previous leg fracture). All participants indicated that they had previously used orthopedic devices, including insoles ($n = 7$), ankle braces or bandages ($n = 3$), and foot orthoses ($n = 2$). Furthermore, four participants already used forearm crutches as medical walking aids. In total, three participants had previous experience with a mobile health app. Participants rated their familiarity with the use of smartphone healthcare apps on a scale from 1 (*not at all familiar*) to 5 (*extremely familiar*), with an average rating of 2.38 ($SD = 1.60$). All participants completed the study and were included in the analysis.

4.2.3 Results

The results section first presents the comparative accuracy validation between the SFO and the instrumented treadmill, followed by the usability assessment of the smartphone application, and concludes with the evaluation of the haptic feedback of both the orthosis and the walking aid.

System Feasibility: Accuracy of Plantar Pressure and Gait Parameters (Smart Orthosis)

Gait Data Measurement To assess the reliability of temporal and spatial gait parameters, the accuracy of the SFO measurements was evaluated against the treadmill system across all trials. The results showed that temporal parameters achieved mean accuracies above 85%, except for swing phase duration, which reached only 60%. For the spatial parameter, the mean accuracy was 83.4%, despite a high standard deviation. The descriptive results are summarized in Table 4.7.

Table 4.7: Mean and standard deviation of gait parameter accuracy of the smart foot orthosis (SFO) compared to treadmill measurements across all participants.

Parameter Type	Parameter	Mean Accuracy	SD Accuracy
Temporal	Stance Phase Duration	96.7%	±3.1
	Swing Phase Duration	60.3%	±12.6
	Cycle Phase Duration	85.2%	±1.6
Spatial	Stride/Cycle Length	83.4%	±13.3

Temporal Parameters The Shapiro–Wilk normality test indicated that both cycle phase duration ($p > 0.095$) and swing phase duration ($p > 0.501$) were normally distributed, while stance phase duration was not ($p = 0.013$). A paired t-test revealed a statistically significant difference between SFO and treadmill for swing phase duration ($t(7) = 9.74$, $p < 0.001$), Cohen's d = 3.44 (large effect size), as well as for cycle phase duration ($t(7) = -18.7$, $p < 0.001$), Cohen's d = 6.59 (large effect size), which was significantly shorter in SFO compared to treadmill. Furthermore, a Wilcoxon signed-rank test on stance duration data showed no significant difference ($W = 8$, $p = 0.195$, with moderate effect $r = 0.50$).

A Pearson correlation analysis showed a strong and significant correlation for cycle duration ($r = 0.973$, $p < 0.001$), and a moderate but not statistically significant correlation for swing duration ($r = 0.673$, $p = 0.067$). Additionally, a Spearman correlation analysis for stance duration, which was not normally distributed, did not reveal a significant correlation ($\rho = -0.167$, $p = 0.693$).

Spatial Parameters The Shapiro–Wilk test indicated that cycle length ($p = 0.259$) was normally distributed. A paired t-test revealed a statistically significant difference in cycle length between the treadmill and orthosis conditions ($t(7) = -3.65$, $p = 0.008$), Cohen's $d = 1.29$ (large effect), indicating that cycle length was significantly shorter when measured with the orthosis compared to the treadmill. Pearson correlation analysis showed no significant correlation for cycle length ($r = 0.678$, $p = 0.065$).

Pressure Data Measurement The force data recorded with the SFO enable the identification of gait cycles and their characteristic phases (initial contact, full contact, and toe-off), as illustrated in Figure 4.12, demonstrating its suitability as a mobile gait analysis system. However, a direct quantitative comparison with the instrumented treadmill is not feasible due to fundamental differences in measurement and data availability. While the Zebris system measures contact forces between the treadmill surface and the bottom of the orthosis, the SFO records plantar forces directly between the foot and the insole. In addition, the Zebris software provides only aggregated outputs with averaged curves rather than continuous raw data, which prevents a synchronized step-wise comparison.

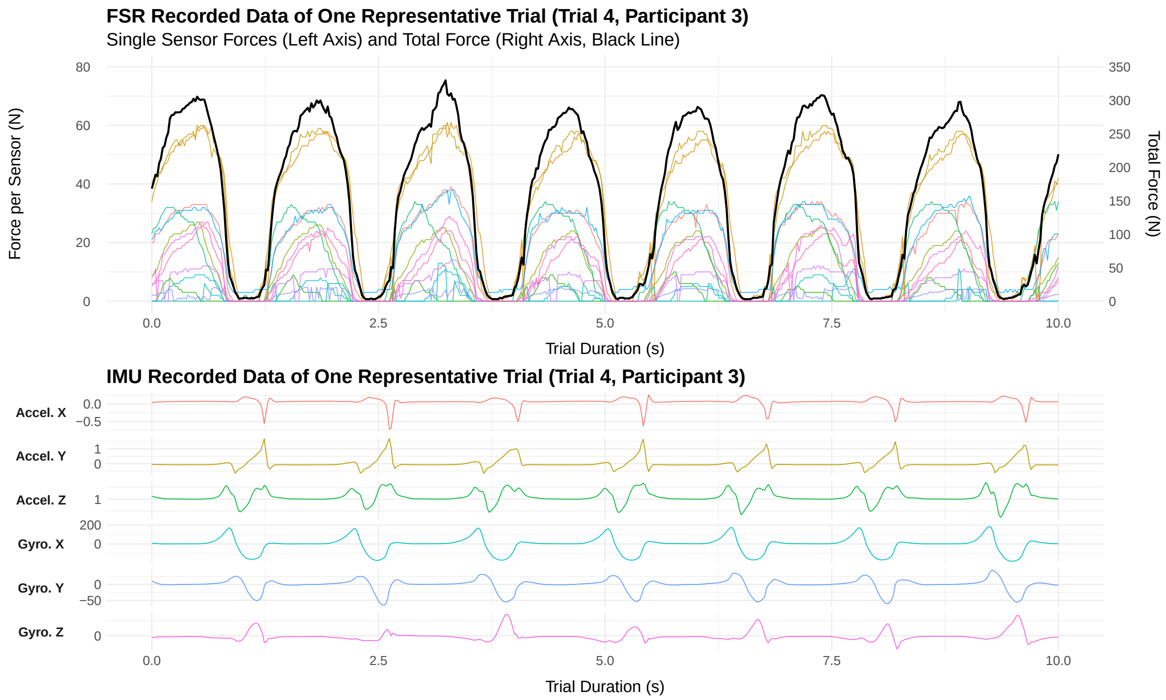


Figure 4.12: (Top): Measured plantar pressure data of Trial 4 (P3), showing force values recorded by multiple FSR sensors over the duration of the task. Distinct gait phases can be identified from the characteristic patterns in the force signals. The left axis represents the force values per sensor, while the right axis indicates the total force (black line) across all included sensors. (Bottom): Measured acceleration and gyroscope data from the IMU for each of the three axes, illustrating gait phase conformity with the recorded pressure data.

Nevertheless, to compare both systems a qualitative comparison of plantar pressure distributions and force curves was conducted using visualizations from both systems for a representative trial (see Figure 4.13). The Zebris data represent averaged force curves for the three foot zones summarized from multiple gait cycles, whereas the orthosis data focus on one representative gait cycle from the same trial, which is also displayed in Figure 4.12.

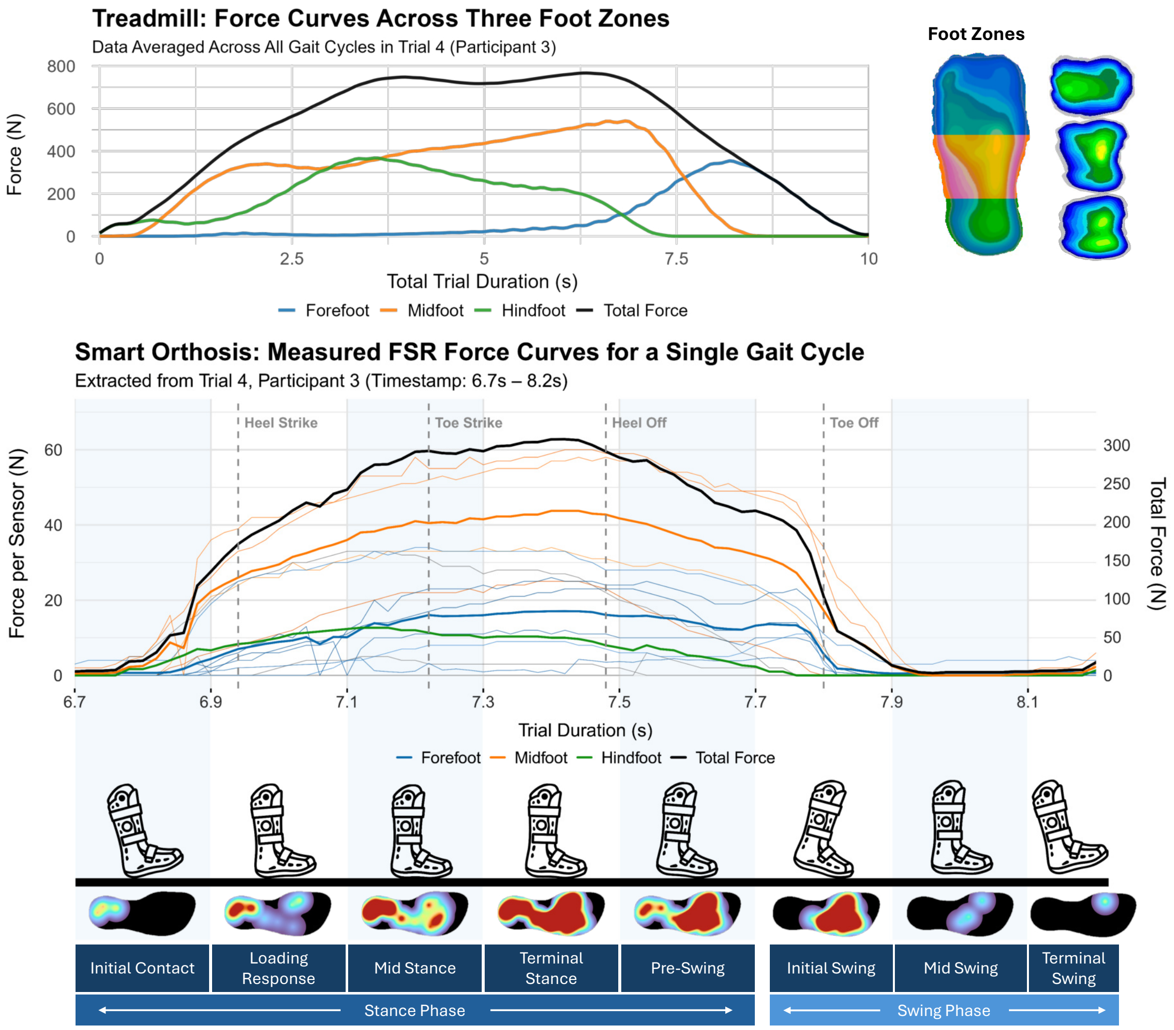


Figure 4.13: The upper plot shows the aggregated plantar force curves of gait cycles recorded in Trial 4 of Participant 3 using the Zebris system. The force curves are color-coded according to three anatomical foot regions, as indicated by the footprint icon on the right. Additionally, a cumulative plantar pressure heatmap illustrates the spatial force distribution over the entire trial. The lower plot displays data from the same trial (Trial 4, Participant 3), focusing on a single gait cycle recorded by the orthosis. The three regional force curves represent the summed signals of the individual FSRs and are color-coded accordingly. Four gait events identified by the SFO are marked in gray. Below the graph, individual heatmaps illustrate the pressure distribution for each gait event.

Although absolute force values are not directly comparable due to differences in sensor configuration, both systems show similar temporal trends across the three anatomical foot zones and in the overall force–time diagram. Compared to the Zebris treadmill, which provides only aggregated contact data, the SFO offers step-wise heatmap visualizations, enabling a detailed analysis of the internal plantar pressure distribution for each gait event.

Application Usability: Evaluation of Smartphone App Usability

The mean SUS score was 80.62 (SD = 10.59), as shown in the boxplot diagram (Figure 4.14). According to the interpretation guidelines by Bangor et al. [294], this score falls within the "Acceptable" range based on the Acceptability Ranges and is considered "Good" according to the Adjective Ratings, indicating a high level of usability. Ratings for the ten individual SUS items are visualized in the radar plot (Figure 4.15), with values ranging from 6.56 to 9.38 (normalized on a 0–10 scale). Furthermore, the raw ratings of the subscales are summarized in Table 4.8 using descriptive statistics.

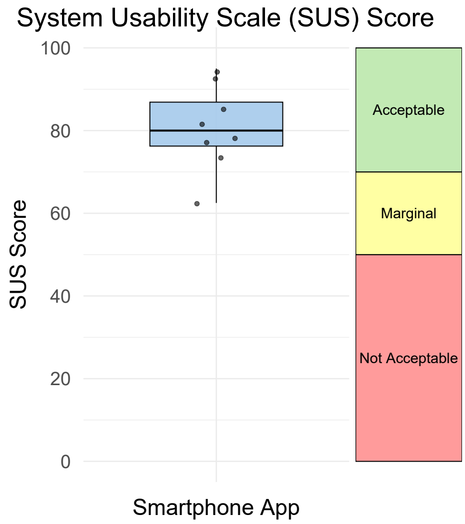


Figure 4.14: Boxplot of overall SUS scores and the corresponding acceptability categories.

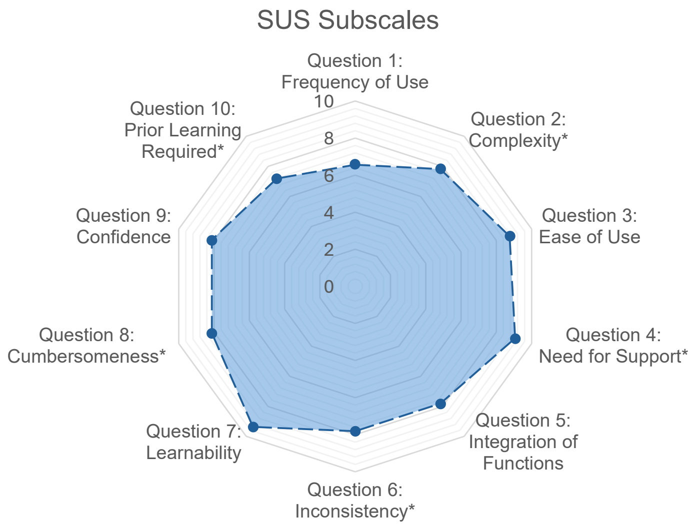


Figure 4.15: Radar plot visualizing the ten individual item scores, normalized on a 0–10 scale. Asterisks (*) indicate reverse-scored items.

Table 4.8: Descriptive statistics of the SUS items.

No.	Item	M	SD
1	Frequency of Use	3.63	0.92
2	Complexity *	1.88	0.83
3	Ease of Use	4.50	0.53
4	Need for Support *	1.38	0.74
5	Integration of Functions	4.13	0.99
6	Inconsistency *	1.88	0.64
7	Learnability	4.75	0.46
8	Cumbersomeness *	1.75	1.04
9	Confidence	4.25	0.71
10	Prior Learning Required *	2.13	1.36

Note. Asterisks (*) indicate reverse-scored items.

Feedback Evaluation: User Perception of Haptic Feedback (Smart Orthosis and Walking Aid)

Shapiro–Wilk tests were conducted for each condition on the questionnaire items *Noticeability*, *Non-Intrusiveness*, *Regular Usage*, and *Usefulness*, and revealed that the data were not normally distributed. Therefore, an Aligned Rank Transform (ART) RM ANOVA was conducted on the four measures, revealing two significant main effects. A statistically significant main effect of *Noticeability* was found for *Device* with $F(1, 21) = 4.49$, $p = 0.046$, $\eta_p^2 = 0.18$ (large effect size). Post hoc pairwise comparisons using Wilcoxon signed-rank tests did not reveal a significant difference between *Orthosis* and *Crutch* ($p = 0.065$). However, the corresponding effect size ($r = 0.483$, moderate-to-large) suggests a potentially meaningful difference that may not have reached significance due to the limited sample size and resulting low statistical power. Furthermore, a statistically significant main effect was found for *Usefulness* on *Feedback Type* with $F(1, 21) = 6.01$, $p = 0.023$, $\eta_p^2 = 0.22$ (large effect size). Post hoc Wilcoxon signed-rank tests revealed a significant difference between *Continuous* and *Pulsed Pattern* ($p = 0.008$).

Results of the ART RM ANOVA and post hoc pairwise comparisons for all conditions are shown in Table 4.9. Additionally, the overall results are presented as a boxplot with descriptive statistics in Figure 4.16.

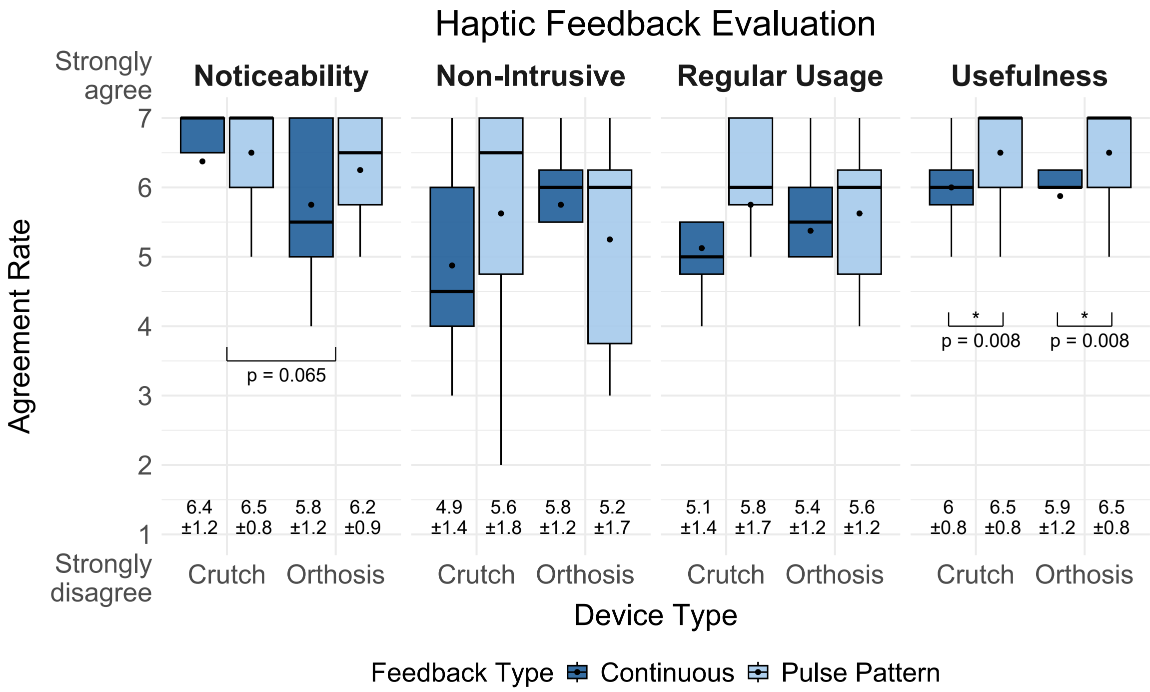


Figure 4.16: Boxplots show agreement ratings for four measures on haptic feedback via the foot orthosis and the forearm crutch, ranging from 1 (strongly disagree) to 7 (strongly agree). It displays the interquartile range (box), 1.5 × IQR range (whiskers), median (line), and mean (black dot). Mean and standard deviation (±) values are shown below each boxplot. Asterisks (*) indicate statistically significant results ($p < 0.05$).

Table 4.9: Results of the ART RM ANOVA and post-hoc Wilcoxon signed-rank tests for the four questionnaire items.

		ART RM ANOVA					Pairwise Comparisons			
Measure	**Factor**	**F**	**Df**	p	**Sig.**	η_p^2	**Stat.**	p	r	**Magnitude**
Noticeability	Device	4.49	1, 21	.046	*	.176	25.0	.065	.483	moderate
	Feedback	1.80	1, 21	.194		.079	22.0	.187	.310	moderate
	Device × Feedback	1.60	1, 21	.220		.071	—	—	—	—
Non-Intrusiveness	Device	0.06	1, 21	.815		.003	27.5	.653	.106	small
	Feedback	0.50	1, 21	.487		.023	42.5	.811	.046	small
	Device × Feedback	2.24	1, 21	.150		.096	—	—	—	—
Regular Usage	Device	0.00	1, 21	.966		.000	42.5	.807	.099	small
	Feedback	2.57	1, 21	.124		.109	50.0	.129	.326	moderate
	Device × Feedback	0.77	1, 21	.390		.035	—	—	—	—
Usefulness	Device	0.00	1, 21	1.00		.000	24.5	.851	.076	small
	Feedback	6.01	1, 21	.023	*	.220	36.0	.008	.703	large
	Device × Feedback	0.00	1, 21	1.00		.000	—	—	—	—

Note. Asterisks (*) indicate statistically significant results ($p < 0.05$). Effect sizes are reported as partial η^2 for ART RM ANOVA and as rank-biserial correlation (r) for Wilcoxon signed-rank tests. Magnitude classification of r: small ($r < 0.3$), moderate ($0.3 \leq r < 0.5$), large ($r \geq 0.5$) [295].

Correlation of Ratings Across Devices To further investigate consistency in user perception across devices, a Spearman rank correlation was conducted between ratings for the same *Feedback Type* on the *Orthosis* and the *Crutch*. Results showed a significant positive correlation for *Noticeability* in the *Pulsed Pattern* feedback condition ($r_s = 0.86$, $p = 0.006$), suggesting a high level of consistency in perceived noticeability across both devices. Furthermore, in the *Continuous* feedback condition, a significant positive correlation was found for *Regular Usage* ($r_s = 0.78$, $p = 0.023$), while the remaining conditions showed positive trends, but did not reach statistical significance: *Noticeability* ($r_s = 0.62$, $p = 0.099$), *Non-Intrusiveness* ($r_s = 0.56$, $p = 0.152$), and *Usefulness* ($r_s = 0.58$, $p = 0.133$).

Post-Experiment Questionnaire Participants rated the feedback types for both devices separately to identify preferences between the different modalities. For the *Orthosis*, the *Pulsed Pattern* feedback was preferred by four participants, while the remaining four participants rated both feedback types as equally acceptable. None of the participants selected the *Continuous* signal or no feedback as their preferred option. For the *Crutch*, six participants preferred the *Pulsed Pattern* feedback. Two participants indicated no preference and rated both options equally. Similar to the orthosis results, no participant selected the *Continuous* signal or no feedback.

4.2.4 Discussion

We developed a sensorized system for monitoring plantar pressure and gait activity within an orthopedic foot orthosis. The system provides haptic feedback through either the orthosis or a corresponding forearm crutch and enables remote monitoring via a smartphone application. In an experimental user study with eight participants, we evaluated the system's feasibility and investigated the influence of device location and feedback modality on user perception of vibrotactile biofeedback. The findings confirm the system's technical feasibility for gait phase detection, the usability of the accompanying mobile application, and provide new insights into haptic feedback perception. The following subsections discuss the results in detail, including implications, limitations, and directions for future research.

System Feasibility and Application Usability

System Feasibility: Accuracy of Plantar Pressure and Gait Parameters (SRQ1) SRQ1 focused on evaluating the accuracy of plantar pressure and gait measurements obtained from the developed SFO compared to a clinical reference system. Our findings showed that the recorded temporal parameters for the swing and cycle phases indicated significantly shorter durations when measured with the SFO compared to the treadmill. However, cycle duration showed a strong correlation between both systems, whereas swing duration did not reach statistical significance ($p = 0.067$). Overall, swing duration showed the lowest accuracy among all temporal parameters, with only 60% accuracy and notably higher standard deviations. This may be explained by the observation that some FSR sensors were constantly triggered and did not return to zero during offloading, which could have impaired the correct detection of events from stance to swing. Although the stance phase showed no significant differences and achieved high accuracy of 96%, the correlation between both systems was not significant. These inconsistencies are likely caused by differences in event detection, as the SFO relies on FSR-based thresholds, whereas the treadmill reference system uses ground reaction forces. Additionally, some triggering issues of certain FSR sensors may have introduced inconsistent deviations in individual measurements, which could explain the reduced correlation, even though the average stance durations between both systems were similar. As highlighted in previous research, lower walking speeds (0.8–1.0 m/s) compared to moderate walking speeds (1.2–1.4 m/s) are associated with higher error rates in temporal parameters (e.g., stance and swing time), as well as spatial parameters (e.g., stride length) [296]. As the walking speed in our experiment was relatively low (0.417 m/s) to ensure participant comfort while walking with the orthosis, this factor may have contributed to reduced accuracy in gait parameter estimation.

The measured accuracy of the spatial parameter (cycle length) indicated that an accuracy above 83% can be achieved. However, the estimation of spatial gait parameters using a single smart insole systematically underestimated cycle length compared to treadmill-based reference measurements, and no significant correlation was found between both measurement systems. One explanation for these systematically lower values is that the insole was worn unilaterally, while the other foot was not equipped with a sensorized insole to provide complementary data. Furthermore, wrong triggering of the algorithm could have led to incorrect gait event detection. By identifying this systematic offset, a post-processing correction factor could be applied to improve the dataset for further analyses.

The evaluation of pressure data showed that gait events can be reliably detected and continuous pressure distribution can be measured during both walking and standing. Although the comparison with the treadmill was performed on a qualitative basis, the results showed that the three foot zones (forefoot, midfoot, and hindfoot) exhibited similar temporal patterns. This highlights the added value of the SFO, as it enables step-wise heatmap visualizations and detailed insights into the internal plantar pressure distribution without post-processing, which are not achievable with stationary treadmill systems. In summary, our results confirm previous findings showing that one IMU is sufficient for gait monitoring [76] and that the recommended configuration of 15 FSRs [71] is suitable for reliable pressure measurement. However, further refinements are recommended to improve measurement accuracy and enhance data reliability.

Compared to related studies on smart insoles and gait monitoring systems, the accuracy levels achieved in this work are within a comparable range. For example, D'arco et al. [69] reported high performance in activity recognition using smart insoles, with accuracies ranging from 70% to 99%. This demonstrates that comparable performance levels can be achieved even without additional post-processing. Despite differences in sensor configuration (type, quantity, and placement), as well as varying application contexts, testing scenarios and data processing, the present study demonstrates that reliable gait analysis can also be achieved using a single insole. With further iterative refinements of the sensing and calibration procedures, even higher accuracy levels may be achieved in future implementations.

Application Usability: Evaluation of Smartphone App Usability (SRQ2) Regarding SRQ2, which addressed the usability of the accompanying mobile application for visualizing gait and pressure data, our results showed an acceptable level of usability. The SUS evaluation reached an average score of 80.62 (SD = 10.59), which is comparable to the results reported by Resch et al. [45] for a smart insole app (type: basic), with a mean SUS score of 77.5 (SD = 16.74). In comparison to other digital health applications, our findings also exceed the average usability benchmark identified by Hyzy et al. [297], who conducted a meta-analysis and reported a mean SUS score of 68.05 (SD = 14.05) across various digital health apps.

Overall, the subscale ratings revealed that item 4 (*need for support*) and item 7 (*learnability*) received the highest scores. This indicates that participants largely agreed with the statement "*most people would learn to use this app very quickly*" (item 7) and disagreed with the statement "*I would need the support of a technical person to use the system*" (item 4), which was reverse-scored. In contrast, item 1 (*frequency of use*) received the lowest average rating (M = 3.62, SD = 0.92 on a 5-point Likert scale), which shows that participants potentially were less willing to use the app frequently. This might be due to the fact that some participants were only temporarily affected by foot-related conditions and therefore did not perceive frequent use of the app as necessary. However, to address the potential limited interest in frequent use, future iterations of the app could incorporate targeted strategies to increase user engagement. For example, personalized notifications and reminders could encourage consistent daily use, particularly for self-monitoring tasks such as the integrated pain-level diary. Previous research suggested that gamification elements in health interventions (e.g., progress tracking or reward systems) can have positive effects on user experience [298]. Therefore, the integration of personalized measurable goals and feedback could support user motivation and foster continued use, especially in long-term rehabilitation contexts.

Feedback Evaluation: User Perception of Haptic Feedback (MRQ)

The MRQ investigated how different haptic feedback types (*Continuous* vs. *Pulsed Pattern*) implemented in an orthosis and a walking aid affect user perception regarding *Noticeability*, *Non-Intrusiveness*, *Regular Usage*, and *Usefulness*. Descriptive results showed that participants generally rated the *Pulsed Pattern* feedback higher than the *Continuous* feedback across all four items for both devices. A likely reason is that the rhythmic structure of the pulsed signal was perceived as more distinct and meaningful, making it easier to recognize and less monotonous compared to a continuous vibration. An exception was found for *Non-Intrusiveness*, where the *Pulsed Pattern* received lower ratings specifically in the orthosis condition.

Inferential analysis using an ART RM ANOVA revealed a significant main effect of *Device* on *Noticeability*, as the crutch was rated significantly higher than the orthosis. One possible explanation is that some participants had reduced sensation in the lower limbs, which made the haptic signal (especially *Continuous* feedback) less perceptible when delivered via the foot. Although previous work has suggested that foot-based feedback may be preferable to hand-based in some applications [299], in this specific use case the hand appears to be more sensitive to the vibration signal using one ERM actuator. This highlights that feedback effectiveness is not only dependent on signal type, but also on feedback location and tactile sensitivity at different body sites and devices.

For *Non-Intrusiveness*, the ratings varied more strongly across conditions. In particular, the *Continuous* feedback was rated lower than the *Pulsed Pattern* when applied via the crutch, which indicates that continuous vibrations may be perceived as more intrusive in this form factor. The *Regular Usage* showed similar ratings, however, without significance. Ratings for perceived *Usefulness* showed a significant effect of feedback type, indicating that the *Pulsed Pattern* was perceived as more useful than the *Continuous* feedback. These findings highlight that temporally modulated vibration signals are more easily interpreted and less annoying than continuous signals in wearable biofeedback applications. This result aligns with the post-study feedback, in which most participants expressed a clear preference for the pulsed signal. Although other trends were observed, they did not reach significance, which is probably due to the small size and limited statistical power of the study.

These findings support previous research on the utility of ERM-based vibration feedback [300]. However, our results extend this work by demonstrating that both pattern and device location influence the perception of vibrotactile feedback in orthopedic assistive devices. While a single ERM actuator ensured sufficient noticeability in the crutch, we recommend using at least two actuators inside the orthosis to enhance perceptibility. Overall, our results underscore the importance of tailoring haptic feedback implementation to the device form factor and user preferences. Therefore, designers should consider not only the feedback modality but also where and how it is delivered, and ensure that the signal is both interpretable and acceptable during everyday use. In line with prior work emphasizing the need to evaluate haptic feedback in assistive systems [105], our study contributes initial insights into user perception, acceptance, and effectiveness.

Implications

The presented system was designed as a general-purpose gait monitoring and feedback platform that can be applied both to individuals with foot conditions and for preventive or rehabilitative use in broader populations. While no clinical validation has yet been conducted, the system's sensing and feedback capabilities hold potential for future application, particularly for monitoring plantar pressure distribution and detecting overload patterns during daily activities.

The primary contribution of this work lies in advancing the understanding of how vibration-based feedback can be applied and perceived in foot-related assistive systems. While the developed hardware and software components served as necessary groundwork, they were mainly used for system verification. Building on this foundation, the central contribution is the investigation of real-time haptic overload notifications driven by the integration of plantar pressure and motion measurements. The fused dataset from pressure and motion sensors enabled a closed-loop feedback mechanism that responds to gait events. Our results demonstrate that such multimodal data can be feasibly integrated to trigger vibration cues. Based on this, the evaluation of haptic feedback provided new insights into the perception and preferences related to vibration-based notifications in assistive systems. The findings show that pattern-based feedback is perceived as more effective for overload alerts, while perceptual differences between the orthosis and the walking aid highlight the need for differentiated actuator placement, intensity modulation, and an increased number of actuators to ensure sufficient noticeability. These insights contribute to a clearer understanding of multimodal feedback perception in foot augmentation systems and can inform design guidelines for future adaptive vibration strategies in assistive devices.

Overall, these findings extend existing knowledge on isolated prototype developments by combining system integration, technical validation, and user-centered evaluation. These implications can guide future research and the development of intelligent footwear, multimodal feedback solutions, and interactive mobile health applications, contributing to the fields of HCI and biomedical engineering.

Limitations

The findings of our study are subject to several limitations. First, the evaluation primarily focused on technical development and validation under controlled laboratory conditions. In particular, the validation was limited to the right foot, as the orthosis was consistently worn on the right side. Moreover, the gait analysis was conducted using short treadmill walking sequences, which may not accurately reflect natural gait behavior during prolonged use or in real-world environments. In addition, calibration of the FSR sensors was performed based on manufacturer specifications without subject-specific mechanical calibration, potentially affecting the accuracy of absolute force estimation. Potential variation in sensor placement and the lack of automated recalibration routines may further impact signal reproducibility and long-term stability in real-world scenarios. Furthermore, comparability with treadmill-derived force values was limited, as only aggregated gait data were available from the treadmill, whereas the insole provided full raw data. Similarly, the developed algorithm for gait event detection was not quantitatively compared with existing data-driven gait phase recognition methods, nor validated on publicly available datasets. Finally, the small sample size limits the statistical power of the study and the generalizability of the observed effects, and no control group was included to provide a baseline comparison between participants with and without foot conditions. Nevertheless, the present work was designed as a proof-of-concept investigation with the primary focus on validating the technical feasibility and feedback functionality of the developed sensor system within the intended target group. Consequently, the study focused on end users with foot conditions to ensure practical relevance. Accordingly, the findings represent an initial technical evaluation conducted under controlled laboratory conditions and serve as a foundation for subsequent large-scale studies aimed at generalizing the results.

Future Work

The modular system architecture, which integrates multimodal sensing and feedback, demonstrates how next-generation orthotic devices can move beyond passive monitoring toward interactive and adaptive support. To address current limitations, future work should aim to evaluate the system under real-world conditions and in clinical contexts. In this context, future testing should include varied environmental conditions, such as uneven outdoor surfaces or stair walking, to evaluate the system's robustness under realistic gait scenarios. Such experiments will be essential for verifying performance outside controlled laboratory settings. Another direction for future work is the extended testing of the smart insole with regard to material specifications and biocompatibility to ensure long-term durability and safe use in rehabilitation contexts. Specifically, future research should include a larger and more diverse participant sample to allow for more comprehensive statistical analyses and improve the generalizability of the findings. Expanding the sample size will also enable subgroup comparisons and allow the assessment of long-term performance and usability across different user populations. Future studies should further include a control group of participants without foot conditions to establish a baseline for gait parameter comparison and system optimization. Including such a reference group with healthy participants will allow for more comprehensive interpretation of gait-related outcomes and support system optimization. Moreover, a case study involving patients with gait impairments could provide valuable insights into the system's robustness, usability, and therapeutic potential.

From a methodological perspective, future evaluations should also move beyond subjective usability scales and include behavioral and performance-based assessments, such as latency, energy consumption, correction rate, or changes in gait symmetry when interacting with the feedback system. Additionally, future work could benchmark the algorithm against state-of-the-art gait phase detection approaches and validate it on open-source datasets to enable objective performance comparison and generalization testing.

From a technical perspective, the integration of additional sensing modalities, such as temperature, GPS, or physiological sensors, could further enhance system functionality and contextual awareness. The modular architecture supports the flexible integration of such components for future developments. Furthermore, the use of AI-based approaches, including machine learning techniques such as support vector machines or artificial neural networks, could improve gait phase classification, anomaly detection, and event prediction. These models may enhance the accuracy and robustness of gait parameter estimation, enable personalized and adaptive feedback mechanisms, and support new functionalities such as fall detection.

At the application level, the smartphone interface may be extended with features such as chatbots or gamified elements to support user-centered interpretation of health data and increase user engagement. Furthermore, the effectiveness of augmented feedback could be examined in user-centered training scenarios to evaluate its potential impact on motor learning and rehabilitation outcomes. As a next step, we plan to test the orthosis for foot augmentation in virtual reality training environments, with a particular focus on investigating participant feedback interactions in immersive scenarios through visual feedback. Additionally, a follow-up study is currently being planned to improve the smartphone application onboarding process using mixed reality, with the goal of enhancing users' initial interaction, comprehension, and engagement.

4.2.5 Conclusion

This work presents the development of an SFO system combining pressure and motion sensing, haptic feedback via a connected walking aid, and an mHealth application for real-time visualization. The proposed architecture enables the acquisition of gait and pressure data while providing immediate feedback to users. Beyond its technical implementation, this work contributes to the growing field of assistive smart systems in healthcare by demonstrating how sensor-based feedback can be integrated into wearable and augmentative medical devices. Despite the limited testing in controlled environments, the system represents a step toward future patient-centered solutions that combine sensing, feedback, and data interpretation in daily-life contexts. We encourage researchers and practitioners to further explore the integration of adaptive feedback and context-aware interaction in orthopedic and rehabilitation technologies to support more personalized and effective gait interventions.

4.3. Chapter Summary

In this chapter, the development and validation of a connected smart device ecosystem for assistive mobility was presented, combining a systematic analysis of existing solutions with the implementation and empirical evaluation of a multimodal smart orthosis and walking aid system.

Section 4.1 addressed RQ1 by conducting a scoping review of intelligent walking aids, focusing on technological trends, target user groups, sensing approaches, and feedback mechanisms. The synthesis of 35 research articles revealed substantial heterogeneity in device concepts and highlighted recurring limitations, including challenges in sensor selection and placement, limited integration of multimodal feedback, predominantly superficial system descriptions, and a lack of comprehensive validation and user-centered evaluation. By systematically identifying these gaps, RQ1 was answered and key implications and research opportunities were derived to inform the development of future systems.

Building on these findings, Section 4.2 addressed RQ2 and RQ3 by presenting the design, implementation, and experimental validation of a comprehensive multimodal smart orthosis and connected walking aid system. The developed orthosis concept demonstrated technical feasibility for measuring plantar pressure and gait parameters, while the integrated system enabled multimodal feedback and real-time interaction across the orthosis, walking aid, and mobile application. An experimental user study with patients evaluated system feasibility, usability, and the perceived impact of feedback, providing empirical evidence on how different feedback modalities are perceived across devices and how they influence user interaction and experience.

Taken together, this chapter answers RQ1–3 by linking identified research gaps to a validated system implementation, moving beyond isolated prototype studies. The results contribute methodological and empirical insights into the design of closed-loop, multi-device assistive technologies and provide guidance for future research and development of user-centered smart mobility aids.

CHAPTER 5

Extended Reality: Interaction and Evaluation of Smart Devices

Chapter 4 focuses on the application of smart devices in XR settings (Figure 5.1). As shown in the related work (Section 3.2.2), immersive training scenarios using MR and VR technologies can contribute to positive performance outcomes, while keeping the users engaged. However, it remains unclear how these technologies can be effectively leveraged to train users with the previously developed smart devices, such as the smart orthosis and companion mobile application (Chapter 4.2).

To address RQ4 and RQ5, this chapter presents two experimental studies that investigate the integration of MR/VR feedback modalities in user-device interactions. Section 5.1 investigates how avatar-based voice and gesture guidance in MR supports usability and tutorial guidance during onboarding of the smart insole mobile app, addressing RQ4. Building on this work, Section 5.2 shifts the focus to the smart foot orthosis and investigates the influence of visual foot pressure feedback and environmental factors during treadmill walking, addressing RQ5. In summary, these studies examine how XR interactions enhance user–device engagement and inform design recommendations for immersive training with smart devices.

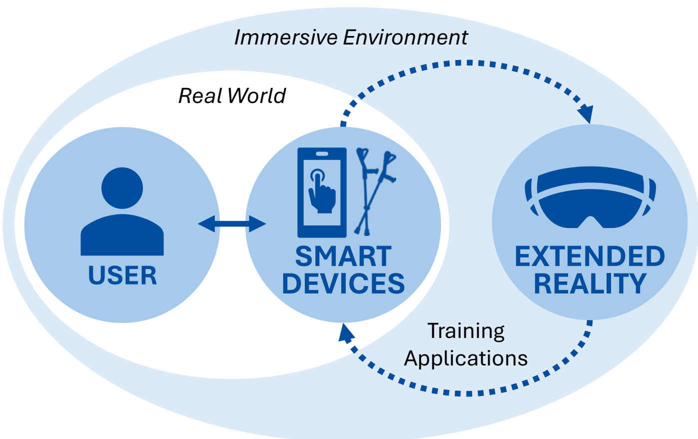


Figure 5.1: Schematic overview of how virtual training applications extend real-world user–device interactions.

5.1. Study I: Mobile Health App Onboarding Using a Multimodal Mixed Reality Avatar

In this section, an experimental user study was conducted to investigate the effects of avatar-based guidance modalities in MR on usability and learning experience during the onboarding process of the previously developed smartphone app. The study examined how different avatar modalities, including voice and gestures, influence user interaction and learning outcomes. The findings indicate that avatar voice and gesture modalities can significantly enhance both usability and the onboarding experience, with each modality contributing differently to navigation and understanding. These results provide concrete design recommendations for improving guidance and interaction in smartphone app onboarding using immersive avatar-based approaches.

 In addition, this chapter provides an extended version of the published paper. The Methods, Results, Discussion, and Conclusion sections have been expanded and supplemented with additional material by the thesis author.

S. Resch, Y. Söyleyici, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Enhancing mHealth App Onboarding Using a Multimodal Mixed Reality Avatar," in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, Barcelona, Spain: Association for Computing Machinery, pp. 1–9, 2026.

The following conference paper provide complementary results:

S. Resch et al., "The Impact of Visual Feedback and Avatar Presence on Balance in Virtual Reality," in *Proceedings of Mensch und Computer 2024 (MuC '24)*, Karlsruhe, Germany: Association for Computing Machinery, pp. 555–559, 2024.

5.1.1 Introduction

The interaction and navigation of novel smartphone applications remain challenging, particularly during first-time use. This issue is especially critical in mHealth applications, where navigating health-related data can be complex, and correct interaction as well as accurate data interpretation are essential for effective use and long-term adoption [301, 45]. Prior work shows that missing or ineffective guidance often leads to misuse, errors, and abandonment, as usability and user satisfaction are strongly linked to successful use [302, 303], highlighting the need for structured onboarding and learning support. In the rapidly growing field of mHealth applications [149, 150], smart insole systems show potential for foot health monitoring and rehabilitation via smartphone apps [69, 155, 156]. However, prior work reports usability challenges, particularly during initial interaction and self-guided learning [161, 162]. Consequently, effective onboarding strategies are a key requirement for successful user–device interaction.

As shown by prior work, targeted onboarding in new applications can support first-time users through tutorials, increasing engagement and learnability [304]. Such onboarding strategies have been tested in various domains, including employee training and consumer apps [305, 306, 307, 163, 165]. For example, Froehlich et al. [307] demonstrated that first-time onboarding in a cryptocurrency app improved usability and informed instructional design. In line with design guidelines proposed by Kascak et al. [308, 309], Ruzic et al. [310] highlight that combining pictorial and verbal presentations helps maintain users' information awareness, an effect also observed for visual and auditory notifications in mHealth apps [311]. However, it remains unclear how these modalities contribute to effective onboarding in the context of mHealth apps.

For this reason, MR offers new opportunities for interactive onboarding through embodied and multimodal guidance [171, 172], and has also been explored in gamified healthcare apps using avatars [170]. Virtual avatar–based approaches have also been explored beyond healthcare, for example in machine-task tutoring, demonstrating their potential for instructional support [312, 313]. Prior work has shown that conversational avatars in MR can support learning, for example in programming education, where virtual avatars enabled natural interaction and improved user experience [314, 315]. Furthermore, combining gesture and voice modalities can enhance user experience in virtual teaching [316] and guidance [173] environments, while visual pointing cues help guide attention to relevant content [317]. Building on this, Weichbroth [181] emphasized that future mobile app usability research should investigate multimodal MR interactions using visual and audio content.

However, while avatar-based learning in MR shows promise, it remains unclear how voice and gesture guidance affect usability and tutorial experience during onboarding in mHealth applications. To address this gap, the following RQ arises: *How does avatar-based voice and gesture guidance in MR influence usability, learning experience, and perceived tutorial guidance during onboarding of a smart insole application?* Therefore, we conducted a within-subject study with 24 participants using a smart insole mHealth app, investigating the effect of an MR avatar combining visual gestures and audible voice guidance on usability and learning/tutorial experiences during app onboarding.

Our findings provide initial insights to inform the future design of onboarding systems for mHealth apps and motivate follow-up investigations. We contribute: (1) empirical insights into multimodal, avatar-based onboarding in mHealth, showing how gesture and voice influence user interactions; and (2) design implications for avatar-supported mHealth onboarding, emphasizing adaptive explanations and interactive steps to enhance learning and usability.

5.1.2 Method

We conducted a mixed-method user study with 24 participants to examine the effects of avatar-based voice explanations and pointing instructions on mHealth app onboarding in MR. The study focused on usability, learning experience, and perceived tutorial guidance during first-time app use. Here, learning experience refers to users' immediate, subjective instructional experience during the tutorial, reflecting their understanding and confidence. The study focused on initial tutorial experiences via avatar-based guidance, not long-term outcomes, as app content and tasks were identical across trials. A smart insole mHealth app was chosen, as such apps require learning both interaction flows and sensor-based data interpretation, creating cognitively demanding onboarding scenarios [45].

We used a 2×2 within-subjects design with the independent variables Avatar Voice (*audible* vs. *non-audible*) and Avatar Gesture (*visible* vs. *non-visible*). The condition with a *non-audible* and *non-visible* avatar served as the control condition. An overview of the experimental conditions is displayed in Figure 5.2. In the baseline condition (no avatar guidance), participants navigated the app based solely on the written task description, without avatar-based audio or gesture guidance. All conditions used identical tasks to isolate the effects of voice, gesture, and their combination. Condition order was counterbalanced using a balanced Latin square to avoid sequencing effects [318].

Based on the specified RQ, we hypothesize that MR avatar-based pointing gestures and voice instructions enhance the learning experience and tutorial experience compared to the condition without avatar support, while not negatively affecting usability.

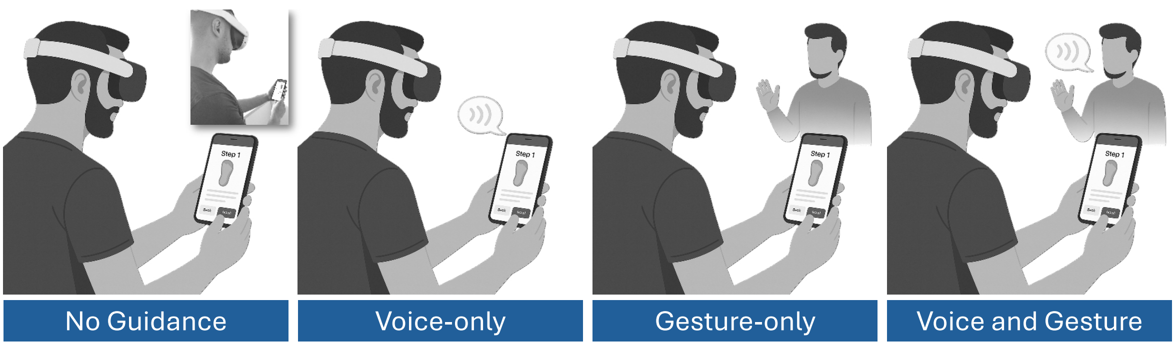


Figure 5.2: Schematic overview of the four MR onboarding conditions: no guidance, voice-only, gesture-only, as well as voice and gesture.

Stimuli

The concept of the mHealth app for smart insoles builds on prior work by Resch et al. [44], which co-designed the app with users and experts. The app has already been tested for usability in related work [45, 46], ensuring comparability across studies. The app included functions for monitoring gait parameters, displaying a foot heatmap, keeping a walking diary to track daily performance and pain behavior, and a search function to find specialists. These functions formed the basis of the tasks tested in the study. Screenshots of the main functions are displayed in Figure 5.3.

Figure 5.3: App screenshots illustrating the main functionalities: (from left to right) main menu navigation, task bar settings, heatmap data understanding, diary overview and data entry, and expert search.

A life-sized, full-body female avatar was used, reflecting common practice in related studies for comparability [180] and to increase acceptance of the avatar's hands [177]. We used the "Kate" model as MR avatar with the "Pointing" animation in its seated version from Mixamo. The avatar held a virtual smartphone in its left hand, matching the real device dimensions and displaying task videos recorded on the actual device. During the tutorial, the avatar pointed with its right hand to relevant interface elements (e.g., buttons, menu icons, text fields), giving the impression of real-time interaction with the device (see Figure 5.4). The animation and video looped, maintaining the avatar's interactive appearance until participants performed the corresponding action on the physical device. In the voice-and-gesture condition, stepwise spoken instructions were synchronized with the pointing gestures using Luvvoice [1] ("Aria" female model).

Apparatus

The mobile app was developed in Android Studio [2] (v. 2024.2.1) and deployed as an Android package (APK) on a Samsung Galaxy S21 5G (Android 15, SDK v. 35). For the visualization of heatmap data within the app, we recorded demo data and

[1] https://luvvoice.com/
[2] https://developer.android.com/studio

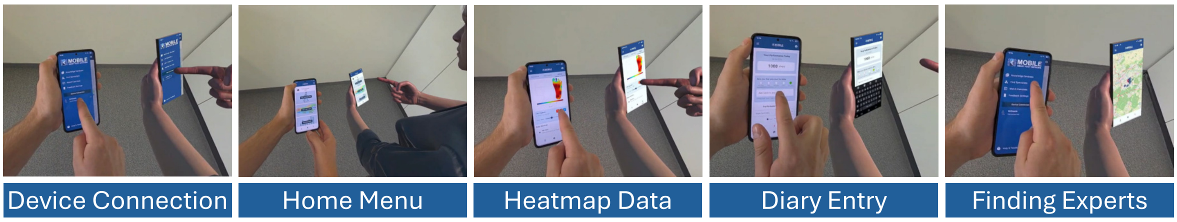


Figure 5.4: Five screenshots from the MR condition with avatar gestures, illustrating different task steps: connecting the device, navigating the home menu, explaining the heatmap using pre-recorded data, entering a diary entry, and locating an expert.

displayed it as a gait trial. To synchronize the avatar's smartphone display in MR with the user-operated smartphone, video clips were recorded using the built-in screen recorder of the Samsung Galaxy S21 5G. The Meta Quest 3 served as HMD with video passthrough for MR, employing the video passthrough function. For MR development, we used the Unity game engine [3] (version 6000.0.45f1) together with the Meta XR All-in-One SDK [319] (v. 72.0.0). The avatar's model, voice, and gestures were implemented according to the specifications described in Section 5.1.2. The interviews were audio-recorded using a Logitech C922 Pro HD Stream webcam and Open Broadcaster Software[4], and subsequently transcribed with Buzz[5]. The software ran on a Windows 11 Pro system (XMG Fusion 15) with an Intel Core i7-11800H 8-core processor (2.3 GHz), NVIDIA GeForce RTX 3070 laptop GPU, and 16 GB RAM.

Procedure

First, the experimental procedure was explained to the participants, including that they could withdraw or discontinue from the study at any time. After providing informed consent, participants were surveyed about demographic data and previous experiences with mobile apps. Subsequently, participants were equipped with the HMD and asked to complete a short reading test to ensure sufficient visibility during the experiment while using the video passthrough function. Furthermore, sound was tested in advance. The avatar was scaled and positioned before the start of the condition to ensure that it was consistently displayed next to the participant. Before starting, they were instructed to focus on tutorial guidance rather than speed, as the usability and learning experience assessments targeted the tutorial type and guidance rather than the app itself. Depending on the condition, participants were guided through the app by the avatar (voice and/or gestures) or explored the app independently based on the predefined tasks. In the control condition, with a *non-audible* and *non-visible* avatar, participants were instructed to read a piece of paper containing a description of each task. After each condition, a semi-structured interview was conducted and audio recorded, followed by the SUS, Learning Experience Questionnaire (LEQ), and Tutorial Experience Questionnaire (TEQ) questionnaires. Afterwards, the same procedure was repeated for the remaining conditions. Finally, a short debriefing session was conducted to gather participants' feedback on the preferred condition. The total study lasted approximately 90 minutes per participant.

[3] https://unity.com/
[4] https://obsproject.com/
[5] https://github.com/chidiwilliams/buzz

Tasks

The app tutorial was structured around predefined tasks to provide a representative onboarding procedure covering the app's core functions. Tasks were selected to reflect typical user goals and involve comparable interaction steps across all conditions, ensuring that differences in performance or perception can be attributed to the tutorial approach rather than the tasks themselves. They included navigating the home menu, connecting the smart insole, testing basic functions, interpreting heatmap data, adding entries to the walking diary, and finding specialists. Participants were also asked to search for specific functions, simulating daily situations requiring navigation of the UI.

The simplified task descriptions are provided below and were read before the first condition to ensure task understanding. For the control condition without guidance, the description was read a second time to ensure a fair comparison.

- Task 1: Connect the smartphone app to the smart insole. Assume that you do not need to leave the app or interact with anything outside of it to complete this task.
- Task 2: Access your recorded foot pressure data, view it as a heatmap, and explore the playback function.
- Task 3: Open your walking diary, record your current pain level, and add a short note.
- Task 4: Use the integrated map to find a specialist in your area and display detailed information about them.

Measures and Data Analysis

Quantitative subjective data were analyzed using descriptive and inferential statistics, including RM ANOVA, in R with the rstatix package [292]. The SUS [158] was used to assess tutorial usability and has been applied in prior onboarding evaluations [307, 169]. Following the standard scoring procedure, item ratings were transformed to yield scores ranging from 0 to 100 and interpreted using established usability benchmarks [158, 294]. The LEQ [320] was used to assess the perceived learning experience of the avatar modalities. The questionnaire consists of 19 items, grouped into four factors (contextuality, effects, design features, explorative nature). Each item was rated on a five-point Likert scale, ranging from 1 (*Fully disagree*) to 5 (*Fully agree*). The TEQ [321] is a standardized questionnaire designed for assessing tutorial experience in digital environments. The questionnaire consists of eight items grouped into two subscales (structural clarity and transparency), each comprising four items. Items were rated on a seven-point Likert scale ranging from -3 (*strongly disagree*) to 3 (*strongly agree*), with 0 as neutral. Overall LEQ/TEQ and subscale scores were computed as the mean per participant and condition.

Qualitative data were collected via semi-structured interviews on participants' perceptions of voice and gesture modalities, allowing flexible follow-up questions. In total, 96 interviews were conducted (182 min of audio, $M = 113$ s, $SD = 60$ s) and transcribed using Buzz [6] (Whisper large-v3). Transcripts were checked and corrected

[6]https://github.com/chidiwilliams/buzz.git

by one researcher, with a second verifying completeness. An inductive thematic analysis [322] was performed independently by two researchers in an iterative process to identify key categories and themes across the dataset. First, the transcribed dataset was coded sentence by sentence (open coding). Following this, axial coding was applied to cluster categories by identifying relationships, and selective coding was used to refine the overarching thematic structure. Differences between coders were resolved through iterative discussion and comparison of coding decisions until consensus was reached, ensuring the final themes accurately reflected the data.

Participants

The study received ethical clearance from the German Society for Nursing Science (No. 23-027). We recruited 24 participants (13 male, 11 female), aged 21–61 ($M = 32.83$, $SD = 12.63$), via personal contacts and institutional mailing lists. The sample size follows the standards for HCI studies [323]. Participants come from five different nationalities (covering Europe, North America, and Asia) and have different educational backgrounds and occupations. All reported prior experience with health-related devices and apps, but none had previously used a smart insole app. Familiarity with mHealth apps averaged 2.71 ($SD = 1.20$) on a five-point Likert scale, indicating a moderate level of familiarity. Half of the participants ($n = 12$) wore glasses. No participants dropped out and all were included in the data analysis.

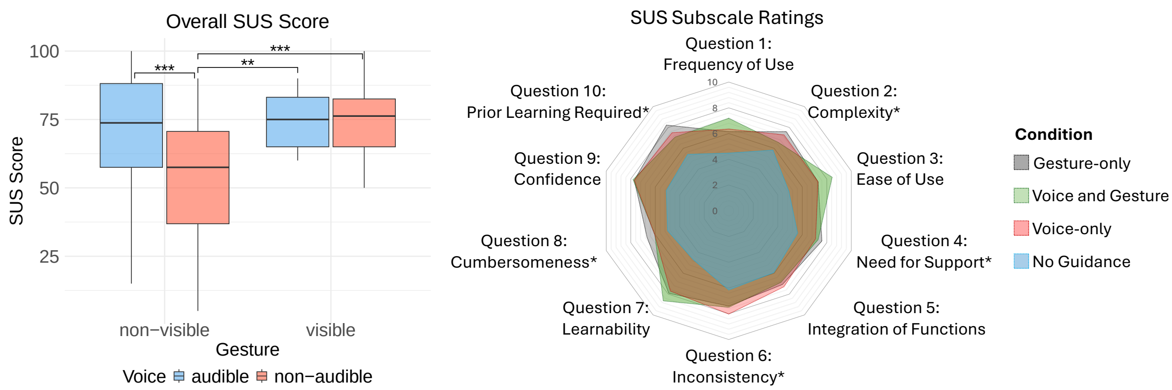
5.1.3 Results

Quantitative Results

System Usability Scale (SUS) Descriptive statistics revealed that a *visible* but *non-audible* avatar achieved the highest SUS score ($M = 73.6$, $SD = 13.1$), followed by the *visible* and audible avatar ($M = 73.4$, $SD = 12.7$), the *non-visible* but audible avatar ($M = 72.3$, $SD = 20.0$), and finally the *non-visible* and non-audible avatar ($M = 53.1$, $SD = 23.4$), see Figure 5.5a.

To investigate significant differences between conditions, a two-way RM ANOVA was conducted. Shapiro–Wilk tests confirmed normality for the overall SUS score ($p > .065$). A two-way RM ANOVA revealed significant main effects of Gesture ($F(1, 23) = 11.05$, $p = .003$, $\eta_p^2 = .32$) and Voice ($F(1, 23) = 6.46$, $p = .018$, $\eta_p^2 = .22$), as well as a significant interaction ($F(1, 23) = 6.78$, $p = .016$, $\eta_p^2 = .23$), all with large effect sizes. Post-hoc tests with Bonferroni correction (Kenward–Roger) showed that Voice was significant only when the Avatar was *non-visible* ($p < .001$), and Avatar was significant only when Voice was *non-audible* ($p < .001$). This indicates that the SUS score increased for avatars with an *audible* voice only when the avatar was *non-visible*.

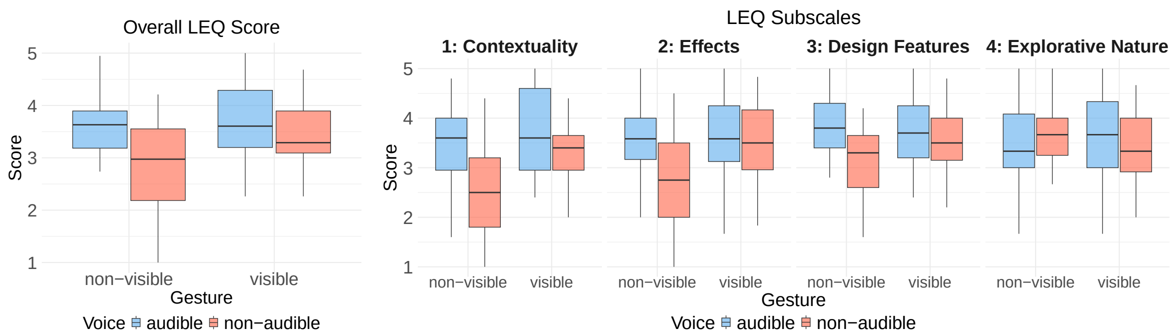
Furthermore, we analyzed the SUS subscales to identify significant differences across the ten dimensions, see Figure 5.5b. Shapiro–Wilk tests indicated that the subscales violated the assumption of normality. Therefore, we applied an ART RM ANOVA for each subscale to test for main and interaction effects across conditions. Post-hoc pairwise comparisons were conducted using Bonferroni-corrected Wilcoxon signed-rank tests. Analysis of SUS subscales confirmed the identified pattern and show significant effects of Gesture and Voice across six dimensions (all $p < .038$): frequency of use, ease of use, inconsistency, learnability, confidence, and prior learning. Detailed results for all SUS subscales are presented in Table C.1 and Figure C.1 (Appendix C.1).



(a) Overall SUS score. (b) Radar plot of the ten SUS items.

Figure 5.5: Plotted results of the SUS ratings: (a) overall SUS score (asterisk indicates statistical significance), and (b) radar plot of the subscale ratings, normalized to a 0–10 scale. In the radar plot, asterisk-marked items were reverse-scored to align scale directionality, such that higher values indicate more positive usability ratings.

Learning Experience Questionnaire (LEQ) LEQ ratings showed the highest learning experience for the *visible and audible* condition ($M = 3.63$, $SD = 0.78$), followed by *non-visible and audible* ($M = 3.52$, $SD = 0.74$), *visible and non-audible* ($M = 3.43$, $SD = 0.64$), and lastly *non-visible and non-audible* ($M = 2.90$, $SD = 0.86$), see Figure 5.6a.



(a) Overall LEQ score. (b) LEQ subscales.

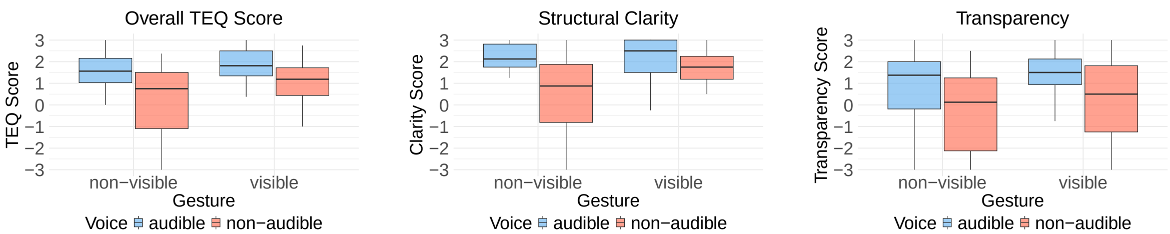
Figure 5.6: Boxplot diagrams of the overall LEQ score (a) and the four corresponding subscales (b), measured on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree).

Shapiro–Wilk tests on LEQ ratings confirmed a normal distribution (all $p > .523$). A two-way RM ANOVA revealed significant main effects of Gesture ($F(1, 23) = 4.57$, $p = .043$, $\eta_p^2 = .166$) and Voice ($F(1, 23) = 14.11$, $p = .001$, $\eta_p^2 = .380$), and a significant interaction ($F(1, 23) = 10.34$, $p = .004$, $\eta_p^2 = .310$), all with large effect sizes. Post-hoc Bonferroni tests (Kenward–Roger) showed that *audible* voice > *non-audible* in the *non-visible* gesture condition ($t(69) = 3.91$, $p < .001$), and *visible* > *non-visible* in the *non-audible* voice condition ($t(69) = -3.30$, $p = .002$). Furthermore, RM ANOVA on subscales revealed significant effects for three dimensions (contextuality, effects, design features, all $p < .011$), but none for explorative nature of learning experience, see Figure 5.6b. The full statistical results of subscale scores are documented in Table C.2 (Appendix C.2).

Tutorial Experience Questionnaire (TEQ) Overall TEQ showed the highest rating for the *visible and audible* condition ($M = 1.75$, $SD = 1.04$), followed by *non-visible and audible* ($M = 1.43$, $SD = 1.19$), *visible and non-audible* ($M = 0.98$, $SD = 1.05$), and lastly *non-visible and non-audible* ($M = 0.15$, $SD = 1.71$), as shown in Figure 5.7a.

Shapiro–Wilk tests indicated non-normality of the overall TEQ score across all conditions. An ART RM ANOVA revealed significant main effects of Gesture ($F(1, 69) = 6.01$, $p = .002$) and Voice ($F(1, 69) = 24.04$, $p < .001$), with no interaction ($F(1, 69) = 0.92$, $p = .341$). Post-hoc Bonferroni tests showed higher ratings for *visible* > *non-visible* ($p = .021$) and *audible* > *non-audible* ($p < .001$), see Figure 5.7a.

Subsequent analyses of two subscales were performed. Shapiro–Wilk tests indicated that both subscales violated the normality assumption. For the structural clarity scale, an ART RM ANOVA with Kenward–Roger correction revealed significant main effects of Gesture ($F(1, 69) = 6.30$, $p = .014$), and Voice ($F(1, 69) = 21.28$, $p < .001$), as well as a significant interaction ($F(1, 69) = 4.71$, $p = .003$). Post-hoc analyses with Bonferroni correction showed that when Voice was *non-audible*, *visible* avatars achieved significantly higher clarity scores compared to *non-visible* avatars ($p = .002$). Conversely, when the Avatar was *non-visible*, the presence of an *audible* voice significantly increased clarity scores compared to the *non-audible* condition ($p < .001$), as shown in Figure 5.7b. For the transparency subscale, an ART RM ANOVA revealed a statistically significant main effect of Voice ($F(1, 69) = 14.70$, $p < .001$). However, there was no main effect of Gesture ($F(1, 69) = 3.03$, $p = .086$), and no interaction effect ($F(1, 69) = 0.06$, $p = .811$). Post-hoc pairwise comparisons using Wilcoxon signed-rank tests revealed a significant difference between *visible* and *non-visible* ($p = .009$), as well as between *audible* and *non-audible* ($p = .001$). The results of the transparency subscale are shown in Figure 5.7c.



(a) Overall TEQ Score (b) Subscale ”Clarity” (c) Subscale ”Transparency”

Figure 5.7: Boxplots of the LEQ ratings (a) overall LEQ score including descriptive statistics, (b) subscale structural clarity, and (c) transparency. Ratings range from -3 (strongly disagree) to +3 (strongly agree), with 0 as a neutral response.

Preferred Conditions (Likert Ratings) After the experiment, participants rated their favored combination of GESTURE and VOICE ($N = 15$), followed by VOICE only ($N = 7$) and NO GUIDANCE ($N = 2$). In contrast, the GESTURE only condition was not preferred by any participant. Participants indicated that a talking avatar in MR could be helpful for explaining and interpreting health data (22 out of 24). Overall satisfaction with the MR app guidance was high ($M = 5.83$, $SD = 1.20$ on a 7-point Likert scale).

Qualitative Results: Semi-Structured Interview

Inductive thematic analysis revealed three key themes: Engagement in Feedback Modalities, Guidance Challenges & Suggestions for Improvement. In the following, these themes are discussed.

1. Engagement in Feedback Modalities: The baseline was described as "convenient" (P8), suitable for self-directed learning (P19), and helpful for people who prefer learning autodidactically (P19). The voice condition was perceived as "clear and understandable" (P11), "easier to follow" (P23), "most effective for learning new information" (P21), and as giving "some instruction to do something" (P15). The combination of voice and avatar modalities was "interactive" (P17), "easy to use" (P13), and "confidence-enhancing" (P5), providing "clear instructions and visual cues" (P9). The participants felt "more confident with the dual instruction" (P12), "because you understand it fast [and] I was more attentive" (P18), which is why it might be "useful for all kinds of people" (P16). Furthermore, the added second modality extended the benefits, as "we don't need to focus more on the voice only, as there was gesture guidance as well, so we could easily do the desired task" (P20).

2. Guidance Challenges: The condition without guidance was criticized as "completely overloaded" (P24) and "difficult to navigate" (P12, P19): "because I couldn't find my way around on my own" (P19), and "it takes longer to find your way around the app. You don't know if you're using the system correctly right away" (P12). In particular, it was highlighted that "new people will have difficulties" (P4), as "to explore all the things by ourselves [would make] it difficult for a nontechnical person to use the app for the first time without any guide" (P20). In the avatar-only condition, participants raised concerns such as "you just clicked without thinking and without knowing why" (P11) and limited learning (P11, P21). One participant further emphasized: "I turned my brain off and just did what the avatar did. Step by step and it worked well, but I didn't learn anything, which is also the negative part." (P21). Furthermore, it was "confusing and difficult to navigate" (P23). Similarly, for the voice-only condition it was mentioned that "I feel the processing from voice to action on the screen was more strenuous or confusing than finding it myself through trial and error or being guided visually by an avatar" (P9). In addition, the combination of the two modalities also led to reduced focus, as it was highlighted that "the voice is distracting" (P19) while using the avatar at the same time.

3. Reflections & Suggestions for Improvement: While the overall experience was perceived as highly positive, participants provided suggestions for future improvements. In general, the avatar was perceived differently, as some participants only focused on the hands and fingers rather than looking at the avatar as a whole. For example, P10 stated: "I didn't really pay attention to him as a person, only to the hand." In line with that, one participant suggested, "I'd prefer a larger display without an avatar and only a hand instead of a whole avatar," (P21) which could be beneficial for improved interaction. This could mean that the type of avatar is not necessarily relevant, as the hand gesture interaction is the main focus. Interestingly, P21 further mentioned that a different placement of the avatar could improve the interaction, saying: "I also tried standing inside the avatar. I only saw the fingers that tapped the screen, and then I quickly repeated that on my own display." Furthermore, some comments addressed device-dependent issues, such as that the vision of MR "should be improved" (P4) and made "clearer" (P23).

5.1.4 Discussion

In this study, we investigated how MR avatar voice and gesture modalities affect usability and tutorial experience during onboarding of a smart insole app. Both modalities, individually and combined, improved SUS, LEQ, and TEQ scores and increased overall satisfaction with MR-based guidance. In line with related work emphasizing the benefits of voice or gesture feedback [166] in multimedia learning, our study extends these findings to the mHealth domain by demonstrating the added value of multimodal avatar interactions for onboarding guidance, usability, and learning experience. Participants consistently perceived the avatar as helpful, with a high level of overall satisfaction. The no-guidance condition was consistently least preferred, while gesture-only was rated lower than voice-only or combined modalities, likely due to its limited contextual information. Our results suggest that onboarding in smart insole apps should be aligned with task type. When users must follow procedural steps (menu navigation, device connection, locating specialists, or diary entry), gesture support helped establish spatial orientation and improves the ability to replicate actions. In contrast, when meaning or context must be communicated, such as interpretative tasks (gait metrics, pressure heatmaps, diary graphs, or health information), voice was perceived as more helpful due to its capacity to convey conceptual and narrative information. Although combined guidance was generally effective, participants indicated that excessive redundancy can become cognitively demanding, suggesting that multimodal support should be applied selectively. Thus, avatar-based onboarding in mHealth apps should distinguish between procedural guidance (gesture-based) and interpretative guidance (voice-based), especially when understanding affects health-related decision-making and long-term adherence.

SUS results indicate that lack of guidance increases perceived complexity, inconsistency, and reliance on prior knowledge. This aligns with prior research showing that self-guided learning in mHealth applications often introduces barriers for novice users [162]. Furthermore, we showed that the addition of avatar modalities enhanced perceived ease of use, learnability, and confidence, supporting frequent app use. The learning experience was consistently rated higher whenever guidance was provided compared to the self-guided condition. While multimodal guidance increased tutorial transparency, adding a second modality did not further improve clarity, suggesting that guidance presence matters more than modality number. Notably, the exploratory aspect of learning was not hindered by guided conditions. Qualitative findings show participants primarily attended to the avatar's hands and gestures, rather than the full body, highlighting the importance of functional embodiment. Taken together, these findings indicate MR avatar onboarding is most effective when modalities match mHealth tasks.

Design Implications

We propose the following context-specific design implications:

- **Modality contextualization:** Guidance in MR mHealth apps should align with task demands. Procedural steps benefit from gestures, interpretative steps from voice. Multimodal support may increase cognitive demand and be toggled by user experience or health literacy.
- **Avatar embodiment:** Our results indicate that participants primarily attended to hands and gestures, suggesting that full-body embodiment may be more relevant for other mHealth tasks. Aligning gesture orientation with the user's viewpoint and providing egocentric augmentations (e.g., rendering virtual hands in the user's field of view) can improve clarity and reduce misinterpretation or distraction.
- **Interactive guidance:** mHealth guidance should let users actively follow and repeat interactive steps, reinforcing health data understanding, supporting confidence and active engagement in self-monitoring. Repeating critical task steps can support procedural memory [324], particularly when learning to interpret gait and pressure parameters. Furthermore, adaptive onboarding (e.g., shortening explanations for experienced users or elaborating for novices) can balance efficiency, learning depth, and health literacy.
- **Device-centered augmentation:** Several participants reported readability issues with the virtual smartphone model, highlighting the need for improved visibility. Although the avatar's virtual smartphone was modeled in realistic proportions, a larger display may improve readability and interaction comfort. MR smartphone onboarding should therefore consider task-specific augmentations, such as enlarged or adaptive displays, to enhance legibility. The choice of HMD also substantially affects visibility and user comfort. While the Meta Quest 3 provides moderate passthrough quality [325], optical see-through AR headsets can mitigate visibility limitations and should be considered when designing avatar-based onboarding.

Limitations and Future Work

Our study is limited by the selected sample and a single mHealth app, constraining generalizability. The selected HMD constituted a major limitation, as reduced screen visibility may have influenced task performance. Therefore, future work should examine optical see-through displays to improve visual perception. Moreover, alternative avatar perspectives (e.g., first-person perspective) should be investigated, as suggested by prior work [326], to further optimize guidance effectiveness. Follow-up studies should replicate and extend these results with a broader range of mobile apps, including commercial mHealth systems, to validate our findings and inform effective onboarding design based on user preferences. Overall, our work provides a foundation for follow-up research on MR avatar-based guidance in mobile apps.

5.1.5 Conclusion

This paper investigated MR avatar-based tutorial guidance using gesture and voice modalities in a smart insole health app. Our findings showed that avatar guidance, whether through gestures, voice, or their combination, led to significantly better outcomes in terms of usability, learning experience, and tutorial experience compared to the absence of an avatar. We further observed that when gestures were visible, adding voice did not provide additional benefit, whereas in the absence of gestures, voice was essential for effective contextualization. The qualitative feedback enriched these findings by providing a deeper understanding of participants' preferences and challenges. Based on these results, we derived design implications for onboarding in health-related and broader mobile contexts. This work contributes to HCI by demonstrating how avatar guidance enhances user experience in mobile and MR apps.

5.2. Study II: Smart Insole Visual Foot Pressure Feedback in Mixed Reality Environments

Building on the first XR study in Chapter 5.1, which investigated smartphone-based onboarding, this study focuses on an immersive gait training application using the previously developed smart foot orthosis (Chapter 4.2). Previous work within this doctoral research developed and successfully tested virtual training scenarios for lower-limb foot placement, demonstrating the potential of incorporating visual feedback in VR [327, 133, 276, 3, 328, 4]. Building on this prior work, the orthosis was integrated into an immersive gait training environment to evaluate the effects of visual foot pressure feedback and environmental factors, addressing RQ5. An experimental user study in MR/VR assessed gait behavior, heart rate, workload, and engagement, revealing that environmental factors interact with visual feedback to influence both objective and subjective outcomes. These findings provide concrete design recommendations for future immersive gait training applications.

 In addition, this chapter provides an extended version of the published paper. The Methods, Results, Discussion, and Conclusion sections have been expanded and supplemented with additional material by the thesis author.

S. Resch, J.-G. Hanania, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Smart Insole Visual Foot Pressure Feedback in Mixed Reality Environments: Impact on Gait, Heart Rate, Workload, and Engagement in Treadmill Walking," in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, Barcelona, Spain: Association for Computing Machinery, 2026.

The following conference papers provide complementary results:

S. Resch and A. Carroll, "Influence of Virtual Footwear on Gait Behavior, Embodiment, and Workload While Walking on a Treadmill in Virtual Reality," in *31st ACM Symposium on Virtual Reality Software and Technology (VRST '25)*, Montreal, QC, Canada: Association for Computing Machinery, pp. 1–2, 2025.

S. Resch et al., "Foot Placement Feedback in Physical Training: Effects of Spatial User Interfaces on Performance and Workload in Virtual Reality," in *ACM Symposium on Spatial User Interaction (SUI '25)*, Montreal, QC, Canada: Association for Computing Machinery, pp. 1–12, 2025.

S. Resch et al., "The Influence of Augmented and Virtual Reality Environments on Foot Positioning Success and Workload in Agility Ladder Exercises," in *Proceedings of the 2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, Saint Malo, France: IEEE, pp. 679–686, 2025.

S. Resch et al., "Correct Foot Positioning in Virtual Reality through Visual Agility Ladder Training," in *Proceedings of Mensch und Computer 2023 (MuC '23)*, Rapperswil, Switzerland: Association for Computing Machinery, pp. 524–528, 2023.

5.2.1 Introduction

Orthopedic footwear, such as a controlled ankle motion boot, is essential for stabilizing lower-limb injuries but substantially alters gait, making walking harder to control [329, 276]. Patients often need to adjust their gait or reduce plantar pressure, but without guidance this is difficult, potentially slowing rehabilitation, prolonging recovery, or causing persistent impairments [330]. Critically, patients rarely receive direct feedback about pressure distribution or gait within the orthosis, which limits self-correction. Conventional treadmill training in laboratories relies on expert supervision or provides only basic cues, often insufficient for gait adaptation, balance, and pressure control [331]. Consequently, this expert-dependent approach restricts patient-device interaction and access to relevant gait information [332].

Therefore, visual and interactive feedback offers potential, enabling users to actively adjust gait and increase independence [333]. For example, sensorized footwear, such as pressure-sensitive insoles, has been used in standard shoes to provide real-time feedback for posture exercises using CoP feedback [103, 127, 136]. Building on their use in lower-limb exercises, pressure-sensing insoles have been integrated as real-time input devices in VR, supporting gait detection and locomotion, such as perceiving ground surfaces while walking [128, 129]. These systems show that visual foot augmentation can guide mobility, balance, and foot placement while supporting task performance. Although solutions for gait-modifying orthoses are lacking, implementing the sensorized concept in interactive training could enable feedback-driven gait adaptation, even when the device substantially alters walking patterns.

In this context, VR and MR enhance this approach by embedding visual feedback into immersive training scenarios [334, 331, 328, 327]. Compared to screen-based feedback, HMD-based VR provides a reliable tool for treadmill training [335], enabling flexible applications beyond stationary labs and supporting gamified training that can influence motivation and physiological responses [8, 135, 130, 131, 132]. While conventional training is usually limited to laboratory settings, some VR studies have used virtual replicas of the same room to improve comparability and generalization [139]. However, immersive VR can increase task-dependent cognitive load [3], which should be managed to preserve training benefits. Beyond traditional laboratory setups, naturalistic virtual environments have shown potential to support stress recovery in other contexts [142], suggesting opportunities for future gait training scenarios.

Despite these advances, most prior work has examined visual feedback or environmental factors in other contexts and typically relies on standardized footwear. It remains unclear how immersive settings and visual pressure augmentation affect movement patterns, physiological responses, and user experience in gait-modifying footwear. To address this gap, our research question asks how smart insole feedback and immersive contexts influence objective gait measures, heart rate, and subjective workload and engagement during treadmill walking. Therefore, we developed a sensorized foot orthosis and conducted a study with healthy participants to investigate the mechanisms of feedback-driven gait modulation under controlled conditions, serving as a basis for future virtual training developments.

In this paper, we present the results of an experimental user study with 24 participants investigating gait adaptation in VR/MR. Our work extends previous findings by focusing on gait-modifying orthopedic footwear, which presents unique challenges for adaptation. We show how foot pressure augmentation interacts with immersive contexts and how naturalistic and stationary VR/MR settings with multiple feedback modalities affect gait, physiological responses, and user experience. Furthermore, we derive design implications for future immersive gait training applications.

5.2.2 Method

We conducted a mixed-method experimental user study to investigate the effects of foot pressure visualization and environmental factors on gait, heart rate, workload, and user engagement during treadmill walking in VR/MR. To address this, the study followed a 4×3 within-subjects design with two independent variables: VISUALIZATION (four levels: *None*, *Heatmap*, *CoP*, *Histogram*) and ENVIRONMENT (three levels: *VR Forest*, *VR Lab*, *MR Lab*). Twelve experimental conditions were counterbalanced using a balanced Latin square to minimize order effects [318, 291], with each condition appearing equally often across 24 participants.

A mixed-method approach was applied, combining quantitative objective and subjective measures with qualitative feedback. Quantitative objective measures included gait (spatial, temporal, and kinetic parameters) and heart rate, while subjective assessments captured workload and engagement. Semi-structured interviews provided qualitative insights, enabling deeper understanding to support quantitative findings.

Stimuli

The stimuli for each condition are shown in Figure 5.8. The *VR Forest* ENVIRONMENT was developed in Unity using assets available from the Unity Asset Store [7]. This environment consists of a straight path in a natural forest, without additional effects such as sound. To avoid potential distraction through sound effects and to maintain a consistent focus on the visual stimuli, no audio was used in the virtual environments. Participant motion in the environment was matched to the continuous treadmill speed to ensure consistency in gait and perception. The *VR Lab* scene was a virtual replica of the study laboratory to ensure comparability. The dimensions matched those of the physical lab, and a virtual treadmill simulated movement synchronized with the real treadmill. The *MR Lab* condition was displayed using the Meta Quest 3 passthrough, to show the real laboratory with a virtual canvas displaying the pressure-feedback visualization.

For the VISUALIZATION of pressure data, three visual stimuli were created. The *None* condition served as the control "baseline" condition to assess the dependent variables without visual feedback. This allowed us to isolate the effect of environmental factors. The three visual stimuli (*Heatmap*, *CoP*, and *Histogram*) were developed for a smart insole prototype using input from 17 FSRs. All visualizations were displayed at a similar size at the participants' eye level, ensuring consistent visibility across conditions.

[7]https://assetstore.unity.com/

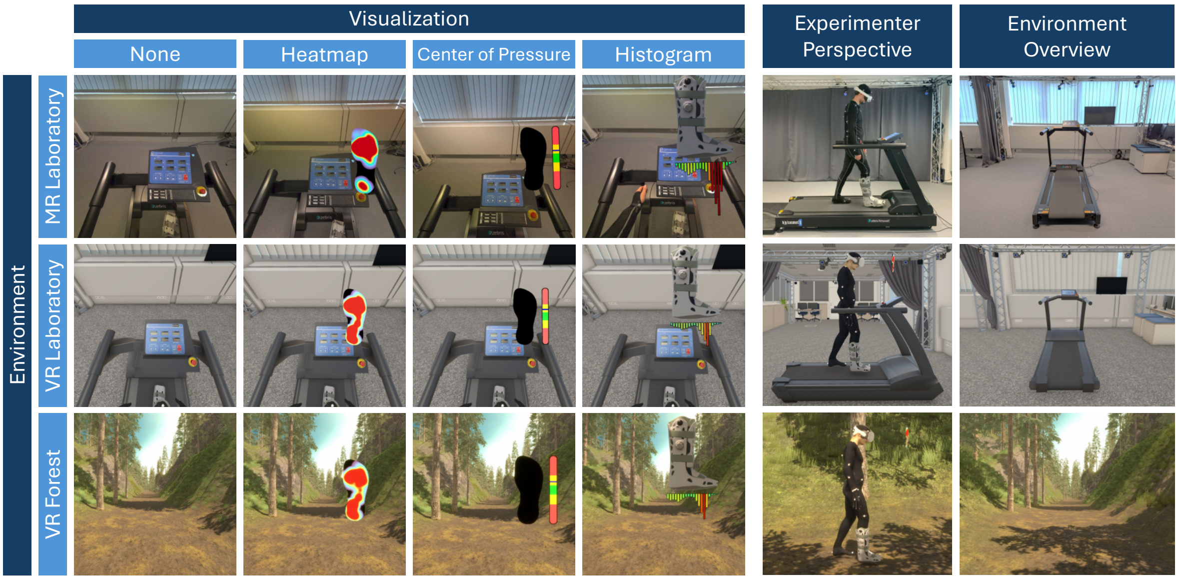
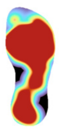


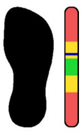
Figure 5.8: Participant perspective (left) and experimenter view (right) of the visualizations and environments.

Heatmap:



Heatmaps are frequently used to provide an intuitive visualization of the plantar pressure distribution. Based on prior work [336, 337, 46], we developed a 2D plantar pressure heatmap using the prototype insole layout. Using input from 17 FSRs, a custom fragment shader mapped the pressure values to colors with the Turbo colormap developed by Google [8]. The pressure values were normalized as a percentage of each participant's body weight.

Center of Pressure (CoP):



We implemented a 1D CoP-based visualization to indicate postural balance, as recommended by Elvitigala et al. [103, 127]. A colored vertical bar represents anterior–posterior balance based on the pressure distribution across the forefoot, midfoot, and hindfoot zones. The bar displays the averaged CoP computed from all FSRs and is segmented into three color-coded regions (green, yellow, and red).

Histogram:

A 2D *histogram* was developed to display cumulative pressure per foot zone using 20 area-scaled bars, scaled according to each sensor's area of effect to ensure balanced spatial coverage. For example, if a sensor is located in the midfoot (three sensors) rather than the forefoot (eight sensors), its area of effect is increased to ensure balanced spatial representation in the bar distribution. Additionally, the virtual orthosis visualized real-time foot rotation during gait.

[8]https://research.google/blog/turbo-an-improved-rainbow-colormap-for-visualization/

Apparatus

The laboratory setup and apparatus are shown in Figure 5.9. We developed a smart insole, placed inside an ankle motion boot orthosis, following Resch et al. [46] (see Chapter 4.2), equipped with 17 FSR-400 sensors (5.6 mm diameter, 0.3 mm thick, 20 N range). A custom printed circuit board (PCB) with an ESP32 microcontroller streamed sensor data via MQTT at 100 Hz. The virtual environments and pressure visualizations were developed in Unity[9] (version 6000.0.46f1) using the Meta XR All-in-One SDK (v. 72.0.0) [319]. Two Mixamo[10] avatars were used to ensure gender representation: the male "Passive Marker Man" and the female "Jody," with textures adapted to the MoCap suit.

A Meta Quest 3 with Elite Strap and battery served as the HMD for VR and MR. An OptiTrack MoCap system with ten Prime^X 13W[11] cameras captured skeleton data at 240 Hz (1280 × 1024 px). The system was calibrated according to the OptiTrack specifications and resulted in a mean ray error of 1.18 mm and a mean wand error of 0.25 mm. The template "Baseline" with 41 markers was used to animate the virtual avatar in Motive[12] (version 3.1.4). Heart rate was recorded using a Polar OH1+[13] optical sensor, validated for measuring physical activity [338]. Gait data were recorded at 100 Hz using an instrumented treadmill (RehaWalk® FDM-THPL-S-2i from Zebris Medical GmbH, Germany) with 3,120 capacitive sensors arranged in an 80 × 39 grid over a 101.6 × 49.5 cm surface (± 0.05% accuracy). Data were exported using the Zebris FDM software (version 2.0.4). All applications ran on a Windows 10 Pro PC with AMD Ryzen 5900X, GeForce RTX 3070, and 32 GB RAM.

Procedure and Tasks

The study was conducted under controlled laboratory conditions. After providing informed consent, participants completed a questionnaire on demographics, gait conditions, and VR/MR experience. Following this, the pulse sensor was attached to the forearm (according to the manufacturer's manual[14]) and verified for data transmission. Afterwards, participants were equipped with the MoCap suit (beanie, jacket, pants, and left foot wrap) and the sensorized orthosis, which was individually fitted using the air pump function. After placing markers on the suit and orthosis according to the template, participants adopted a T-pose to calibrate the skeleton. Participants were then positioned on the treadmill, secured with a magnetic safety clip and safety mats. After receiving the HMD, the visualization was aligned to eye level, the virtual avatar scaled to body height, and pressure data normalized to body weight. They were instructed to walk naturally while aiming to control their gait based on the visual feedback, maintaining balanced weight distribution and reducing localized pressure. Treadmill speed was set to 1.5km/h. Five gait trials of ten seconds each were recorded per condition, always starting with right-foot contact. After each condition, participants completed

[9] https://unity.com/
[10] https://www.mixamo.com/
[11] https://optitrack.com/cameras/primex-13w/
[12] https://optitrack.com/software/motive/
[13] https://www.polar.com/en/sensors/oh1-optical-heart-rate-sensor
[14] https://support.polar.com/e_manuals/OH1/Polar_OH1_user_manual_English/Content/Wearing-your-OH1.htm

Figure 5.9: Study setup of a participant walking on an instrumented treadmill wearing an orthosis with a prototyped smart insole, including real-time plantar pressure visualization in virtual/mixed reality, full-body motion capture, and heart rate monitoring.

NASA RTLX and User Engagement Scale – Short Form (UES-SF) questionnaires in VR [339], followed by a semi-structured interview for qualitative feedback. This procedure was repeated for all conditions. After completing all conditions, participants removed the HMD and tracking suit, were debriefed, and provided post-study feedback.

Measures and Data Analysis

Quantitative Objective Spatial, temporal, and kinetic gait parameters were measured using the instrumented Zebris treadmill, appropriate for clinical gait analysis [340]. Spatial gait parameters included cycle and step characteristics (e.g., step length, width, path length, foot rotation), while temporal parameters covered timing-related metrics such as cadence, contact time, and stance/swing phases. Kinetic measures comprised maximum plantar pressure values for hind-, mid-, and forefoot. All parameters were obtained in standard biomechanical units (cm, degrees, s, steps/min, N/cm^2). Heart rate served as a secondary physiological control, reflecting general arousal during treadmill walking, and was recorded continuously in Beats Per Minute (bpm) and aggregated per condition.

Quantitative Subjective Two subjective questionnaires were used after each condition to assess workload and user engagement. In this study, workload refers to the perceived cognitive demand of processing visual pressure feedback while simultaneously controlling gait performance during the immersive task. The NASA RTLX [146] questionnaire was used, as it is widely applied in HCI for workload assessment. Participants rated six subscales (mental, physical, and temporal demand, performance, effort, and frustration) to assess the perceived workload after each condition. For each participant, an overall workload score was derived, and the individual subscales were analyzed separately. The UES-SF [341] was used to measure perceived user engagement for each task. The short form of the standardized questionnaire was applied following the authors' guidelines. It comprises twelve items, with three items for each of the four dimensions: Focused Attention (FA), Perceived Usability (PU), Aesthetic Appeal (AE), and Reward Factor (RW). The items were presented in randomized order and rated on a 5-point Likert scale. Subscale scores were calculated as the mean of the corresponding items and an overall engagement score was also derived. In a debriefing session, participants rated preferred and least liked conditions and provided final feedback. Quantitative data were analyzed using descriptive and inferential statistics, including two-way RM ANOVA.

Qualitative Data Qualitative feedback was collected via semi-structured interviews, focusing on participants' experience, gait influence, and perceived pros and cons of the visualizations. The semi-structured format enabled targeted questions while allowing follow-up questions to obtain clarification and deeper insights. Each post-condition interview was audio-recorded, resulting in 288 recordings of approximately two minutes each. The recordings were subsequently transcribed using Buzz [15] (Whisper large-v3). The transcripts were manually reviewed, corrected, and verified for accuracy by the first researcher, then cross-checked by a second researcher. An inductive thematic analysis [322] was conducted to identify patterns across the dataset. The transcripts were open coded sentence by sentence, followed by axial coding to identify relationships within and across categories, and selective coding to refine the overarching thematic structure. A second reviewer cross-checked the coding and analysis, which followed an iterative process to ensure reliability and data accuracy.

Participants

In total, 24 participants (14 male, 10 female) were recruited via institutional mailing lists. The age of participants was between 21 and 34 (M = 27.54, SD = 2.55). Six participants wore glasses during the study. Participants' familiarity with VR/MR was moderate (5-point Likert, M = 2.5, SD = 1.1). All participants were healthy, free of balance disorders or pain, and had right-leg or bilateral dominance to ensure consistent use of the right-foot orthosis. Foot dominance was assessed by participants' preferred kicking leg and a short single-leg stance test. We applied this criterion to ensure consistent use of the orthosis on the right foot across all participants, while previous work indicates that leg dominance does not affect stance or balance performance in healthy young adults [342, 343].

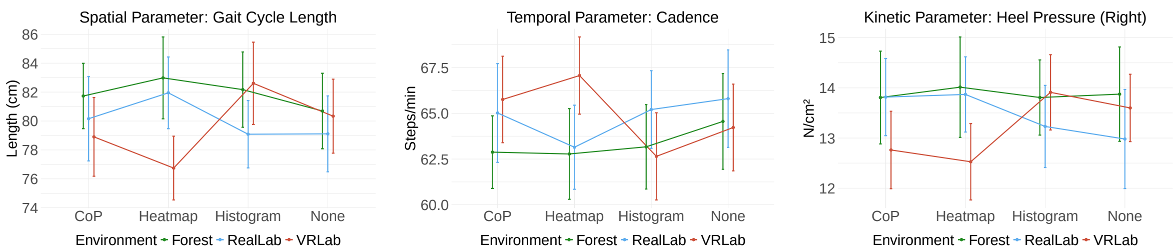
The study followed institutional data privacy and hygiene protocols and received ethical approval by the German Society for Nursing Science (No. 23-027). All participants completed the study without interruption and were included in the analysis.

5.2.3 Results

Quantitative Objective Results

Gait Data (Spatial, Temporal, Kinetic Parameters) Gait parameters were analyzed using RM ANOVA (Greenhouse–Geisser for normal, ART for non-normal data), revealing main effects of *Environment* and interactions across metrics. The main results of spatial (Figure 5.10a), temporal (Figure 5.10b), and kinetic (Figure 5.10c) measures are displayed as plots in Figure 5.10.

[15] https://github.com/chidiwilliams/buzz.git



(a) Spatial Parameter (b) Temporal Parameter (c) Kinetic Parameter

Figure 5.10: Interaction plots for spatial (gait cycle length), temporal (cadence), and kinetic (heel pressure) gait parameters.

Spatial Parameters The Shapiro–Wilk test indicated that path length for both feet violated normality ($p < .05$) Therefore, an ART RM ANOVA with Kenward–Roger correction revealed a significant main effect of ENVIRONMENT on path length for the right foot $F(2, 253) = 5.759$, $p = .004$, and the left foot, $F(2, 253) = 7.778$, $p < .001$. Bonferroni-adjusted post-hoc comparisons showed a significant difference between *VR Forest* and *MR Lab* for both feet ($p < .004$). For normally distributed data, a Greenhouse–Geisser corrected RM ANOVA revealed significant interaction effects for gait cycle length and step width. However, no significant main or interaction effects were found for single leg stance or foot rotation. Bonferroni-adjusted post-hoc tests showed that in the *VR Lab*, gait cycle length was significantly higher in the *Heatmap* condition compared to both other environments. Step width was also significantly affected in the *VR Lab*, with differences among *Heatmap*, *Histogram*, and *None*. For foot rotation, the orthosis foot (right) showed higher values in the *VR Forest* compared to both lab environments ($p = .023$), while the left foot did not show this trend. However, interaction effects did not reach significance after correction (left: $p = .078$, right: $p = .092$).

Temporal Parameters Shapiro-Wilk tests indicated that all temporal parameters were normally distributed, except cadence (eleven conditions with $p < .047$). A Greenhouse–Geisser corrected RM ANOVA revealed significant main effects of ENVIRONMENT and interaction effects for temporal parameters, and an ART RM ANOVA confirmed the same effects for cadence. Bonferroni-adjusted post-hoc tests showed that *VR Forest* reduced cadence ($p = .030$) and increased cycle/contact times ($p = .001$–$.004$) vs. both labs. Interaction effects indicated that in the *VR Lab*, the *Heatmap* condition increased cadence and reduced cycle duration, swing, and stance times ($p = .001$–$.046$). In contrast, *VR Forest* increased cycle duration and contact time while reducing cadence compared to both lab environments.

Kinetic Parameters Shapiro–Wilk tests indicated non-normality for pressure and force values across all foot regions; therefore, an ART RM ANOVA was applied. The analysis revealed significant main effects of ENVIRONMENT and several VISUALIZATION $\times$ ENVIRONMENT interactions. Bonferroni-adjusted post-hoc tests revealed higher heel, mid-, and forefoot pressures in *VR Forest* (all $p \leq .008$) than in both lab environments. Interaction effects were driven mainly by the *Histogram*, where *VR Lab* showed increased forces, while the *Heatmap* partly reversed this pattern.

Heart Rate The Shapiro-Wilk test indicated that the assumption of normality was violated in several conditions ($p < .05$). An ART RM ANOVA with Kenward–Roger correction revealed a statistically significant main effect of ENVIRONMENT, $F(2, 253) = 4.43$, $p = .013$, but no main effect of VISUALIZATION, $F(3, 253) = 1.01$, $p = .389$, and no interaction effect, $F(6, 253) = 0.53$, $p = .785$. Post-hoc Wilcoxon signed-rank tests with Holm correction showed a significant difference between *VR Forest* and *VR Lab* ($p = .008$), while *MR Lab* vs. *VR Forest* ($p = .065$) and all other pairwise differences were not significant, see Figure 5.11.

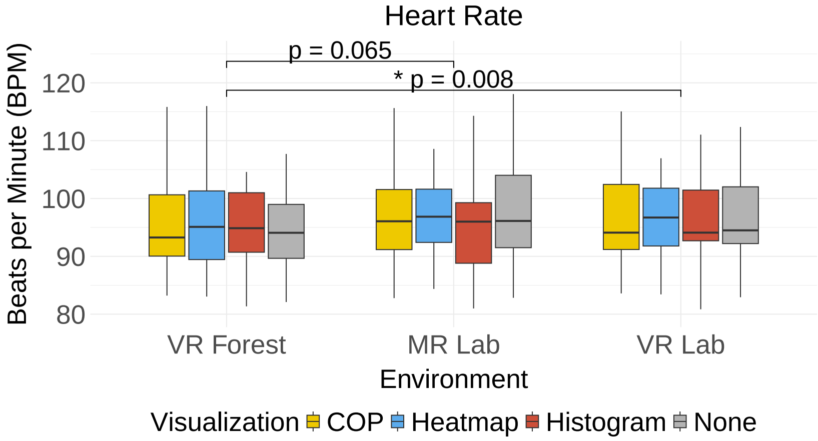
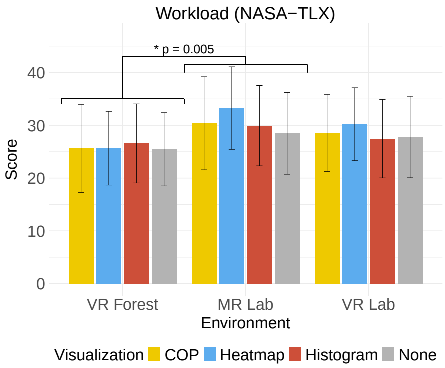


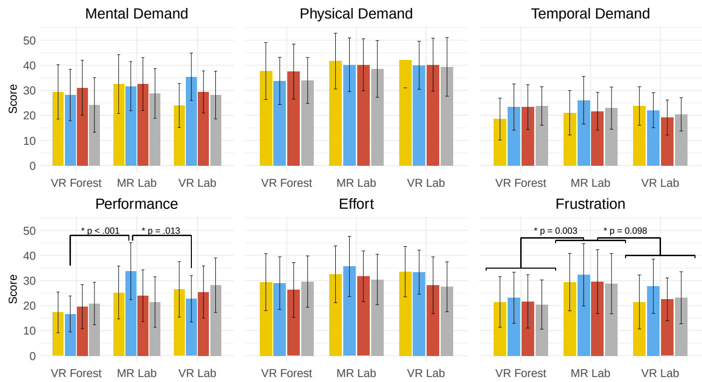
Figure 5.11: Boxplots of measured heart rates in bpm (y-axis) across three environments (x-axis), grouped by four visualization conditions.

Quantitative Subjective Results

Workload The Shapiro-Wilk test indicated normality across all conditions ($p > .069$). A two-way RM ANOVA on the overall NASA RTLX score showed no significant main effects of VISUALIZATION, $F(2.23, 51.22) = 0.969$, $p = .394$, $\eta_p^2 = .040$, or ENVIRONMENT, $F(2, 46) = 2.844$, $p = .068$, $\eta_p^2 = .110$, and no interaction $F(3.96, 91.13) = 0.425$, $p = .789$, $\eta_p^2 = .018$. Furthermore, we analyzed the subscales to identify trends across the workload dimensions. The Shapiro-Wilk test indicated that all subscales were not normally distributed. An ART RM ANOVA revealed significant effects only for performance and frustration. For performance, there were main effects of ENVIRONMENT, $F(2, 253) = 11.06$, $p < .001$, and an interaction effect VISUALIZATION $\times$ ENVIRONMENT, $F(6, 253) = 2.634$, $p = .017$. Participants reported lower workload in *VR Forest* than in both *MR Lab* ($p = .002$) and *VR Lab* ($p < .001$). Post-hoc pairwise comparisons (Bonferroni-adjusted) revealed a significant effect in the *Heatmap* condition, with higher scores in the *MR Lab* compared to *VR Forest* ($p < .001$) and *VR Lab* ($p = .013$) All other comparisons were non-significant after Bonferroni correction ($p > .05$). For frustration, ENVIRONMENT had a significant main effect of ENVIRONMENT, $F(2, 253) = 4.70$, $p = .009$, with higher scores in *MR Lab* than in *VR Forest* ($p = .003$), while the comparison to *VR Lab* did not reach significance ($p = .098$) after correction. However, no statistically significant effects were found for the remaining subscales. Figure 5.12 shows overall workload (Figure 5.12a) and subscale scores (Figure 5.12b).



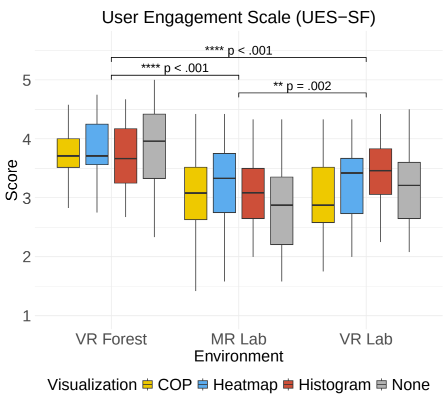
(a) Overall NASA-TLX score



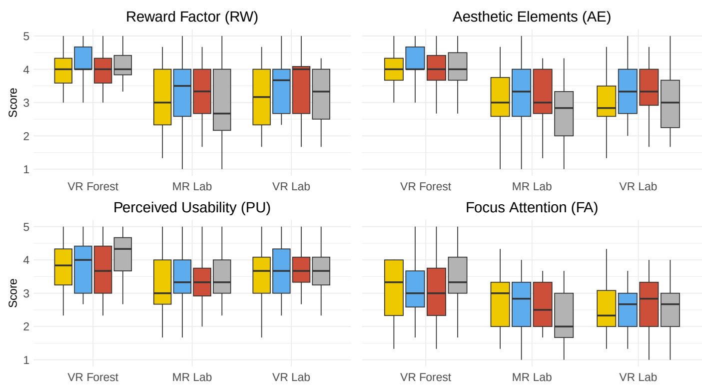
(b) NASA-RTLX subscale scores

Figure 5.12: Quantitative subjective workload: (a) the overall NASA-RTLX score and (b) the six subscale scores

User Engagement Shapiro-Wilk test on the overall UES-SF score indicated normal distribution across all conditions $p > 0.211$. An RM ANOVA with Greenhouse-Geisser correction revealed a significant main effect of ENVIRONMENT, $F(1.45, 33.45) = 22.020$, $p < .001$, $\eta_p^2 = .489$ (large effect size). However, no significant main effect of VISUALIZATION was identified, $F(3, 69) = 1.876$, $p = .142$, $\eta_p^2 = .075$ (medium effect size), as well as no interaction effect of VISUALIZATION × ENVIRONMENT, $F(6, 138) = 1.962$, $p = .750$, $\eta_p^2 = .079$ (medium effect size). Post-hoc pairwise comparisons with Bonferroni correction revealed a significant difference between *Forest* vs. *MR Lab* ($p < .001$), *Forest* vs. *VR Lab* ($p < .001$), and *MR Lab* vs. *VR Lab* ($p = .002$). Furthermore, the four subscales were normally distributed and showed main and interaction effects. All significant effects are reported in Table D.1 (see Appendix D). The results are plotted in Figure 5.13, including the overall UES-SF score (Figure 5.13a), and the four subscales (Figure 5.13b).



(a) Overall user engagement score



(b) UES-SF Subscale Scores

Figure 5.13: Quantitative subjective engagement: (a) the overall user engagement score and (b) the four subscale scores

Post-Study Feedback

Debriefing indicated *VR Forest* as the preferred environment ($N = 19$) and *MR Lab* as the least preferred ($N = 12$). *Histogram* was rated the most liked visualization ($N = 9$). Most disliked were *Center of Pressure* ($N = 11$) and *None* ($N = 10$). Furthermore, participants rated their agreement with the statement that the use of VR/MR increased their engagement with the study on a 5-point Likert scale with an average of 3.92 ($SD = 1.39$), indicating general agreement. Additionally, they rated that the use of the technology has enhanced the ability to maintain correct foot pressure distribution with an average of 3.38 ($SD = 1.38$).

Qualitative: Semi-Structured Interview

Using an inductive approach of the thematic analysis, we identified three main themes across the dataset: *Environmental Influence of Naturalness & Realism*, *Visual Feedback Interpretability & Informativeness*, and *Emotional and Physical Experience*. In the following the themes are discussed.

Environmental Influence of Naturalness & Realism Participants highlighted naturalness and realism as key environmental factors influencing overall experience and performance. For example, the *VR Forest* condition was perceived as "relaxing" (P14, P18), because "I feel good in walking in a forest or green environment which feels more natural" (P14) and "it feels like a real and beautiful scenery" (P21). One participant stated that the forest positively influenced the task, as "it felt natural and less mental pressuring" (P7). Furthermore, it was mentioned that *VR Forest* is an "appealing environment, which has a relaxing effect on the nervous system. It is less tedious, more interesting, and more distracting from monotonous walking." (P15). However, it was also mentioned that "it doesn´t feel real" (P13), which can be "boring" (P20). To achieve higher realism, it was added that "if I can hear some natural sound it could be more attractive" (P6). In contrast, in the *VR Lab*, one participant mentioned that "I felt the most safe, calm, and focused" (P1). However, it was also associated negatively, as it feels like "I am in a fake place" (P14), it is "boring and monotonous" (P19), "irritating" (P18), and "confusing" (P24), and participants "only see the empty lab and not anything else" (P2). Furthermore, the *MR Lab* was perceived as "it felt more real and therefore more comfortable" (P23), "comparing to VR with forest and virtual lab, it was pleasant in augmented reality with real lab." (P22). In contrast, the passthrough function also affected visual appearance; it was "not visually appealing (because of) the pixelated quality was the most difficult of all environments to hide" (P15), which may have impacted gait behaviour, as one participant stated, "I had to walk very closely" (P17). In general, this environment was also "not interesting" (P11, P12) and "more disorienting" (P1).

Visual Feedback Interpretability & Informativeness Regarding the visual feedback of pressure information, participants noted positive and negative aspects of distinct visualizations in terms of interpretability and informativeness. For example, the *Heatmap* received only positive remarks, as it provides a "clear representation of data" (P11) and shows "how pressure is spread" (P13). Furthermore, one participant also mentioned spatial compactness and informativeness: "took up the least space on the screen and visually showed exactly where I was, thus providing the most information" (P19). Data interpretability was positively noted for the *Histogram*: "because it's clear to anyone" (P2), "it seems more attractive" (P6), and "I can measure and realize in my foot where I am given more pressure" (P23). Compared to other visualizations, participants felt the *Histogram* "values can be more predictable" (P9) due to the visualized orthosis movement. However, it also led to a change in attention, as one participant noted, "I was trying to be careful with histogram" (P17). One reason for disliking this visualization was that "the footprint was bigger [and] taking more space in visual" (P20). The *CoP* was rarely mentioned positively, as one participant stated that it was "most specificity of where max pressure distributed" (P1). In contrast, it was perceived as "not understandable" (P12) and "distracting" (P7), as well as "not enough information, lack of understanding of what this means to me" (P19). Furthermore, it might be harder to control, as participants mentioned: "I did not understand what changing" (P3), "it's confusing and I can't do my task properly" (P23), and "doesn't provide feedback to correct within the same step, only to correct the next step due to the nature of the accumulation" (P1). The absence of visual feedback in the control condition (*None*) was mentioned positively, because "without any visualization, i could walk freely" (P17) and "it helped me more to focus on feeling" (P15). However, missing information was also cited as a critical aspect, as "every pressure visualization has a significance" (P8).

Pressure-based visualizations were perceived as less effective in the naturalistic *VR Forest* environment, being described as "confusing" (P18), "annoying" (P15), and distracting from natural walking: "I can't concentrate ... we want to walk naturally" (P6). Furthermore, it was emphasized that "data with the forest walking is not a good combination. So I cannot enjoy my walking" (P6) and "I feel like the forest is very immersive, but ... looking at the visualization of pressure couldn't do both as much as I wanted to at the same time" (P1). In contrast, the *Heatmap* in the *VR Lab* was evaluated positively, as it "helped me to understand where I actually put my pressure when I was walking" (P5) and "because the VR environment is static, it is easy to concentrate on the visualization" (P4).

Emotional and Physical Experience Participants reported a wide range of affective experiences during the study, ranging from relaxation to fatigue and discomfort. For example, one participant described the experience as relaxing: "I feel(ed) relaxed, it was like walking for an hour" (P19). However, several participants also reported fatigue and reduced motivation, noting that the study was "more mentally and physically demanding" (P12). Physical discomfort was frequently mentioned in relation to both the HMD and the foot orthosis. The footwear was described as "heavy" (P22) and "should be more comfortable," with one participant reporting that "sometimes I felt a bit of strain on my ankle" (P7). Similar concerns were raised regarding the headset, which was considered "too heavy" (P8). The resolution in the MR video see-through condition was negatively perceived, with one participant reporting "motion sickness [...] and the weight of the headset gave me a slight headache" (P7). In addition, another participant confirmed issues with MR, noting that "shaking in MR made it more difficult to maintain balance" (P4). Beyond physical aspects, some participants were less satisfied with the virtual environments, describing them as "an artificial environment [and] not real life" (P14). To improve engagement, participants suggested adding more engaging environments: ”please make more enjoyable sceneries like a beach” (P3). Overall, these findings highlight the importance of both emotional and physical factors for user experience: while feelings of relaxation may foster acceptance, fatigue and discomfort can hinder engagement.

5.2.4 Discussion

We conducted a study with 24 participants to examine how visual foot pressure feedback and environmental context affect objective and subjective measures in VR/MR treadmill walking. Results showed both factors influenced these outcomes.

Influence of Visualized Foot Pressure Feedback Overall, the quantitative gait results revealed primarily interaction effects with the environment rather than main effects of the visualizations alone. In particular, the *Heatmap* visualization led to the largest changes in the *VR Lab* condition, while it performed differently in other environments, indicating environmental influences. In general, the *Heatmap* was associated with a more focused gait, as participants aimed to control their steps. In contrast to previous work which highlighted the benefits of CoP visualizations [136], we could not find such an effect, which may be task-dependent and may not occur during fixed treadmill speed. The *CoP* visualization performed similarly to the *None* baseline condition, suggesting that it had little to no influence on gait. This finding is consistent with qualitative feedback, where participants frequently mentioned difficulties in understanding how to use *CoP* during gait, which often led them to ignore its functionality. The strongest contrast emerged between *Histogram* and *Heatmap*, reflecting distinct pressure regulation strategies. The *Heatmap* visualization led to slightly higher perceived effort in the *MR* and *VR Lab* compared to the *VR Forest*, where all visualizations were rated similarly low, suggesting a more relaxing experience in the naturalistic environment. Across conditions, visual feedback was generally rated as more demanding than the absence of feedback. Interestingly, in the *VR Lab*, participants rated their own performance with the *Heatmap* higher than with other visualizations, confirming the effects observed in gait measures. However, in the *MR Lab*, the *Heatmap* was associated with lower perceived performance compared to its use in the *VR Forest* or *VR Lab*. This could be explained by participants' attempts to control their gait more intensively in the *MR Lab* condition, where the limitations of the video see-through display reduced visual fidelity and thus led to lower perceived performance compared to the *VR Forest* or *VR Lab*. Finally, UES-SF results showed a distinct pattern in the *VR Forest*. Without visual feedback, participants rated usability higher and reported better focus of attention. This was also confirmed by qualitative comments, where they mentioned more enjoyment when focusing on the natural environment, without visuals. In contrast, in the *VR Lab* and MR conditions they preferred having visual feedback rather than none.

Overall, our findings highlight that *Heatmap* visualization influenced gait in a stationary lab environment, leading to more controlled movement, while also affecting perceived engagement and performance. Other visualizations showed no measurable effects, likely due to control challenges and environmental factors.

Influence of Immersive Environment Our findings showed that the immersive environment significantly influenced both objective measures (gait parameters, heart rate) and subjective measures (workload, engagement), confirming that the environment is not a neutral background variable but can actively shape gait outcomes and the training experience. The *VR Forest* environment was preferred by almost 80% of participants. In line with prior work, natural environments reduced cognitive load and influenced task performance [142]. This preference was accompanied by lower heart rate and supported by qualitative feedback, describing the environment as relaxing and natural. Participants reported lower mental demand, reduced frustration, and consistently higher engagement ratings. These findings highlight the value of naturalistic immersive environments in supporting both physiological relaxation and subjective engagement. Despite treadmill speed constraints, the natural environment promoted longer step lengths and lower cadence, indicating a gait pattern closer to natural walking. However, pressure visualization was perceived as less useful, as participants preferred to enjoy the environment and walk naturally. In contrast, the *VR Lab* environment performed differently. While comparable to MR in terms of heart rate and workload, it was rated as more engaging, suggesting that a controlled virtual setting without real-world elements can still foster involvement. Notably, the addition of the *Heatmap* visualization in the *VR Lab* led to significant changes in spatial, temporal, and kinetic gait parameters, with shorter gait cycles and higher step frequency, indicating increased attention and deliberate task control. This demonstrates that static virtual environments, when combined with targeted feedback, can channel user attention, as also reflected in qualitative data. The contrasting performance of the forest and lab environments indicates a trade-off: naturalistic scenes promote relaxation and natural gait, whereas controlled laboratory scenes enhance focus and task precision. Thus, VR environments can be designed to encourage either natural movement or attentional control, depending on training objectives. The *MR Lab* results were less promising for immersive gait training. Consistent with related work [3, 148], the video passthrough function negatively affected task execution. Visibility challenges led to higher frustration and lower engagement compared to both VR conditions. Overall, immersive environments led to distinct physiological and behavioral outcomes. These findings highlight a trade-off: a naturalistic environment (*VR Forest*) promote relaxed, comfortable walking, while a controlled virtual lab (*VR Lab*) with *Heatmap* feedback support attentional focus and performance. Moreover, the environment can modulate the perception of visual cues, suggesting an interaction between context and feedback in shaping gait and user experience. In contrast, the *MR Lab* offered limited benefit, as reflected in both quantitative and qualitative data.

Implications and Design Recommendations

Our study demonstrates that visual foot pressure feedback does not merely support task awareness, but can actively influence orthotic gait regulation strategies when embedded in specific virtual environments. By examining their combined effects, we extend previous work that focused on isolated measures. Although conducted with healthy participants, similar to related studies [141], the findings remain relevant for gait-impaired users and the design of future gait training systems. Based on these insights, we propose design recommendations for immersive orthotic gait training:

- **Support natural gait through naturalistic environments:** Based on our findings, naturalistic environments (e.g., the *VR Forest*) foster relaxed and unconstrained walking patterns, which is particularly valuable when orthotic footwear alters gait or when trust in the device is still developing. We recommend using such environments in recovery-oriented training phases to support natural movement and reduce workload. Visual pressure feedback should be avoided in these scenarios, as it disrupts immersion and shifts attention away from intuitive gait adjustment.

- **Enable controlled gait through targeted feedback in stationary VR settings:** In contrast to natural environments, controlled VR settings (e.g., a virtual lab) in combination with *Heatmap* feedback enabled deliberate regulation of plantar pressure and step timing, an effect directly relevant for corrective orthotic gait training. This suggests that stationary environments are particularly suited for precision-oriented training phases.

- **Adapt and align visual feedback to the training context:** Visual pressure feedback increases cognitive demand and should therefore be applied selectively. It is most effective in precision-oriented phases, where users regulate plantar loading or step timing, but should be minimized when the goal is relaxation or orthosis familiarization. In such corrective phases, feedback should be compact, spatially direct, and interpretable as actionable guidance (e.g., heatmap or histogram with orthosis rotation), rather than abstract displays (e.g., CoP), which were often disregarded. To avoid visual overload, feedback elements should be visually minimal and aligned with the virtual scene to maintain immersion and support gait adjustment.

- **Consider device dependency in MR gait training:** The MR condition was associated with lower usability and higher frustration due to passthrough fidelity limitations affecting depth perception and visual clarity of foot placement. These aspects are particularly critical in orthotic gait training, where precise perception of foot–surface interaction is essential. Although the Meta Quest 3 offers moderate video passthrough quality [325], it is currently less suited for this training application, whereas optical see-through devices provide more reliable visual conditions for corrective training. Therefore, designers should carefully evaluate the display technology when selecting MR devices for gait rehabilitation.

Limitations and Future Work

The study was restricted to healthy participants, limiting generalizability, and a fixed treadmill speed constrained natural gait adaptations. Furthermore, the MR setup with the Meta Quest 3 may have introduced confounding factors such as altered visual flow or passthrough fidelity, potentially influencing gait adaptation. Hardware limitations include the prototype insole being restricted to the right foot, and the system's end-to-end latency was not analyzed. Future work should evaluate and minimize such delays to ensure reliable real-time biofeedback. Follow-up studies should include patient populations, different footwear, and extended tasks.

5.2.5 Conclusion

In this paper, we investigated the effects of visual foot pressure feedback from a smart insole and immersive environments during treadmill walking in VR and MR. Our findings show that the effectiveness of pressure-based feedback depends on its interaction with the virtual environment, leading either to exploratory gait behavior or to corrective, attention-focused strategies. Based on these insights, we propose design recommendations for adapting foot pressure feedback to natural and laboratory contexts and outline future research directions. This work advances the understanding of foot augmentation in immersive settings and informs the design of smart insole systems within HCI.

5.3. Chapter Summary

Chapter 4 investigated the use of MR and VR applications for interacting with smart devices, specifically the previously developed smartphone app and smart foot orthosis. Building on these devices, this chapter explored their application in immersive training environments, leveraging MR and VR technologies to enhance user interaction and experience.

Section 5.1 addressed RQ4, focusing on avatar-based feedback modalities during smartphone app onboarding and their effects on usability and learning experience. The study demonstrated that additional visual or auditory guidance can improve tutorial support and usability, while the combined use of modalities yielded limited additional benefits. Qualitative feedback highlighted both challenges and advantages of immersive guidance compared to unguided interaction. These results provide empirical insights and inform design recommendations for future guidance in mHealth applications, while identifying directions for further research.

Section 5.2 addressed RQ5 by examining the perception of visual foot pressure feedback and the role of environmental factors in MR/VR during treadmill walking with the smart orthosis. The experimental study revealed that environmental context interacts with visual feedback to influence gait behavior, mental workload, and user engagement. Participants preferred VR over MR while walking, and visual feedback was most effective when the training environment (naturalistic vs. static lab) aligned with the task. These findings contribute new empirical knowledge and provide concrete design recommendations for immersive gait training applications.

Taken together, both studies advance understanding of how XR interactions can enhance user–device engagement, from onboarding mobile apps to gait training with smart orthoses. They highlight the strengths of immersive training while acknowledging technological limitations, and offer actionable insights for designing effective MR/VR-based interventions. These results extend previous findings to new applications and provide a foundation for future work in immersive smart device training.

CHAPTER 6

AI in Design: User-Centered Design of Smart Foot Orthoses

While the development and validation of smart assistive devices demonstrate technological potential, their successful adoption in daily use remains strongly dependent on aesthetic design factors. Related work has highlighted persistent discrepancies between user preferences and conventional orthotic designs, often resulting in limited acceptance, reduced wearing compliance, and consequently unrealized health benefits. Therefore, prior research has emphasized the importance of user-centered design approaches that actively integrate users into the design process. In this context, recent advances in AI offer new opportunities to rapidly generate design concepts based on user input, enabling participatory design processes that are well aligned with user-centered development principles. However, it remains largely unclear to what extent AI can be effectively used to generate orthotic designs that influence social acceptance, and how such approaches can be systematically integrated into established design workflows.

To address RQ6 and RQ7, this chapter presents the results of one journal article and one conference paper focusing on the role of AI in the design of smart foot orthoses. Section 6.1 investigates RQ6 by examining how AI-supported design approaches influence the perceived social acceptability and potential adoption of personalized smart orthoses. Two empirical studies were conducted to compare AI-generated orthotic designs with conventional orthopedic products across different device categories. The findings provide evidence that design characteristics significantly influence user perception and demonstrate how generative AI can shape aesthetic outcomes and perceived acceptance. Building on these insights, Section 6.2 addresses RQ7 by examining how AI can be systematically integrated into a user-centered design process to simultaneously consider functional, aesthetic, and usability requirements. A patient-centered AI-driven design framework is proposed and validated through a case study, offering initial insights into how generative models can be embedded within a structured product development workflow.

Finally, the studies presented in this chapter demonstrate the potential of AI to bridge the gap between technical functionality and user acceptance in orthotic product design.

The integration of AI into the design process is illustrated in Figure 6.1.

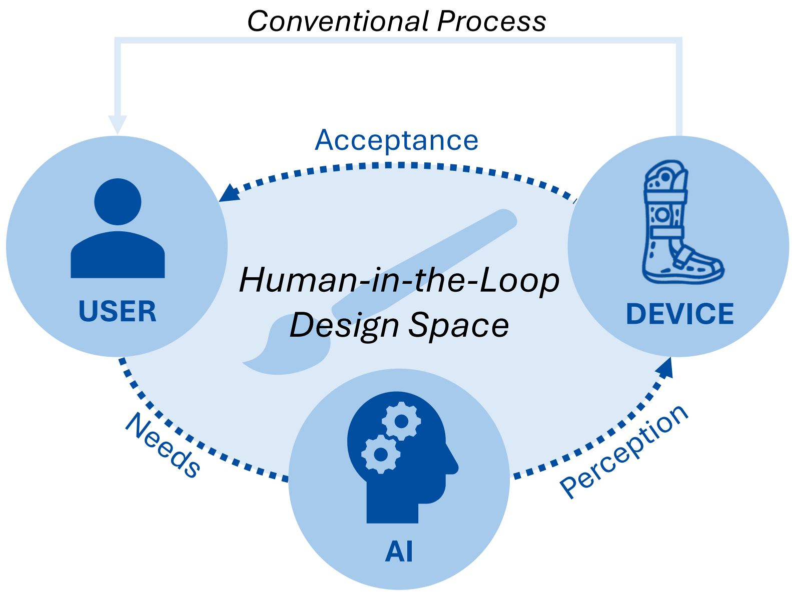


Figure 6.1: Schematic overview of the AI-supported user-centered design approach.

In this context, the term AI refers specifically to image-generative AI models, used as a design tool to support the rapid creation of visual stimuli based on user input. The approach focuses on leveraging existing knowledge embedded in pre-trained models to generate design concepts from text prompts using tools such as DALL-E (ChatGPT, OpenAI). It is important to note that the use of AI in this work does not replace the technical or functional design requirements of medical devices, which remain essential in orthotic development. Instead, generative AI is used to accelerate early-stage design exploration and to support collaboration with end-users, while expert knowledge remains central to decision-making. Accordingly, the following studies aim to investigate the impact of AI-generated design outcomes on user perception and social acceptability, as well as the influence of prompt structures and GPT personalization on these perceptions. This approach is intended as a methodological provocation, highlighting the potential of generative AI to support user-centered design and to inform future research on AI-assisted design processes.

6.1. Study I: Influence of Generative-AI on Social Acceptance

This section presents one article comprising two studies (online surveys) investigating the impact of generative AI on the design of foot orthoses and its influence on social acceptance. Figure 6.2 shows an AI-generated orthosis as an illustrative example, serving as a graphical abstract for RQ6 on how design characteristics can influence perceived acceptance. The results highlight the effect of AI-generated orthotic footwear on perceived acceptance across various device categories and indicate how specific keywords may influence these outcomes. By providing empirical evidence, this study contributes new knowledge to the field, offers implications, and identifies directions for future implementation.

Figure 6.2: AI-generated image (via DALL-E, 2024) illustrating a foot orthosis, serving as a visual representation of the research question on social acceptability [5].

S. Resch, J. Schauer, V. Schwind, D. Völz, and D. Sanchez-Morillo, “Improving Social Acceptance of Orthopedic Foot Orthoses Through Image-Generative AI in Product Design,” Applied Sciences, vol. 15, no. 8, Art. no. 4132, 2025.

6.1.1 Introduction

Orthopedic foot orthoses are standardized devices that are essential for the rehabilitation of patients recovering from injuries or managing chronic foot diseases [344]. The designs of these assistive devices differ, featuring orthoses with varying leg lengths, bandages, and foot drop orthoses, each customized for particular medical purposes [345]. Each category of orthotic products is designed to support patients with different medical needs and mobility requirements [346]. This product diversity requires highly personalized designs to meet the specific demands of different health conditions and patient preferences [347]. While technical function is essential for medical effectiveness, aesthetics and usability are also key elements in the design of orthopedic foot orthoses to promote patient acceptance and support successful use and rehabilitation outcomes [345]. For example, a study by Ghoseiri and Bahramian [23] investigated user satisfaction with orthotic devices among 293 participants, revealing that concerns related to the appearance of the devices often resulted in low satisfaction rates. Furthermore, a systematic literature review by Swinnen and Kerckhofs [25] analyzed ten studies involving 1576 patients and highlighted the critical role of aesthetics in the compliance of wearing lower limb orthotic devices. The review found that a notable number of patients avoid wearing their devices regularly due to their "cosmetically unacceptable" appearance, highlighting the "large importance" of aesthetics in patients' decisions to wear these devices. This underscores the significant impact of aesthetics on device adoption and user perception [348]. However, the design processes of these standardized devices frequently rely on traditional methods that limit adaptability and patient involvement, leading to compliance issues [27]. This dissatisfaction can lead to inconsistent use of standardized orthopedic aids and missed health benefits, as highlighted by previous work [349].

Current advancements in sensor technology offer the potential to transform these conventional orthoses into smart products, commonly known as wearable electronic devices, which can monitor specific health parameters [15]. In theory, the creation of smart orthotic footwear could enhance the rehabilitation process by enabling the continuous monitoring of various conditions while also extending health benefits [350]. However, the conceptual development and integration of such advanced technologies also bring new challenges for product design that must be addressed to ensure patient acceptance in line with medical requirements [351].

A systematic literature review by Orlando et al. [352] highlighted that there is a significant need for improvement in lower extremity orthotic devices. A list of five main user needs (function, expression, aesthetics, accessibility, and other) and the corresponding sub-items for the use of lower limb orthoses was compiled. It was concluded that improvements in the design, prescription, and implementation of these devices are crucial to improve utilization and achieve greater user satisfaction. Therefore, incorporating user perspectives and identifying their needs in the development process is essential to ensure a high level of user acceptance for smart technologies. For example, Van der Wilk et al. [353] applied a user-centered design approach and conducted a qualitative study with eight participants in focus groups to investigate patients' perspectives on improving the design of an AFO. The three main themes identified by the patients, *walking and standing ability*, *activities*, and *AFO characteristics*, were highlighted as the most relevant aspects for future developments.

To address these design challenges, the use of image-generative AI applications in the design process of future developments presents a novel approach [354]. As shown by Suessmuth et al. [28], generative AI can support concept creation in footwear design by actively involving users in the design process and tailoring products to their preferences, highlighting valuable potential for the industry. Building on this idea, the emerging trend of rapid image creation with tools such as DALL-E [355] from OpenAI provides new opportunities for generating design concepts based on user-defined prompts. The use of such technologies could support the iterative design process within the structured product development of orthopedic orthoses. However, the impact of AI-generated wearable designs on social acceptance remains an open question. To address this research gap, we formulate the following three RQs:

- RQ1: How does the use of AI-generated designs impact the perceived social acceptability of foot orthoses compared to conventional orthopedic products or research developments?
- RQ2: How does the category of orthotic products influence the social acceptance of foot orthoses, comparing categories such as short-leg orthoses, high-leg orthoses, foot bandages/textiles, and foot drop orthoses?
- RQ3: How does a custom GPT, trained on data for personalized orthoses, impact social acceptance, and what effect do specific keywords in the prompt structure (such as "social acceptance" or "sporty design") have on this acceptance?

While aspects such as biocompatibility and manufacturability are critical in the development of orthopedic devices, our study focuses on the visual perception and social acceptance of AI-generated designs. These factors are particularly relevant for end users, whereas material and production requirements fall within the domain of technical experts, who were not the target group of this study. In addition, current generative AI models are primarily suitable for the conceptual phase of designs, as they are generally unable to take technical manufacturability into account, which is why specialized technical expertise is required [356].

Therefore, we conducted two studies to investigate the influence of AI-generated orthoses designs on social acceptance. The first study aimed to assess the potential of AI integration by evaluating whether AI-generated orthoses achieved higher acceptance ratings compared to conventional products. The second study compared the output of a custom GPT and ChatGPT-4 and investigated the effects of specific keywords in the prompts on acceptability ratings. Based on the findings, we provide recommendations for refining text prompts and discuss the implications of these enhancements for future developments in wearable design.

We aim to contribute to the field of human-centered design using generative AI by expanding the knowledge of socially acceptable SFO designs and investigating the role of image-generative AI in the product development process.

Accordingly, this paper highlights the following three main contributions:

1. We introduce a novel approach that leverages image-generative AI to enhance the customization and aesthetic appeal of foot orthosis designs, increasing their alignment with user requirements and visual attractiveness.

2. Our research provides empirical evidence demonstrating that AI-generated orthosis designs significantly enhance social acceptability among users. This potential could be utilized to create widely accepted design solutions for other products as well.
3. We offer a comprehensive set of future recommendations for integrating AI into the orthotic design, aiming to refine the prompt design process to ensure that the resulting designs meet product requirements and patient design expectations.

6.1.2 Method

Study 1: The Impact of AI-Generated and Conventional Orthotic Designs Across Device Categories on Social Acceptability

To examine the influence of AI-generated orthotic designs on social acceptability, this study compared different AI-created designs with conventional orthopedic products and research-based developments. Additionally, we analyzed how various orthotic device categories affect acceptance. Social acceptability was quantitatively assessed using the WEAR scale, while stereotypical perceptions were evaluated through the SCM model, measuring warmth and competence ratings. To complement these findings, qualitative feedback provided deeper insights into user needs and preferences.

Study Design A total of 161 participants were recruited through mailing lists via our institution and within personal networks. We conducted an online survey using a two-factorial within-subject design to investigate how different design types and product categories of orthopedic foot orthoses influence social acceptability. The two independent variables were Design Type and Orthotic Category. The variable Design Type includes three individual levels: *Generative AI*, *Orthopedic products*, and *Development concepts*. The Orthotic Category includes four levels: *Short-Leg Orthoses*, *High-Leg Orthoses*, *Foot Bandages/Textiles*, and *Foot Drop Orthoses*. We hypothesized that the perceived social acceptability of foot orthoses will be positively influenced by *Generative AI* concepts compared to the traditional *Orthopedic products* and *Development* concepts. We also hypothesized that the perception of acceptability would vary strongly within the Orthotic Category, as different device types may elicit different user reactions depending on their design and visibility.

Stimuli Four different categories of orthopedic foot orthoses (*Short-Leg Orthoses*, *High-Leg Orthoses*, *Foot Bandages/Textiles*, and *Foot Drop Orthoses*) were chosen to cover the medical needs of different foot conditions. We selected six different images for each Orthotic Category, resulting in a total of twenty-four stimuli (see Figure 6.3).

Each Orthotic Category included two images of *Generative AI* orthoses, two *Orthopedic products*, and two orthoses from *Development concepts*. The selection of two different orthoses for each Design Type ensured a diverse representation of design variations within each Orthotic Category. All images were presented with a white background and consistent positioning of the orthosis to ensure comparability. Adobe Photoshop version 2023 24.0.1 was used for manual adjustments, such as rotating the orthosis and creating a transparent background where necessary. Each image contained

Figure 6.3: Overview of the 24 stimuli used in the online survey, categorized by the two independent variables DESIGN TYPE and ORTHOTIC CATEGORY. The stimuli are arranged horizontally by the four orthotic categories, *Short-Leg Orthoses*, *High-Leg Orthoses*, *Foot Bandages/Textiles*, and *Foot Drop Orthoses*, and vertically by the three design types, *Generative AI*, *Orthopedic products*, and *Development concepts*.

only the orthosis and some visible parts of the lower leg to maintain focus on the orthotic device itself.

Two approaches were used for *Generative AI* designs: text-to-image and image-to-image generation. Text-to-image generation was applied using DALL-E 3, which is based on a transformer architecture, from ChatGPT-4 [357] (OpenAI). Text prompts were created to design smart orthoses based on the type of orthosis and its medical functions. These typically feature a gray device color on a white background. Additionally, the prompts were extended to include small black boxes on the sides of the orthoses, which serve as placeholders for electronic components to enable smart functions. For image-to-image generation, images of medical orthoses were used as prompts to create realistic designs. The following prompt was used and iteratively refined during re-prompting: "*Generate an image of a foot orthosis based on the shape of the reference image. The orthosis should provide full protection for the lower leg and foot, consisting of two rigid housing parts that enclose the leg and are secured with Velcro straps. The orthosis should be gray and displayed against a white background. Additionally, a small box should be integrated into the orthosis to house sensors for smart functionalities*". The images of *Orthopedic products* are based on standardized medical devices that are available in online shops or medical supply stores: *VACOpedes* [358], *AIRCAST® AIRSELECT™ Short Walker* [359], *VACOcast Diabetic Boot* [360], *AIRCAST® AIRSELECT™ Achilles Walker* [361], *FastProtect Malleo* [362], *The BetterGuard* [363], *WalkOn Reaction Lateral* [364], and *ofa Push ortho Fußheberorthese AFO* [365]. These products are currently used by patients with foot conditions and represent state-of-the-art solutions. The *Development concepts* stimuli include various devices proposed in research projects [366, 367, 368, 369], 3D-printed prototypes published on the web [370, 371, 364], and a self-created 3D CAD-design (C4-DEV2).

Procedure The study was conducted in accordance with the ethical standards for user studies as required by our institution and received ethical approval from the German Society for Nursing Science (No. 23-027). Participants were informed about the study's purpose, ethical procedures, data privacy, data anonymity, and the intended use of their data before participation. Informed consent was obtained via the online platform before the start of the survey. Participants received a web link to our landing page, where they could access the online survey, which was created using LimeSurvey. After signing the informed consent form, participants were asked about demographic data, information about past conditions, and prior knowledge of orthopedic footwear or smart orthoses. Following this, participants received a brief introductory text for each orthotic category to prevent major misinterpretations and to provide context for the device and medical application. Participants were then presented with the 24 stimuli in a randomized order. For each stimulus, the corresponding picture was shown, and participants rated two statements about social acceptance and aesthetic design expectations on a 7-point Likert scale. The questions were "*To which extent do you agree with the statement that the device is socially accepted?*" and "*To what extent does the design meet your aesthetic expectations?*", with responses ranging from *totally unacceptable* (1) to *perfectly acceptable* (7). In addition to the quantitative questions, qualitative feedback was obtained via a free text entry question for each stimulus: "*What positive/negative aspects of the orthosis do you notice?*". Following this, the standardized SCM and WEAR questionnaires were completed for each stimulus. After answering all 24 conditions, participants responded to concluding questions regarding any changes in their perspective on orthoses, noting positive aspects, negative aspects, prior knowledge of AI image generation, and any additional feedback. The average completion time for this survey was 75 min.

Measures and Data Analysis In this study, we used a mixed methods approach based on quantitative and qualitative feedback. Quantitative data were measured using the SCM and WEAR scales, complemented by two seven-point Likert items assessing social acceptance and aesthetic design expectations. These data were analyzed through descriptive and inferential statistics. Additionally, qualitative feedback derived from participants' free-text responses was assessed using thematic analysis. The methods and data analysis procedures are described in detail in the following paragraphs.

Quantitative: Stereotype Content Model (SCM) The SCM questionnaire by Fiske et al. [205] was used to assess stereotypes and social perceptions of various foot orthoses based on two perceived dimensions: warmth and competence. This standardized questionnaire consists of nine items rated on a 5-point scale ranging from *not at all* (1) to *extremely* (5). The competence dimension was evaluated using five items (*competent*, *confident*, *independent*, *competitive*, and *intelligent*), each assessed by the question "*As viewed by society, how ... are members of this group?*". Additionally, the same question was used to assess the warmth dimension, consisting of four items:

tolerant, *warm*, *good-natured*, and *sincere*. For the data analysis, a two-factorial RM ANOVA was used.

Quantitative: Wearable Acceptability Range (WEAR) To measure the social acceptability of wearable technology, the WEAR scale by Kelly [206] was used. The WEAR scale (version 3) consists of 14 items, each rated on a 6-point scale ranging from *strongly disagree* (1) to *strongly agree* (6). Five of these 14 items are reverse-scored. The total score was divided by 14 to obtain an average score, which can be graded from *extremely low social acceptance* (1) to *extremely high acceptance* (6). An RM ANOVA was used for data analysis, similar to the procedure used for the SCM scale.

Quantitative: 7-Point Likert Scales As described previously in the procedure paragraph, two questions were used to rate statements about social acceptance and aesthetic design expectations on 7-point Likert scales. For the evaluation, a non-parametric ART RM ANOVA was applied.

Qualitative Feedback Qualitative results from the free text entries were assessed using inductive thematic analysis [322]. Anonymized data were transcribed, coded, and analyzed paragraph-wise to identify categories and main themes. Two researchers were involved in the data analysis process: one researcher transcribed and coded the data set independently, while the second researcher reviewed the results for consistency.

Study 2: The Impact of GPT Customization and Prompt Keywords on the Social Acceptability of AI-Generated Orthotic Designs

Based on our first study, which showed that AI-generated designs can increase the social acceptance of orthopedic foot wearables, this study further explored the role of prompt customization in shaping user perception. In particular, the first study revealed that high-leg orthoses received the lowest acceptance ratings. Therefore, we examined which factors in AI image generation could positively influence the acceptance of this device category. While the initial study confirmed the potential of generative AI for orthotic design, it also raised questions regarding the influence of specific keywords in the prompts used and the application of customized GPTs. Furthermore, it remained unclear how the inclusion of keywords related to user preferences would affect the perceived acceptability of designs. To address this, our second study investigated the impact of various keywords and the use of a personalized GPT tailored to specific patient and product requirements. Previous research has demonstrated the influence of prompt structures and provided recommendations for different user roles [185]. Building on this knowledge, we take one step further and contribute to user-centered prompting strategies by incorporating keywords that enhance the alignment between generated designs and user needs.

Study Design In our second study, we conducted an online survey to investigate the effects of different prompt keywords and GPT personalization on the social acceptability of AI-generated high-leg orthosis designs. In total, 133 participants were recruited.

The recruitment of participants was conducted similarly to the first study, using mailing lists of our institution and personal networks. A two-factorial within-subject design was carried out with the two independent variables: GPT and KEYWORD. The variable GPT has two levels: *ChatGPT* and *OrthoticFootGPT*. The variable KEYWORD has four levels: *No Keyword*, *Usability*, *Social Acceptability*, and *Sporty Design*. The hypothesis of our second study is that perceived acceptability will significantly vary depending on the specific keywords used in the prompts and that the inclusion of new keywords will positively influence this acceptability. Furthermore, we hypothesize that a customized GPT will lead to more acceptable perceived outcomes compared to the standardized model.

Stimuli The model GPT-4 from ChatGPT was used to create the stimuli of high-leg orthoses, which allows the creation of a custom GPT that can be configured according to the desired response and outputs. DALL-E 3 image generation was activated in the configuration interface of the customized GPT, and specific knowledge and instructions were provided as input. However, OpenAI does not disclose specific details about its training configuration or model architecture. The customized GPT named *OrthoticFootGPT* was described as a "generative design assistant to support the creation of socially acceptable foot orthoses". To generate a user-specific output, the knowledge base for the *OrthoticFootGPT* was customized with a comprehensive set of data tailored to high-leg orthoses. This dataset included

- Product specifications: Detailed information on the aesthetic and functional aspects of high-leg orthoses, such as material types, sizes and weight from data sheets and guidelines.
- Usage instructions: Documents with instructions and user manuals for the use and maintenance of these devices.
- Product images: Sample images of high-leg orthoses from various suppliers as a visual reference for a realistic design.
- Research articles: Related articles providing insights into product design rules, the concepts of accessibility/usability, and orthotic development were searched on Google Scholar.
- Keyword definitions: Documents based on international standards and dictionary definitions provided explanations about the three keywords (usability, social acceptability, and sport design) to guide the AI in generating contextually relevant content.

For both *ChatGPT* and *OrthoticFootGPT*, the following prompt was used as the foundation for image generation: "*Create a realistic image of a white-grey foot orthosis with a high shaft. The orthosis should have stabilizing properties so that it can be worn for longer periods during rehabilitation. The background should be white and only show the leg with the orthosis*". In addition, a sentence was added to the basic query for each keyword to tailor the images to the specific definition. For example, for social acceptability: "*The orthosis should ensure a high level of social user acceptance by offering an attractive aesthetic design and high wearing comfort*".

To ensure realistic product images and avoid misinterpretation or misleading details, the DALL-E editor interface was applied manually to ensure consistency across all images (e.g., by removing the second leg or unrealistic features). Similar to the first study, each image was manually adjusted using Adobe Photoshop to create transparent backgrounds and ensure consistent foot rotation angles. All generated images are shown in Figure 6.4 and the workflow for generating the visual stimuli is illustrated in Figure 6.5.

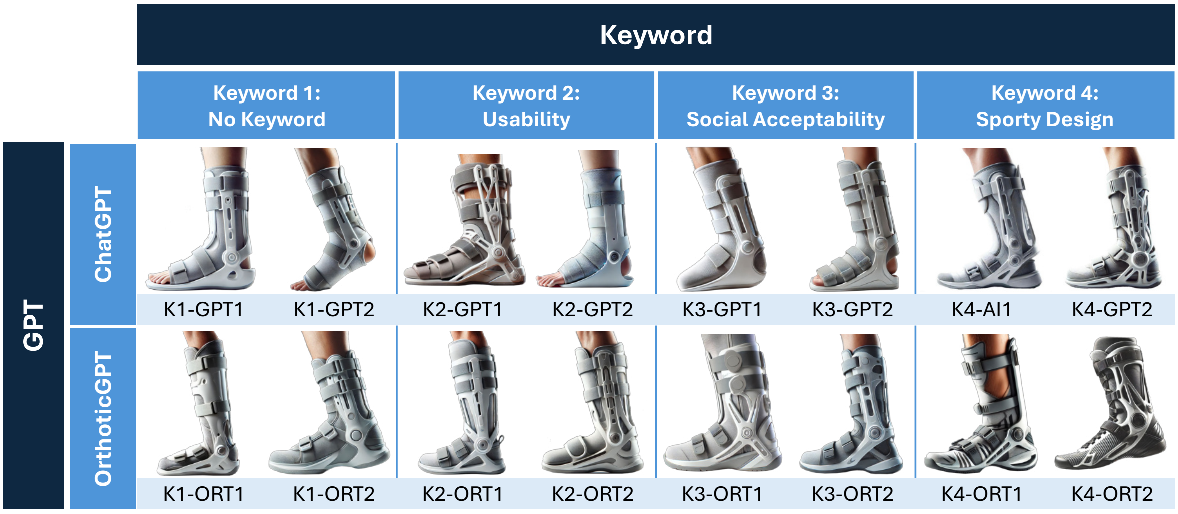


Figure 6.4: Overview of the 16 stimuli used in the online survey, categorized by the two independent variables, GPT and Keyword. The stimuli are arranged horizontally by the four keywords, *No Keyword*, *Usability*, *Social Acceptability*, and *Sporty Design*, and vertically by the two GPTs: *ChatGPT* and *OrthoticFootGPT*.

Procedure The study procedure was identical to that of the previous study, including ethical approval by the German Society for Nursing Science (No. 23-027) and compliance with institutional ethical standards. As in the previous study, participants were informed about data privacy, anonymity, and the intended use of their data before providing informed consent via the online platform. Participants signed an informed consent form and provided demographic information along with details about their previous experience with image-generative AI. The 16 conditions were then presented in a randomized order. At the end of the survey, participants were asked to answer some open-ended questions about positive and negative aspects of the designs, as well as additional feedback. The average time to complete this survey was approximately 64 min.

Measures and Data Analysis Similar to the first study, the quantitative data were collected using the standardized SCM [205] questionnaire, the WEAR [206] scale, and two 7-point Likert items to assess the social acceptance and aesthetic design expectations. Additionally, participants provided qualitative feedback through free-text responses about positive and negative aspects of the designs. The quantitative data analysis was conducted using descriptive and inferential statistics. Qualitative feedback was analyzed using inductive thematic analysis [322].

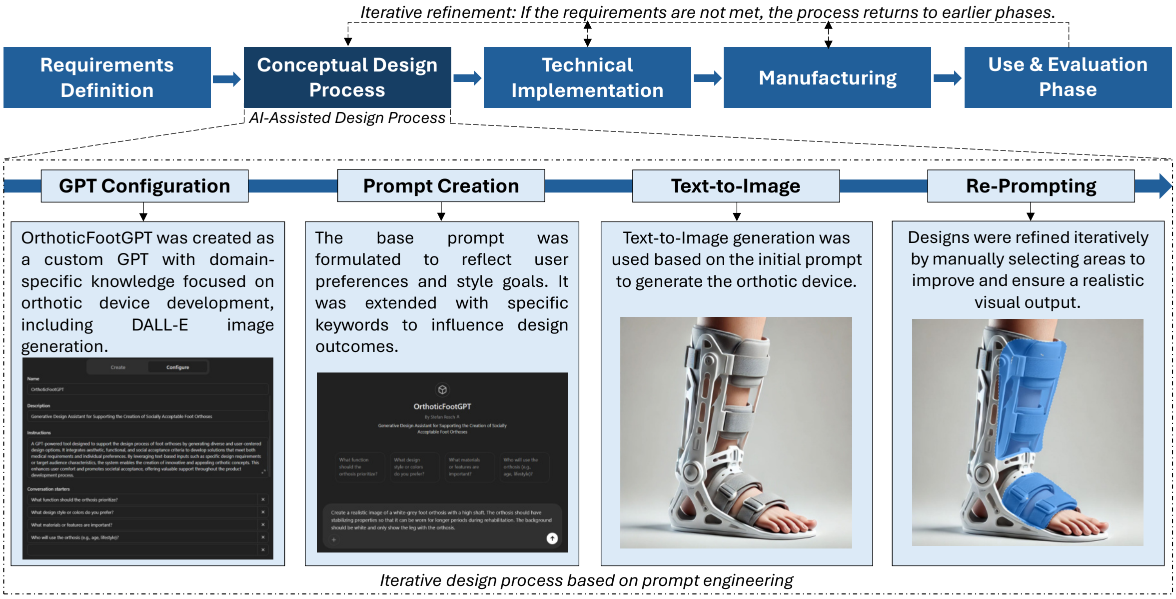


Figure 6.5: Schematic overview of the design generation workflow, including GPT customization, prompt creation, and text-to-image generation.

6.1.3 Results

Study 1

This section presents the characteristics of the study group, followed by the evaluation of quantitative results. This includes the correlation of SCM and WEAR Ratings, assessments of SCM and WEAR data, as well as the analysis of seven-point Likert scale ratings on device acceptance and aesthetic design expectations. The statistical analysis was performed in R using the package rstatix [292]. Finally, qualitative findings obtained through thematic analysis are presented.

Participants A total of 80 out of 161 participants (31 female, 48 male, and 1 divers) completed the online survey and were included in the data analysis. Participants' ages ranged from 19 to 77 years (M = 30.62, SD = 13.69). The participants had diverse educational backgrounds and came from seven different nationalities. Occupations of participants included (N = 1) each in natural sciences, healthcare, and crafts; (N = 2) each in law and administration; (N = 3) each in management and social work; (N = 5) in services; (N = 20) in other professions; and (N = 42) in engineering. Twenty-five participants reported having foot conditions, with sixteen of these in the past and nine currently. In total, 14 had chronic issues and eleven had acute conditions. Overall, 21 participants reported wearing orthopedic aids, including insoles (N = 11), bandages (N = 5), high-leg orthoses (N = 4), and short-leg orthoses (N = 1). In response to the question of whether the participants had previously heard of the term "smart orthoses", 51 responded "no", and 29 responded "yes". Twenty-nine participants were aware of generative AI image generation, twenty-eight had actively used it, and twenty-three had no prior knowledge of the technology.

To investigate whether participant demographics influenced the results, we conducted an ART ANOVA, which showed no statistically significant main or interaction effects of gender, prior AI knowledge, or past orthopedic conditions on SCM, WEAR, and Likert scores (all $p > 0.05$). These results show that demographic factors did not systematically influence the ratings of the participants.

Quantitative Results

Correlation of SCM and WEAR Ratings A Pearson correlation analysis was conducted to assess the relationship between SCM (means of warm and competence data) and WEAR ratings. A statistically significant moderate positive correlation was found between the Euclidean length of the warmth–competence vector and the WEAR scores, $r(958) = 0.453$, $p < 0.001$, with a 95% confidence interval of (0.402, 0.502). This result confirms the findings from Schwind and Henze [372] that social acceptance ratings correlate with combined competence and warmth data.

Stereotype Content Model (SCM) The Shapiro–Wilk normality test indicated no normal distribution (only selected SCM competence data showed $p \succeq 0.05$, while all other stimuli for SCM competence and warmth had $p < 0.05$). Therefore, a non-parametric two-factorial ART RM-ANOVA was conducted to determine the effects of Design Type and Orthotic Category on SCM competence and warmth.

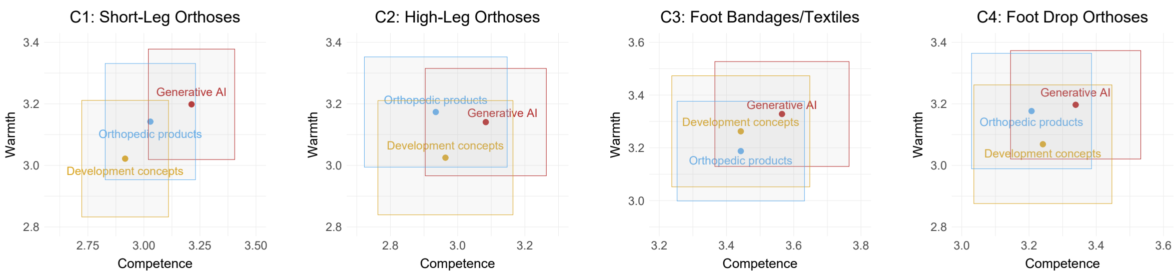
For the SCM competence data, statistically significant main effects were found for the Orthotic Category, $F(3, 869) = 56.752$, $p < 0.001$, $\eta_p^2 = 0.464$ (large effect size), and Design Type, $F(2, 869) = 9.931$, $p < 0.001$, $\eta_p^2 = 0.092$ (medium effect size). However, no interaction effect was found between Orthotic Category $\times$ Design Type, $F(6, 869) = 1.038$, $p = 0.399$, $\eta_p^2 = 0.031$ (small effect size).

For the SCM warmth data, statistically significant main effects were identified for the Orthotic Category, $F(3, 869) = 3.201$, $p = 0.023$, $\eta_p^2 = 0.278$, and Design Type, $F(2, 869) = 4.493$, $p = 0.012$, $\eta_p^2 = 0.265$, both with large effect sizes. However, the interaction effect of the Orthotic Category $\times$ Design Type was not significant, with $F(6, 869) = 1.055$, $p = 0.388$, and $\eta_p^2 = 0.202$ (large effect size).

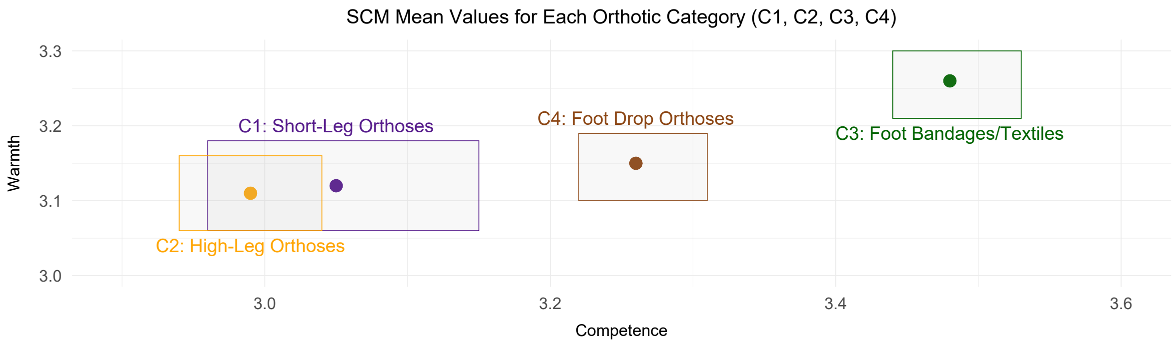
Post hoc comparisons using Wilcoxon signed-rank tests with Bonferroni correction revealed significant effects in SCM competence and warmth data, as documented in Table 6.1. Figure 6.6 displays the mean values of perceived warmth and competence for each of the three Design Types within each of the four Orthotic Categories. Additionally, the figure also presents a comparison across the four Orthotic Categories within the SCM model to highlight differences in the stereotypical perceptions of these devices.

Table 6.1: Pairwise comparisons of SCM competence and warmth ratings within orthotic categories and design types. Significance codes: * ($p < .05$), ** ($p < .01$), *** ($p < .001$), **** ($p < .0001$), ns = not significant.

			SCM Competence				SCM Warmth			
Pairwise Comparisons	**n1**	**n2**	***Stat.***	p	$p_{adj.}$	***Sig.***	***Stat.***	p	$p_{adj.}$	***Sig.***
Short Orth.—High Orth.	240	240	10094	.057	.343	ns	6820	.612	1	ns
Short Orth.—Bandage/Tex.	240	240	2997	< .001	< .001	****	5401	.005	.03	*
Short Orth.—Foot Drop Orth.	240	240	5367	< .001	< .001	****	6286	.993	1	ns
High Orth.—Bandage/Textile	240	240	2846	< .001	< .001	****	5724	.002	.013	*
High Orth.—Foot Drop Orth.	240	240	4640	< .001	< .001	****	5904	.360	1	ns
Bandage/Tex.—Foot Drop O.	240	240	11938	< .001	< .001	****	7802	.044	.265	ns
Gen. AI—Development	320	320	22421	< .001	< .001	****	15445	< .001	< .001	***
Gen. AI—Orthopedic prod.	320	320	21927	< .001	< .001	****	13268	.191	.573	ns
Development—Orth. product	320	320	16596	.928	1	ns	9968	.003	.008	**



(a) (b) (c) (d)



(e)

Figure 6.6: SCM ratings of perceived competence (x-axis) and warmth (y-axis) for each of the four Orthotic Categories. In the upper row, the four quadrants are assigned as follows, from left to right: (**a**) *Short-Leg Orthoses*, (**b**) *High-Leg Orthoses*, (**c**) *Foot Bandages/Textiles*, and (**d**) *Foot Drop Orthoses*. The mean values for the three Design Types are shown in separate colors: red (*Generative AI*), blue (*Orthopedic products*), and yellow (*Development concepts*). The lower quadrant (**e**) contains the SCM mean values for each Orthotic Category: *Short-Leg Orthoses* (purple), *High-Leg Orthoses* (orange), *Foot Bandages/Textiles* (brown), and *Foot Drop Orthoses* (green). The rectangles represent the 95% confidence interval.

Wearable Acceptability Range (WEAR) The Shapiro–Wilk test for normality indicated that the data are not normally distributed ($p = 0.003$). An ART RM-ANOVA showed statistically significant main effects for the ORTHOTIC CATEGORY, $F(3, 869) = 39.949$, $p < 0.001$, $\eta_p^2 = 0.423$ (large effect size), and DESIGN TYPE, $F(2, 869) = 17.677$, $p < 0.001$, $\eta_p^2 = 0.178$ (large effect size), but no interaction effect for ORTHOTIC CATEGORY × DESIGN TYPE, $F(6, 869) = 1.372$, $p = 0.223$, $\eta_p^2 = 0.048$ (small effect size).

Bonferroni corrected pairwise comparisons were conducted using Wilcoxon signed-rank tests and revealed significant effects between *Generative AI* × *Development concepts* ($p_{\text{adj.}} < 0.001$), *Generative AI* × *Orthopedic products* ($p_{\text{adj.}} < 0.001$), and *Development concepts* × *Orthopedic products* ($p_{\text{adj.}} = 0.021$). Furthermore, pairwise comparisons between ORTHOTIC CATEGORIES revealed five significant effects, which are summarized in Figure 6.7.

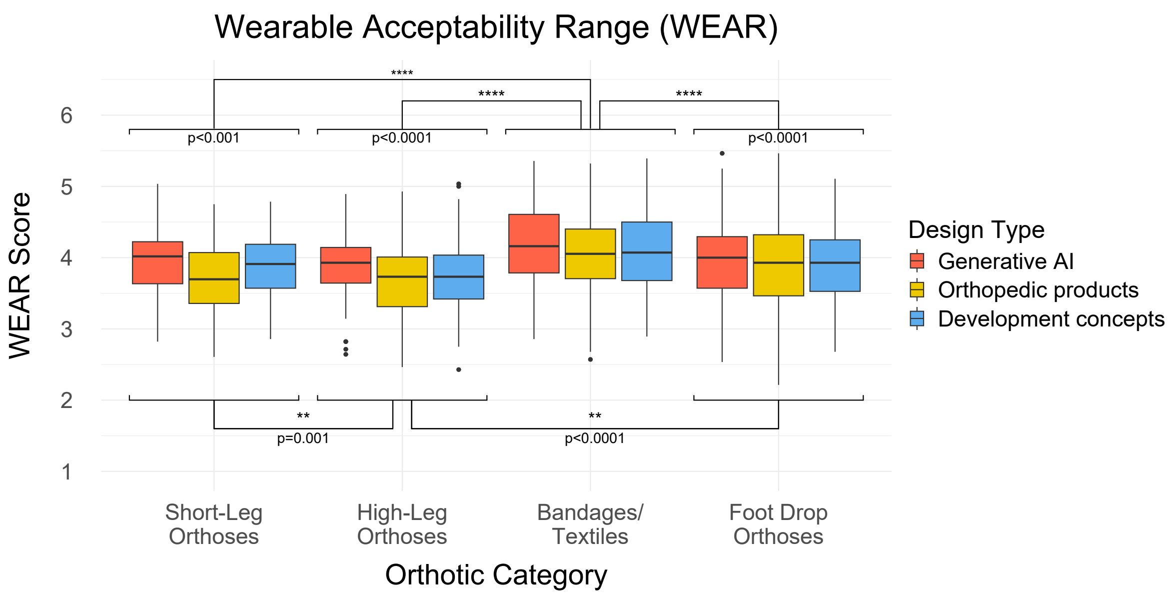


Figure 6.7: This graph shows the results of the WEAR ratings. The four categories are listed on the x-axis: *short-leg orthoses*, *high-leg orthoses*, *foot supports/textiles* and *foot drop orthoses*, and the y-axis represents the corresponding scores on the WEAR scale. The boxplots are color-coded to differentiate the design types: *Generative AI* in red, *Orthopedic products* in yellow, and *Development concepts* in blue. Asterisks indicate significance levels: * ($p < .05$), ** ($p < .01$), *** ($p < .001$), **** ($p < .0001$).

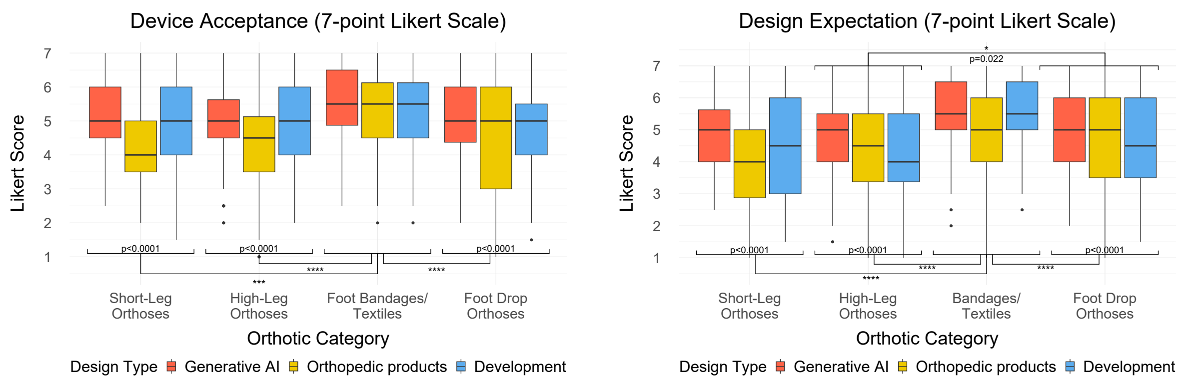
Seven-Point Likert Scale The evaluations of the two statements regarding device acceptance and aesthetic design expectations were not normally distributed ($p < 0.05$).

A non-parametric two-factorial ART-ANOVA on rated device acceptance statements revealed a significant main effect for the ORTHOTIC CATEGORY, $F(3, 869) = 24.717$, $p < 0.001$, $\eta_p^2 = 0.336$, and DESIGN TYPE, $F(2, 869) = 24.077$, $p < 0.001$, $\eta_p^2 = 0.247$, as well as a significant interaction effect for ORTHOTIC CATEGORY × DESIGN TYPE, $F(6, 869) = 4.068$, $p < 0.001$, $\eta_p^2 = 0.142$ (each with a large effect size).

Furthermore, the two-factorial ART-ANOVA on rated aesthetic design expectations showed a significant main effect for the Orthotic Category, $F(3, 869) = 43.730$, $p < 0.001$, $\eta_p^2 = 0.407$, and Design Type, $F(2, 869) = 21.613$, $p < 0.001$, $\eta_p^2 = 0.185$ (both with large effect size), as well as a significant interaction effect for Orthotic Category × Design Type, $F(6, 869) = 32.743$, $p = 0.012$, and $\eta_p^2 = 0.079$ (medium effect size). Pairwise comparisons using Bonferroni-corrected Wilcoxon signed-rank tests revealed statistically significant differences within Orthotic Categories and Design Types, as documented in Table 6.2. The results of the Likert scale evaluations are displayed in Figure 6.8.

Table 6.2: Post hoc pairwise comparisons within orthotic categories and design types for Likert scores of device acceptance and design expectations. Significance codes: * ($p < .05$), ** ($p < .01$), *** ($p < .001$), **** ($p < .0001$), ns = not significant.

			Device Acceptance				Design Expectation			
Pairwise Comparisons	**n1**	**n2**	***Stat.***	p	$p_{adj.}$	***Sig.***	***Stat.***	p	$p_{adj.}$	***Sig.***
Short Orth.—High Orth.	240	240	7901	.148	.888	ns	7804	.771	1	ns
Short Orth.—Bandage/Tex.	240	240	4630	< .001	< .001	****	2854	< .001	< .001	****
Short Orth.—Foot Drop Orth.	240	240	7456	.568	1	ns	7370	.024	.143	ns
High Orth.—Bandage/Textile	240	240	3583	< .001	< .001	****	3426	< .001	< .001	****
High Orth.—Foot Drop O.	240	240	7032	.712	1	ns	5688	.004	.022	*
Bandage/Tex.—Foot Drop O.	240	240	11454	< .001	< .001	****	13938	< .001	< .001	****
Gen. AI—Development	320	320	21972	< .001	< .001	****	24376	< .001	< .001	****
Gen. AI—Orthopedic prod.	320	320	13612	.237	.711	ns	19850	< .001	< .001	****
Development—Orth. product	320	320	7405	< .001	< .001	****	12490	.004	.011	*



(a) (b)

Figure 6.8: This figure contains two boxplot diagrams, including the rated device acceptance (**a**) and aesthetic design expectation (**b**) for the three Design Types in each Orthotic Category. The x-axis lists the four categories: *Short-Leg Orthoses*, *High-Leg Orthoses*, *Foot Bandages/Textiles*, and *Foot Drop Orthoses*. The y-axis represents the corresponding scores from the Likert rating. The boxplots are color-coded to distinguish the design types: *Generative AI* in red, *Orthopedic products* in yellow, and *Development concepts* in blue. Asterisks indicate significance levels: * ($p < .05$), ** ($p < .01$), *** ($p < .001$), **** ($p < .0001$).

Qualitative Feedback The axial coding procedure within inductive thematic analysis revealed three main themes: *Aesthetic Design*, *Technical Properties*, and *Usability and Comfort*. Each theme includes both positive and negative perspectives and reflects participant feedback in four Orthotic Categories and three Design Types.

Overall, perceptions of all four Orthotic Categories vary strongly concerning design, technical aspects, usability, and comfort. In general, the Orthotic Categories of *Short-Leg Orthoses* and *High-Leg Orthoses* are often associated with serious injuries. *Short-Leg Orthoses* focus on “stabilization” (P83, P56) and “facilitates mobility for users with lower leg injuries” (P110) but offer limited practicality in daily life. *High-Leg Orthoses* offer “robust protection for serious injuries” (P27, P101, P110), but are “bulky” (P5, P24, P26, P81, P163), “heavy” (P7, P22, P32, P119) and “restrictive” (P39, P83, P113), so their design “looks like a ski boot” (P144). Participants mentioned that these designs “could not be worn with regular shoes” (P75), “leading to clothing restrictions” (P101). In contrast, *Foot Bandages/Textiles* are perceived as “sporty” (P26, P39), “modern” (P34, P156), and “simple” (P56, P84) because of their “unobtrusive design” (P8) and “can be worn with shoes” (P13, P46), but they are also viewed as less suitable for serious injuries. Lastly, *Foot Drop Orthoses* have a “functional design” (P161) which is “not conspicuous” (P56, P84, P163) and directly addresses medical needs but is also seen as “too mechanical” (P8, P156), “not stable” (P21, P53, P73), and “old-fashioned” (P13, P22).

Aesthetic Design: The aesthetic perception varies considerably across the three Design Types. The designs of *Generative AI* were perceived as “simple and unobtrusive” (P26, P56, P66), with “subtle colors” (P43, P81), a “clean” (P156) and “modern design [that] could be accepted as a replacement for shoes” (P75), which in some cases appear “futuristic” (P66). However, some participants mentioned that the designs look “too technical” (P63) because “the box looks too extreme” (P156). *Orthopedic products* were associated with a “minimalistic” (P63, P142) and “classic design” (P8, P26), characterized by “neutral colors” (P81) that “looks like a conventional product” (P162). However, these designs were criticized for being visually “unesthetic” (P16, P56, P63)—described as “old-fashioned” (P13), “unfashionable” (P161), “crude” (P107), “strange” (P13, P39), and “too big” (P32, P66)—and therefore perceived as “less socially acceptable” (P16) because of their “robotic appearance” (P32, P119, P144). The medical appearance of some orthoses could affect the wearer’s self-confidence because the “impairment attracts attention” (P43). The *Development concepts* were generally recognized as “innovative” (P75), “stylish” (P144), “futuristic” (P13, P34, P60, P64, P156), and “it is nice that the color is different from other design” (P130), which makes them appear “like a fashion item” (P13) and “do not look like a medical product” (P22). In contrast, the bright color was also perceived negatively, along with the “cheap-looking quality” (P63, P66, P72) and “ugly holes” (P84). Additionally, some designs were criticized for their overly “mechanical appearance” (P8, P156) and the “excessive size of the box” (P63), which could lead to “aesthetic acceptance issues” (P24).

Technical Properties: The technical properties of *Generative AI* designs were recognized for “technology integration” (P152) as an “electronic device” (P119). However, some of these technical functions were also negatively considered for their “complexity and the need for regular maintenance or charging” (P152). *Orthopedic products* are recognized for "longevity and low-maintenance" (P109), which "provides strong support and stability for the foot and ankle" (P120) because they "immobilize the ankle and foot" (P27), making them reliable for medical use and effective in the rehabilitation of severe injuries. Although the *Development concepts* featured "lighter" (P22, P25) and "simpler designs" (P5, P161) for "improved mobility" (P110), they were "not stable" (P53, P160) enough to support more serious medical needs. Participants expressed concerns about "durability under heavy loads" (P110), "lack of strength/stiffness" (P101), and the technical efficiency of these designs.

Usability and Comfort: The usability and comfort of the three DESIGN TYPES resulted in contrary user reactions, revealing strengths and weaknesses in product design. *Generative AI* designs faced criticism for certain elements, such as "visible toes" (P39, P72), and raised concerns about "the longevity of materials ... [because the] material could wear out more quickly" (P152). Despite the technical advantages of the *Orthopedic products*, they "can be bulky" (P141) "which can be uncomfortable during everyday activities" (P152), and can reduce comfort over time because it "restricts freedom of movement" (P101). Although some models are described as "comfortable" (P142, P162), there is criticism due to insufficient air circulation because the "feet could sweat" (P84) which can lead to "moisture and skin irritation" (P113). In contrast, it was noted that *Development concepts* are "easy to wear" (P164) and "suitable for everyday use" (P22), offering a practical and "less intrusive" (P8, P161) alternative. However, participants also mentioned that it "can be uncomfortable during long-term use" (P110).

Study 2

First, the study group characteristics are described. Next, the quantitative analysis is presented, including the correlation of SCM and WEAR ratings, SCM ratings, WEAR data, and the analysis of the two seven-point Likert scale ratings. Finally, the thematic analysis of the qualitative responses is reported.

Participants In total, 54 out of 133 participants (27 female, 26 male, and 1 diverse) completed the survey and were included in the data analysis. The age of the participants ranged from 18 to 49 years (M = 23.30, SD = 5.02). The participants had different educational backgrounds and came from nine countries. The participants' occupations varied from (N = 1) in healthcare; (N = 2) each in other services and natural sciences; (N = 3) each in administration, social services, and management; (N = 5) in law; (N = 16) inother professions; and (N = 19) in engineering. Twenty-six participants reported having foot conditions; twenty had conditions in the past and six have current conditions. Of these, 17 were acute diseases and nine were chronic conditions. Eight of them used insoles, seven used bandages, three used a low-leg orthosis, two used a plaster cast, two used a low-leg orthosis, and one used a high-leg orthosis. A total of 20 have already used image-generative AI, 19 participants have no experience with such tools, and 15 are only familiar with the tools from hearsay.

Similarly to our first study, we conducted an ART ANOVA to examine the influence of demographic factors. The results showed no statistically significant main or interaction effects on SCM, WEAR, and Likert scores (all $p > 0.05$), indicating that demographic variables did not influence participant ratings.

Quantitative Results

Correlation of SCM and WEAR Ratings The Pearson correlation analysis showed a statistically significant correlation between the Euclidean length of the SCM warmth-competence vector (means of warm and competence data) and the WEAR scores: $r(430) = 0.800$, $p < 0.001$, with a 95% confidence interval of (0.763, 0.831).

Stereotype Content Model (SCM) The Shapiro–Wilk normality test indicated a significant deviation from normality for SCM competence and warmth (both $p < 0.0001$). A non-parametric two-factorial ART RM-ANOVA for the SCM competence data showed a statistically significant main effect for GPT, $F(1, 371) = 124.871$, $p < 0.0001$, $\eta_p^2 = 0.448$, and for Keyword, $F(3, 371) = 8.499$, $p < 0.0001$, $\eta_p^2 = 0.142$ (both with large effect size). However, no significant interaction effect was found for GPT $\times$ Keyword, $F(3, 371) = 1.198$, $p = 0.310$, $\eta_p^2 = 0.023$ (small effect size). For the SCM warmth data, a significant main effect was identified for GPT, $F(1, 371) = 62.461$, $p < 0.0001$, and $\eta_p^2 = 0.471$ (large effect size). However, no significant effect was found for Keyword, $F(3, 371) = 1.282$, $p = 0.280$, $\eta_p^2 = 0.052$, nor for the interaction effect of GPT $\times$ Keyword, $F(3, 371) = 1.267$, $p = 0.285$, $\eta_p^2 = 0.051$ (both with a small effect size). Post hoc comparisons using Wilcoxon signed-rank tests with Bonferroni correction revealed significant effects in SCM competence and warmth data, as documented in Table 6.3. The mean values of perceived warmth and competence are depicted in Figure 6.9.

Table 6.3: Pairwise comparisons of SCM competence and warmth ratings within keywords and GPTs. Significance codes: * ($p < .05$), ** ($p < .01$), *** ($p < .001$), **** ($p < .0001$), ns = not significant.

			SCM Competence				**SCM Warmth**			
Pairwise Comparisons	**n1**	**n2**	***Stat.***	***p***	***p_adj.***	***Sig.***	***Stat.***	***p***	***p_adj.***	***Sig.***
No Keyword—Usability	108	108	1482	.312	1	ns	868	.337	1	ns
No Keyword—Social Acceptab.	108	108	1943	.482	1	ns	1331	.605	1	ns
No Keyword—Sporty Design	108	108	748	< .001	< .001	****	850	.146	.876	ns
Usability—Social Acceptability	108	108	2135	.075	.450	ns	1508	.186	1	ns
Usability—Sporty Design	108	108	725	< .001	< .001	****	914	.221	1	ns
Social Acceptab.—Sporty Design	108	108	764	< .001	< .001	****	854	.107	.642	ns
No Keyword: ChatGPT—OrthoticFootGPT	54	54	942	.002	.002	**	1135	.046	.046	*
Usability: ChatGPT—OrthoticFootGPT	54	54	828	< .001	< .001	***	994	.004	.004	**
Social Accept.: ChatGPT—OrthoticFootGPT	54	54	777	< .001	< .001	****	875	< .001	< .001	***
Sporty Design: ChatGPT—OrthoticFootGPT	54	54	830	< .001	< .001	***	1069	.016	.016	*

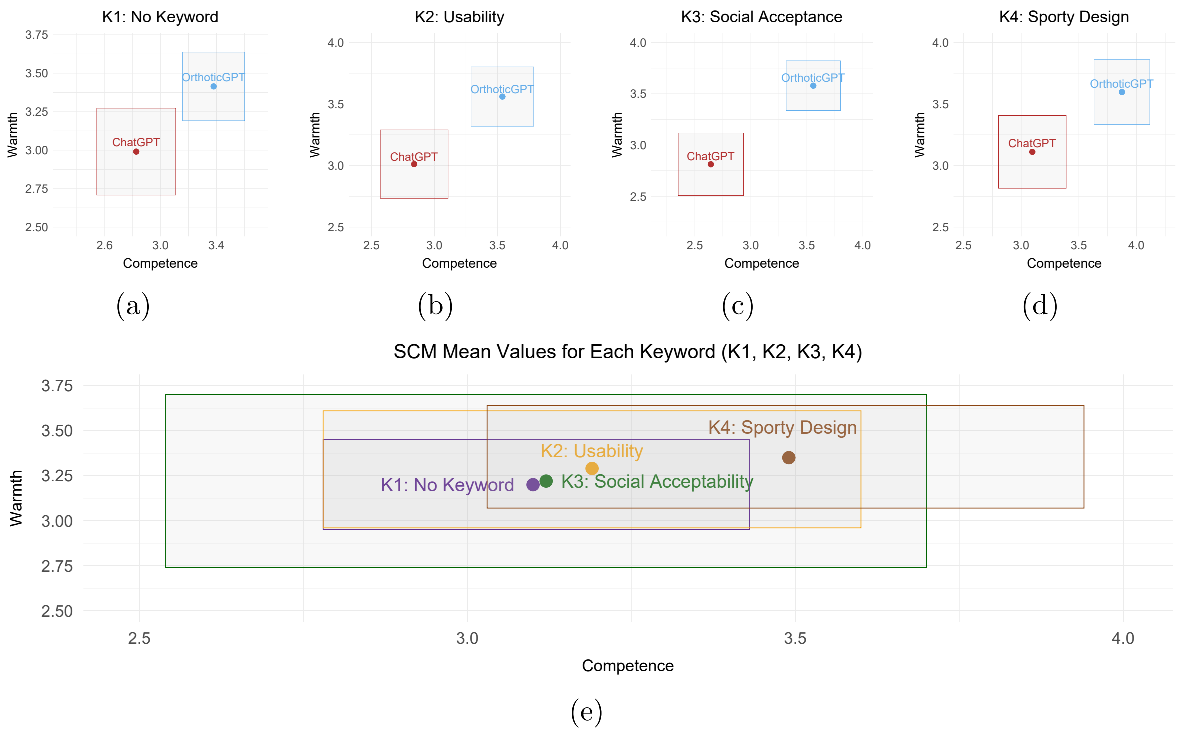


Figure 6.9: SCM ratings of perceived competence (x-axis) and warmth (y-axis) for each of the four KEYWORDS. In the upper row, the four quadrants are assigned as follows, from left to right: (**a**) *No Keyword*, (**b**) *Usability*, (**c**) *Social Acceptability*, and (**d**) *Sporty Design*. The mean values for two GPTs are shown in separate colors: red (*ChatGPT*) and blue (*OrthoticFootGPT*). The lower quadrant (**e**) contains the SCM mean values for each KEYWORD: *No Keyword* (purple), *Usability* (orange), *Social Acceptability* (green), and *Sporty Design* (brown). The rectangles represent the 95% confidence interval.

Wearable Acceptability Range (WEAR) The Shapiro–Wilk normality test indicated that the data is not normally distributed ($p < 0.0001$). An ART RM-ANOVA showed statistically significant main effects for the GPT, $F(1, 371) = 109.325$, $p < 0.0001$, $\eta_p^2 = 0.450$ (large effect size), and for KEYWORD, $F(3, 371) = 6.270$, $p = 0.0003$, $\eta_p^2 = 0.123$ (medium effect size). However, no interaction effect was found for GPT $\times$ KEYWORD, $F(3, 371) = 1.948$, $p = 0.121$, and $\eta_p^2 = 0.042$ (small effect size).

Bonferroni-corrected pairwise comparisons were conducted using Wilcoxon signed-rank tests, which revealed statistically significant differences between *No Keyword* $\times$ *Sporty Design* ($p_{\text{adj}} < 0.0001$), *Usability* $\times$ *Sporty Design* ($p_{\text{adj}} = 0.0002$), and *Social Acceptability* $\times$ *Sporty Design* ($p_{\text{adj}} < 0.0001$). The pairwise comparison between *ChatGPT* and *OrthoticFootGPT* was statistically significant, with $p < 0.0001$. Pairwise comparisons between the GPT and KEYWORD demonstrated statistically significant differences across all keywords: *No Keywords* ($p_{\text{adj.}} = 0.0004$), *Usability* ($p_{\text{adj.}} = 0.0002$), *Social Acceptability* ($p_{\text{adj.}} < 0.0001$), and *Sporty Design* ($p_{\text{adj.}} = 0.009$). The results of the WEAR scale are summarized in Figure 6.10.

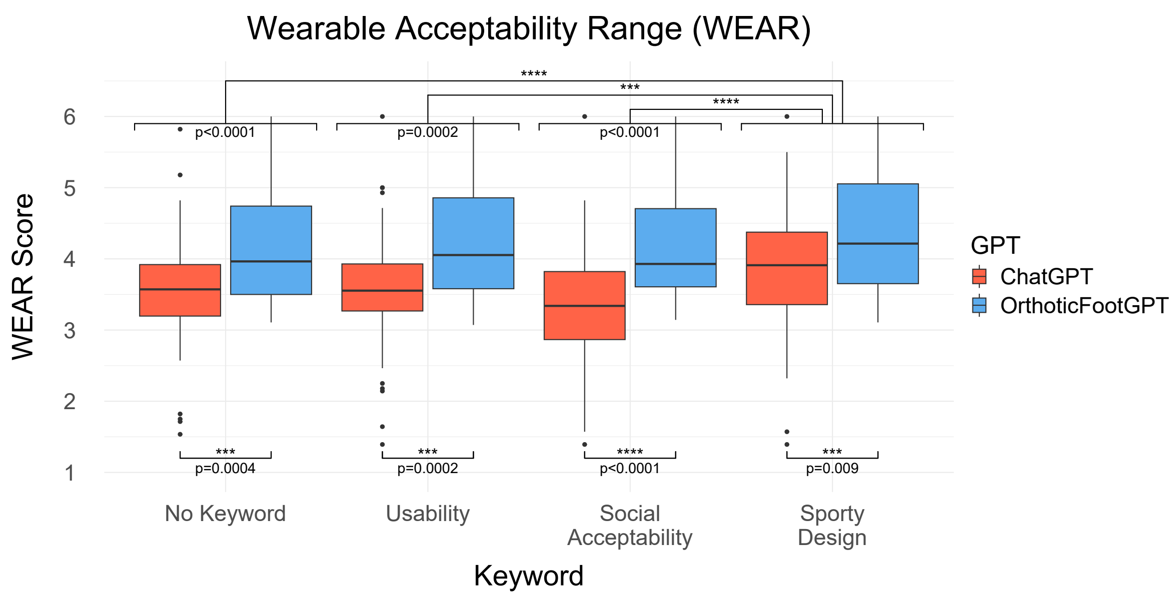


Figure 6.10: This graph shows the results of the WEAR ratings. The four Keywords are listed on the x-axis, *No Keyword*, *Usability*, *Social Acceptability*, and *Sporty Design*, and the y-axis represents the corresponding scores on the WEAR scale. The boxplots are color-coded to distinguish the GPT: *ChatGPT* in red and *OrthoticFootGPT* in blue. Asterisks indicate significance levels: * ($p < .05$), ** ($p < .01$), *** ($p < .001$), **** ($p < .0001$).

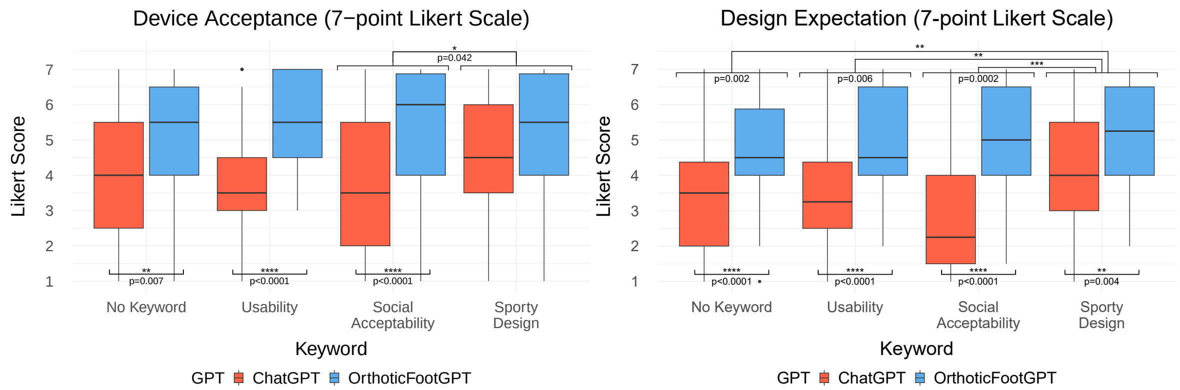
Seven-Point Likert Scale The Shapiro–Wilk normality test for the two statements on device acceptance and aesthetic design expectations indicated no normal distribution ($p < 0.0001$).

A non-parametric two-factorial ART-ANOVA on rated device acceptance statements revealed a significant main effect for the GPT, $F(1, 371) = 90.864$, $p < 0.0001$, $\eta_p^2 = 0.442$ (large effect size), and for Keyword, $F(3, 371) = 3.081$, $p = 0.027$, $\eta_p^2 = 0.074$ (medium effect size), as well as an interaction effect for GPT × Keyword, $F(3, 371) = 4.834$, $p = 0.002$, and $\eta_p^2 = 0.111$ (medium effect size).

The two-factorial ART-ANOVA on rated aesthetic design expectations showed a significant main effect for GPT, $F(1, 371) = 129.196$, $p < 0.0001$, $\eta_p^2 = 0.442$, and Keyword, $F(3, 371) = 7.397$, $p < 0.0001$, $\eta_p^2 = 0.119$ (both with a large effect size), as well as a significant interaction effect for GPT × Keyword, $F(3, 371) = 4.021$, $p = 0.007$, and $\eta_p^2 = 0.068$ (medium effect size). Post hoc pairwise comparisons using the Wilcoxon signed-rank test with Bonferroni correction revealed statistically significant differences and notable trends in both device acceptance and design expectation data, as documented in Table 6.4. The results are presented in Figure 6.11.

Table 6.4: Post hoc pairwise comparisons within keywords and GPTs for Likert scores of device acceptance and design expectation. Significance codes: * ($p < .05$), ** ($p < .01$), *** ($p < .001$), **** ($p < .0001$), ns = not significant.

			Device Acceptance				Design Expectation			
Pairwise Comparisons	**n1**	**n2**	***Stat.***	p	$p_{adj.}$	***Sig.***	***Stat.***	p	$p_{adj.}$	***Sig.***
No Keyword—Usability	108	108	1382	.548	1	ns	1545	.282	1	ns
No Keyword—Social Acceptab.	108	108	1382	.199	1	ns	1833	.304	1	ns
No Keyword—Sporty Design	108	108	1110	.066	.398	ns	1093	< .001	.002	**
Usability—Social Acceptability	108	108	1529	.325	1	ns	2022	.030	.179	ns
Usability—Sporty Design	108	108	732	.011	.064	ns	934	< .001	.006	**
Social Acc.—Sporty Design	108	108	1030	.007	.042	*	836	< .001	< .001	***
No Keyword: ChatGPT—OrthoticFootGPT	54	54	1022	.007	.007	**	819	< .001	< .001	****
Usability: ChatGPT—OrthoticFootGPT	54	54	692	< .001	< .001	****	712	< .001	< .001	****
Social Accept.: ChatGPT—OrthoticFootGPT	54	54	744	< .001	< .001	****	540	< .001	< .001	****
Sporty Design: ChatGPT—OrthoticFootGPT	54	54	1212	.129	.129	ns	997	.004	.004	**



(a) (b)

Figure 6.11: This figure contains two boxplot diagrams, including the rated device acceptance (**a**) and aesthetic design expectation (**b**) for the four KEYWORDS in each GPT. The x-axis lists the four Keywords: *No Keyword*, *Usability*, *Social Acceptability*, and *Sporty Design*. The y-axis represents the corresponding scores from the Likert rating. The boxplots are color-coded to distinguish the GPT: *ChatGPT* in red and *OrthoticFootGPT* in blue. Asterisks indicate significance levels: * ($p < .05$), ** ($p < .01$), *** ($p < .001$), **** ($p < .0001$).

Qualitative Feedback An inductive thematic analysis on transcribed data revealed three main themes: *Aesthetic Design*, *Functionality*, as well as *Comfort and Fit*. These three themes include both positive and negative perspectives and reflect the participants' feedback on the two GPTs and four KEYWORDS.

Aesthetic Design: The designs with *No Keyword* were generally described as "standard" (P141), "simple" (P56), and "not visually appealing" (P61) when generated by *ChatGPT*. In contrast, designs generated by *OrthoticFootGPT* were characterized as "modern" (P83, P90), "aesthetic" (P9), and "compact" (P60), with a "positive shape and design" (P74) resembling a "mixture of a boot and a sports shoe" (P102). For the keyword *Usability*, *ChatGPT* designs were recognized as "similar to shoes" (P33) with a "good color" (P14) but were also criticized for being "old-fashioned" (P21) and "unaesthetic" (P61). Meanwhile, *OrthoticFootGPT* designs were praised as "aesthetic" (P56), "modern and sporty" (P21), and also "similar to shoes" (P41, P69). Furthermore, one participant stated that "the eye-catching design ensures that the orthosis is easily recognized by others, which can lead to more consideration from others" (P108). Images generated using the keyword *Social Acceptability* via *ChatGPT* were described as "minimalistic" (P24) with a "positive color" (P14), but their perception was negatively impacted by a "dubious shape" (P21) and "incomprehensible design" (P47). On the other hand, designs from *OrthoticFootGPT* were regarded as "very aesthetic and modern" (P41), "sporty" (P24), and featuring a "nice design" (P83), due to "less conspicuous Velcro fasteners" (P90) and a resemblance to "winter boots" (P159). In general, the *Sporty Design* keyword led to designs perceived as modern and sporty, which were positively received by participants. The *ChatGPT* outputs were described as a "nice and modern design" (P24) contributing to a "self-confident appearance" (P74). In comparison, *OrthoticFootGPT* designs were associated with a "sporty design" (P9), resembling "sport shoes" (P74, P102, P138) and perceived as designs that "remind one of a fashion brand" (P89). It was also positively noted that "the toes are not visible" (P94).

Functionality: The use of *No Keyword* with *ChatGPT* raised concerns about functionality, as the designs were perceived as "not effective" (P157) and "not stable" (P90, P150) due to the "missing insole" (P9, P14) and because they "offer less protection (e.g., no padding)" (P36). In contrast, the *OrthoticFootGPT* designs "offer good stability and fulfill [their] functional requirements effectively" (P61) because they are "easy to attach and remove" (P89, P90, P152). For the keyword *Usability*, *ChatGPT* designs were criticized for being "too mechanical" (P150) and "complicated to use" (P105, P111, P149). Meanwhile, *OrthoticFootGPT* outputs were associated with "simple handling" (P36, P56), due to "a good balance of stability and comfort" (P36) and because "it also gives the feeling that it has very useful functions" (P21). In the context of *Social Acceptability*, *ChatGPT* designs were described as "too bulky, seems heavy" (P14), "unrealistic" (P60), and "impractical" (P37), primarily due to "too many straps" (P9, P89) and "components are mainly made of textile. Water ... can be absorbed" (P79). In contrast, *OrthoticFootGPT* designs were perceived as "very innovative" (P93), featuring "a good sole, similar to a shoe" (P14), making them an "eye-catcher for everyone" (P21). Regarding *Sporty Design*, *ChatGPT* was noted for its "functional and rather simple design ... suitable for a broad target group" (P90). Similarly, *OrthoticFootGPT* designs were defined as "robust" (P94), "stable" (P36, P105), and "breathable" (P33), with functionality that is "easy to integrate for sports activities" (P89).

Comfort and Fit: The primary issues with *ChatGPT* designs using *No Keyword* were identified as being “uncomfortable” (P21) and “impractical due to open toes” (P37), concerns that were similarly reflected in the keywords *Usability* and *Social Acceptability*. In contrast, the *OrthoticFootGPT* designs with *No Keyword* were perceived more positively, as “the orthosis is easily adjustable and can be customized” (P61) and it “has good padding on the foot” (P36). For the keyword *Usability*, the *OrthoticFootGPT* was highlighted for being ”easy to use” (P36) and it “is light and comfortable to wear, even for long periods” (P61), which could be “suitable for thicker calves” (P74). In the context of *Social Acceptability*, designs generated by *OrthoticFootGPT* were described as resembling a “sport shoe” (P21) or “winter boots” (P159), with positive feedback emphasizing that “no toes are visible” (P94). Regarding *Sporty Design*, a participant described the *ChatGPT* output as follows: “It looks like a boot. It does not attract attention and is therefore great for everyday wear. It makes you feel more confident” (P74). Meanwhile, the *OrthoticFootGPT* designs were seen as both “comfortable” (P157) and “sporty” (P47, P60, P92, P101, P117), and they “look like an orthopedic hiking/running/football shoe” (P157) which is “great for social integration” (P74). Furthermore, one participant noted that “the orthosis appears to be easy to put on with the fasteners, which could potentially make everyday life easier for the wearer”, but also mentioned negatively that “the unobtrusive design could lead to people taking less notice of the wearer” (P108).

6.1.4 Discussion

The Impact of AI-Generated and Conventional Orthotic Designs Across Device Categories on Social Acceptability

In our first study, we conducted a mixed-method online survey to explore the impact of three Design Types and four Orthotic Categories on social acceptance. The quantitative analyses of Likert scales, as well as the SCM and WEAR models, indicated that Orthotic Categories such as *Foot Bandages/Textiles* achieved significantly higher acceptance ratings, followed by *Foot Drop Orthoses* and then *Short-Leg* and *High-Leg Orthoses*. Furthermore, the *Generative AI* Design Type had a significant impact on perceived acceptability, as these devices were associated with stereotypes of greater competence and warmth compared to medical or developmental designs, which confirms our hypothesis. These findings were supported by the qualitative feedback, which highlighted the potential of using *Generative AI* to consider user preferences and increase acceptance of orthopedic footwear.

Overall, these results confirm the potential of integrating AI into the design process for wearable foot devices. However, the significantly lower acceptance rates for short and high-leg orthoses represent a consistent challenge that reflects the concerns reported in previous research. By identifying the challenges of user acceptance associated with different categories of orthoses, this study contributes to a deeper understanding of the factors that influence the social acceptance of orthotic devices. Our findings are valuable for future developments aimed at enhancing the aesthetic design of these devices to better meet user needs.

The Impact of GPT Customization and Prompt Keywords on the Social Acceptability of AI-Generated Orthotic Designs

The second study examined the impact of high-leg orthosis designs generated using *OrthoticFootGPT* and *ChatGPT*, as well as the influence of four different prompt structures incorporating specific keywords on social acceptability. This extends the results of the first study to explore if a personalized GPT and specific KEYWORDS can have a systematic positive influence on user perception. The quantitative results indicate that the customized *OrthoticFootGPT* model can enhance acceptance ratings compared to *ChatGPT*. Additionally, the study showed that different keywords evoke different stereotypical perceptions of the wearable device. In particular, the keyword *Sporty Design* led to significantly better ratings compared to other keyword variations, suggesting that aesthetic framing plays a crucial role in user perception. These findings are consistent with the qualitative feedback, which revealed a discrepancy between the perception of medical orthoses and more socially integrated designs, such as those inspired by the aesthetics of footwear or sports. Participants expressed contrasting preferences regarding the visual appearance of orthotic designs: while some preferred an inconspicuous design to avoid attracting attention, others emphasized that a more visible, eye-catching design could facilitate social recognition and consideration by others. This contrast illustrates the complexity of the relationship between aesthetics as well as functional and social expectations.

These results highlight the potential of generative AI to enhance the design process of wearable medical devices. Furthermore, our results align with previous studies on wearable device perception, emphasizing that design plays a crucial role in acceptance [198, 197]. In addition, these results support earlier findings that highlight a discrepancy in the literature regarding the stigmatization of medical devices, where functional necessity often conflicts with social acceptability, leading to lower acceptance [198].

Similar to findings by Sehrt et al. [24], who observed that wearables positioned on the ankle received lower social acceptance scores, our study showed that high-leg orthoses face significant acceptance challenges compared to other orthotic device categories. By demonstrating that customized GPT models and prompt engineering can systematically shape social acceptability, our study builds on prior work [185, 186] that highlights the importance of structured input in AI-assisted design.

Therefore, we recommend that future developments prioritize the use of customized GPT models in combination with personalized prompting strategies that explicitly address user preferences and social perception factors.

Implications

The results obtained from the two studies conducted show the potential of generative AI to improve the social acceptability of foot wearable devices and demonstrate its applicability in product design. Building on existing work, such as the use of DALL-E for designing hand orthoses [30], our study extends this approach to foot orthoses, with a specific focus on social acceptance factors. The integration of image-generative AI in the design phase can significantly improve acceptance and lead to more user-centered orthosis designs. In particular, the consideration of user preferences with prompting alongside the technical functionalities of wearables offers a promising approach for personalization in product development. Furthermore, the findings from this research extend beyond orthopedic footwear, affecting broader applications within the fields of HCI, User Experience (UX) design, and the integration of social factors in product development. Understanding how AI-generated designs influence product acceptance offers valuable insights for improving design processes across various applications and product categories. For example, designers can integrate user feedback directly into aesthetic and functional design aspects to enhance user engagement and satisfaction. Moreover, this study highlights the potential of generative AI as a valuable tool for inspiring the design of socially driven consumer products and wearable technology. We recommend the use of customized, fine-tuned GPT models to generate user-specific outputs that align with product specifications. Furthermore, incorporating specific keywords related to functionality, target user groups, and design preferences directly within the prompt can enhance the quality and relevance of the generated results.

Beyond improving social acceptability, AI-generated designs also introduce new opportunities for streamlining the product development process. While traditional design approaches rely entirely on manual modeling, which can be time-consuming and expensive, especially when adapting to individual user preferences, AI-generated designs still require manual expert input in the post-processing phase. This includes converting 2D images into 3D data, adapting the design to specific medical requirements, and preparing the data for manufacturing. Related work has demonstrated that integrating AI into the shoe design process can reduce design time by 59% compared to traditional methods while also lowering production costs [29]. These findings indicate that AI-supported workflows have the potential to increase efficiency in orthotic design while maintaining a high degree of individualization.

Limitations

Our studies have some limitations that affect the generalizability of the results. Although our studies included participants from various backgrounds, the lack of gender balance in the first study and the inclusion of healthy individuals in both studies could potentially affect the external validity. However, our analysis showed that demographic factors did not systematically influence participants' ratings, which indicates that the observed effects are not due to the sample population. Nevertheless, a more patient-centered focus could provide deeper insights into the specific needs and perceptions of individuals with orthopedic conditions. While some results did not reach statistical significance, they indicate notable trends that suggest potential effects worth further investigation. Furthermore, the results are limited to the specific stimuli and orthotic categories that were used in the study. Additionally, the AI-generated designs, which rely on individual prompts or selected pictures of orthopedic products, cannot be consistently reproduced with identical inputs. This inconsistency requires multiple iterations of prompts to achieve the desired outcomes.

Another notable limitation is that AI-generated designs do not adhere to any medical guidelines or standards, as they are primarily visual concepts rather than functional prototypes. While the prompts and images were adjusted to the medical functions and specific requirements of the selected orthosis categories, DALL-E image generation is not specialized for medical functions and does not take manufacturing constraints or biocompatibility into account. The results are limited to the perspective of end users and do not consider technical aspects that are relevant from an expert perspective, such as material feasibility or regulatory compliance.

The generative AI tool DALL-E was limited to its current version during both studies. Furthermore, biases in the training datasets of AI models could reinforce stereotypes (e.g., related to ethnicity, gender, or other factors) [373]. These factors are critical as they could influence the social acceptance of the designs, particularly if they fail to address the needs and preferences of various user groups or if they misrepresent them. Therefore, any application of such AI-generated designs in real-world medical products must carefully consider and comply with strict regulatory standards to ensure their safety and effectiveness.

Future Work

The growing field of AI applications will contribute to opening new possibilities for user-centered design solutions. The potential for generating 3D models that are compatible with CAD software could significantly enhance the development process, particularly in the area of rapid prototyping. Future developments should focus on training device-specific GPTs to generate tailored image outputs that meet user needs. Moreover, refining and validating prompt structures will be crucial to achieving more personalized results. Furthermore, to better understand the context and impact of AI-generated designs on social acceptability, future research should extend beyond orthopedic footwear to include diverse product categories. This extension would provide a comprehensive overview of how generative AI can influence product acceptance in different fields. By addressing these areas, follow-up work can make an important contribution to the further development of AI in product design and potentially lead to more effective and socially accepted solutions. Follow-up studies should include a more diverse participant sample, focused on individuals with orthopedic conditions, to better understand user-specific requirements. An exemplary case study could demonstrate how AI-generated design can be integrated into a patient-centric design process that covers the entire workflow, including patient requirements, AI-based design generation, post-processing of 3D models, product personalization based on anatomical needs, manufacturing aspects, and product testing. While long-term biocompatibility testing may be required for regulatory approval, initial usability studies can focus on short-term comfort and usability. Additionally, human intervention by expert designers remains essential to test, validate, and refine AI-generated outputs to ensure practical applicability and manufacturability.

To ensure that AI-generated orthotic designs consider different cultural factors, AI models should be trained on datasets that are representative of diverse user demographics. Additionally, engaging patients with orthopedic conditions in co-design sessions can help align AI-generated outputs with real-life needs and prevent the reinforcement of stereotypes in training data. Involving clinicians, orthopedic experts, and diverse patient representatives in the AI design evaluation process ensures that biased or stereotypical outputs are identified and corrected early. Future research must critically evaluate these biases and develop strategies to ensure that AI-assisted design workflows align with ethical principles and medical standards.

6.1.5 Conclusion

In this paper, we explored the influence of image-generative AI on the social acceptance of footwear designs through two studies involving 134 participants. Our first study demonstrated that AI-generated designs can significantly enhance perceived user acceptance compared to conventional designs. The second study showed that customizing prompts with specific keywords and using a personalized GPT can better tailor these designs to user preferences and therefore improve acceptance. These findings suggest that integrating image-generative AI with personalized prompting strategies in the design process could transform how these devices are perceived and accepted, aligning product functionality with user expectations more effectively. This approach introduces a new framework for user-centric design processes, leveraging AI-driven systems to adapt designs based on specific user inputs. Our findings are not limited to medical wearables and could be extended to various applications and products. Overall, this study enhances our understanding of how AI-generated outputs impact user perception, setting the groundwork for future research to explore new GPT models and expand these findings across different contexts.

6.2. Study II: A Novel AI-Based Framework for Product Development

Building on the findings from Chapter 6.1, which demonstrated the potential impact of generative AI on social acceptability, this study extends these insights by implementing AI within a systematic framework for product development. This section proposes an eight-stage framework for designing aesthetically and functionally optimized devices based on user and expert requirements, addressing the current lack of a structured approach to integrating AI in the design workflow. To address RQ7, a case study was conducted in which a personalized foot wearable device was designed, manufactured, and evaluated through feedback from an end-user and an expert. The results indicate that the framework can achieve desired aesthetic outcomes and is perceived as innovative, while also revealing limitations related to regulatory constraints and practical implementation challenges. This article contributes theoretical and methodological guidance, as well as new insights for integrating generative AI into user-centered product development workflows.

S. Resch, "Designing Customized Wearable Devices: A Novel AI-Based Framework for Product Development," in *9th International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*, Ankara, Turkiye: IEEE, pp. 1–6, 2025.

One conference abstract was published on this topic, serving as a preliminary dissemination of the work to a scientific audience:

S. Resch, L. Damrau, D. Völz, and D. Sanchez-Morillo, "Personalized Foot Orthoses Through a Patient-Centered Design Approach Using Generative AI," *Abstracts of the 2025 Joint Annual Conference of the Austrian (ÖGBMT), German (VDE DGBMT) and Swiss (SSBE) Societies for Biomedical Engineering, Biomedical Engineering / Biomedizinische Technik*, vol. 70, no. s1, pp. 1–374, De Gruyter, 2025.

Additionally, the following book chapter serves as a follow-up publication, addressing the topic from a broader perspective:

D. Völz, J. Schneider, and S. Resch, "Generatives Produktdesign orthopädischer Hilfsmittel: Potenziale und Herausforderungen von KI im Produktentwicklungsprozess," in *Künstliche Intelligenz im Einsatz für die erfolgreiche Patientenreise: Innovation, Integration und Stärkung*, Wiesbaden: Springer Gabler, pp. 219–242, 2025.

6.2.1 Introduction

The development of wearable devices like personalized orthopedic products has to consider individual customization to ensure long-term use and therapeutic effectiveness. Therefore, their development needs to address both aesthetic and functional aspects to achieve high user satisfaction. Traditional design methods rely on manual customization, which is highly time-consuming and requires expert knowledge at every stage of the process. To make personalized solutions more accessible and efficient, new approaches are needed that streamline the design process while preserving individual adaptability.

Previous research has explored user-centered design approaches to improve product personalization [27, 374]. For example, Dabnichki and Pang [27] proposed a framework for personalized ankle-foot orthoses using additive manufacturing. However, this process still requires highly specific user input throughout the entire design process. Furthermore, Slegers et al. [374] compared three workflows for designing and manufacturing custom-made 3D-printed assistive devices, highlighting the value of interdisciplinary design teams consisting of clinicians, clients, and 3D printing experts. They further emphasized that 3D printing aligns well with a user-centered design approach, as it enables the creation of individual solutions tailored to specific user needs. Nevertheless, the process is always based on personal input for the design process.

Recent advancements in the field of AI, particularly in text-to-image generation, offer new opportunities to incorporate AI into the design process, for example to improve the personalization of footwear design [28, 375, 376]. A study by Resch et al. [377] demonstrated that AI-generated orthosis designs using DALL-E from OpenAI can significantly improve perceived social acceptability compared to conventionally designed devices. Additionally, they found that a customized GPT trained with orthopedic data and specific prompting keywords further improved acceptance scores. These results highlight the potential for AI-driven design assistance in personalized medical device development. Moreover, a study by Popescu [30] highlighted the feasibility of AI-assisted hand orthosis design, where AI-generated 2D images were manually converted into 3D models for subsequent additive manufacturing.

Similarly, AI has been used in footwear design, where AI-generated shoe concepts rely on designers and specialized software to transform 2D outputs into functional 3D models [378]. While recent advancements enable direct text-to-3D generation, these outputs often lack the necessary quality and require manual refinement by experts to meet design and manufacturing standards. Taking this a step further, a study by Minaoglou et al. [29] demonstrated that integrating AI into the shoe design process can reduce design time by 59% compared to traditional methods, while also reducing costs. Furthermore, frameworks such as the proposed design process by Sengupta [379] highlight the potential of AI tools to accelerate conceptual design workflows in the fashion footwear domain. However, existing approaches focus predominantly on aesthetic aspects of shoe design, while functional and anatomical considerations, which are essential for orthopedic foot wearables, are not sufficiently addressed. In particular, the methodological integration of AI into the development of personalized orthopedic orthoses, which addresses both medical functionality and individual user needs, remains an open challenge.

This leads to the following research question: *How can a patient-centered AI-driven design framework for personalized foot orthoses be developed to meet both aesthetic and functional requirements while ensuring high user acceptance?*

Therefore, a new methodological approach is introduced that incorporates generative AI tools into the development of personalized wearable devices. The resulting framework aims to fulfill both aesthetic and functional requirements while enhancing patient acceptance.

6.2.2 Method

To implement the proposed framework, an iterative design process was developed and applied for the creation of a customized foot orthosis. The process is structured as an iterative eight-phase workflow that integrates image-based generative AI, digital 3D modeling, and additive manufacturing (see Figure 6.12). The iterative approach allows for modifications at any stage if requirements are not met or further improvements are necessary. Each step of this process is illustrated through a case study in which a personalized orthosis was developed based on patient-specific input.

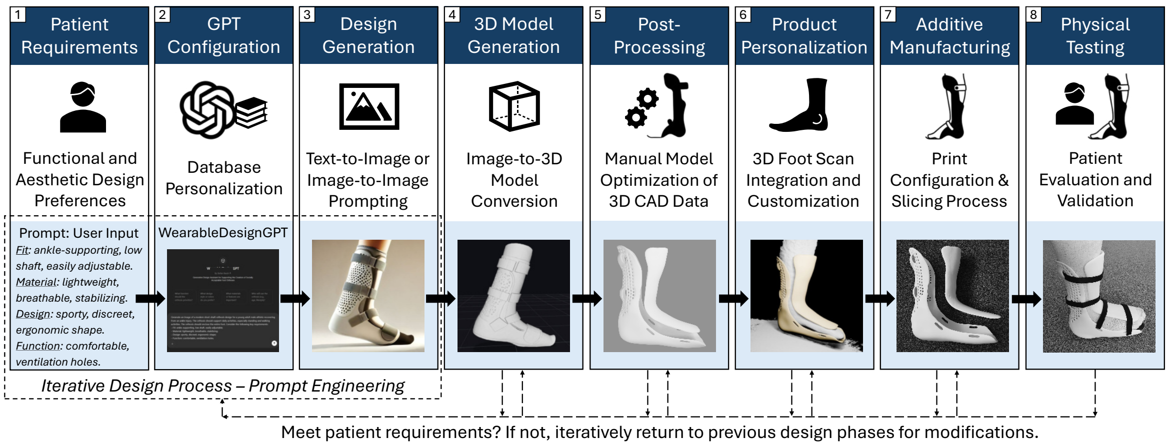


Figure 6.12: Schematic overview of the eight-stage development process for personalized orthopedic devices using generative AI. The framework integrates patient-specific input, AI-assisted design generation, digital modeling, and additive manufacturing. It supports iterative refinement across all stages, from requirements analysis to physical testing.

Initially, individual user needs were systematically analyzed and the resulting requirements were translated into keyword-based prompts for generative design. These prompts were used to generate a visual design concept via a tailored GPT model. The generated concept served as input for 3D digital modeling, which was further refined using CAD software to meet functional and aesthetic criteria. Anatomical personalization was achieved by integrating 3D foot scan data, allowing the digital model to be precisely adapted to the patient's foot morphology. The finalized model was fabricated using 3D printing and the resulting prototype subsequently underwent a physical testing procedure.

The following subsections describe the technical implementation as well as the execution of the physical testing.

GPT Configuration

A custom GPT, named "WearableDesignGPT", was created using the ChatGPT-4 Pro interface from OpenAI (version as of March 2025), to support the generation of image outputs that align with orthopedic design requirements. As part of this configuration, the integrated image generation function (DALL-E) was activated to create visual design concepts. DALL-E utilizes a transformer-based diffusion architecture to generate images based on natural language prompts. The knowledge base of "WearableDesignGPT" was filled with domain-specific documents, including orthosis design guidelines, manufacturer manuals, product specifications (e.g., materials, mechanical properties), and reference images of existing products, as recommended by related works [377, 30]. These resources were provided in PDF and text format, reflecting both clinical standards and commercially available solutions.

Patient Requirements and Design Generation

Medical and aesthetic requirements were previously recorded and documented to address patient needs. Aesthetic preferences include color, design style (whether traditional, sporty, or casual), as well as other aspects of visibility. These requirements are then summarized in the prompt as keywords to achieve suitable design outcomes. Following this, an iterative prompt engineering process is essential to refine the generated results. While earlier versions of ChatGPT generated entirely new image variations upon re-prompting, the latest version enables manual selection of specific areas within an image to modify them based on an updated prompt, preserving the overall concept. This iterative re-prompting approach enables targeted modifications to achieve the desired design solution. Alternatively, image-to-image prompting based on desired product designs or sketches can be used in combination with text-based prompts to guide the creation process.

In this case study, the following prompt was used to design a short-shaft orthosis for a sporty 28-year-old male participant: *"Generate an image of a modern short-shaft orthosis design for a young adult male athlete recovering from an ankle injury. The orthosis should support daily activities, especially standing and walking activities. The orthosis should enclose the entire foot. Consider the following key requirements:*

- *Fit: ankle-supporting, low shaft, easily adjustable;*
- *Material: lightweight, breathable, stabilizing;*
- *Design: sporty, discreet, ergonomic shape;*
- *Function: comfortable, ventilation holes."*

The initial design concept was iteratively refined through four re-prompting steps, resulting in a final design that aligned with the required technical functionalities (confirmed by one expert) and matched participants' preferences. The design process is documented in Figure 6.13.

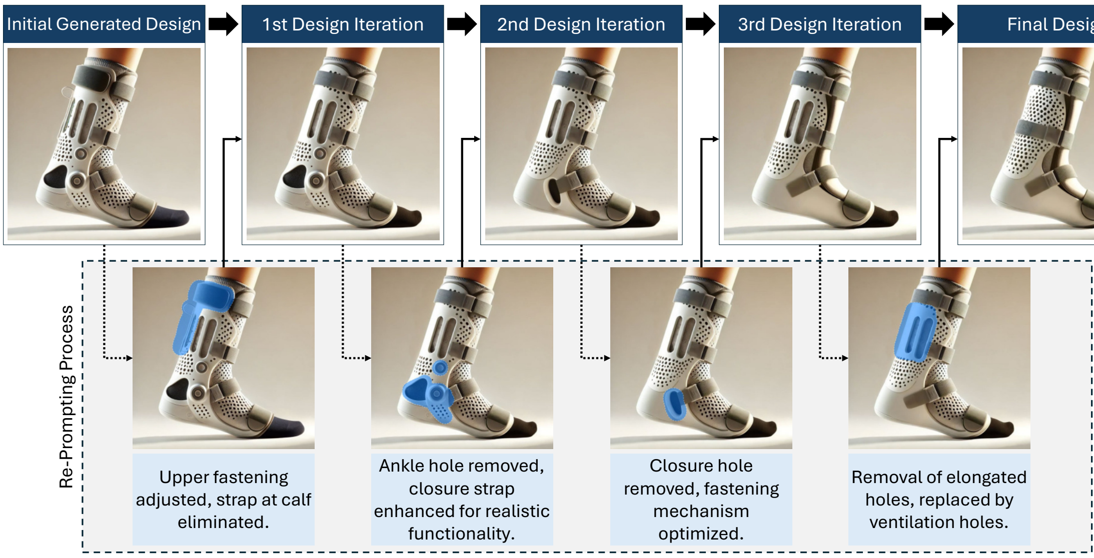


Figure 6.13: Iterative design process of the orthosis using WearableDesignGPT. From left to right: initial design, three targeted modifications, and final design. The lower row illustrates the four-step re-prompting process, involving manual selection of orthosis components for targeted refinement.

3D-Model Generation

The transformation from the 2D-generated image to a virtual 3D model can be achieved through various approaches. Traditionally, this process requires a manual redesign by a designer, which can be time-consuming. Recent advancements in AI have introduced tools that can generate simple 3D objects directly from text prompts or images. However, text-to-3D output is still primarily applied in simple 3D modeling or avatar creation, rather than in the design of functional CAD components. To obtain a suitable 3D representation of the generated image, an image-to-3D reconstruction method was used. This method allows for a rapid transformation of the outer orthosis contour, similar to a 3D scan of an object. Backflip.ai [1] was applied, which converts the image into a 3D mesh model and enables subsequent modifications. The generated part of the orthosis was exported in Stereolithography (STL) format, commonly used in CAD software and 3D printing, containing only surface geometry without texture or color information. This approach enables rapid 3D model generation for subsequent post-processing and personalization.

Post-Processing and Product Personalization

For manual post-processing, the 3D model was imported into Blender [2] (software version 4.1). The outer contour of the orthosis was adjusted using an offset operation to separate key components, such as the posterior and anterior corpus, straps, and inner padding. The surfaces were further refined to ensure that all components had the desired shape and contour. For the personalization of the orthosis, a 3D scan of the participant's leg was used. The scan data were obtained using the "Heges 3D Scanner"

[1] https://www.backflip.ai/
[2] https://www.blender.org/

app [3] on an iPhone 13 (iOS 18.3.2), utilizing the TrueDepth camera for depth sensing. To ensure that the scan captured the entire foot geometry, the foot was placed on a small rectangular object positioned at midfoot level, simulating a stance-phase scan. This method enables a rapid scan of the outer leg contour in approximately 30 seconds, requiring only minor post-processing adjustments. Although this freely available app offers only moderate scan quality, it is sufficient to capture the outer leg contour and dimensions. The scanned STL data were imported into Blender, where the orthosis model was scaled to match the participant's foot size. Further refinements were applied to adapt the shape to the individual's foot morphology, ensuring an optimal fit and providing targeted support in key areas. After finalizing the personalization and manual adjustments, the 3D model was prepared for subsequent additive manufacturing.

Additive Manufacturing

For rapid prototyping, the Fused Filament Fabrication (FFF) process was used. The final 3D model was imported into FlashPrint [4] slicer software (version 5.8.7) to prepare the parts for 3D printing. A Creator 4A printer (Flashforge) equipped with an HT extruder was used to print with white Acrylonitrile Butadiene Styrene (ABS) thermoplastic material. Printing settings followed recommended ABS material specifications: 0.4 mm nozzle, 1.75 mm filament diameter, and 230°C extruder temperature. A 0.3 mm layer height, 30% infill density, and a treelike support structure were applied. The total print duration for the main corpus and the front shell was 40 hours. After printing, manual post-processing steps were performed, including the removal of support material and surface finishing. The printed prototype (see Figure 6.12), was prepared with Velcro strips for fixation and attachment. The total weight of the assembled parts was 375 grams.

Physical Testing

To evaluate the proposed framework from an end-user perspective, a short-duration physical test was conducted with one 28-year-old male participant. Additionally, the development process was validated through a semi-structured interview with an orthopedic technician (29, male) from a medical supply store, who has practical experience in the production of foot and hand orthoses and insoles. He reported no prior experience with AI-based tools in this context. The expert was recruited via personal contacts and the participant through institutional mailing lists. The study received ethical clearance and was conducted in accordance with the privacy regulations of our institution. Informed consent was obtained from both the participant and the expert.

The test was conducted in a controlled laboratory setting, following the described procedure. First, the participant was introduced to the orthosis, followed by a visual inspection and a fitting process. Following this, the participant performed representative daily activities, including walking, climbing stairs, sitting, and standing. Afterwards, the participant completed the AttrakDiff Mini [380] questionnaire. This standardized questionnaire is commonly used in UX research to assess the subjective perception of usability and visual appeal. The questionnaire was used to assess whether the final

[3] https://apps.apple.com/app/id1382310112
[4] https://enterprise.flashforge.com/pages/flashprint

design met the participant's visual expectations. Descriptive statistics were performed to analyze the results.

After physical testing, a semi-structured expert interview was conducted in person to validate the overall design framework and final prototype from a professional perspective. The interview was guided by open-ended questions focusing on general feasibility, suitability, and clinical integration. A semi-structured approach was applied to address key themes while allowing for additional inquiries to gain deeper insights into the expert's perspective. The interview was documented with manual notes, which were transcribed and thematically coded sentence-wise by one researcher and analyzed using an inductive thematic analysis [322].

6.2.3 Results

The results section presents findings from quantitative patient feedback (see Subsection 6.2.3) and qualitative expert feedback obtained through a semi-structured interview (see Subsection 6.2.3).

Quantitative Patient Evaluation

The AttrakDiff mini questionnaire consists of ten items (five reversed scored), which were rated on a 7-point Likert scale, and covers four dimensions: Attractiveness (ATT), Pragmatic Quality (PQ), Hedonic Quality (HQ), which is categorized into Hedonic Quality Identity (HQ-I) and Hedonic Quality Stimulation (HQ-S). ATT describes the product's overall appeal, PQ refers to perceived usability and functionality, HQ-I captures aspects of self-expression; and HQ-S reflects perceived novelty and stimulation. The scale ranges from +3 (positive rating) to –3 (negative rating), with 0 representing a neutral midpoint. The results show positive ratings across all four dimensions, indicating high user satisfaction. Based on the participant's responses, PQ and HQ-I received the highest rating (1.5), followed by ATT (1.0), and HQ-S (0.5). Figure 6.14 summarizes the ratings of the four dimensions. Furthermore, Figure 6.15 presents the PQ-HQ portfolio visualization. The rating ($PQ = 1.5$ and $HQ = 1.0$) placed the product between the "desired" and "task-oriented" quadrant of the PQ-HQ matrix.

Qualitative Expert Evaluation

An inductive thematic analysis revealed three key themes: *alignment with patient needs*, *perceived innovativeness*, and *challenges for practical use*. In the following paragraphs each theme is documented.

Alignment with patient needs
In general, the expert stated that the exemplified process is *"targeting patient needs."* He reported an example of how aesthetic appearance can negatively influence the willingness to wear the device, aligning with existing literature [23, 25]. Therefore, he concluded that *"if it looks appealing, the patient will wear it ... as it goes hand in hand with increasing user acceptance."* However, it was also emphasized that the workflow requires a technical expert to ensure optimal alignment with the patient's needs, as *"a technician must take over the adjustment step regarding technical specifications (e.g., ideal force absorption)."*

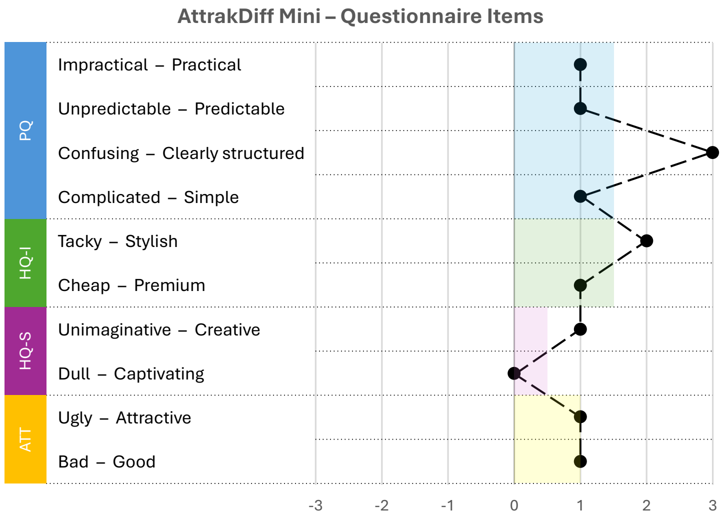


Figure 6.14: Vertical line chart showing participant ratings of the ten AttrakDiff Mini items. Colored bars represent the mean values of each dimension: PQ (1.5), HQ-I (1.5), HQ-S (0.5), ATT (1.0).

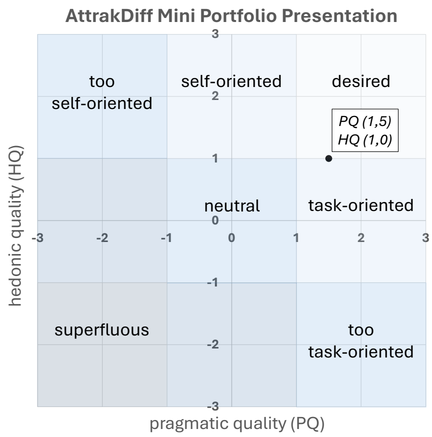


Figure 6.15: Portfolio presentation of HQ and PQ dimensions, placing the product between the "desired" and "task-oriented" quadrant of the PQ-HQ matrix ($PQ = 1.5$, $HQ = 1.0$).

Perceived innovativeness

The presented workflow of an AI-assisted design process was generally perceived as *"innovative,"* because conventional approaches lack a systematic method for incorporating patient preferences. Currently, products are typically adapted only to anatomical shape, and in some cases, basic color preferences are considered for personalization. However, patients are usually not involved in the design process itself. The expert noted that the new approach is *"extremely practical for medical supply facilities, as the process can begin without the technician having to be present."* Furthermore, he suggested that a standardized form could be envisioned in which patients indicate their preferences, enabling rapid generation and adaptation of design solutions through automation. From his point of view, this approach targets patient needs and is *"valuable for future development."* Additionally, *"time savings could play a role."*

Challenges for practical use

In addition to the perceived innovativeness and future potential, the expert raised several concerns regarding current challenges, limitations, and risks associated with the integration of AI into orthopedic product development workflows. One central point was the issue of regulatory compliance: *"Regulatory hurdles and medical approvals need to be considered, as there are strict certification procedures for medical products, and orthotic boots are commonly certified."*

In general, certified professionals at medical supply stores are able to individually adapt solutions for personalized products such as insoles. However, implementing a user-centered design process with AI tools *"requires AI expertise, which is currently lacking."* Furthermore, a limiting factor is that *"healthcare insurers typically do not cover additional costs."* Therefore, it can be challenging to integrate such solutions into traditional workflows, as facilities must be prepared accordingly and *"require the necessary software and an established 3D printing process."*

From an ethical perspective, the expert expressed no fundamental concerns. However, data privacy remains an important issue, as *"the protection of patient data (e.g., anatomically recorded data) must be strictly ensured to comply with data protection regulations."* Finally, regarding potential limitations of AI it was noted that *"AI data set filtering (one-sided feeding) could potentially limit creativity."*

6.2.4 Discussion

A single-participant case study was conducted to evaluate the proposed generative AI workflow for orthosis design. Following the framework, a foot orthosis was generated, modeled, and 3D-printed for short-term testing and expert validation.

Overall, the participant's ratings of the AttrakDiff mini questionnaire suggest that PQ was successfully addressed, indicating sufficient usability, clarity, and efficiency, while HQ scores reflected positive perceptions of attractiveness, stimulation, and identification. However, the results lack comparability, as they are based on a single, subjective short-term evaluation. The design was positioned between the desired and task-oriented quadrants, showing that the workflow balances functional and preference-oriented aspects. The trend toward the task-oriented side likely reflects the medical context, as such products are primarily designed for functionality and utility rather than self-expression. By contrast, more fashion-oriented items could shift the evaluation toward the self-oriented quadrant.

Qualitative expert feedback emphasized the potential of this workflow as an innovative alternative to traditional methods by directly involving patients in the design process. However, several challenges remain, including the reliance on specialized orthopedic software, limited transparency and data privacy in current AI solutions, and the integration of AI expertise into clinical workflows. A promising future direction is the integration of prompt-to-3D-model functionality in professional CAD software, supported by custom AI modules, for a seamless workflow from user input to 3D modeling and printing.

Implications

This case study demonstrates the integration of AI-assisted design processes in the medical domain, particularly for the development of personalized orthopedic devices. The proposed framework illustrates how generative AI can be embedded in early design phases to support patient-centered development. Although exemplified for the foot, the applied approach is not limited to this body region and can be extended to other types of wearable devices. The evaluation with a user and an expert highlights both the potential and challenges of such workflows in addressing individual user needs while streamlining the customization process. Building on previous work that demonstrated the benefits of AI-integrated workflows [381, 382], the findings suggest that AI can support the aesthetic dimension of product development, but functional validation must remain under the supervision of qualified professionals. This opens new perspectives for future applications, especially in environments where expert time and resources are limited.

Moreover, the approach could be extended by integrating diagnostic data to inform the design process, as shown by Rajagopal et al. [383], who used machine learning for foot condition classification and personalized footwear recommendation. Therefore, future research could use this framework as a basis to develop and evaluate AI-assisted workflows in other domains of wearable device design.

Limitations and Future Work

This study is subject to several limitations, including the AI software used, as the training data is non-public which reduces transparency and adaptability. Furthermore, the validation is based on a single use case with short-term testing, which limits the generalizability and representativeness of the results for broader user groups. To address these limitations, future research should validate the approach using larger samples and extended testing periods. In addition, future implementations of AI-assisted workflows could help overcome the limitations of isolated AI tools and improve overall efficiency and applicability. Future work could adapt the design process beyond orthoses and footwear, extending it to consumer products where personalization and adaptability are essential.

6.2.5 Conclusion

A novel framework for human-AI-driven product development was introduced and exemplified through a case study. Our workflow demonstrates the potential of AI in rapid design generation for the development of personalized foot orthoses. The results highlighted both the strengths and current limitations of the proposed approach. While challenges remain, the proposed approach provides a foundation for future research and can be adapted to application areas beyond orthotic and footwear design. Overall, this work demonstrates the broader potential of integrating AI into user-centered design processes, particularly for addressing individual user needs.

6.3. Chapter Summary

This chapter investigated the role of generative AI in the design of orthopedic footwear, with a focus on its influence on perceived social acceptability and its integration into a structured, user-centered design process for wearable assistive devices. Building on the development of smart assistive technologies presented in Chapter 4.2, this chapter shifted the perspective toward design creation and evaluation from an end-user viewpoint, addressing aesthetic and functional aspects that are critical for long-term adoption.

Section 6.1 addressed RQ6 by investigating how AI-supported design approaches influence the perceived social acceptability of personalized smart orthoses. Two empirical user studies with a total of 134 participants were conducted. The first study compared four orthopedic device categories to examine differences in perceived acceptance between AI-generated designs and conventional orthopedic products. The second study focused on AI-specific design parameters, analyzing how prompt keywords and customized GPTs configurations influenced generated design characteristics. Across both studies, the results demonstrated that AI-generated designs systematically affected perceived social acceptability, supported by quantitative ratings and qualitative user feedback. While these results demonstrate the potential of generative AI for design creation, they also reveal limitations in addressing functional and technical requirements from a product development perspective.

Addressing these limitations, Section 6.2 answered RQ7 by investigating how generative AI can be systematically integrated into a user-centered design process to simultaneously consider functional, aesthetic, and usability requirements of wearable devices. An eight-stage iterative design framework was proposed and exemplified through a case study ranging from AI-supported concept generation to additive manufacturing of a personalized foot orthosis. The combined evaluation involving an end user and expert provided insights into the strengths and current constraints of the proposed framework, highlighting its potential to support participatory design while identifying challenges for practical implementation.

Taken together, the studies presented in this chapter address RQ6 and RQ7 by providing empirical evidence on how generative AI affects design perception and by demonstrating how such technologies can be embedded within a structured, user-centered product development workflow. The findings contribute to an improved understanding of AI-supported design creation for assistive devices and offer methodological guidance for future research aiming to bridge functional performance and user acceptance. Building on these results, subsequent work has further explored the potential and challenges of AI in product development, providing a broader perspective on generative design approaches for orthopedic assistive devices [384].

CHAPTER 7

Discussion and Conclusions

This thesis focused on the development of a smart orthotic system with feedback capabilities and the testing of it across physical and virtual contexts, as well as the investigation of AI-generated orthotic designs on user perception. This chapter summarizes the main research contributions, presents implications, and discusses limitations and future work. Finally, the chapter concludes with a reflection on the dissertation in a broader context.

7.1. Summary of Research Contributions

The thesis motivation introduced assistive technologies and supportive feedback systems, with a particular focus on sensorized solutions for foot augmentation. Although recent developments in smart insoles have enabled health monitoring and biofeedback applications, no comprehensive system has been established that integrates sensorized technologies within standardized orthotic gait devices. From a technical perspective, a lack of health monitoring and biofeedback integration in gait-affecting footwear, such as foot orthoses, was identified.

In addition to technical aspects, the integration of XR technologies opens new possibilities for immersive training and rehabilitation scenarios for patients. While XR has demonstrated potential in learning and training contexts, its application in combination with smart orthotic devices remains largely unexplored. In particular, foot augmentation studies are underrepresented in the context of immersive environments.

Beyond the technical and application-based perspective, critical user-related challenges were also highlighted. The aesthetic design of orthotic footwear can negatively influence social acceptance, potentially leading users to avoid wearing such devices despite their therapeutic benefits. This indicates that future technological developments must incorporate a strong user-centered perspective to ensure acceptance and effective use. While related work offers new opportunities to actively involve patients in the design process, such as generative AI and participatory design methods, the systematic integration of such approaches into orthotic product development remains limited. Based on these observations, the thesis spans three central themes: *Smart Devices*, *Extended Reality*, and *AI in Design*.

Together, these themes form the conceptual framework of this work, combining technical system development, immersive application contexts, and user-centered design approaches. They structure the thesis and are operationalized through three overarching research goals and corresponding sub-objectives. In total, seven RQs were formulated to address the identified research gaps. Building upon the theoretical background, which establishes key definitions and conceptual foundations, and the related work, which systematically identifies research gaps, the following sections outline the specific contributions of this thesis.

Chapter 4 addressed the primary objective of developing and evaluating an integrated smart foot orthosis system capable of measuring plantar pressure and gait characteristics, providing haptic feedback, and communicating with external devices such as a walking aid and a smartphone application. By answering RQ1–3, this chapter contributes to the field of smart device development. First, a scoping review on smart walking aids synthesized findings from 35 research articles, categorizing device types, sensor technologies, and target user groups. Through this synthesis, current technological challenges and research gaps have been identified, and future research directions were outlined. Building upon these highlighted gaps, the second study presented the development of a comprehensive orthotic system consisting of a smart foot orthosis, an instrumented walking aid, and a smartphone application. The system enables haptic feedback through both the orthosis and a forearm crutch, based on plantar pressure and gait parameters. The study demonstrated the feasibility of accurate gait and pressure measurement, as well as the usability of the accompanying smartphone application. Furthermore, haptic feedback was empirically evaluated, providing evidence regarding feedback perception in the foot orthosis and the forearm crutch, as well as for different feedback patterns. These findings introduce a novel integrated system that extends previous smart insole developments to standardized foot orthoses capable of multi-device communication. Overall, this chapter establishes a technical and empirical foundation for next-generation smart foot orthoses and provides initial evidence for device-dependent haptic feedback perception. This work serves as a basis for follow-up investigations by combining technical development with an end-user perspective.

Building on the successful development of the smart foot orthosis system, Chapter 5 focuses on the main objective of exploring the potential of XR technologies for immersive training applications with smart devices. To address this objective, two experimental user studies were conducted to answer RQ4–5 and examine how XR environments support training and interaction with the smart foot orthosis. The first study examined how avatar-based pointing gestures and voice cues in MR impact smartphone app onboarding in terms of usability, learning, and tutorial experience. Both modalities were found to improve the overall onboarding experience compared to no guidance. Concrete recommendations were derived for modality contextualization: visual pointing gestures enhance navigation guidance, while auditory cues improve comprehension of foot pressure health data. The second study investigated the effects of environmental context in MR/VR and different visual foot pressure feedback types on objective gait performance, subjective workload, and engagement during walking with a smart foot orthosis. Results indicated that VR provides a more immersive gait training experience due to technical limitations of MR passthrough. A natural forest environment supported more natural gait, lower cognitive workload, and higher engagement, whereas a stationary laboratory environment using heatmap visualizations promoted more cautious and controlled gait, reducing localized foot pressure. These findings highlight the importance of presenting visual feedback cues in appropriate environmental contexts to ensure functional support and avoid user frustration. Design guidelines derived from this study contribute to a better understanding of immersive gait training for gait-affecting footwear. Overall, both studies demonstrate the potential of immersive training environments for enhancing smart device interaction through feedback modalities. They provide empirical evidence and concrete design recommendations to inform future research and development in this field.

Chapter 6 shifts the focus from system functionality and training applications to user-centered design. It specifically investigates how design factors influence the social acceptability of orthopedic footwear and how generative AI can be integrated into the design process. To ensure patient adoption of the developed smart foot orthosis and address current concerns regarding aesthetic design perception, this chapter examines RQ6–7. The first article presents the results of two user studies, including a total of 134 participants. We found that AI-generated designs impacted the perceived warmth and competence of the wearer, and the device itself strongly influenced the perception of design outcomes. Based on these findings, the second study proposed a framework for integrating generative AI as a supportive design generation tool within a user-centered design workflow. A case study was conducted to exemplify the iterative development process from user input to additive manufacturing of a personalized foot orthosis. User validation and expert feedback indicated that this approach successfully addressed user preferences while also highlighting concerns regarding regulatory hurdles for implementation. Taken together, both articles contributed new insights into the impact of generative AI on design outcomes as well as on the design and development process.

In summary, the thesis results are grounded in five different types of contributions (empirical, artifact, theoretical, methodological, and survey). They primarily address the field of HCI, but are also relevant to assistive technology, XR research, and AI interaction across applied sciences, such as engineering and computer science, as well as health-related research. The main contributions of the thesis are summarized by answering the following research questions:

Smart Devices

- **RQ1:** *What are the current technological trends, target user groups, and functional capabilities of smart walking aids, and which gaps exist in existing solutions that need to be addressed to improve mobility support and user-centered design?*

The scoping review in Chapter 4.1 identified a broad spectrum of device types (including crutches, canes/sticks, walkers, and rollators) with various embedded sensors, supporting different functions for specific user groups. We identified a set of technological gaps, such as lack of system consistency in sensor placement and configurations, missing experimental validations, unknown reliability and usability factors, as well as unclear impact of implemented feedback due to the wide range of early-stage prototypes without user testing.

- **RQ2:** *How can a smart orthosis and walking aid be developed as a comprehensive system, to ensure feasible sensor data acquisition and effective user feedback, while also enabling user-centered mobile application usability?*

In Chapter 4.2, a novel concept of a smart foot orthosis, forearm crutch, and smartphone application was developed. The orthosis enabled pressure and motion sensing and was connected with the walking aid, capable of providing closed-loop biofeedback across both devices. A lab study validated the system feasibility for pressure and motion sensing, as well as feedback perception. In addition, the usability of the smartphone app was positively assessed, demonstrating the potential of the entire system.

- **RQ3:** *How do different types of haptic feedback delivered via a smart orthosis and walking aid influence user perception?*

The experimental validation revealed that pattern-based haptic feedback was significantly preferred compared to standard vibrational cues, and that users perceived differences in noticeability and usefulness between the smart orthosis and walking aid. These findings provide insights into haptic feedback perception and offer recommendations for researchers, essential to avoid implementing feedback without considering user perspectives.

Extended Reality

- **RQ4:** *How can avatar-based feedback modalities in MR enhance usability and learning experience during smartphone app onboarding for a smart orthosis system?*

Chapter 5.1 demonstrated the potential of MR-based avatar voice and pointing-gesture modalities to enhance usability and learning experience during onboarding of an mHealth app. Concrete design implications were provided for better contextualization of these modalities, aiming to improve user guidance while interacting with the smart device.

- **RQ5:** *How do MR and VR environmental factors and visual foot pressure feedback influence objective gait parameters and subjective experience when walking with a smart foot orthosis?*

The study in Chapter 5.2 highlighted the benefits of VR for immersive gait training and showed that natural versus laboratory environments can impact the usefulness of foot pressure feedback. Heatmap data in stationary lab settings ensure controlled gait, while natural gait in a forest environment can increase engagement and reduce workload. Concrete design recommendations were provided to account for both environmental and visualization factors, aiming to optimize both objective gait parameters and subjective user experience.

AI in Design

- **RQ6:** *How can AI-supported design approaches influence the perceived social acceptability of orthopedic footwear?*

Chapter 6.1 showed that AI-designed footwear enhances the perceived social warmth and competence of the wearer compared to conventional products and prototypes. Although the type of orthopedic footwear significantly influences acceptance outcomes, the positive effect of AI-designed footwear is consistent across different footwear categories. In addition, customized GPT models with orthopedic knowledge can further improve design outcomes, and specific prompt keywords may also play a role.

- **RQ7:** *How can AI be incorporated into a user-centered design framework to support the development of foot orthoses that address functional and aesthetic requirements?*

In Chapter 6.2, a novel framework consisting of eight stages was presented, incorporating AI as a supportive concept creation tool in the iterative development process of orthopedic foot orthoses. The framework was validated through a case study and successfully addressed aesthetic user preferences, functional needs, and usability requirements, while also revealing current limitations in practical implementation.

7.2. Implications

This chapter discusses the overall theoretical and practical implications. Detailed design recommendations are provided in the corresponding articles.

Integrate Technical and User Perspectives in Orthotic Device Development
We present a comprehensive orthotic system comprising an orthosis, walking aid, and smartphone app, extending previous work on isolated prototypes. Our findings confirm prior research [71], demonstrating that 15 FSRs combined with a single IMU are effective for gait monitoring. Beyond feasibility and usability, the system has both theoretical and practical implications. Theoretically, the developed system provides a foundation for hardware and software extensions, which can be adapted to specific functional requirements. The system architecture allows sensor extensions to capture individual foot morphology or provide more detailed measurements for patients with foot deformities. Similarly, the system can support software modifications and be adapted for applications such as fall detection or other monitoring tasks. The system is not limited to forearm crutches and can be adapted to other walking aids, such as sticks or walkers, through handle modifications. Moreover, the controlled ankle motion boot concept is adaptable to diverse orthotic footwear, facilitating broader applicability. Practically, the system supports real-time communication across devices to enable gait monitoring with multiple feedback modalities. From a technical perspective, next-generation orthotic footwear development should consider sensor configuration, data processing, and usability, in line with the ten research challenges identified by Subramaniam et al. [13]. These outcomes highlight the importance of designing for optimal feedback modalities, sensor placement, and usability considerations.

Optimize Haptic Feedback Placement Across Devices
As highlighted in the scoping review, a gap exists in evaluating the effectiveness of haptic feedback in previously developed systems. While some studies report feedback implementations, none systematically assessed their impact from a user perspective, resulting in limited understanding of perception and user preferences. Study 4.2 addressed this gap by empirically validating the impact of vibratory feedback in both a foot orthosis and a forearm crutch on user perception. The findings show that the implementation of vibratory feedback does not automatically create the intended value, as its effectiveness is influenced by multiple interacting factors and must be evaluated with

users to identify preferences. Notably, haptic feedback cannot be seamlessly transferred across devices while achieving comparable perceptual effects. The vibration pattern, intensity, and number of actuators all influence effectiveness. Furthermore, sensory sensitivity differs between devices: perception within the orthosis was lower than in the crutch handle, suggesting that two or more ERM actuators may be required to achieve comparable outcomes. These findings provide concrete guidance for the practical implementation of haptic feedback in smart orthotic systems. We recommend systematic evaluation of feedback placement, intensity, and vibration pattern to ensure alignment with user preferences and effective implementation. Consequently, the implementation of haptic feedback must take into account device-specific and individual differences in perception, rather than assuming uniform effectiveness across devices and users.

Adapt Avatar-Based Modalities to Context and User Needs
We demonstrate that visual and auditory cues in MR can effectively support the onboarding of a smartphone app developed to monitor gait data. In particular, older adults and first-time users who experience difficulties during initial app interaction may benefit from guided avatar-based assistance. Consistent with prior research showing that avatars in MR can enhance learning in various domains [314, 315], our findings extend these insights to the domain-specific context of mHealth app onboarding. From a practical perspective, our findings translate into several design implications for MR-based onboarding, particularly regarding modality contextualization and avatar embodiment in line with onboarding task demands [385]. While MR-based guidance is more effective than no support, visual and auditory cues should be applied selectively rather than simultaneously, depending on the intended explanatory goal. Moreover, for augmented onboarding tasks that primarily involve hand–smartphone interactions, full-body avatars appear less relevant, since the focus is on hands and gestures. In addition, smartphone augmentation through virtual overlays with enlarged screens could improve task flow, interaction clarity, and readability. To support users' interpretation of health-related data, such as gait metrics and plantar pressure measures, interactive guidance mechanisms incorporating step repetition and active task involvement are recommended to enhance comprehension and engagement. Together, these findings contribute to a more differentiated understanding of MR-based guidance mechanisms and provide actionable design strategies for mHealth onboarding.

Consider Environmental Context When Designing Visual Gait Feedback
Our findings demonstrate the potential of VR and heatmap-based foot pressure visualization for immersive gait training, confirming recently published results [386]. We extend these insights by investigating the interaction between environmental context and visual feedback cues while walking with a sensorized foot orthosis on a treadmill. The results reveal a clear interaction between environmental setting and feedback modality. Visual feedback effects cannot be generalized across contexts, as heatmap visualizations interact with environmental conditions and influence both objective gait parameters and subjective perceptions. We identify notable differences between the stationary laboratory setting and a natural forest environment. While a controlled laboratory context combined with heatmap visualization supported more focused and controlled gait adaptations, immersive natural environments in VR promoted more natural walking behavior and reduced cautious movement patterns. In such environments, additional visual overlays may not always be beneficial. These findings have direct im-

plications for the design of gait training applications. Visual feedback should be aligned with the specific training goal and environmental demands. Context-dependent implementation of visual feedback is essential, as environmental factors primarily influence workload, engagement, and movement outcomes. Therefore, the combined use of environmental context and visual feedback should be carefully considered, as it can result in fundamentally different training effects.

Understand the Mechanisms and User Impact of Generative AI in Design
Our studies provide empirical insights into how users perceive AI-generated designs for orthotic footwear, including the effects on acceptability and aesthetic preferences. To the best of our knowledge, this is the first study investigating both the mechanisms of generative tools and their impact on users in the context of orthotic footwear. These findings advance understanding of human–AI interaction by highlighting the relevance of GPT model selection and keyword input strategies. From a practical perspective, our results recommend the targeted use of domain-specific models for orthotic footwear design rather than general-purpose models, as this can improve design outcomes and better align with user preferences. Furthermore, prompt keywords and input strategies should be systematically considered to optimize the quality and acceptability of generated designs. The study also contributed theoretical knowledge by showing that acceptability of AI-generated designs can vary across device-specific characteristics, suggesting that these factors should be considered individually. Overall, understanding generated outputs and their effects on user perception is critical for the successful and responsible use of generative design tools in orthotic footwear.

Systematize the Use of Generative AI as a Supportive Design Tool
The proposed framework [387, 388] introduces a novel approach for integrating generative AI into orthotic footwear design, extending previous approaches [27]. This framework has several implications. From a theoretical perspective, it demonstrates that systematic, user-centered incorporation of generative AI is feasible within iterative design processes and can serve as a model for replication or adaptation in related contexts. From a practical perspective, the framework enables designers to incorporate user preferences early in the process, accelerate ideation and visualization, and strengthen user involvement without substituting technical expertise. While acceptable designs can also be developed without AI, its systematic and targeted use ensures that early-stage ideation accounts for domain-specific constraints under expert supervision. Integrating AI into orthotic footwear design requires careful consideration of technical, ethical, and regulatory constraints to ensure feasible and responsible implementation. By embedding AI into a structured framework rather than using it as a short-term visual gadget, user involvement can be strengthened without replacing engineering expertise. This approach offers a new perspective on cooperative design and could be extended beyond orthotic footwear to wearable prototyping in combination with additive manufacturing.

7.3. Limitations and Perspectives

The presented research contributions and implications must be considered in terms of the limitations of the underlying studies. While these limitations were discussed in detail in the respective chapters, they are synthesized in this chapter from a broader perspective. All limitations also highlight opportunities for follow-up investigations and future research. Therefore, the three overarching themes are discussed in terms of their overall limitations and potential for future work.

Smart Devices
The technical development of the presented system focuses on a proof-of-concept study conducted in a controlled setting, demonstrating internal validity. However, the system has not been evaluated under real-world conditions or for long-term use, which limits external validity. While the study results are constrained by a small sample size, follow-up investigations could further extend our insights into user perception of feedback. Therefore, the presented results provide a foundation for future studies and extensions. As previously highlighted, several aspects should be considered for follow-up investigations, focusing on methodological, technical, or application-level perspectives. In particular, it remains an open question how adaptive feedback strategies should be applied to individual gait dynamics across device types and body locations. Addressing this is essential to understand the impact of haptic feedback outside controlled laboratory settings and to ensure therapeutic relevance. As this research field is driven by new innovations, emerging sensor technologies and refined algorithms for accurate gait measurement could further enhance practical applications. In addition, system and feedback extensions could provide new functionalities, such as gait or fall prediction. Based on the provided insights, next-generation device development should not only focus on technical efficiency but also address usability and feedback perception factors.

Extended Reality
The application of XR has been evaluated as an immersive training and interaction environment in combination with the smart orthosis and the smartphone application. Both studies [389, 385] are limited in terms of generalizability due to the selected tasks and the inclusion of healthy participants. The results also highlight limitations associated with the HMD. Future development of MR devices with improved resolution, reduced weight, and enhanced depth perception could further improve feasibility. Follow-up studies of mobile app onboarding should also investigate avatar voice and gesture modalities in broader contexts beyond mHealth, such as learning applications. The immersive treadmill training with the smart orthosis is constrained by the fixed treadmill speed, which may limit the identification of larger gait modifications. Experiments focusing on natural gait without a treadmill would provide valuable insights into human behavior when interacting with gait-affecting footwear. Future research should also explore training applications with patient populations, which is particularly relevant for extending applications beyond the XR research and HCI domains into rehabilitation contexts.

AI in Design
The exploration of generative AI in orthotic footwear design is intentionally scoped to support aesthetic concept generation and address user needs at an early development stage. Consequently, AI is used as a supportive tool for rapid text-to-image design generation. However, the tool itself represents a limiting factor, as its applicability is strictly restricted to creating visual stimuli. This is a key limitation, as the separation of aesthetic generation from functional and biomechanical constraints does not follow medical product standards. Based on the proposed framework, future work could use generative AI to inspire new concepts and investigate the feasibility of such approaches. Consequently, the rapid evolution of generative AI raises new questions that must be considered in future implementations. The use of AI tools should not be limited to accelerating design creation alone, as it remains essential to understand the mechanisms and effects of design generation on end users. While orthotic device–dependent acceptability factors should be investigated independently to identify underlying user mechanisms, the impact of AI requires individual evaluation. A better understanding of these factors can guide future applications of generative tools to more effectively address user needs. In particular, long-term efforts should focus on the development and implementation of domain-specific trained models for orthopedic devices. To avoid reliance on current LLMs with unknown training data, customized models integrated into orthopedic workflows and CAD software should be prioritized. Regulatory and ethical considerations also need to be addressed in this context. To better understand the perception of dedicated design aspects, future research should investigate these factors independently, which could contribute to establishing design principles relevant for both human and AI-assisted design. In general, systematic approaches for co-creative implementation should be explored, involving not only patients and clinicians but also designers to capture the full perspective of collaborative workflows.

7.4. Final Remarks

This thesis examined the technological affordances of smart systems by integrating sensing technologies, immersive XR environments, and generative AI within the context of assistive devices. Beyond applying existing technologies, it contributed to their advancement through the development and evaluation at the intersection of engineering and HCI in the health context. Additionally, this thesis also identified new research gaps and perspectives that call for further investigation. Future implementations should not only build on these insights and extend them from an interdisciplinary perspective. Beyond technological factors, this could also be a critical reflection on ethical aspects of AI integration and responsible technology deployment.

A key takeaway of this work is that developing successful smart assistive systems requires a deep understanding of end users from multiple perspectives. As stated in the preamble of this dissertation, ”learning is an experience” rather than just information. This principle applies in two directions: users understand intelligent devices primarily through experience rather than abstract instruction, and researchers can only understand this interaction by systematically examining those experiences. Therefore, the design of smart technologies requires careful investigation of how experience shapes user perception, behavior, and adoption.

Bibliography

[1] Enovis, "Aircast® airselect™ elite walker." https://www.enovis-medtech.de/AIRCAST-AIRSELECT-Elite-Walker. (accessed on 15 August 2025).

[2] P. Milgram, H. Takemura, A. Utsumi, and F. Kishino, "Augmented reality: A class of displays on the reality-virtuality continuum," in *Telemanipulator and telepresence technologies*, vol. 2351, pp. 282–292, Spie, 1995.

[3] S. Resch, J.-G. Hanania, V. Schwind, D. Völz, and D. Sanchez-Morillo, "The influence of augmented and virtual reality environments on foot positioning success and workload in agility ladder exercises," in *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 679–686, IEEE, 2025.

[4] S. Resch, V. Schwind, and D. Völz, "Body position in virtual reality-balancing support for patients with neuropathy," *Proceedings on Automation in Medical Engineering*, vol. 2, no. 1, pp. 744–744, 2023.

[5] ChatGPT/DALL-E, "Ai-generated foot orthosis:"how to design"." https://chat.openai.com, Nov. 2024. Generated using DALL·E via ChatGPT on 26.11.2024.

[6] W. H. Organization and U. N. C. Fund, *Global report on assistive technology.* World Health Organization, 2022.

[7] J. Walker, A. Thomas, M. L. Stolycia, R. A. Wilkins, and D. E. Lunn, "Acute biomechanical responses to wearing a controlled ankle motion (cam) walker boot during walking," *Gait & Posture*, vol. 108, pp. 208–214, 2024.

[8] D. S. Elvitigala, J. Huber, and S. Nanayakkara, "Augmented foot: A comprehensive survey of augmented foot interfaces," in *Proceedings of the Augmented Humans International Conference 2021*, AHs '21, (New York, NY, USA), p. 228–239, Association for Computing Machinery, 2021.

[9] J. J. Ferreira, C. I. Fernandes, H. G. Rammal, and P. M. Veiga, "Wearable technology and consumer interaction: A systematic review and research agenda," *Computers in human behavior*, vol. 118, p. 106710, 2021.

[10] L. Brognara, A. Mazzotti, S. O. Zielli, A. Arceri, E. Artioli, F. Traina, and C. Faldini, "Wearable technology applications and methods to assess clinical outcomes in foot and ankle disorders: Achievements and perspectives," *Sensors*, vol. 24, no. 21, 2024.

[11] P. B. Shull and D. D. Damian, "Haptic wearables as sensory replacement, sensory augmentation and trainer–a review," *Journal of neuroengineering and rehabilitation*, vol. 12, no. 1, p. 59, 2015.

[12] O. M. Giggins, U. M. Persson, and B. Caulfield, "Biofeedback in rehabilitation," *Journal of neuroengineering and rehabilitation*, vol. 10, no. 1, p. 60, 2013.

[13] S. Subramaniam, S. Majumder, A. I. Faisal, and M. J. Deen, "Insole-based systems for health monitoring: Current solutions and research challenges," *Sensors*, vol. 22, no. 2, p. 438, 2022.

[14] R. C. Moroni and K. Majewska, "Innovations in orthotic devices: Additive manufacturing, auxetic materials and smart sensors for enhanced rehabilitation," *Applied Sciences*, vol. 15, no. 18, 2025.

[15] M. G. Finco, G. Cay, M. Lee, J. Garcia, E. Salazar, T.-W. Tan, D. G. Armstrong, and B. Najafi, "Taking a load off: User perceptions of smart offloading walkers for diabetic foot ulcers using the technology acceptance model," *Sensors*, vol. 23, no. 5, 2023.

[16] F. R. Shefa, F. H. Sifat, S. C. Shah, and M. G. Kibria, "Iot-based smart ankle-foot orthosis for patients with gait imbalance," *International Conference on Control, Automation and Systems*, pp. 969–974, 2023.

[17] C. Andrews, M. K. Southworth, J. N. Silva, and J. R. Silva, "Extended reality in medical practice," *Current treatment options in cardiovascular medicine*, vol. 21, no. 4, p. 18, 2019.

[18] L. Y. Liu, S. Sangani, K. K. Patterson, J. Fung, and A. Lamontagne, "Real-time avatar-based feedback to enhance the symmetry of spatiotemporal parameters after stroke: Instantaneous effects of different avatar views," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 28, no. 4, pp. 878–887, 2020.

[19] G. Duque, D. Boersma, G. Loza-Diaz, S. Hassan, H. Suarez, D. Geisinger, P. Suriyaarachchi, A. Sharma, and O. Demontiero, "Effects of balance training using a virtual-reality system in older fallers," *Clinical interventions in aging*, pp. 257–263, 2013.

[20] I. Rakkolainen, A. Farooq, J. Kangas, J. Hakulinen, J. Rantala, M. Turunen, and R. Raisamo, "Technologies for multimodal interaction in extended reality—a scoping review," *Multimodal Technologies and Interaction*, vol. 5, no. 12, 2021.

[21] S. Okuno, T. Ito, M. Suzuki, and S. Tani, "Gait diagnosis for elderly people to prevent fall and development of training assist tool using augmented reality," in *2017 International Symposium on Micro-NanoMechatronics and Human Science (MHS)*, pp. 1–6, 2017.

[22] B. Butz, A. Jussen, A. Rafi, G. Lux, and J. Gerken, "A taxonomy for augmented and mixed reality applications to support physical exercises in medical rehabilitation—a literature review," *Healthcare*, vol. 10, no. 4, 2022.

[23] K. Ghoseiri and H. B. and, "User satisfaction with orthotic and prosthetic devices and services of a single clinic," *Disability and Rehabilitation*, vol. 34, no. 15, pp. 1328–1332, 2012. PMID: 22200105.

[24] J. Sehrt, B. Braams, N. Henze, and V. Schwind, "Social acceptability in context: Stereotypical perception of shape, body location, and usage of wearable devices," *Big Data and Cognitive Computing*, vol. 6, no. 4, p. 100, 2022.

[25] E. Swinnen and E. Kerckhofs, "Compliance of patients wearing an orthotic device or orthopedic shoes: A systematic review," *Journal of Bodywork and Movement Therapies*, vol. 19, no. 4, pp. 759–770, 2015.

[26] F. Holtkamp, E. Wouters, J. van Hoof, Y. van Zaalen, and M. Verkerk, "Use of and satisfaction with ankle foot orthoses," *Clinical research on foot & ankle*, vol. 3, no. 1, p. 167, 2015.

[27] P. Dabnichki and T. Y. Pang, "User-centered design framework for personalized ankle–foot orthoses," *Prosthesis*, vol. 7, no. 1, 2025.

[28] J. Suessmuth, F. Fick, and S. Van Der Vossen, "Generative ai for concept creation in footwear design," in *ACM SIGGRAPH 2023 Talks*, SIGGRAPH '23, (New York, NY, USA), Association for Computing Machinery, 2023.

[29] P. Minaoglou, A. Tzotzis, N. Efkolidis, and P. Kyratsis, "Integrating artificial intelligence into the shoe design process," *Engineering Proceedings*, vol. 72, no. 1, 2024.

[30] D. Popescu, "Exploring the use of chatgpt and dall-e for the preliminary design of 3d-printed wrist-hand orthoses," *Proceedings in Manufacturing Systems*, vol. 18, no. 3, pp. 97–102, 2023.

[31] J. O. Wobbrock and J. A. Kientz, "Research contributions in human-computer interaction," *Interactions*, vol. 23, p. 38–44, Apr. 2016.

[32] NISO CRediT Working Group and others, "Credit, contributor roles taxonomy," *National Information Standards Organization*, Feb. 2022. ANSI/NISO Z39.104-2022; Approved January 14, 2022 by the American National Standards Institute.

[33] International Organization for Standardization, "Assistive products — classification and terminology." ISO Standard 9999:2022, 7th edition, 05 2022. Technical Committee: ISO/TC 173/SC 2; ICS: 11.180.01.

[34] World Health Organization, *Priority Assistive Products List*. Geneva: World Health Organization, 2016. Handbook; Accessed: 2026-02-05.

[35] International Organization for Standardization (ISO), "Prosthetics and orthotics - vocabulary - part 3: Terms relating to orthoses," Sept. 2020. Published in Switzerland.

[36] D. Yao, L. Meyer-Kobbe, S. Ettinger, L. Claassen, A. Altemeier-Sasse, C. Sturm, A. Kerling, C. Stukenborg-Colsman, and C. Plaass, “Functional, spiroergometric, and subjective comparisons between forearm crutches and hands-free single crutches in a crossover study,” *Foot & Ankle Orthopaedics*, vol. 8, no. 2, p. 24730114231172734, 2023.

[37] R. T. Azuma, “A survey of augmented reality,” *Presence: teleoperators & virtual environments*, vol. 6, no. 4, pp. 355–385, 1997.

[38] G. C. Burdea and P. Coiffet, *Virtual reality technology.* John Wiley & Sons, 2003.

[39] D. A. Norman and S. W. Draper, *User centered system design; new perspectives on human-computer interaction.* L. Erlbaum Associates Inc., 1986.

[40] J. D. Gould and C. Lewis, “Designing for usability: key principles and what designers think,” *Commun. ACM*, vol. 28, p. 300–311, Mar. 1985.

[41] International Organization for Standardization, “Ergonomics of human-system interaction — part 210: Human-centred design for interactive systems.” ISO Standard 9241-210:2019, 2nd edition, 07 2019. Technical Committee: ISO/TC 159/SC 4; ICS: 13.180, 35.180.

[42] T. Brown *et al.*, “Design thinking,” *Harvard business review*, vol. 86, no. 6, p. 84, 2008.

[43] Verein Deutscher Ingenieure, “Design of technical products and systems — model of product design.” VDI Guideline 2221 Part 1:2019-11, 11 2019. VDI 2221 Blatt 1; Publisher: VDI Verein Deutscher Ingenieure e.V., Düsseldorf.

[44] S. Resch, K. Zoufal, I. Akhouaji, M. A. Abbou, V. Schwind, and D. Völz, “Augmented smart insoles - prototyping a mobile application: Usage preferences of healthcare professionals and people with foot deformities,” *Current Directions in Biomedical Engineering*, vol. 9, pp. 698–701, 9 2023.

[45] S. Resch, L. Zimmermann, A. Kunkel, D. Gerstung, V. Schwind, D. Völz, and D. Sanchez-Morillo, “Usability evaluation of a mobile application for foot health monitoring of smart insoles: A mixed methods study,” *MobileHCI 2024 Adjunct Proceedings - Publication of the 26th International Conference on Mobile Human-Computer Interaction*, 9 2024.

[46] S. Resch, A. Kousha, A. Carroll, N. Severinghaus, F. Rehberg, M. Zatschker, Y. Söyleyici, and D. Sanchez-Morillo, “Smart device development for gait monitoring: Multimodal feedback in an interactive foot orthosis, walking aid, and mobile application,” *Technologies*, vol. 13, no. 12, 2025.

[47] B. Najafi, H. Mohseni, G. S. Grewal, T. K. Talal, R. A. Menzies, and D. G. Armstrong, “An optical-fiber-based smart textile (smart socks) to manage biomechanical risk factors associated with diabetic foot amputation,” *Journal of diabetes science and technology*, vol. 11, no. 4, pp. 668–677, 2017.

[48] T. Isezaki, H. Kadone, A. Niijima, R. Aoki, T. Watanabe, T. Kimura, and K. Suzuki, "Sock-type wearable sensor for estimating lower leg muscle activity using distal emg signals," *Sensors*, vol. 19, no. 8, 2019.

[49] Z. Zhang, T. He, M. Zhu, Z. Sun, Q. Shi, J. Zhu, B. Dong, M. R. Yuce, and C. Lee, "Deep learning-enabled triboelectric smart socks for iot-based gait analysis and vr applications," *npj Flexible Electronics*, vol. 4, no. 1, p. 29, 2020.

[50] A. Drăgulinescu, A.-M. Drăgulinescu, G. Zincă, D. Bucur, V. Feieș, and D.-M. Neagu, "Smart socks and in-shoe systems: State-of-the-art for two popular technologies for foot motion analysis, sports, and medical applications," *Sensors*, vol. 20, no. 15, p. 4316, 2020.

[51] G. Semjonova, A. Davidovica, N. Kozlovskis, A. Okss, and A. Katashevs, "Smart textile sock system for athletes' self-correction during functional tasks: Formative usability evaluation," *Sensors 2022, Vol. 22, Page 4779*, vol. 22, p. 4779, 6 2022.

[52] R. Morley, E. Richter, J. Klaesner, K. Maluf, and M. Mueller, "In-shoe multisensory data acquisition system," *IEEE Transactions on Biomedical Engineering*, vol. 48, no. 7, pp. 815–820, 2001.

[53] B. M. Eskofier, S. I. Lee, M. Baron, A. Simon, C. F. Martindale, H. Gaßner, and J. Klucken, "An overview of smart shoes in the internet of health things: gait and mobility assessment in health promotion and disease monitoring," *Applied Sciences*, vol. 7, no. 10, p. 986, 2017.

[54] U. Sunarya, Y. Sun Hariyani, T. Cho, J. Roh, J. Hyeong, I. Sohn, S. Kim, and C. Park, "Feature analysis of smart shoe sensors for classification of gait patterns," *Sensors*, vol. 20, no. 21, p. 6253, 2020.

[55] D. Gokalgandhi, L. Kamdar, N. Shah, and N. Mehendale, "A review of smart technologies embedded in shoes," *Journal of medical systems*, vol. 44, pp. 1–9, 2020.

[56] M. Ullrich, A. Kuderle, J. Hannink, S. D. Din, H. Gasner, F. Marxreiter, J. Klucken, B. M. Eskofier, and F. Kluge, "Detection of gait from continuous inertial sensor data using harmonic frequencies," *IEEE Journal of Biomedical and Health Informatics*, vol. 24, pp. 1869–1878, 7 2020.

[57] F. Lin, A. Wang, Y. Zhuang, M. R. Tomita, and W. Xu, "Smart insole: A wearable sensor device for unobtrusive gait monitoring in daily life," *IEEE Transactions on Industrial Informatics*, vol. 12, no. 6, pp. 2281–2291, 2016.

[58] I. Almuteb, R. Hua, and Y. Wang, "Smart insoles review (2008-2021): Applications, potentials, and future," *Smart Health*, vol. 25, p. 100301, 2022.

[59] W. Xu, M.-C. Huang, N. Amini, J. J. Liu, L. He, and M. Sarrafzadeh, "Smart insole: A wearable system for gait analysis," in *Proceedings of the 5th international conference on pervasive technologies related to assistive environments*, pp. 1–4, 2012.

[60] S. I. Okoduwa, B. E. Igiri, J. I. Tagang, U. J. Okoduwa, and A. O. Adeyi, "Therapeutic smart-footwear approach for management of neuropathic diabetic foot ulcers: Current challenges and focus for future perspective," *Medicine in Novel Technology and Devices*, vol. 23, p. 100311, 2024.

[61] T. T. Duong, S. Goldman, H. Zhang, R. Salazar, S. Beenders, K. M. Cornett, J. M. Bain, J. Montes, and D. Zanotto, "Validation of insole-based gait analysis system in young children with a neurodevelopmental disorder and autism traits," *Proceedings of the IEEE RAS and EMBS International Conference on Biomedical Robotics and Biomechatronics*, vol. 2020-November, pp. 715–720, 11 2020.

[62] C. Park, R. Mishra, D. Vigano, M. Macagno, S. Rossotti, K. D'Huyvetter, J. Garcia, D. G. Armstrong, and B. Najafi, "Smart offloading boot system for remote patient monitoring: toward adherence reinforcement and proper physical activity prescription for diabetic foot ulcer patients," *Journal of Diabetes Science and Technology*, vol. 17, no. 1, pp. 42–51, 2023.

[63] F. R. Shefa, F. H. Sifat, J. Uddin, Z. Ahmad, J. M. Kim, and M. G. Kibria, "Deep learning and iot-based ankle–foot orthosis for enhanced gait optimization," *Healthcare 2024*, vol. 12, p. 2273, 11 2024.

[64] P. Han, K. Wang, C. Yan, and L. Wang, "Design and experimental research of an intelligent ankle-foot orthosis system for strephenopodia," in *Proceedings of the 2023 4th International Symposium on Artificial Intelligence for Medicine Science*, ISAIMS '23, (New York, NY, USA), p. 985–991, Association for Computing Machinery, 2024.

[65] Y. S. Mustufa, J. Barton, B. O'Flynn, R. Davies, P. McCullagh, and H. Zheng, "Design of a smart insole for ambulatory assessment of gait," *2015 IEEE 12th International Conference on Wearable and Implantable Body Sensor Networks, BSN 2015*, 10 2015.

[66] N. Roth, C. F. Martindale, H. Gaßner, Z. Kohl, J. Klucken, and B. M. Eskofer, "Synchronized sensor insoles for clinical gait analysis in home-monitoring applications," *Current Directions in Biomedical Engineering*, vol. 4, pp. 433–437, 9 2018.

[67] L. Farid, D. Jacobs, J. D. Santos, O. Simon, J. M. Gracies, and E. Hutin, "Feetme® monitor-connected insoles are a valid and reliable alternative for the evaluation of gait speed after stroke," *Topics in Stroke Rehabilitation*, pp. 1–8, 2 2021.

[68] D. Chen, G. Asaeikheybari, H. Chen, W. Xu, and M. C. Huang, "Ubiquitous fall hazard identification with smart insole," *IEEE journal of biomedical and health informatics*, vol. 25, pp. 2768–2776, 7 2021.

[69] L. D'arco, G. Mccalmont, H. Wang, and H. Zheng, "Application of smart insoles for recognition of activities of daily living: A systematic review," *ACM Trans. Comput. Healthcare*, vol. 5, Jan. 2024.

[70] B. Djamaa, M. M. Bessa, B. Diaf, A. Rouigueb, and A. Yachir, "Boostsole: Design and realization of a smart insole for automatic human gait classification," *Proceedings of the 2020 Federated Conference on Computer Science and Information Systems, FedCSIS 2020*, pp. 35–43, 9 2020.

[71] A. H. A. Razak, A. Zayegh, R. K. Begg, and Y. Wahab, "Foot plantar pressure measurement system: A review," *Sensors 2012, Vol. 12, Pages 9884-9912*, vol. 12, pp. 9884–9912, 7 2012.

[72] L. Shu, T. Hua, Y. Wang, Q. Li, D. D. Feng, and X. Tao, "In-shoe plantar pressure measurement and analysis system based on fabric pressure sensing array," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 3, pp. 767–775, 2010.

[73] C. M. Senanayake and S. M. N. A. Senanayake, "Computational intelligent gait-phase detection system to identify pathological gait," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 5, pp. 1173–1179, 2010.

[74] T. Gujarathi and K. Bhole, "Gait analysis using imu sensor," *2019 10th International Conference on Computing, Communication and Networking Technologies, ICCCNT 2019*, 7 2019.

[75] A. Aggarwal, R. Gupta, and R. Agarwal, "Design and development of integrated insole system for gait analysis," *2018 11th International Conference on Contemporary Computing, IC3 2018*, 11 2018.

[76] A. M. Sabatini, C. Martelloni, S. Scapellato, and F. Cavallo, "Assessment of walking features from foot inertial sensing," *IEEE Transactions on Biomedical Engineering*, vol. 52, pp. 486–494, 3 2005.

[77] A. Küderle, N. Roth, J. Zlatanovic, M. Zrenner, B. Eskofier, and F. Kluge, "The placement of foot-mounted imu sensors does affect the accuracy of spatial parameters during regular walking," *PLOS ONE*, vol. 17, p. e0269567, 6 2022.

[78] Z. Zhang and S. Poslad, "Improved use of foot force sensors and mobile phone gps for mobility activity recognition," *IEEE Sensors Journal*, vol. 14, pp. 4340–4347, 12 2014.

[79] J. Grech, S. Mizzi, and O. Falzon, "A technical review of foot temperature measurement systems," *Journal of the American Podiatric Medical Association*, vol. 112, no. 2, 2022.

[80] E. Matijevich, E. Minty, E. Bray, C. Bachus, M. Hajizadeh, and B. Liden, "A multi-faceted digital health solution for monitoring and managing diabetic foot ulcer risk: A case series," *Sensors 2024, Vol. 24, Page 2675*, vol. 24, p. 2675, 4 2024.

[81] S. Hong and K. S. Park, "Unobtrusive photoplethysmographic monitoring under the foot sole while in a standing posture," *Sensors*, vol. 18, no. 10, p. 3239, 2018.

[82] A. M. Joseph, A. Kian, and R. Begg, "State-of-the-art review on wearable obstacle detection systems developed for assistive technologies and footwear," *Sensors 2023, Vol. 23, Page 2802*, vol. 23, p. 2802, 3 2023.

[83] S. K. Manna, M. A. H. B. Azhar, and A. Greace, "Optimal locations and computational frameworks of fsr and imu sensors for measuring gait abnormalities," *Heliyon*, vol. 9, p. e15210, 4 2023.

[84] H. Prasanth, M. Caban, U. Keller, G. Courtine, A. Ijspeert, H. Vallery, and J. von Zitzewitz, "Wearable sensor-based real-time gait detection: A systematic review," *Sensors 2021, Vol. 21, Page 2727*, vol. 21, p. 2727, 4 2021.

[85] Y. C. Han, K. I. Wong, and I. Murray, "Gait phase detection for normal and abnormal gaits using imu," *IEEE Sensors Journal*, vol. 19, pp. 3439–3448, 5 2019.

[86] H. Kim, Y. Kang, D. R. Valencia, and D. Kim, "An integrated system for gait analysis using fsrs and an imu," *Proceedings - 2nd IEEE International Conference on Robotic Computing, IRC 2018*, vol. 2018-January, pp. 347–351, 4 2018.

[87] U. Manupibul, R. Tanthuwapathom, W. Jarumethitanont, P. Kaimuk, W. Limroongreungrat, and W. Charoensuk, "Integration of force and imu sensors for developing low-cost portable gait measurement system in lower extremities," *Scientific Reports 2023 13:1*, vol. 13, pp. 1–12, 6 2023.

[88] M. Benocci, L. Rocchi, E. Farella, L. Chiari, and L. Benini, "A wireless system for gait and posture analysis based on pressure insoles and inertial measurement units," *3rd International Conference on Pervasive Computing Technologies for Healthcare - Pervasive Health 2009*, 2009.

[89] Y. Hutabarat, D. Owaki, and M. Hayashibe, "Recent advances in quantitative gait analysis using wearable sensors: A review," *IEEE Sensors Journal*, vol. 21, pp. 26470–26487, 12 2021.

[90] V. Jakob, A. Küderle, F. Kluge, J. Klucken, B. M. Eskofier, J. Winkler, M. Winterholler, and H. Gassner, "Validation of a sensor-based gait analysis system with a gold-standard motion capture system in patients with parkinson's disease," *Sensors*, vol. 21, no. 22, 2021.

[91] L. Donath, O. Faude, E. Lichtenstein, C. Nuesch, and A. Mündermann, "Validity and reliability of a portable gait analysis system for measuring spatiotemporal gait characteristics: Comparison to an instrumented treadmill," *Journal of NeuroEngineering and Rehabilitation*, vol. 13, pp. 1–9, 1 2016.

[92] H. Jagos, K. Pils, M. Haller, C. Wassermann, C. Chhatwal, D. Rafolt, and F. Rattay, "Mobile gait analysis via eshoes instrumented shoe insoles: a pilot study for validation against the gold standard gaitrite®," *Journal of Medical Engineering & Technology*, vol. 41, pp. 375–386, 7 2017.

[93] A. Raghuvanshi, P. Mitra, D. Rane, S. D. Kumar, and U. Lahiri, "Gaitwear: An augmented wearable system for gait quantification," *IEEE Sensors Journal*, 2024.

[94] V. M. Santos, B. B. Gomes, M. A. Neto, and A. M. Amaro, "A systematic review of insole sensor technology: Recent studies and future directions," *Applied Sciences*, vol. 14, no. 14, 2024.

[95] A. Saboor, T. Kask, A. Kuusik, M. M. Alam, Y. L. Moullec, I. K. Niazi, A. Zoha, and R. Ahmad, "Latest research trends in gait analysis using wearable sensors and machine learning: A systematic review," *IEEE Access*, vol. 8, pp. 167830–167864, 2020.

[96] S. Crea, C. Cipriani, M. Donati, M. C. Carrozza, and N. Vitiello, "Providing time-discrete gait information by wearable feedback apparatus for lower-limb amputees: Usability and functional validation," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 23, no. 2, pp. 250–257, 2015.

[97] A. R. Oates, L. Hauck, R. Moraes, and K. M. Sibley, "The effects of haptic input on biomechanical and neurophysiological parameters of walking: A scoping review," *Gait & Posture*, vol. 58, pp. 232–239, 10 2017.

[98] J. Berengueres, M. Fritschi, and R. McClanahan, "A smart pressure-sensitive insole that reminds you to walk correctly: An orthotic-less treatment for over pronation," *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC 2014*, pp. 2488–2491, 11 2014.

[99] S. Biesmans and P. Markopoulos, "Design and evaluation of sonis, a wearable biofeedback system for gait retraining," *Multimodal Technologies and Interaction*, vol. 4, no. 3, 2020.

[100] A. Pacilli, I. Mileti, M. Germanotta, E. Di Sipio, I. Imbimbo, I. Aprile, L. Padua, S. Rossi, E. Palermo, and P. Cappa, "A wearable setup for auditory cued gait analysis in patients with parkinson's disease," in *2016 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*, pp. 1–6, 2016.

[101] C. C. Ma, S. Y. Wei, and J. H. Lin, "Smart footwear feedback interface analysis and design," *Proceedings - 2015 International Conference on Computational Science and Computational Intelligence, CSCI 2015*, pp. 766–770, 3 2016.

[102] C. B. Redd and S. J. Bamberg, "A wireless sensory feedback device for real-time gait feedback and training," *IEEE/ASME Transactions on Mechatronics*, vol. 17, pp. 425–433, 2012.

[103] D. S. Elvitigala, D. J. Matthies, L. David, C. Weerasinghe, and S. Nanayakkara, "Gymsoles: Improving squats and dead-lifts by visualizing the user's center of pressure," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, (New York, NY, USA), p. 1–12, Association for Computing Machinery, 2019.

[104] Q. Yan, J. Huang, C. Tao, X. Chen, and W. X. and, "Intelligent mobile walking-aids: perception, control and safety," *Advanced Robotics*, vol. 34, no. 1, pp. 2–18, 2020.

[105] S. Resch, A. Zirari, T. D. Q. Tran, L. M. Bauer, and D. Sanchez-Morillo, "Smart walking aids with sensor technology for gait support and health monitoring: A scoping review," *Technologies*, vol. 13, no. 8, 2025.

[106] S. Srinivasan and M. Rajesh, "Smart walking stick," *Proceedings of the International Conference on Trends in Electronics and Informatics, ICOEI 2019*, pp. 576–579, 4 2019.

[107] Z. Yang, M. Gao, and J. Choi, "Smart walking cane based on triboelectric nanogenerators for assisting the visually impaired," *Nano Energy*, vol. 124, p. 109485, 6 2024.

[108] G. Chamorro-Moriana, J. L. Sevillano, and C. Ridao-Fernández, "A compact forearm crutch based on force sensors for aided gait: Reliability and validity," *Sensors 2016, Vol. 16, Page 925*, vol. 16, p. 925, 6 2016.

[109] M. M. Shah, M. N. R. Khan, M. R. A. Khan, M. M. H. Plabon, and M. A. Razzak, "A cost-effective smart walking stick for visually impaired people," *Proceedings of the 6th International Conference on Communication and Electronics Systems, ICCES 2021*, pp. 1582–1585, 7 2021.

[110] M. Bansal, S. Malik, M. Kumar, and N. Meena, "Arduino based smart walking cane for visually impaired people," *Proceedings of the 4th International Conference on Inventive Systems and Control, ICISC 2020*, pp. 462–465, 1 2020.

[111] M. A. Ikbal, F. Rahman, and M. H. Kabir, "Microcontroller based smart walking stick for visually impaired people," *4th International Conference on Electrical Engineering and Information and Communication Technology, iCEEiCT 2018*, pp. 255–259, 7 2018.

[112] S. Gallo, D. Chapuis, L. Santos-Carreras, Y. Kim, P. Retornaz, H. Bleuler, and R. Gassert, "Augmented white cane with multimodal haptic feedback," *2010 3rd IEEE RAS and EMBS International Conference on Biomedical Robotics and Biomechatronics, BioRob 2010*, pp. 149–155, 2010.

[113] P. Di, Y. Hasegawa, S. Nakagawa, K. Sekiyama, T. Fukuda, J. Huang, and Q. Huang, "Fall detection and prevention control using walking-aid cane robot," *IEEE/ASME Transactions on Mechatronics*, vol. 21, no. 2, pp. 625–637, 2016.

[114] N. Tsuda, A. Hayashi, M. Tounai, and S. Akutagawa, "Visualization system of crutch walking based on internal sensors," *IEEE/ASME International Conference on Advanced Intelligent Mechatronics, AIM*, pp. 19–24, 2010.

[115] A. Trujillo-León, A. de Guzmán-Manzano, R. Velázquez, and F. Vidal-Verdú, "Generation of gait events with a fsr based cane handle," *Sensors*, vol. 21, no. 16, 2021.

[116] S. Murali, R. Shrivatsan, V. Sreenivas, S. Vijjappu, S. J. Gladwin, and R. Rajavel, "Smart walking cane for the visually challenged," *IEEE Region 10 Humanitarian Technology Conference 2016, R10-HTC 2016 - Proceedings*, 4 2017.

[117] G. V. Merrett, M. A. Ettabib, C. Peters, G. Hallett, and N. M. White, "Augmenting forearm crutches with wireless sensors for lower limb rehabilitation," *Measurement Science and Technology*, vol. 21, p. 124008, 10 2010.

[118] E. Schuster, R. L. Routson, M. Hinchcliff, K. Benoff, P. Suri, C. Richburg, B. C. Muir, J. M. Czerniecki, and P. M. Aubin, "A novel walking cane with haptic biofeedback reduces knee adduction moment in the osteoarthritic knee," *Journal of Biomechanics*, vol. 114, p. 110150, 1 2021.

[119] S. Mohapatra, S. H. Rout, V. Tripathi, T. Saxena, and Y. Karuna, "Smart walking stick for blind integrated with sos navigation system," *Proceedings of the 2nd International Conference on Trends in Electronics and Informatics, ICOEI 2018*, pp. 441–447, 11 2018.

[120] S. Suman, S. Mishra, K. S. Sahoo, and A. Nayyar, "Vision navigator: a smart and intelligent obstacle recognition model for visually impaired users," *Mobile Information Systems*, vol. 2022, no. 1, p. 9715891, 2022.

[121] M. Lancini, M. Serpelloni, S. Pasinetti, and E. Guanziroli, "Healthcare sensor system exploiting instrumented crutches for force measurement during assisted gait of exoskeleton users," *IEEE Sensors Journal*, vol. 16, pp. 8228–8237, 12 2016.

[122] S. Resch, A. Röll, A. Malech, V. Schwind, D. Sanchez-Morillo, and D. Völz, "Walking aid with haptic feedback for combined use with a smart foot orthosis," *Gerontechnology*, vol. 23, pp. 1–1, 2024.

[123] D. S. Elvitigala, D. J. C. Matthies, and S. Nanayakkara, "Stressfoot: Uncovering the potential of the foot for acute stress sensing in sitting posture," *Sensors*, vol. 20, no. 10, 2020.

[124] J. A. Paradiso, S. J. Morris, A. Y. Benbasat, and E. Asmussen, "Interactive therapy with instrumented footwear," in *CHI '04 Extended Abstracts on Human Factors in Computing Systems*, CHI EA '04, (New York, NY, USA), p. 1341–1343, Association for Computing Machinery, 2004.

[125] D. S. Elvitigala, Y. Wang, Y. Hu, and A. J. Quigley, "Radarfoot: Fine-grain ground surface context awareness for smart shoes," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, UIST '23, (New York, NY, USA), Association for Computing Machinery, 2023.

[126] Y. Wang, T. E. Truong, S. W. Chesebrough, P. Willemsen, K. B. Foreman, A. S. Merryweather, J. M. Hollerbach, and M. A. Minor, "Augmenting virtual reality terrain display with smart shoe physical rendering: A pilot study," *IEEE Transactions on Haptics*, vol. 14, no. 1, pp. 174–187, 2020.

[127] D. S. Elvitigala, D. J. Matthies, C. Weerasinghe, and S. Nanayakkara, "Gymsoles++: Combining google glass with smart insoles to improve body posture when performing squats," in *Proceedings of the 14th PErvasive Technologies Related to Assistive Environments Conference*, PETRA '21, (New York, NY, USA), p. 48–54, Association for Computing Machinery, 2021.

[128] H. Son, H. Gil, S. Byeon, S.-Y. Kim, and J. R. Kim, “Realwalk: Feeling ground surfaces while walking in virtual reality,” in *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI EA ’18, (New York, NY, USA), p. 1–4, Association for Computing Machinery, 2018.

[129] H. Son, I. Hwang, T.-H. Yang, S. Choi, S.-Y. Kim, and J. R. Kim, “Realwalk: Haptic shoes using actuated mr fluid for walking in vr,” in *2019 IEEE World Haptics Conference (WHC)*, pp. 241–246, IEEE, 2019.

[130] S. Ghai, I. Ghai, and A. Lamontagne, “Virtual reality training enhances gait poststroke: a systematic review and meta-analysis,” *Annals of the New York Academy of Sciences*, vol. 1478, no. 1, pp. 18–42, 2020.

[131] H. Oagaz, A. Sable, M.-H. Choi, W. Xu, and F. Lin, “Vrinsole: An unobtrusive and immersive mobility training system for stroke rehabilitation,” in *2018 IEEE 15th International Conference on Wearable and Implantable Body Sensor Networks (BSN)*, pp. 5–8, IEEE, 2018.

[132] M. Villiger, J. Liviero, L. Awai, R. Stoop, P. Pyk, R. Clijsen, A. Curt, K. Eng, and M. Bolliger, “Home-based virtual reality-augmented training improves lower limb muscle strength, balance, and functional mobility following chronic incomplete spinal cord injury,” *Frontiers in neurology*, vol. 8, p. 635, 2017.

[133] S. Resch, M. Rafati, A. Altomare, O. Raddi, A. Tahmas, V. Schwind, and D. Völz, “Correct foot positioning in virtual reality through visual agility ladder training,” in *Proceedings of Mensch Und Computer 2023*, MuC ’23, (New York, NY, USA), p. 524–528, Association for Computing Machinery, 2023.

[134] J. Feasel, M. C. Whitton, L. Kassler, F. P. Brooks, and M. D. Lewek, “The integrated virtual environment rehabilitation treadmill system,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 19, no. 3, pp. 290–297, 2011.

[135] Y. Jiao, P. F. Dajime, Q. Zhang, S. Reading, M.-C. Smith, and Y. Zhang, “A clinically oriented vr-based treadmill training system for post-stroke rehabilitation: integration of motor learning and control principles,” *Virtual Reality*, vol. 29, no. 3, p. 112, 2025.

[136] S. Kim, Y. Lee, and K. Kim, “Gait training with virtual reality-based real-time feedback for chronic post-stroke patients: A pilot study,” *Healthcare*, vol. 13, no. 2, p. 203, 2025.

[137] Z. Y. Chan, A. J. MacPhail, I. P. Au, J. H. Zhang, B. M. Lam, R. Ferber, and R. T. Cheung, “Walking with head-mounted virtual and augmented reality devices: effects on position control and gait biomechanics,” *PLoS one*, vol. 14, no. 12, p. e0225972, 2019.

[138] L. Sloot, M. Van der Krogt, and J. Harlaar, “Effects of adding a virtual reality environment to different modes of treadmill walking,” *Gait & posture*, vol. 39, no. 3, pp. 939–945, 2014.

[139] S. Kalantari, A. Mostafavi, T. B. Xu, A. S. Lee, and Q. Yang, "Comparing spatial navigation in a virtual environment vs. an identical real environment across the adult lifespan," *Computers in Human Behavior*, vol. 157, p. 108210, 2024.

[140] W. Jiang, M. Windl, B. Tag, Z. Sarsenbayeva, and S. Mayer, "An immersive and interactive vr dataset to elicit emotions," *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 11, pp. 7343–7353, 2024.

[141] A. Mucha, A. Pollak, and E. Wojtyna, "Effect of virtual walk in green or urban spaces on pain perception among healthy adults," *Scientific Reports*, vol. 15, no. 1, p. 27586, 2025.

[142] P. Suppakittpaisarn, C.-C. Wu, Y.-H. Tung, Y.-c. Yeh, C. Wanitchayapaisit, M. H. Browning, C.-Y. Chang, and W. C. Sullivan, "Durations of virtual exposure to built and natural landscapes impact self-reported stress recovery: evidence from three countries," *Landscape and ecological engineering*, vol. 19, no. 1, pp. 95–105, 2023.

[143] H.-G. Kim, E.-J. Cheon, D.-S. Bai, Y. H. Lee, and B.-H. Koo, "Stress and heart rate variability: a meta-analysis and review of the literature," *Psychiatry investigation*, vol. 15, no. 3, p. 235, 2018.

[144] J. Taelman, S. Vandeput, A. Spaepen, and S. Van Huffel, "Influence of mental stress on heart rate and heart rate variability," in *4th European Conference of the International Federation for Medical and Biological Engineering*, (Berlin, Heidelberg), pp. 1366–1369, Springer Berlin Heidelberg, 2009.

[145] K. Olsson, J. Salier Eriksson, H. Rosdahl, and P. Schantz, "Are heart rate methods based on ergometer cycling and level treadmill walking interchangeable?," *PLoS One*, vol. 15, no. 8, p. e0237388, 2020.

[146] S. G. Hart and L. E. Staveland, "Development of nasa-tlx (task load index): Results of empirical and theoretical research," in *Advances in psychology*, vol. 52, pp. 139–183, Elsevier, 1988.

[147] C. Winter, F. Kern, D. Gall, M. E. Latoschik, P. Pauli, and I. Käthner, "Immersive virtual reality during gait rehabilitation increases walking speed and motivation: a usability evaluation with healthy participants and patients with multiple sclerosis and stroke," *Journal of neuroengineering and rehabilitation*, vol. 18, no. 1, p. 68, 2021.

[148] T. Li, E. Velloso, A. Withana, and Z. Sarsenbayeva, "Estimating the effects of encumbrance and walking on mixed reality interaction," in *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, CHI '25, (New York, NY, USA), Association for Computing Machinery, 2025.

[149] I. Maramba, A. Chatterjee, and C. Newman, "Methods of usability testing in the development of ehealth applications: a scoping review," *International journal of medical informatics*, vol. 126, pp. 95–104, 2019.

[150] M. Hensher, P. Cooper, S. W. A. Dona, M. R. Angeles, D. Nguyen, N. Heynsbergh, M. L. Chatterton, and A. Peeters, "Scoping review: development and assessment of evaluation frameworks of mobile health apps for recommendations to consumers," *Journal of the American Medical Informatics Association*, vol. 28, no. 6, pp. 1318–1329, 2021.

[151] Y.-H. Chou, C. Lin, S.-H. Lee, Y.-W. C. Chien, and L.-C. Cheng, "Potential mobile health applications for improving the mental health of the elderly: A systematic review," *Clinical Interventions in Aging*, vol. 18, pp. 1523–1534, 2023. PMID: 37727447.

[152] R. Nussbaum, C. Kelly, E. Quinby, A. Mac, B. Parmanto, and B. E. Dicianno, "Systematic review of mobile health applications in rehabilitation," *Archives of physical medicine and rehabilitation*, vol. 100, no. 1, pp. 115–127, 2019.

[153] S. R. Steinhubl, E. D. Muse, and E. J. Topol, "The emerging field of mobile health," *Science translational medicine*, vol. 7, no. 283, pp. 283rv3–283rv3, 2015.

[154] S. Sadler, J. Gerrard, A. Searle, S. Lanting, M. West, R. Wilson, A. Ginige, K. Y. Fang, and V. Chuter, "The use of mhealth apps for the assessment and management of diabetes-related foot health outcomes: systematic review," *Journal of Medical Internet Research*, vol. 25, p. e47608, 2023.

[155] A. Piau, Y. Charlon, E. Campo, B. Vellas, F. Nourhashemi, *et al.*, "A smart insole to promote healthy aging for frail elderly individuals: specifications, design, and preliminary results," *JMIR rehabilitation and assistive technologies*, vol. 2, no. 1, p. e4084, 2015.

[156] N. Hegde, E. Melanson, and E. Sazonov, "Development of a real time activity monitoring android application utilizing smartstep," in *2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1886–1889, IEEE, 2016.

[157] B. C. Zapata, J. L. Fernández-Alemán, A. Idri, and A. Toval, "Empirical studies on usability of mhealth apps: a systematic literature review," *Journal of medical systems*, vol. 39, no. 2, p. 1, 2015.

[158] J. Brooke *et al.*, "Sus-a quick and dirty usability scale," *Usability evaluation in industry*, vol. 189, no. 194, pp. 4–7, 1996.

[159] W. Septiani, N. Rahmawati, D. M. Safitri, and M. Luis, "Usability evaluation for mobile health application: Systematic literature review," *Sinergi*, vol. 28, no. 2, pp. 287–304, 2024.

[160] Q. Wang, J. Liu, L. Zhou, J. Tian, X. Chen, W. Zhang, H. Wang, W. Zhou, and Y. Gao, "Usability evaluation of mhealth apps for elderly individuals: a scoping review," *BMC medical informatics and decision making*, vol. 22, no. 1, p. 317, 2022.

[161] K. Moulaei and A. Sheikhtaheri, "Usability evaluation of wearable technology: A pilot study on a smart diabetic shoe for foot care," *Clinical eHealth*, vol. 8, pp. 94–102, 2025.

[162] R. Ogrin, R. Viswanathan, T. Aylen, F. Wallace, J. Scott, and D. Kumar, "Co-design of an evidence-based health education diabetes foot app to prevent serious foot complications: a feasibility study," *Practical diabetes*, vol. 35, no. 6, pp. 203–209d, 2018.

[163] B. Strahm, C. M. Gray, and M. Vorvoreanu, "Generating mobile application onboarding insights through minimalist instruction," in *Proceedings of the 2018 Designing Interactive Systems Conference*, DIS '18, (New York, NY, USA), p. 361–372, Association for Computing Machinery, 2018.

[164] J. M. Carroll, *The Nurnberg funnel: Designing minimalist instruction for practical computer skill.* MIT press, 1990.

[165] A. Palmquist, "Design elements of conflict: A design study of a gamified smartphone application for employee onboarding," *Technology, Knowledge and Learning*, vol. 28, no. 3, pp. 1133–1173, 2023.

[166] R. E. Mayer, "The promise of multimedia learning: using the same instructional design methods across different media," *Learning and instruction*, vol. 13, no. 2, pp. 125–139, 2003.

[167] F. Eriksson, "Onboarding users to a voice user interface: Comparing different teaching methods for onboarding new users to intelligent personal assistants," 2018.

[168] C. Stoiber, M. Wagner, F. Grassinger, M. Pohl, H. Stitz, M. Streit, B. Potzmann, and W. Aigner, "Visualization onboarding grounded in educational theories," in *Visualization psychology*, pp. 139–164, Cham: Springer International Publishing, 2023.

[169] C.-y. Yu, "Enhancing usability and user experience through user onboarding in an ecological information app," in *Proceedings of the Future Technologies Conference* (K. Arai, ed.), (Cham), pp. 376–389, Springer, Springer Nature Switzerland, 2024.

[170] Y. Fu, Y. Hu, and V. Sundstedt, "A systematic literature review of virtual, augmented, and mixed reality game applications in healthcare," *ACM Trans. Comput. Healthcare*, vol. 3, Mar. 2022.

[171] L. Whittaker, "Onboarding and offboarding in virtual reality: A user-centred framework for audience experience across genres and spaces," *Convergence*, vol. 31, no. 1, pp. 87–106, 2025.

[172] E. Chauvergne, M. Hachet, and A. Prouzeau, "User onboarding in virtual reality: An investigation of current practices," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23, (New York, NY, USA), Association for Computing Machinery, 2023.

[173] F. Wu, J. Thomas, S. Chinnola, and E. S. Rosenberg, "Exploring communication modalities to support collaborative guidance in virtual reality," in *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 79–86, IEEE, 2020.

[174] B. Aruanno and F. Garzotto, “Memholo: mixed reality experiences for subjects with alzheimer’s disease,” *Multimedia Tools and Applications*, vol. 78, no. 10, pp. 13517–13537, 2019.

[175] T. Combe, R. Fribourg, L. Detto, and J.-M. Normand, “Exploring the influence of virtual avatar heads in mixed reality on social presence, performance and user experience in collaborative tasks,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 5, pp. 2206–2216, 2024.

[176] S. Hajahmadi, P. Cascarano, F. Mostajeran, K. Heuer, A. Lux, G. O. Mends-Cole, F. Steinicke, and G. Marfia, “Investigating the impact of voice-only and embodied conversational virtual agents on mixed reality puzzle solving,” in *2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, pp. 602–612, IEEE, 2025.

[177] V. Schwind, P. Knierim, C. Tasci, P. Franczak, N. Haas, and N. Henze, “”these are not my hands!”: Effect of gender on the perception of avatar hands in virtual reality,” in *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, CHI ’17, (New York, NY, USA), p. 1577–1582, Association for Computing Machinery, 2017.

[178] V. Schwind, L. Lin, M. Di Luca, S. Jörg, and J. Hillis, “Touch with foreign hands: the effect of virtual hand appearance on visual-haptic integration,” in *Proceedings of the 15th ACM Symposium on Applied Perception*, SAP ’18, (New York, NY, USA), Association for Computing Machinery, 2018.

[179] K. Ma and B. Hommel, “The virtual-hand illusion: effects of impact and threat on perceived ownership and affective resonance,” *Frontiers in psychology*, vol. 4, p. 604, 2013.

[180] F.-C. Yang, P. Acevedo, S. Guo, M. Choi, and C. Mousas, “Embodied conversational agents in extended reality: A systematic review,” *IEEE Access*, vol. 13, pp. 79805–79824, 2025.

[181] P. Weichbroth, “Usability issues with mobile applications: Insights from practitioners and future research directions,” *IEEE Access*, vol. 13, pp. 91301–91311, 2025.

[182] K. M. Edwards, B. Man, and F. Ahmed, “Sketch2prototype: Rapid conceptual design exploration and prototyping with generative ai,” *Proceedings of the Design Society*, vol. 4, pp. 1989–1998, May 2024.

[183] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Proceedings of the 31st International Conference on Neural Information Processing Systems*, NIPS’17, (Red Hook, NY, USA), p. 6000–6010, Curran Associates Inc., 2017.

[184] R. Brisco, L. Hay, and S. Dhami, “Exploring the role of text-to-image ai in concept generation,” *Proceedings of the Design Society*, vol. 3, pp. 1835–1844, June 2023.

[185] J. White, Q. Fu, S. Hays, M. Sandborn, C. Olea, H. Gilbert, A. Elnashar, J. Spencer-Smith, and D. C. Schmidt, “A prompt pattern catalog to enhance prompt engineering with chatgpt,” *arXiv*, 2023.

[186] V. Liu and L. B. Chilton, “Design guidelines for prompt engineering text-to-image generative models,” in *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, CHI ’22, (New York, NY, USA), Association for Computing Machinery, 2022.

[187] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, “Generative adversarial networks,” *Communications of the ACM*, vol. 63, p. 139–144, October 2020.

[188] L.-Y. Chiou, P.-K. Hung, R.-H. Liang, and C.-T. Wang, “Designing with ai: An exploration of co-ideation with image generators,” in *Proceedings of the 2023 ACM Designing Interactive Systems Conference*, DIS ’23, (New York, NY, USA), p. 1941–1954, Association for Computing Machinery, 2023.

[189] M. Barros and Q. Ai, “Designing with words: exploring the integration of text-to-image models in industrial design,” *Digital Creativity*, vol. 35, pp. 378–391, October 2024.

[190] Y.-H. Lee and C.-Y. Chiu, “The impact of ai text-to-image generator on product styling design,” in *Human Interface and the Management of Information* (H. Mori and Y. Asahi, eds.), (Cham), pp. 502–515, Springer Nature Switzerland, 2023.

[191] A. Cai, S. R. Rick, J. L. Heyman, Y. Zhang, A. Filipowicz, M. Hong, M. Klenk, and T. Malone, “Designaid: Using generative ai and semantic diversity for design inspiration,” in *Proceedings of The ACM Collective Intelligence Conference*, CI ’23, (New York, NY, USA), p. 1–11, Association for Computing Machinery, 2023.

[192] M. Elasri, O. Elharrouss, S. Al-Maadeed, and H. Tairi, “Image generation: A review,” *Neural Processing Letters*, vol. 54, no. 5, pp. 4609–4646, 2022.

[193] K. A. Bartlett and J. D. Camba, “Generative artificial intelligence in product design education: Navigating concerns of originality and ethics,” *International Journal of Interactive Multimedia and Artificial Intelligence*, vol. 8, pp. 55–64, March 2024.

[194] L. E. Dunne, H. Profita, C. Zeagler, J. Clawson, S. Gilliland, E. Y.-L. Do, and J. Budd, “The social comfort of wearable technology and gestural interaction,” in *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, (Washington, DC, USA), pp. 4159–4162, IEEE Computer Society, 2014.

[195] L. Dunne, H. Profita, and C. Zeagler, “Chapter 1.2 - social aspects of wearability and interaction,” in *Wearable Sensors* (E. Sazonov and M. R. Neuman, eds.), pp. 25–43, Oxford: Academic Press, 2014.

[196] H. Yang, J. Yu, H. Zo, and M. Choi, “User acceptance of wearable devices: An extended perspective of perceived value,” *Telematics and Informatics*, vol. 33, no. 2, pp. 256–269, 2016.

[197] H. Profita, R. Albaghli, L. Findlater, P. Jaeger, and S. K. Kane, "The at effect: How disability affects the perceived social acceptability of head-mounted display use," in *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, CHI '16, (New York, NY, USA), p. 4884–4895, Association for Computing Machinery, 2016.

[198] N. Kelly and S. B. Gilbert, "The wearer, the device, and its use: advances in understanding the social acceptability of wearables," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 62, pp. 1027–1031, September 2018.

[199] A. I. Canhoto and S. Arp, "Exploring the factors that support adoption and sustained use of health and fitness wearables," *Journal of Marketing Management*, vol. 33, pp. 32–60, October 2017.

[200] O. El-Gayar and A. Elnoshokaty, "Factors and design features influencing the continued use of wearable devices," *Journal of Healthcare Informatics Research*, vol. 7, pp. 359–385, July 2023.

[201] A. Adapa, F. F.-H. Nah, R. H. Hall, K. Siau, and S. N. Smith, "Factors influencing the adoption of smart wearable devices," *International Journal of Human–Computer Interaction*, vol. 34, pp. 399–409, September 2018.

[202] T. Liao, K. Tanner, and E. F. MacDonald, "Revealing insights of users' perception: an approach to evaluate wearable products based on emotions," *Design Science*, vol. 6, p. e14, July 2020.

[203] F. D. Davis, R. P. Bagozzi, and P. R. Warshaw, "User acceptance of computer technology: A comparison of two theoretical models," *Management science*, vol. 35, no. 8, pp. 982–1003, 1989.

[204] V. Venkatesh, M. G. Morris, G. B. Davis, and F. D. Davis, "User acceptance of information technology: Toward a unified view," *MIS Quarterly*, vol. 27, pp. 425–478, September 2003.

[205] S. T. Fiske, A. J. Cuddy, P. Glick, and J. Xu, *A model of (often mixed) stereotype content: Competence and warmth respectively follow from perceived status and competition*. London: Routledge, January 2018.

[206] N. Kelly, "All the world's a stage: what makes a wearable socially acceptable," *Interactions*, vol. 24, p. 56–60, October 2017.

[207] V. Venkatesh and H. Bala, "Technology acceptance model 3 and a research agenda on interventions," *Decision sciences*, vol. 39, pp. 273–315, May 2008.

[208] N. Kelly and S. Gilbert, "The wear scale: Developing a measure of the social acceptability of a wearable device," in *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*, CHI EA '16, (New York, NY, USA), p. 2864–2871, Association for Computing Machinery, 2016.

[209] A. J. Cuddy, S. T. Fiske, and P. Glick, *Warmth and Competence as Universal Dimensions of Social Perception: The Stereotype Content Model and the BIAS Map*, vol. 40. Academic Press, 2008.

[210] V. Schwind, N. Deierlein, R. Poguntke, and N. Henze, "Understanding the social acceptability of mobile devices using the stereotype content model," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, (New York, NY, USA), p. 1–12, Association for Computing Machinery, 2019.

[211] A. J. Cuddy, S. T. Fiske, V. S. Kwan, P. Glick, S. Demoulin, J.-P. Leyens, M. H. Bond, J.-C. Croizet, N. Ellemers, E. Sleebos, *et al.*, "Stereotype content model across cultures: Towards universal similarities and some differences," *British journal of social psychology*, vol. 48, no. 1, pp. 1–33, 2009.

[212] H. P. Profita, J. Clawson, S. Gilliland, C. Zeagler, T. Starner, J. Budd, and E. Y.-L. Do, "Don't mind me touching my wrist: a case study of interacting with on-body technology in public," in *Proceedings of the 2013 International Symposium on Wearable Computers*, ISWC '13, (New York, NY, USA), p. 89–96, Association for Computing Machinery, 2013.

[213] C. Harrison, B. Y. Lim, A. Shick, and S. E. Hudson, "Where to locate wearable displays? reaction time performance of visual alerts from tip to toe," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI '09, (New York, NY, USA), p. 941–944, Association for Computing Machinery, 2009.

[214] F. Gemperle, C. Kasabach, J. Stivoric, M. Bauer, and R. Martin, "Design for wearability," in *Proceedings of the 2nd IEEE International Symposium on Wearable Computers*, ISWC '98, (Washington, DC, USA), pp. 116–122, IEEE Computer Society, 1998.

[215] O. Postolache, J. M. Dias Pereira, V. Viegas, L. Pedro, P. S. Girao, R. Oliveira, and G. Postolache, "Smart walker solutions for physical rehabilitation," *IEEE Instrumentation & Measurement Magazine*, vol. 18, no. 5, pp. 21–30, 2015.

[216] M. Martins, C. P. Santos, A. Frizera, A. Matias, T. Pereira, M. Cotter, and F. Pereira, "Smart walker use for ataxia's rehabilitation: Case study," in *2015 IEEE International Conference on Rehabilitation Robotics (ICORR)*, pp. 852–857, IEEE, 8/11/2015 - 8/14/2015.

[217] M. M. Martins, C. P. Santos, A. Frizera-Neto, and R. Ceres, "Assistive mobility devices focusing on smart walkers: Classification and review," *Robotics and Autonomous Systems*, vol. 60, no. 4, pp. 548–562, 2012.

[218] E. Costamagna, S. Thies, L. Kenney, D. Howard, A. Liu, and D. Ogden, "A generalisable methodology for stability assessment of walking aid users," *Medical Engineering & Physics*, vol. 47, pp. 167–175, 2017.

[219] C.-E. Panazan and E.-H. Dulf, "Intelligent cane for assisting the visually impaired," *Technologies*, vol. 12, no. 6, p. 75, 2024.

[220] M. P. Menikdiwela, K. Dharmasena, and A. H. S. Abeykoon, “Haptic based walking stick for visually impaired people,” in *2013 International conference on Circuits, Controls and Communications (CCUBE)*, pp. 1–6, 2013.

[221] M. Huang and E. Clancy, “Smart walker: an imu-based device for patient activity logging and fall detection,” in *2022 IEEE Signal Processing in Medicine and Biology Symposium (SPMB)*, pp. 1–4, 2022.

[222] A. Frizera, R. Ceres, J. L. Pons, A. Abellanas, and R. Raya, “The smart walkers as geriatric assistive device. the simbiosis purpose,” in *Proceedings of the 6th International Conference of the International Society for Gerontechnology*, vol. 7, pp. 1–6, 2008.

[223] V. Kaushalya, K. Premarathne, H. Shadir, P. Krithika, and S. Fernando, “Automated help aid for visually impaired people using obstacle detection and gps technology: Akshi,” *Symposium on Statistical & Computational Modelling with Applications*, 2016.

[224] M. M. Islam, M. Sheikh Sadi, K. Z. Zamli, and M. M. Ahmed, “Developing walking assistants for visually impaired people: A review,” *IEEE Sensors Journal*, vol. 19, no. 8, pp. 2814–2828, 2019.

[225] J. Engel, A. Amir, E. Messer, and I. Caspi, “Walking cane designed to assist partial weight bearing.,” *Archives of physical medicine and rehabilitation*, vol. 64, no. 8, pp. 386–388, 1983.

[226] P. J. McCandless, B. J. Evans, J. Janssen, J. Selfe, A. Churchill, and J. Richards, “Effect of three cueing devices for people with parkinson’s disease with gait initiation difficulties,” *Gait & Posture*, vol. 44, pp. 7–11, 2016.

[227] D. A. Oladele, E. D. Markus, and A. M. Abu-Mahfouz, “Adaptability of assistive mobility devices and the role of the internet of medical things: Comprehensive review,” *JMIR Rehabilitation and Assistive Technologies*, vol. 8, no. 4, p. e29610, 2021.

[228] T.-N. Doan, E. Schroter, and T.-B. Phan, “Fall detection using intelligent walking-aids and machine learning methods,” in *Intelligent Systems and Data Science* (N. Thai-Nghe, T.-N. Do, and P. Haddawy, eds.), (Singapore), pp. 95–109, Springer Nature Singapore, 2024.

[229] J. Wang, X. Jiang, Q. Meng, M. Saada, and H. Cai, “Walking motion real-time detection method based on walking stick, iot, copod and improved lightgbm,” *Applied Intelligence*, vol. 52, no. 14, pp. 16398–16416, 2022.

[230] A. C. Tricco, E. Lillie, W. Zarin, K. K. O’Brien, H. Colquhoun, D. Levac, D. Moher, M. D. Peters, T. Horsley, L. Weeks, *et al.*, “Prisma extension for scoping reviews (prisma-scr): Checklist and explanation,” *Annals of Internal Medicine*, vol. 169, no. 7, pp. 467–473, 2018. PMID: 30178033.

[231] Zotero, “Free and open source reference manager - your personal research assistant.” `https://www.zotero.org/`, 2025. Accessed: 10 July 2025.

[232] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.

[233] P. Mayring, "Qualitative content analysis," *Forum Qualitative Sozialforschung / Forum: Qualitative Social Research*, vol. 1, Jun. 2000.

[234] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, *et al.*, "The prisma 2020 statement: an updated guideline for reporting systematic reviews," *bmj*, vol. 372, 2021.

[235] N. R. Haddaway, M. J. Page, C. C. Pritchard, and L. A. McGuinness, "Prisma2020: An r package and shiny app for producing prisma 2020-compliant flow diagrams, with interactivity for optimised digital transparency and open synthesis," *Campbell systematic reviews*, vol. 18, no. 2, p. e1230, 2022.

[236] M. Ojeda, A. Cortés, J. Béjar, and U. Cortés, "Automatic classification of gait patterns using a smart rollator and the boss model," in *Proceedings of the 11th PErvasive Technologies Related to Assistive Environments Conference*, PETRA '18, (New York, NY, USA), p. 384–390, Association for Computing Machinery, 2018.

[237] E. Sardini, M. Serpelloni, and M. Lancini, "Wireless instrumented crutches for force and movement measurements for gait monitoring," *IEEE Transactions on Instrumentation and Measurement*, vol. 64, no. 12, pp. 3369–3379, 2015.

[238] V. Viegas, J. M. Dias Pereira, O. Postolache, and P. S. Girão, "Spy walker: A convenient way to assess gait in walker assistive devices," in *2018 IEEE International Instrumentation and Measurement Technology Conference (I2MTC)*, pp. 1–6, 2018.

[239] S. Gill, S. Nssk, N. Seth, and E. Scheme, "Design of a smart iot-enabled walker for deployable activity and gait monitoring," in *2018 IEEE Life Sciences Conference (LSC)*, pp. 183–186, 2018.

[240] J. Ballesteros, C. Urdiales, A. B. Martinez, and M. Tirado, "Automatic assessment of a rollator-user's condition during rehabilitation using the i-walker platform," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 11, pp. 2009–2017, 2017.

[241] J. Wade, M. Beccani, A. Myszka, E. Bekele, P. Valdastri, P. Flemming, M. de Riesthal, T. Withrow, and N. Sarkar, "Design and implementation of an instrumented cane for gait recognition," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 5904–5909, 2015.

[242] F. Mekki, M. Borghetti, E. Sardini, and M. Serpelloni, "Wireless instrumented cane for walking monitoring in parkinson patients," in *2017 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*, pp. 414–419, 2017.

[243] M. Narváez, M. Salazar, and J. Aranda, "Identification of gait patterns in walking with crutches through the selection of significant spatio-temporal parameters," in *2022 International Conference on Rehabilitation Robotics (ICORR)*, pp. 1–6, 2022.

[244] J. W. Wade, R. Boyles, P. Flemming, A. Sarkar, M. de Riesthal, T. J. Withrow, and N. Sarkar, "Feasibility of automated mobility assessment of older adults via an instrumented cane," *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 4, pp. 1631–1638, 2019.

[245] O. Postolache, V. Viegas, and J. M. D. Pereira, "Gait assesment using microwave radars mounted on standard walker," in *2024 IEEE International Conference And Exposition On Electric And Power Engineering (EPEi)*, pp. 687–692, 2024.

[246] J.-L. Zhou, W.-F. Li, Q. Zhang, F. Xie, and Q. Wang, "A smart walking stick for gait analysis of elderly and people with disabilities," *IEEE Sensors Journal*, vol. 22, no. 9, pp. 9035–9045, 2022.

[247] S. Gill, J. Hearn, G. Powell, and E. Scheme, "Design of a multi-sensor iot-enabled assistive device for discrete and deployable gait monitoring," in *2017 IEEE Healthcare Innovations and Point of Care Technologies (HI-POCT)*, pp. 216–220, 2017.

[248] O. Postolache, "Physical rehabilitation assessment based on smart training equipment and mobile apps," in *2015 E-Health and Bioengineering Conference (EHB)*, pp. 1–6, 2015.

[249] J. Ballesteros, C. Urdiales, A. B. Martinez, and M. Tirado, "Online estimation of rollator user condition using spatiotemporal gait parameters," in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 3180–3185, 2016.

[250] P. Batoca, O. Postolache, and A. Correia, "Physical therapy gait assessment based on smart sensing and cloud services," in *2022 International Symposium on Sensing and Instrumentation in 5G and IoT Era (ISSI)*, pp. 138–143, 2022.

[251] J. Ballesteros, A. Tudela, J. R. Caro-Romero, and C. Urdiales, "A cane-based low cost sensor to implement attention mechanisms in telecare robots," in *2019 International Conference on Robotics and Automation (ICRA)*, pp. 1473–1478, 2019.

[252] D. G. Domingues, S. Funghetto, M. R. Miranda, P. K. C. M. Batista, P. R. F. de Oliveira, G. A. Assis, A. F. da Rocha, and R. d. S. Torres, "Mobility and freedom: Affective cane for expanded sensorium and embodied cognition," in *2017 23rd International Conference on Virtual System & Multimedia (VSMM)*, pp. 1–7, 2017.

[253] P. R. Culmer, P. C. Brooks, D. N. Strauss, D. H. Ross, M. C. Levesley, R. J. O'Connor, and B. B. Bhakta, "An instrumented walking aid to assess and retrain gait," *IEEE/ASME Transactions on Mechatronics*, vol. 19, no. 1, pp. 141–148, 2014.

[254] A. Valsangkar, P. Kumar, and E. Scheme, “Automated segmentation of a timed up and go test using an instrumented cane,” in *2021 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, pp. 1–5, 2021.

[255] G. G. Chalvatzaki, G. Pavlakos, K. Maninis, X. S. Papageorgiou, V. Pitsikalis, C. S. Tzafestas, and P. Maragos, “Towards an intelligent robotic walker for assisted living using multimodal sensorial data,” in *2014 4th International Conference on Wireless Mobile Communication and Healthcare - Transforming Healthcare Through Innovations in Mobile and Wireless Technologies (MOBIHEALTH)*, pp. 156–159, 2014.

[256] A. B. Mesanza, S. Lucas, A. Zubizarreta, I. Cabanes, E. Portillo, and A. Rodriguez-Larrad, “A machine learning approach to perform physical activity classification using a sensorized crutch tip,” *IEEE Access*, vol. 8, pp. 210023–210034, 2020.

[257] C. Seylan and U. Saranlı, “Estimation of ground reaction forces using low-cost instrumented forearm crutches,” *IEEE Transactions on Instrumentation and Measurement*, vol. 67, no. 6, pp. 1308–1316, 2018.

[258] A. D. C. Chan and J. R. Green, “Smart rollator prototype,” in *2008 IEEE International Workshop on Medical Measurements and Applications*, pp. 97–100, 2008.

[259] J. Ballesteros, C. Urdiales, A. B. Martinez, and M. Tirado, “Gait analysis for challenged users based on a rollator equipped with force sensors,” in *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 5587–5592, 2015.

[260] P. M. V. L. Frango and O. A. Postolache, “Mobile application based on wireless sensor network for physical rehabilitation,” in *2018 International Symposium in Sensing and Instrumentation in IoT Era (ISSI)*, pp. 1–6, 2018.

[261] V. Arcobelli, M. Zauli, L. De Marchi, L. Chiari, and S. Mellone, “Assessment of crutch-assisted walking with sensorized crutches in a 6-minute walk test,” in *2023 IEEE EMBS Special Topic Conference on Data Science and Engineering in Healthcare, Medicine and Biology*, pp. 11–12, 2023.

[262] M. S. Cavaglià, S. D. Sierra M, L. Palmerini, S. Orlandi, M. Múnera, and C. A. Cifuentes, “Towards safer mobility: Developing and evaluating a fall detection system for a smart walker,” in *2024 10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, pp. 1623–1628, 2024.

[263] E. O. Zambrano, K. R. Muñoz, J. Armas-Aguirre, and P. A. González, “Technological architecture with low cost sensors to improve physical therapy monitoring,” in *2020 15th Iberian Conference on Information Systems and Technologies (CISTI)*, pp. 1–6, 2020.

[264] N. F. Ribeiro and C. P. Santos, "Two fall-related and kinematic data-based approaches for an instrumented conventional cane," *IEEE Transactions on Human-Machine Systems*, vol. 51, no. 5, pp. 554–563, 2021.

[265] A. Anushree, D. Nesakumar, R. Ajeet, S. Gopika, V. Haritha, and K. Pavithra, "Posturesense intelligent body support crutch tool," in *2024 International Conference on Computing and Data Science (ICCDS)*, pp. 1–6, 2024.

[266] T. Kikuchi, T. Tanaka, K. Anzai, S. Kawakami, M. Hosaka, and K. N. and, "Evaluation of line-tracing controller of intelligently controllable walker," *Advanced Robotics*, vol. 27, no. 7, pp. 493–502, 2013.

[267] J. Ballesteros, J. M. Peula, A. B. Martinez, and C. Urdiales, "Automatic fall risk assessment for challenged users obtained from a rollator equipped with force sensors and a rgb-d camera," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 7356–7361, 2018.

[268] B. Padmavathi, S. L. Y. Sahithi, V. Sudarshan, and K. N. Nandini, "An effective body posture management system using novel intelligent crutch tool mechanism," in *2023 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI)*, pp. 1–6, 2023.

[269] S. Koziol, P. A. Friedman, B. Schultze, and K. Schubert, "The reminding walker cognitive orthotic device," in *2017 International Symposium on Wearable Robotics and Rehabilitation (WeRob)*, pp. 1–2, 2017.

[270] R. Gerena, P. VanDeventer, A. Vistamehr, C. Conroy, P. Freeborn, H. Govin, E. Fox, and J. Aceros, "Wireless instrumented walker for remote rehabilitation monitoring," in *2015 IEEE Virtual Conference on Applications of Commercial Sensors (VCACS)*, pp. 1–7, 2015.

[271] A. M. Barbara and A. Grobelna, "Orthopedic footwear for people with lower-limb conditions," *Canadian Journal of Health Technologies*, vol. 2, no. 1, 2022.

[272] K. Bertrand, M.-H. Raymond, W. C. Miller, K. A. M. Ginis, and L. Demers, "Walking aids for enabling activity and participation: a systematic review," *American Journal of Physical Medicine & Rehabilitation*, vol. 96, no. 12, pp. 894–903, 2017.

[273] J. Burns, C. Wegener, L. Begg, M. Vicaretti, and J. Fletcher, "Randomized trial of custom orthoses and footwear on foot pain and plantar pressure in diabetic peripheral arterial disease," *Diabetic medicine*, vol. 26, no. 9, pp. 893–899, 2009.

[274] L. A. de Jong, Y. L. Kerkum, V. C. Altmann, A. C. Geurts, and N. L. Keijsers, "Orthopedic footwear has a positive influence on gait adaptability in individuals with hereditary motor and sensory neuropathy," *Gait & Posture*, vol. 106, pp. 18–22, 2023.

[275] E. S. Arch and S. J. Stanhope, *Orthotic Device Research*, pp. 99–116. New York, NY: Springer New York, 2017.

[276] S. Resch and A. Carroll, "Influence of virtual footwear on gait behavior, embodiment, and workload while walking on a treadmill in virtual reality," in *Proceedings of the 2025 31st ACM Symposium on Virtual Reality Software and Technology*, VRST '25, (New York, NY, USA), Association for Computing Machinery, 2025.

[277] F. Salis, S. Bertuletti, T. Bonci, M. Caruso, K. Scott, L. Alcock, E. Buckley, E. Gazit, C. Hansen, L. Schwickert, *et al.*, "A multi-sensor wearable system for the assessment of diseased gait in real-world conditions," *Frontiers in Bioengineering and Biotechnology*, vol. 11, p. 1143248, 2023.

[278] R. G. Frykberg, L. A. Lavery, H. Pham, C. Harvey, L. Harkless, and A. Veves, "Role of neuropathy and high foot pressures in diabetic foot ulceration," *Diabetes care*, vol. 21, no. 10, pp. 1714–1719, 1998.

[279] U. Alam, D. R. Riley, R. S. Jugdey, S. Azmi, S. Rajbhandari, K. D'Août, and R. A. Malik, "Diabetic neuropathy and gait: a review," *Diabetes therapy*, vol. 8, pp. 1253–1264, 2017.

[280] L. C. Rogers, R. G. Frykberg, D. G. Armstrong, A. J. Boulton, M. Edmonds, G. H. Van, A. Hartemann, F. Game, W. Jeffcoate, A. Jirkovska, *et al.*, "The charcot foot in diabetes," *Journal of the American Podiatric Medical Association*, vol. 101, no. 5, pp. 437–446, 2011.

[281] C. Zhou, L. Yang, H. Liao, B. Liang, and X. Ye, "Ankle foot motion recognition based on wireless wearable semg and acceleration sensors for smart afo," *Sensors and Actuators A: Physical*, vol. 331, p. 113025, 2021.

[282] Deutsches Institut für Normung e.V., "Assistive products for walking manipulated by one arm – requirements and test methods – part 1: Elbow crutches (iso 11334-1:2007)." `https://dx.doi.org/10.31030/9777139`, 2007. German version EN ISO 11334-1:2007.

[283] I. Electronics, "Fsr® 400 short." `https://www.interlinkelectronics.com/fsr-400-short`. (accessed on 15 August 2025).

[284] Interlink Electronics, *FSR® Integration Guide*. Interlink Electronics, Camarillo, CA, USA, 2013. Document Part Number 94-00004 Rev. C. Available upon request from Interlink Electronics.

[285] B. S. GmbH, "Small. low power. optimized for wearables -inertial measurement unit: Bmi270." `https://www.bosch-sensortec.com/products/motion-sensors/imus/bmi270/`. (accessed on 15 August 2025).

[286] Bosch Sensortec GmbH, *BMI270 Datasheet*. Bosch Sensortec GmbH, Gerhard-Kindler-Straße 9, 72770 Reutlingen, Germany, revision 1.5 ed., Mar. 2023. Document number: BST-BMI270-DS000-07.

[287] B. Mariani, H. Rouhani, X. Crevoisier, and K. Aminian, "Quantitative estimation of foot-flat and stance phase of gait using foot-worn inertial sensors," *Gait & posture*, vol. 37, no. 2, pp. 229–234, 2013.

[288] A. M. (Google), "Turbo, an improved rainbow colormap for visualization." `https://research.google/blog/turbo-an-improved-rainbow-colormap-for-visualization/`, August 2019. (accessed on 15 August 2025).

[289] A. Studio, "Android studio – the official ide for android development." `https://developer.android.com/studio`. (accessed on 15 August 2025).

[290] A. Developers, "Build better apps faster with jetpack compose." `https://developer.android.com/compose`. (accessed on 15 August 2025).

[291] V. Schwind, S. Resch, and J. Sehrt, "The hci user studies toolkit: Supporting study designing and planning for undergraduates and novice researchers in human-computer interaction," in *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI EA '23, (New York, NY, USA), Association for Computing Machinery, 2023.

[292] A. Kassambara, *rstatix: Pipe-Friendly Framework for Basic Statistical Tests*. The R Foundation, 2023. R package version 0.7.2.

[293] R. Likert, "A technique for the measurement of attitudes.," *Archives of psychology*, 1932.

[294] A. Bangor, P. Kortum, and J. Miller, "Determining what individual sus scores mean: Adding an adjective rating scale," *Journal of usability studies*, vol. 4, no. 3, pp. 114–123, 2009.

[295] J. Cohen, *Statistical power analysis for the behavioral sciences*. routledge, 2013.

[296] A. Ganguly, B. A. Olmanson, C. B. Knowlton, M. A. Wimmer, and C. Ferrigno, "Accuracy of the fully integrated insole3's estimates of spatiotemporal parameters during walking," *Medical Engineering & Physics*, vol. 111, p. 103925, 2023.

[297] M. Hyzy, R. Bond, M. Mulvenna, L. Bai, A. Dix, S. Leigh, and S. Hunt, "System usability scale benchmarking for digital health apps: Meta-analysis," *JMIR Mhealth Uhealth*, vol. 10, p. e37290, Aug 2022.

[298] B. Alzghoul, "The effectiveness of gamification in changing health-related behaviors: A systematic review and meta-analysis," *The Open Public Health Journal*, vol. 17, no. 1, 2024.

[299] N. Gurari, K. Smith, M. Madhav, and A. M. Okamura, "Environment discrimination with vibration feedback to the foot, arm, and fingertip," in *2009 IEEE International Conference on Rehabilitation Robotics*, pp. 343–348, 2009.

[300] S. Choi and K. J. Kuchenbecker, "Vibrotactile display: Perception, technology, and applications," *Proceedings of the IEEE*, vol. 101, no. 9, pp. 2093–2104, 2013.

[301] S. Jin, B. Kim, and K. Han, ""i don't know why i should use this app": Holistic analysis on user engagement challenges in mobile mental health," in *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, CHI '25, (New York, NY, USA), Association for Computing Machinery, 2025.

[302] M. N. Islam, M. M. Karim, T. T. Inan, and A. N. Islam, "Investigating usability of mobile health applications in bangladesh," *BMC medical informatics and decision making*, vol. 20, no. 1, p. 19, 2020.

[303] O. Oyebode, F. Alqahtani, and R. Orji, "Using machine learning and thematic analysis methods to evaluate mental health apps based on user reviews," *IEEE Access*, vol. 8, pp. 111141–111158, 2020.

[304] A. Chiappetta, "Designing effective user onboarding experiences for mobile applications." PhD thesis, Politecnico di Milano, 2019.

[305] H. J. Klein, B. Polin, and K. Leigh Sutton, "Specific onboarding practices for the socialization of new employees," *International Journal of Selection and Assessment*, vol. 23, no. 3, pp. 263–283, 2015.

[306] A. Snell, "Researching onboarding best practice: Using research to connect onboarding processes with employee satisfaction," *Strategic HR Review*, vol. 5, no. 6, pp. 32–35, 2006.

[307] M. Froehlich, C. Kobiella, A. Schmidt, and F. Alt, "Is it better with onboarding? improving first-time cryptocurrency app experiences," in *Proceedings of the 2021 ACM Designing Interactive Systems Conference*, DIS '21, (New York, NY, USA), p. 78–89, Association for Computing Machinery, 2021.

[308] L. R. Kascak, C. B. Rébola, and J. A. Sanford, "Integrating universal design (ud) principles and mobile design guidelines to improve design of mobile health applications for older adults," in *2014 IEEE International Conference on Healthcare Informatics*, pp. 343–348, IEEE, 2014.

[309] L. R. Kascak, S. T. Lee, E. Y. Liu, and J. A. Sanford, "Universal design (ud) guidelines for interactive mobile voting interfaces for older adults," in *Universal Access in Human-Computer Interaction. Access to the Human Environment and Culture* (M. Antona and C. Stephanidis, eds.), (Cham), pp. 215–225, Springer International Publishing, 2015.

[310] L. Ruzic, C. Harrington, and J. Sanford, "Universal design mobile interface guidelines for mobile health and wellness apps for an aging population including people aging with disabilities," *Int J Adv Softw*, vol. 10, no. 3, pp. 372–84, 2017.

[311] J. Woodward, Y.-P. Chen, K. Jurczyk, K. M. Ross, L. Anthony, and J. Ruiz, "A survey of notification designs in commercial mhealth apps," in *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI EA '21, (New York, NY, USA), Association for Computing Machinery, 2021.

[312] Y. Cao, X. Qian, T. Wang, R. Lee, K. Huo, and K. Ramani, "An exploratory study of augmented reality presence for tutoring machine tasks," in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, CHI '20, (New York, NY, USA), p. 1–13, Association for Computing Machinery, 2020.

[313] G. Huang, X. Qian, T. Wang, F. Patel, M. Sreeram, Y. Cao, K. Ramani, and A. J. Quinn, "Adaptutar: An adaptive tutoring system for machine tasks in augmented reality," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI '21, (New York, NY, USA), Association for Computing Machinery, 2021.

[314] G. Manfredi, U. Erra, and G. Gilio, "A mixed reality approach for innovative pair programming education with a conversational ai virtual avatar," in *Proceedings of the 27th International Conference on Evaluation and Assessment in Software Engineering*, EASE '23, (New York, NY, USA), p. 450–454, Association for Computing Machinery, 2023.

[315] J. Kim, S. Agarwal, K. Marotta, S. Li, J. Leo, and D. H. Chau, "Mixed reality for learning programming," in *Proceedings of the 18th ACM International Conference on Interaction Design and Children*, IDC '19, (New York, NY, USA), p. 574–579, Association for Computing Machinery, 2019.

[316] K. Fang and J. Wang, "Interactive design with gesture and voice recognition in virtual teaching environments," *IEEE Access*, vol. 12, pp. 4213–4224, 2024.

[317] Z. Pi, J. Hong, and J. Yang, "Effects of the instructor's pointing gestures on learning performance in video lectures," *British Journal of Educational Technology*, vol. 48, no. 4, pp. 1020–1029, 2017.

[318] J. V. Bradley, "Complete counterbalancing of immediate sequential effects in a latin square design," *Journal of the American Statistical Association*, vol. 53, no. 282, pp. 525–528, 1958.

[319] Meta, "Meta xr all-in-one sdk." Unity Asset Store, `https://assetstore.unity.com/packages/tools/integration/meta-xr-all-in-one-sdk-269657`, 2025. Accessed: January 19, 2026.

[320] T. Yang and H. Luo, "Learning experience questionnaire for multimedia case-based instruction: Development, validation and implications for design," in *Blended Learning. New Challenges and Innovative Practices*, (Cham), pp. 400–411, Springer, Springer International Publishing, 2017.

[321] M. Schrepp and H. Santoso, "Construction of a questionnaire to measure the learner experience in online tutorials," in *Proceedings of the 20th International Conference on Web Information Systems and Technologies, WEBIST 2024* (F. Penalvo, K. Aberer, and M. Marchiori, eds.), International Conference on Web Information Systems and Technologies, WEBIST - Proceedings, (Portugal), pp. 315–322, Science and Technology Publications, Lda, 2024.

[322] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative research in psychology*, vol. 3, no. 2, pp. 77–101, 2006.

[323] K. Caine, "Local standards for sample size at chi," in *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, CHI '16, (New York, NY, USA), p. 981–992, Association for Computing Machinery, 2016.

[324] M. D. Cohen and P. Bacdayan, "Organizational routines are stored as procedural memory: Evidence from a laboratory study," *Organization science*, vol. 5, no. 4, pp. 554–568, 1994.

[325] F. Vona, J. Schorlemmer, M. Stern, N. Ashrafi, M. Vergari, T. Kojic, and J.-N. Voigt-Antons, "Comparing pass-through quality of mixed reality devices: A user experience study during real-world tasks," in *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1432–1433, 2025.

[326] C. Sayffaerth, A. Köhler, J. Rasch, A. Schmidt, and F. Müller, "Through the expert's eyes: Exploring asynchronous expert perspectives and gaze visualizations in xr," in *2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, (Daejeon, Korea), pp. 1311–1321, IEEE, 2025.

[327] S. Resch, J.-G. Hanania, D. Schor, E. Johannes, B.-J. Kovac, V. Schwind, D. Völz, and D. Sanchez-Morillo, "Foot placement feedback in physical training: Effects of spatial user interfaces on performance and workload in virtual reality," in *Proceedings of the 2025 ACM Symposium on Spatial User Interaction*, SUI '25, (New York, NY, USA), Association for Computing Machinery, Nov. 2025.

[328] S. Resch, R. B. Tiwari, H. R. Vankawala, P. Singh, M. Rafati, V. Schwind, D. Völz, and D. Sanchez-Morillo, "The impact of visual feedback and avatar presence on balance in virtual reality," in *Proceedings of Mensch Und Computer 2024*, MuC '24, (New York, NY, USA), p. 555–559, Association for Computing Machinery, 2024.

[329] R. van Deursen, "Footwear for the neuropathic patient: offloading and stability," *Diabetes/metabolism research and reviews*, vol. 24, no. S1, pp. S96–S100, 2008.

[330] J. Perry, J. M. Burnfield, and L. M. Cabico, "Gait analysis: Normal and pathological function," *Journal of Sports Science & Medicine*, vol. 9, p. 353, 2010.

[331] S.-H. Han, H. J. Jang, J. W. Lee, J. W. Cheong, Y. D. Kim, H. S. Nam, and D. Y. Kim, "The effect of virtual reality-based treadmill gait training on functional mobility and balance in chronic stroke patients: a randomized controlled trial," *Frontiers in Neurology*, vol. 16, p. 1603233, 2025.

[332] S. Tashiro, N. Gotou, Y. Oku, T. Sugano, T. Nakamura, H. Suzuki, N. Otomo, S. Yamada, T. Tsuji, Y. Asato, *et al.*, "Relationship between plantar pressure and sensory disturbance in patients with hansen's disease—preliminary research and review of the literature," *Sensors*, vol. 20, no. 23, p. 6976, 2020.

[333] C. Silva-Batista, G. Harker, R. Vitorio, M. Studer, B. Whetten, J. Lapidus, P. Carlson-Kuhta, S. Pearson, J. VanDerwalker, F. B. Horak, *et al.*, "Mobility rehab visual feedback system for gait rehabilitation in older adults," *Journal of NeuroEngineering and Rehabilitation*, vol. 20, no. 1, p. 144, 2023.

[334] M. Kržišnik, B. Horvat Rauter, and N. Bizovčar, "Effects of virtual reality-based treadmill training on the balance and gait ability in patients after stroke: A randomized controlled trial," *Hrvatska revija za rehabilitacijska istraživanja*, vol. 57, no. 2, pp. 92–102, 2021.

[335] E. B. Wilson, J. S. Bergquist, W. G. Wright, and D. A. Jacobs, "Gait stability in virtual reality: effects of vr display modality in the presence of visual perturbations," *Journal of NeuroEngineering and Rehabilitation*, vol. 22, no. 1, p. 32, 2025.

[336] H. Zhao, Y. Han, D. Yue, M. She, Z. Wang, F. Yang, H. Tu, and S. Qiu, "Recognition of lower limb movements based on multiregional plantar pressure and foot posture," *IEEE Sensors Journal*, vol. 25, no. 10, pp. 16754–16763, 2025.

[337] J. Lee, J. Lee, Y. J. Lee, H. Kim, Y. Kwon, Y. Huang, M. Kuczajda, I. Soltis, and W.-H. Yeo, "Flexible smart insole and plantar pressure monitoring using screen-printed nanomaterials and piezoresistive sensors," *ACS Applied Materials & Interfaces*, vol. 17, no. 33, pp. 47153–47161, 2025. PMID: 40729702.

[338] I. T. Hettiarachchi, S. Hanoun, D. Nahavandi, and S. Nahavandi, "Validation of polar oh1 optical heart rate sensor for moderate and high intensity physical activities," *PloS one*, vol. 14, no. 5, p. e0217288, 2019.

[339] V. Schwind, P. Knierim, N. Haas, and N. Henze, "Using presence questionnaires in virtual reality," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, (New York, NY, USA), p. 1–12, Association for Computing Machinery, 2019.

[340] A. Kalron, Z. Dvir, L. Frid, and A. Achiron, "Quantifying gait impairment using an instrumented treadmill in people with multiple sclerosis," *International Scholarly Research Notices*, vol. 2013, no. 1, p. 867575, 2013.

[341] H. L. O'Brien, P. Cairns, and M. Hall, "A practical approach to measuring user engagement with the refined user engagement scale (ues) and new ues short form," *International Journal of Human-Computer Studies*, vol. 112, pp. 28–39, 2018.

[342] C. Schorderet, R. Hilfiker, and L. Allet, "The role of the dominant leg while assessing balance performance. a systematic review and meta-analysis," *Gait & posture*, vol. 84, pp. 66–78, 2021.

[343] T. Muehlbauer, C. Mettler, R. Roth, and U. Granacher, "One-leg standing performance and muscle activity: are there limb differences?," *Journal of applied biomechanics*, vol. 30, no. 3, pp. 407–414, 2014.

[344] M. Mohaddis, S. A. Maqsood, E. Ago, S. Singh, Z. Naim, and S. Prasad, "Enhancing functional rehabilitation through orthotic interventions for foot and ankle conditions: A narrative review," *Cureus*, vol. 15, no. 11, 2023.

[345] G. Slater and T. Slater, "Current concepts review: orthotics in post-operative foot and ankle surgery," *J Regen Biol Med*, vol. 4, no. 1, pp. 1–4, 2022.

[346] H. K. Surmen, N. E. Akalan, and Y. Z. Arslan, “Design, manufacture, and selection of ankle-foot-orthoses,” in *Advanced Methodologies and Technologies in Artificial Intelligence, Computer Simulation, and Human-Computer Interaction*, pp. 250–266, IGI Global, 2019.

[347] M. Leite, B. Soares, V. Lopes, S. Santos, and M. T. Silva, “Design for personalized medicine in orthotics and prosthetics,” *Procedia CIRP*, vol. 84, pp. 457–461, 2019. 29th CIRP Design Conference 2019, 08-10 May 2019, Póvoa de Varzim, Portgal.

[348] A. Faux-Nightingale, M. Kelemen, and C. Stewart, “Ankle-foot orthosis adherence in children and adolescents with cerebral palsy: a scoping review,” *Prosthetics and Orthotics International*, vol. 46, no. 4, pp. 351–356, 2022.

[349] S. H. Exterkate, M. Jongebloed-Westra, P. M. Ten Klooster, H. Koffijberg, C. Bode, J. E. van Gemert-Pijnen, J. G. van Baal, and J. J. van Netten, “Objectively assessed long-term wearing patterns and predictors of wearing orthopaedic footwear in people with diabetes at moderate-to-high risk of foot ulceration: a 12 months observational study,” *Journal of Foot and Ankle Research*, vol. 16, p. 60, September 2023.

[350] G. Mageshkumar, G. Prasanna, G. Sakthi Priya, and A. Samkilbart, “Design of iot-based orthotic footwear for lower limb correction and alignment,” in *2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS)*, vol. 1, pp. 633–638, IEEE, 2024.

[351] S. Imbesi and S. Scataglini, “A user centered methodology for the design of smart apparel for older users,” *Sensors*, vol. 21, no. 8, 2021.

[352] J. M. Orlando, B. Li, B. Bodt, and M. A. Lobo, “Users’ perceptions about lower extremity orthotic devices: a systematic review,” *Archives of Physical Medicine and Rehabilitation*, vol. 104, no. 4, pp. 645–655, 2023.

[353] D. van der Wilk, J. M. Hijmans, K. Postema, and G. J. Verkerke, “A user-centered qualitative study on experiences with ankle-foot orthoses and suggestions for improved design,” *Prosthetics and orthotics international*, vol. 42, no. 2, pp. 121–128, 2018.

[354] M. S. Adhikari, Y. K. Verma, M. Sindhwani, and S. Sachdeva, “Generative ai tools for product design and engineering,” *Generative Artificial Intelligence in Finance: Large Language Models, Interfaces, and Industry Use Cases to Transform Accounting and Finance Processes*, pp. 299–325, 2025.

[355] OpenAI, “DALLE 3.” `https://openai.com/index/dall-e-3/`, 2023. Accessed: 2025-01-10.

[356] M. F. Alam, A. Lentsch, N. Yu, S. Barmack, S. Kim, D. Acemoglu, J. Hart, S. Johnson, and F. Ahmed, “From Automation to Augmentation: Redefining Engineering Design and Manufacturing in the Age of NextGen-AI,” *An MIT Exploration of Generative AI*, mar 27 2024. https://mit-genai.pubpub.org/pub/9s6690gd.

[357] OpenAI, "GPT-4 Is OpenAI's Most Advanced System, Producing Safer and More Useful Responses." https://openai.com/index/gpt-4/, 2023. Accessed: 2025-01-10.

[358] OPED-AG, "Vacopedes - orthese mit optimaler druckentlastung: Für mittel- und vorfuss." https://oped.ch/vacopedes. (accessed on 10 January 2025).

[359] Enovis, "Aircast® airselect™ short walker." https://www.enovis-medtech.de/AIRCAST-AIRSELECT-Short-Walker. (accessed on 10 January 2025).

[360] OPED-Medical-Inc., "Vacocast diabetic boot." https://opedmedical.com/products/vacocast-diabetic/. (accessed on 10 January 2025).

[361] Enovis, "Aircast® airselect™ achilles walker." https://www.enovis-medtech.de/AIRCAST-AIRSELECT-Achilles-Walker. (accessed on 10 January 2025).

[362] Berglingstyle, "Fastprotect malleo: Der sicherheitsgurt für dein sprunggelenk." https://www.berglingstyle.com/products/fastprotect-malleo-5. (accessed on 10 January 2025).

[363] Betterguards-Technology, "The betterguard 1.0: Best ankle brace." https://betterguards.de/en/products/the-betterguard-1. (accessed on 10 January 2025).

[364] Otto-Bock-HealthCare, "Walkon reaction lateral." https://www.ottobock.com/en-us/product/28U34. (accessed on 10 January 2025).

[365] Vitego, "ofa push ortho fußheberorthese afo." https://www.vitego-shop.de/bandagen-fuss-ofa-push-ortho-fussheberorthese-afo. (accessed on 10 January 2025).

[366] M. Pillai, "Revolutionizing healing: The era of 3d-printed orthotics." https://amchronicle.com/case-studies/revolutionizing-healing-the-era-of-3d-printed-orthotics/, October 2023. (accessed on 10 January 2025).

[367] HfG Schwäbisch Gmünd, "exojack - eine smarte orthese für sprunggelenksbrüche." https://ausstellung.hfg-gmuend.de/w-2021/projekte/exojack-eine-smarte-orthese-fuer-sprunggelenksbrueche, 2021. (accessed on 10 January 2025).

[368] Fraunhofer-Institut für Werkzeugmaschinen und Umformtechnik IWU, "Senso-feet – sensorisch aktive sprunggelenk-orthese mit taktiler feedback-technologie." https://www.iwu.fraunhofer.de/de/projekte/senso-feet-sensorisch-aktive-sprunggelenk-orthese-mit-taktiler-feedback-technologie.html. (accessed on 10 January 2025).

[369] T. Taylor, "Sensoria's smart socks measure your running patterns, improving gait." https://www.si.com/edge/2015/03/06/sensoria-smart-socks-measure-running-patterns, 2015. (accessed on 10 January 2025).

[370] X. P. . Orthotics, "Ankle & foot." https://xtremity.ae/ankle-and-foot/. (accessed on 10 January 2025).

[371] Free3D, “3d-gedruckter orthopädischer gipsverband am bein 3d-modell.” `https://free3d.com/de/3d-model/3d-printed-orthopedic-cast-on-leg-7449.html`, 2024. (accessed on 10 January 2025).

[372] V. Schwind and N. Henze, “Anticipated user stereotypes systematically affect the social acceptability of mobile devices,” in *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society*, NordiCHI ’20, (New York, NY, USA), Association for Computing Machinery, 2020.

[373] K.-Q. Zhou and H. Nabus, “The ethical implications of dall-e: Opportunities and challenges,” *Mesopotamian Journal of Computer Science*, vol. 2023, pp. 16–21, February 2023.

[374] K. Slegers, T. Delien, V. Bettelli, M. Lexis, T. Saey, D. Banes, and R. Daniëls, “Designing and manufacturing custom-made 3d printed assistive devices: A comparison of three workflows,” *Technology and Disability*, vol. 35, no. 3, pp. 171–182, 2023.

[375] H.-J. Kim, J. Kim, S. Jeong, M. Lee, J. Choo, and S.-H. Kim, “Shoegenai: A creativity support tool for high-feasible shoe product design,” in *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, pp. 1–11, 2025.

[376] S.-H. Cheng, “Impact of generative artificial intelligence on footwear design concept and ideation,” *Engineering Proceedings*, vol. 55, no. 1, p. 32, 2023.

[377] S. Resch, J. Schauer, V. Schwind, D. Völz, and D. Sanchez-Morillo, “Improving social acceptance of orthopedic foot orthoses through image-generative ai in product design,” *Applied Sciences*, vol. 15, no. 8, 2025.

[378] Z.-h. Chiang, W.-w. Su, and M.-h. Chuang, “Innovative ai-drive product design: Leveraging language models and ai image generators-a case study on insole design for high-heeled shoes,” *Human Factors in Design, Engineering, and Computing*, vol. 159, no. 159, 2024.

[379] K. SenGupta, “A framework of augmented design process: Development of footwear concept designs using artificial intelligence,” in *Emerging Trends in Leather Science and Technology*, pp. 251–273, Springer, 2024.

[380] M. Hassenzahl and A. Monk, “The inference of perceived usability from beauty,” *Human–Computer Interaction*, vol. 25, no. 3, pp. 235–260, 2010.

[381] M. A. Kuhail, J. Berengueres, F. Taher, and S. Khan, “Integrating generative ai into the design process: A case study on space engineering,” *Human Interaction and Emerging Technologies (IHIET-AI 2024): Artificial Intelligence and Future Applications*, vol. 120, no. 120, 2024.

[382] C. Ryu and Y. K. Lee, “Effective fashion design collection implementation with generative ai: Chatgpt and dall-e,” *Clothing and Textiles Research Journal*, vol. 44, no. 1, pp. 125–140, 2026.

[383] S. Rajagopal, M. Mani, S. Venkatraman, and R. Suganya, “Personalized solutions for foot health: Machine learning-based foot condition detection, classification, and recommendation of customized footwear,” *IEEE Access*, 2025.

[384] D. Völz, J. Schneider, and S. Resch, *Generatives Produktdesign orthopädischer Hilfsmittel: Potenziale und Herausforderungen von KI im Produktentwicklungsprozess*, pp. 219–242. Wiesbaden: Springer Fachmedien Wiesbaden, 2025.

[385] S. Resch, Y. Söyleyici, V. Schwind, D. Völz, and D. Sanchez-Morillo, “Enhancing mhealth app onboarding using a multimodal mixed reality avatar,” in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI EA ’26, (New York, NY, USA), Association for Computing Machinery, 2026.

[386] F. Zaman, N. Pantidi, J. Young, R. Kuffner dos Anjos, T. Trengrove, J. Drown, and J. Lee, “Xrgait: Immersive gait training visualization with integrated sensing,” in *Proceedings of the 37th Australian Conference on Human-Computer Interaction*, OzCHI ’25, (New York, NY, USA), p. 414–425, Association for Computing Machinery, 2025.

[387] S. Resch, L. Damrau, D. Völz, and D. Sanchez-Morillo, “Personalized foot orthoses through a patient-centered design approach using generative ai,” *Biomedical Engineering / Biomedizinische Technik*, vol. 70, no. s1, pp. 1–374, 2025.

[388] S. Resch, “Designing customized wearable devices: A novel ai-based framework for product development,” in *2025 9th International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*, pp. 1–6, 2025.

[389] S. Resch, J.-G. Hanania, V. Schwind, D. Völz, and D. Sanchez-Morillo, “Smart insole visual foot pressure feedback in mixed reality environments: Impact on gait, heart rate, workload, and engagement in treadmill walking,” in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI EA ’26, (New York, NY, USA), Association for Computing Machinery, 2026.

[390] T. Nikodelis, D. Moscha, D. Metaxiotis, and I. Kollias, “Commercial video frame rates can produce reliable results for both normal and cp spastic gait’s spatiotemporal, angular, and linear displacement variables,” *Journal of applied biomechanics*, vol. 27, no. 3, pp. 266–271, 2011.

[391] J. P. Santos, J. P. Ferreira, M. Crisóstomo, and A. Paulo Coimbra, “Instrumented shoes for 3d grf analysis and characterization of human gait,” in *Bioinformatics and Biomedical Engineering: 7th International Work-Conference, IWBBIO 2019, Granada, Spain, May 8-10, 2019, Proceedings, Part II 7*, pp. 51–62, Springer, 2019.

[392] S. Majumder and M. J. Deen, “Wearable imu-based system for real-time monitoring of lower-limb joints,” *IEEE sensors journal*, vol. 21, no. 6, pp. 8267–8275, 2020.

[393] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *Journal of Basic Engineering*, vol. 82, pp. 35–45, 3 1960.

[394] P. S. U. Shankar, N. Raveendranathan, N. R. Gans, and R. Jafari, “Towards power optimized kalman filter for gait assessment using wearable sensors,” *Proceedings - Wireless Health 2010*, pp. 137–144, 2010.

[395] T. Bennett, R. Jafari, and N. Gans, “An extended kalman filter to estimate human gait parameters and walking distance,” *Proceedings of the American Control Conference*, pp. 752–757, 2013.

[396] R. Mahony, T. Hamel, and J.-M. Pflimlin, “Nonlinear complementary filters on the special orthogonal group,” *IEEE Transactions on Automatic Control*, vol. 53, no. 5, pp. 1203–1218, 2008.

[397] V. G. (aster94), “Sensorfusion: A simple implementation of some complex sensor fusion algorithms.” `https://github.com/aster94/SensorFusion`, 2023. GitHub repository, accessed 2025-07-05.

[398] I. Skog, P. Handel, J.-O. Nilsson, and J. Rantakokko, “Zero-velocity detection—an algorithm evaluation,” *IEEE Transactions on Biomedical Engineering*, vol. 57, no. 11, pp. 2657–2666, 2010.

[399] N. Lovecchio, M. Zago, and C. Sforza, *Gait Analysis in the Rehabilitation Process*, pp. 109–118. Cham: Springer International Publishing, 2021.

Appendix

APPENDIX A

Scoping Review – Search Query

A.1. ACM Digital Library

[[Full Text: “assistive device”] OR [Full Text: “assistive system”] OR [Full Text: “assistive technolog*”] OR [Full Text: “walking aid”]] AND [Full Text: sensor*] AND [[Full Text: walker*] OR [Full Text: cane] OR [Full Text: “walking stick”] OR [Full Text: crutch]] AND [[Full Text: “walking impairments”] OR [Full Text: “mobility support”] OR [Full Text: “gait rehabilitation”] OR [Full Text: “patient support”] OR [Full Text: “foot conditions”]]

A.2. Web of Science

(((ALL=(“assistive device” OR “assistive system” OR “assistive technolog*” OR “walking aid”)) AND ALL=(sensor*)) AND ALL=(walker* OR cane OR “walking stick” OR crutch)) AND ALL=(“walking impairments” OR “mobility support” OR “gait rehabilitation” OR “patient support” OR “foot conditions”)

APPENDIX B

Software Architecture and Data Processing – Smart Orthosis

B.1. Raw Data Acquisition

Accurate and synchronized data acquisition is essential to ensure reliable downstream signal processing and gait detection. To ensure synchronization, data from the FSRs and the IMU were recorded at a sampling rate of 100 Hz, consistent with values reported in related studies [390, 391, 392, 83]. Analog pressure signals from the FSRs were acquired via a 32-channel analog multiplexer. Each sensor was connected in a voltage divider circuit with a 10 kΩ resistor and the resulting signals were digitized using the microcontroller's 12-bit ADC. IMU data was acquired via the I^2C interface and processed into triaxial acceleration (m/s^2) and angular velocity (rad/s) using the BMI270's internal processing unit. To ensure temporal synchronization, each sample was assigned a unified timestamp. All sensor readings were subsequently organized into structured data packets for further processing.

B.2. Pre-Processing

Sensor data pre-processing was performed to improve signal quality and prepare the data for subsequent segmentation and analysis. Due to manufacturing tolerances and environmental influences, FSR signals required individual calibration to obtain accurate force measurements. Each FSR sensor provides a non-linear voltage output that decreases with increasing applied force. The analog readings were first converted to voltage values, from which the resistance R_{FSR} was calculated using the voltage divider principle (Equation (B.1)). Here, R_M is the reference resistor ($10\,\text{k}\Omega$), V^+ is the supply voltage, and V_{out} is the measured voltage across the FSR.

$$V_{\text{out}} = \frac{R_M \cdot V^+}{R_M + R_{\text{FSR}}} \tag{B.1}$$

Following the manufacturer's specification [284], the raw sensor output was used to estimate relative plantar forces without applying sensor-specific calibration. To reduce high-frequency noise, a moving average filter was applied to the estimated force signals (Equation (B.2)).

$$\text{filteredForce}[i] = \frac{1}{N}\sum_{k=0}^{N-1} \text{fsrForce}[i-k] \tag{B.2}$$

The BMI270 includes internal configuration options for filtering and calibration, which were utilized to enhance signal quality. The sensor was configured with a measurement range of ±4 g for the accelerometer and ±2000°/s for the gyroscope to capture typical gait-related movements as well as rapid rotational changes. Offset correction was automatically performed during initialization by fixing the IMU in a stationary position to determine the static bias for each axis. These offset values were removed from all subsequent accelerometer and gyroscope samples to ensure base-corrected measurements (Equations (B.3) and (B.4)).

$$\text{accelCorrected}[i] = \text{accelRaw}[i] - \text{accelOffset} \tag{B.3}$$

$$\text{gyroCorrected}[i] = \text{gyroRaw}[i] - \text{gyroOffset} \tag{B.4}$$

Additionally, separate internal low-pass filters were applied to the accelerometer and gyroscope axes to remove high-frequency noise (Equation (B.5)).

$$\text{filteredValue}[i] = \alpha \cdot \text{filteredValue}[i-1] + (1-\alpha) \cdot \text{rawValue}[i] \tag{B.5}$$

A filter coefficient of $\alpha = 0.9$ was chosen to effectively reduce high-frequency noise. Pre-processing of the IMU data through offset correction and low-pass filtering ensures that the acceleration and rotation data are precise and reliable.

B.3. Signal Processing

In the signal processing stage, pre-processed sensor data were further refined to reduce residual noise and enable robust detection of gait events. To improve signal quality, Kalman filters [393] were applied to both FSR and IMU signals. The Kalman filter was selected due to its proven suitability for real-time gait analysis in wearable systems, including IMU-based orientation estimation, walking distance calculation, and general gait assessment [84, 394, 395]. For the FSR signals, individual Kalman filters were assigned to each of the sensor channels, with parameters set to $Q = 1.0$, $R = 1.0$, and $P = 0.01$. The filter parameters were adjusted to maintain responsiveness while reducing high-frequency fluctuations. For the IMU, each axis of the accelerometer and gyroscope data was filtered individually, with configuration parameters adjusted to the sensor's noise characteristics. Filter parameters were empirically set to $Q = 0.1$, $R = 0.1$, and $P = 0.3$ for accelerometer data, and to $Q = 1.0$, $R = 1.0$, and $P = 0.01$ for gyroscope data to balance signal smoothness and responsiveness.

The filtered FSR signals served as the basis for gait event detection. A custom rule-based algorithm was implemented to identify key gait phases using threshold values defined as relative percentages of the user's body weight. The algorithm evaluated pressure variations within specific sensor zones corresponding to the forefoot and hindfoot regions to detect events such as heel strike, toe strike, heel off, and toe off. This individualized approach enabled the detection of characteristic loading patterns and supported the identification of critical gait events and potential overload conditions. Thresholds were empirically determined through pre-testing across multiple gait cycles and optimized to minimize detection errors. A finite state machine was implemented to manage gait phase transitions based on these events. This approach ensured a consistent event sequence and enabled accurate temporal segmentation. A baseline correction algorithm was used to counteract the hysteresis effects of the FSR sensors and to ensure zero-base alignment after the load is removed. This adjustment was essential for maintaining continuous and reliable gait event detection across steps. The detected gait events, including their timestamps, were subsequently used to compute spatiotemporal gait parameters such as step duration, stance/swing phase duration, and cadence.

B.4. Sensor Fusion

Filtered FSR and IMU data were fused to extract both temporal and spatial characteristics of the gait cycle. While FSR data provide reliable information on foot-ground contact timing, the IMU captures the foot's three-dimensional orientation and motion dynamics. A Mahony filter [396] was employed to estimate foot orientation in terms of roll, pitch, and yaw. The Mahony filter was implemented via the `fusion.MahonyUpdate()` function from the SensorFusion library [397]. This approach combines gyroscope data for short-term tracking with accelerometer data for long-term correction, effectively compensating for drift and high-frequency noise. To improve the accuracy of position estimation, zero-velocity updates [398] were applied during the stance phase, identified from FSR signals. These updates reset the velocity to zero between the heel contact and the heel off, thereby limiting integration drift and reducing cumulative error in position calculation.

B.5. Extraction of Gait Parameters

Based on the fused sensor data, key gait parameters were extracted. The segmentation of the gait cycle follows the definition of Perry [330], which distinguishes two main phases: the stance phase (60% of the gait cycle) and the swing phase (40%), as well as eight sub-phases (including foot strike, foot flat, heel off, and foot off). These gait parameters can be classified into spatial, temporal, spatio-temporal, and kinetic categories [399]. Stride length was calculated by integrating acceleration data, corrected for foot orientation and stabilized using zero-velocity updates during stance phases. Gait speed was subsequently derived as the ratio of stride (cycle) length to gait cycle duration. As the measurement setup included only a single foot-mounted IMU, step length and step width could not be directly calculated. However, when the instrumented insoles are worn on both feet, these parameters can be extracted. In addition to standard parameters, a stability index was introduced to quantify gait stability.

This metric integrates two normalized components:

- Pressure variability: Temporal fluctuations in total plantar pressure across all FSR channels.
- Anterior-posterior pressure ratio: Relative distribution of pressure between forefoot and rearfoot regions.

The final stability index was computed as a weighted sum of both components and ranges from 0 (low stability) to 1 (high stability). Higher values indicate low pressure variability and a balanced anterior-posterior distribution, whereas lower values suggest unsteady gait characteristics. By combining FSR and IMU data, the system enables accurate, real-time estimation of gait events and parameters. The integration of Mahony filtering and zero-velocity detection effectively mitigates drift and noise, while the proposed stability index provides an interpretable measure for detecting variations in gait control.

The extracted gait parameters serve as the basis for subsequent data analysis and interpretation. Quantitative measured variables such as stride length, cadence, stance time and gait speed can be statistically evaluated. Furthermore, the parameters enable the classification of gait patterns and the detection of anomalies, which may be indicative of pathological gait or early-stage motor impairments.

APPENDIX C

XR Study I – Quantitative Results

C.1. System Usability Scale (SUS)

Table C.1: Significant main and interaction effects for SUS subscales with post-hoc comparisons.

SUS Item	**Effect**	**F(1,69)**	p	**Post-hoc comparisons**
1: Frequency of Use	Gesture	4.97	.029	non-visible < visible ($p = .056$)
	Voice	5.51	.022	audible < non-audible ($p = .038$)
3: Ease of Use	Gesture	11.99	$< .001$	non-visible < visible ($p = .006$)
	Voice	11.20	.001	audible < non-audible ($p = .008$)
4: Need for Support	Gesture	4.56	.036	–
6: Inconsistency	Gesture×Voice	4.20	.044	No Gesture: audible < non-audible ($p = .012$)
7: Learnability	Gesture	11.77	.001	–
	Voice	11.11	.001	–
	Gesture×Voice	4.23	.043	No Voice: non-visible < visible ($p < .001$), No Gesture: audible > non-audible ($p < .001$)
9: Confidence	Gesture	7.35	.008	–
	Voice	7.97	.006	–
	Gesture×Voice	7.32	.009	No Voice: non-visible < visible ($p = .002$), No Gesture: audible > non-audible ($p = .002$)
10: Prior Learning	Gesture	5.01	.028	–
	Gesture×Voice	7.55	.008	No Voice: non-visible > visible ($p = .003$), No Gesture: audible < non-audible ($p = .028$)

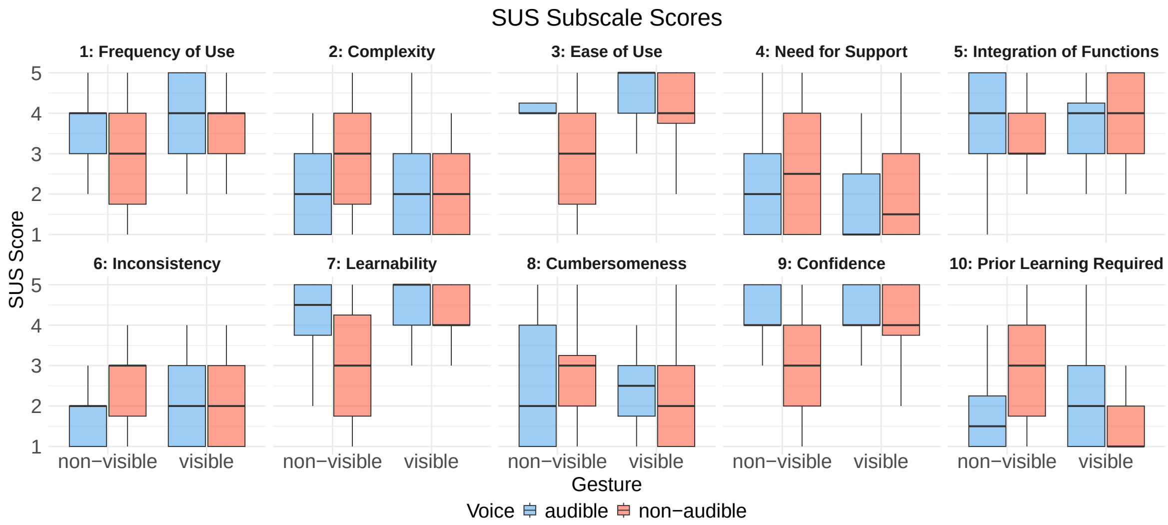


Figure C.1: Boxplots of the ten SUS items. Ratings range from 1 (strongly disagree) to 5 (strongly agree).

C.2. Learning Experience Questionnaire (LEQ)

Table C.2: Overview of significant main and interaction effects for LEQ subscales with Bonferroni-corrected post-hoc t-tests.

Subscale	**Effect**	**F(1,23)**	p	η_p^2	**Post-hoc comparisons**
Contextuality	Gesture	10.15	.004	.31	–
	Voice	13.28	.001	.37	–
	Gesture×Voice	5.41	.029	.19	No Voice: non-visible < visible ($p < .001$) No Gesture: audible > non-audible ($p < .001$)
Effects	Gesture	5.42	.029	.19	–
	Voice	7.72	.011	.25	–
	Gesture×Voice	11.78	.002	.34	No Voice: non-visible < visible ($p < .001$) No Gesture: audible > non-audible ($p < .001$)
Design Features	Voice	20.14	$< .001$	.47	–
	Gesture×Voice	6.49	.018	.22	No Voice: non-visible < visible ($p = .011$) No Gesture: audible > non-audible ($p < .001$)

APPENDIX D

XR Study II – Quantitative Results

User Engagement Subscales

Table D.1: ANOVA and post-hoc results for UES subscales (Bonferroni-adjusted).

Effect	F	p	η_p^2	Condition	Comparison	p-adj	Sig.
Focus Attention							
Environment	10.02	$< .001$	.303	–	–	–	–
Vis×Env	2.89	.024	.112	Heatmap	VR Forest > MR Lab	.029	*
				Heatmap	VR Forest > VR Lab	.024	*
				Histogram	VR Forest > MR Lab	.037	*
				CoP	VR Forest > VR Lab	.012	*
				None	VR Forest > MR Lab	$< .001$	***
				None	VR Forest > VR Lab	$< .001$	***
Perceived Usability							
Environment	9.57	$< .001$	.294	–	VR Forest > MR Lab	$< .001$	***
				–	VR Forest > VR Lab	.034	*
				–	MR Lab < VR Lab	.001	**
Aesthetic Elements							
Visualization	3.48	.020	.132	–	–	–	–
Environment	22.96	$< .001$	.500	–	VR Forest > MR Lab	$< .001$	***
				–	VR Forest > VR Lab	$< .001$	***
				–	MR Lab < VR Lab	.012	*
Reward Factor							
Visualization	4.15	.020	.153	–	–	–	–
Environment	18.57	$< .001$	.447	–	VR Forest > MR Lab	$< .001$	***
				–	VR Forest > VR Lab	$< .001$	***

APPENDIX E

Ethical Approval

Ethikkommission der DGP • Alfred-Herrhausen-Straße 50 • 58455 Witten

Frankfurt University of Applied Sciences
Frau Prof. Dr. Diana Völz
Professur für Produktentwicklung und CAD
Fachbereich 2: Informatik und Ingenieurwissenschaften
Nibelungenplatz 1
60318 Frankfurt am Main
voelz@fb2.fra-uas.de

DEUTSCHE GESELLSCHAFT FÜR PFLEGEWISSENSCHAFT e.V.

Deutsche Gesellschaft für Pflegewissenschaft
(DGP) e. V.
Geschäftsstelle
Bürgerstr. 47
47057 Duisburg

Telefon: 0203 – 356793
info@dg-pflegewissenschaft.de

Ethikkommission der Deutschen Gesellschaft
für Pflegewissenschaft
(DGP) e.V.
Vorsitz: Christine Dunger, Ph.D.
Alfred-Herrhausen-Straße 50
58455 Witten

Ethikkommission@dg-pflegewissenschaft.de

Ihr Antrag an die Ethikkommission vom 01.09.2023
(Amendment zum Antrag 23-027)

Sehr geehrte Frau Völz,

die Ethikkommission der DG Pflegewissenschaft hat Ihren Antrag/ das Amendment zu Ihrem Antrag

Entwicklung von Lern- und Erfahrungsräumen in der pflegerischen und medizinischen Versorgung durch Mixed Reality (Hybrid Thinking)

geprüft und erteilt Ihnen hiermit das ethische Clearing.

Wir wünschen Ihnen viel Erfolg!

12.02.2024

Christine Dunger

Christine Dunger, Ph.D.
Vorsitzende der Ethikkommission der DGP

APPENDIX F

Consent of Co-Authors

The signed consent forms from all co-authors of the included publications are provided below.

Appendix F.1: Study 4.1 – Scoping Review

- S. Resch, A. Zirari, T. D. Q. Tran, L. M. Bauer, and D. Sanchez-Morillo, “Smart Walking Aids with Sensor Technology for Gait Support and Health Monitoring: A Scoping Review,” Technologies, vol. 13, no. 8, Art. no. 346, 2025.

Appendix F.2: Study 4.2 – Smart Device Development

- S. Resch, A. Kousha, A. Carroll, N. Severinghaus, F. Rehberg, M. Zatschker, Y. Söyleyici, and D. Sanchez-Morillo, “Smart Device Development for Gait Monitoring: Multimodal Feedback in an Interactive Foot Orthosis, Walking Aid, and Mobile Application,” Technologies, vol. 13, no. 12, Art. no. 588, 2025.

Appendix F.3: Study 5.1 – mHealth App Onboarding in MR

- S. Resch, Y. Söyleyici, V. Schwind, D. Völz, and D. Sanchez-Morillo, ”Enhancing mHealth App Onboarding Using a Multimodal Mixed Reality Avatar,” in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA ’26)*, Barcelona, Spain: Association for Computing Machinery, pp. 1–9, 2026.

Appendix F.4: Study 5.2 – Smart Insole Feedback in MR/VR

- S. Resch, J.-G. Hanania, V. Schwind, D. Völz, and D. Sanchez-Morillo, ”Smart Insole Visual Foot Pressure Feedback in Mixed Reality Environments: Impact on Gait, Heart Rate, Workload, and Engagement in Treadmill Walking,” in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA ’26)*, Barcelona, Spain: Association for Computing Machinery, pp. 1–8, 2026.

Appendix F.5: Study 6.1 – Social Acceptance

- S. Resch, J. Schauer, V. Schwind, D. Völz, and D. Sanchez-Morillo, “Improving Social Acceptance of Orthopedic Foot Orthoses Through Image-Generative AI in Product Design,” Applied Sciences, vol. 15, no. 8, Art. no. 4132, 2025.

F.1. Study 4.1 – Scoping Review

Universidad de Cádiz

Thesis by compendium of publications.
Document of agreement and resignation of co-authors.

D. /Dª.: Aya Zirari, Thi Diem Quynh Tran, Luca Marco Bauer, and Daniel Sanchez-Morillo

co-author of the following publication:
Smart Walking Aids with Sensor Technology for Gait Support and Health Monitoring:
A Scoping Review

In accordance with the **Article 23.4** of the **Regulation UCA/CG12/2023, September 29, 2012, of doctoral studies at the University of Cádiz (BOUCA No. 394).**

States his/her approval to the presentation of this publication as a part of the doctoral thesis written by D. Stefan Resch, entitled
User-Centered Design and Evaluation of Smart Devices for Foot Augmentation: Interactive Mixed Reality Feedback and Training Applications

And expresses his/her resignation to present said publication as a part of another doctoral thesis at any other University.

In Cádiz, 10 of December, 2025

Signed (Aya Ziari): ____________

Signed (Thi Diem Quynh Tran): ____________

Signed (Luca Marco Bauer): ____________

Signed (Daniel Sanchez-Morillo): ____________

F.2. Study 4.2 – Smart Device Development

UCA Universidad de Cádiz

Thesis by compendium of publications.
Document of agreement and resignation of co-authors.

D. /Dª.: Andre Kousha, Anna Carroll, Noah Severinghaus, Felix Rehberg, Marco Zatschker, Yunus Söyleyici, Daniel Sanchez-Morillo

co-author of the following publication:
Smart Device Development for Gait Monitoring: Multimodal Feedback in an Interactive Foot Orthosis, Walking Aid, and Mobile Application

In accordance with the **Article 23.4** of the **Regulation UCA/CG12/2023, September 29, 2012, of doctoral studies at the University of Cádiz (BOUCA No. 394).**

States his/her approval to the presentation of this publication as a part of the doctoral thesis written by D. Stefan Resch, entitled
User-Centered Design and Evaluation of Smart Devices for Foot Augmentation: Interactive Mixed Reality Feedback and Training Applications

And expresses his/her resignation to present said publication as a part of another doctoral thesis at any other University.

In Cádiz, 15 of December, 2025

Signatures:

Andre Kousha: ______ Anna Carroll: ______

Noah Severinghaus: ______ Felix Rehberg: ______

Marco Zatschker: ______ Yunus Söyleyici: ______

Daniel Sanchez-Morillo: ______

F.3. Study 5.1 – mHealth App Onboarding in MR

UCA Universidad de Cádiz

Thesis by compendium of publications.
Document of agreement and resignation of co-authors.

D. /Dª.: Yunus Söyleyici, Valentin Schwind, Diana Völz, Daniel Sanchez-Morillo

co-author of the following publication:
Enhancing mHealth App Onboarding Using a Multimodal Mixed Reality Avatar

In accordance with the **Article 23.4** of the **Regulation UCA/CG12/2023, September 29, 2012, of doctoral studies at the University of Cádiz (BOUCA No. 394).**

States his/her approval to the presentation of this publication as a part of the doctoral thesis written by D. Stefan Resch, entitled User-Centered Design and Evaluation of Smart Devices for Foot Augmentation: Interactive Mixed Reality Feedback and Training Applications

And expresses his/her resignation to present said publication as a part of another doctoral thesis at any other University.

In Cádiz, 20 of January, 2026

Signed (Yunus Söyleyici): Y. Söyleyici

Signed (Valentin Schwind):

Signed (Diana Völz): D. Völz

Signed (Daniel Sanchez-Morillo):

F.4. Study 5.2 – Smart Insole Feedback in MR/VR

UCA Universidad de Cádiz

Thesis by compendium of publications.
Document of agreement and resignation of co-authors.

D. /Dª.: Jean Gabriel-Hanania, Valentin Schwind, Diana Völz, Daniel Sanchez-Morillo

co-author of the following publication:
Smart Insole Visual Foot Pressure Feedback in Mixed Reality Environments: Impact on Gait, Heart Rate, Workload, and Engagement in Treadmill Walking

In accordance with the **Article 23.4** of the **Regulation UCA/CG12/2023, September 29, 2012, of doctoral studies at the University of Cádiz (BOUCA No. 394).**

States his/her approval to the presentation of this publication as a part of the doctoral thesis written by D. Stefan Resch, entitled
User-Centered Design and Evaluation of Smart Devices for Foot Augmentation: Interactive Mixed Reality Feedback and Training Applications

And expresses his/her resignation to present said publication as a part of another doctoral thesis at any other University.

In Cádiz, 20 of January, 2026

Signed (Jean Gabriel-Hanania): __________

Signed (Valentin Schwind): __________

Signed (Diana Völz): __________

Signed (Daniel Sanchez-Morillo): __________

F.5. Study 6.1 – Social Acceptance

UCA | Universidad de Cádiz

Thesis by compendium of publications.
Document of agreement and resignation of co-authors.

D. /Dª.: Jakob Schauer, Valentin Schwind, Diana Völz, and Daniel Sanchez-Morillo

co-author of the following publication:
Improving Social Acceptance of Orthopedic Foot Orthoses Through Image-Generative AI in Product Design

In accordance with the **Article 23.4** of the **Regulation UCA/CG12/2023, September 29, 2012, of doctoral studies at the University of Cádiz (BOUCA No. 394).**

States his/her approval to the presentation of this publication as a part of the doctoral thesis written by D. Stefan Resch, entitled User-Centered Design and Evaluation of Smart Devices for Foot Augmentation: Interactive Mixed Reality Feedback and Training Applications

And expresses his/her resignation to present said publication as a part of another doctoral thesis at any other University.

In Cádiz, 10 of December, 2025

Signed (Jakob Schauer): J. Scha

Signed (Valentin Schwind): Valentin Schwind

Signed (Diana Völz): D. Völz

Signed (Daniel Sanchez-Morillo):

APPENDIX G

Report on the Publications Included in the Compendium

The Doctoral thesis developed by Mr. Stefan Resch, titled "User-Centered Design and Evaluation of Smart Devices for Foot Augmentation: Interactive Mixed Reality Training and Feedback Applications," is submitted as a thesis comprising a collection of articles. The thesis consists of three journal publications in the following pages and is supplemented by a series of contributions presented at international conferences.
This report includes:

- Citation metrics and indicators of quality for each journal article, including the impact factor and quartile from the Journal Citation Reports for the best field in which the submitted articles appear.
- A description of each author's contribution, detailing the doctoral candidate's role. In all the articles, the doctoral candidate was responsible for writing the manuscripts, reviewing and summarizing the relevant recent literature, planning and conducting the experiments, developing the applications, validating the associated statistical analyses, discussing the results, and drawing the most relevant conclusions. Further details are provided in Section 1.4.

G.1. Journal Publication 1

Citation
S. Resch, A. Zirari, T. D. Q. Tran, L. M. Bauer, and D. Sanchez-Morillo, “Smart Walking Aids with Sensor Technology for Gait Support and Health Monitoring: A Scoping Review,” Technologies, vol. 13, no. 8, Art. no. 346, 2025.
DOI: https://doi.org/10.3390/technologies13080346

Journal

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- Publisher: MDPI
- eISSN: 2227-7080
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- 5-Year Impact Factor: 4.2
- Citescore: 8.5
- JCR Category: Engineering, Multidisciplinary
  - Category Quartile: Q1
  - Rank by Journal Impact Factor: 35 / 179

Metrics

- Altmetrics: 1
- Citations: 3 (Google Scholar)
- Article Views: 6844

Recognition
The article was selected as “Editor’s Choice” by the journal, based on a recommendation from the editorial board.

Author Contributions
Conceptualization, S.R.; methodology, S.R., A.Z., T.D.Q.T. and L.M.B.; software, S.R.; validation, S.R. and D.S.-M.; formal analysis, S.R.; investigation, S.R., A.Z., T.D.Q.T. and L.M.B.; resources, S.R.; data curation, S.R.; writing—original draft preparation, S.R.; writing—review and editing, S.R. and D.S.-M.; visualization, S.R.; supervision, D.S.-M.; project administration, D.S.-M.; funding acquisition, D.S.-M. All authors have read and agreed to the published version of the manuscript.

technologies

Review

Smart Walking Aids with Sensor Technology for Gait Support and Health Monitoring: A Scoping Review

Stefan Resch [1,2,*], Aya Zirari [2], Thi Diem Quynh Tran [2], Luca Marco Bauer [2] and Daniel Sanchez-Morillo [3,4]

[1] Doctorate Program in Manufacturing, Materials and Environmental Engineering, Doctorate School (EDUCA), University of Cadiz, 11003 Cádiz, Spain
[2] Faculty of Computer Science and Engineering, Frankfurt University of Applied Sciences, 60318 Frankfurt am Main, Germany
[3] Department of Automation Engineering, Electronics and Computer Architecture and Networks, School of Engineering, University of Cadiz, 11519 Cádiz, Spain
[4] Instituto de Investigación e Innovación Biomédica de Cádiz (INiBICA), 11009 Cádiz, Spain
* Correspondence: stefan.resch@alum.uca.es

Abstract

Smart walking aids represent a growing trend in assistive technologies designed to support individuals with mobility impairments in their daily lives and rehabilitation. Previous research has introduced sensor-integrated systems that provide user feedback to enhance safety and functional mobility. However, a comprehensive overview of their technological and functional characteristics is lacking. To address this gap, this scoping review systematically mapped the current state of research in sensor-based walking aids, focusing on device types, sensor technologies, application contexts, target populations, and reported outcomes. In addition, integrated artificial intelligence (AI)-based approaches for functional support and health monitoring were examined. Following PRISMA-ScR guidelines, 35 peer-reviewed articles were identified from three databases: ACM Digital Library, IEEE Xplore, and Web of Science. Extracted data were thematically analyzed and synthesized across device types (e.g., walking canes, crutches, walkers, rollators) and use cases, including gait training, fall prevention, and daily support. Findings show that, while many prototypes show promising features, few have been evaluated in clinical settings or over extended periods. A lack of standardized methods for sensor location assessment, often the superficial implementation of feedback modalities, and limited integration with other assistive technologies were identified. In addition, system validation and user testing lack consensus, with few long-term studies and often incomplete demographic data. Diversity in data communication approaches and the heterogeneous use of AI algorithms were also notable. The review highlights key challenges and research opportunities to guide the future development of intelligent, user-centered mobility systems.

Keywords: smart walking aids; assistive devices; mobility support; health monitoring; sensor technology; rehabilitation; prevention; scoping review

check for updates

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1. Introduction

Mobility impairments, particularly among older adults and individuals with long-term health conditions, are a growing global concern. The Global Report on Assistive Technology from the World Health Organization (WHO) and UNICEF [1] highlights that over 2.5 billion people (approximately one in three individuals globally) currently require at least one assistive device, a number that is expected to rise due to the aging population.

The report emphasizes the urgent demand for accessible, technology-driven solutions and provides recommendations to foster research and innovation in this field.

Assistive technologies cover a wide range of products designed to maintain independence and improve health across diverse populations [1]. In the domain of mobility support, body-worn devices such as orthoses are often used together with external devices like canes, crutches, or walkers. Although these conventional walking aids are well-established in clinical and everyday settings, they are typically passive and lack adaptability to the individual needs of users.

Recent advancements in sensor technology, embedded systems, and artificial intelligence (AI) have enabled the development of a new generation of smart walking aids that extend beyond basic support. Overall, current research highlights the benefits of smart walkers for monitoring balance and stability conditions, aiming to improve quality of life and prevent falls and injuries [2,3]. A review by Martins et al. [4] emphasized the substantial potential of intelligent walkers for rehabilitation, as these systems can support various functions such as gait monitoring [5], obstacle detection [6], and real-time feedback [7]. Depending on the intended function, different sensor and actuator technologies are employed, such as force sensing resistors (FSRs) for gait event detection [8], inertial measurement units (IMUs) for motion sensing and fall detection [9], ultrasonic sensors for obstacle detection [10], and Global Positioning System (GPS) modules for localization [11].

Based on the measured sensor input, active user feedback can be provided through a range of output modalities, including acoustic, haptic, and visual cues. Auditory and tactile feedback combinations are frequently used to support visually impaired users [12]. For example, haptic feedback can be provided via vibration in cane handles to guide postoperative weight-bearing [13]. In another case, visual cues such as a projected 2D laser line from a modified cane have been shown to improve gait initiation in individuals with Parkinson's disease [14].

Beyond local feedback, smart walking aids can be embedded into Internet of Things (IoT) infrastructures [15] to enable data transmission for enhanced gait monitoring and real-time alerts. In this context, AI-based methods such as machine learning (ML) have been applied for fall detection and motion classification based on accelerometer and gyroscope data [16]. For example, Wang et al. [17] proposed an intelligent walking stick capable of human motion analysis and imbalance classification, which communicates with a mobile application. Additionally, the interaction with other wearable systems, such as smart insoles [18] or foot orthoses [19], enables multimodal data fusion and cross-device feedback. For instance, Resch et al. [19] proposed a forearm crutch system that delivers vibration alerts triggered by pressure thresholds detected in a connected smart foot orthosis.

Although numerous studies have investigated specific aspects of smart walking aids for several use cases, there is currently no comprehensive synthesis of the field. In particular, there is a lack of systematic analysis regarding the types of sensors used, functional purposes, health-related contributions, user-specific applications, and the role of AI in extending device capabilities. However, such an overview is essential to inform evidence-based development, support cross-disciplinary innovation, and guide future implementation strategies. To address this gap and provide a structured overview, we conducted a systematic review of the literature using Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) [20].

The central aim of this review is to provide a structured overview of new trends in sensor-based walking aids developed to support the mobility of individuals with walking impairments. In particular, we aim to categorize the types of sensors used, identify their key functionalities, and examine the targeted user populations and domains of application.

The main research question (MRQ) guiding this review is

- MRQ: What types of intelligent walking aids equipped with sensor technologies have been developed to support the mobility of individuals with walking impairments, and in which domains of application are they used?

In addition, we define two specific research questions (SRQs):

- SRQ1: Which sensor technologies and AI-based approaches are integrated into smart walking aids to enable personalized functional support, health monitoring, and user-specific feedback mechanisms?
- SRQ2: Which target user groups are addressed by these devices, and what key functionalities do they offer to meet the specific mobility-related needs of these populations?

By synthesizing findings from the existing literature, we aim to identify technological trends, current advancements, and existing limitations in intelligent walking aids. We provide a comprehensive overview of sensor-based assistive systems and outline future directions to enhance mobility support for individuals with physical impairments. Based on our findings, we derive recommendations for researchers and practitioners. This work contributes to the fields of assistive technology, sensor technologies, and human–computer interaction (HCI).

2. Materials and Methods

We conducted a systematic literature review using the PRISMA-ScR approach. The PRISMA-ScR checklist with 22 items was used to guide the identification, selection and synthesis of relevant articles.

2.1. Eligibility Criteria

As eligibility criteria, the following inclusion criteria were defined: (1) The study must investigate smart, intelligent, or instrumented walking aids (e.g., canes, sticks, walkers, or crutches) with integrated sensor technology; (2) The walking aid must be user-controlled and serve a health-related purpose (e.g., fall prevention, gait rehabilitation, mobility or balance assistance); (3) Only English-language journal and conference papers were considered.

Exclusion criteria were as follows: (1) Non-original articles (e.g., literature reviews, meta-analyses, or editorials) and non-peer-reviewed conference papers (e.g., abstracts, posters) were excluded; (2) Studies focused solely on comfort, transportation, or navigation assistance and that did not directly relate to health outcomes were excluded. In addition, studies that exclusively targeted people with visual impairments were not included; (3) Autonomous systems without active user guidance (such as exoskeletons or humanoid robots), motorized systems, or wearable devices (such as orthotics or prosthetics) were excluded.

2.2. Database and Search Strategy

Three scientific databases were selected as information sources: ACM Digital Library, IEEE Xplore, and Web of Science. These were chosen because they comprehensively cover technical and interdisciplinary peer-reviewed research domains related to the development and evaluation of assistive technologies, particularly in engineering, computer science, and HCI. This focus aligns with the objective of this review, which is to investigate technological innovations and sensor-integrated system developments in the context of smart walking aids. The literature search in all three databases was conducted on 9 January 2025.

The search strategy was designed to capture studies at the intersection of assistive systems, sensor technology, walking aid devices, and health-related applications. To ensure thematic relevance, the search was explicitly restricted to health-related terms (e.g., walking impairments, mobility support, gait rehabilitation, patient support, foot conditions), as well as to device types such as walkers, canes, sticks, and crutches. In addition, keywords such as "sensor" and specific assistive technology terms (such as assistive device, assistive system, assistive technologies, and walking aid) were incorporated to identify studies focusing on these key categories of devices. Boolean operators were used to logically combine all relevant search terms. The search was applied to the full-text fields in all three databases. The specific search query for IEEE Xplore is documented below:

("Full Text Only": "assistive device" OR "assistive system" OR "assistive technolog" OR "walking aid") AND ("Full Text Only": sensor*) AND ("Full Text Only": walker* OR cane OR "walking stick" OR crutch) AND ("Full Text Only": "walking impairments" OR "mobility support" OR "gait rehabilitation" OR "patient support" OR "foot conditions").*

Full search strategies for ACM Digital Library and Web of Science are provided in the Appendix A.

2.3. Data Charting and Synthesis

All retrieved records were imported into the Zotero reference management software [21]. Duplicate entries were identified automatically and removed manually. The selection and data collection process was conducted by three reviewers (A.Z., T.D.Q.T., and L.M.B.), who divided the records into three equal parts. Each record was independently screened by two of the three reviewers, following the four-eyes principle, to ensure consistency and minimize bias. In cases of disagreement, the third reviewer counter-checked the record to resolve the conflict and reach consensus. After each screening phase, the first author (S.R.) independently reviewed all records to ensure methodological consistency, validate screening decisions, and document the process for transparency and accuracy. The inter-rater reliability was assessed using Cohen's κ and reached a substantial level ($\kappa = 0.72$), according to the definition by Landis and Koch [22].

The screening procedure followed a three-phase approach: First, titles and abstracts were screened based on the predefined eligibility criteria, with a focus on keyword relevance and study scope. Second, the included articles were screened for full text and examined in detail to determine whether they met the eligibility criteria. Third, relevant data were extracted from the included studies and cross-checked for accuracy.

For each included study, the following data items were extracted to enable a structured synthesis: device type, sensor technology (including sensor type, placement, and specifications), device communication, use of AI, target population, key functionality, sample size, objective, and results. In addition, general metadata were collected to support the descriptive analysis, including the first author, year of publication, and the country of origin of the study. No assumptions or simplifications were made beyond those explicitly stated in the original articles. Where necessary, unclear or missing data were noted accordingly in the table. No formal critical appraisal was conducted, as the aim of this scoping review was to map existing research rather than assess study quality.

For data synthesis, a deductive thematic analysis [23] was conducted. The initial synthesis process was performed by the first author (S.R.) and subsequently discussed and cross-checked with a second reviewer (D.S.-M.) to ensure consistency in the thematic interpretation. The predefined themes were derived from the research objectives and corresponded to the data items identified during extraction. The discussion follows a narrative synthesis structure, integrating the findings within this thematic framework. Structured summaries were provided for each theme and key study characteristics were presented in tabular format to facilitate comparison across studies. Descriptive statistics were applied to identify trends and to provide an overview of relevant study characteristics. However, no statistical synthesis was performed, following the methodological framework of scoping reviews.

3. Results

The selection process was documented using the PRISMA flow diagram [24], generated with the R package PRISMA2020 (version 1.1.1) proposed by Haddaway et al. [25], see Figure 1.

A total of 829 records were identified through database searches across the ACM Digital Library, IEEE Xplore, and Web of Science. After the removal of one duplicate, 828 records remained for title and abstract screening. During this screening phase, 719 records were excluded because they did not match the study scope or lacked relevant keywords in their titles or abstracts. Subsequently, 109 full-text articles were assessed for eligibility. A total of 74 articles were excluded for various reasons, as they did not meet the inclusion criteria, as shown in Figure 1. Finally, 35 studies met all inclusion criteria and were included in the synthesis.

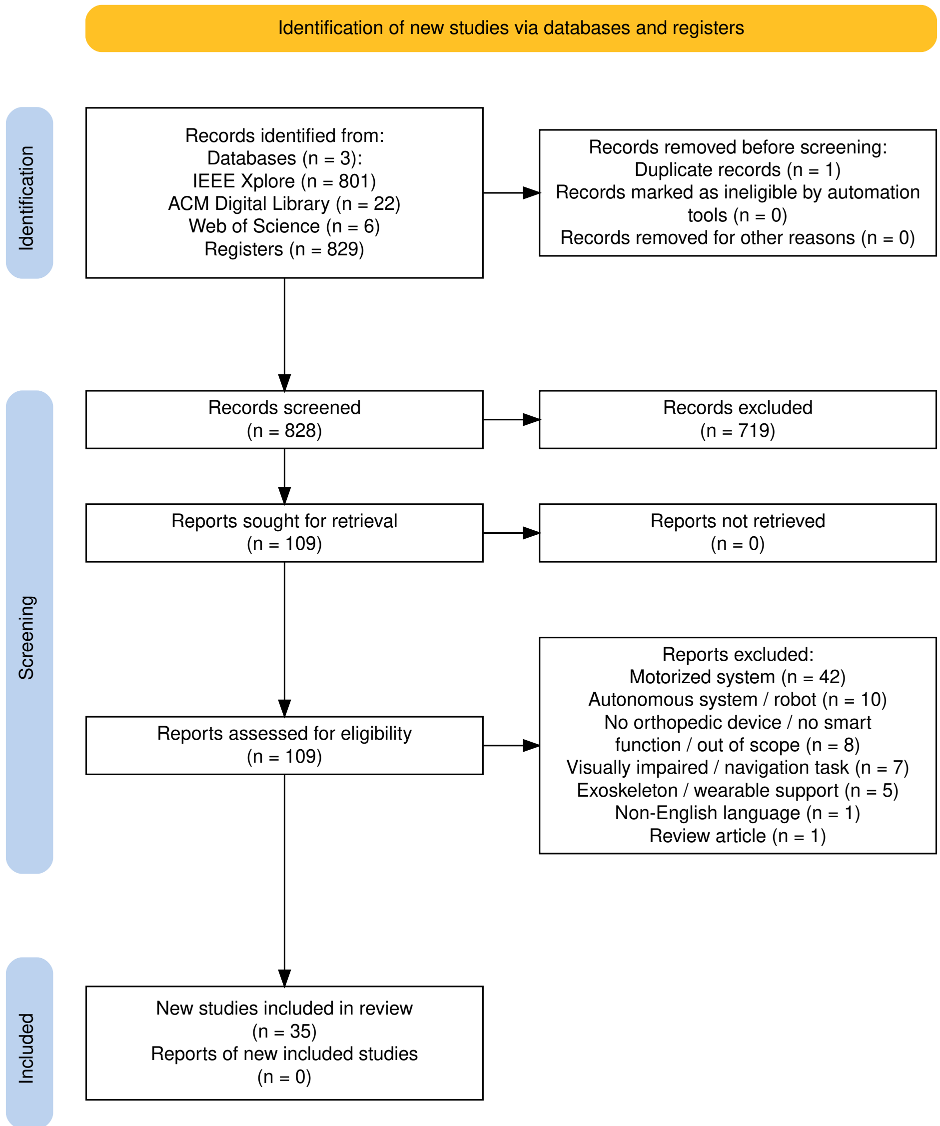


Figure 1. PRISMA flowchart of the article selection process, covering the identification, screening, and inclusion phases.

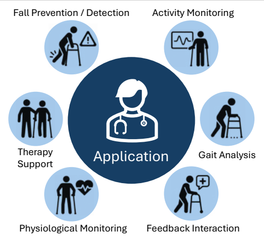
3.1. Study Characteristics

This section presents the main characteristics of the included studies, beginning with an overview of their application contexts, followed by a detailed synthesis of study populations, aims, and evaluation settings.

3.1.1. Application Context

The main application areas covered by the included studies are summarized in Table 1. Several studies address multiple application domains and therefore appear in more than one category. The frequencies are presented descriptively, including the percentage of studies in which each application context was identified.

Table 1. Overview of application areas, their frequency in the included studies, and corresponding references, grouped by main application categories.



Application	Frequency	%	References
Gait analysis	32	91.4	[26–56]
Fall prevention/detection	11	31.4	[29,34,35,42,52,54–59]
Therapy support	8	22.9	[30,31,38,40,43,50,53,55]
Feedback interaction	7	20.0	[27,34,43,55,58–60]
Activity monitoring	6	17.1	[29,31,37,44–46]
Physiological monitoring	5	14.3	[28,38,42,48,55]
Total	**69**		

Note: Some studies cover multiple application areas and appear in more than one category.

3.1.2. Synthesis of Study Characteristics

The 35 included publications come from 16 different countries and were published between 2008 and 2024. Most studies were conducted in Europe, North America, and Asia. Table 2 summarizes the key characteristics of the selected studies, such as target population, key functionality, sample size, objective, and outcomes.

The primary aims across studies included gait monitoring and analysis, fall detection, force or pressure measurement, and the integration of feedback modalities such as haptic or remote monitoring. The most frequently targeted groups were older adults (10 studies) [26,28,29,34,36,41,45,48,54,56], followed by a broad group of rehabilitation patients (9 studies) [27,30,33,35,39,49,55,58,60], and users with mobility or gait impairments (8 studies) [31,37,42–44,46,47,51]. Additionally, specific use for physical therapy patients was reported in four studies [38,40,50,53], and users at risk of falling were targeted in three articles [52,57,59]. Parkinson's patients were addressed in one study [32].

Sample sizes varied widely, ranging from single-case studies to group studies with up to 42 participants or technical validation without participants. The majority of studies included between 7 and 20 participants, often healthy, but also patients with several conditions. Demographic data varied widely, with mean participant ages spanning from 22 to over 80 years.

To further characterize the reviewed studies, we identified four types of research focus across the included studies:

- System development: [32,35,38,42,48,50,55,58,59]
  These articles present systems targeting a specific user group, but without comprehensive empirical validation involving participants. They typically focus on functionality demonstration and proof-of-concept implementations.
- Technical validation studies: [26,27,29,31,33,40,43,45–47,51,54,60]
  These studies focus on the development and validation of prototypes, often conducted in laboratory settings with healthy participants or a small number of users.
- Application-oriented studies: [28,36,37,41,44,52,53,56,57]
  These articles evaluate the developed systems with the intended user population, aiming to demonstrate feasibility and functionality in context.
- Patient-centered pilot studies: [30,34,39,49]
  These studies assess the impact of the system within structured interventions involving affected users, and often serve as preliminary investigations toward clinical integration.

Table 2. Overview of study characteristics grouped by publication year.

Year	Authors	Country	Target Population	Key Functionality	Sample Size	Objective	Results
2024	Anushree et al. [55]	India	Rehabilitation patients	To support balance, detect falls, monitor physiology, and provide emergency feedback	n = not specified	Develop a smart walker to support balance, posture, and feedback during physical therapy.	Prototype developed; no formal validation
2024	Postolache et al. [35]	Portugal	Rehabilitation patients	To detect and classify gait abnormalities	n = not specified (healthy volunteers)	Analyze gait patterns (normal/abnormal) using spectrogram-based metrics.	Successful gait abnormality classification using spectrogram-based features
2024	Cavagila et al. [52]	Italy, UK	Fall risk users	To detect falls, activate walker brakes, and alert caregivers	n = 6 (healthy, mean age 30 ± 8.3 y)	Design a multi-sensor system for fall and near-fall detection in smart walkers.	High fall detection accuracy; pre-fall prediction less reliable
2023	Arcobelli et al. [51]	Italy	Gait-impaired users	To detect stance phases and visualize data in real time	n = 1 (male, 29 y)	Demonstrate stance and force analysis with the mCrutch during walk tests.	Crutch stance segmentation accuracy: 94%
2023	Padmavathi et al. [58]	India	Rehabilitation patients	To correct posture, detect falls, and avoid obstacles via haptic feedback	n = 1 (not specified)	Develop a smart crutch with sensors for posture and safety monitoring.	Proof-of-concept; sensor data successfully validated
2022	Zhou et al. [36]	China	Older adults	To analyze gait (step count, stride, length, speed)	n = 6 (2 elderly with assistance; 4 healthy: 2 young, 2 elderly)	Implement a smart stick with 9-axis sensor for gait analysis.	Step count 100%; stride/step metrics > 94% accuracy
2022	Batoca et al. [40]	Portugal	Physical therapy patients	To assess gait and store data in the cloud with web-based visualization	n = 2 (healthy, 1m/1f, ages 24 and 25 y)	Collect and store gait and balance data via smart crutch during rehab.	Successful collection of force and orientation data
2022	Narváez et al. [33]	Spain	Rehabilitation patients	To identify and classify crutch gait patterns	n = 20 (healthy, 8f/12m)	Recognize gait patterns in crutch users using sensors and ML.	Gait classification accuracy: 88–89% (GPS, ANN)
2021	Valsangkar et al. [44]	Canada	Mobility-impaired users	To segment and analyze the Timed Up and Go (TUG) test	n = 16 (musculoskeletal injuries)	Assess sensor-based cane data for clinical mobility assessment.	High segmentation accuracy; low classification errors (LDA/ANN)
2021	Ribeiro and Santos [54]	Portugal	Older adults	To detect abnormal gait and fall-related instability	n = 11 (healthy, mean age 24.2 ± 2.6 y, range 22–29 y)	Detect falls in real-time using instrumented cane with ML and FSM.	Fall detection accuracy > 99%; phase classification ≈ 96.5%
2020	Mesanza et al. [46]	Spain	Gait-impaired users	To classify physical activities (e.g., walking, standing, stairs)	n = 11 (healthy, 4f/7m, age 24–48 y)	Classify physical activities from wearable sensor data using ML.	Feature selection achieved 92–97% classification accuracy
2020	Zambrano et al. [53]	Peru, Canada	Physical therapy patients	To monitor gait speed and handle pressure in real time	n = 10 (patients, not further specified)	Develop a low-cost walker to monitor gait parameters.	Session duration reduced from 25 to 5.2 min
2019	Ballesteros et al. [41]	Sweden, Spain	Older adults	To detect dynamic weight-bearing trends	n = 8 (elderly, cane users, 6m/2f, mean age 82.1 ± 6.0 y)	Validate sensor system for anomaly detection in cane load.	Sensor outputs correlated with user physical status
2018	Wade et al. [34]	USA	Older adults	To monitor axial load, grip pressure, and object proximity in real time	n = 9 + 18 (Study 1: 3m/6f; Study 2: 8m/10f; active cane users)	Assess feasibility of fall risk estimation via sensorized cane.	Significant correlation between grip pressure and gait performance
2018	Frango and Postolache [50]	Portugal	Physical therapy patients	To analyze force and orientation, and support therapists via mobile app	n = not specified (several healthy volunteers, various ages)	Support therapists with a mobile app for rehab progress monitoring.	Functional system; no formal validation conducted
2018	Seylan et al. [47]	Turkey	Gait-impaired users	To estimate 3D ground reaction forces	n = not specified	Estimate ground reaction forces using low-cost sensor system.	Prediction errors < 7% (static), <8% (dynamic)
2018	Viegas et al. [28]	Portugal	Older adults	To assess gait using load and heart rate data	n = 1 (impaired gait, right lower limb injury)	Validate "Spy Walker" for gait monitoring during rehabilitation.	Step classification successful; potential for rehab assessment
2018	Ojeda et al. [26]	Mexico, Spain	Older adults	To classify gait patterns and estimate walking age from force data	n = 42 (age range 22–94 y)	Predict walking age via unsupervised learning from gait data.	Clustering revealed four distinct gait types linked to age/speed/force

Table 2. *Cont.*

Year	Authors	Country	Target Population	Key Functionality	Sample Size	Objective	Results
2018	Ballesteros et al. [57]	Spain	Fall risk users	To estimate fall risk from spatial and force sensor data	n = 10 (physical/neurological disabilities, 3m/7f, mean age 61.4 y, range 46–74 y)	Assess feasibility of smart rollator for fall risk prediction.	Fall risk score correlated significantly with Tinetti scores and speed
2018	Gill et al. [29]	Canada, India	Older adults	To monitor gait, activity levels, and walking environment	n = 10 (healthy, 8m/2f, mean age 22.9 ± 2.3 y)	Design IoT smart walker for monitoring gait and environmental data.	System identified gait events and periods of rest/activity
2017	Mekki et al. [32]	Italy, France	Parkinson patients	To extract gait parameters in real time (e.g., duration, asymmetry)	n = not specified (Parkinson's participants)	Create an instrumented cane for gait analysis in Parkinson's patients.	Prototype developed; no formal validation
2017	Gill et al. [37]	Canada	Mobility-impaired users	To measure loading, mobility, and stability	n = 30 (healthy adults, 20m/10f, age 20–60 y, mean 24.2 ± 7.1 y)	Design smart cane for unobtrusive gait monitoring and event detection.	Gait event differences identified between normal and perturbed conditions
2017	Domingues et al. [42]	Brazil	Mobility-impaired users	To detect falls, monitor gait, and assess physiological signals	n = not specified	Develop sensor-integrated cane for mobility and health tracking	Strong correlation between behavior and sensor data
2017	Ballesteros et al. [30]	Spain	Rehabilitation patients	To estimate balance scales and extract spatiotemporal gait features	n = 19 (physical/cognitive disabilities, 6m/13f, mean age 68 ± 9.3 y, range 50–80 y)	Predict Tinetti scores using rollator sensor data.	Predicted vs. actual Tinetti scores: high correlation
2017	Koziol et al. [59]	USA	Fall risk users	To deliver reminders and reduce fall risk	n = 0 (no subjects involved)	Evaluate "Reminding Walker" for posture detection and cueing	Proof-of-concept confirmed; reminder cues detected successfully
2016	Ballesteros et al. [39]	Spain	Rehabilitation patients	To estimate gait and balance using spatiotemporal parameters	n = 19 (physical/cognitive disabilities, 6m/13f, mean age 67.5 ± 9.7 y, range 46–80 y)	Estimate Tinetti scores in real-time from gait parameters.	PCA + Ridge regression: $R^2 = 0.92$ (gait), 0.88 (balance)
2015	Wade et al. [31]	USA	Gait-impaired users	To assess timing, speed, acceleration, and angular velocity; therapist feedback	n = 7 (4f, mean age 27 ± 3.9 y; 3m, mean age 27.3 ± 4.5 y)	Support gait classification via sensor-based instrumented cane.	Gait classification accuracy > 95% (decision tree), 84% (ANN)
2015	Sardini et al. [27]	Italy	Rehabilitation patients	To measure axial/shear forces and provide vibratory feedback	n = 10 (healthy, mean age 30 y, range 28–55 y)	Quantify upper-limb input using wireless instrumented crutches.	Axial force error ∼9 N; shear ∼5 N; angular error ≈ 1°
2015	Ballesteros et al. [49]	Spain	Rehabilitation patients	To monitor gait parameters and detect abnormalities	n = 9 (physical/cognitive disabilities, 6f/3m, mean age 68.2 ± 14.6 y, range 45–86 y)	Estimate clinical gait parameters using sensorized systems.	Accurate extraction of cadence, gait time/length, and load metrics
2015	Postolache [38]	Portugal	Physical therapy patients	To monitor gait, posture, and force interaction via smart aids	n = not specified	Enable real-time rehab assessment using smart walking aids.	Feasibility of multi-sensor integration successfully demonstrated
2015	Gerena et al. [60]	USA	Rehabilitation patients	To measure upper-limb loading	n = 3 (2 healthy, 1 patient)	Monitor upper-limb load with low-cost instrumented walker.	±1 lb force accuracy; synchronized with motion capture and EMG
2014	Culmer et al. [43]	UK	Gait-impaired users	To assess kinematic and kinetic gait properties	n = 1 (female, 43 y, multiple sclerosis, walking aid on right)	Provide quantitative gait feedback via smart walking aid.	Accurate orientation/load data; aligned with clinical observations
2014	Chalvatzaki et al. [45]	Greece	Older adults	To recognize actions, gestures, and gait cycles	n = 10 (healthy subjects)	Design walker with user localization, pose, and intention recognition.	Accurate recognition of actions and gait segmentation (HHM)
2013	Kikuchi et al. [56]	Japan	Older adults	To support passive motion control via line-tracing navigation	n = 3 (disorders, ages 82, 90, 100; 2m/1f)	Evaluate line-tracing controller in an intelligent walker.	Stride width/length improved; positional errors reduced by controller
2008	Chan and Green [48]	Canada	Older adults	To monitor distance, speed, acceleration, handle force, and vital signs	n = 0 (no empirical study reported)	Enhance rollators with sensors for real-world gait monitoring.	Subsystems showed monitoring potential; complexity varied

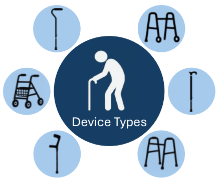
3.2. Technological Aspects

This section introduces the categorized device types and corresponding sensor technologies, detailed in the subsequent subsections and concluded by a synthesis of key technological aspects.

3.2.1. Device Types

The included studies covered a diverse range of smart assistive devices, which are descriptively summarized in Table 3.

Table 3. Overview of device types, their frequency in the included studies, and corresponding references.



Device Type	Frequency	%	References
Walkers	11	31.4	[28,29,35,38,52,53,55,56,58–60] *
Canes	8	22.9	[31,32,34,37,41,42,44,54]
Rollators	8	22.9	[26,30,38,39,45,48,49,57] *
Forearm crutches	8	22.9	[27,33,38,40,46,47,50,51] *
Sticks	2	5.7	[36,43]
Total	**37**		

* *Note: Article [38] addresses three types of devices and is therefore listed in multiple categories.*

For analytical clarity, the devices were grouped into three main categories: walkers/rollators (n = 19), canes/sticks (n = 10), and forearm crutches (n = 8). Based on their form factor and intended use, walkers and rollators were combined into one category due to their comparable use cases and technical components. Similarly, canes and sticks were grouped based on shared sensor configurations and form factors. This categorization enables a structured comparison of technological components, feedback modalities, and target populations across different device types. The following paragraphs synthesize each category in detail, with a focus on the application context, technological components, and the use of AI for data processing.

Walkers and Rollators

Eighteen of the included studies [26,28–30,35,38,39,45,48,49,52,53,55–60] published between 2008 and 2024 focused on smart walkers or rollators designed to support individuals reliant on walking aids such as older adults, individuals with mobility limitations, or an increased risk of falling. The primary functions of these systems include gait monitoring, weight distribution analysis, and fall detection.

Ten studies investigated various walker configurations: five studies examined four-leg walkers [28,35,38,55,58], four studies focused on front-wheel walkers [29,53,59,60], and one study investigated a four-wheel walker [52]. These devices are typically equipped with a combination of sensors, most commonly force or pressure sensors paired with IMUs. In addition, various other components are used, such as infrared or ultrasonic sensors, accelerometers, rotary encoders, GPS modules, or dry electrocardiogram (ECG) electrodes integrated into the handgrips. Force sensors and load cells are commonly used to quantify physical load, grip forces, and weight distribution during ambulation. These signals are frequently combined with IMUs to generate a more holistic gait analysis. For instance, the combination of force sensors and accelerometers enables step time, stride length, and stop movements to be derived from linear acceleration signals. Such configurations facilitate basic gait event detection and activity classification and may be extended to assess gait symmetry, turning behavior, postural transitions, and stability.

All these systems enable wireless communication, using either Wi-Fi or Bluetooth-based solutions. Seven systems offer user feedback via visual, acoustic, or haptic cues. Visual feedback was provided in six systems (e.g., via smartphone apps, indicator lights, or liquid crystal display (LCD) displays showing force distribution and pulse rate) [28,38,55,58–60], acoustic feedback (e.g., sound alerts for nearby assistance) in three [52,55,59], and haptic feedback (e.g., vibration alerts for excessive handle force) in two [55,58].

Six studies [26,30,39,49,56,57] focused on the iWalker platform, which is based on previous publications and often referenced as a rollator-based system. These devices typically included rotary encoders mounted in the wheels and force sensors on the handlebars, as well as additional components depending on the specific application. Five of these devices did not incorporate user feedback mechanisms and did not specify the type of system communication. Only one article [56] referred to a cable-based connection to a mobile PC, which was also used to provide visual feedback. None of the iWalker-based systems applied AI methods, but instead relied on conventional techniques such as regression or clustering for basic prediction or classification tasks.

Two additional studies [45,48] investigated rollators equipped with more extensive sensor arrays. One system [48] utilized force sensors, an accelerometer, a hall sensor, a pressure sensor, and a photoplethysmogram (PPG) sensor. The second system [45] integrated laser range sensors, MEMS microphones, a GoPro HD camera, two Kinect modules, rotary encoders, and 6-degrees of freedom (DOF) force/torque sensors. This device is the only walker-based system that explicitly employed ML, specifically HHMs and support vector machines (SVMs), for automatic activity recognition. In addition, three articles [35,55,58] mentioned the potential application of ML or deep learning (DL) methods in future work. In total, four studies [26,30,39,52] have explored the use of predictive or regression-based methods for assessing gait patterns and estimating fall risk.

Forearm Crutches

Smart crutches were described in eight [27,33,38,40,46,47,50,51] of the included studies published between 2015 and 2024. These devices primarily focus on gait support within physical rehabilitation for patients with lower-limb impairments. The predominant functionality identified is gait assessment, aimed at enhancing mobile health monitoring for users. Five studies integrated IMU sensors, including two devices [46,50] equipped with 9-DOF sensors, one [40] with a 10-DOF sensor, and two [33,51] with unspecified DOF. Each study reported different sensor placements: two at the ground tip, one along the shaft, and two did not specify the exact positioning. Additionally, accelerometers were mentioned in two studies [27,47], with placements at the ground tip and mid-shaft.

Furthermore, five studies utilized force sensors, with three [38,40,50] placing them on the handles and two at the ground tip [46,47], with sensor counts varying from one to four. Load cells were consistently incorporated at the ground tip in two studies [40,50]. Additionally, strain gauges were utilized in two studies: one employing twelve sensors positioned on the lower shaft for measuring shear and axial forces [27], and another using eight sensors on the handles to measure axial forces [33]. Other sensor technologies, such as radio-frequency identification (RFID) and ultrasonic sensors, were each mentioned once.

Four studies explicitly incorporated real-time user feedback: three [38,50,51] applied visual feedback via smartphone apps and light-emitting diode (LED) lights, while one [27] implemented haptic vibratory biofeedback. Wireless communication methods were primarily based on Bluetooth, with only one study [40] reporting the use of Wi-Fi and MQTT protocols. Three articles explicitly mentioned connectivity to dedicated Android applications [38,50,51]. Machine learning algorithms, including SVM, ANN, Random Forest (RF), and K-Nearest Neighbors (kNN), were used in two studies [33,46] for IMU-based physical activity classification.

Walking Canes and Sticks

A total of eight studies [31,32,34,37,41,42,44,54] published between 2015 and 2021 addressed smart walking canes, while two [36,43] additional studies focused on walking sticks (published in 2014 and 2022). These devices primarily target mobility-impaired users, including older adults and individuals with Parkinson's disease. The main purposes include the assessment of gait parameters, pressure distribution analysis, fall detection, and monitoring changes in gait behavior. Inertial sensors were the most commonly integrated technology, used in seven [31,32,34,36,37,43,54] of the ten studies. Two further studies [42,44] employed discrete accelerometer and gyroscope sensors. Except for one study using two IMUs [31], all others utilized a single unit, positioned either at the handle or the ground tip. IMUs enable the measurement of linear acceleration, angular velocity, and orientation to support functionalities such as gait analysis and fall detection.

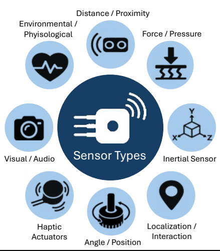
Force sensors were mentioned in four articles to detect grip force and ground contact. Two devices [31,34] were equipped with eight FSRs each, placed in the handle and at the ground tip. One device [41] implemented two FSR sensors positioned at different heights in the shaft, while another [42] used a single sensor without specifying its placement. Additional sensor technologies included capacitive touch, ultrasonic, load cells, strain gauges, humidity sensors, and thermistors, each implemented to support specific applications.

Two devices applied visual feedback: one via a smartphone app [43], and the other via an OLED display combined with a real-time clock [42]. Furthermore, one study explicitly incorporated vibratory feedback via the cane handle for excessive force [34]. Communication methods were predominantly wireless via Bluetooth and Wi-Fi, though two studies also included a USB connection, and one did not specify the protocol. ML methods were reported in three studies [31,44,54], with algorithms such as ANN, SVM, and naive Bayes used for classifying gait-related activities. One study mentioned plans to incorporate ML in future work [34]. Moreover, two studies [36,43] employed Kalman filtering for sensor data fusion, and one utilized a custom sensor fusion algorithm [37].

3.2.2. Sensor Technologies

The reviewed studies highlight a diversity of sensor technologies integrated into smart walking aids. Table 4 presents an overview of the sensor types, grouped by their main categories, which are described in detail in the following paragraphs.

Table 4. Overview of sensor/actuator types, their frequency in the included studies, and corresponding references, grouped by sensor main categories.



Sensor/Actuator Type	Frequency	%	References
Force/pressure	29	82.9	[26–28,30–34,37–53,55,57,58,60]
Inertial sensors	25	71.4	[26–29,31–34,36–38,40,42–44,46–48,50–52,54,55,58,59]
Distance/proximity	13	37.1	[29,34,35,38,42,45,49,52,53,55,56,58,59]
Angle/position	7	20.0	[30,39,45,48,49,52,57]
Environmental/physiological	6	17.1	[28,42,46,48,55,58]
Localization/interaction	5	14.3	[37,40,42,55,58]
Haptic actuators	4	11.4	[27,34,55,58]
Visual/audio	4	11.4	[45,52,56,57]
Total	93		

Note: The frequency indicates the number of studies that employed at least one sensor of the respective type, not the total number of sensors per device.

- **Force/Pressure Sensors, Load Cells, and Strain Gauges:** The most used sensor types across the device categories are those for measuring grip force, axial and shear forces, ground reaction forces, and weight-bearing compliance. In total, the use of force sensors is reported in twelve articles [26,30,38–40,46,48–50,55,57,58]. Furthermore, seven articles explicitly mention the use of FSR integrated into the devices [31,34,41,42,47,52,53], while two others employed pressure sensors [48,60], and one mentioned the use of a force/torque sensor [45]. Load cells were used in six studies [28,34,40,43,50,51], where they play a crucial role in assessing ground reaction forces and ensuring correct weight distribution during gait training and rehabilitation.

Strain gauges were reported in five articles [27,32,33,37,44]; three of these addressed canes (each equipped with four strain gauges), while the other two addressed forearm crutches (equipped with either eight or twelve strain gauges) for measuring shear and/or axial forces.

- **Inertial Sensors:** The second most frequently used sensor types across the device categories are IMUs, which typically comprise accelerometers, gyroscopes, and magnetometers. IMUs were used in 16 studies [28,29,31–34,36,37,40,43,46,50–52,54,59]. IMU data were used for gait phase detection, posture monitoring, and movement analysis. Accelerometers (whether 3-DOF, 6-DOF, or 9-DOF) were mentioned in ten articles [26,27,34,38,42,44,47,48,55,58], and gyroscopes were explicitly reported in three articles [26,42,44].
- **Distance and Proximity Sensors:** Distance and proximity sensors were incorporated in several studies to enable obstacle or user detection, enhance environmental awareness, and support navigation. A total of six studies used ultrasonic sensors [29,34,42,55,58,59], three used laser rangefinders [45,53,56], and one applied a 2D laser scanner [49]. Additionally, two studies utilized microwave Doppler radar [35,38], and two implemented infrared (IR) sensors for user detection [59] and speed monitoring [53].
- **Angle and position sensor:** Rotary encoders for wheel position tracking were reported in six studies, all of which focused on rollators or walkers [30,39,45,49,52,57]. A tilt sensor was also mentioned in one article [49], as well as one Hall effect sensor for rotary angle measurement [48].
- **Environmental and physiological sensors:** Environmental parameters were recorded using light-dependent resistor (LDR) for light intensity in three studies [42,55,58], and a barometer for atmospheric pressure sensing in one study [46]. Furthermore, one device [42] included humidity and temperature sensors for ambient monitoring and embedded a thermistor in the handle to measure local temperature. Physiological sensors were each used once and included dry ECG electrodes [28], PPG sensors [48], and IR-based pulse monitoring [55].
- **Localization, Identification and Interaction Sensors:** GPS modules for outdoor localization were used in three studies [42,55,58]. Additionally, one device used RFID tags [40] for user identification, and one article implemented capacitive touch sensors [37] to detect user interaction.
- **Haptic Actuators:** Haptic feedback was implemented via vibration motors in four articles [27,34,55,58]. The placement and number of vibration motors varied, including configurations such as four motors, a single motor in one handle, motors in both handles, or unspecified locations.
- **Visual and Audio Sensors:** Four articles mentioned the use of various camera systems, such as an RGB camera [57], a GoPro camera and Kinect [45], a webcam [56], and one unspecified camera [52]. As an audio sensor, MEMS microphones were employed in one article [45]. While not explicitly listed or categorized as sensors or actuators in this review, various studies implemented visual and auditory output interfaces to support user interaction, information display, or alert mechanisms. These include LCD [55,58,60] and OLED screens [42], LED indicators [50,59], smartphone applications [38,43,50,51] for feedback display, and audio components such as speakers [59] and alarms [52,55].

3.2.3. Synthesis of Technological Aspects

Table 5 presents the extracted technical data of the included studies, highlighting the device type, used sensor technology, and corresponding placement, feedback modality, communication, and use of AI/ML methods or conventional algorithms.

Table 5. Technical overview of smart walking aid systems, including device type, sensor specifications, feedback modality, communication, and AI use.

Ref.	Device Type	Sensor Type(s)	Sensor Placement	Feedback Modality	Communication	Algorithm, AI Method
[55]	Walker (four-legs)	∘ 2 × Force ∘ 1 × Ultrasonic ∘ 1 × Accelerometer ∘ 1 × LDR ∘ 1 × GPS ∘ 1 × IR sensor	∘ Handle ∘ Cross bar (bottom) ∘ Top platform ∘ Top platform ∘ Top platform ∘ Handle	Haptic: Vibration motor Visual: LCD display, Panic SOS button, SMS with GPS on fall Acoustic: Sound alert	Global system for mobile communications (GSM) module, IoT (unspecified)	Algorithm: Fuzzy logic (fall detection, haptic feedback); Planned: AI/ML extension
[35]	Walker (four-legs)	∘ 2 × Doppler radar	∘ Cross bar	–	Bluetooth to PC (LabView GUI)	Planned: DL classifiers (based on WVD spectrograms, MIF/MIB features)
[52]	Walker (Four-Wheels)	∘ 2 × Force (FSR) ∘ 2 × Rotary encoders ∘ 1 × IMU (9-DOF) ∘ 1 × IMU (6-DOF) ∘ 1 × Camera ∘ 1 × Laser rangefinder	∘ Handle ∘ Rear wheels ∘ Seat (bottom) ∘ Waist ∘ Backrest ∘ Cross bar	Acoustic: Alarm alert, caregiver notifications	Wireless (unspecified)	Algorithm: threshold-based (e.g., Kangas, vertical velocity); fall/near-fall detection
[51]	Forearm Crutch	∘ 1 × Load cell ∘ 1 × IMU	∘ Ground tip	Visual: Smartphone app	Bluetooth to Android app (mCrutch)	Algorithm: Threshold-based segmentation; crutch stance detection
[58]	Walker (four-legs)	∘ 2 × Force ∘ 1 × Ultrasonic ∘ 1 × Accelerometer ∘ 1 × LDR ∘ 1 × GPS	∘ Handle ∘ Bottom ∘ Top platform ∘ Top platform ∘ Top platform	Haptic: Vibration motor Visual: LCD screen	Wi-Fi and Bluetooth	Algorithm: Fuzzy logic (fall detection); Planned: DL for camera module
[36]	Stick	∘ 1 × IMU (9-DOF)	∘ Handle	–	Bluetooth to PC	Algorithm: Extended Kalman filter
[40]	Forearm crutch	∘ 1 × Load cell ∘ 1 × Force ∘ 1 × IMU (10-DOF) ∘ 1 × RFID	∘ Bottom ∘ Handle ∘ Not specified ∘ Not specified	–	Wi-Fi and MQTT to Cloud (IoT)	Planned: ML methods
[33]	Forearm Crutch	∘ 8 × Strain gauges (axial) ∘ 1 × IMU	∘ Handle ∘ Not specified	–	Bluetooth to PC	Algorithm: Feature selection (Relief-F, mRMR, CFS); AI/ML: kNN, RF, SVM
[44]	Cane	∘ 1 × Strain gauge ∘ 1 × Accelerometer (3-DOF) ∘ 1 × Gyroscope (3-DOF)	∘ Not specified	–	Not specified	AI/ML: LDA, SVM
[54]	Cane	∘ 1 × IMU (9-DOF)	∘ Handle (box-mounted)	–	Local logging (USB)	AI/ML: LSTM, kNN, FSM, CNN, etc.

Table 5. *Cont.*

Ref.	Device Type	Sensor Type(s)	Sensor Placement	Feedback Modality	Communication	Algorithm, AI Method
[46]	Forearm Crutch	◦ 1 × IMU (9-DOF) ◦ * × Barometer ◦ * × Force	◦ Ground tip ◦ Ground tip ◦ Ground tip	–	BLE to mobile phone	AI/ML: SVM, ANN, kNN
[53]	Walker (front-wheels)	◦ 1 × IR sensor ◦ * × Force (FSR)	◦ Not specified ◦ Handle	–	Wireless to Android app	Not specified
[41]	Cane	◦ 2 × Force (FSR)	◦ Shaft (low, multiple depths)	–	Bluetooth to external systems	Not specified
[34]	Cane	◦ 8 × Force (FSR) ◦ 1 × IMU (9-DOF) ◦ 1 × Ultrasonic ◦ 1 × Accelerometer (3-DOF) ◦ 1 × Load cell	◦ Handle ◦ Handle ◦ Shaft (middle) ◦ Ground tip ◦ Ground tip	Haptic: Vibration motor	USB to PC	Planned: ML in future work (applied in previous work)
[50]	Forearm Crutch	◦ 3 × Force (FlexiForce) ◦ 1 × IMU (9-DOF) ◦ 1 × Load cell	◦ Handle ◦ Shaft ◦ Ground tip	Visual: Smartphone app, LED light alert	Bluetooth to Android app	Algorithm: Kalman filter Planned: Prediction models
[47]	Forearm crutch	◦ 4 × Force (FSR) ◦ 1 × Accelerometer	◦ Ground tip ◦ Shaft (middle)	–	Bluetooth	Algorithm: Quadratic regression
[28]	Walker (four-legs)	◦ 4 × Load cells ◦ 2 × Dry ECG electrodes ◦ 1 × IMU (9-DOF)	◦ Ground tip ◦ Handle ◦ Cross bar	Visual: Spy Walker application	Bluetooth to PC (Spy Walker app)	Not specified
[26]	Rollator (i-Walker)	◦ 1 × Accelerometer ◦ 1 × Gyroscope ◦ * × Force	◦ Central box ◦ Central box ◦ Handle	–	Not specified	AI/ML: BOSS (feature extraction), Bayesian Gaussian Mixture model (clustering)
[57]	Rollator (i-Walker)	◦ 2 × Rotary encoders ◦ 1 × RGB-D camera ◦ * × Force	◦ Wheels ◦ Cross bar ◦ Handle	–	Not specified	Not specified
[29]	Walker (front-wheels)	◦ 1 × IMU (9-DOF) ◦ 1 × Ultrasonic	◦ Cross bar ◦ Cross bar	–	BLE to external device (IoT)	Not specified
[32]	Cane	◦ 4 × Strain gauges (axial) ◦ 1 × IMU (9-DOF)	◦ Shaft (low) ◦ Shaft (low)	–	Bluetooth to PC (LabView GUI)	Not specified
[37]	Cane	◦ 4 × Strain gauges ◦ 1 × IMU (9-DOF) ◦ 1 × Touch (capacitive)	◦ 2 × Cane curve, 2 × Handle ◦ Handle ◦ Handle	–	BLE 4.2 to Tablet app (later Bluetooth 5.0)	Algorithm: Gait segmentation algorithm
[42]	Cane	◦ 1 × Force (FSR) ◦ 1 × Accelerometer and Gyroscope (6-DOF) ◦ 1 × Ultrasonic ◦ 1 × Ambient temperature and humidity ◦ 1 × Thermistor ◦ 1 × LDR ◦ 1 × GPS	◦ Not specified ◦ Shaft (external housing) ◦ Shaft (external housing) ◦ Shaft (external housing) ◦ Handle ◦ Shaft (external housing) ◦ Shaft (external housing)	Visual: OLED monitor, Real-time clock	Wi-Fi and Bluetooth	Planned: ML (intended but not implemented)

Table 5. *Cont.*

Ref.	Device Type	Sensor Type(s)	Sensor Placement	Feedback Modality	Communication	Algorithm, AI Method
[30]	Rollator (i-Walker)	∘ 2 × Rotary encoder ∘ * × Force	∘ Wheels ∘ Handle	–	Not specified	Algorithm: Regression models for prediction
[59]	Walker	∘ 1 × Ultrasonic ∘ 1 × IR sensor ∘ 1 × IMU (3-DOF)	∘ Cross bar ∘ Cross bar ∘ Cross bar	Visual: LED lights Acoustic: Speaker output	Wireless (unspecified)	Not specified
[39]	Rollator (i-Walker)	∘ 2 × Rotary encoder ∘ * × Force	∘ Wheels ∘ Handle	–	Not specified	Algorithm: Regression models for prediction
[31]	Cane	∘ 8 × Force (FSR) ∘ 2 × IMU (9-DOF)	∘ 7 × Handle, 1 × Shaft (base) ∘ 1 × Handle, 1 × Shaft (base)	–	Wireless to PC application	AI/ML: C4.5, ANN, SVM, naive Bayes
[27]	Forearm crutch	∘ 12 × Strain gauges (axial/shear) ∘ 1 × Accelerometer (3-DOF)	∘ Shaft (low) ∘ Ground tip	Haptic: Vibratory biofeedback	Bluetooth to PC (LabView GUI)	Not specified
[49]	Rollator (i-Walker)	∘ 2 × Force (3-axis) ∘ 2 × Force (1-axis) ∘ 2 × Rotary encoders ∘ 1 × Tilt sensor ∘ 1 × 2D laser scanner	∘ Handle ∘ Hind legs ∘ Wheels ∘ Not specified ∘ Not specified	–	Not specified	Not specified
[38]	Walker/rollator and crutches	∘ 2 × Doppler radar ∘ 1 × Accelerometer ∘ * × Force	∘ Cross bar ∘ Cross bar ∘ Handle, ground tip	Visual: Smartphone app	Bluetooth to mHealth app	Not specified
[60]	Walker	∘ 2 × Pressure	∘ Handle	Visual: LCD screen	XBee RF to receiver	Not specified
[43]	Cane/stick	∘ 1 × Load cell (1-DOF) ∘ 1 × IMU (5-DOF)	∘ Ground tip ∘ Shaft	Visual: Smartphone app	Bluetooth to smartphone (LabView GUI)	Algorithm: Kalman filter
[45]	Rollator	∘ 2 × Laser rangefinder ∘ 2 × Kinect ∘ 2 × Rotary encoders ∘ 2 × F/T sensors (6-DOF) ∘ 1 × 8-mic MEMS ∘ 1 × GoPro HD camera	∘ Back and front side ∘ Top cross bar ∘ Rear wheels ∘ Handle ∘ Top cross bar ∘ Top cross bar	–	Not specified	AI/ML: HHM, SVM
[56]	Walker (i-Walker)	∘ 1 × Webcam ∘ 1 × Laser rangefinder	∘ On the top ∘ Lower chassis	Visual: Mobile PC	USB to mobile PC	Not specified
[48]	Rollator	∘ 2 × Force (6-DOF) ∘ 1 × Accelerometer (3-DOF) ∘ 1 × Hall effect sensor ∘ 1 × Pressure ∘ 1 × PPG	∘ Handle ∘ Not specified ∘ Wheel ∘ Rollator seat ∘ Finger	–	Bluetooth	Not specified

* *Note: Asterisks (*) indicate missing or unspecified numerical data in the original source.*

4. Discussion

In this PRISMA-ScR review, we systematically examined 35 articles focusing on intelligent walking aids equipped with sensor technologies to support health monitoring and mobility in individuals with impairments. Our synthesis revealed a diverse range of device types (including canes, walkers, and crutches) augmented with various sensor modalities to enable gait assessment, fall detection, and user feedback through real-time data processing and communication strategies. These systems addressed a wide spectrum of application areas, including rehabilitation, fall prevention, and activity monitoring, and were targeted at different user groups such as older adults, physical therapy patients, or fall risk users. In the following sections, we discuss the current challenges and technological gaps (Section 4.1) identified across the included studies. We also highlight opportunities for future research directions (Section 4.2), followed by implications for researchers and practitioners (Section 4.3), and a reflection on the limitations of this review (Section 4.4).

4.1. Current Challenges and Technological Gaps

Besides the demonstrated potential of the systems, some challenges and technological gaps were also identified, which are discussed below.

4.1.1. Sensor Selection and Placement

A wide range of sensors was used in the included studies, which varied in terms of specifications, placement, and calibration approaches. However, inconsistent sensor placement practices represent a challenge for data comparability, as there is currently no consensus on recommended placement strategies that ensure sufficient and reliable data collection. Ballesteros et al. [41] highlighted that sensor placement can present a number of challenges: Integrating sensors into the handle may require major modifications to canes, while components attached to the shaft could alter ergonomics, potentially requiring extensive validation or additional certification processes. For example, the use of IMU sensor varied strongly across the studies, in terms of the used DOF and in placement (e.g., at the ground tip, handle, or shaft). This variability directly affects the type and quality of motion data captured, and thus influences the detection of gait parameters or fall events. Similarly, the number and positioning of force and pressure sensors differed widely. Some systems integrated a single sensor, while others used multiple sensors, positioned on the handle or ground tip, to capture load distribution or grip force. These differences complicate the direct comparison of outcomes between studies and underscore the need for standardized sensor configurations. Furthermore, several articles did not specify the exact type of sensors used or failed to provide detailed system information, making it difficult to reproduce study outcomes and assess the validity of reported results. These findings indicate the need for more transparent reporting practices and the creation of standardized design guidelines to ensure data quality, reproducibility, and clinical utility.

4.1.2. Feedback Modalities

This review showed that most systems provide visual feedback, usually via smartphone apps [38,43,50,51], mobile PCs [56], or LED indicators [50,59]. Haptic feedback was typically integrated into device handles via vibration motors. It was primarily used as biofeedback to alert users when predefined thresholds were exceeded, such as in cases of excessive handle force [27,34,55,58]. However, no consistent procedure for setting these thresholds was described across studies. Sardini et al. [27] highlighted that such threshold parameters should be individually defined by physiotherapists in order to address patient-specific conditions. One prototype provided multimodal feedback by combining visual displays (e.g., LCD screens for force distribution and pulse rate) with haptic and acoustic alerts [55]. However, the description of feedback implementations was often superficial: details on actuator specifications, placement, or feedback timing were frequently missing, limiting reproducibility and comparability. Furthermore, the effectiveness of these feedback modalities on user experience or behavioral outcomes was not evaluated, indicating a gap in current research. No comparative analysis using quantitative measures to assess the impact of different feedback modalities was found. This limits the generalizability of recommendations regarding suitable feedback types for specific tasks or user populations. In particular, the implementation of such features should be guided by user needs and integrated in a manner that enhances both device usability and functionality. Therefore, future research should not only report feedback implementations more transparently but also systematically evaluate their effectiveness to ensure both user acceptance and clinical relevance.

4.1.3. Data Communication

Data communication approaches varied depending on the application context, but most systems relied on wireless transmission. Bluetooth or BLE were used in around half of the studies and are therefore the most frequently used wireless standard. In addition, a smaller number of systems employed Wi-Fi to transmit sensor data to external devices, including personal computers, tablets, or smartphones. One study [40] incorporated cloud connectivity through protocols such as MQTT to IoT platforms, enabling remote monitoring and data analysis. In several cases, communication protocols and data transfer details were not explicitly specified [26,30,39,44,45,49,57] or relied on cable-based connections [34,54,56]. The diversity of communication approaches reflects the varying use cases for the data, ranging from real-time visualization on a corresponding smartphone app to a simple Graphical User Interface (GUI) for laboratory validation. Only one study [55] explicitly addressed data security and privacy concerns. They used the Thingspeak IoT platform with enhanced protection features, like a private mode that restricts access to authorized people only. This approach enables data sharing with trusted parties, which is particularly relevant for the approval procedures and regulatory compliance. In general, topics such as data security and privacy were not mentioned in the other reviewed studies, but should be considered in future developments to ensure safe integration with digital health platforms and mobile applications.

4.1.4. Use of Artificial Intelligence/Algorithm

The use of AI, ML, and predictive algorithms was heterogeneous across studies to analyze gait parameters. Several systems [31,33,44–46,54] employed established methods for data processing and prediction, including SVM, ANN, RF, kNN, or planned for future use [34,35,40,42,55,58]. For example, Narváez et al. [33] tested three ML techniques (kNN, RF, and SVM) for gait pattern identification with smart crutches. The results showed that all three could be used to identify gait patterns with similar results: 88% accuracy for both kNN and SVM, as well as 89% accuracy for RF. Additionally, Mesanza et al. [46] applied an ML approach for physical activity classification using a forearm crutch and highlighted that SVM, ANN, and kNN can reach a success rate of 92–97%, which was higher than the results of previous studies (91%). In contrast, Wade et al. [31] developed an instrumented cane and found that SVM performed worse for user activity prediction with just 64% accuracy, while the C4.5 algorithm achieved the highest accuracy with over 95%, followed by ANN with 84%. Some approaches utilized more advanced methods, such as Long short-term memory (LSTM) [54], HHM [45], Bayesian Gaussian Mixture Model for clustering [26], or regression techniques for prediction. For example, Riberio and Santos [54] investigated the most suitable ML models for three fall classification problems (fall event, fall phase, and fall category) using an instrumented cane. They evaluated eight classifiers (LSTM, kNN, SVM, CNN, etc.) and demonstrated that LSTM was the most appropriate for fall event classification (accuracy: 99.26%) and fall phase classification (accuracy: 95.91%), due to its better performance compared to kNN. In contrast, the kNN model achieved the best performance for fall category classification (accuracy: 74.54%). The authors claimed that ML can lead to better performance than an FSM. However, further studies are needed to evaluate this in the context of real-time classification.

Feature extraction and sensor fusion algorithms, such as Kalman or extended Kalman filters, were also reported [36,43,50]. For example, Zhou et al. [36] applied an extended Kalman filter for gait analysis using data from a walking stick for older adults and individuals with disabilities. They reported a step count accuracy of 100% and average accuracies between 94% and 96% for stride duration, step length, and step speed. Notably, most systems do not use AI or ML but rely on traditional algorithmic approaches or do not use predictive analytics at all.

Nevertheless, AI methods or advanced algorithms hold potential for future implementations, as they have been successfully applied to the identification of gait events [31], gait abnormality classification [35], as well as fall detection and prevention [52,54]. Building on these findings, follow-up studies can refine these approaches and implement them across diverse application domains. In particular, current implementations are often limited to post-processed data analysis, highlighting that real-time integration requires further development.

4.1.5. System Validation and User Testing

Overall, there was no clear consensus regarding system validation and user testing across the included studies. In several articles, the testing procedures and the characteristics of involved participants were not specified, and demographic data were partially missing [32,35,38,42,47,50,55]. Furthermore, some articles focused solely on presenting prototypes without any technical validation or without involving user testing [48,59]. In general, most studies involved participants (both healthy and with impairments) of varying ages, with partially gender-balanced samples. However, some studies were conducted with only one participant [28,43,51,58] or exclusively with healthy participants, which did not cover the target user group [26,27,29,33,37,40,45,46,52,54]. In total, one long-term study was identified, which covered a 10-week rehabilitation process with patients [30]. Overall, the review revealed a lack of long-term studies or in-depth case studies that could provide deeper insights into real-world usage and long-term effects. For better comparability between systems, there is a need for standardized procedures in system validation and user testing. Such standardization would facilitate the evaluation of study outcomes and support the refinement of assistive devices for broader clinical and practical application.

4.2. Opportunities and Future Research Directions

Future research should systematically investigate the influence of sensor placement and calibration on measurement quality, including signal shifts and interpretation errors. This is essential for the development of practical guidelines that support the selection and placement of sensors and therefore improve the comparability, reproducibility, and generalizability of the results of different studies. In addition, the device tests should be conducted with the actual target groups under real conditions rather than in a laboratory environment. Moreover, larger sample sizes, as well as case studies and long-term evaluations, are needed to verify the reliability, usability, and clinical effectiveness of these systems over time. Beyond addressing current limitations, expanding sensor configurations and feedback modalities offers the opportunity to extend device functionalities and embed them within broader ecosystems of smart assistive technologies. In this context, future research should also explore the integration of intelligent walking aids with other connected devices (e.g., wearables) or mobile health platforms. Furthermore, the increasing use of ML holds potential to improve the accuracy of gait classification and the prediction of clinically relevant parameters, enabling broader application across various domains and advancing the analytical capabilities of such systems.

4.3. Implications

The findings of this review emphasize the need to further develop sensor-based walking aids beyond the early prototype stage. For researchers and practitioners, this involves conducting user-centered validation studies that address not only technical performance but also long-term usability, clinical effectiveness, and patient-specific needs. Several developments demonstrated the potential for modular sensor architectures, enabling adaptable configurations across various types of assistive devices. However, sensor systems must be integrated without compromising ergonomics and user comfort, and the measured health data should be interpretable for both patients and therapists. To enable wider adoption, future systems should be developed with consideration for interoperability and in line with ethical, legal, and data protection requirements. By overcoming these challenges and limitations, the development and implementation of smart walking aids can take a step forward in both clinical and home care settings.

4.4. Limitations of the Review

This scoping review is limited to the three selected databases and the applied keyword search strategy, which covered literature published until January 2025. While these sources were selected to capture studies on sensor-based assistive technologies, the inclusion of additional health-focused databases, particularly those indexing clinical trials, as well as gray literature containing unpublished innovations, could potentially have expanded the scope of our findings. As a result, certain perspectives may not be fully reflected in this review, which could introduce a selection bias. Another limitation is the targeted device categories and contexts of use that were included, as expanding to additional areas such as exoskeletons, wearable devices, or visually impaired users would lead to a broader range of studies. The quality and risk of bias of the included studies were not assessed, as the aim was to provide a comprehensive overview rather than a critical appraisal. Finally, heterogeneity in study designs, reported outcomes, and user testing procedures limit the comparability and synthesis of findings across studies.

5. Conclusions

This scoping review provides a structured overview of intelligent walking aids equipped with sensor technologies developed to support individuals with mobility impairments. Addressing the MRQ, we identified a broad spectrum of device types (including crutches, canes/sticks, walkers, and rollators) and examined their domains of application across rehabilitation, fall prevention, and gait monitoring.

In response to SRQ1, we found that various sensor technologies such as IMUs, force/pressure sensors, and environmental sensors are integrated to enable functional support, health monitoring, and interactive feedback. However, sensor placement and configuration varied widely across studies, which challenges the comparability of results and underscores the need for standardization. Regarding SRQ2, target user groups ranged from older adults to individuals with Parkinson’s disease or physical therapy patients. While the intended functionalities often align with user-specific needs, empirical evidence on system reliability and clinical effectiveness remains limited.

Our synthesis revealed several technological and methodological gaps. Although many studies propose innovative approaches, most systems remain in early-stage prototyping or proof-of-concept phases. In particular, few studies included system validation with the target population or long-term evaluation. As a result, the practical impact, reliability, and usability of these technologies are still unknown. Additionally, the heterogeneity in sensor placement, configuration, and system objectives complicates direct comparison and limits generalizability.

To address these challenges, we offer recommendations for researchers and practitioners. Future research should focus on rigorous system validation and on extending the functionality of adaptive feedback mechanisms tailored to individual patient needs, to overcome current limitations and strengthen the evidence base. The insights gained from this review could serve as a basis for the development of next-generation assistive technologies that are personalized, interactive, and directly support health outcomes. Overall, this review contributes to the broader fields of assistive technology, sensor-based rehabilitation, and HCI by synthesizing current developments and outlining future research directions.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	artificial intelligence
ANN	artificial neural network
BLE	Bluetooth low energy
DL	deep learning
DOF	degrees of freedom
ECG	electrocardiogram
EMG	electromyography
FSM	finite-state machine
FMCW	frequency modulated continuous wave
FSR	force sensing resistor
GPS	Global Positioning System
GSM	Global System for Mobile Communications
GUI	graphical user interface
HCI	human–computer interaction
HMM	hidden Markov model
IMU	inertial measurement unit
IoT	Internet of Things

IR	infrared
kNN	k-nearest neighbors
LCD	liquid crystal display
LDA	linear discriminant analysis
LED	light-emitting diode
LDR	light-dependent resistor
LSTM	long short-term memory
ML	machine learning
MRQ	main research question
PCA	principal component analysis
PPG	photoplethysmogram
PRISMA-ScR	Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews
RF	random forest
RFID	radio-frequency identification
SRQ	specific research question
SVM	support vector machine
WHO	World Health Organization

Appendix A. Search Query

Appendix A.1. ACM Digital Library

[[Full Text: "assistive device"] OR [Full Text: "assistive system"] OR [Full Text: "assistive technolog*"] OR [Full Text: "walking aid"]] AND [Full Text: sensor*] AND [[Full Text: walker*] OR [Full Text: cane] OR [Full Text: "walking stick"] OR [Full Text: crutch]] AND [[Full Text: "walking impairments"] OR [Full Text: "mobility support"] OR [Full Text: "gait rehabilitation"] OR [Full Text: "patient support"] OR [Full Text: "foot conditions"]]

Appendix A.2. Web of Science

(((ALL=("assistive device" OR "assistive system" OR "assistive technolog*" OR "walking aid")) AND ALL=(sensor*)) AND ALL=(walker* OR cane OR "walking stick" OR crutch)) AND ALL=("walking impairments" OR "mobility support" OR "gait rehabilitation" OR "patient support" OR "foot conditions")

References

1. World Health Organization; United Nations Children's Fund. *Global Report on Assistive Technology*; World Health Organization: Geneva, Switzerland, 2022.
2. Postolache, O.; Dias Pereira, J.M.; Viegas, V.; Pedro, L.; Girao, P.S.; Oliveira, R.; Postolache, G. Smart walker solutions for physical rehabilitation. *IEEE Instrum. Meas. Mag.* **2015**, *18*, 21–30. [CrossRef]
3. Martins, M.; Santos, C.P.; Frizera, A.; Matias, A.; Pereira, T.; Cotter, M.; Pereira, F. Smart walker use for ataxia's rehabilitation: Case study. In Proceedings of the 2015 IEEE International Conference on Rehabilitation Robotics (ICORR), Singapore, 11–14 August 2015; IEEE: Piscataway, NJ, USA, 2015; pp. 852–857. [CrossRef]
4. Martins, M.M.; Santos, C.P.; Frizera-Neto, A.; Ceres, R. Assistive mobility devices focusing on Smart Walkers: Classification and review. *Robot. Auton. Syst.* **2012**, *60*, 548–562. [CrossRef]
5. Costamagna, E.; Thies, S.; Kenney, L.; Howard, D.; Liu, A.; Ogden, D. A generalisable methodology for stability assessment of walking aid users. *Med Eng. Phys.* **2017**, *47*, 167–175. [CrossRef] [PubMed]
6. Panazan, C.E.; Dulf, E.H. Intelligent Cane for Assisting the Visually Impaired. *Technologies* **2024**, *12*, 75. [CrossRef]
7. Menikdiwela, M.P.; Dharmasena, K.; Abeykoon, A.H.S. Haptic based walking stick for visually impaired people. In Proceedings of the 2013 International conference on Circuits, Controls and Communications (CCUBE), Bengaluru, India, 27–28 December 2013; pp. 1–6. [CrossRef]
8. Trujillo-León, A.; de Guzmán-Manzano, A.; Velázquez, R.; Vidal-Verdú, F. Generation of Gait Events with a FSR Based Cane Handle. *Sensors* **2021**, *21*, 5632. [CrossRef]

9. Huang, M.; Clancy, E. Smart Walker: An IMU-Based Device for Patient Activity Logging and Fall Detection. In Proceedings of the 2022 IEEE Signal Processing in Medicine and Biology Symposium (SPMB), Philadelphia, PA, USA, 3 December 2022; pp. 1–4. [CrossRef]
10. Frizera, A.; Ceres, R.; Pons, J.L.; Abellanas, A.; Raya, R. The smart walkers as geriatric assistive device. The simbiosis purpose. In Proceedings of the 6th International Conference of the International Society for Gerontechnology, Pisa, Italy, 4–6 June 2008; Volume 7, pp. 1–6.
11. Kaushalya, V.; Premarathne, K.; Shadir, H.; Krithika, P.; Fernando, S. Automated Help Aid for Visually Impaired People using Obstacle Detection and GPS Technology: AKSHI. *Int. J. Sci. Res. Publ.* **2016**, *6*, 579–583.
12. Islam, M.M.; Sheikh Sadi, M.; Zamli, K.Z.; Ahmed, M.M. Developing Walking Assistants for Visually Impaired People: A Review. *IEEE Sens. J.* **2019**, *19*, 2814–2828. [CrossRef]
13. Engel, J.; Amir, A.; Messer, E.; Caspi, I. Walking cane designed to assist partial weight bearing. *Arch. Phys. Med. Rehabil.* **1983**, *64*, 386–388.
14. McCandless, P.J.; Evans, B.J.; Janssen, J.; Selfe, J.; Churchill, A.; Richards, J. Effect of three cueing devices for people with Parkinson's disease with gait initiation difficulties. *Gait Posture* **2016**, *44*, 7–11. [CrossRef]
15. Oladele, D.A.; Markus, E.D.; Abu-Mahfouz, A.M. Adaptability of assistive mobility devices and the role of the internet of medical things: Comprehensive review. *JMIR Rehabil. Assist. Technol.* **2021**, *8*, e29610. [CrossRef] [PubMed]
16. Doan, T.N.; Schroter, E.; Phan, T.B. Fall Detection Using Intelligent Walking-Aids and Machine Learning Methods. In Proceedings of the Intelligent Systems and Data Science, CanTho, Vietnam, 11–12 November 2023; Thai-Nghe, N., Do, T.N., Haddawy, P., Eds.; Springer: Singapore, 2023; pp. 95–109.
17. Wang, J.; Jiang, X.; Meng, Q.; Saada, M.; Cai, H. Walking motion real-time detection method based on walking stick, IoT, COPOD and improved LightGBM. *Appl. Intell.* **2022**, *52*, 16398–16416. [CrossRef]
18. Di, P.; Hasegawa, Y.; Nakagawa, S.; Sekiyama, K.; Fukuda, T.; Huang, J.; Huang, Q. Fall Detection and Prevention Control Using Walking-Aid Cane Robot. *IEEE/ASME Trans. Mechatron.* **2016**, *21*, 625–637. [CrossRef]
19. Resch, S.; Röll, A.; Malech, A.; Schwind, V.; Sanchez-Morillo, D.; Völz, D. Walking aid with haptic feedback for combined use with a smart foot orthosis. *Gerontechnology* **2024**, *23*, 1. [CrossRef]
20. Tricco, A.C.; Lillie, E.; Zarin, W.; O'Brien, K.K.; Colquhoun, H.; Levac, D.; Moher, D.; Peters, M.D.; Horsley, T.; Weeks, L.; et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. *Ann. Intern. Med.* **2018**, *169*, 467–473. [CrossRef] [PubMed]
21. Zotero. Free and Open Source Reference Manager—Your Personal Research Assistant. 2025. Available online: https://www.zotero.org/ (accessed on 10 July 2025).
22. Landis, J.R.; Koch, G.G. The Measurement of Observer Agreement for Categorical Data. *Biometrics* **1977**, *33*, 159–174. [CrossRef]
23. Mayring, P. Qualitative Content Analysis. *Forum Qual. Sozialforschung/Forum Qual. Soc. Res.* **2000**, *1*. [CrossRef]
24. Page, M.J.; McKenzie, J.E.; Bossuyt, P.M.; Boutron, I.; Hoffmann, T.C.; Mulrow, C.D.; Shamseer, L.; Tetzlaff, J.M.; Akl, E.A.; Brennan, S.E.; et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* **2021**, *372*. [CrossRef]
25. Haddaway, N.R.; Page, M.J.; Pritchard, C.C.; McGuinness, L.A. PRISMA2020: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis. *Campbell Syst. Rev.* **2022**, *18*, e1230. [CrossRef]
26. Ojeda, M.; Cortés, A.; Béjar, J.; Cortés, U. Automatic classification of gait patterns using a smart rollator and the BOSS model. In Proceedings of the 11th PErvasive Technologies Related to Assistive Environments Conference, PETRA '18, Corfu, Greece, 26–29 June 2018; ACM: New York, NY, USA, 2018; pp. 384–390. [CrossRef]
27. Sardini, E.; Serpelloni, M.; Lancini, M. Wireless Instrumented Crutches for Force and Movement Measurements for Gait Monitoring. *IEEE Trans. Instrum. Meas.* **2015**, *64*, 3369–3379. [CrossRef]
28. Viegas, V.; Dias Pereira, J.M.; Postolache, O.; Girão, P.S. Spy walker: A convenient way to assess gait in walker assistive devices. In Proceedings of the 2018 IEEE International Instrumentation and Measurement Technology Conference (I2MTC), Houston, TX, USA, 14–17 May 2018; pp. 1–6. [CrossRef]
29. Gill, S.; Nssk, S.; Seth, N.; Scheme, E. Design of a Smart IoT-Enabled Walker for Deployable Activity and Gait Monitoring. In Proceedings of the 2018 IEEE Life Sciences Conference (LSC), Montreal, QC, Canada, 28–30 October 2018; pp. 183–186. [CrossRef]
30. Ballesteros, J.; Urdiales, C.; Martinez, A.B.; Tirado, M. Automatic Assessment of a Rollator-User's Condition During Rehabilitation Using the i-Walker Platform. *IEEE Trans. Neural Syst. Rehabil. Eng.* **2017**, *25*, 2009–2017. [CrossRef]
31. Wade, J.; Beccani, M.; Myszka, A.; Bekele, E.; Valdastri, P.; Flemming, P.; de Riesthal, M.; Withrow, T.; Sarkar, N. Design and implementation of an instrumented cane for gait recognition. In Proceedings of the 2015 IEEE International Conference on Robotics and Automation (ICRA), Seattle, WA, USA, 26–30 May 2015; pp. 5904–5909. [CrossRef]

32. Mekki, F.; Borghetti, M.; Sardini, E.; Serpelloni, M. Wireless instrumented cane for walking monitoring in Parkinson patients. In Proceedings of the 2017 IEEE International Symposium on Medical Measurements and Applications (MeMeA), Rochester, MN, USA, 7–10 May 2017; pp. 414–419. [CrossRef]
33. Narváez, M.; Salazar, M.; Aranda, J. Identification of gait patterns in walking with crutches through the selection of significant spatio-temporal parameters. In Proceedings of the 2022 International Conference on Rehabilitation Robotics (ICORR), Rotterdam, The Netherlands, 25–29 July 2022; pp. 1–6. [CrossRef]
34. Wade, J.W.; Boyles, R.; Flemming, P.; Sarkar, A.; de Riesthal, M.; Withrow, T.J.; Sarkar, N. Feasibility of Automated Mobility Assessment of Older Adults via an Instrumented Cane. *IEEE J. Biomed. Health Inform.* **2019**, *23*, 1631–1638. [CrossRef]
35. Postolache, O.; Viegas, V.; Pereira, J.M.D. Gait Assesment using Microwave Radars mounted on Standard Walker. In Proceedings of the 2024 IEEE International Conference And Exposition On Electric And Power Engineering (EPEi), Iasi, Romania, 17–19 October 2024; pp. 687–692. [CrossRef]
36. Zhou, J.L.; Li, W.F.; Zhang, Q.; Xie, F.; Wang, Q. A Smart Walking Stick for Gait Analysis of Elderly and People with Disabilities. *IEEE Sens. J.* **2022**, *22*, 9035–9045. [CrossRef]
37. Gill, S.; Hearn, J.; Powell, G.; Scheme, E. Design of a multi-sensor IoT-enabled assistive device for discrete and deployable gait monitoring. In Proceedings of the 2017 IEEE Healthcare Innovations and Point of Care Technologies (HI-POCT), Bethesda, MD, USA, 6–8 November 2017; pp. 216–220. [CrossRef]
38. Postolache, O. Physical rehabilitation assessment based on smart training equipment and mobile APPs. In Proceedings of the 2015 E-Health and Bioengineering Conference (EHB), Iasi, Romania, 19–21 November 2015; pp. 1–6. [CrossRef]
39. Ballesteros, J.; Urdiales, C.; Martinez, A.B.; Tirado, M. Online estimation of rollator user condition using spatiotemporal gait parameters. In Proceedings of the 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Daejeon, Republic of Korea, 9–14 October 2016; pp. 3180–3185. [CrossRef]
40. Batoca, P.; Postolache, O.; Correia, A. Physical Therapy Gait Assessment based on Smart Sensing and Cloud Services. In Proceedings of the 2022 International Symposium on Sensing and Instrumentation in 5G and IoT Era (ISSI), Shanghai, China, 17–18 November 2022; pp. 138–143. [CrossRef]
41. Ballesteros, J.; Tudela, A.; Caro-Romero, J.R.; Urdiales, C. A cane-based low cost sensor to implement attention mechanisms in telecare robots. In Proceedings of the 2019 International Conference on Robotics and Automation (ICRA), Montreal, QC, Canada, 20–24 May 2019; pp. 1473–1478. [CrossRef]
42. Domingues, D.G.; Funghetto, S.; Miranda, M.R.; Batista, P.K.C.M.; de Oliveira, P.R.F.; Assis, G.A.; da Rocha, A.F.; Torres, R.d.S. Mobility and freedom: Affective cane for expanded sensorium and embodied cognition. In Proceedings of the 2017 23rd International Conference on Virtual System & Multimedia (VSMM), Dublin, Ireland, 31 October–4 November 2017; pp. 1–7. [CrossRef]
43. Culmer, P.R.; Brooks, P.C.; Strauss, D.N.; Ross, D.H.; Levesley, M.C.; O'Connor, R.J.; Bhakta, B.B. An Instrumented Walking Aid to Assess and Retrain Gait. *IEEE/ASME Trans. Mechatron.* **2014**, *19*, 141–148. [CrossRef]
44. Valsangkar, A.; Kumar, P.; Scheme, E. Automated Segmentation of a Timed Up and Go Test Using an Instrumented Cane. In Proceedings of the 2021 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI), Athens, Greece, 27–30 July 2021; pp. 1–5. [CrossRef]
45. Chalvatzaki, G.G.; Pavlakos, G.; Maninis, K.; Papageorgiou, X.S.; Pitsikalis, V.; Tzafestas, C.S.; Maragos, P. Towards an intelligent robotic walker for assisted living using multimodal sensorial data. In Proceedings of the 2014 4th International Conference on Wireless Mobile Communication and Healthcare—Transforming Healthcare Through Innovations in Mobile and Wireless Technologies (MOBIHEALTH), Athens, Greece, 3–5 November 2014; pp. 156–159. [CrossRef]
46. Mesanza, A.B.; Lucas, S.; Zubizarreta, A.; Cabanes, I.; Portillo, E.; Rodriguez-Larrad, A. A Machine Learning Approach to Perform Physical Activity Classification Using a Sensorized Crutch Tip. *IEEE Access* **2020**, *8*, 210023–210034. [CrossRef]
47. Seylan, C.; Saranlı, U. Estimation of Ground Reaction Forces Using Low-Cost Instrumented Forearm Crutches. *IEEE Trans. Instrum. Meas.* **2018**, *67*, 1308–1316. [CrossRef]
48. Chan, A.D.C.; Green, J.R. Smart Rollator Prototype. In Proceedings of the 2008 IEEE International Workshop on Medical Measurements and Applications, Ottawa, ON, Canada, 9–10 May 2008; pp. 97–100. [CrossRef]
49. Ballesteros, J.; Urdiales, C.; Martinez, A.B.; Tirado, M. Gait analysis for challenged users based on a rollator equipped with force sensors. In Proceedings of the 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hamburg, Germany, 28 September–2 October 2015; pp. 5587–5592. [CrossRef]
50. Frango, P.M.V.L.; Postolache, O.A. Mobile Application based on Wireless Sensor Network for Physical Rehabilitation. In Proceedings of the 2018 International Symposium in Sensing and Instrumentation in IoT Era (ISSI), Shanghai, China, 6–7 September 2018; pp. 1–6. [CrossRef]
51. Arcobelli, V.; Zauli, M.; De Marchi, L.; Chiari, L.; Mellone, S. Assessment of crutch-assisted walking with sensorized crutches in a 6-Minute Walk Test. In Proceedings of the 2023 IEEE EMBS Special Topic Conference on Data Science and Engineering in Healthcare, Medicine and Biology, Portomaso, Malta, 7–9 December 2023; pp. 11–12. [CrossRef]

52. Cavaglià, M.S.; Sierra M, S.D.; Palmerini, L.; Orlandi, S.; Múnera, M.; Cifuentes, C.A. Towards Safer Mobility: Developing and Evaluating a Fall Detection System for a Smart Walker. In Proceedings of the 2024 10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob), Heidelberg, Germany, 1–4 September 2024; pp. 1623–1628. [CrossRef]
53. Zambrano, E.O.; Muñoz, K.R.; Armas-Aguirre, J.; González, P.A. Technological Architecture with Low Cost Sensors to Improve Physical Therapy Monitoring. In Proceedings of the 2020 15th Iberian Conference on Information Systems and Technologies (CISTI), Seville, Spain, 24–27 June 2020; pp. 1–6. [CrossRef]
54. Ribeiro, N.F.; Santos, C.P. Two Fall-Related and Kinematic Data-Based Approaches for an Instrumented Conventional Cane. *IEEE Trans. Hum.-Mach. Syst.* **2021**, *51*, 554–563. [CrossRef]
55. Anushree, A.; Nesakumar, D.; Ajeet, R.; Gopika, S.; Haritha, V.; Pavithra, K. Posturesense Intelligent Body Support Crutch Tool. In Proceedings of the 2024 International Conference on Computing and Data Science (ICCDS), Chennai, India, 26–27 April 2024; pp. 1–6. [CrossRef]
56. Kikuchi, T.; Tanaka, T.; Anzai, K.; Kawakami, S.; Hosaka, M.; and, K.N. Evaluation of line-tracing controller of intelligently controllable walker. *Adv. Robot.* **2013**, *27*, 493–502. [CrossRef]
57. Ballesteros, J.; Peula, J.M.; Martinez, A.B.; Urdiales, C. Automatic Fall Risk Assessment for Challenged Users Obtained from a Rollator Equipped with Force Sensors and a RGB-D Camera. In Proceedings of the 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Madrid, Spain, 1–5 October 2018; pp. 7356–7361. [CrossRef]
58. Padmavathi, B.; Sahithi, S.L.Y.; Sudarshan, V.; Nandini, K.N. An Effective Body Posture Management System using Novel Intelligent Crutch Tool Mechanism. In Proceedings of the 2023 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI), Chennai, India, 19–21 April 2023; pp. 1–6. [CrossRef]
59. Koziol, S.; Friedman, P.A.; Schultze, B.; Schubert, K. The reminding walker cognitive orthotic device. In Proceedings of the 2017 International Symposium on Wearable Robotics and Rehabilitation (WeRob), Houston, TX, USA, 5–8 November 2017; pp. 1–2. [CrossRef]
60. Gerena, R.; VanDeventer, P.; Vistamehr, A.; Conroy, C.; Freeborn, P.; Govin, H.; Fox, E.; Aceros, J. Wireless instrumented walker for remote rehabilitation monitoring. In Proceedings of the 2015 IEEE Virtual Conference on Applications of Commercial Sensors (VCACS), Raleigh, NC, USA, 15 March–15 October 2015; pp. 1–7. [CrossRef]

G.2. Journal Publication 2

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Smart Device Development for Gait Monitoring: Multimodal Feedback in an Interactive Foot Orthosis, Walking Aid, and Mobile Application

Stefan Resch [1,2,*], André Kousha [1], Anna Carroll [3], Noah Severinghaus [4], Felix Rehberg [1], Marco Zatschker [1], Yunus Söyleyici [1] and Daniel Sanchez-Morillo [5,6]

1 Faculty of Computer Science and Engineering, Frankfurt University of Applied Sciences, 60318 Frankfurt am Main, Germany
2 Doctorate Program in Manufacturing, Materials and Environmental Engineering, Doctorate School (EDUCA), University of Cadiz, 11003 Cádiz, Spain
3 Department of Biomedical Engineering, University of Illinois Chicago, Chicago, IL 60607, USA
4 School for Engineering of Matter, Transport, and Energy, Arizona State University, Tempe, AZ 85287, USA
5 Department of Automation Engineering, Electronics and Computer Architecture and Networks, School of Engineering, University of Cadiz, 11519 Cádiz, Spain
6 Instituto de Investigación e Innovación Biomédica de Cádiz (INiBICA), 11009 Cádiz, Spain
* Correspondence: stefan.resch@alum.uca.es

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Abstract

Smart assistive technologies such as sensor-based footwear and walking aids offer promising opportunities for gait rehabilitation through real-time feedback and patient-centered monitoring. While biofeedback applications show great potential, current research rarely explores integrated closed-loop systems with device- and modality-specific feedback. In this work, we present a modular sensor-based system combining a smart foot orthosis and an instrumented forearm crutch to deliver real-time vibrotactile biofeedback. The system integrates plantar pressure and motion sensing, vibrotactile feedback, and wireless communication via a smartphone application. We conducted a user study with eight participants to validate the system's feasibility for mobile gait detection and app usability, and to evaluate different vibrotactile feedback types across the orthosis and forearm crutch. The results indicate that pattern-based vibrotactile feedback was rated as more useful and suitable for regular use than simple vibration alerts. Moreover, participants reported clear perceptual differences between feedback delivered via the orthosis and the forearm crutch, indicating device-dependent feedback perception. The findings highlight the relevance of feedback strategy design beyond hardware implementation and inform the development of user-centered haptic biofeedback systems.

Keywords: assistive technology; foot health; gait analysis; haptic feedback; instrumented walking aid; mHealth application; rehabilitation; smart insole; smart orthosis; wearable sensors

1. Introduction

Gait rehabilitation plays an important role in the recovery process after acute foot injuries and in the management of chronic foot-related conditions. A variety of assistive devices are used to support these patients in their daily mobility, including orthopedic devices such as foot orthoses, insoles, bandages, and pressure-relieving footwear [1]. In addition, mobility aids such as crutches, walking sticks, or canes are commonly used for temporary or long-term walking impairments [2]. While these devices are essential in stabilizing gait and reducing mechanical stress [3,4], they are typically standardized and require customization to achieve optimal recovery outcomes [5], as they can measurably affect gait [6]. These limitations underscore the need for more individualized and adaptive orthopedic solutions that combine health monitoring with interactive feedback.

In recent years, the integration of wearable electronics has contributed to various developments such as smart insoles, socks, and shoes equipped with sensor technologies for gait analysis and health monitoring [7–9]. In contrast to conventional stationary systems, these mobile solutions enable gait analysis outside clinical settings and open up new possibilities for continuous monitoring of spatial, temporal, and spatio-temporal parameters [10]. These systems frequently rely on force sensing resistors (FSRs) and inertial measurement units (IMUs) to assess parameters such as plantar pressure distribution and gait characteristics (e.g., step cadence and foot strike patterns) [11]. Such technologies show promise in areas such as fall risk assessment, biofeedback interventions, and the prevention of foot ulcers [12]. In particular, monitoring plantar pressure is crucial for patients with diabetic foot syndrome (DFS), where elevated sole pressure can impair wound healing and increase the risk of ulcer formation [13]. They also hold potential for injury prevention and early pathology detection [14].

However, orthopedic devices such as ankle-foot orthoses (AFOs), which are used to stabilize gait and support recovery, still rarely combine embedded sensing and feedback within an interactive system. Integrating feedback modalities (such as haptic alerts triggered by pressure overload) could enhance the benefits of smart foot orthosis (SFO) development, particularly for individuals with foot conditions [15]. For example, diabetic neuropathy in the lower extremities impairs the sensation of pain and limits the detection of excessive plantar pressure [16], which can lead to unnoticed injuries and increase the risk of serious deformities such as Charcot foot [17]. Providing patients with intrinsic feedback could therefore positively impact gait recovery. While several conceptual prototypes have integrated electronics into orthoses [18–20], most solutions remain technically focused, without addressing how haptic feedback should be designed or delivered. There is a lack of empirical evidence on how feedback modality in smart footwear and walking aids influences user perception, perceived effectiveness, and intended use. This is critical, as unclear or intrusive feedback can reduce acceptance even when technically accurate.

To address the identified research gap, we developed a modular sensor system to investigate the influence of haptic feedback modality and device-dependent feedback on user perception in smart footwear and walking aids. The system comprises three key components: (1) an SFO equipped with IMU and FSR sensors as well as a vibration feedback module, (2) a sensor-integrated walking aid that communicates with the orthosis and includes its own IMU and haptic actuators, and (3) a mobile smartphone application for health monitoring and feedback interaction. The system architecture enables real-time communication across all components and provides the basis for evaluating closed-loop vibrotactile biofeedback strategies. To ensure a valid basis for user perception testing, we examined the system's feasibility for gait detection and the usability of the mobile application from a user interaction perspective. These validation objectives guided the following supporting research questions (SRQs):

- SRQ1 (System Feasibility): How accurately does the developed SFO measure plantar pressure and gait parameters compared to a clinical reference system, as a foundation for reliable real-time multimodal feedback?
- SRQ2 (Application Usability): How do users evaluate the usability of the accompanying mobile application for interpreting gait and pressure information within the integrated system?

Based on this, the core objective of this work is to investigate feedback strategy design for user-centered biofeedback, addressing the main research question (MRQ):

- MRQ (Feedback Evaluation): How do different vibrotactile feedback types (continuous vs. pattern-based), delivered through the orthosis and the walking aid, influence user perception in terms of noticeability, intrusiveness, perceived usefulness, and intended use?

This work contributes to the field of assistive technology by introducing a novel closed-loop, sensor-driven haptic feedback mechanism in a sensorized foot orthosis and instrumented forearm crutch, enabling real-time feedback that responds to gait dynamics and pressure distribution. Our key research findings provide empirical insights that users significantly prefer pattern-based over simple vibrotactile feedback notifications and that device location influences user perception. These results offer concrete recommendations for designing effective, user-centered haptic rehabilitation systems and can serve as a basis for future patient-centered evaluations of multimodal feedback in human-computer interaction (HCI).

2. Related Work

This section reviews related research in the areas of smart footwear for mobile gait analysis and health monitoring (Section 2.1), foot augmentation with multimodal feedback for patient–device interaction (Section 2.2), and sensor-integrated walking aids for mobility support (Section 2.3). We conclude by summarizing current research gaps and outlining our contributions to the field.

2.1. Smart Footwear for Mobile Gait Analysis and Monitoring Health Parameters

The research field of sensor technology demonstrates that different types of sensors are used for measuring continuous physiological data depending on the application [14]. Based on the device type, sensor technologies can be integrated into various systems such as socks/textiles [21–25], shoes [26–30], insoles [8,9,31–33], or orthoses [18,19,34–36]. In particular, smart insoles are developed for a broad range of applications, ranging from ambulatory gait assessment and home-based monitoring [37,38], to gait rehabilitation after stroke [39], fall risk detection [40], and activity classification using machine learning techniques [41,42].

A systematic literature review by Subramaniam et al. [14] identified two primary health-related functions of smart insoles: (1) the measurement of plantar pressure distribution and (2) the analysis of gait and activity patterns. Plantar pressure data are typically acquired using integrated FSRs for the assessment of foot stability, balance, body weight distribution, and early detection of pressure overload conditions. A review by Razak et al. [43] recommended the use of 15 FSR sensors to ensure sufficient coverage of plantar pressure distribution according to the individual foot anatomy. This recommendation is based on an anatomical segmentation of the foot sole into 15 regions: heel (areas 1–3), midfoot (4–5), metatarsals (6–10), and toes (11–15), which support most of the body load [44]. However, the individual shoe size and foot anatomy need to be considered for proper sensor placement to prevent measurement errors [45].

Furthermore, the use of IMU sensors (a combination of accelerometers and gyroscopes) enables the measurement of gait characteristics such as step length, cadence, walking speed, and swing phase duration [46]. These systems are used for performance tracking of athletes [14] or to enable mobile gait analysis [47]. A single IMU sensor embedded in the insole can provide sufficient data for this purpose [48]. However, the literature reports inconsistencies regarding the optimal sensor placement for precise data extraction, as signal drift and measurement errors may occur depending on the location of the sensor [49].

As a result, calibration remains a critical challenge in smart insole development [14]. To minimize mechanical load and improve signal quality, placement in the midfoot region is generally recommended [14].

Beyond IMU and FSR-based gait analysis, additional sensing modalities have been explored to extend the functionality of wearable systems. These include Global Positioning System (GPS) tracking for spatial gait analysis [50], foot temperature monitoring for early detection of DFS [12,51], and physiological measurements such as heart rate via photoplethysmography (PPG) sensors [52] and muscle activity using electromyography (EMG) electrodes in textile-based wearables [22]. However, PPG and EMG-based approaches remain largely limited to research prototypes and have not yet been adopted in commercial devices. Other studies have also investigated sensing technologies for obstacle detection, including ultrasonic sensors, ToF cameras, and radar-based systems, particularly in applications for visually impaired users [53]. While these alternative sensing modalities offer promising extensions for specific use cases, the majority of current smart insole systems rely on the integration of IMU and FSR sensors. Due to its efficiency and widespread adoption, the combination is frequently used in smart insole systems [54]. This configuration enables accurate gait detection [55,56] and simultaneous acquisition of kinetic (pressure) and kinematic (motion) data [11,57]. Typically, data from IMU and FSR sensors are acquired using a portable microcontroller and transmitted wirelessly (e.g., via Bluetooth) to an external graphical user interface (GUI) [37,58].

To ensure the reliability of gait-related measurements derived from wearable systems, their performance must be validated against established reference standards. A review by Hutabarat et al. [10] emphasized that such wearable gait systems need to be validated against proven standard gait measurement systems to ensure data accuracy. For this purpose, instrumented treadmills and motion capture systems (MoCaps) represent the gold standard for validating gait analysis systems [10,59]. This approach has already been applied in several validation studies, for instance by comparing wearable systems to a stationary Zebris treadmill [60] or GAITRite force platforms [61,62]. Overall, recent research highlights the importance of comparative validation studies with established gait analysis systems to strengthen the credibility and reliability of findings in wearable gait assessment [63]. In conclusion, despite considerable advances in this research field, key challenges remain regarding data accuracy, system validation, and participant-centered evaluation.

2.2. Foot Augmentation with Multimodal Feedback for Patient Device Interaction

Smart wearable systems such as insoles, socks, or orthoses are often connected to external devices (e.g., smartphone applications, smartwatches, or other interfaces) to enable continuous mobile health monitoring and interactive feedback [14]. A systematic review by Saboor et al. [64] highlighted the growing potential of smartphone applications for mobile gait analysis, particularly in remote health contexts, while also noting ongoing challenges related to accessible and user-friendly user interface (UI) design. In this context, Resch et al. [65] developed a mobile Health (mHealth) application for smart insoles on the basis of focus groups with patients and experts. In a follow-up study, functional mockups were evaluated in a usability test with 30 participants, which confirmed the user-friendliness and provided recommendations for future design improvements [66]. The resulting design recommendations, integrated features, and overall app concept provide a valuable basis for the continued development of smart insole applications and the integration of feedback modalities.

Beyond interface design, wearable systems increasingly incorporate real-time feedback mechanisms (through visual, acoustic, or haptic signals) typically via smartphone apps or connected devices to support user interaction. For instance, vibration cues have been applied to indicate pressure overload [67]. It was shown that vibration feedback applied within the shoe can significantly influence gait behavior in healthy participants [68], and vibrotactile biofeedback triggered by FSR input reduced overpronation duration by 50% [69]. Furthermore, auditory feedback has also been investigated, such as acoustic signals triggered by in-sock pressure sensors for gait correction [70], or auditory stimulation via insoles for patients with Parkinson's disease [71]. These approaches are particularly relevant for patients with impaired peripheral sensation, such as individuals with DFS, for whom smartwatch-based alerts and real-time gait correction via smart insoles have been proposed [72]. Several studies have applied multimodal feedback approaches and shown that they influence gait symmetry, posture and pressure distribution [68,69,73,74]. In an experimental study, Redd et al. [73] compared visual, acoustic, and vibrotactile feedback modalities in healthy participants and found that both visual and haptic cues significantly improved gait symmetry. Despite the demonstrated potential of smartphone applications and feedback modalities, further validation is needed regarding user perception and system effectiveness.

2.3. Smart Walking Aids and Applied Sensor Technologies

Walking aids play a crucial role in daily mobility support for patients with walking impairments. Recent research has explored how these devices can be enhanced through sensor technologies to support health monitoring, user interaction, and feedback-based interventions [75,76]. Various smart walking aid prototypes have been proposed, including instrumented sticks [77], canes [78], and forearm crutches [79]. Many of these systems have been developed specifically for users with visual impairments [80–83]. These systems typically contain sensors such as IMUs for fall detection or for motion tracking [84,85], FSRs embedded in handles to detect gait events [86], GPS modules for positional tracking [87], and ultrasonic sensors for obstacle detection [80–82].

Beyond data acquisition, several studies have explored the integration of feedback mechanisms to enhance user support [83,88,89]. For example, Merrett et al. [88] demonstrated the potential of biofeedback in forearm crutches using acoustic signals based on FSR data to support optimal weight distribution. Schuster et al. [89] showed that haptic feedback delivered via the handle of a cane can reduce knee adduction moments in individuals with knee osteoarthritis. In addition, a smart stick with integrated SOS functionality has been developed that connects to smartphone applications [90].

While individual sensor-based walking aids have demonstrated promising results, integrated systems that combine smart walking aids with intelligent footwear for foot augmentation remain rare and are typically limited to specific application scenarios. For instance, Suman et al. [91] proposed a system in which a smart cane communicates with smart footwear, using ultrasonic sensors in the shoe and auditory feedback via the cane handle. Another example is the use of force-sensitive crutches to assist lower-limb exoskeleton users by capturing ground force data to enable active support [92]. Furthermore, a robotic walking cane was developed that receives center of pressure (COP) data from four FSRs integrated in an insole to support fall detection [84]. However, direct communication between walking aids and orthopedic devices, such as sensor-based foot orthoses, has not yet been realized. In this context, Resch et al. [15] proposed a conceptual system that enables a walking aid to communicate with a smart AFO through vibration feedback in the handle to indicate plantar pressure overload. Nevertheless, the proposed system remains a concept without implementation or empirical evaluation.

Overall, while sensor-based walking aids and feedback modalities offer high potential, most systems remain in the prototype stage and lack systematic validation or integration into broader rehabilitation frameworks [76].

2.4. Research Gap

In summary, various systems for mobile foot health monitoring have been developed, primarily based on IMUs and FSRs, often in combination with wireless communication to external devices such as smartphone applications. While smart insoles have been extensively studied, their integration into orthopedic devices such as AFOs remains limited. Similarly, sensor-based walking aids show promising results for gait support and biofeedback, but most remain standalone prototypes without integration into more comprehensive assistive systems. Although haptic feedback has proven to be beneficial for improving user performance, there is no tested approach to date that combines it with sensor-based footwear and walking aids in a unified system.

Related work lacks concepts and empirical evidence for the combined use of smart walking aids and foot monitoring technologies. However, this integration offers future potential for improving patient monitoring, supporting real-time gait analysis and enabling multimodal feedback in healthcare and everyday use. Building on these findings, our work addresses these gaps by developing and validating an integrated system that combines a smart foot orthosis, an instrumented walking aid, and a smartphone application for real-time gait monitoring and interactive feedback. By addressing these gaps, we contribute to the advancement of assistive device technologies, mobile gait analysis, and user-centered feedback systems.

3. Method

This section describes the overall system architecture and the technical implementation of the smart orthosis, the instrumented walking aid, and the smartphone application. It is structured as follows: Section 3.1 provides a system overview, Section 3.2 presents the hardware architecture and concept development, and Section 3.3 outlines the software architecture and data processing for each subsystem. Finally, Section 3.4 presents the experimental setup of the prototype validation study, including apparatus, procedure, and measurements with data analysis.

3.1. System Overview

Figure 1 provides an overview of the system architecture, showing the interaction between the SFO, the instrumented walking aid, and the mobile application for data visualization and feedback control.

The SFO is based on a standardized lower-leg orthosis (AIRCAST® AIRSELECT™ Elite Walker [93] by ORMED GmbH, a company of Enovis, Freiburg, Germany), also known as a controlled ankle motion boot, which can be worn universally on either foot. A sensor concept consisting of multiple FSRs and a single IMU is integrated into a 3D-printed insole to capture plantar pressure and motion data. The insole is designed as a modular component, with adjustable size and shape to accommodate different orthotic devices and foot sizes.

Sensor fusion and real-time data processing enable the extraction of gait parameters and the detection of critical loading patterns. To support active user guidance, haptic feedback is provided via a vibration motor embedded in the orthosis, which is triggered in response to excessive plantar pressure. A custom-designed printed circuit board (PCB), equipped with an on-board microcontroller unit (MCU), enables wireless communication via Bluetooth Low Energy (BLE) with both the mobile application and the instrumented walking aid. The forearm crutch is based on a standardized design (ISO 11334-1:2007) [94] and includes a custom-designed PCB equipped with an MCU, an IMU sensor, two vibration modules, and a rechargeable battery. All components are embedded in the handle to enable synchronized haptic feedback and motion sensing. In addition, a corresponding mHealth application developed in Android Studio provides a graphical interface for visualizing gait parameters and plantar pressure data.

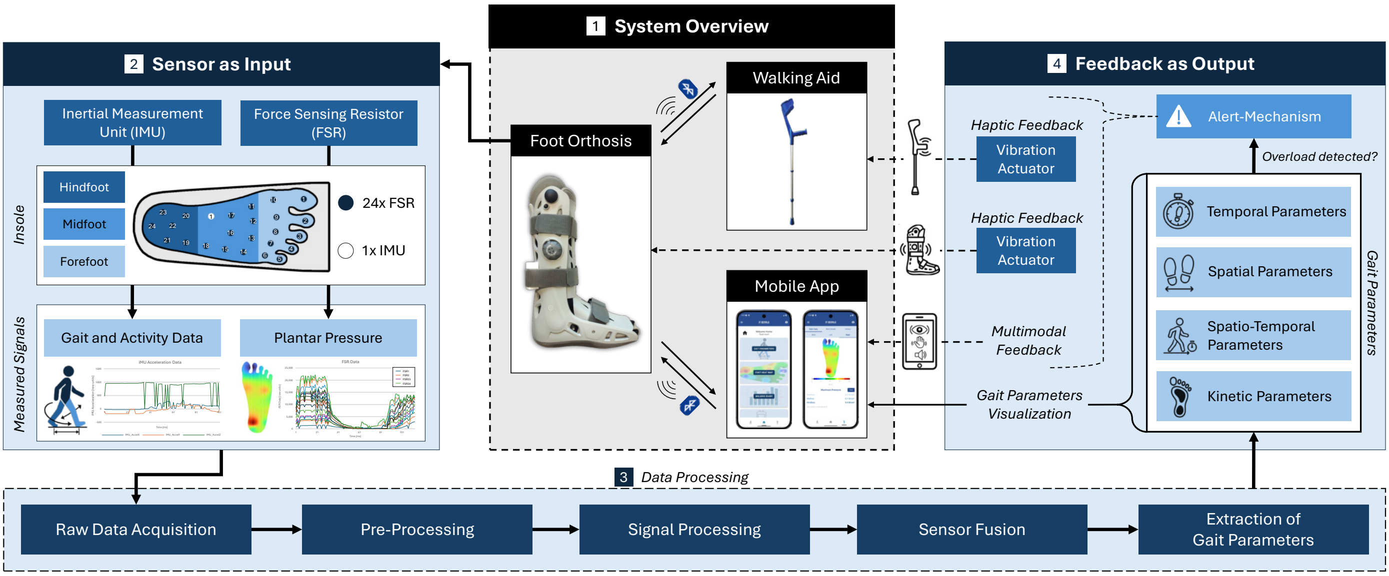


Figure 1. Overview of the sensor-based assistive system consisting of a smart orthosis with integrated FSR and IMU sensors, vibration feedback, and its communication with a walking aid and an mHealth application for interactive gait monitoring.

3.2. Hardware Architecture and Concept Development

3.2.1. Smart Orthosis

A schematic overview of the SFO prototype is presented in Figure 2a and the assembled insole with integrated sensor components is shown in Figure 2b. The SFO is based on a modular insole integrated into a standard lower-leg orthosis. An optional external vibration module can be placed at varying heights along the calf using the holes of the orthosis, allowing flexible positioning for haptic feedback. The insole includes multiple FSRs and an IMU sensor, connected via a flat cable to an external PCB, which is enclosed in a 3D-printed housing mounted on the posterior strap of the orthosis. The enclosure was designed using the computer-aided design (CAD) software Autodesk Fusion 360 (v.2603.0.86) and fabricated from polylactid (PLA) with a 0.4 mm nozzle (HT extruder) using the Flashforge Creator 4 3D printer. The insole consists of a two-part structure, comprising a bottom and a top layer, both fabricated from flexible thermoplastic polyurethane (TPU) using the extruder F with 0.4 mm nozzle size. The geometry replicates the original insole of the AIRCAST® orthosis to ensure proper fit and alignment, without affecting wearing comfort. The design enables adaptation for the contralateral healthy foot by modifying the insole shape and enclosure mounting to fit standard footwear. It also supports future integration into other orthotic systems tailored to individual foot anatomies.

The modular insole was designed to include the recommended minimum of 15 FSRs [43] (Model 400 short tail, Interlink Electronics [95]) to capture plantar pressure distribution. The placement of the FSRs is based on the anatomical structure of the foot and covers key plantar regions, including the heel, forefoot, metatarsal arch, and toes. The layout supports extensions up to 24 FSRs, which may be beneficial for applications involving foot deformities or more detailed regional pressure mapping. Each sensor has an active area of 5.6 mm in diameter, a thickness of 0.3 mm, and a force sensitivity range of 0–20 N, offering fast response times and high reliability for dynamic pressure measurements [96].

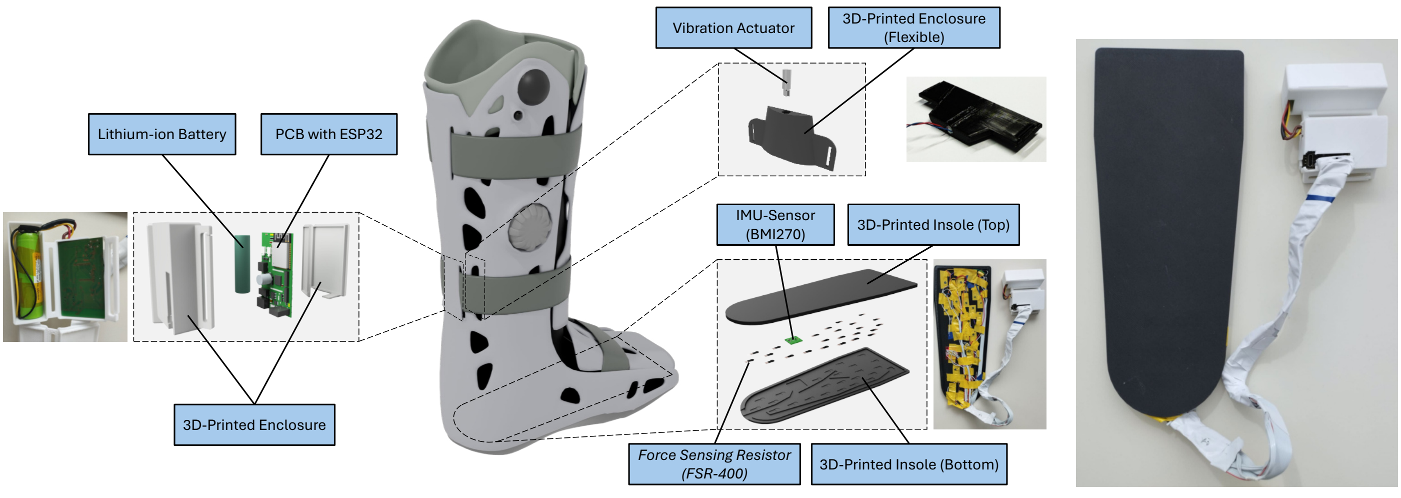


Figure 2. (**a**) Overview of the orthosis prototype and assembled components; (**b**) Final assembled prototype of the smart insole.

For motion tracking, a 6-axis IMU (BMI270, Bosch [97]) was integrated into the insole. The IMU was positioned near the metatarsal arch to minimize pressure exposure. This sensor was selected based on key technical requirements for wearable gait analysis systems and is commonly used in fitness trackers and smart wearables, such as smart shoes for step counting and activity recognition [98]. The ultra-low power sensor combines a 3-axis accelerometer (sampling frequencies from 25 Hz to 6.4 kHz) and a 3-axis gyroscope (0.78 Hz to 1.6 kHz) within compact dimensions of 0.8 mm $\times$ 3 mm $\times$ 2.5 mm.

A custom PCB (1.55 mm thickness, 35 µm copper) with a total size of 60 $\times$ 35 mm was designed using Altium Designer (v.25.1.2) as Electronic Computer-Aided Design (ECAD) software. A four-layer design was chosen to achieve a compact and reliable layout. The top layer was used for signals and components, the inner layers for ground planes, and the bottom layer for additional signal routing, allowing clear separation of signal paths, power supply, and grounding. The SFO is powered by a 3.7 V, 2600 mAh lithium-ion battery (Emmerich ICR-18650NQ-SP) integrated with a fuse, a shutdown switch, and a battery management system (MCP73838). A buck-boost converter regulates the supply voltage to a stable 3.3 V. To enable Wi-Fi, Bluetooth, and BLE communication, a 2.4 GHz ESP32-WROOM-32E-N8 MCU with an integrated PCB antenna from Espressif Systems was used, which is widely utilized in internet of things (IoT) applications. Additional components include a USB-C charging port and a multicolor light emitting diode (LED), which indicates the connection state, battery level, and charging status. Electrostatic discharge protection for electronic components was implemented in accordance with IEC 61000-4-2 and IEC 61340-5, using protection diodes on critical inputs such as the USB interface. A 32-channel analog multiplexer was implemented to enable the integration of up to 24 FSRs, while keeping signal paths short and maintaining robust ground routing to preserve signal integrity.

An optional interface for vibration feedback was implemented using a transistor-based low-side switch controlled via pulse-width modulation. This simple and efficient solution was chosen over more complex driver ICs or inter-integrated circuit communication (I^2C)-based vibration modules to reduce system complexity and footprint. This approach allows for flexible use of an eccentric rotating mass motor (ERM). To generate a sufficiently strong tactile signal, an ERM vibration actuator (Model TC-13324656, TRU Components) operating at 2.4 V and 250 mA was selected.

To minimize space, components were placed and routed according to system requirements and manufacturer guidelines. The final design was verified using electrical rule check (ERC) and design rule checks (DRC). Board assembly was performed manually using solder paste and a heating plate for surface-mounted device (SMD) soldering, followed by manual mounting of through-hole technology (THT) components. The assembled board was visually inspected and functionally tested to ensure proper soldering and reliable system performance. Additionally, detailed battery tests confirmed the functionality of the integrated battery management system, including discharge and charging profiles. The estimated runtime without vibration feedback was approximately 35.5 h, with a full recharge time of 6–7 h depending on the initial state of discharge.

3.2.2. Smart Walking Aid

To enable active haptic feedback through an assistive device, a forearm crutch was selected as the basis for sensor integration, as such walking aids are commonly used in combination with AFOs and offer a practical interface for feedback delivery. We developed a modular integration concept that can be transferred to various walking aids, including sticks, canes, or walkers, provided the handle design allows for modification without compromising ergonomics. This is essential, as major modifications to the handgrip require careful consideration of weight distribution and ergonomic design to avoid negative impact. For example, sensor placement along the shaft could alter the center of gravity, which may require new validation/certification of the walking aid [99]. Therefore, our design specifically targeted integration without major modifications to the handgrip. In this implementation, we used a standardized aluminum forearm crutch with a plastic grip and an integrated reflector. After removing the reflector, the inner cavity of the handle provided a usable space of 25 mm in diameter and 115 mm in length for the integration of the electronic components.

A custom-designed four-layer PCB (1.55 mm thickness, 35 µm copper) with a total size of 90 × 21 mm was developed to ensure compact placement and signal integrity. The board integrates a BMI270 IMU for motion sensing, an ESP32-WROOM-32E MCU (2.4 GHz Wi-Fi, Bluetooth) for wireless communication via BLE with the SFO, and a Type-C USB connector for programming and battery charging. Power is supplied by an IFR 14500J AA lithium iron phosphate LiFePO4 battery (600 mAh, 3.2 V) from Keeppower, managed by a Microchip MCP73123 battery charge controller. A charge status LED was connected to a status pin for visual indication. To enable vibrotactile feedback, two DRV2605L haptic driver integrated circuits (ICs) from Texas Instruments were integrated, allowing the control of two ERM or linear resonant actuator (LRA) vibration modules.

The USB connector, MCU, and battery holder were symmetrically arranged along the central axis of the PCB to ensure balanced layout and optimal use of space. The MCU and USB connector were placed on the front side, while the battery was positioned on the back side in accordance with the manufacturer's mounting guidelines. A universal asynchronous receiver transmitter (UART) interface was included to allow communication with an external PC and GUI during development. Before fabrication, ERC and DRC were performed to verify the circuit design and layout. The PCB was then assembled using a manual pick-and-place process, followed by reflow soldering in a three-zone convection oven (Mistral 260) for SMD components. Additional THT components were soldered manually in a subsequent step. After assembly, the boards underwent functional testing and visual inspection to ensure proper soldering and system performance. To ensure optimal positioning and mechanical stability within the handle, the assembled PCB was mounted inside a custom 3D-printed enclosure made of rigid PLA. To enable charging without disassembly, an access was added to the back of the crutch, providing external access to the USB-C port. An overview of the hardware architecture is provided in Figure 3a, and the assembled prototype is shown in Figure 3b.

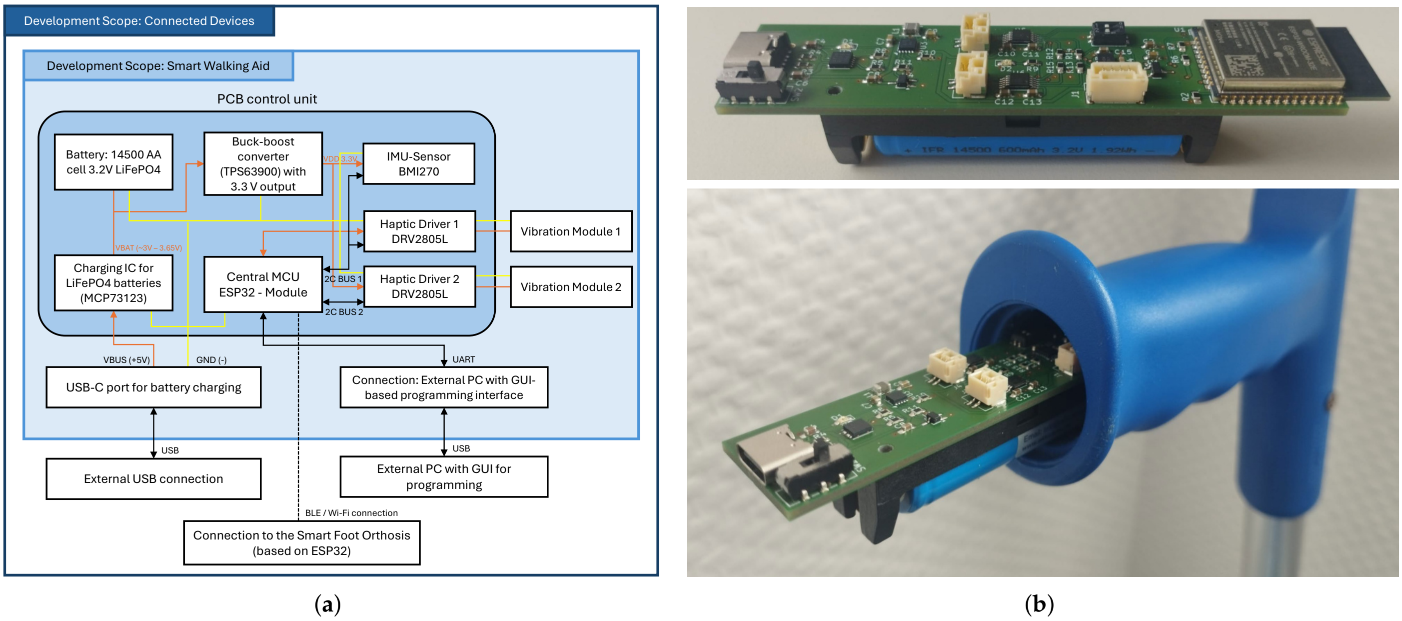


Figure 3. (**a**) Hardware block diagram of the smart walking aid; (**b**) Final assembled prototype with integrated electronics.

3.3. Software Architecture and Data Processing

3.3.1. Smart Orthosis—Mobile Gait Analysis

Figure 4 presents the SFO data processing architecture, structured into five stages: (1) raw data acquisition, (2) pre-processing, (3) signal processing, (4) sensor fusion, and (5) extraction of gait parameters. Each step is described in detail; see Appendix A.

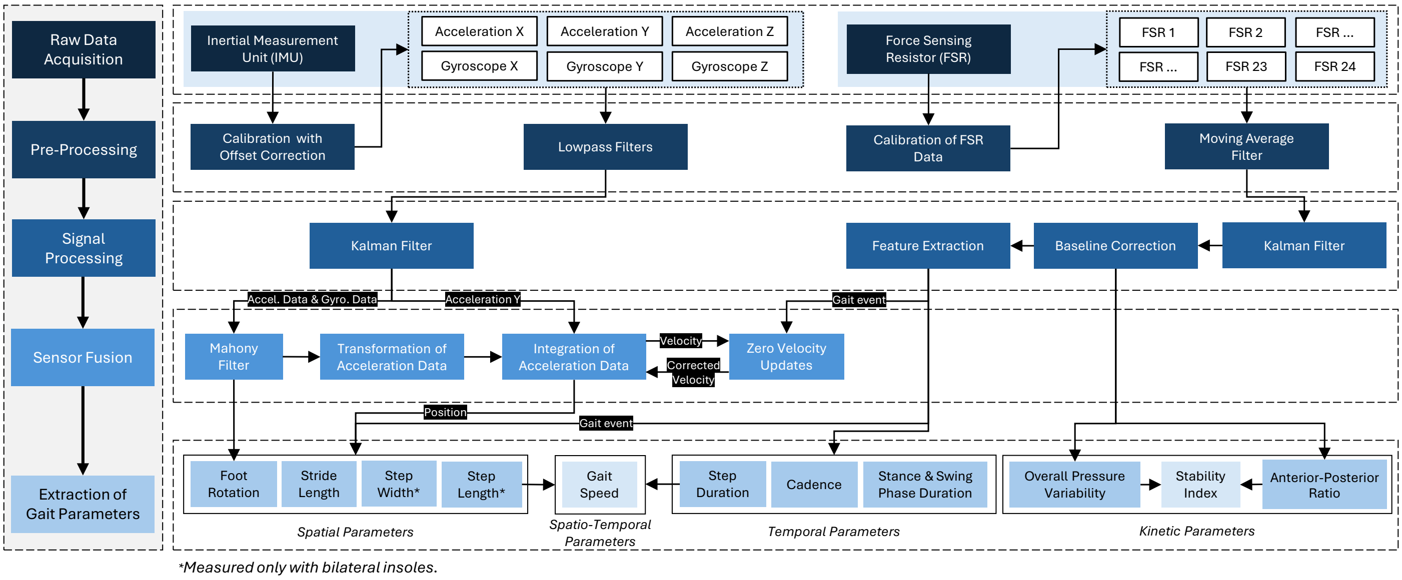


Figure 4. Overview of the data processing for gait data extraction of the smart foot orthosis.

3.3.2. Smart Walking Aid—Haptic Feedback and Motion Data

The ESP32 MCU receives processed gait event data from the SFO via wireless communication using either BLE or Wi-Fi. The firmware was developed in C/C++ using the Arduino integrated development environment (IDE) and enables the control of two actuators to generate vibration patterns depending on pre-defined overloads. In the current prototype implementation, relative pressure thresholds were defined as fixed percentage values of the user's body weight to ensure consistent triggering conditions during system testing, similar to

Mariani et al. [100], who used a threshold of 5% of body weight for gait phase detection. This constant threshold design served to demonstrate the feedback functionality under controlled conditions and was not intended to represent clinically calibrated loads. In future implementations, these thresholds could be dynamically personalized and adapted per gait phase (stance vs. swing) based on baseline load measurements obtained during standing and normal walking. However, such thresholds are individual to each patient and thresholds should be defined and validated by medical experts to ensure the detection of clinically relevant overload conditions [101]. Vibration signals are emitted using a configurable pulse-based pattern. Depending on the detected parameter both the intensity and interval of the vibration can be dynamically adapted. This customization allows targeted feedback for specific gait abnormalities, such as in-toeing or out-toeing patterns, where intensity can be increased with foot rotation angles. The integrated IMU, initialized with the same configuration as in the SFO, continuously records motion data for activity recognition. This data is also streamed to the smartphone application. To extend the functionality, these data could be used in future implementations for activity classification or fall detection.

3.3.3. Smartphone Application

The design of the smartphone application builds upon the main functions proposed by Resch et al. [65] for a smart insole app dedicated to foot health monitoring. Building on these proposed functions, our application implements key features such as a heatmap visualization of plantar pressure based on the FSR sensors, real-time gait characteristics, and a walking diary displaying step counts and activity-related health status ratings. Additional modules include a knowledge database with a search function for articles on foot health, a training area featuring video tutorials for foot-related exercises, and a map for locating local medical specialists. The app also enables monitoring of the communication status between the SFO and the walking aid.

For pressure data visualization, a custom heatmap fragment shader was implemented. Its output was based on the number of FSRs used and colored using the Turbo colormap developed by Google [102]. The application was developed in Android Studio [103] (version 2024.2.1) using Kotlin as the main programming language. We utilized Jetpack Compose [104] and its integrated UI libraries from Material2 and Material3, enabling flexible design of forms, buttons, and icons. Wireless communication was implemented via Wi-Fi, Bluetooth, and MQTT protocols to ensure real-time data transmission between devices. The prototype was deployed as an android package kit (APK) on an Android 14 device using software development kit (SDK) version 35. Figure 5 shows selected screenshots of the main functionalities of the developed mHealth application.

Figure 5. Screenshots of the developed mHealth app showing key interface components, including account creation, main menu navigation, plantar pressure heatmap, gait diary, exercise tutorials, and access to foot health resources.

3.4. Prototype Validation Study: Experimental Setup

A technical feasibility and validation study was conducted with eight participants to evaluate the integrated system in terms of both objective measurement accuracy and subjective user experience. The primary goal of this proof-of-concept investigation was to verify the system's functional reliability and multimodal feedback performance under controlled laboratory conditions, rather than to focus on algorithmic novelty or benchmark testing of individual algorithms, or to conduct a clinical study. The study comprised two supporting validation objectives (system feasibility and application usability) and one primary objective (feedback evaluation).

1. System Feasibility: Validation of the SFO plantar pressure and spatial/temporal gait data against an instrumented treadmill.
2. Application Usability: User experience evaluation of the smartphone app based on the System Usability Scale (SUS) [105].
3. Feedback Evaluation: Assessment of two feedback mechanisms applied within the SFO and forearm crutch using a customized questionnaire.

In the following subsections, the study design, apparatus, procedure and tasks, as well as the applied measures and data analysis, are documented.

3.4.1. Study Design

The study was conducted in three parts, each with a specific validation objective. An overview of the experimental validation and evaluation setup is shown in Figure 6.

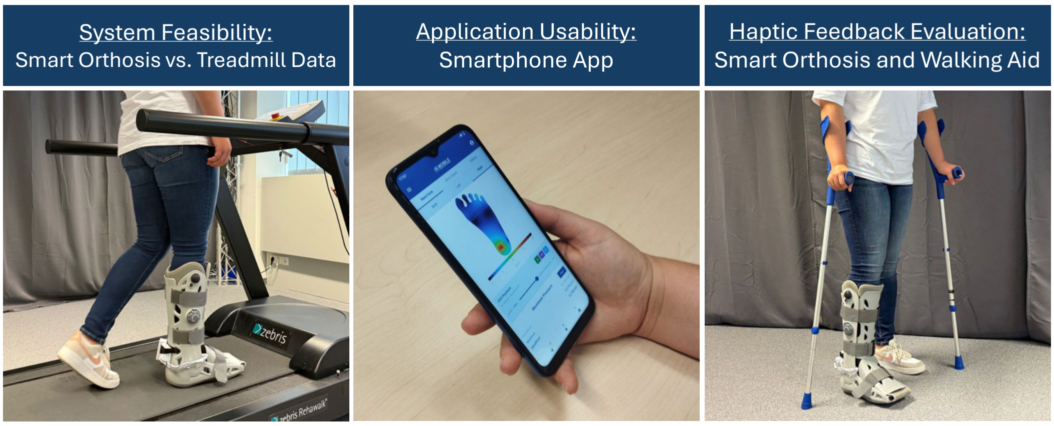


Figure 6. Visual overview of the experimental validation process including orthosis testing, usability assessment, and haptic feedback evaluation.

To evaluate the measurement accuracy of the SFO, an experimental validation was conducted against a clinical gait analysis reference system (Zebris FDM-T treadmill). The independent within-subject variable *Measurement System* included two levels: *SFO* and *Zebris Treadmill*. Dependent variables included key spatial (step and cycle length), temporal (stance, swing, and cycle phase durations), and kinetic (plantar pressure distribution) parameters.

The smartphone app was tested to assess perceived usability using the standardized SUS questionnaire.

To evaluate the influence of haptic feedback on user perception, participants tested both the SFO and the crutch, each equipped with one vibration module. The study was based on a 2×2 factorial within-subject design with two independent variables: *System* (two levels: *Orthosis* and *Crutch*) and *Feedback Type* (two levels: *Continuous* and *Pulsed Pattern*). Dependent variables were based on subjective ratings regarding noticeability, intrusiveness, regular usage, and perceived usefulness. A 2×2 balanced Latin square design was applied to counterbalance the conditions and avoid sequence effects [106].

3.4.2. Apparatus

The prototype systems (SFO, instrumented forearm crutch, and smartphone app) were used according to the previously described specifications and deployed in the study using MQTT for real-time data communication between the devices. The 3D-printed insole was designed to fit EU shoe sizes 38–43, covering the medium shoe size range in both women and men. To meet the minimum sensing requirements with optimal sensor placement, the insole of the orthosis was equipped with 15 FSRs. Furthermore, a single ERM 3.7V DC motor (model: JYCL0610R2540) from Jie Yi Electronics Limited was used as vibration actuator, integrated both into the orthosis and the handle of the crutch. This configuration was chosen to assess whether a single actuator can provide sufficient haptic feedback to be reliably perceived. The app was executed on a Motorola Moto G8 Power Lite 64 GB (Android 10).

For the evaluation of motion and pressure data acquired from the IMU and FSR sensors, we used an instrumented treadmill, which is considered the gold standard for validating gait analysis systems. Specifically, a RehaWalk® FDM-THPL-S-2i instrumented treadmill (Zebris Medical GmbH, Isny, Germany) was used to measure spatial and temporal gait parameters. The system is based on an h/p/cosmos pluto treadmill and integrates 3120 capacitive sensors ($\pm$0.05% measurement accuracy), arranged in an 80 $\times$ 39 grid layout, covering a sensor area of 101.6 × 49.5 cm. Data were recorded at a sampling rate of 100 Hz (similar to the SFO), within a sensor range of 1–120 N/cm². Zebris FDM software (version 2.0.4) was used to run gait analysis and export CSV raw data. The software ran on a Windows 10 Pro system with an AMD Ryzen 5900X 12-core processor, 3.70 GHz, a GeForce RTX 3070 graphics card, and 32 GB RAM.

3.4.3. Procedure and Tasks

The study was conducted under controlled laboratory conditions without external influences. First, participants were informed about the study procedures and that they could withdraw or discontinue participation at any time. After providing informed consent, they completed a survey about their demographic data, foot conditions, previous experience with assistive devices, and mHealth applications. Furthermore, participants were asked to rate their familiarity with the use of smartphone healthcare apps. Following this, the SFO was provided to each participant as a fully preassembled system. An integrated vacuum-based mechanism was used to ensure optimal adaptation to the individual leg shape. After a brief instruction, each participant completed five walking trials on the Zebris treadmill at a fixed speed of 1.5 km/h. Each trial lasted 10 s to ensure the recording of at least one full gait cycle, typically capturing multiple cycles for averaging and comparison across multiple measurements. Participants were instructed to walk as normally as possible without holding onto the treadmill handles.

After the walking trials, the walking aid prototype was introduced and tested both while standing and walking, in combination with the SFO. Haptic feedback was evaluated for both systems under predefined overload conditions to demonstrate feedback alerts and to ensure identical vibration triggering conditions across all participants. Two feedback type modalities (continuous and pulsed patterns) were activated in randomized order for each participant. Following this, participants completed a questionnaire consisting of quantitative Likert items to assess noticeability, intrusiveness, regular usage, and perceived usefulness. Additionally, participants were asked to indicate their preferred vibration alert type in a single-choice question.

Finally, participants completed a smartphone usability task and assessed the app using the SUS questionnaire. An optional certificate of participation was offered after the study.

3.4.4. Measurement and Data Analysis

A quantitative measurement approach was chosen to evaluate the overall system and its individual subsystems. Descriptive and inferential statistical analyses were conducted using R (version 4.2.2) with the rstatix package [107]. The following paragraphs describe the corresponding measurements and data analysis in detail.

System Feasibility: Accuracy of Plantar Pressure and Gait Parameters (Smart Orthosis)

Both systems recorded gait events and pressure-related data simultaneously. Manual triggering was performed during the swing phase of the right leg (foot orthosis) to synchronize both systems at the moment of initial right foot contact. Each trial lasted for 10 s, recorded at a sampling rate of 100 Hz in both systems, resulting in five datasets per participant. The Zebris FDM software enabled the export of pressure data and gait characteristics were aggregated as mean and standard deviation values per trial. The definitions of the respective gait parameters are provided in Table 1.

To enable a valid comparison, the raw data from the SFO were processed equally: gait characteristics were computed per trial and then aggregated on the participant level into mean and standard deviation values. Descriptive statistics were used to summarize the data, and inferential statistics were applied using paired *t*-tests to examine significant differences between both measurement systems.

Table 1. Overview of analyzed gait parameters grouped by parameter type.

Parameter Type	Parameter	Unit
Temporal	Stance Duration	s
	Swing Duration	s
	Cycle Duration	s
Spatial	Stride/Cycle Length	m
Kinetic	Mean Plantar Pressure	N/cm^2

For pressure data, the full time-series dataset was used for the SFO, while the treadmill provided aggregated data for one step. However, a direct quantitative comparison of force or pressure values between the treadmill and orthosis systems is not possible, due to fundamental differences in sensor configuration and measurement principles. The treadmill measures plantar pressure at the interface between the orthosis sole and the treadmill surface, while the orthosis records forces between the foot and the insole. As a result, the absolute values and spatial resolution of the pressure data differ between systems. Therefore, pressure data were visualized as heatmaps and force–time curves to qualitatively assess whether gait events could be reliably identified from the force signal progression.

Application Usability: Evaluation of Smartphone App Usability

Usability of the smartphone application was assessed using the standardized SUS, which is frequently used to assess user experience of interactive systems such as smartphone apps. The questionnaire consists of ten items rated on a five-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Even-numbered items were reverse-scored and calculated according to the specifications by Brooke [105]. The total score was calculated and converted into a value between 0 and 100 as a quantifiable measure of overall usability. Descriptive statistics were used to analyze the SUS scores.

Feedback Evaluation: User Perception of Haptic Feedback (Orthosis and Walking Aid)

To assess the user experience and perceived effectiveness of the haptic feedback system, the participants rated four statements on a 7-point Likert scale [108]. The items addressed noticeability, intrusiveness, regular usage, and perceived usefulness, with response options ranging from 1 (*strongly disagree*) to 7 (*strongly agree*).

The questionnaire items included:

- Noticeability: "*The haptic feedback was clearly noticeable during use.*"
- Non-Intrusiveness "*I found the feedback comfortable and non-intrusive.*"
- Regular Usage "*I could imagine using such feedback during everyday mobility tasks.*"
- Usefulness "*The feedback would be useful in real-life situations (e.g., alerting me of incorrect foot placement or walking posture).*"

Descriptive statistics were applied to summarize the participants' responses and identify trends in agreement levels. A two-factorial repeated-measures (RM) Analysis of Variance (ANOVA) was used to assess the effect of system and feedback type on noticeability, intrusiveness, regular usage, and perceived usefulness.

3.5. Participants

The study was conducted in accordance with the data privacy regulations of our institution and received ethical approval from the German Society for Nursing Science (No. 23-027). Eight participants were recruited through institutional mailing channels and provided informed consent before participation. To participate in the study, individuals were required to meet the following inclusion criteria:

- Footwear size: European shoe size between 38 and 43.
- Current or past foot-related conditions, including either acute or chronic conditions
- Current or previous use of foot-related orthopedic products, such as shoe insoles, foot orthoses, bandages, or foot casts.
- Right foot affected or bilateral condition (but not solely left foot), as the sensor insole was configured specifically for the right foot.

Prior to study participation, all individuals completed a brief pre-screening to confirm that these criteria were met. In total, 8 participants (4 female, 4 male) were recruited. The age of the participants ranged between 24 and 72 years ($M = 39.75$, $SD = 17.92$). Participants' educational backgrounds included four with vocational training, three with a bachelor's degree, and one with a master's degree. Occupations were distributed as follows: services and sales ($n = 2$), engineering ($n = 2$), other fields ($n = 2$), clerical ($n = 1$), and health-related professions ($n = 1$). The shoe size ranged from EU 38 to EU 43 ($M = 40.37$, $SD = 1.85$). All participants reported having a current or past foot condition. Four had previous conditions (such as a broken leg, sprained ankle, pes planus, or torn ligament). Two participants reported chronic conditions (foot malalignment and arthrosis), one participant had an acute condition (sprained ankle), and one reported both an acute (sprained ankle) and chronic condition (torn ligament and previous leg fracture). All participants indicated that they had previously used orthopedic devices, including insoles ($n = 7$), ankle braces or bandages ($n = 3$), and foot orthoses ($n = 2$). Furthermore, four participants already used forearm crutches as medical walking aids. In total, three participants had previous experience with a mobile health app. Participants rated their familiarity with the use of smartphone healthcare apps on a scale from 1 (*not at all familiar*) to 5 (*extremely familiar*), with an average rating of 2.38 ($SD = 1.60$). All participants completed the study and were included in the analysis.

4. Results

The results section first presents the comparative accuracy validation between the SFO and the instrumented treadmill (Section 4.1), followed by the usability assessment of the smartphone application (Section 4.2), and concludes with the evaluation of the haptic feedback of both the orthosis and the walking aid (Section 4.3).

4.1. System Feasibility: Accuracy of Plantar Pressure and Gait Parameters (Smart Orthosis)

4.1.1. Gait Data Measurement

To assess the reliability of temporal and spatial gait parameters, the accuracy of the SFO measurements was evaluated against the treadmill system across all trials. The results showed that temporal parameters achieved mean accuracies above 85%, except for swing phase duration, which reached only 60%. For the spatial parameter, the mean accuracy was 83.4%, despite a high standard deviation. The descriptive results are summarized in Table 2.

Table 2. Mean and standard deviation of gait parameter accuracy of the smart foot orthosis (SFO) compared to treadmill measurements across all participants.

Parameter Type	Parameter	Mean Accuracy	SD Accuracy
Temporal	Stance Phase Duration	96.7%	±3.1
	Swing Phase Duration	60.3%	±12.6
	Cycle Phase Duration	85.2%	±1.6
Spatial	Stride/Cycle Length	83.4%	±13.3

Temporal Parameters

The Shapiro–Wilk normality test indicated that both cycle phase duration ($p > 0.095$) and swing phase duration ($p > 0.501$) were normally distributed, while stance phase duration was not ($p = 0.013$). A paired t-test revealed a statistically significant difference between SFO and treadmill for swing phase duration ($t(7) = 9.74$, $p < 0.001$), Cohen's d = 3.44 (large effect size), as well as for cycle phase duration ($t(7) = -18.7$, $p < 0.001$), Cohen's d = 6.59 (large effect size), which was significantly shorter in SFO compared to treadmill. Furthermore, a Wilcoxon signed-rank test on stance duration data showed no significant difference ($W = 8$, $p = 0.195$, with moderate effect $r = 0.50$).

A Pearson correlation analysis showed a strong and significant correlation for cycle duration ($r = 0.973$, $p < 0.001$), and a moderate but not statistically significant correlation for swing duration ($r = 0.673$, $p = 0.067$). Additionally, a Spearman correlation analysis for stance duration, which was not normally distributed, did not reveal a significant correlation ($\rho = -0.167$, $p = 0.693$).

Spatial Parameters

The Shapiro–Wilk test indicated that cycle length ($p = 0.259$) was normally distributed. A paired t-test revealed a statistically significant difference in cycle length between the treadmill and orthosis conditions ($t(7) = -3.65$, $p = 0.008$), Cohen's $d = 1.29$ (large effect), indicating that cycle length was significantly shorter when measured with the orthosis compared to the treadmill. Pearson correlation analysis showed no significant correlation for cycle length ($r = 0.678$, $p = 0.065$).

4.1.2. Pressure Data Measurement

The force data recorded with the SFO enable the identification of gait cycles and their characteristic phases (initial contact, full contact, and toe-off), as illustrated in Figure 7, demonstrating its suitability as a mobile gait analysis system. However, a direct quantitative comparison with the instrumented treadmill is not feasible due to fundamental differences in measurement and data availability. While the Zebris system measures contact forces between the treadmill surface and the bottom of the orthosis, the SFO records plantar forces directly between the foot and the insole. In addition, the Zebris software provides only aggregated outputs with averaged curves rather than continuous raw data, which prevents a synchronized step-wise comparison.

Nevertheless, to compare both systems a qualitative comparison of plantar pressure distributions and force curves was conducted using visualizations from both systems for a representative trial (see Figure 8). The Zebris data represent averaged force curves for the three foot zones summarized from multiple gait cycles, whereas the orthosis data focus on one representative gait cycle from the same trial, which is also displayed in Figure 7.

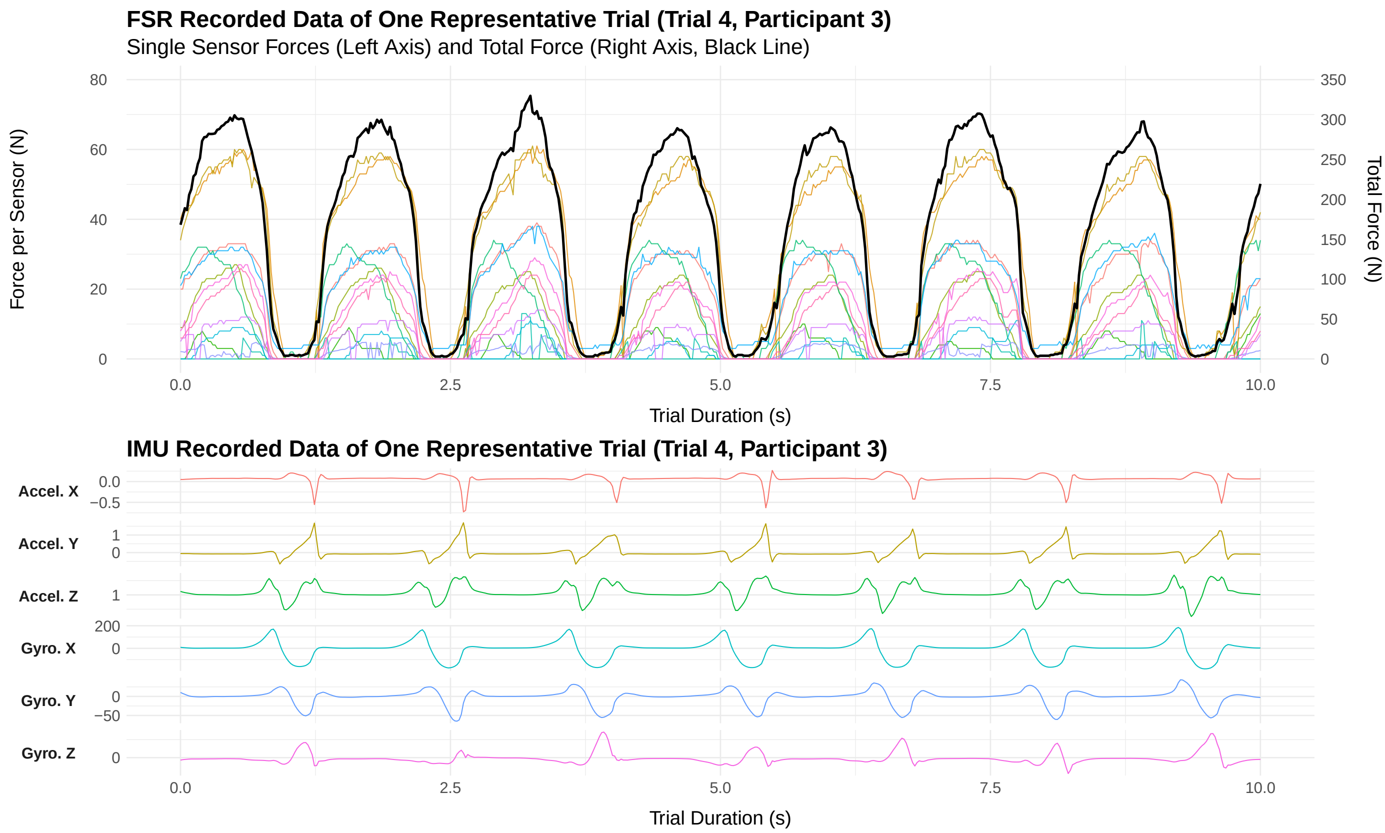


Figure 7. (**Top**): Measured plantar pressure data of Trial 4 (P3), showing force values recorded by multiple FSR sensors over the duration of the task. Distinct gait phases can be identified from the characteristic patterns in the force signals. The left axis represents the force values per sensor, while the right axis indicates the total force (black line) across all included sensors. (**Bottom**): Measured acceleration and gyroscope data from the IMU for each of the three axes, illustrating gait phase conformity with the recorded pressure data.

Although absolute force values are not directly comparable due to differences in sensor configuration, both systems show similar temporal trends across the three anatomical foot zones and in the overall force–time diagram. Compared to the Zebris treadmill, which provides only aggregated contact data, the SFO offers step-wise heatmap visualizations, enabling a detailed analysis of the internal plantar pressure distribution for each gait event.

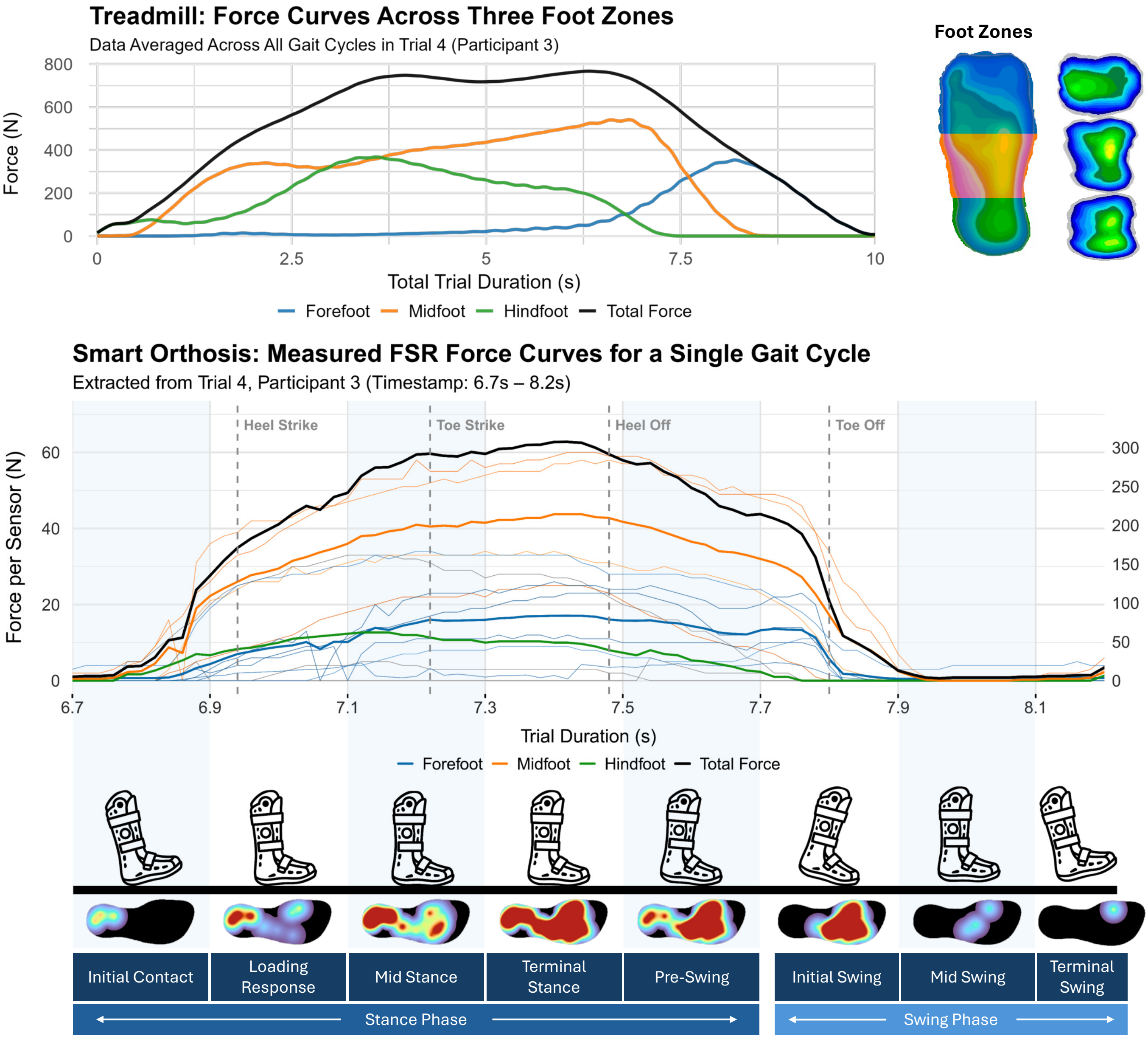
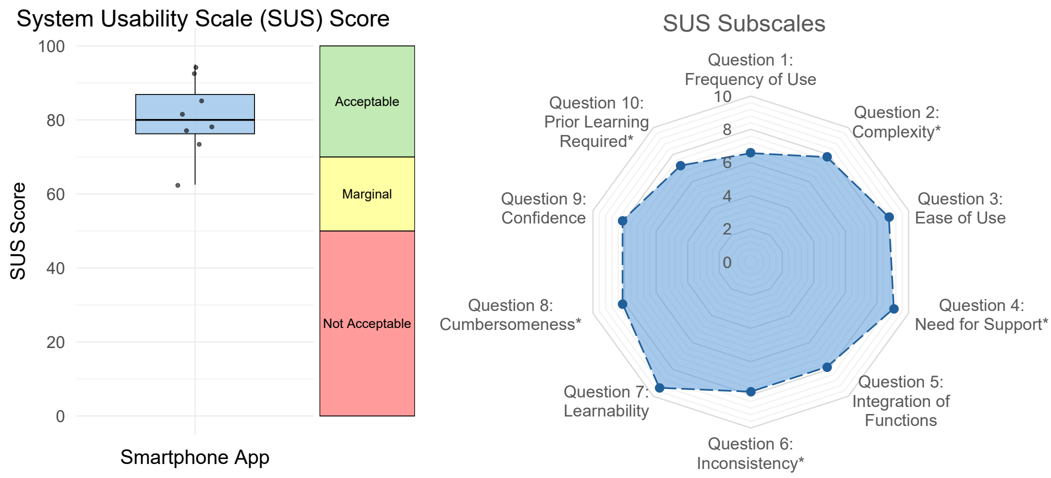


Figure 8. The upper plot shows the aggregated plantar force curves of gait cycles recorded in Trial 4 of Participant 3 using the Zebris system. The force curves are color-coded according to three anatomical foot regions, as indicated by the footprint icon on the right. Additionally, a cumulative plantar pressure heatmap illustrates the spatial force distribution over the entire trial. The lower plot displays data from the same trial (Trial 4, Participant 3), focusing on a single gait cycle recorded by the orthosis. The three regional force curves represent the summed signals of the individual FSRs and are color-coded accordingly. Four gait events identified by the SFO are marked in gray. Below the graph, individual heatmaps illustrate the pressure distribution for each gait event.

4.2. Application Usability: Evaluation of Smartphone App Usability

The mean SUS score was 80.62 (SD = 10.59), as shown in the boxplot diagram (Figure 9a). According to the interpretation guidelines by Bangor et al. [109], this score falls within the "Acceptable" range based on the Acceptability Ranges and is considered "Good" according to the Adjective Ratings, indicating a high level of usability. Ratings for the ten individual SUS items are visualized in the radar plot (Figure 9b), with values ranging from 6.56 to 9.38 (normalized on a 0–10 scale). Furthermore, the raw ratings of the subscales are summarized in Table 3 using descriptive statistics.



(**a**) (**b**)

Figure 9. (**a**) Boxplot of overall SUS scores and the corresponding acceptability categories; (**b**) Radar plot visualizing the ten individual item scores, normalized on a 0–10 scale. Asterisks (*) indicate reverse-scored items.

Table 3. Descriptive statistics of the SUS items (M = Mean, SD = Standard Deviation).

No.	Item	M	SD
1	Frequency of Use	3.63	0.92
2	Complexity *	1.88	0.83
3	Ease of Use	4.50	0.53
4	Need for Support *	1.38	0.74
5	Integration of Functions	4.13	0.99
6	Inconsistency *	1.88	0.64
7	Learnability	4.75	0.46
8	Cumbersomeness *	1.75	1.04
9	Confidence	4.25	0.71
10	Prior Learning Required *	2.13	1.36

Note. Asterisks (*) indicate reverse-scored items.

4.3. Feedback Evaluation: User Perception of Haptic Feedback (Smart Orthosis and Walking Aid)

Shapiro–Wilk tests were conducted for each condition on the questionnaire items *Noticeability*, *Non-Intrusiveness*, *Regular Usage*, and *Usefulness*, and revealed that the data were not normally distributed. Therefore, an Aligned Rank Transform (ART) RM ANOVA was conducted on the four measures, revealing two significant main effects. A statistically significant main effect of *Noticeability* was found for *Device* with $F(1,21) = 4.49$, $p = 0.046$, $\eta_p^2 = 0.18$ (large effect size). Post hoc pairwise comparisons using Wilcoxon signed-rank tests did not reveal a significant difference between *Orthosis* and *Crutch* ($p = 0.065$). However, the corresponding effect size ($r = 0.483$, moderate-to-large) suggests a potentially meaningful difference that may not have reached significance due to the limited sample size and resulting low statistical power. Furthermore, a statistically significant main effect was found for *Usefulness* on *Feedback Type* with $F(1,21) = 6.01$, $p = 0.023$, $\eta_p^2 = 0.22$ (large effect size). Post hoc Wilcoxon signed-rank tests revealed a significant difference between *Continuous* and *Pulsed Pattern* ($p = 0.008$).

Results of the ART RM ANOVA and post hoc pairwise comparisons for all conditions are shown in Table 4. Additionally, the overall results are presented as a boxplot with descriptive statistics in Figure 10.

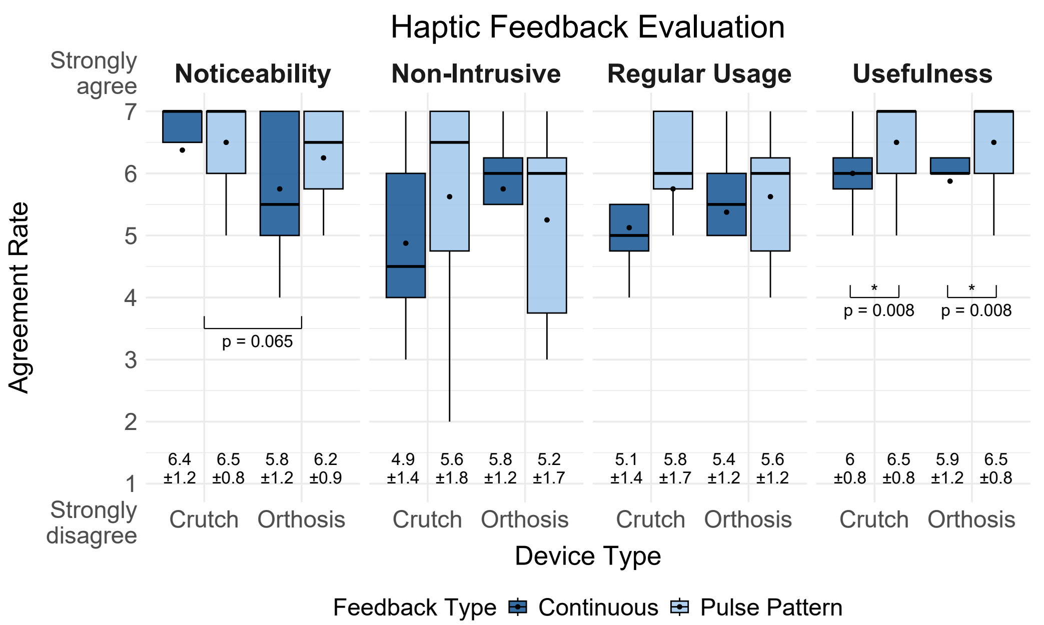


Figure 10. Boxplots show agreement ratings for four measures on haptic feedback via the foot orthosis and the forearm crutch, ranging from 1 (strongly disagree) to 7 (strongly agree). It displays the interquartile range (box), 1.5 × IQR range (whiskers), median (line), and mean (black dot). Mean and standard deviation (±) values are shown below each boxplot. Asterisks (*) indicate statistically significant results ($p < 0.05$).

Table 4. Results of the ART RM ANOVA and post-hoc Wilcoxon signed-rank tests for the four questionnaire items.

		ART RM ANOVA					Pairwise Comparisons (Wilcoxon)			
Measure	**Factor**	**F**	**Df**	p	**Sig.**	η_p^2	**Statistic**	p	r	**Magnitude**
Noticeability	Device	4.49	1, 21	0.046	*	0.176	25.0	0.065	0.483	moderate
	Feedback	1.80	1, 21	0.194		0.079	22.0	0.187	0.310	moderate
	Device × Feedback	1.60	1, 21	0.220		0.071	—	—	—	—
Non-Intrusiveness	Device	0.06	1, 21	0.815		0.003	27.5	0.653	0.106	small
	Feedback	0.50	1, 21	0.487		0.023	42.5	0.811	0.046	small
	Device × Feedback	2.24	1, 21	0.150		0.096	—	—	—	—
Regular Usage	Device	0.00	1, 21	0.966		0.000	42.5	0.807	0.099	small
	Feedback	2.57	1, 21	0.124		0.109	50.0	0.129	0.326	moderate
	Device × Feedback	0.77	1, 21	0.390		0.035	—	—	—	—
Usefulness	Device	0.00	1, 21	1.000		0.000	24.5	0.851	0.076	small
	Feedback	6.01	1, 21	0.023	*	0.220	36.0	0.008	0.703	large
	Device × Feedback	0.00	1, 21	1.000		0.000	—	—	—	—

Note. Asterisks (*) indicate statistically significant results ($p < 0.05$). Effect sizes are reported as partial η^2 for ART RM ANOVA and as rank-biserial correlation (r) for Wilcoxon signed-rank tests. Magnitude classification of r: small ($r < 0.3$), moderate ($0.3 \leq r < 0.5$), large ($r \geq 0.5$) [110].

4.3.1. Correlation of Ratings Across Devices

To further investigate consistency in user perception across devices, a Spearman rank correlation was conducted between ratings for the same *Feedback Type* on the *Orthosis* and the *Crutch*. Results showed a significant positive correlation for *Noticeability* in the *Pulsed Pattern* feedback condition ($r_s = 0.86$, $p = 0.006$), suggesting a high level of consistency in perceived noticeability across both devices. Furthermore, in the *Continuous* feedback condition, a significant positive correlation was found for *Regular Usage* ($r_s = 0.78$, $p = 0.023$), while the remaining conditions showed positive trends, but did not reach statistical significance: *Noticeability* ($r_s = 0.62$, $p = 0.099$), *Non-Intrusiveness* ($r_s = 0.56$, $p = 0.152$), and *Usefulness* ($r_s = 0.58$, $p = 0.133$).

4.3.2. Post-Experiment Questionnaire

Participants rated the feedback types for both devices separately to identify preferences between the different modalities. For the *Orthosis*, the *Pulsed Pattern* feedback was preferred by four participants, while the remaining four participants rated both feedback types as equally acceptable. None of the participants selected the *Continuous* signal or no feedback as their preferred option. For the *Crutch*, six participants preferred the *Pulsed Pattern* feedback. Two participants indicated no preference and rated both options equally. Similar to the orthosis results, no participant selected the *Continuous* signal or no feedback.

5. Discussion

We developed a sensorized system for monitoring plantar pressure and gait activity within an orthopedic foot orthosis. The system provides haptic feedback through either the orthosis or a corresponding forearm crutch and enables remote monitoring via a smartphone application. In an experimental user study with eight participants, we evaluated the system's feasibility and investigated the influence of device location and feedback modality on user perception of vibrotactile biofeedback. The findings confirm the system's technical feasibility for gait phase detection, the usability of the accompanying mobile application, and provide new insights into haptic feedback perception. The following subsections discuss the results in detail, including implications, limitations, and directions for future research.

5.1. System Feasibility and Application Usability

5.1.1. System Feasibility: Accuracy of Plantar Pressure and Gait Parameters (SRQ1)

SRQ1 focused on evaluating the accuracy of plantar pressure and gait measurements obtained from the developed SFO compared to a clinical reference system. Our findings showed that the recorded temporal parameters for the swing and cycle phases indicated significantly shorter durations when measured with the SFO compared to the treadmill. However, cycle duration showed a strong correlation between both systems, whereas swing duration did not reach statistical significance ($p = 0.067$). Overall, swing duration showed the lowest accuracy among all temporal parameters, with only 60% accuracy and notably higher standard deviations. This may be explained by the observation that some FSR sensors were constantly triggered and did not return to zero during offloading, which could have impaired the correct detection of events from stance to swing. Although the stance phase showed no significant differences and achieved high accuracy of 96%, the correlation between both systems was not significant. These inconsistencies are likely caused by differences in event detection, as the SFO relies on FSR-based thresholds, whereas the treadmill reference system uses ground reaction forces. Additionally, some triggering issues of certain FSR sensors may have introduced inconsistent deviations in individual measurements, which could explain the reduced correlation, even though the average stance durations between both systems were similar. As highlighted in previous research, lower walking speeds (0.8–1.0 m/s) compared to moderate walking speeds (1.2–1.4 m/s) are associated with higher error rates in temporal parameters (e.g., stance and swing time), as well as spatial parameters (e.g., stride length) [111]. As the walking speed in our experiment was relatively low (0.417 m/s) to ensure participant comfort while walking with the orthosis, this factor may have contributed to reduced accuracy in gait parameter estimation.

The measured accuracy of the spatial parameter (cycle length) indicated that an accuracy above 83% can be achieved. However, the estimation of spatial gait parameters using a single smart insole systematically underestimated cycle length compared to treadmill-based reference measurements, and no significant correlation was found between both measurement systems. One explanation for these systematically lower values is that the insole was worn unilaterally, while the other foot was not equipped with a sensorized insole to provide complementary data. Furthermore, wrong triggering of the algorithm could have led to incorrect gait event detection. By identifying this systematic offset, a post-processing correction factor could be applied to improve the dataset for further analyses.

The evaluation of pressure data showed that gait events can be reliably detected and continuous pressure distribution can be measured during both walking and standing. Although the comparison with the treadmill was performed on a qualitative basis, the results showed that the three foot zones (forefoot, midfoot, and hindfoot) exhibited similar temporal patterns. This highlights the added value of the SFO, as it enables step-wise heatmap visualizations and detailed insights into the internal plantar pressure distribution without post-processing, which are not achievable with stationary treadmill systems. In summary, our results confirm previous findings showing that one IMU is sufficient for gait monitoring [48] and that the recommended configuration of 15 FSRs [43] is suitable for reliable pressure measurement. However, further refinements are recommended to improve measurement accuracy and enhance data reliability.

Compared to related studies on smart insoles and gait monitoring systems, the accuracy levels achieved in this work are within a comparable range. For example, D'arco et al. [41] reported high performance in activity recognition using smart insoles, with accuracies ranging from 70% to 99%. This demonstrates that comparable performance levels can be achieved even without additional post-processing. Despite differences in sensor configuration (type, quantity, and placement), as well as varying application contexts, testing scenarios and data processing, the present study demonstrates that reliable gait analysis can also be achieved using a single insole. With further iterative refinements of the sensing and calibration procedures, even higher accuracy levels may be achieved in future implementations.

5.1.2. Application Usability: Evaluation of Smartphone App Usability (SRQ2)

Regarding SRQ2, which addressed the usability of the accompanying mobile application for visualizing gait and pressure data, our results showed an acceptable level of usability. The SUS evaluation reached an average score of 80.62 (*SD* = 10.59), which is comparable to the results reported by Resch et al. [66] for a smart insole app (type: basic), with a mean SUS score of 77.5 (*SD* = 16.74). In comparison to other digital health applications, our findings also exceed the average usability benchmark identified by Hyzy et al. [112], who conducted a meta-analysis and reported a mean SUS score of 68.05 (*SD* = 14.05) across various digital health apps.

Overall, the subscale ratings revealed that item 4 (*need for support*) and item 7 (*learnability*) received the highest scores. This indicates that participants largely agreed with the statement "*most people would learn to use this app very quickly*" (item 7) and disagreed with the statement "*I would need the support of a technical person to use the system*" (item 4), which was reverse-scored. In contrast, item 1 (*frequency of use*) received the lowest average rating (*M* = 3.62, *SD* = 0.92 on a 5-point Likert scale), which shows that participants potentially were less willing to use the app frequently. This might be due to the fact that some participants were only temporarily affected by foot-related conditions and therefore did not perceive frequent use of the app as necessary. However, to address the potential limited interest in frequent use, future iterations of the app could incorporate targeted strategies to increase user engagement. For example, personalized notifications and reminders could encourage consistent daily use, particularly for self-monitoring tasks such as the integrated pain-level diary. Previous research suggested that gamification elements in health interventions (e.g., progress tracking or reward systems) can have positive effects on user experience [113]. Therefore, the integration of personalized measurable goals and feedback could support user motivation and foster continued use, especially in long-term rehabilitation contexts.

5.2. Feedback Evaluation: User Perception of Haptic Feedback (MRQ)

The MRQ investigated how different haptic feedback types (*Continuous* vs. *Pulsed Pattern*) implemented in an orthosis and a walking aid affect user perception regarding *Noticeability*, *Non-Intrusiveness*, *Regular Usage*, and *Usefulness*. Descriptive results showed that participants generally rated the *Pulsed Pattern* feedback higher than the *Continuous* feedback across all four items for both devices. A likely reason is that the rhythmic structure of the pulsed signal was perceived as more distinct and meaningful, making it easier to recognize and less monotonous compared to a continuous vibration. An exception was found for *Non-Intrusiveness*, where the *Pulsed Pattern* received lower ratings specifically in the orthosis condition.

Inferential analysis using an ART RM ANOVA revealed a significant main effect of *Device* on *Noticeability*, as the crutch was rated significantly higher than the orthosis. One possible explanation is that some participants had reduced sensation in the lower limbs, which made the haptic signal (especially *Continuous* feedback) less perceptible when delivered via the foot. Although previous work has suggested that foot-based feedback may be preferable to hand-based in some applications [114], in this specific use case the hand appears to be more sensitive to the vibration signal using one ERM actuator. This highlights that feedback effectiveness is not only dependent on signal type, but also on feedback location and tactile sensitivity at different body sites and devices.

For *Non-Intrusiveness*, the ratings varied more strongly across conditions. In particular, the *Continuous* feedback was rated lower than the *Pulsed Pattern* when applied via the crutch, which indicates that continuous vibrations may be perceived as more intrusive in this form factor. The *Regular Usage* showed similar ratings, however, without significance. Ratings for perceived *Usefulness* showed a significant effect of feedback type, indicating that the *Pulsed Pattern* was perceived as more useful than the *Continuous* feedback. These findings highlight that temporally modulated vibration signals are more easily interpreted and less annoying than continuous signals in wearable biofeedback applications. This result aligns with the post-study feedback, in which most participants expressed a clear preference for the pulsed signal. Although other trends were observed, they did not reach significance, which is probably due to the small size and limited statistical power of the study.

These findings support previous research on the utility of ERM-based vibration feedback [115]. However, our results extend this work by demonstrating that both pattern and device location influence the perception of vibrotactile feedback in orthopedic assistive devices. While a single ERM actuator ensured sufficient noticeability in the crutch, we recommend using at least two actuators inside the orthosis to enhance perceptibility. Overall, our results underscore the importance of tailoring haptic feedback implementation to the device form factor and user preferences. Therefore, designers should consider not only the feedback modality but also where and how it is delivered, and ensure that the signal is both interpretable and acceptable during everyday use. In line with prior work emphasizing the need to evaluate haptic feedback in assistive systems [76], our study contributes initial insights into user perception, acceptance, and effectiveness.

5.3. Implications

The presented system was designed as a general-purpose gait monitoring and feedback platform that can be applied both to individuals with foot conditions and for preventive or rehabilitative use in broader populations. While no clinical validation has yet been conducted, the system's sensing and feedback capabilities hold potential for future application, particularly for monitoring plantar pressure distribution and detecting overload patterns during daily activities.

The primary contribution of this work lies in advancing the understanding of how vibration-based feedback can be applied and perceived in foot-related assistive systems. While the developed hardware and software components served as necessary groundwork, they were mainly used for system verification. Building on this foundation, the central contribution is the investigation of real-time haptic overload notifications driven by the integration of plantar pressure and motion measurements. The fused dataset from pressure and motion sensors enabled a closed-loop feedback mechanism that responds to gait events. Our results demonstrate that such multimodal data can be feasibly integrated to trigger vibration cues. Based on this, the evaluation of haptic feedback provided new insights into the perception and preferences related to vibration-based notifications in assistive systems. The findings show that pattern-based feedback is perceived as more effective for overload alerts, while perceptual differences between the orthosis and the walking aid highlight the need for differentiated actuator placement, intensity modulation, and an increased number of actuators to ensure sufficient noticeability. These insights contribute to a clearer understanding of multimodal feedback perception in foot augmentation systems and can inform design guidelines for future adaptive vibration strategies in assistive devices.

Overall, these findings extend existing knowledge on isolated prototype developments by combining system integration, technical validation, and user-centered evaluation. These implications can guide future research and the development of intelligent footwear, multimodal feedback solutions, and interactive mobile health applications, contributing to the fields of HCI and biomedical engineering.

5.4. Limitations

The findings of our study are subject to several limitations. First, the evaluation primarily focused on technical development and validation under controlled laboratory conditions. In particular, the validation was limited to the right foot, as the orthosis was consistently worn on the right side. Moreover, the gait analysis was conducted using short treadmill walking sequences, which may not accurately reflect natural gait behavior during prolonged use or in real-world environments. In addition, calibration of the FSR sensors was performed based on manufacturer specifications without subject-specific mechanical calibration, potentially affecting the accuracy of absolute force estimation. Potential variation in sensor placement and the lack of automated recalibration routines may further impact signal reproducibility and long-term stability in real-world scenarios. Furthermore, comparability with treadmill-derived force values was limited, as only aggregated gait data were available from the treadmill, whereas the insole provided full raw data. Similarly, the developed algorithm for gait event detection was not quantitatively compared with existing data-driven gait phase recognition methods, nor validated on publicly available datasets. Finally, the small sample size limits the statistical power of the study and the generalizability of the observed effects, and no control group was included to provide a baseline comparison between participants with and without foot conditions. Nevertheless, the present work was designed as a proof-of-concept investigation with the primary focus on validating the technical feasibility and feedback functionality of the developed sensor system within the intended target group. Consequently, the study focused on end users with foot conditions to ensure practical relevance. Accordingly, the findings represent an initial technical evaluation conducted under controlled laboratory conditions and serve as a foundation for subsequent large-scale studies aimed at generalizing the results.

5.5. Future Work

The modular system architecture, which integrates multimodal sensing and feedback, demonstrates how next-generation orthotic devices can move beyond passive monitoring toward interactive and adaptive support. To address current limitations, future work should aim to evaluate the system under real-world conditions and in clinical contexts. In this context, future testing should include varied environmental conditions, such as uneven outdoor surfaces or stair walking, to evaluate the system's robustness under realistic gait scenarios. Such experiments will be essential for verifying performance outside controlled laboratory settings. Another direction for future work is the extended testing of the smart insole with regard to material specifications and biocompatibility to ensure long-term durability and safe use in rehabilitation contexts. Specifically, future research should include a larger and more diverse participant sample to allow for more comprehensive statistical analyses and improve the generalizability of the findings. Expanding the sample size will also enable subgroup comparisons and allow the assessment of long-term performance and usability across different user populations. Future studies should further include a control group of participants without foot conditions to establish a baseline for gait parameter comparison and system optimization. Including such a reference group with healthy participants will allow for more comprehensive interpretation of gait-related outcomes and support system optimization. Moreover, a case study involving patients with gait impairments could provide valuable insights into the system's robustness, usability, and therapeutic potential.

From a methodological perspective, future evaluations should also move beyond subjective usability scales and include behavioral and performance-based assessments, such as latency, energy consumption, correction rate, or changes in gait symmetry when interacting with the feedback system. Additionally, future work could benchmark the algorithm against state-of-the-art gait phase detection approaches and validate it on open-source datasets to enable objective performance comparison and generalization testing.

From a technical perspective, the integration of additional sensing modalities, such as temperature, GPS, or physiological sensors, could further enhance system functionality and contextual awareness. The modular architecture supports the flexible integration of such components for future developments. Furthermore, the use of artificial intelligence (AI)-based approaches, including machine learning techniques such as support vector machines or artificial neural networks, could improve gait phase classification, anomaly detection, and event prediction. These models may enhance the accuracy and robustness of gait parameter estimation, enable personalized and adaptive feedback mechanisms, and support new functionalities such as fall detection.

At the application level, the smartphone interface may be extended with features such as chatbots or gamified elements to support user-centered interpretation of health data and increase user engagement. Furthermore, the effectiveness of augmented feedback could be examined in user-centered training scenarios to evaluate its potential impact on motor learning and rehabilitation outcomes. As a next step, we plan to test the orthosis for foot augmentation in virtual reality training environments, with a particular focus on investigating participant feedback interactions in immersive scenarios through visual feedback. Additionally, a follow-up study is currently being planned to improve the smartphone application onboarding process using mixed reality, with the goal of enhancing users' initial interaction, comprehension, and engagement.

6. Conclusions

This work presents the development of an SFO system combining pressure and motion sensing, haptic feedback via a connected walking aid, and an mHealth application for real-time visualization. The proposed architecture enables the acquisition of gait and pressure data while providing immediate feedback to users. Beyond its technical implementation, this work contributes to the growing field of assistive smart systems in healthcare by demonstrating how sensor-based feedback can be integrated into wearable and augmentative medical devices. Despite the limited testing in controlled environments, the system represents a step toward future patient-centered solutions that combine sensing, feedback, and data interpretation in daily-life contexts. We encourage researchers and practitioners to further explore the integration of adaptive feedback and context-aware interaction in orthopedic and rehabilitation technologies to support more personalized and effective gait interventions.

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Appendix A. Software Architecture and Data Processing—Smart Orthosis

Appendix A.1. Raw Data Acquisition

Accurate and synchronized data acquisition is essential to ensure reliable downstream signal processing and gait detection. To ensure synchronization, data from the FSRs and the IMU were recorded at a sampling rate of 100 Hz, consistent with values reported in related studies [54,116–118]. Analog pressure signals from the FSRs were acquired via a 32-channel analog multiplexer. Each sensor was connected in a voltage divider circuit with a 10 kΩ resistor and the resulting signals were digitized using the microcontroller's 12-bit ADC. IMU data was acquired via the I²C interface and processed into triaxial acceleration (m/s^2) and angular velocity (rad/s) using the BMI270's internal processing unit. To ensure temporal synchronization, each sample was assigned a unified timestamp. All sensor readings were subsequently organized into structured data packets for further processing.

Appendix A.2. Pre-Processing

Sensor data pre-processing was performed to improve signal quality and prepare the data for subsequent segmentation and analysis. Due to manufacturing tolerances and environmental influences, FSR signals required individual calibration to obtain accurate force measurements. Each FSR sensor provides a non-linear voltage output that decreases with increasing applied force. The analog readings were first converted to voltage values, from which the resistance R_{FSR} was calculated using the voltage divider principle (Equation (A1)). Here, R_M is the reference resistor (10 kΩ), V^+ is the supply voltage, and V_{out} is the measured voltage across the FSR.

$$V_{\text{out}} = \frac{R_M \cdot V^+}{R_M + R_{\text{FSR}}} \tag{A1}$$

Following the manufacturer's specification [96], the raw sensor output was used to estimate relative plantar forces without applying sensor-specific calibration. To reduce high-frequency noise, a moving average filter was applied to the estimated force signals (Equation (A2)).

$$\text{filteredForce}[i] = \frac{1}{N} \sum_{k=0}^{N-1} \text{fsrForce}[i-k] \tag{A2}$$

The BMI270 includes internal configuration options for filtering and calibration, which were utilized to enhance signal quality. The sensor was configured with a measurement range of ±4 g for the accelerometer and ±2000°/s for the gyroscope to capture typical gait-related movements as well as rapid rotational changes. Offset correction was automatically performed during initialization by fixing the IMU in a stationary position to determine the static bias for each axis. These offset values were removed from all subsequent accelerometer and gyroscope samples to ensure base-corrected measurements (Equations (A3) and (A4)).

$$\text{accelCorrected}[i] = \text{accelRaw}[i] - \text{accelOffset} \tag{A3}$$

$$\text{gyroCorrected}[i] = \text{gyroRaw}[i] - \text{gyroOffset} \tag{A4}$$

Additionally, separate internal low-pass filters were applied to the accelerometer and gyroscope axes to remove high-frequency noise (Equation (A5)).

$$\text{filteredValue}[i] = \alpha \cdot \text{filteredValue}[i-1] + (1-\alpha) \cdot \text{rawValue}[i] \tag{A5}$$

A filter coefficient of $\alpha = 0.9$ was chosen to effectively reduce high-frequency noise. Preprocessing of the IMU data through offset correction and low-pass filtering ensures that the acceleration and rotation data are precise and reliable.

Appendix A.3. Signal Processing

In the signal processing stage, pre-processed sensor data were further refined to reduce residual noise and enable robust detection of gait events. To improve signal quality, Kalman filters [119] were applied to both FSR and IMU signals. The Kalman filter was selected due to its proven suitability for real-time gait analysis in wearable systems, including IMU-based orientation estimation, walking distance calculation, and general gait assessment [55,120,121]. For the FSR signals, individual Kalman filters were assigned to each of the sensor channels, with parameters set to $Q = 1.0$, $R = 1.0$, and $P = 0.01$. The filter parameters were adjusted to maintain responsiveness while reducing high-frequency fluctuations. For the IMU, each axis of the accelerometer and gyroscope data was filtered individually, with configuration parameters adjusted to the sensor's noise characteristics. Filter parameters were empirically set to $Q = 0.1$, $R = 0.1$, and $P = 0.3$ for accelerometer data, and to $Q = 1.0$, $R = 1.0$, and $P = 0.01$ for gyroscope data to balance signal smoothness and responsiveness.

The filtered FSR signals served as the basis for gait event detection. A custom rule-based algorithm was implemented to identify key gait phases using threshold values defined as relative percentages of the user's body weight. The algorithm evaluated pressure variations within specific sensor zones corresponding to the forefoot and hindfoot regions to detect events such as heel strike, toe strike, heel off, and toe off. This individualized approach enabled the detection of characteristic loading patterns and supported the identification of critical gait events and potential overload conditions. Thresholds were empirically determined through pre-testing across multiple gait cycles and optimized to minimize detection errors. A finite state machine was implemented to manage gait phase transitions based on these events. This approach ensured a consistent event sequence and enabled accurate temporal segmentation. A baseline correction algorithm was used to counteract the hysteresis effects of the FSR sensors and to ensure zero-base alignment after the load is removed. This adjustment was essential for maintaining continuous and reliable gait event detection across steps. The detected gait events, including their timestamps, were subsequently used to compute spatiotemporal gait parameters such as step duration, stance/swing phase duration, and cadence.

Appendix A.4. Sensor Fusion

Filtered FSR and IMU data were fused to extract both temporal and spatial characteristics of the gait cycle. While FSR data provide reliable information on foot-ground contact timing, the IMU captures the foot's three-dimensional orientation and motion dynamics. A Mahony filter [122] was employed to estimate foot orientation in terms of roll, pitch, and yaw. The Mahony filter was implemented via the `fusion.MahonyUpdate()` function from the SensorFusion library [123]. This approach combines gyroscope data for short-term tracking with accelerometer data for long-term correction, effectively compensating for drift and high-frequency noise. To improve the accuracy of position estimation, zero-velocity updates [124] were applied during the stance phase, identified from FSR signals. These updates reset the velocity to zero between the heel contact and the heel off, thereby limiting integration drift and reducing cumulative error in position calculation.

Appendix A.5. Extraction of Gait Parameters

Based on the fused sensor data, key gait parameters were extracted. The segmentation of the gait cycle follows the definition of Perry [125], which distinguishes two main phases: the stance phase (60% of the gait cycle) and the swing phase (40%), as well as eight sub-phases (including foot strike, foot flat, heel off, and foot off). These gait parameters can be classified into spatial, temporal, spatio-temporal, and kinetic categories [126]. Stride length was calculated by integrating acceleration data, corrected for foot orientation and stabilized using zero-velocity updates during stance phases. Gait speed was subsequently derived as the ratio of stride (cycle) length to gait cycle duration. As the measurement setup included only a single foot-mounted IMU, step length and step width could not be directly calculated. However, when the instrumented insoles are worn on both feet, these parameters can be extracted. In addition to standard parameters, a stability index was introduced to quantify gait stability. This metric integrates two normalized components:

- Pressure variability: Temporal fluctuations in total plantar pressure across all FSR channels.
- Anterior-posterior pressure ratio: Relative distribution of pressure between forefoot and rearfoot regions.

The final stability index was computed as a weighted sum of both components and ranges from 0 (low stability) to 1 (high stability). Higher values indicate low pressure variability and a balanced anterior-posterior distribution, whereas lower values suggest unsteady gait characteristics. By combining FSR and IMU data, the system enables accurate, real-time estimation of gait events and parameters. The integration of Mahony filtering and zero-velocity detection effectively mitigates drift and noise, while the proposed stability index provides an interpretable measure for detecting variations in gait control.

The extracted gait parameters serve as the basis for subsequent data analysis and interpretation. Quantitative measured variables such as stride length, cadence, stance time and gait speed can be statistically evaluated. Furthermore, the parameters enable the classification of gait patterns and the detection of anomalies, which may be indicative of pathological gait or early-stage motor impairments.

References

1. Barbara, A.M.; Grobelna, A. Orthopedic Footwear for People with Lower-Limb Conditions. *Can. J. Health Technol.* **2022**, *2*, 1–42. [CrossRef]
2. Bertrand, K.; Raymond, M.H.; Miller, W.C.; Ginis, K.A.M.; Demers, L. Walking aids for enabling activity and participation: A systematic review. *Am. J. Phys. Med. Rehabil.* **2017**, *96*, 894–903. [CrossRef]
3. Burns, J.; Wegener, C.; Begg, L.; Vicaretti, M.; Fletcher, J. Randomized trial of custom orthoses and footwear on foot pain and plantar pressure in diabetic peripheral arterial disease. *Diabet. Med.* **2009**, *26*, 893–899. [CrossRef]
4. de Jong, L.A.; Kerkum, Y.L.; Altmann, V.C.; Geurts, A.C.; Keijsers, N.L. Orthopedic footwear has a positive influence on gait adaptability in individuals with hereditary motor and sensory neuropathy. *Gait Posture* **2023**, *106*, 18–22. [CrossRef] [PubMed]
5. Arch, E.S.; Stanhope, S.J., Orthotic Device Research. In *Full Stride: Advancing the State of the Art in Lower Extremity Gait Systems*; Springer: New York, NY, USA, 2017; pp. 99–116. [CrossRef]
6. Resch, S.; Carroll, A. Influence of Virtual Footwear on Gait Behavior, Embodiment, and Workload While Walking on a Treadmill in Virtual Reality. In Proceedings of the 2025 31st ACM Symposium on Virtual Reality Software and Technology, Montreal, QC, Canada, 12–14 November 2025; VRST '25. [CrossRef]
7. Salis, F.; Bertuletti, S.; Bonci, T.; Caruso, M.; Scott, K.; Alcock, L.; Buckley, E.; Gazit, E.; Hansen, C.; Schwickert, L.; et al. A multi-sensor wearable system for the assessment of diseased gait in real-world conditions. *Front. Bioeng. Biotechnol.* **2023**, *11*, 1143248. [CrossRef]
8. Almuteb, I.; Hua, R.; Wang, Y. Smart insoles review (2008–2021): Applications, potentials, and future. *Smart Health* **2022**, *25*, 100301. [CrossRef]
9. Okoduwa, S.I.; Igiri, B.E.; Tagang, J.I.; Okoduwa, U.J.; Adeyi, A.O. Therapeutic smart-footwear approach for management of neuropathic diabetic foot ulcers: Current challenges and focus for future perspective. *Med. Nov. Technol. Devices* **2024**, *23*, 100311. [CrossRef]
10. Hutabarat, Y.; Owaki, D.; Hayashibe, M. Recent Advances in Quantitative Gait Analysis Using Wearable Sensors: A Review. *IEEE Sens. J.* **2021**, *21*, 26470–26487. [CrossRef]
11. Kim, H.; Kang, Y.; Valencia, D.R.; Kim, D. An integrated system for gait analysis using FSRs and an IMU. In Proceedings of the 2018 Second IEEE International Conference on Robotic Computing (IRC), Laguna Hills, CA, USA, 31 January–2 February 2018; pp. 347–351. [CrossRef]
12. Grech, J.; Mizzi, S.; Falzon, O. A technical review of foot temperature measurement systems. *J. Am. Podiatr. Med. Assoc.* **2022**, *112*, 21–65. [CrossRef]
13. Frykberg, R.G.; Lavery, L.A.; Pham, H.; Harvey, C.; Harkless, L.; Veves, A. Role of neuropathy and high foot pressures in diabetic foot ulceration. *Diabetes Care* **1998**, *21*, 1714–1719. [CrossRef]
14. Subramaniam, S.; Majumder, S.; Faisal, A.I.; Deen, M.J. Insole-based systems for health monitoring: Current solutions and research challenges. *Sensors* **2022**, *22*, 438. [CrossRef]
15. Resch, S.; Röll, A.; Malech, A.; Schwind, V.; Sanchez-Morillo, D.; Völz, D. Walking aid with haptic feedback for combined use with a smart foot orthosis. *Gerontechnology* **2024**, *23*, 2. [CrossRef]
16. Alam, U.; Riley, D.R.; Jugdey, R.S.; Azmi, S.; Rajbhandari, S.; D'Août, K.; Malik, R.A. Diabetic neuropathy and gait: A review. *Diabetes Ther.* **2017**, *8*, 1253–1264. [CrossRef] [PubMed]
17. Rogers, L.C.; Frykberg, R.G.; Armstrong, D.G.; Boulton, A.J.; Edmonds, M.; Van, G.H.; Hartemann, A.; Game, F.; Jeffcoate, W.; Jirkovska, A.; et al. The Charcot foot in diabetes. *J. Am. Podiatr. Med. Assoc.* **2011**, *101*, 437–446. [CrossRef]
18. Finco, M.G.; Cay, G.; Lee, M.; Garcia, J.; Salazar, E.; Tan, T.W.; Armstrong, D.G.; Najafi, B. Taking a Load Off: User Perceptions of Smart Offloading Walkers for Diabetic Foot Ulcers Using the Technology Acceptance Model. *Sensors* **2023**, *23*, 2768. [CrossRef]
19. Park, C.; Mishra, R.; Vigano, D.; Macagno, M.; Rossotti, S.; D'Huyvetter, K.; Garcia, J.; Armstrong, D.G.; Najafi, B. Smart offloading boot system for remote patient monitoring: Toward adherence reinforcement and proper physical activity prescription for diabetic foot ulcer patients. *J. Diabetes Sci. Technol.* **2023**, *17*, 42–51. [CrossRef]

20. Zhou, C.; Yang, L.; Liao, H.; Liang, B.; Ye, X. Ankle foot motion recognition based on wireless wearable sEMG and acceleration sensors for smart AFO. *Sens. Actuators A Phys.* **2021**, *331*, 113025. [CrossRef]
21. Najafi, B.; Mohseni, H.; Grewal, G.S.; Talal, T.K.; Menzies, R.A.; Armstrong, D.G. An optical-fiber-based smart textile (smart socks) to manage biomechanical risk factors associated with diabetic foot amputation. *J. Diabetes Sci. Technol.* **2017**, *11*, 668–677. [CrossRef]
22. Isezaki, T.; Kadone, H.; Niijima, A.; Aoki, R.; Watanabe, T.; Kimura, T.; Suzuki, K. Sock-Type Wearable Sensor for Estimating Lower Leg Muscle Activity Using Distal EMG Signals. *Sensors* **2019**, *19*, 1954. [CrossRef]
23. Zhang, Z.; He, T.; Zhu, M.; Sun, Z.; Shi, Q.; Zhu, J.; Dong, B.; Yuce, M.R.; Lee, C. Deep learning-enabled triboelectric smart socks for IoT-based gait analysis and VR applications. *npj Flex. Electron.* **2020**, *4*, 29. [CrossRef]
24. Drăgulinescu, A.; Drăgulinescu, A.M.; Zincă, G.; Bucur, D.; Feieș, V.; Neagu, D.M. Smart socks and in-shoe systems: State-of-the-art for two popular technologies for foot motion analysis, sports, and medical applications. *Sensors* **2020**, *20*, 4316. [CrossRef]
25. Semjonova, G.; Davidovica, A.; Kozlovskis, N.; Okss, A.; Katashevs, A. Smart Textile Sock System for Athletes' Self-Correction during Functional Tasks: Formative Usability Evaluation. *Sensors* **2022**, *22*, 4779. [CrossRef] [PubMed]
26. Morley, R.; Richter, E.; Klaesner, J.; Maluf, K.; Mueller, M. In-shoe multisensory data acquisition system. *IEEE Trans. Biomed. Eng.* **2001**, *48*, 815–820. [CrossRef]
27. Eskofier, B.M.; Lee, S.I.; Baron, M.; Simon, A.; Martindale, C.F.; Gaßner, H.; Klucken, J. An overview of smart shoes in the internet of health things: Gait and mobility assessment in health promotion and disease monitoring. *Appl. Sci.* **2017**, *7*, 986. [CrossRef]
28. Sunarya, U.; Sun Hariyani, Y.; Cho, T.; Roh, J.; Hyeong, J.; Sohn, I.; Kim, S.; Park, C. Feature analysis of smart shoe sensors for classification of gait patterns. *Sensors* **2020**, *20*, 6253. [CrossRef] [PubMed]
29. Gokalgandhi, D.; Kamdar, L.; Shah, N.; Mehendale, N. A review of smart technologies embedded in shoes. *J. Med. Syst.* **2020**, *44*, 150. [CrossRef] [PubMed]
30. Ullrich, M.; Kuderle, A.; Hannink, J.; Din, S.D.; Gasner, H.; Marxreiter, F.; Klucken, J.; Eskofier, B.M.; Kluge, F. Detection of Gait from Continuous Inertial Sensor Data Using Harmonic Frequencies. *IEEE J. Biomed. Health Inform.* **2020**, *24*, 1869–1878. [CrossRef] [PubMed]
31. Lin, F.; Wang, A.; Zhuang, Y.; Tomita, M.R.; Xu, W. Smart insole: A wearable sensor device for unobtrusive gait monitoring in daily life. *IEEE Trans. Ind. Inform.* **2016**, *12*, 2281–2291. [CrossRef]
32. Xu, W.; Huang, M.C.; Amini, N.; Liu, J.J.; He, L.; Sarrafzadeh, M. Smart insole: A wearable system for gait analysis. In Proceedings of the 5th International Conference on Pervasive Technologies Related to Assistive Environments, Heraklion, Crete, Greece, 6–8 June 2012; pp. 1–4. [CrossRef]
33. Duong, T.T.; Goldman, S.; Zhang, H.; Salazar, R.; Beenders, S.; Cornett, K.M.; Bain, J.M.; Montes, J.; Zanotto, D. Validation of Insole-based Gait Analysis System in Young Children with a Neurodevelopmental Disorder and Autism Traits. In Proceedings of the IEEE RAS and EMBS International Conference on Biomedical Robotics and Biomechatronics, New York, NY, USA, 29 November–1 December 2020; pp. 715–720. [CrossRef]
34. Shefa, F.R.; Sifat, F.H.; Shah, S.C.; Kibria, M.G. IoT-Based Smart Ankle-Foot Orthosis for Patients with Gait Imbalance. In Proceedings of the International Conference on Control, Automation and Systems, Yeosu, Republic of Korea, 17–20 October 2023; pp. 969–974. [CrossRef]
35. Shefa, F.R.; Sifat, F.H.; Uddin, J.; Ahmad, Z.; Kim, J.M.; Kibria, M.G. Deep Learning and IoT-Based Ankle–Foot Orthosis for Enhanced Gait Optimization. *Healthcare* **2024**, *12*, 2273. [CrossRef]
36. Han, P.; Wang, K.; Yan, C.; Wang, L. Design and Experimental Research of an Intelligent Ankle-Foot Orthosis System for Strephenopodia. In Proceedings of the 2023 4th International Symposium on Artificial Intelligence for Medicine Science, Chengdu, China, 20–22 October 2023; ISAIMS '23, pp. 985–991. [CrossRef]
37. Mustufa, Y.S.; Barton, J.; O'Flynn, B.; Davies, R.; McCullagh, P.; Zheng, H. Design of a smart insole for ambulatory assessment of gait. In Proceedings of the 2015 IEEE 12th International Conference on Wearable and Implantable Body Sensor Networks, BSN 2015, Cambridge, MA, USA, 9–12 June 2015. [CrossRef]
38. Roth, N.; Martindale, C.F.; Gaßner, H.; Kohl, Z.; Klucken, J.; Eskofer, B.M. Synchronized sensor insoles for clinical gait analysis in home-monitoring applications. *Curr. Dir. Biomed. Eng.* **2018**, *4*, 433–437. [CrossRef]
39. Farid, L.; Jacobs, D.; Santos, J.D.; Simon, O.; Gracies, J.M.; Hutin, E. FeetMe® Monitor-connected insoles are a valid and reliable alternative for the evaluation of gait speed after stroke. *Top. Stroke Rehabil.* **2021**, *28*, 127–134. [CrossRef]
40. Chen, D.; Asaeikheybari, G.; Chen, H.; Xu, W.; Huang, M.C. Ubiquitous Fall Hazard Identification with Smart Insole. *IEEE J. Biomed. Health Inform.* **2021**, *25*, 2768–2776. [CrossRef] [PubMed]
41. D'arco, L.; Mccalmont, G.; Wang, H.; Zheng, H. Application of Smart Insoles for Recognition of Activities of Daily Living: A Systematic Review. *ACM Trans. Comput. Healthc.* **2024**, *5*, 1–34. [CrossRef]

42. Djamaa, B.; Bessa, M.M.; Diaf, B.; Rouigueb, A.; Yachir, A. BoostSole: Design and Realization of a Smart Insole for Automatic Human Gait Classification. In Proceedings of the 2020 Federated Conference on Computer Science and Information Systems, FedCSIS 2020, Sofia, Bulgaria, 6–9 September 2020; pp. 35–43. [CrossRef]
43. Razak, A.H.A.; Zayegh, A.; Begg, R.K.; Wahab, Y. Foot Plantar Pressure Measurement System: A Review. *Sensors* **2012**, *12*, 9884–9912. [CrossRef] [PubMed]
44. Shu, L.; Hua, T.; Wang, Y.; Li, Q.; Feng, D.D.; Tao, X. In-Shoe Plantar Pressure Measurement and Analysis System Based on Fabric Pressure Sensing Array. *IEEE Trans. Inf. Technol. Biomed.* **2010**, *14*, 767–775. [CrossRef]
45. Senanayake, C.M.; Senanayake, S.M.N.A. Computational Intelligent Gait-Phase Detection System to Identify Pathological Gait. *IEEE Trans. Inf. Technol. Biomed.* **2010**, *14*, 1173–1179. [CrossRef] [PubMed]
46. Gujarathi, T.; Bhole, K. GAIT ANALYSIS USING IMU SENSOR. In Proceedings of the 2019 10th International Conference on Computing, Communication and Networking Technologies, ICCCNT 2019, Kanpur, India, 6–8 July 2019. [CrossRef]
47. Aggarwal, A.; Gupta, R.; Agarwal, R. Design and Development of Integrated Insole System for Gait Analysis. In Proceedings of the 2018 11th International Conference on Contemporary Computing, IC3 2018, Noida, India, 2–4 August 2018. [CrossRef]
48. Sabatini, A.M.; Martelloni, C.; Scapellato, S.; Cavallo, F. Assessment of walking features from foot inertial sensing. *IEEE Trans. Biomed. Eng.* **2005**, *52*, 486–494. [CrossRef]
49. Küderle, A.; Roth, N.; Zlatanovic, J.; Zrenner, M.; Eskofier, B.; Kluge, F. The placement of foot-mounted IMU sensors does affect the accuracy of spatial parameters during regular walking. *PLoS ONE* **2022**, *17*, e0269567. [CrossRef]
50. Zhang, Z.; Poslad, S. Improved use of foot force sensors and mobile phone GPS for mobility activity recognition. *IEEE Sens. J.* **2014**, *14*, 4340–4347. [CrossRef]
51. Matijevich, E.; Minty, E.; Bray, E.; Bachus, C.; Hajizadeh, M.; Liden, B. A Multi-Faceted Digital Health Solution for Monitoring and Managing Diabetic Foot Ulcer Risk: A Case Series. *Sensors* **2024**, *24*, 2675. [CrossRef]
52. Hong, S.; Park, K.S. Unobtrusive photoplethysmographic monitoring under the foot sole while in a standing posture. *Sensors* **2018**, *18*, 3239. [CrossRef] [PubMed]
53. Joseph, A.M.; Kian, A.; Begg, R. State-of-the-Art Review on Wearable Obstacle Detection Systems Developed for Assistive Technologies and Footwear. *Sensors* **2023**, *23*, 2802. [CrossRef] [PubMed]
54. Manna, S.K.; Azhar, M.A.H.B.; Greace, A. Optimal locations and computational frameworks of FSR and IMU sensors for measuring gait abnormalities. *Heliyon* **2023**, *9*, e15210. [CrossRef] [PubMed]
55. Prasanth, H.; Caban, M.; Keller, U.; Courtine, G.; Ijspeert, A.; Vallery, H.; von Zitzewitz, J. Wearable Sensor-Based Real-Time Gait Detection: A Systematic Review. *Sensors* **2021**, *21*, 2727. [CrossRef]
56. Han, Y.C.; Wong, K.I.; Murray, I. Gait Phase Detection for Normal and Abnormal Gaits Using IMU. *IEEE Sens. J.* **2019**, *19*, 3439–3448. [CrossRef]
57. Manupibul, U.; Tanthuwapathom, R.; Jarumethitanont, W.; Kaimuk, P.; Limroongreungrat, W.; Charoensuk, W. Integration of force and IMU sensors for developing low-cost portable gait measurement system in lower extremities. *Sci. Rep.* **2023**, *13*, 10653. [CrossRef]
58. Benocci, M.; Rocchi, L.; Farella, E.; Chiari, L.; Benini, L. A wireless system for gait and posture analysis based on pressure insoles and inertial measurement units. In Proceedings of the 3rd International Conference on Pervasive Computing Technologies for Healthcare-Pervasive Health 2009, London, UK, 1–3 April 2009. [CrossRef]
59. Jakob, V.; Küderle, A.; Kluge, F.; Klucken, J.; Eskofier, B.M.; Winkler, J.; Winterholler, M.; Gassner, H. Validation of a Sensor-Based Gait Analysis System with a Gold-Standard Motion Capture System in Patients with Parkinson's Disease. *Sensors* **2021**, *21*, 7680. [CrossRef]
60. Donath, L.; Faude, O.; Lichtenstein, E.; Nuesch, C.; Mündermann, A. Validity and reliability of a portable gait analysis system for measuring spatiotemporal gait characteristics: Comparison to an instrumented treadmill. *J. Neuroeng. Rehabil.* **2016**, *13*, 6. [CrossRef] [PubMed]
61. Jagos, H.; Pils, K.; Haller, M.; Wassermann, C.; Chhatwal, C.; Rafolt, D.; Rattay, F. Mobile gait analysis via eSHOEs instrumented shoe insoles: A pilot study for validation against the gold standard GAITRite®. *J. Med. Eng. Technol.* **2017**, *41*, 375–386. [CrossRef]
62. Raghuvanshi, A.; Mitra, P.; Rane, D.; Kumar, S.D.; Lahiri, U. GaitWEAR: An Augmented Wearable System for Gait Quantification. *IEEE Sens. J.* **2024**, *24*, 35673–35685. [CrossRef]
63. Santos, V.M.; Gomes, B.B.; Neto, M.A.; Amaro, A.M. A Systematic Review of Insole Sensor Technology: Recent Studies and Future Directions. *Appl. Sci.* **2024**, *14*, 6085. [CrossRef]
64. Saboor, A.; Kask, T.; Kuusik, A.; Alam, M.M.; Moullec, Y.L.; Niazi, I.K.; Zoha, A.; Ahmad, R. Latest research trends in gait analysis using wearable sensors and machine learning: A systematic review. *IEEE Access* **2020**, *8*, 167830–167864. [CrossRef]
65. Resch, S.; Zoufal, K.; Akhouaji, I.; Abbou, M.A.; Schwind, V.; Völz, D. Augmented Smart Insoles-Prototyping a Mobile Application: Usage Preferences of Healthcare Professionals and People with Foot Deformities. *Curr. Dir. Biomed. Eng.* **2023**, *9*, 698–701. [CrossRef]

66. Resch, S.; Zimmermann, L.; Kunkel, A.; Gerstung, D.; Schwind, V.; Völz, D.; Sanchez-Morillo, D. Usability Evaluation of a Mobile Application for Foot Health Monitoring of Smart Insoles: A Mixed Methods Study. In Proceedings of the MobileHCI 2024 Adjunct Proceedings-Publication of the 26th International Conference on Mobile Human-Computer Interaction, Melbourne, VIC, Australia, 30 September–3 October 2024. [CrossRef]
67. Crea, S.; Cipriani, C.; Donati, M.; Carrozza, M.C.; Vitiello, N. Providing Time-Discrete Gait Information by Wearable Feedback Apparatus for Lower-Limb Amputees: Usability and Functional Validation. *IEEE Trans. Neural Syst. Rehabil. Eng.* **2015**, *23*, 250–257. [CrossRef] [PubMed]
68. Oates, A.R.; Hauck, L.; Moraes, R.; Sibley, K.M. The effects of haptic input on biomechanical and neurophysiological parameters of walking: A scoping review. *Gait Posture* **2017**, *58*, 232–239. [CrossRef] [PubMed]
69. Berengueres, J.; Fritschi, M.; McClanahan, R. A smart pressure-sensitive insole that reminds you to walk correctly: An orthotic-less treatment for over pronation. In Proceedings of the 2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC 2014, Chicago, IL, USA, 26–30 August 2014; pp. 2488–2491. [CrossRef]
70. Biesmans, S.; Markopoulos, P. Design and Evaluation of SONIS, a Wearable Biofeedback System for Gait Retraining. *Multimodal Technol. Interact.* **2020**, *4*, 60. [CrossRef]
71. Pacilli, A.; Mileti, I.; Germanotta, M.; Di Sipio, E.; Imbimbo, I.; Aprile, I.; Padua, L.; Rossi, S.; Palermo, E.; Cappa, P. A wearable setup for auditory cued gait analysis in patients with Parkinson's Disease. In Proceedings of the 2016 IEEE International Symposium on Medical Measurements and Applications (MeMeA), Benevento, Italy, 15–18 May 2016; pp. 1–6. [CrossRef]
72. Ma, C.C.; Wei, S.Y.; Lin, J.H. Smart footwear feedback interface analysis and design. In Proceedings of the 2015 International Conference on Computational Science and Computational Intelligence, CSCI 2015, Las Vegas, NV, USA, 7–9 December 2015; pp. 766–770. [CrossRef]
73. Redd, C.B.; Bamberg, S.J. A wireless sensory feedback device for real-time gait feedback and training. *IEEE/ASME Trans. Mechatronics* **2012**, *17*, 425–433. [CrossRef]
74. Elvitigala, D.S.; Matthies, D.J.; David, L.; Weerasinghe, C.; Nanayakkara, S. GymSoles: Improving Squats and Dead-Lifts by Visualizing the User's Center of Pressure. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, New York, NY, USA, 4–9 May 2019; CHI '19, pp. 1–12. [CrossRef]
75. Yan, Q.; Huang, J.; Tao, C.; Chen, X.; Xu, W.X. Intelligent mobile walking-aids: Perception, control and safety. *Adv. Robot.* **2020**, *34*, 2–18. [CrossRef]
76. Resch, S.; Zirari, A.; Tran, T.D.Q.; Bauer, L.M.; Sanchez-Morillo, D. Smart Walking Aids with Sensor Technology for Gait Support and Health Monitoring: A Scoping Review. *Technologies* **2025**, *13*, 346. [CrossRef]
77. Srinivasan, S.; Rajesh, M. Smart walking stick. In Proceedings of the International Conference on Trends in Electronics and Informatics, ICOEI 2019, Tirunelveli, India, 23–25 April 2019; pp. 576–579. [CrossRef]
78. Yang, Z.; Gao, M.; Choi, J. Smart walking cane based on triboelectric nanogenerators for assisting the visually impaired. *Nano Energy* **2024**, *124*, 109485. [CrossRef]
79. Chamorro-Moriana, G.; Sevillano, J.L.; Ridao-Fernández, C. A Compact Forearm Crutch Based on Force Sensors for Aided Gait: Reliability and Validity. *Sensors* **2016**, *16*, 925. [CrossRef]
80. Shah, M.M.; Khan, M.N.R.; Khan, M.R.A.; Plabon, M.M.H.; Razzak, M.A. A Cost-Effective Smart Walking Stick for Visually Impaired People. In Proceedings of the 6th International Conference on Communication and Electronics Systems, ICCES 2021, Coimbatre, India, 8–10 July 2021; pp. 1582–1585. [CrossRef]
81. Bansal, M.; Malik, S.; Kumar, M.; Meena, N. Arduino based Smart Walking Cane for Visually Impaired People. In Proceedings of the 4th International Conference on Inventive Systems and Control, ICISC 2020, Coimbatore, India, 8–10 January 2020; pp. 462–465. [CrossRef]
82. Ikbal, M.A.; Rahman, F.; Kabir, M.H. Microcontroller based smart walking stick for visually impaired people. In Proceedings of the 4th International Conference on Electrical Engineering and Information and Communication Technology, iCEEiCT 2018, Dhaka, Bangladesh, 13–15 September 2018; pp. 255–259. [CrossRef]
83. Gallo, S.; Chapuis, D.; Santos-Carreras, L.; Kim, Y.; Retornaz, P.; Bleuler, H.; Gassert, R. Augmented white cane with multimodal haptic feedback. In Proceedings of the 2010 3rd IEEE RAS and EMBS International Conference on Biomedical Robotics and Biomechatronics, BioRob 2010, Tokyo, Japan, 26–29 September 2010; pp. 149–155. [CrossRef]
84. Di, P.; Hasegawa, Y.; Nakagawa, S.; Sekiyama, K.; Fukuda, T.; Huang, J.; Huang, Q. Fall Detection and Prevention Control Using Walking-Aid Cane Robot. *IEEE/ASME Trans. Mechatron.* **2016**, *21*, 625–637. [CrossRef]
85. Tsuda, N.; Hayashi, A.; Tounai, M.; Akutagawa, S. Visualization system of crutch walking based on internal sensors. In Proceedings of the IEEE/ASME International Conference on Advanced Intelligent Mechatronics, AIM, Montreal, QC, Canada, 6–9 July 2010; pp. 19–24. [CrossRef]
86. Trujillo-León, A.; de Guzmán-Manzano, A.; Velázquez, R.; Vidal-Verdú, F. Generation of gait events with a fsr based cane handle. *Sensors* **2021**, *21*, 5632. [CrossRef]

87. Murali, S.; Shrivatsan, R.; Sreenivas, V.; Vijjappu, S.; Gladwin, S.J.; Rajavel, R. Smart walking cane for the visually challenged. In Proceedings of the IEEE Region 10 Humanitarian Technology Conference 2016, R10-HTC 2016-Proceedings, Agra, India, 21–23 December 2016. [CrossRef]
88. Merrett, G.V.; Ettabib, M.A.; Peters, C.; Hallett, G.; White, N.M. Augmenting forearm crutches with wireless sensors for lower limb rehabilitation. *Meas. Sci. Technol.* **2010**, *21*, 124008. [CrossRef]
89. Schuster, E.; Routson, R.L.; Hinchcliff, M.; Benoff, K.; Suri, P.; Richburg, C.; Muir, B.C.; Czerniecki, J.M.; Aubin, P.M. A novel walking cane with haptic biofeedback reduces knee adduction moment in the osteoarthritic knee. *J. Biomech.* **2021**, *114*, 110150. [CrossRef] [PubMed]
90. Mohapatra, S.; Rout, S.H.; Tripathi, V.; Saxena, T.; Karuna, Y. Smart Walking Stick for Blind Integrated with SOS Navigation System. In Proceedings of the 2nd International Conference on Trends in Electronics and Informatics, ICOEI 2018, Tirunelveli, India, 11–12 May 2018; pp. 441–447. [CrossRef]
91. Suman, S.; Mishra, S.; Sahoo, K.S.; Nayyar, A. Vision navigator: A smart and intelligent obstacle recognition model for visually impaired users. *Mob. Inf. Syst.* **2022**, *2022*, 9715891. [CrossRef]
92. Lancini, M.; Serpelloni, M.; Pasinetti, S.; Guanziroli, E. Healthcare Sensor System Exploiting Instrumented Crutches for Force Measurement during Assisted Gait of Exoskeleton Users. *IEEE Sens. J.* **2016**, *16*, 8228–8237. [CrossRef]
93. Enovis. AIRCAST® AIRSELECT™ Elite Walker. Available online: https://www.enovis-medtech.de/AIRCAST-AIRSELECT-Elite-Walker (accessed on 15 August 2025).
94. *ISO 11334-1:2007*; Assistive Products for Walking Manipulated by One Arm—Requirements and Test Methods—Part 1: Elbow Crutches. Deutsches Institut für Normung e.V. (DIN): Berlin, Germany, 2007. [CrossRef]
95. Interlink Electronics. FSR® 400 Short. Available online: https://www.interlinkelectronics.com/fsr-400-short (accessed on 15 August 2025).
96. Interlink Electronics. *FSR® Integration Guide*. Interlink Electronics: Camarillo, CA, USA, 2013.
97. Bosch Sensortec GmbH. Small. Low Power. Optimized for Wearables—Inertial Measurement Unit: BMI270. Available online: https://www.bosch-sensortec.com/products/motion-sensors/imus/bmi270/ (accessed on 15 August 2025).
98. Bosch Sensortec GmbH. *BMI270 Datasheet*, revision 1.5 ed.; Bosch Sensortec GmbH: Reutlingen, Germany, 2023.
99. Ballesteros, J.; Tudela, A.; Caro-Romero, J.R.; Urdiales, C. A cane-based low cost sensor to implement attention mechanisms in telecare robots. In Proceedings of the 2019 International Conference on Robotics and Automation (ICRA), Montreal, QC, Canada, 20–24 May 2019; pp. 1473–1478. [CrossRef]
100. Mariani, B.; Rouhani, H.; Crevoisier, X.; Aminian, K. Quantitative estimation of foot-flat and stance phase of gait using foot-worn inertial sensors. *Gait Posture* **2013**, *37*, 229–234. [CrossRef] [PubMed]
101. Sardini, E.; Serpelloni, M.; Lancini, M. Wireless Instrumented Crutches for Force and Movement Measurements for Gait Monitoring. *IEEE Trans. Instrum. Meas.* **2015**, *64*, 3369–3379. [CrossRef]
102. Mikhailov, A. Turbo, An Improved Rainbow Colormap for Visualization. Google. 2019. Available online: https://research.google/blog/turbo-an-improved-rainbow-colormap-for-visualization/ (accessed on 15 August 2025).
103. Android Studio. Android Studio–The Official IDE for Android Development. Available online: https://developer.android.com/studio (accessed on 15 August 2025).
104. Developers. Build Better Apps Faster with Jetpack Compose. Available online: https://developer.android.com/compose (accessed on 15 August 2025).
105. Brooke, J. SUS-A Quick and Dirty Usability Scale. In *Usability Evaluation In Industry*; CRC Press: Boca Raton, FL, USA, 1996; Volume 189, pp. 4–7.
106. Schwind, V.; Resch, S.; Sehrt, J. The HCI User Studies Toolkit: Supporting Study Designing and Planning for Undergraduates and Novice Researchers in Human-Computer Interaction. In Proceedings of the Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems, New York, NY, USA, 23–28 April 2023; CHI EA '23. [CrossRef]
107. Kassambara, A. rstatix: Pipe-Friendly Framework for Basic Statistical Tests, R Package Version 0.7.2. 2023. Available online: https://rpkgs.datanovia.com/rstatix/ (accessed on 15 August 2025).
108. Likert, R. A technique for the measurement of attitudes. *Arch. Psychol.* **1932**, *140*, 1–55..
109. Bangor, A.; Kortum, P.; Miller, J. Determining what individual SUS scores mean: Adding an adjective rating scale. *J. Usability Stud.* **2009**, *4*, 114–123.
110. Cohen, J. *Statistical Power Analysis for the Behavioral Sciences*; Routledge: London, UK, 2013.
111. Ganguly, A.; Olmanson, B.A.; Knowlton, C.B.; Wimmer, M.A.; Ferrigno, C. Accuracy of the fully integrated Insole3's estimates of spatiotemporal parameters during walking. *Med. Eng. Phys.* **2023**, *111*, 103925. [CrossRef]
112. Hyzy, M.; Bond, R.; Mulvenna, M.; Bai, L.; Dix, A.; Leigh, S.; Hunt, S. System Usability Scale Benchmarking for Digital Health Apps: Meta-analysis. *JMIR mHealth uHealth* **2022**, *10*, e37290. [CrossRef] [PubMed]
113. Alzghoul, B. The Effectiveness of Gamification in Changing Health-related Behaviors: A Systematic Review and Meta-analysis. *Open Public Health J.* **2024**, *17*. [CrossRef]

114. Gurari, N.; Smith, K.; Madhav, M.; Okamura, A.M. Environment discrimination with vibration feedback to the foot, arm, and fingertip. In Proceedings of the 2009 IEEE International Conference on Rehabilitation Robotics, Kyoto, Japan, 23–26 June 2009; pp. 343–348. [CrossRef]
115. Choi, S.; Kuchenbecker, K.J. Vibrotactile Display: Perception, Technology, and Applications. *Proc. IEEE* **2013**, *101*, 2093–2104. [CrossRef]
116. Nikodelis, T.; Moscha, D.; Metaxiotis, D.; Kollias, I. Commercial video frame rates can produce reliable results for both normal and CP spastic gait's spatiotemporal, angular, and linear displacement variables. *J. Appl. Biomech.* **2011**, *27*, 266–271. [CrossRef]
117. Santos, J.P.; Ferreira, J.P.; Crisóstomo, M.; Paulo Coimbra, A. Instrumented Shoes for 3D GRF Analysis and Characterization of Human Gait. In Proceedings of the Bioinformatics and Biomedical Engineering: 7th International Work-Conference, IWBBIO 2019, Granada, Spain, 8–10 May 2019, Proceedings, Part II 7; Springer: Berlin/Heidelberg, Germany, 2019; pp. 51–62.
118. Majumder, S.; Deen, M.J. Wearable IMU-based system for real-time monitoring of lower-limb joints. *IEEE Sens. J.* **2020**, *21*, 8267–8275. [CrossRef]
119. Kalman, R.E. A new approach to linear filtering and prediction problems. *J. Basic Eng.* **1960**, *82*, 35–45. [CrossRef]
120. Shankar, P.S.U.; Raveendranathan, N.; Gans, N.R.; Jafari, R. Towards power optimized kalman filter for gait assessment using wearable sensors. In *Proceedings-Wireless Health 2010*; Association for Computing Machinery: New York, NY, USA, 2010; pp. 137–144. [CrossRef]
121. Bennett, T.; Jafari, R.; Gans, N. An extended Kalman filter to estimate human gait parameters and walking distance. In Proceedings of the American Control Conference, Washington, DC, USA, 17–19 June 2013; pp. 752–757. [CrossRef]
122. Mahony, R.; Hamel, T.; Pflimlin, J.M. Nonlinear Complementary Filters on the Special Orthogonal Group. *IEEE Trans. Autom. Control* **2008**, *53*, 1203–1218. [CrossRef]
123. SensorFusion: A Simple Implementation of Some Complex Sensor Fusion Algorithms. 2023. GitHub Repository. Available online: https://github.com/aster94/SensorFusion (accessed on 5 July 2025).
124. Skog, I.; Handel, P.; Nilsson, J.O.; Rantakokko, J. Zero-Velocity Detection—An Algorithm Evaluation. *IEEE Trans. Biomed. Eng.* **2010**, *57*, 2657–2666. [CrossRef]
125. Perry, J.; Burnfield, J.M.; Cabico, L.M. Gait Analysis: Normal and Pathological Function. *J. Sport. Sci. Med.* **2010**, *9*, 353.
126. Lovecchio, N.; Zago, M.; Sforza, C., Gait Analysis in the Rehabilitation Process. In *Rehabilitation After Limb Salvage Surgery*; Springer International Publishing: Cham, Switzerland, 2021; pp. 109–118. [CrossRef]

G.3. Journal Publication 3

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Article

Improving Social Acceptance of Orthopedic Foot Orthoses Through Image-Generative AI in Product Design

Stefan Resch [1,2,*], Jakob Schauer [2], Valentin Schwind [2], Diana Völz [2] and Daniel Sanchez-Morillo [3,4]

1 Doctorate Program in Manufacturing, Materials and Environmental Engineering, Doctorate School (EDUCA), University of Cadiz, 11003 Cádiz, Spain
2 Faculty of Computer Science and Engineering, Frankfurt University of Applied Sciences, 60318 Frankfurt am Main, Germany
3 Department of Automation Engineering, Electronics and Computer Architecture and Networks, School of Engineering, University of Cadiz, 11519 Cádiz, Spain
4 Instituto de Investigación e Innovación Biomédica de Cádiz (INiBICA), 11009 Cádiz, Spain
* Correspondence: stefan.resch@alum.uca.es

Abstract: The lack of social acceptability for wearable devices such as orthopedic foot orthoses can lead to irregular usage and missed health benefits, as shown in prior studies. While AI-generated designs have been explored for prototyping aesthetic hand orthoses, their impact on social acceptability, particularly for foot orthoses, remains unknown. The current state of research is limited, as no empirical evidence exists on whether AI-designed orthoses influence acceptance, nor has the role of customized generative pre-trained transformers (GPTs) and specific prompting strategies been examined in this context. To address these gaps, we conducted two mixed-methods studies to investigate (1) the impact of AI-generated orthosis designs on social acceptability compared to existing orthopedic products and development concepts and (2) how a customized GPT and different prompt keywords influence acceptance. Our results show that AI-generated designs significantly enhance social acceptance across orthotic categories. Furthermore, we found that personalized GPTs and targeted prompt keywords significantly influence user perception. Overall, our findings highlight the potential of using AI to create socially acceptable design solutions for wearable technology and offer new applications for future smart devices. We contribute to generative AI in product design and provide concrete recommendations for optimizing prompting strategies to enhance social acceptance.

Keywords: artificial intelligence; generative AI; ChatGPT; text-to-image; human-centered design; human–computer interaction; product design; social acceptance; foot orthoses; wearables

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1. Introduction

Orthopedic foot orthoses are standardized devices that are essential for the rehabilitation of patients recovering from injuries or managing chronic foot diseases [1]. The designs of these assistive devices differ, featuring orthoses with varying leg lengths, bandages, and foot drop orthoses, each customized for particular medical purposes [2]. Each category of orthotic products is designed to support patients with different medical needs and mobility requirements [3]. This product diversity requires highly personalized designs to meet the specific demands of different health conditions and patient preferences [4]. While technical function is essential for medical effectiveness, aesthetics and usability are also key elements in the design of orthopedic foot orthoses to promote patient acceptance and support successful use and rehabilitation outcomes [2]. For example, a study by

Ghoseiri and Bahramian [5] investigated user satisfaction with orthotic devices among 293 participants, revealing that concerns related to the appearance of the devices often resulted in low satisfaction rates. Furthermore, a systematic literature review by Swinnen and Kerckhofs [6] analyzed ten studies involving 1576 patients and highlighted the critical role of aesthetics in the compliance of wearing lower limb orthotic devices. The review found that a notable number of patients avoid wearing their devices regularly due to their "cosmetically unacceptable" appearance, highlighting the "large importance" of aesthetics in patients' decisions to wear these devices. This underscores the significant impact of aesthetics on device adoption and user perception [7]. However, the design processes of these standardized devices frequently rely on traditional methods that limit adaptability and patient involvement, leading to compliance issues [8]. This dissatisfaction can lead to inconsistent use of standardized orthopedic aids and missed health benefits, as highlighted by previous work [9].

Current advancements in sensor technology offer the potential to transform these conventional orthoses into smart products, commonly known as wearable electronic devices, which can monitor specific health parameters [10]. In theory, the creation of smart orthotic footwear could enhance the rehabilitation process by enabling the continuous monitoring of various conditions while also extending health benefits [11]. However, the conceptual development and integration of such advanced technologies also bring new challenges for product design that must be addressed to ensure patient acceptance in line with medical requirements [12].

A systematic literature review by Orlando et al. [13] highlighted that there is a significant need for improvement in lower extremity orthotic devices. A list of five main user needs (function, expression, aesthetics, accessibility, and other) and the corresponding subitems for the use of lower limb orthoses was compiled. It was concluded that improvements in the design, prescription, and implementation of these devices are crucial to improve utilization and achieve greater user satisfaction. Therefore, incorporating user perspectives and identifying their needs in the development process is essential to ensure a high level of user acceptance for smart technologies. For example, Van der Wilk et al. [14] applied a user-centered design approach and conducted a qualitative study with eight participants in focus groups to investigate patients' perspectives on improving the design of an ankle foot orthosis (AFO). The three main themes identified by the patients, *walking and standing ability*, *activities*, and *AFO characteristics*, were highlighted as the most relevant aspects for future developments.

To address these design challenges, the use of image-generative artificial intelligence (AI) applications in the design process of future developments presents a novel approach [15]. As shown by Suessmuth et al. [16], generative AI can support concept creation in footwear design by actively involving users in the design process and tailoring products to their preferences, highlighting valuable potential for the industry. Building on this idea, the emerging trend of rapid image creation with tools such as DALL-E [17] from OpenAI provides new opportunities for generating design concepts based on user-defined prompts. The use of such technologies could support the iterative design process within the structured product development of orthopedic orthoses. However, the impact of AI-generated wearable designs on social acceptance remains an open question. To address this research gap, we formulate the following three research questions (RQs):

- RQ1: How does the use of AI-generated designs impact the perceived social acceptability of foot orthoses compared to conventional orthopedic products or research developments?

- RQ2: How does the category of orthotic products influence the social acceptance of foot orthoses, comparing categories such as short-leg orthoses, high-leg orthoses, foot bandages/textiles, and foot drop orthoses?
- RQ3: How does a custom generative pre-trained transformer (GPT), trained on data for personalized orthoses, impact social acceptance, and what effect do specific keywords in the prompt structure (such as "social acceptance" or "sporty design") have on this acceptance?

While aspects such as biocompatibility and manufacturability are critical in the development of orthopedic devices, our study focuses on the visual perception and social acceptance of AI-generated designs. These factors are particularly relevant for end users, whereas material and production requirements fall within the domain of technical experts, who were not the target group of this study. In addition, current generative AI models are primarily suitable for the conceptual phase of designs, as they are generally unable to take technical manufacturability into account, which is why specialized technical expertise is required [18].

Therefore, we conducted two studies to investigate the influence of AI-generated orthoses designs on social acceptance. The first study aimed to assess the potential of AI integration by evaluating whether AI-generated orthoses achieved higher acceptance ratings compared to conventional products. The second study compared the output of a custom GPT and ChatGPT-4 and investigated the effects of specific keywords in the prompts on acceptability ratings. Based on the findings, we provide recommendations for refining text prompts and discuss the implications of these enhancements for future developments in wearable design.

We aim to contribute to the field of human-centered design using generative AI by expanding the knowledge of socially acceptable smart foot orthosis (SFO) designs and investigating the role of image-generative AI in the product development process. Accordingly, this paper highlights the following three main contributions:

1. We introduce a novel approach that leverages image-generative AI to enhance the customization and aesthetic appeal of foot orthosis designs, increasing their alignment with user requirements and visual attractiveness.
2. Our research provides empirical evidence demonstrating that AI-generated orthosis designs significantly enhance social acceptability among users. This potential could be utilized to create widely accepted design solutions for other products as well.
3. We offer a comprehensive set of future recommendations for integrating AI into the orthotic design, aiming to refine the prompt design process to ensure that the resulting designs meet product requirements and patient design expectations.

2. Related Work

This section examines generative AI techniques, particularly text-to-image and image-to-image generation, and explores their potential applications in product design. Following this, we discuss how design influences the social perception of wearable devices. Finally, we summarize the identified research gaps and outline how our studies intend to contribute to this research field.

2.1. Generative AI: Image-to-Image and Text-to-Image Creation

Generative AI is a subset of AI that uses trained models from deep learning, a key component of machine learning, to generate new content from collected data. It employs trained multi-layer artificial neural networks that can identify patterns in datasets and make decisions for data generation. Therefore, the use of AI image-generation tools enables rapid prototyping from a simple sketch to a 3D prototype, whether virtual or physical [19].

One example is the image generation tool DALL-E from OpenAI, which is a separate tool from a GPT chatbot—ChatGPT—based on an large language model (LLM). DALL-E is a transformer model [20] and enables text-to-image or image-to-image generation via text or image prompts.

Text-to-image generation transforms textual information into pixel-based image data. This method can be used in product development to create initial concepts based on specific product requirements [21]. However, achieving appropriate outputs requires an iterative approach to prompt engineering. Therefore, White et al. [22] proposed a prompt pattern catalog to improve prompt engineering with ChatGPT. They stated that specifying personas and templates allows the model to adopt specific roles and ensures that output quality and consistency are tailored to user needs. Furthermore, they discussed prompt patterns for visualizations that enable ChatGPT to generate specific textual outputs, which can subsequently be used to create visualizations with tools like DALL-E. Additionally, a study by Liu and Chilton [23] highlighted the importance of prompt engineering in text-to-image generation and proposed guidelines to achieve better outcomes.

While text-to-image generation uses text input to create a visual output, image-to-image generation requires visual data from an existing image to create a visual output. The use of image-to-image generation is based on Generative Adversarial Networks (GAN) [24], which enable image generation through the interaction of two neural networks: a generator and a discriminator. This technology can be used to refine initial sketches or create design variations, improving the iterative design process.

2.2. Generative AI in Product Design

Image-generative AI enables new possibilities for designers in product development through creativity-expanding iterations of design sketches [25]. These tools could be used in the styling process for various applications [26–29]. However, the application of image-generative AI tools in the development of orthotic devices has not yet been realized.

Nonetheless, initial research projects demonstrate promising results from using AI throughout the design process, from initial concept to final manufacturing via 3D printing [30,31]. For example, Popescu [30] proposed *orthosis_GPT*, a custom version of ChatGPT trained with predefined configurations to support the preliminary design of a wrist–hand orthosis. Based on an iterative design process driven by prompt engineering, the wrist–hand orthosis was finally 3D-printed. This workflow illustrates the potential of using AI in the creation of orthosis designs and subsequent processing for 3D printing. Furthermore, Bartlett and Camba [31] presented examples of generative AI in product design, showcasing the creation of shoes with DALL-E. The generated image output was used by designer Kedar Benjamin to overlay the topology onto this image with additional software and to create a 3D model for 3D printing. Finally, a significant limitation identified was that although the printed versions appeared similar, manual input from designers was necessary to create a topologically optimized 3D model. Additionally, Chiou et al. [25] conducted a study with six designers to explore the collaboration between designers and generative AI in creating new ideas. It was shown that AI image generation offers new possibilities for artistic expression and inspiration with great potential for future contributions in the design field. However, the study also highlighted future challenges and raised questions about copyright and social justice. In conclusion, these studies established a novel approach for creating AI-based designs that meet user requirements and enable the progression from initial design to final 3D printed prototypes.

2.3. Influence of Design on the Social Perception of Wearable Devices

In the field of Human-Computer Interaction (HCI), the influence of factors such as social perception and comfort on wearable devices is extensively researched [32–35]. Research demonstrates across multiple studies that design is crucial for the acceptance and continued use of wearables [36–38]. For instance, a focus group study by Canhoto and Arp [37] investigated the influence of factors for the sustainable use of various Internet of Things (IoT) health and fitness wearables. The study showed that designing distinctive devices enhances visibility, which influences user acceptance and encourages them to continue wearing the device. In addition, Adapda et al. [39] found that social acceptance can vary greatly depending on the type of smart device as well as the user group. Furthermore, Liao et al. [40] showed that smart insoles placed in shoes received the highest comfort ratings compared to other electronic wearable devices that were directly attached to the body. However, social acceptance can be influenced by various factors, including gender and cultural aspects [33]. Therefore, designers must consider a wide range of user preferences to develop widely accepted product designs.

In HCI research, several standardized questionnaires exist for the measurement of social impact from interactive and ubiquitous wearable technologies (e.g., Technology Acceptance Model (TAM) [41], Unified Theory for Acceptance and Use of Technology (UTAUT) [42], Stereotype Content Model (SCM) [43], or the Wearable Acceptability Range (WEAR) [44]). For instance, Davis et al. [41] proposed the TAM model, which was later updated to TAM 3.0 by Venkatesh and Bala [45], an enhanced version for a more comprehensive user understanding of technology adoption. In order to summarize various competing models with differing acceptance factors, Venkatesh et al. [42] compared eight models for measuring technology acceptance and validated UTAUT as a unified theoretical model. However, the literature indicates that while acceptance models such as TAM or UTAUT are suitable for evaluating technological acceptance itself, they are not "clearly applicable" to technologies that are directly worn on the body [36].

For this reason, the WEAR scale developed by Kelly [44,46] serves as an effective tool for measuring the social acceptability of wearable devices or prototypes. For example, Kelly and Gilbert [36] applied the WEAR scale in their study, demonstrating that medical necessity enhances the social acceptability of wearable devices, as it underscores the wearer's dependence on the device. However, they also highlighted the discrepancy in the literature between the stigmatization of medical devices and the recognized medical necessity of such assistive technologies. To address existing stereotypes and social perceptions, the SCM questionnaire by Fiske et al. [43] could be utilized, as it assesses these factors across two perceived dimensions: competence and warmth [47]. Previous work has applied this model to evaluate mobile devices [48] and has confirmed its effectiveness across different cultures [49]. A study by Sehrt et al. [50] utilized the SCM and WEAR scales to investigate the social acceptability of wearable devices based on their body location. The results showed that placement on the ankle received the lowest SCM ratings, indicating poorer acceptability compared to upper body locations. These findings support the statement that the social perception of wearable devices varies for different body locations [51–53]. Overall, the use of these questionnaires revealed current problems with the social acceptance of various devices and that better design is needed to achieve greater acceptance in the future.

2.4. Research Gaps

AI image generation opens up new possibilities for creativity, assisting designers in progressing from sketches to 3D models through text-to-image and image-to-image generation techniques. Initial studies have demonstrated that AI can be used effectively in product design by enabling the development of concepts such as wrist–hand orthoses and the subsequent 3D printing of prototypes. However, the impact of AI-generated aesthetic designs on the social acceptance of orthopedic foot orthoses remains unknown.

To evaluate stereotypes and social acceptance of wearables, the SCM and WEAR scales are often used as validated questionnaires. Previous research using these questionnaires has shown that wearables attached to the ankle typically receive lower acceptance rates compared to those worn on other body parts. Consequently, it remains unclear how orthopedic footwear is perceived by users and which aesthetic design adjustments are necessary to enhance acceptance for future smart developments.

By addressing these research gaps, we aim to contribute to the field of socially acceptable wearable orthopedic footwear design. Additionally, we aim to explore the potential of AI in future design processes to develop aesthetically acceptable design concepts. To this end, we conducted two studies to investigate the effects of AI-generated designs on the social acceptance of orthopedic footwear.

3. Materials and Methods

3.1. Study 1: The Impact of AI-Generated and Conventional Orthotic Designs Across Device Categories on Social Acceptability

To examine the influence of AI-generated orthotic designs on social acceptability, this study compared different AI-created designs with conventional orthopedic products and research-based developments. Additionally, we analyzed how various orthotic device categories affect acceptance. Social acceptability was quantitatively assessed using the WEAR scale, while stereotypical perceptions were evaluated through the SCM model, measuring warmth and competence ratings. To complement these findings, qualitative feedback provided deeper insights into user needs and preferences.

3.1.1. Study Design

A total of 161 participants were recruited through mailing lists via our institution and within personal networks. We conducted an online survey using a two-factorial within-subject design to investigate how different design types and product categories of orthopedic foot orthoses influence social acceptability. The two independent variables were DESIGN TYPE and ORTHOTIC CATEGORY. The variable DESIGN TYPE includes three individual levels: *Generative AI*, *Orthopedic products*, and *Development concepts*. The ORTHOTIC CATEGORY includes four levels: *Short-Leg Orthoses*, *High-Leg Orthoses*, *Foot Bandages/Textiles*, and *Foot Drop Orthoses*. We hypothesized that the perceived social acceptability of foot orthoses will be positively influenced by *Generative AI* concepts compared to the traditional *Orthopedic products* and *Development* concepts. We also hypothesized that the perception of acceptability would vary strongly within the ORTHOTIC CATEGORY, as different device types may elicit different user reactions depending on their design and visibility.

3.1.2. Stimuli

Four different categories of orthopedic foot orthoses (*Short-Leg Orthoses*, *High-Leg Orthoses*, *Foot Bandages/Textiles*, and *Foot Drop Orthoses*) were chosen to cover the medical needs of different foot conditions. We selected six different images for each ORTHOTIC CATEGORY, resulting in a total of twenty-four stimuli (see Figure 1).

Each ORTHOTIC CATEGORY included two images of *Generative AI* orthoses, two *Orthopedic products*, and two orthoses from *Development concepts*. The selection of two different orthoses for each DESIGN TYPE ensured a diverse representation of design variations within each ORTHOTIC CATEGORY. All images were presented with a white background and consistent positioning of the orthosis to ensure comparability. Adobe Photoshop version 2023 24.0.1 was used for manual adjustments, such as rotating the orthosis and creating a transparent background where necessary. Each image contained only the orthosis and some visible parts of the lower leg to maintain focus on the orthotic device itself.

Two approaches were used for *Generative AI* designs: text-to-image and image-to-image generation. Text-to-image generation was applied using DALL-E 3, which is based on a transformer architecture, from ChatGPT-4 [54] (OpenAI). Text prompts were created to design smart orthoses based on the type of orthosis and its medical functions. These typically feature a gray device color on a white background. Additionally, the prompts were extended to include small black boxes on the sides of the orthoses, which serve as placeholders for electronic components to enable smart functions. For image-to-image generation, images of medical orthoses were used as prompts to create realistic designs. The following prompt was used and iteratively refined during re-prompting: "*Generate an image of a foot orthosis based on the shape of the reference image. The orthosis should provide full protection for the lower leg and foot, consisting of two rigid housing parts that enclose the leg and are secured with Velcro straps. The orthosis should be gray and displayed against a white background. Additionally, a small box should be integrated into the orthosis to house sensors for smart functionalities*". The images of *Orthopedic products* are based on standardized medical devices that are available in online shops or medical supply stores: *VACOpedes* [55], *AIRCAST® AIRSELECT™ Short Walker* [56], *VACOcast Diabetic Boot* [57], *AIRCAST® AIRSELECT™ Achilles Walker* [58], *FastProtect Malleo* [59], *The BetterGuard* [60], *WalkOn Reaction Lateral* [61], and *ofa Push ortho Fußheberorthese AFO* [62]. These products are currently used by patients with foot conditions and represent state-of-the-art solutions. The *Development concepts* stimuli include various devices proposed in research projects [63–66], 3D-printed prototypes published on the web [61,67,68], and a self-created 3D CAD-design (C4-DEV2).

Figure 1. Overview of the 24 stimuli used in the online survey, categorized by the two independent variables DESIGN TYPE and ORTHOTIC CATEGORY. The stimuli are arranged horizontally by the four orthotic categories, *Short-Leg Orthoses*, *High-Leg Orthoses*, *Foot Bandages/Textiles*, and *Foot Drop Orthoses*, and vertically by the three design types, *Generative AI*, *Orthopedic products*, and *Development concepts*.

3.1.3. Procedure

The study was conducted in accordance with the ethical standards for user studies as required by our institution and received ethical approval from the German Society for Nursing Science (No. 23-027). Participants were informed about the study's purpose, ethical procedures, data privacy, data anonymity, and the intended use of their data before participation. Informed consent was obtained via the online platform before the start of the survey. Participants received a web link to our landing page, where they could access the online survey, which was created using LimeSurvey. After signing the informed consent form, participants were asked about demographic data, information about past conditions, and prior knowledge of orthopedic footwear or smart orthoses. Following this, participants received a brief introductory text for each orthotic category to prevent major misinterpretations and to provide context for the device and medical application. Participants were then presented with the 24 stimuli in a randomized order. For each stimulus, the corresponding picture was shown, and participants rated two statements about social acceptance and aesthetic design expectations on a 7-point Likert scale. The questions were "*To which extent do you agree with the statement that the device is socially accepted?*" and "*To what extent does the design meet your aesthetic expectations?*", with responses ranging from *totally unacceptable* (1) to *perfectly acceptable* (7). In addition to the quantitative questions, qualitative feedback was obtained via a free text entry question for each stimulus: "*What positive/negative aspects of the orthosis do you notice?*". Following this, the standardized SCM and WEAR questionnaires were completed for each stimulus. After answering all 24 conditions, participants responded to concluding questions regarding any changes in their perspective on orthoses, noting positive aspects, negative aspects, prior knowledge of AI image generation, and any additional feedback. The average completion time for this survey was 75 min.

3.1.4. Measures and Data Analysis

In this study, we used a mixed methods approach based on quantitative and qualitative feedback. Quantitative data were measured using the SCM and WEAR scales, complemented by two seven-point Likert items assessing social acceptance and aesthetic design expectations. These data were analyzed through descriptive and inferential statistics. Additionally, qualitative feedback derived from participants' free-text responses was assessed using thematic analysis. The methods and data analysis procedures are described in detail in the following paragraphs.

Quantitative: Stereotype Content Model (SCM)

The SCM questionnaire by Fiske et al. [43] was used to assess stereotypes and social perceptions of various foot orthoses based on two perceived dimensions: warmth and competence. This standardized questionnaire consists of nine items rated on a 5-point scale ranging from *not at all* (1) to *extremely* (5). The competence dimension was evaluated using five items (*competent*, *confident*, *independent*, *competitive*, and *intelligent*), each assessed by the question "*As viewed by society, how . . . are members of this group?*". Additionally, the same question was used to assess the warmth dimension, consisting of four items: *tolerant*, *warm*, *good-natured*, and *sincere*. For the data analysis, a two-factorial repeated measures (RM) ANOVA was used.

Quantitative: Wearable Acceptability Range (WEAR)

To measure the social acceptability of wearable technology, the WEAR scale by Kelly [44] was used. The WEAR scale (version 3) consists of 14 items, each rated on a 6-point scale ranging from *strongly disagree* (1) to *strongly agree* (6). Five of these 14 items are reverse-scored. The total score was divided by 14 to obtain an average score, which can be graded from *extremely low social acceptance* (1) to *extremely high acceptance* (6). An RM ANOVA was used for data analysis, similar to the procedure used for the SCM scale.

Quantitative: 7-Point Likert Scales

As described previously in Section 3.1.3, two questions were used to rate statements about social acceptance and aesthetic design expectations on 7-point Likert scales. For the evaluation, a non-parametric Aligned Rank Transform (ART) RM-ANOVA was applied.

Qualitative Feedback

Qualitative results from the free text entries were assessed using inductive thematic analysis [69]. Anonymized data were transcribed, coded, and analyzed paragraph-wise to identify categories and main themes. Two researchers were involved in the data analysis process: one researcher transcribed and coded the data set independently, while the second researcher reviewed the results for consistency.

3.2. Study 2: The Impact of GPT Customization and Prompt Keywords on the Social Acceptability of AI-Generated Orthotic Designs

Based on our first study, which showed that AI-generated designs can increase the social acceptance of orthopedic foot wearables, this study further explored the role of prompt customization in shaping user perception. In particular, the first study revealed that high-leg orthoses received the lowest acceptance ratings. Therefore, we examined which factors in AI image generation could positively influence the acceptance of this device category. While the initial study confirmed the potential of generative AI for orthotic design, it also raised questions regarding the influence of specific keywords in the prompts used and the application of customized GPTs. Furthermore, it remained unclear how the inclusion of keywords related to user preferences would affect the perceived acceptability of designs. To address this, our second study investigated the impact of various keywords and the use of a personalized GPT tailored to specific patient and product requirements. Previous research has demonstrated the influence of prompt structures and provided recommendations for different user roles [22]. Building on this knowledge, we take one step further and contribute to user-centered prompting strategies by incorporating keywords that enhance the alignment between generated designs and user needs.

3.2.1. Study Design

In our second study, we conducted an online survey to investigate the effects of different prompt keywords and GPT personalization on the social acceptability of AI-generated high-leg orthosis designs. In total, 133 participants were recruited. The recruitment of participants was conducted similarly to the first study, using mailing lists of our institution and personal networks. A two-factorial within-subject design was carried out with the two independent variables: GPT and KEYWORD. The variable GPT has two levels: *ChatGPT* and *OrthoticFootGPT*. The variable KEYWORD has four levels: *No Keyword*, *Usability*, *Social Acceptability*, and *Sporty Design*. The hypothesis of our second study is that perceived acceptability will significantly vary depending on the specific keywords used in the prompts and that the inclusion of new keywords will positively influence this acceptability. Furthermore, we hypothesize that a customized GPT will lead to more acceptable perceived outcomes compared to the standardized model.

3.2.2. Stimuli

The model GPT-4 from ChatGPT was used to create the stimuli of high-leg orthoses, which allows the creation of a custom GPT that can be configured according to the desired response and outputs. DALL-E 3 image generation was activated in the configuration interface of the customized GPT, and specific knowledge and instructions were provided as input. However, OpenAI does not disclose specific details about its training configuration or model architecture. The customized GPT named *OrthoticFootGPT* was described as a "generative design assistant to support the creation of socially acceptable foot orthoses". To generate a user-specific output, the knowledge base for the *OrthoticFootGPT* was customized with a comprehensive set of data tailored to high-leg orthoses. This dataset included

- Product specifications: Detailed information on the aesthetic and functional aspects of high-leg orthoses, such as material types, sizes and weight from data sheets and guidelines.
- Usage instructions: Documents with instructions and user manuals for the use and maintenance of these devices.
- Product images: Sample images of high-leg orthoses from various suppliers as a visual reference for a realistic design.
- Research articles: Related articles providing insights into product design rules, the concepts of accessibility/usability, and orthotic development were searched on Google Scholar.
- Keyword definitions: Documents based on international standards and dictionary definitions provided explanations about the three keywords (usability, social acceptability, and sport design) to guide the AI in generating contextually relevant content.

For both *ChatGPT* and *OrthoticFootGPT*, the following prompt was used as the foundation for image generation: "*Create a realistic image of a white-grey foot orthosis with a high shaft. The orthosis should have stabilizing properties so that it can be worn for longer periods during rehabilitation. The background should be white and only show the leg with the orthosis*". In addition, a sentence was added to the basic query for each keyword to tailor the images to the specific definition. For example, for social acceptability: "*The orthosis should ensure a high level of social user acceptance by offering an attractive aesthetic design and high wearing comfort*".

To ensure realistic product images and avoid misinterpretation or misleading details, the DALL-E editor interface was applied manually to ensure consistency across all images (e.g., by removing the second leg or unrealistic features). Similar to the first study, each image was manually adjusted using Adobe Photoshop to create transparent backgrounds and ensure consistent foot rotation angles. All generated images are shown in Figure 2 and the workflow for generating the visual stimuli is illustrated in Figure 3.

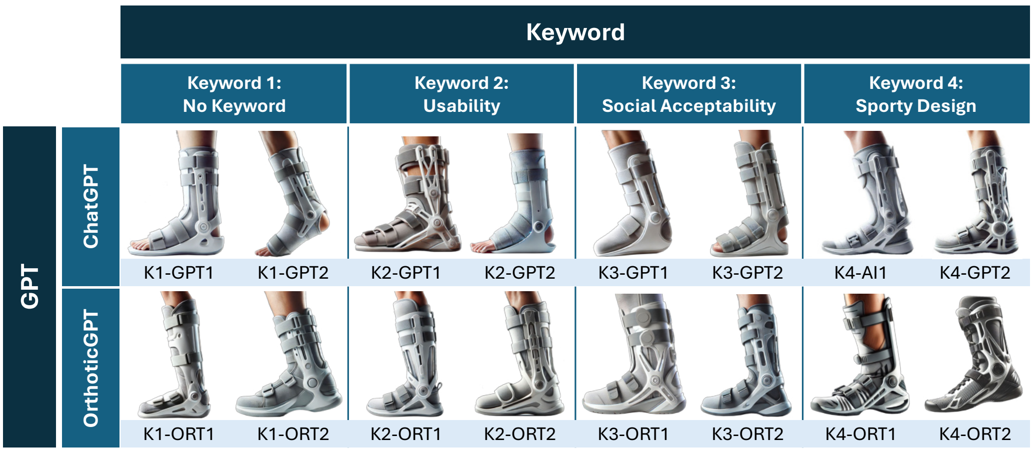


Figure 2. Overview of the 16 stimuli used in the online survey, categorized by the two independent variables, GPT and KEYWORD. The stimuli are arranged horizontally by the four keywords, *No Keyword*, *Usability*, *Social Acceptability*, and *Sporty Design*, and vertically by the two GPTs: *ChatGPT* and *OrthoticFootGPT*.

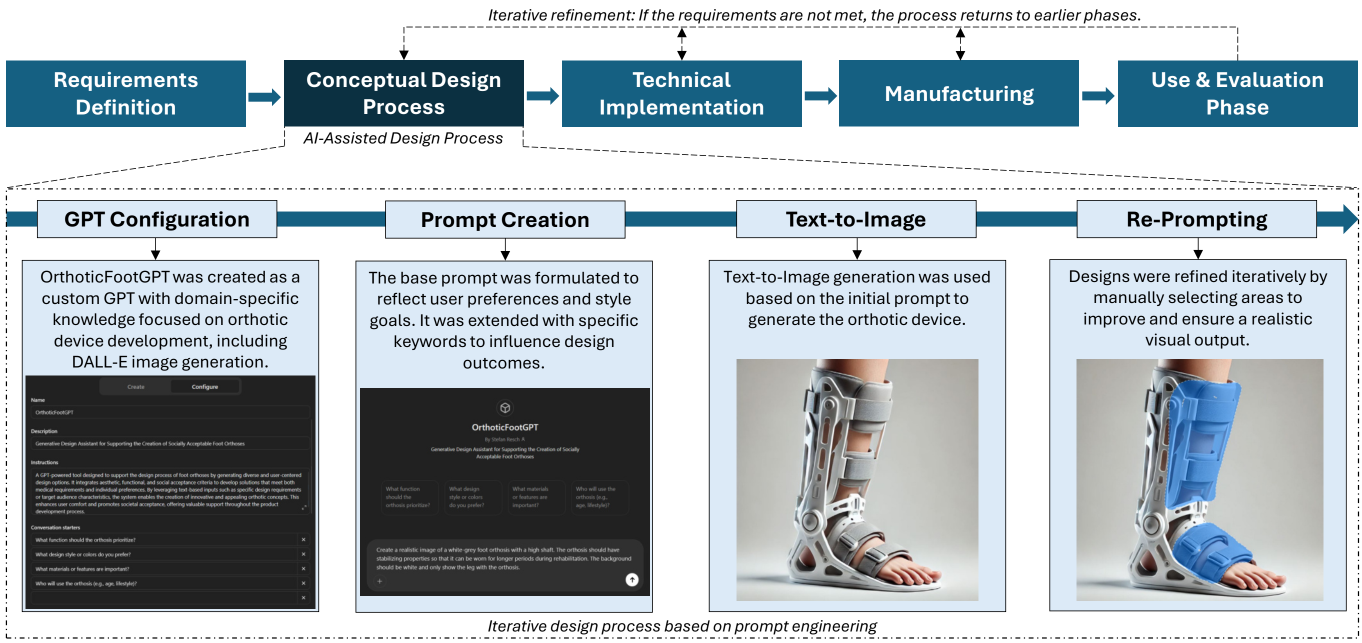


Figure 3. Schematic overview of the design generation workflow, including GPT customization, prompt creation, and text-to-image generation.

3.2.3. Procedure

The study procedure was identical to that of the previous study, including ethical approval by the German Society for Nursing Science (No. 23-027) and compliance with institutional ethical standards. As in the previous study, participants were informed about data privacy, anonymity, and the intended use of their data before providing informed consent via the online platform. Participants signed an informed consent form and provided demographic information along with details about their previous experience with image-generative AI. The 16 conditions were then presented in a randomized order. At the end of the survey, participants were asked to answer some open-ended questions about positive and negative aspects of the designs, as well as additional feedback. The average time to complete this survey was approximately 64 min.

3.2.4. Measures and Data Analysis

Similar to the first study, the quantitative data were collected using the standardized SCM [43] questionnaire, the WEAR [44] scale, and two 7-point Likert items to assess the social acceptance and aesthetic design expectations. Additionally, participants provided qualitative feedback through free-text responses about positive and negative aspects of the designs. The quantitative data analysis was conducted using descriptive and inferential statistics. Qualitative feedback was analyzed using inductive thematic analysis [69].

4. Results

4.1. Study 1

This section presents the characteristics of the study group (Section 4.1.1), followed by the evaluation of quantitative results (Section 4.1.2). This includes the Correlation of SCM and WEAR Ratings, assessments of Stereotype Content Model (SCM) and Wearable Acceptability Range (WEAR) data, as well as the analysis of Seven-Point Likert Scale ratings on device acceptance and aesthetic design expectations. The statistical analysis was performed in R using the package rstatix [70]. Qualitative results obtained through thematic analysis are documented in Section 4.1.3.

4.1.1. Participants

A total of 80 out of 161 participants (31 female, 48 male, and 1 divers) completed the online survey and were included in the data analysis. Participants' ages ranged from 19 to 77 years (*M* = 30.62, *SD* = 13.69). The participants had diverse educational backgrounds and came from seven different nationalities. Occupations of participants included (N = 1) each in natural sciences, healthcare, and crafts; (N = 2) each in law and administration; (N = 3) each in management and social work; (N = 5) in services; (N = 20) in other professions; and (N = 42) in engineering. Twenty-five participants reported having foot conditions, with sixteen of these in the past and nine currently. In total, 14 had chronic issues and eleven had acute conditions. Overall, 21 participants reported wearing orthopedic aids, including insoles (N = 11), bandages (N = 5), high-leg orthoses (N = 4), and short-leg orthoses (N = 1). In response to the question of whether the participants had previously heard of the term "smart orthoses", 51 responded "no", and 29 responded "yes". Twenty-nine participants were aware of generative AI image generation, twenty-eight had actively used it, and twenty-three had no prior knowledge of the technology.

To investigate whether participant demographics influenced the results, we conducted an ART ANOVA, which showed no statistically significant main or interaction effects of gender, prior AI knowledge, or past orthopedic conditions on SCM, WEAR, and Likert scores (all $p > 0.05$). These results show that demographic factors did not systematically influence the ratings of the participants.

4.1.2. Quantitative Results

Correlation of SCM and WEAR Ratings

A Pearson correlation analysis was conducted to assess the relationship between SCM (means of warm and competence data) and WEAR ratings. A statistically significant moderate positive correlation was found between the Euclidean length of the warmth–competence vector and the WEAR scores, $r(958) = 0.453$, $p < 0.001$, with a 95% confidence interval of (0.402, 0.502). This result confirms the findings from Schwind and Henze [71] that social acceptance ratings correlate with combined competence and warmth data.

Stereotype Content Model (SCM)

The Shapiro–Wilk normality test indicated no normal distribution (only selected SCM competence data showed $p \succeq 0.05$, while all other stimuli for SCM competence and warmth had $p < 0.05$). Therefore, a non-parametric two-factorial ART RM-ANOVA was conducted to determine the effects of DESIGN TYPE and ORTHOTIC CATEGORY on SCM competence and warmth.

For the SCM competence data, statistically significant main effects were found for the ORTHOTIC CATEGORY, $F(3, 869) = 56.752$, $p < 0.001$, $\eta_p^2 = 0.464$ (large effect size), and DESIGN TYPE, $F(2, 869) = 9.931$, $p < 0.001$, $\eta_p^2 = 0.092$ (medium effect size). However, no interaction effect was found between ORTHOTIC CATEGORY $\times$ DESIGN TYPE, $F(6, 869) = 1.038$, $p = 0.399$, $\eta_p^2 = 0.031$ (small effect size).

For the SCM warmth data, statistically significant main effects were identified for the ORTHOTIC CATEGORY, $F(3, 869) = 3.201$, $p = 0.023$, $\eta_p^2 = 0.278$, and DESIGN TYPE, $F(2, 869) = 4.493$, $p = 0.012$, $\eta_p^2 = 0.265$, both with large effect sizes. However, the interaction effect of the ORTHOTIC CATEGORY $\times$ DESIGN TYPE was not significant, with $F(6, 869) = 1.055$, $p = 0.388$, and $\eta_p^2 = 0.202$ (large effect size).

Post hoc comparisons using Wilcoxon signed-rank tests with Bonferroni correction revealed significant effects in SCM competence and warmth data, as documented in Table 1. Figure 4 displays the mean values of perceived warmth and competence for each of the three DESIGN TYPES within each of the four ORTHOTIC CATEGORIES. Additionally, the figure also presents a comparison across the four ORTHOTIC CATEGORIES within the SCM model to highlight differences in the stereotypical perceptions of these devices.

Table 1. Pairwise comparisons of SCM competence and warmth ratings within orthotic categories and design types. Significance codes: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$), **** ($p < 0.0001$), ns = not significant.

			SCM Competence				SCM Warmth			
Pairwise Comparisons	**n1**	**n2**	***Stat.***	p	$p_{adj.}$	***Sig.***	***Stat.***	p	$p_{adj.}$	***Sig.***
Short-Leg Orthoses—High-Leg Orthoses	240	240	10094	0.057	0.343	ns	6820	0.612	1	ns
Short-Leg Orthoses—Bandages/Textiles	240	240	2997	<0.001	<0.001	****	5401	0.005	0.03	*
Short-Leg Orthoses—Foot Drop Orthoses	240	240	5367	<0.001	<0.001	****	6286	0.993	1	ns
High-Leg Orthoses—Bandages/Textiles	240	240	2846	<0.001	<0.001	****	5724	0.002	0.013	*
High-Leg Orthoses—Foot Drop Orthoses	240	240	4640	<0.001	<0.001	****	5904	0.360	1	ns
Bandages/Textiles—Foot Drop Orthoses	240	240	11938	<0.001	<0.001	****	7802	0.044	0.265	ns
Generative AI—Development	320	320	22421	<0.001	<0.001	****	15445	<0.001	<0.001	***
Generative AI—Orthopedic products	320	320	21927	<0.001	<0.001	****	13268	0.191	0.573	ns
Development—Orthopedic products	320	320	16596	0.928	1	ns	9968	0.003	0.008	**

Figure 4. SCM ratings of perceived competence (x-axis) and warmth (y-axis) for each of the four ORTHOTIC CATEGORIES. In the upper row, the four quadrants are assigned as follows, from left to right: (**a**) *Short-Leg Orthoses*, (**b**) *High-Leg Orthoses*, (**c**) *Foot Bandages/Textiles*, and (**d**) *Foot Drop Orthoses*. The mean values for the three DESIGN TYPES are shown in separate colors: red (*Generative AI*), blue (*Orthopedic products*), and yellow (*Development concepts*). The lower quadrant (**e**) contains the SCM mean values for each ORTHOTIC CATEGORY: *Short-Leg Orthoses* (purple), *High-Leg Orthoses* (orange), *Foot Bandages/Textiles* (brown), and *Foot Drop Orthoses* (green). The rectangles represent the 95% confidence interval.

Wearable Acceptability Range (WEAR)

The Shapiro–Wilk test for normality indicated that the data are not normally distributed ($p = 0.003$). An ART RM-ANOVA showed statistically significant main effects for the ORTHOTIC CATEGORY, $F(3, 869) = 39.949$, $p < 0.001$, $\eta_p^2 = 0.423$ (large effect size), and DESIGN TYPE, $F(2, 869) = 17.677$, $p < 0.001$, $\eta_p^2 = 0.178$ (large effect size), but no interaction effect for ORTHOTIC CATEGORY $\times$ DESIGN TYPE, $F(6, 869) = 1.372$, $p = 0.223$, $\eta_p^2 = 0.048$ (small effect size).

Bonferroni corrected pairwise comparisons were conducted using Wilcoxon signed-rank tests and revealed significant effects between *Generative AI* $\times$ *Development concepts* ($p_{\text{adj.}} < 0.001$), *Generative AI* $\times$ *Orthopedic products* ($p_{\text{adj.}} < 0.001$), and *Development concepts* $\times$ *Orthopedic products* ($p_{\text{adj.}} = 0.021$). Furthermore, pairwise comparisons between ORTHOTIC CATEGORIES revealed five significant effects, which are summarized in Figure 5.

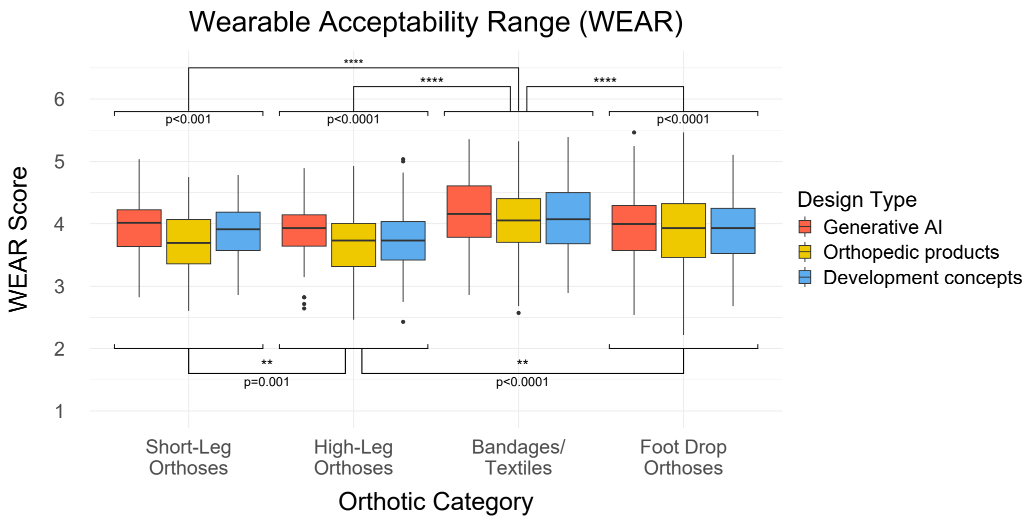


Figure 5. This graph shows the results of the WEAR ratings. The four categories are listed on the x-axis: *short-leg orthoses*, *high-leg orthoses*, *foot supports/textiles* and *foot drop orthoses*, and the y-axis represents the corresponding scores on the WEAR scale. The boxplots are color-coded to differentiate the design types: *Generative AI* in red, *Orthopedic products* in yellow, and *Development concepts* in blue. Asterisks indicate significance levels: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$), **** ($p < 0.0001$).

Seven-Point Likert Scale

The evaluations of the two statements regarding device acceptance and aesthetic design expectations were not normally distributed ($p < 0.05$).

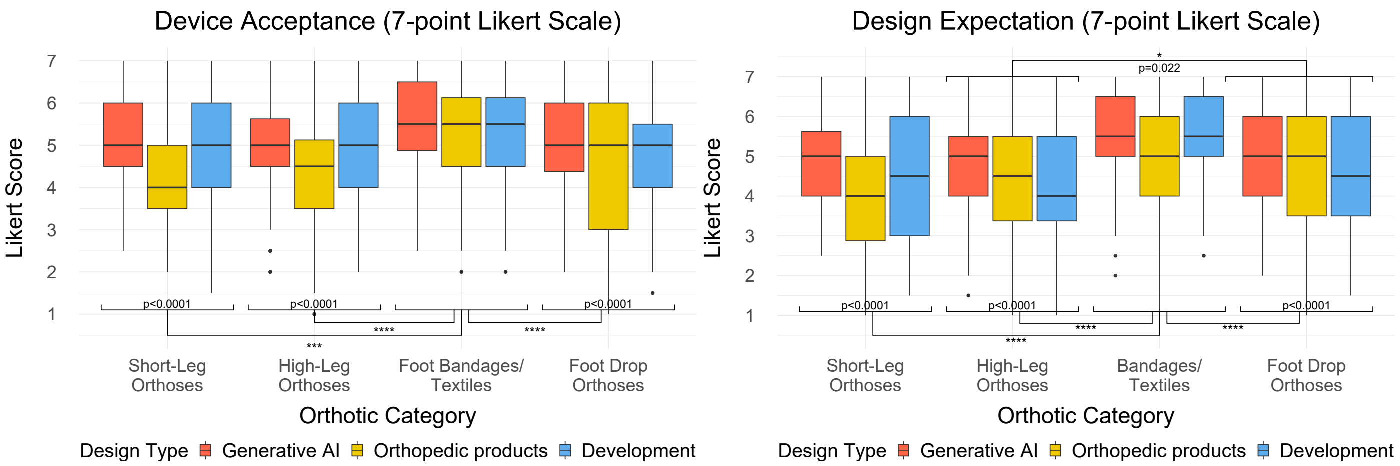
A non-parametric two-factorial ART-ANOVA on rated device acceptance statements revealed a significant main effect for the ORTHOTIC CATEGORY, $F(3, 869) = 24.717$, $p < 0.001$, $\eta_p^2 = 0.336$, and DESIGN TYPE, $F(2, 869) = 24.077$, $p < 0.001$, $\eta_p^2 = 0.247$, as well as a significant interaction effect for ORTHOTIC CATEGORY $\times$ DESIGN TYPE, $F(6, 869) = 4.068$, $p < 0.001$, $\eta_p^2 = 0.142$ (each with a large effect size).

Furthermore, the two-factorial ART-ANOVA on rated aesthetic design expectations showed a significant main effect for the ORTHOTIC CATEGORY, $F(3, 869) = 43.730$, $p < 0.001$, $\eta_p^2 = 0.407$, and DESIGN TYPE, $F(2, 869) = 21.613$, $p < 0.001$, $\eta_p^2 = 0.185$ (both with large effect size), as well as a significant interaction effect for ORTHOTIC CATEGORY $\times$ DESIGN TYPE, $F(6, 869) = 32.743$, $p = 0.012$, and $\eta_p^2 = 0.079$ (medium effect size).

Pairwise comparisons using Bonferroni-corrected Wilcoxon signed-rank tests revealed statistically significant differences within ORTHOTIC CATEGORIES and DESIGN TYPES, as documented in Table 2. The results of the Likert scale evaluations are displayed in Figure 6.

Table 2. Post hoc pairwise comparisons within orthotic categories and design types for Likert scores of device acceptance and design expectations. Significance codes: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$), **** ($p < 0.0001$), ns = not significant.

			Device Acceptance				Design Expectation			
Pairwise Comparisons	**n1**	**n2**	***Stat.***	***p***	***$p_{adj.}$***	***Sig.***	***Stat.***	***p***	***$p_{adj.}$***	***Sig.***
Short-Leg Orth.—High-Leg Orth.	240	240	7901	0.148	0.888	ns	7804	0.771	1	ns
Short-Leg Orth.—Bandages/Textiles	240	240	4630	<0.001	<0.001	****	2854	<0.001	<0.001	****
Short-Leg Orth.—Foot Drop Orth.	240	240	7456	0.568	1	ns	7370	0.024	0.143	ns
High-Leg Orth.—Bandages/Textiles	240	240	3583	<0.001	<0.001	****	3426	<0.001	<0.001	****
High-Leg Orth.—Foot Drop Orth.	240	240	7032	0.712	1	ns	5688	0.004	0.022	*
Bandages/Textiles—Foot Drop Orth.	240	240	11454	<0.001	<0.001	****	13938	<0.001	<0.001	****
Generative AI—Development	320	320	21972	<0.001	<0.001	****	24376	<0.001	<0.001	****
Generative AI—Orthopedic products	320	320	13612	0.237	0.711	ns	19850	<0.001	<0.001	****
Development—Orthopedic products	320	320	7405	< 0.001	< 0.001	****	12490	0.004	0.011	*



(**a**) (**b**)

Figure 6. This figure contains two boxplot diagrams, including the rated device acceptance (**a**) and aesthetic design expectation (**b**) for the three DESIGN TYPES in each ORTHOTIC CATEGORY. The x-axis lists the four categories: *Short-Leg Orthoses*, *High-Leg Orthoses*, *Foot Bandages/Textiles*, and *Foot Drop Orthoses*. The y-axis represents the corresponding scores from the Likert rating. The boxplots are color-coded to distinguish the design types: *Generative AI* in red, *Orthopedic products* in yellow, and *Development concepts* in blue. Asterisks indicate significance levels: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$), **** ($p < 0.0001$).

4.1.3. Qualitative Feedback

The axial coding procedure within inductive thematic analysis revealed three main themes: *Aesthetic Design*, *Technical Properties*, and *Usability and Comfort*. Each theme includes both positive and negative perspectives and reflects participant feedback in four ORTHOTIC CATEGORIES and three DESIGN TYPES.

Overall, perceptions of all four ORTHOTIC CATEGORIES vary strongly concerning design, technical aspects, usability, and comfort. In general, the ORTHOTIC CATEGORIES of *Short-Leg Orthoses* and *High-Leg Orthoses* are often associated with serious injuries. *Short-Leg Orthoses* focus on “stabilization” (P83, P56) and “facilitates mobility for users with lower leg injuries” (P110) but offer limited practicality in daily life. *High-Leg Orthoses* offer “robust protection for serious injuries” (P27, P101, P110), but are “bulky” (P5, P24, P26, P81, P163), “heavy” (P7, P22, P32, P119) and “restrictive” (P39, P83, P113), so their design “looks like a ski boot" (P144). Participants mentioned that these designs "could not be worn with regular shoes" (P75), "leading to clothing restrictions" (P101). In contrast, *Foot Bandages/Textiles* are perceived as "sporty" (P26, P39), "modern" (P34, P156), and "simple" (P56, P84) because of their "unobtrusive design" (P8) and "can be worn with shoes" (P13, P46), but they are also viewed as less suitable for serious injuries. Lastly, *Foot Drop Orthoses* have a "functional design" (P161) which is "not conspicuous" (P56, P84, P163) and directly addresses medical needs but is also seen as "too mechanical" (P8, P156), "not stable" (P21, P53, P73), and "old-fashioned" (P13, P22).

Aesthetic Design: The aesthetic perception varies considerably across the three DESIGN TYPES. The designs of *Generative AI* were perceived as "simple and unobtrusive" (P26, P56, P66), with "subtle colors" (P43, P81), a "clean" (P156) and "modern design [that] could be accepted as a replacement for shoes" (P75), which in some cases appear "futuristic" (P66). However, some participants mentioned that the designs look "too technical" (P63) because "the box looks too extreme" (P156). *Orthopedic products* were associated with a "minimalistic" (P63, P142) and "classic design" (P8, P26), characterized by "neutral colors" (P81) that "looks like a conventional product" (P162). However, these designs were criticized for being visually "unesthetic" (P16, P56, P63)—described as "old-fashioned" (P13), "unfashionable" (P161), "crude" (P107), "strange" (P13, P39), and "too big" (P32, P66)—and therefore perceived as "less socially acceptable" (P16) because of their "robotic appearance" (P32, P119, P144). The medical appearance of some orthoses could affect the wearer's self-confidence because the "impairment attracts attention" (P43). The *Development concepts* were generally recognized as "innovative" (P75), "stylish" (P144), "futuristic" (P13, P34, P60, P64, P156), and "it is nice that the color is different from other design" (P130), which makes them appear "like a fashion item" (P13) and "do not look like a medical product" (P22). In contrast, the bright color was also perceived negatively, along with the "cheap-looking quality" (P63, P66, P72) and "ugly holes" (P84). Additionally, some designs were criticized for their overly "mechanical appearance" (P8, P156) and the "excessive size of the box" (P63), which could lead to "aesthetic acceptance issues" (P24).

Technical Properties: The technical properties of *Generative AI* designs were recognized for "technology integration" (P152) as an "electronic device" (P119). However, some of these technical functions were also negatively considered for their "complexity and the need for regular maintenance or charging" (P152). *Orthopedic products* are recognized for "longevity and low-maintenance" (P109), which "provides strong support and stability for the foot and ankle" (P120) because they "immobilize the ankle and foot" (P27), making them reliable for medical use and effective in the rehabilitation of severe injuries. Although the *Development concepts* featured "lighter" (P22, P25) and "simpler designs" (P5, P161) for "improved mobility" (P110), they were "not stable" (P53, P160) enough to support more serious medical needs. Participants expressed concerns about "durability under heavy loads" (P110), "lack of strength/stiffness" (P101), and the technical efficiency of these designs.

Usability and Comfort: The usability and comfort of the three DESIGN TYPES resulted in contrary user reactions, revealing strengths and weaknesses in product design. *Generative AI* designs faced criticism for certain elements, such as "visible toes" (P39, P72), and raised concerns about "the longevity of materials ... [because the] material could wear out more quickly" (P152). Despite the technical advantages of the *Orthopedic products*, they "can be bulky" (P141) "which can be uncomfortable during everyday activities" (P152), and can reduce comfort over time because it "restricts freedom of movement" (P101). Although some models are described as "comfortable" (P142, P162), there is criticism due to insufficient air circulation because the "feet could sweat" (P84) which can lead to "moisture and skin irritation" (P113). In contrast, it was noted that *Development concepts* are "easy to wear" (P164) and "suitable for everyday use" (P22), offering a practical and "less intrusive" (P8, P161) alternative. However, participants also mentioned that it "can be uncomfortable during long-term use" (P110).

4.2. Study 2

The study group characteristics are described in the Section 4.2.1. The quantitative analysis is documented in Section 4.2.2, including the Correlation of SCM and WEAR Ratings, Stereotype Content Model (SCM) ratings, Wearable Acceptability Range (WEAR) data, and the analysis of the two Seven-Point Likert Scale ratings. The thematic analysis of the qualitative responses is presented in Section 4.2.3.

4.2.1. Participants

In total, 54 out of 133 participants (27 female, 26 male, and 1 diverse) completed the survey and were included in the data analysis. The age of the participants ranged from 18 to 49 years (*M* = 23.30, *SD* = 5.02). The participants had different educational backgrounds and came from nine countries. The participants' occupations varied from (N = 1) in healthcare; (N = 2) each in other services and natural sciences; (N = 3) each in administration, social services, and management; (N = 5) in law; (N = 16) inother professions; and (N = 19) in engineering. Twenty-six participants reported having foot conditions; twenty had conditions in the past and six have current conditions. Of these, 17 were acute diseases and nine were chronic conditions. Eight of them used insoles, seven used bandages, three used a low-leg orthosis, two used a plaster cast, two used a low-leg orthosis, and one used a high-leg orthosis. A total of 20 have already used image-generative AI, 19 participants have no experience with such tools, and 15 are only familiar with the tools from hearsay.

Similarly to our first study, we conducted an ART ANOVA to examine the influence of demographic factors. The results showed no statistically significant main or interaction effects on SCM, WEAR, and Likert scores (all $p > 0.05$), indicating that demographic variables did not influence participant ratings.

4.2.2. Quantitative Results

Correlation of SCM and WEAR Ratings

The Pearson correlation analysis showed a statistically significant correlation between the Euclidean length of the SCM warmth-competence vector (means of warm and competence data) and the WEAR scores: $r(430) = 0.800$, $p < 0.001$, with a 95% confidence interval of (0.763, 0.831).

Stereotype Content Model (SCM)

The Shapiro–Wilk normality test indicated a significant deviation from normality for SCM competence and warmth (both $p < 0.0001$). A non-parametric two-factorial ART RM-ANOVA for the SCM competence data showed a statistically significant main effect for GPT, $F(1,371) = 124.871$, $p < 0.0001$, $\eta_p^2 = 0.448$, and for KEYWORD, $F(3,371) = 8.499$, $p < 0.0001$, $\eta_p^2 = 0.142$ (both with large effect size). However, no significant interaction effect was found for GPT × KEYWORD, $F(3,371) = 1.198$, $p = 0.310$, $\eta_p^2 = 0.023$ (small effect size). For the SCM warmth data, a significant main effect was identified for GPT, $F(1,371) = 62.461$, $p < 0.0001$, and $\eta_p^2 = 0.471$ (large effect size). However, no significant effect was found for KEYWORD, $F(3,371) = 1.282$, $p = 0.280$, $\eta_p^2 = 0.052$, nor for the interaction effect of GPT × KEYWORD, $F(3,371) = 1.267$, $p = 0.285$, $\eta_p^2 = 0.051$ (both with a small effect size). Post hoc comparisons using Wilcoxon signed-rank tests with Bonferroni correction revealed significant effects in SCM competence and warmth data, as documented in Table 3. The mean values of perceived warmth and competence are depicted in Figure 7.

Table 3. Pairwise comparisons of SCM competence and warmth ratings within keywords and GPTs. Significance codes: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$), **** ($p < 0.0001$), ns = not significant.

			SCM Competence				SCM Warmth			
Pairwise Comparisons	**n1**	**n2**	***Stat.***	p	$p_{adj.}$	***Sig.***	***Stat.***	p	$p_{adj.}$	***Sig.***
No Keyword—Usability	108	108	1482	0.312	1	ns	868	0.337	1	ns
No Keyword—Social Acceptability	108	108	1943	0.482	1	ns	1331	0.605	1	ns
No Keyword—Sporty Design	108	108	748	<0.001	<0.001	****	850	0.146	0.876	ns
Usability—Social Acceptability	108	108	2135	0.075	0.450	ns	1508	0.186	1	ns
Usability—Sporty Design	108	108	725	<0.001	<0.001	****	914	0.221	1	ns
Social Acceptability—Sporty Design	108	108	764	<0.001	<0.001	****	854	0.107	0.642	ns
No Keyword: ChatGPT—Orth.FootGPT	54	54	942	0.002	0.002	**	1135	0.046	0.046	*
Usability: ChatGPT—OrthoticFootGPT	54	54	828	<0.001	<0.001	***	994	0.004	0.004	**
Social Accept.: ChatGPT—Orth.GPT	54	54	777	<0.001	<0.001	****	875	<0.001	<0.001	***
Sporty Design: ChatGPT—Orth.FootGPT	54	54	830	<0.001	<0.001	***	1069	0.016	0.016	*

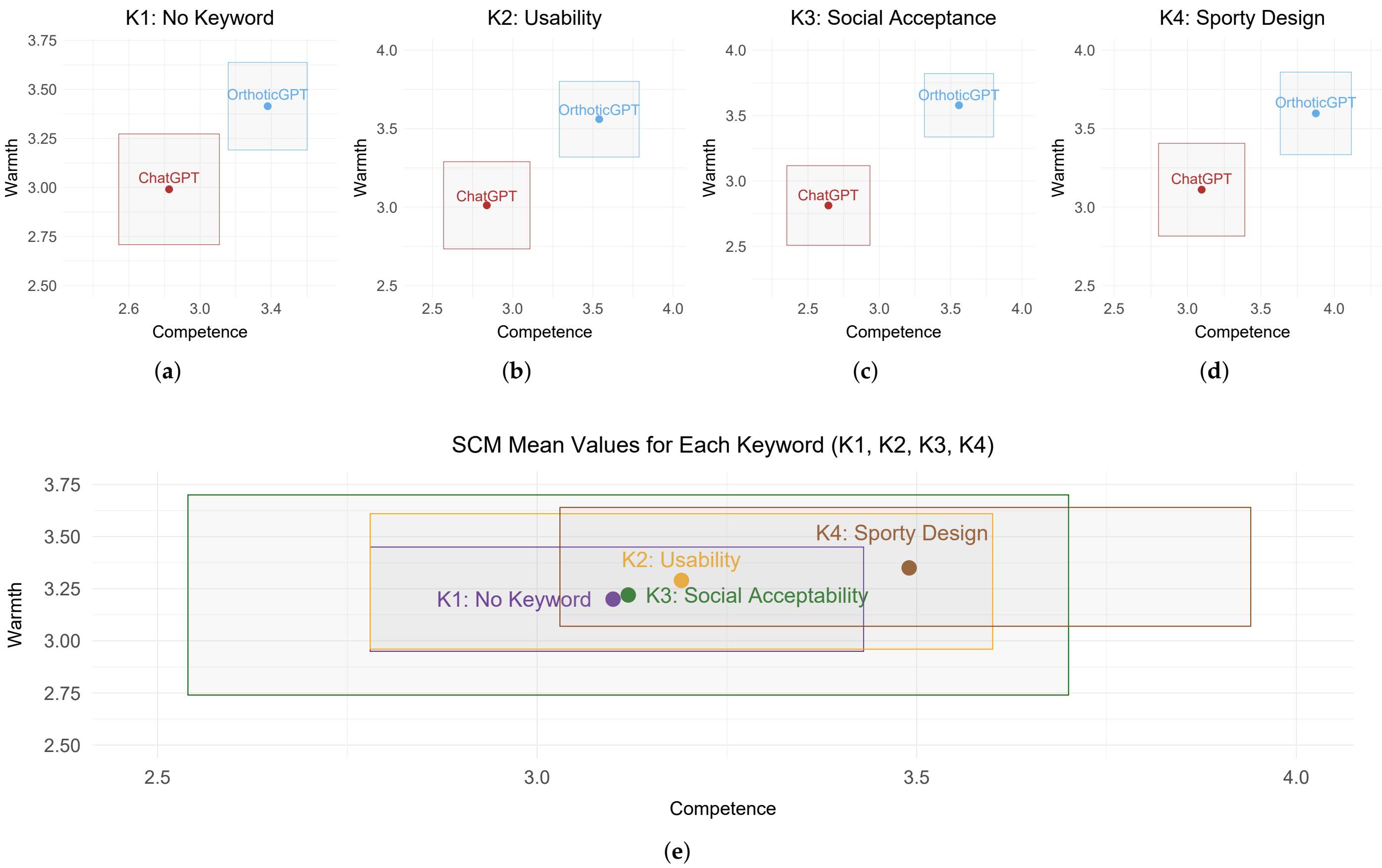


Figure 7. SCM ratings of perceived competence (x-axis) and warmth (y-axis) for each of the four KEYWORDS. In the upper row, the four quadrants are assigned as follows, from left to right: (**a**) *No Keyword*, (**b**) *Usability*, (**c**) *Social Acceptability*, and (**d**) *Sporty Design*. The mean values for two GPTs are shown in separate colors: red (*ChatGPT*) and blue (*OrthoticFootGPT*). The lower quadrant (**e**) contains the SCM mean values for each KEYWORD: *No Keyword* (purple), *Usability* (orange), *Social Acceptability* (green), and *Sporty Design* (brown). The rectangles represent the 95% confidence interval.

Wearable Acceptability Range (WEAR)

The Shapiro–Wilk normality test indicated that the data is not normally distributed ($p < 0.0001$). An ART RM-ANOVA showed statistically significant main effects for the GPT, $F(1, 371) = 109.325$, $p < 0.0001$, $\eta_p^2 = 0.450$ (large effect size), and for KEYWORD,

$F(3, 371) = 6.270$, $p = 0.0003$, $\eta_p^2 = 0.123$ (medium effect size). However, no interaction effect was found for GPT × KEYWORD, $F(3, 371) = 1.948$, $p = 0.121$, and $\eta_p^2 = 0.042$ (small effect size).

Bonferroni-corrected pairwise comparisons were conducted using Wilcoxon signed-rank tests, which revealed statistically significant differences between *No Keyword* × *Sporty Design* ($p_{adj} < 0.0001$), *Usability* × *Sporty Design* ($p_{adj} = 0.0002$), and *Social Acceptability* × *Sporty Design* ($p_{adj} < 0.0001$). The pairwise comparison between *ChatGPT* and *OrthoticFootGPT* was statistically significant, with $p < 0.0001$. Pairwise comparisons between the GPT and KEYWORD demonstrated statistically significant differences across all keywords: *No Keywords* ($p_{adj.} = 0.0004$), *Usability* ($p_{adj.} = 0.0002$), *Social Acceptability* ($p_{adj.} < 0.0001$), and *Sporty Design* ($p_{adj.} = 0.009$). The results of the WEAR scale are summarized in Figure 8.

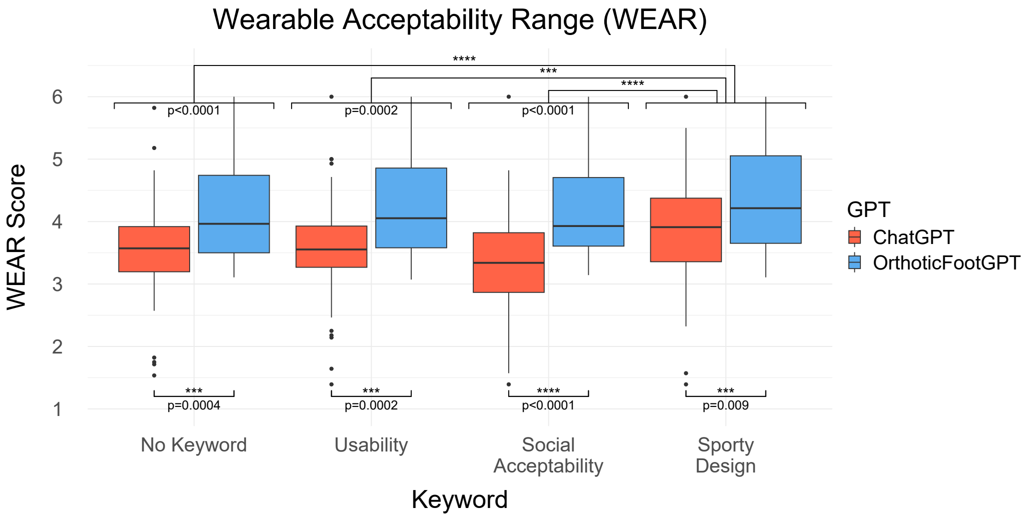


Figure 8. This graph shows the results of the WEAR ratings. The four Keywords are listed on the x-axis, *No Keyword*, *Usability*, *Social Acceptability*, and *Sporty Design*, and the y-axis represents the corresponding scores on the WEAR scale. The boxplots are color-coded to distinguish the GPT: *ChatGPT* in red and *OrthoticFootGPT* in blue. Asterisks indicate significance levels: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$), **** ($p < 0.0001$).

Seven-Point Likert Scale

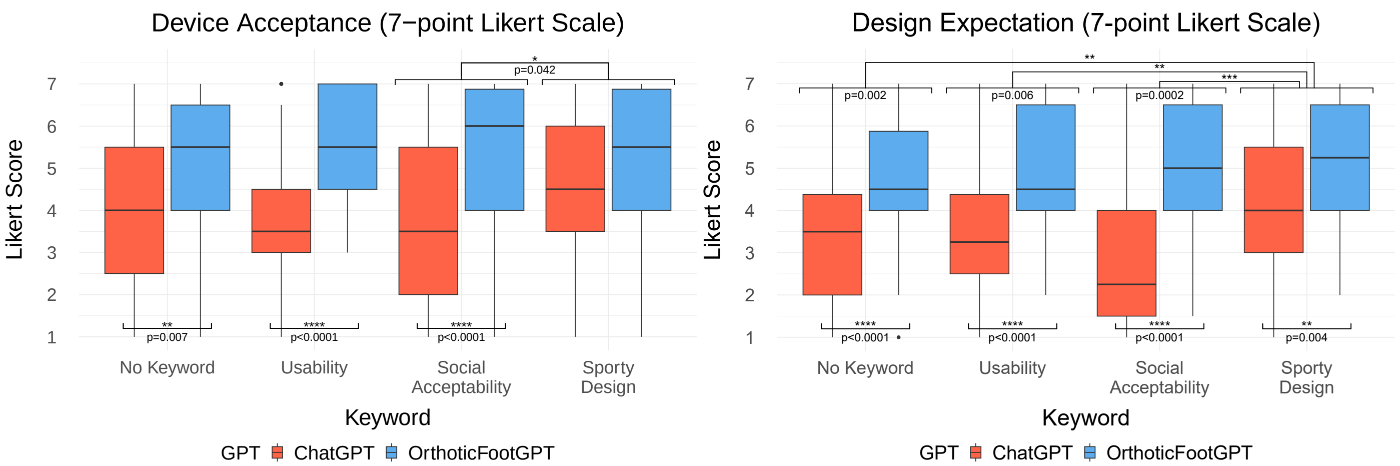
The Shapiro–Wilk normality test for the two statements on device acceptance and aesthetic design expectations indicated no normal distribution ($p < 0.0001$).

A non-parametric two-factorial ART-ANOVA on rated device acceptance statements revealed a significant main effect for the GPT, $F(1, 371) = 90.864$, $p < 0.0001$, $\eta_p^2 = 0.442$ (large effect size), and for KEYWORD, $F(3, 371) = 3.081$, $p = 0.027$, $\eta_p^2 = 0.074$ (medium effect size), as well as an interaction effect for GPT × KEYWORD, $F(3, 371) = 4.834$, $p = 0.002$, and $\eta_p^2 = 0.111$ (medium effect size).

The two-factorial ART-ANOVA on rated aesthetic design expectations showed a significant main effect for GPT, $F(1, 371) = 129.196$, $p < 0.0001$, $\eta_p^2 = 0.442$, and KEYWORD, $F(3, 371) = 7.397$, $p < 0.0001$, $\eta_p^2 = 0.119$ (both with a large effect size), as well as a significant interaction effect for GPT × KEYWORD, $F(3, 371) = 4.021$, $p = 0.007$, and $\eta_p^2 = 0.068$ (medium effect size). Post hoc pairwise comparisons using the Wilcoxon signed-rank test with Bonferroni correction revealed statistically significant differences and notable trends in both device acceptance and design expectation data, as documented in Table 4. The results are presented in Figure 9.

Table 4. Post hoc pairwise comparisons within keywords and GPTs for Likert scores of device acceptance and design expectation. Significance codes: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$), **** ($p < 0.0001$), ns = not significant.

			Device Acceptance				Design Expectation			
Pairwise Comparisons	**n1**	**n2**	***Stat.***	p	$p_{adj.}$	***Sig.***	***Stat.***	p	$p_{adj.}$	***Sig.***
No Keyword—Usability	108	108	1382	0.548	1	ns	1545	0.282	1	ns
No Keyword—Social Acceptability	108	108	1382	0.199	1	ns	1833	0.304	1	ns
No Keyword—Sporty Design	108	108	1110	0.066	0.398	ns	1093	<0.001	0.002	**
Usability—Social Acceptability	108	108	1529	0.325	1	ns	2022	0.030	0.179	ns
Usability—Sporty Design	108	108	732	0.011	0.064	ns	934	<0.001	0.006	**
Social Acceptability—Sporty Design	108	108	1030	0.007	0.042	*	836	<0.001	<0.001	***
No Keyword: ChatGPT—Orth.FootGPT	54	54	1022	0.007	0.007	**	819	<0.001	<0.001	****
Usability: ChatGPT—OrthoticFootGPT	54	54	692	<0.001	<0.001	****	712	<0.001	<0.001	****
Social Accept.: ChatGPT—Orth.GPT	54	54	744	<0.001	<0.001	****	540	<0.001	<0.001	****
Sporty Design: ChatGPT—Orth.FootGPT	54	54	1212	0.129	0.129	ns	997	0.004	0.004	**



(a) (b)

Figure 9. This figure contains two boxplot diagrams, including the rated device acceptance (**a**) and aesthetic design expectation (**b**) for the four KEYWORDS in each GPT. The x-axis lists the four Keywords: *No Keyword*, *Usability*, *Social Acceptability*, and *Sporty Design*. The y-axis represents the corresponding scores from the Likert rating. The boxplots are color-coded to distinguish the GPT: *ChatGPT* in red and *OrthoticFootGPT* in blue. Asterisks indicate significance levels: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$), **** ($p < 0.0001$).

4.2.3. Qualitative Feedback

An inductive thematic analysis on transcribed data revealed three main themes: *Aesthetic Design*, *Functionality*, as well as *Comfort and Fit*. These three themes include both positive and negative perspectives and reflect the participants' feedback on the two GPTS and four KEYWORDS.

Aesthetic Design: The designs with *No Keyword* were generally described as "standard" (P141), "simple" (P56), and "not visually appealing" (P61) when generated by *ChatGPT*. In contrast, designs generated by *OrthoticFootGPT* were characterized as "modern" (P83, P90), "aesthetic" (P9), and "compact" (P60), with a "positive shape and design" (P74) resembling a "mixture of a boot and a sports shoe" (P102). For the keyword *Usability*, *ChatGPT* designs were recognized as "similar to shoes" (P33) with a "good color" (P14) but were also criticized for being "old-fashioned" (P21) and "unaesthetic" (P61). Meanwhile, *OrthoticFootGPT* designs were praised as "aesthetic" (P56), "modern and sporty" (P21), and also “similar to shoes” (P41, P69). Furthermore, one participant stated that “the eye-catching design ensures that the orthosis is easily recognized by others, which can lead to more consideration from others” (P108). Images generated using the keyword *Social Acceptability* via *ChatGPT* were described as “minimalistic” (P24) with a “positive color” (P14), but their perception was negatively impacted by a “dubious shape” (P21) and “incomprehensible design” (P47). On the other hand, designs from *OrthoticFootGPT* were regarded as “very aesthetic and modern” (P41), “sporty” (P24), and featuring a “nice design” (P83), due to “less conspicuous Velcro fasteners” (P90) and a resemblance to “winter boots” (P159). In general, the *Sporty Design* keyword led to designs perceived as modern and sporty, which were positively received by participants. The *ChatGPT* outputs were described as a “nice and modern design” (P24) contributing to a “self-confident appearance” (P74). In comparison, *OrthoticFootGPT* designs were associated with a “sporty design” (P9), resembling “sport shoes” (P74, P102, P138) and perceived as designs that “remind one of a fashion brand” (P89). It was also positively noted that “the toes are not visible” (P94).

Functionality: The use of *No Keyword* with *ChatGPT* raised concerns about functionality, as the designs were perceived as “not effective” (P157) and “not stable” (P90, P150) due to the “missing insole” (P9, P14) and because they “offer less protection (e.g., no padding)” (P36). In contrast, the *OrthoticFootGPT* designs “offer good stability and fulfill [their] functional requirements effectively” (P61) because they are “easy to attach and remove” (P89, P90, P152). For the keyword *Usability*, *ChatGPT* designs were criticized for being “too mechanical” (P150) and “complicated to use” (P105, P111, P149). Meanwhile, *OrthoticFootGPT* outputs were associated with “simple handling” (P36, P56), due to “a good balance of stability and comfort” (P36) and because “it also gives the feeling that it has very useful functions” (P21). In the context of *Social Acceptability*, *ChatGPT* designs were described as “too bulky, seems heavy” (P14), “unrealistic” (P60), and “impractical” (P37), primarily due to “too many straps” (P9, P89) and “components are mainly made of textile. Water . . . can be absorbed” (P79). In contrast, *OrthoticFootGPT* designs were perceived as “very innovative” (P93), featuring “a good sole, similar to a shoe” (P14), making them an “eye-catcher for everyone” (P21). Regarding *Sporty Design*, *ChatGPT* was noted for its “functional and rather simple design . . . suitable for a broad target group” (P90). Similarly, *OrthoticFootGPT* designs were defined as “robust” (P94), “stable” (P36, P105), and “breathable” (P33), with functionality that is “easy to integrate for sports activities” (P89).

Comfort and Fit: The primary issues with *ChatGPT* designs using *No Keyword* were identified as being “uncomfortable” (P21) and “impractical due to open toes” (P37), concerns that were similarly reflected in the keywords *Usability* and *Social Acceptability*. In contrast, the *OrthoticFootGPT* designs with *No Keyword* were perceived more positively, as “the orthosis is easily adjustable and can be customized” (P61) and it “has good padding on the foot” (P36). For the keyword *Usability*, the *OrthoticFootGPT* was highlighted for being ”easy to use” (P36) and it “is light and comfortable to wear, even for long periods” (P61), which could be “suitable for thicker calves” (P74). In the context of *Social Acceptability*, designs generated by *OrthoticFootGPT* were described as resembling a “sport shoe” (P21) or “winter boots” (P159), with positive feedback emphasizing that “no toes are visible” (P94). Regarding *Sporty Design*, a participant described the *ChatGPT* output as follows: “It looks like a boot. It does not attract attention and is therefore great for everyday wear. It makes you feel more confident” (P74). Meanwhile, the *OrthoticFootGPT* designs were seen as both “comfortable” (P157) and “sporty” (P47, P60, P92, P101, P117), and they “look like an orthopedic hiking/running/football shoe” (P157) which is “great for social integration” (P74). Furthermore, one participant noted that “the orthosis appears to be easy to put on with the fasteners, which could potentially make everyday life easier for the wearer”, but also mentioned negatively that "the unobtrusive design could lead to people taking less notice of the wearer" (P108).

5. Discussion

5.1. The Impact of AI-Generated and Conventional Orthotic Designs Across Device Categories on Social Acceptability

In our first study, we conducted a mixed-method online survey to explore the impact of three DESIGN TYPES and four ORTHOTIC CATEGORIES on social acceptance. The quantitative analyses of Likert scales, as well as the SCM and WEAR models, indicated that ORTHOTIC CATEGORIES such as *Foot Bandages/Textiles* achieved significantly higher acceptance ratings, followed by *Foot Drop Orthoses* and then *Short-Leg* and *High-Leg Orthoses*. Furthermore, the *Generative AI* DESIGN TYPE had a significant impact on perceived acceptability, as these devices were associated with stereotypes of greater competence and warmth compared to medical or developmental designs, which confirms our hypothesis. These findings were supported by the qualitative feedback, which highlighted the potential of using *Generative AI* to consider user preferences and increase acceptance of orthopedic footwear.

Overall, these results confirm the potential of integrating AI into the design process for wearable foot devices. However, the significantly lower acceptance rates for short and high-leg orthoses represent a consistent challenge that reflects the concerns reported in previous research. By identifying the challenges of user acceptance associated with different categories of orthoses, this study contributes to a deeper understanding of the factors that influence the social acceptance of orthotic devices. Our findings are valuable for future developments aimed at enhancing the aesthetic design of these devices to better meet user needs.

5.2. The Impact of GPT Customization and Prompt Keywords on the Social Acceptability of AI-Generated Orthotic Designs

The second study examined the impact of high-leg orthosis designs generated using *OrthoticFootGPT* and *ChatGPT*, as well as the influence of four different prompt structures incorporating specific keywords on social acceptability. This extends the results of the first study to explore if a personalized GPT and specific KEYWORDS can have a systematic positive influence on user perception. The quantitative results indicate that the customized *OrthoticFootGPT* model can enhance acceptance ratings compared to *ChatGPT*. Additionally, the study showed that different keywords evoke different stereotypical perceptions of the wearable device. In particular, the keyword *Sporty Design* led to significantly better ratings compared to other keyword variations, suggesting that aesthetic framing plays a crucial role in user perception. These findings are consistent with the qualitative feedback, which revealed a discrepancy between the perception of medical orthoses and more socially integrated designs, such as those inspired by the aesthetics of footwear or sports. Participants expressed contrasting preferences regarding the visual appearance of orthotic designs: while some preferred an inconspicuous design to avoid attracting attention, others emphasized that a more visible, eye-catching design could facilitate social recognition and consideration by others. This contrast illustrates the complexity of the relationship between aesthetics as well as functional and social expectations.

These results highlight the potential of generative AI to enhance the design process of wearable medical devices. Furthermore, our results align with previous studies on wearable device perception, emphasizing that design plays a crucial role in acceptance [35,36]. In addition, these results support earlier findings that highlight a discrepancy in the literature regarding the stigmatization of medical devices, where functional necessity often conflicts with social acceptability, leading to lower acceptance [36].

Similar to findings by Sehrt et al. [50], who observed that wearables positioned on the ankle received lower social acceptance scores, our study showed that high-leg orthoses face significant acceptance challenges compared to other orthotic device categories. By demonstrating that customized GPT models and prompt engineering can systematically shape social acceptability, our study builds on prior work [22,23] that highlights the importance of structured input in AI-assisted design.

Therefore, we recommend that future developments prioritize the use of customized GPT models in combination with personalized prompting strategies that explicitly address user preferences and social perception factors.

5.3. Implications

The results obtained from the two studies conducted show the potential of generative AI to improve the social acceptability of foot wearable devices and demonstrate its applicability in product design. Building on existing work, such as the use of DALL-E for designing hand orthoses [30], our study extends this approach to foot orthoses, with a specific focus on social acceptance factors. The integration of image-generative AI in the design phase can significantly improve acceptance and lead to more user-centered orthosis designs. In particular, the consideration of user preferences with prompting alongside the technical functionalities of wearables offers a promising approach for personalization in product development. Furthermore, the findings from this research extend beyond orthopedic footwear, affecting broader applications within the fields of HCI, user experience (UX) design, and the integration of social factors in product development. Understanding how AI-generated designs influence product acceptance offers valuable insights for improving design processes across various applications and product categories. For example, designers can integrate user feedback directly into aesthetic and functional design aspects to enhance user engagement and satisfaction. Moreover, this study highlights the potential of generative AI as a valuable tool for inspiring the design of socially driven consumer products and wearable technology. We recommend the use of customized, fine-tuned GPT models to generate user-specific outputs that align with product specifications. Furthermore, incorporating specific keywords related to functionality, target user groups, and design preferences directly within the prompt can enhance the quality and relevance of the generated results.

Beyond improving social acceptability, AI-generated designs also introduce new opportunities for streamlining the product development process. While traditional design approaches rely entirely on manual modeling, which can be time-consuming and expensive, especially when adapting to individual user preferences, AI-generated designs still require manual expert input in the post-processing phase. This includes converting 2D images into 3D data, adapting the design to specific medical requirements, and preparing the data for manufacturing. Related work has demonstrated that integrating AI into the shoe design process can reduce design time by 59% compared to traditional methods while also lowering production costs [72]. These findings indicate that AI-supported workflows have the potential to increase efficiency in orthotic design while maintaining a high degree of individualization.

5.4. Limitations

Our studies have some limitations that affect the generalizability of the results. Although our studies included participants from various backgrounds, the lack of gender balance in the first study and the inclusion of healthy individuals in both studies could potentially affect the external validity. However, our analysis showed that demographic factors did not systematically influence participants' ratings, which indicates that the observed effects are not due to the sample population. Nevertheless, a more patient-centered focus could provide deeper insights into the specific needs and perceptions of individuals with orthopedic conditions. While some results did not reach statistical significance, they indicate notable trends that suggest potential effects worth further investigation. Furthermore, the results are limited to the specific stimuli and orthotic categories that were used in the study. Additionally, the AI-generated designs, which rely on individual prompts or selected pictures of orthopedic products, cannot be consistently reproduced with identical inputs. This inconsistency requires multiple iterations of prompts to achieve the desired outcomes.

Another notable limitation is that AI-generated designs do not adhere to any medical guidelines or standards, as they are primarily visual concepts rather than functional prototypes. While the prompts and images were adjusted to the medical functions and specific requirements of the selected orthosis categories, DALL-E image generation is not specialized for medical functions and does not take manufacturing constraints or biocompatibility into account. The results are limited to the perspective of end users and do not consider technical aspects that are relevant from an expert perspective, such as material feasibility or regulatory compliance.

The generative AI tool DALL-E was limited to its current version during both studies. Furthermore, biases in the training datasets of AI models could reinforce stereotypes (e.g., related to ethnicity, gender, or other factors) [73]. These factors are critical as they could influence the social acceptance of the designs, particularly if they fail to address the needs and preferences of various user groups or if they misrepresent them. Therefore, any application of such AI-generated designs in real-world medical products must carefully consider and comply with strict regulatory standards to ensure their safety and effectiveness.

5.5. Future Work

The growing field of AI applications will contribute to opening new possibilities for user-centered design solutions. The potential for generating 3D models that are compatible with Computer-Aided Design (CAD) software could significantly enhance the development process, particularly in the area of rapid prototyping. Future developments should focus on training device-specific GPTs to generate tailored image outputs that meet user needs. Moreover, refining and validating prompt structures will be crucial to achieving more personalized results. Furthermore, to better understand the context and impact of AI-generated designs on social acceptability, future research should extend beyond orthopedic footwear to include diverse product categories. This extension would provide a comprehensive overview of how generative AI can influence product acceptance in different fields. By addressing these areas, follow-up work can make an important contribution to the further development of AI in product design and potentially lead to more effective and socially accepted solutions. Follow-up studies should include a more diverse participant sample, focused on individuals with orthopedic conditions, to better understand user-specific requirements. An exemplary case study could demonstrate how AI-generated design can be integrated into a patient-centric design process that covers the entire workflow, including patient requirements, AI-based design generation, post-processing of 3D models, product personalization based on anatomical needs, manufacturing aspects, and product testing. While long-term biocompatibility testing may be required for regulatory approval, initial usability studies can focus on short-term comfort and usability. Additionally, human intervention by expert designers remains essential to test, validate, and refine AI-generated outputs to ensure practical applicability and manufacturability.

To ensure that AI-generated orthotic designs consider different cultural factors, AI models should be trained on datasets that are representative of diverse user demographics. Additionally, engaging patients with orthopedic conditions in co-design sessions can help align AI-generated outputs with real-life needs and prevent the reinforcement of stereotypes in training data. Involving clinicians, orthopedic experts, and diverse patient representatives in the AI design evaluation process ensures that biased or stereotypical outputs are identified and corrected early. Future research must critically evaluate these biases and develop strategies to ensure that AI-assisted design workflows align with ethical principles and medical standards.

6. Conclusions

In this paper, we explored the influence of image-generative AI on the social acceptance of footwear designs through two studies involving 134 participants. Our first study demonstrated that AI-generated designs can significantly enhance perceived user acceptance compared to conventional designs. The second study showed that customizing prompts with specific keywords and using a personalized GPT can better tailor these designs to user preferences and therefore improve acceptance. These findings suggest that integrating image-generative AI with personalized prompting strategies in the design process could transform how these devices are perceived and accepted, aligning product functionality with user expectations more effectively. This approach introduces a new framework for user-centric design processes, leveraging AI-driven systems to adapt designs based on specific user inputs. Our findings are not limited to medical wearables and could be extended to various applications and products. Overall, this study enhances our understanding of how AI-generated outputs impact user perception, setting the groundwork for future research to explore new GPT models and expand these findings across different contexts.

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References

1. Mohaddis, M.; Maqsood, S.A.; Ago, E.; Singh, S.; Naim, Z.; Prasad, S. Enhancing Functional Rehabilitation Through Orthotic Interventions for Foot and Ankle Conditions: A Narrative Review. *Cureus* **2023**, *15*, e49103. [CrossRef]
2. Slater, G.; Slater, T. Current concepts review: Orthotics in post-operative foot and ankle surgery. *J. Regen. Biol. Med.* **2022**, *4*, 1–4. [CrossRef] [PubMed]
3. Surmen, H.K.; Akalan, N.E.; Arslan, Y.Z. Design, manufacture, and selection of ankle-foot-orthoses. In *Advanced Methodologies and Technologies in Artificial Intelligence, Computer Simulation, and Human-Computer Interaction*; IGI Global: Hershey, PA, USA, 2019; pp. 250–266.
4. Leite, M.; Soares, B.; Lopes, V.; Santos, S.; Silva, M.T. Design for personalized medicine in orthotics and prosthetics. *Procedia CIRP* **2019**, *84*, 457–461.
5. Ghoseiri, K.; Bahramian, H. User satisfaction with orthotic and prosthetic devices and services of a single clinic. *Disabil. Rehabil.* **2012**, *34*, 1328–1332.
6. Swinnen, E.; Kerckhofs, E. Compliance of patients wearing an orthotic device or orthopedic shoes: A systematic review. *J. Bodyw. Mov. Ther.* **2015**, *19*, 759–770. [CrossRef]
7. Faux-Nightingale, A.; Kelemen, M.; Stewart, C. Ankle-foot orthosis adherence in children and adolescents with cerebral palsy: A scoping review. *Prosthet. Orthot. Int.* **2022**, *46*, 351–356. [CrossRef] [PubMed]
8. Dabnichki, P.; Pang, T.Y. User-Centered Design Framework for Personalized Ankle–Foot Orthoses. *Prosthesis* **2025**, *7*, 11. [CrossRef]
9. Exterkate, S.H.; Jongebloed-Westra, M.; Ten Klooster, P.M.; Koffijberg, H.; Bode, C.; van Gemert-Pijnen, J.E.; van Baal, J.G.; van Netten, J.J. Objectively assessed long-term wearing patterns and predictors of wearing orthopaedic footwear in people with diabetes at moderate-to-high risk of foot ulceration: A 12 months observational study. *J. Foot Ankle Res.* **2023**, *16*, 60. [CrossRef]
10. Finco, M.G.; Cay, G.; Lee, M.; Garcia, J.; Salazar, E.; Tan, T.W.; Armstrong, D.G.; Najafi, B. Taking a Load Off: User Perceptions of Smart Offloading Walkers for Diabetic Foot Ulcers Using the Technology Acceptance Model. *Sensors* **2023**, *23*, 2768. [CrossRef]
11. Mageshkumar, G.; Prasanna, G.; Sakthi Priya, G.; Samkilbart, A. Design of IoT-Based Orthotic Footwear for Lower Limb Correction and Alignment. In Proceedings of the 2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 14–15 March 2024; Volume 1, pp. 633–638.
12. Imbesi, S.; Scataglini, S. A User Centered Methodology for the Design of Smart Apparel for Older Users. *Sensors* **2021**, *21*, 2804. [CrossRef]
13. Orlando, J.M.; Li, B.; Bodt, B.; Lobo, M.A. Users' perceptions about lower extremity orthotic devices: A systematic review. *Arch. Phys. Med. Rehabil.* **2023**, *104*, 645–655. [CrossRef] [PubMed]
14. van der Wilk, D.; Hijmans, J.M.; Postema, K.; Verkerke, G.J. A user-centered qualitative study on experiences with ankle-foot orthoses and suggestions for improved design. *Prosthet. Orthot. Int.* **2018**, *42*, 121–128. [CrossRef]
15. Adhikari, M.S.; Verma, Y.K.; Sindhwani, M.; Sachdeva, S. Generative AI Tools for Product Design and Engineering. In *Generative Artificial Intelligence in Finance: Large Language Models, Interfaces, and Industry Use Cases to Transform Accounting and Finance Processes*; Wiley: Hoboken, NJ, USA, 2025; pp. 299–325.
16. Suessmuth, J.; Fick, F.; Van Der Vossen, S. Generative AI for Concept Creation in Footwear Design. In *Proceedings of the ACM SIGGRAPH 2023 Talks*; Association for Computing Machinery: New York, NY, USA, 2023. [CrossRef]
17. DALL·E 3. OpenAI. Available online: https://openai.com/index/dall-e-3/ (accessed on 10 January 2025).
18. Alam, M.F.; Lentsch, A.; Yu, N.; Barmack, S.; Kim, S.; Acemoglu, D.; Hart, J.; Johnson, S.; Ahmed, F. From Automation to Augmentation: Redefining Engineering Design and Manufacturing in the Age of NextGen-AI. An MIT Exploration of Generative AI. 2024. Available online: https://mit-genai.pubpub.org/pub/9s6690gd (accessed on 10 February 2025).
19. Edwards, K.M.; Man, B.; Ahmed, F. Sketch2Prototype: Rapid Conceptual Design Exploration and Prototyping with Generative AI. *Proc. Des. Soc.* **2024**, *4*, 1989–1998. [CrossRef]
20. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, L.; Polosukhin, I. Attention is all you need. In Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17), Red Hook, NY, USA, 4–9 December 2017; pp. 6000–6010.
21. Brisco, R.; Hay, L.; Dhami, S. Exploring the role of text-to-image AI in concept generation. *Proc. Des. Soc.* **2023**, *3*, 1835–1844. [CrossRef]
22. White, J.; Fu, Q.; Hays, S.; Sandborn, M.; Olea, C.; Gilbert, H.; Elnashar, A.; Spencer-Smith, J.; Schmidt, D.C. A prompt pattern catalog to enhance prompt engineering with chatgpt. *arXiv* **2023**, arXiv:2302.11382.
23. Liu, V.; Chilton, L.B. Design Guidelines for Prompt Engineering Text-to-Image Generative Models. In Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22), New Orleans, LA, USA, 29 April–5 May 2022. [CrossRef]

24. Goodfellow, I.; Pouget-Abadie, J.; Mirza, M.; Xu, B.; Warde-Farley, D.; Ozair, S.; Courville, A.; Bengio, Y. Generative adversarial networks. *Commun. ACM* **2020**, *63*, 139–144. [CrossRef]
25. Chiou, L.Y.; Hung, P.K.; Liang, R.H.; Wang, C.T. Designing with AI: An Exploration of Co-Ideation with Image Generators. In Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23), Pittsburgh, PA, USA, 10–14 July 2023; pp. 1941–1954. [CrossRef]
26. Barros, M.; Ai, Q. Designing with words: Exploring the integration of text-to-image models in industrial design. *Digit. Creat.* **2024**, *35*, 378–391. [CrossRef]
27. Lee, Y.H.; Chiu, C.Y. The Impact of AI Text-to-Image Generator on Product Styling Design. In *Proceedings of the Human Interface and the Management of Information*; Mori, H., Asahi, Y., Eds.; Springer: Cham, Switzerland, 2023; pp. 502–515. [CrossRef]
28. Cai, A.; Rick, S.R.; Heyman, J.L.; Zhang, Y.; Filipowicz, A.; Hong, M.; Klenk, M.; Malone, T. DesignAID: Using Generative AI and Semantic Diversity for Design Inspiration. In Proceedings of the ACM Collective Intelligence Conference (CI '23), Delft, The Netherlands, 6–9 November 2023; Association for Computing Machinery: New York, NY, USA, 2023; pp. 1–11. [CrossRef]
29. Elasri, M.; Elharrouss, O.; Al-Maadeed, S.; Tairi, H. Image generation: A review. *Neural Process. Lett.* **2022**, *54*, 4609–4646. [CrossRef]
30. Popescu, D. Exploring the use of ChatGPT and Dall-E for the preliminary design of 3D-printed wrist-hand orthoses. *Proc. Manuf. Syst.* **2023**, *18*, 97–102.
31. Bartlett, K.A.; Camba, J.D. Generative Artificial Intelligence in Product Design Education: Navigating Concerns of Originality and Ethics. *Int. J. Interact. Multimed. Artif. Intell.* **2024**, *8*, 55–64. [CrossRef]
32. Dunne, L.E.; Profita, H.; Zeagler, C.; Clawson, J.; Gilliland, S.; Do, E.Y.L.; Budd, J. The social comfort of wearable technology and gestural interaction. In Proceedings of the 2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, Chicago, IL, USA, 26–30 August 2014; pp. 4159–4162. [CrossRef]
33. Dunne, L.; Profita, H.; Zeagler, C. Chapter 1.2—Social Aspects of Wearability and Interaction. In *Wearable Sensors*; Sazonov, E., Neuman, M.R., Eds.; Academic Press: Oxford, UK, 2014; pp. 25–43. [CrossRef]
34. Yang, H.; Yu, J.; Zo, H.; Choi, M. User acceptance of wearable devices: An extended perspective of perceived value. *Telemat. Inform.* **2016**, *33*, 256–269. [CrossRef]
35. Profita, H.; Albaghli, R.; Findlater, L.; Jaeger, P.; Kane, S.K. The AT Effect: How Disability Affects the Perceived Social Acceptability of Head-Mounted Display Use. In Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16), San Jose, CA, USA, 7–12 May 2016; Association for Computing Machinery: New York, NY, USA, 2016; pp. 4884–4895. [CrossRef]
36. Kelly, N.; Gilbert, S.B. The wearer, the device, and its use: Advances in understanding the social acceptability of wearables. *Proc. Hum. Factors Ergon. Soc. Annu. Meet.* **2018**, *62*, 1027–1031. [CrossRef]
37. Canhoto, A.I.; Arp, S. Exploring the factors that support adoption and sustained use of health and fitness wearables. *J. Mark. Manag.* **2017**, *33*, 32–60. [CrossRef]
38. El-Gayar, O.; Elnoshokaty, A. Factors and design features influencing the continued use of wearable devices. *J. Healthc. Inform. Res.* **2023**, *7*, 359–385. [CrossRef]
39. Adapa, A.; Nah, F.F.H.; Hall, R.H.; Siau, K.; Smith, S.N. Factors Influencing the Adoption of Smart Wearable Devices. *Int. J. Hum. Comput. Interact.* **2018**, *34*, 399–409. [CrossRef]
40. Liao, T.; Tanner, K.; MacDonald, E.F. Revealing insights of users' perception: An approach to evaluate wearable products based on emotions. *Des. Sci.* **2020**, *6*, e14. [CrossRef]
41. Davis, F.D.; Bagozzi, R.P.; Warshaw, P.R. User acceptance of computer technology: A comparison of two theoretical models. *Manag. Sci.* **1989**, *35*, 982–1003. [CrossRef]
42. Venkatesh, V.; Morris, M.G.; Davis, G.B.; Davis, F.D. User Acceptance of Information Technology: Toward a Unified View. *MIS Q.* **2003**, *27*, 425–478. [CrossRef]
43. Fiske, S.T.; Cuddy, A.J.; Glick, P.; Xu, J. *A Model of (Often Mixed) Stereotype Content: Competence and Warmth Respectively Follow from Perceived Status and Competition*; Routledge: London, UK, 2018; pp. 162–214.
44. Kelly, N. All the world's a stage: What makes a wearable socially acceptable. *Interactions* **2017**, *24*, 56–60. [CrossRef]
45. Venkatesh, V.; Bala, H. Technology acceptance model 3 and a research agenda on interventions. *Decis. Sci.* **2008**, *39*, 273–315. [CrossRef]
46. Kelly, N.; Gilbert, S. The WEAR Scale: Developing a Measure of the Social Acceptability of a Wearable Device. In Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems, San Jose, CA, USA, 7–12 May 2016; Association for Computing Machinery: New York, NY, USA, 2016; pp. 2864–2871. [CrossRef]
47. Cuddy, A.J.; Fiske, S.T.; Glick, P. *Warmth and Competence as Universal Dimensions of Social Perception: The Stereotype Content Model and the BIAS Map*; Association for Computing Machinery: New York, NY, USA, 2008; Volume 40, pp. 61–149. [CrossRef]

48. Schwind, V.; Deierlein, N.; Poguntke, R.; Henze, N. Understanding the Social Acceptability of Mobile Devices using the Stereotype Content Model. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, Glasgow, UK, 4–9 May 2019; Association for Computing Machinery: New York, NY, USA, 2019; pp. 1–12. [CrossRef]
49. Cuddy, A.J.; Fiske, S.T.; Kwan, V.S.; Glick, P.; Demoulin, S.; Leyens, J.P.; Bond, M.H.; Croizet, J.C.; Ellemers, N.; Sleebos, E.; et al. Stereotype content model across cultures: Towards universal similarities and some differences. *Br. J. Soc. Psychol.* **2009**, *48*, 1–33. [CrossRef] [PubMed]
50. Sehrt, J.; Braams, B.; Henze, N.; Schwind, V. Social Acceptability in Context: Stereotypical Perception of Shape, Body Location, and Usage of Wearable Devices. *Big Data Cogn. Comput.* **2022**, *6*, 100. [CrossRef]
51. Profita, H.P.; Clawson, J.; Gilliland, S.; Zeagler, C.; Starner, T.; Budd, J.; Do, E.Y.L. Don't mind me touching my wrist: A case study of interacting with on-body technology in public. In Proceedings of the 2013 International Symposium on Wearable Computers, Zurich, Switzerland, 8–12 September 2013; Association for Computing Machinery: New York, NY, USA, 2013; pp. 89–96. . [CrossRef]
52. Harrison, C.; Lim, B.Y.; Shick, A.; Hudson, S.E. Where to locate wearable displays?: Reaction time performance of visual alerts from tip to toe. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*; Association for Computing Machinery: New York, NY, USA, 2009; pp. 941–944. [CrossRef]
53. Gemperle, F.; Kasabach, C.; Stivoric, J.; Bauer, M.; Martin, R. Design for Wearability. In Proceedings of the 2nd IEEE International Symposium on Wearable Computers (ISWC '98), Pittsburgh, PA, USA, 19–20 October 1998; pp. 116–122. [CrossRef]
54. GPT-4 Is OpenAI's Most Advanced System, Producing Safer and More Useful Responses. Available online: https://openai.com/index/gpt-4/ (accessed on 10 January 2025).
55. OPED-AG. VACOpedes—Orthese Mit Optimaler Druckentlastung: Für Mittel- und Vorfuß. Available online: https://oped.ch/vacopedes (accessed on 10 January 2025).
56. Enovis. AIRCAST® AIRSELECT™ Short Walker. Available online: https://www.enovis-medtech.de/AIRCAST-AIRSELECT-Short-Walker (accessed on 10 January 2025).
57. OPED-Medical-Inc. VACOcast Diabetic Boot. Available online: https://opedmedical.com/products/vacocast-diabetic/ (accessed on 10 January 2025).
58. Enovis. AIRCAST® AIRSELECT™ Achilles Walker. Available online: https://www.enovis-medtech.de/AIRCAST-AIRSELECT-Achilles-Walker (accessed on 10 January 2025).
59. Berglingstyle. FastProtect Malleo: Der Sicherheitsgurt für Dein Sprunggelenk. Available online: https://www.berglingstyle.com/products/fastprotect-malleo-5 (accessed on 10 January 2025).
60. Betterguards-Technology. The BetterGuard 1.0: Best Ankle Brace. Available online: https://betterguards.de/en/products/the-betterguard-1 (accessed on 10 January 2025).
61. Otto-Bock-HealthCare. WalkOn Reaction Lateral. Available online: https://www.ottobock.com/en-us/product/28U34 (accessed on 10 January 2025).
62. Vitego. ofa Push Ortho Fußheberorthese AFO. Available online: https://www.vitego-shop.de/bandagen-fuss-ofa-push-ortho-fussheberorthese-afo (accessed on 10 January 2025).
63. Pillai, M. Revolutionizing Healing: The Era of 3D-Printed Orthotics. 2023. Available online: https://amchronicle.com/case-studies/revolutionizing-healing-the-era-of-3d-printed-orthotics/ (accessed on 10 January 2025).
64. HfG Schwäbisch Gmünd. exoJack—Eine Smarte Orthese für Sprunggelenksbrüche. 2021. Available online: https://ausstellung.hfg-gmuend.de/w-2021/projekte/exojack-eine-smarte-orthese-fuer-sprunggelenksbrueche (accessed on 10 January 2025).
65. Fraunhofer-Institut für Werkzeugmaschinen und Umformtechnik IWU. SensO-FeeT—Sensorisch Aktive Sprunggelenk-Orthese Mit Taktiler Feedback-Technologie. Available online: https://www.iwu.fraunhofer.de/de/projekte/senso-feet-sensorisch-aktive-sprunggelenk-orthese-mit-taktiler-feedback-technologie.html (accessed on 10 January 2025).
66. Taylor, T. Sensoria's Smart Socks Measure Your Running Patterns, Improving Gait. 2015. Available online: https://www.si.com/edge/2015/03/06/sensoria-smart-socks-measure-running-patterns (accessed on 10 January 2025).
67. Orthotics, X.P. Ankle & Foot. Available online: https://xtremity.ae/ankle-and-foot/ (accessed on 10 January 2025).
68. Free3D. 3D-Gedruckter Orthopädischer Gipsverband am Bein 3D-Modell. 2024. Available online: https://free3d.com/de/3d-model/3d-printed-orthopedic-cast-on-leg-7449.html (accessed on 10 January 2025).
69. Braun, V.; Clarke, V. Using thematic analysis in psychology. *Qual. Res. Psychol.* **2006**, *3*, 77–101. [CrossRef]
70. Kassambara, A. rstatix: Pipe-Friendly Framework for Basic Statistical Tests. Available online: https://rpkgs.datanovia.com/rstatix/ (accessed on 10 January 2025).
71. Schwind, V.; Henze, N. Anticipated User Stereotypes Systematically Affect the Social Acceptability of Mobile Devices. In Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences (NordiCHI '20), Tallinn, Estonia, 25–29 October 2020; Shaping Society, Association for Computing Machinery: New York, NY, USA, 2020. [CrossRef]

72. Minaoglou, P.; Tzotzis, A.; Efkolidis, N.; Kyratsis, P. Integrating Artificial Intelligence into the Shoe Design Process. *Eng. Proc.* **2024**, *72*, 7. [CrossRef]
73. Zhou, K.Q.; Nabus, H. The ethical implications of DALL-E: Opportunities and challenges. *Mesop. J. Comput. Sci.* **2023**, *2023*, 16–21. [CrossRef]

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